



Chemistry 9701

Paper 2 · AS Level Structured Questions

Part-level Topical Workbook

with Mark Scheme

- Every question part filed under its own syllabus topic (22 AS units)
- Each item carries the official context it needs: stem, tables, figures, earlier parts
- Answer lines removed; tables to complete kept
- Newest papers first; mark scheme in an Answers Section after each unit
- Index of every part with its unit, marks and page; Periodic Table appendix

Coverage

35 papers: MAR 22/P22 to M/J 26/P24 (newest first in each unit)

809 items, 1962 marks across 22 units

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Unit 1

ATOMIC STRUCTURE

62 items · 182 marks

Syllabus sections in this unit

- 1.1 Particles in the atom and atomic radius
- 1.2 Isotopes
- 1.3 Electrons, energy levels and atomic orbitals
- 1.4 Ionisation energy

Unit 1: Atomic structure**1. M/J 26/P21/Q1(b)**

(b) Table 1.1 shows information about sulfur and chlorine.

Table 1.1

element	electronic configuration	first ionisation energy /kJ mol ⁻¹
sulfur	[Ne] 3s ² 3p ⁴	1000
chlorine	[Ne] 3s ² 3p ⁵	1260

(i) Name the orbital that contains the lowest energy electron in an atom of sulfur. Draw the shape of this orbital.

name

shape

[1]

(ii) Construct an equation to represent the first ionisation energy of chlorine.

[1]

(iii) Explain the difference in the values of first ionisation energy of sulfur and chlorine in Table 1.1.

[3]

2. M/J 26/P22/Q1(a)

Context

1 Oxygen reacts with many elements.

(a) (i) State the total number of electrons that occupy the 2p sub-shell in an oxygen atom.

[1]

(ii) Name the orbital that contains the highest energy electron in an oxygen atom. Draw the shape of this orbital.

name

shape

[1]

3. M/J 26/P22/Q2(a)

- 2 (a)** Protons, neutrons and electrons are released with the same velocity from a source.

Figure 2.1 shows the movement of two of these types of particle when they travel between oppositely charged plates.

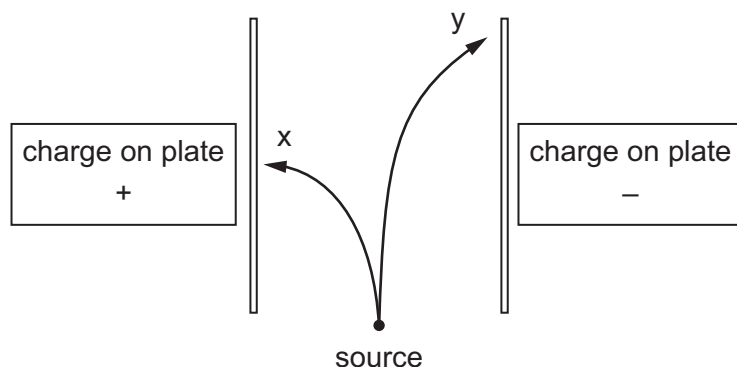


Figure 2.1

- (i) Name each type of particle that produces the deflection shown by curves x and y.

curve x =

curve y =

[1]

- (ii) Suggest why the particles producing curve y are deflected less than the particles producing curve x.

[1]

4. M/J 26/P22/Q2(b)

- (b) A student considers the information in Table 2.1 and proposes hypothesis 1.

hypothesis 1 'Atoms with a higher nuclear charge are always larger.'

Table 2.1

atom	atomic number	mass number	atomic radius/nm
A	6	12	0.077
B	17	37	0.099
C	14	28	0.117

- (i) Identify a pair of atoms from Table 2.1 that supports hypothesis 1.

[1]

- (ii) Identify a pair of atoms from Table 2.1 that shows hypothesis 1 is false.

[1]

- (iii) Explain why atom **A** is smaller than atom **C**.

[2]

5. M/J 26/P23/Q2(a)

Context

- 2** Group 18 elements exist as colourless gases at room temperature and pressure.

Table 2.1

atom	relative volume of atom
neon	172
argon	287
krypton	355
xenon	446

- (a)** Use Table 2.1 to describe and explain the variation in atomic radius down Group 18.

[2]

6. M/J 26/P23/Q5(b)(i)

Context

- 5** Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

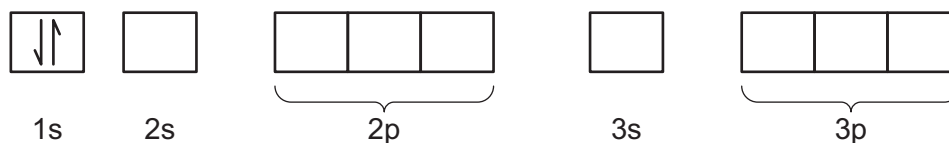
Context: part (b) (introduction)

- (b)** The mechanism for this reaction involves three stages.

stage 1 initiation**stage 2** propagation**stage 3** termination

During the initiation reaction, a chlorine free radical is produced.

- (i)** Complete Figure 5.1 to describe the arrangement of electrons in a chlorine free radical.

**Figure 5.1**

[1]

7. M/J 26/P24/Q1(a)

Context

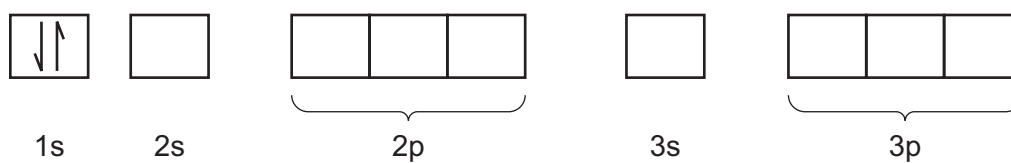
- 1** Aluminium is a Group 13 element that forms a number of compounds.

Table 1.1

atom or ion	number of protons	number of neutrons
^{27}Al		
$^{28}\text{Al}^{3+}$		

- (a) (i) Complete Table 1.1 to show the number of particles in each atom or ion. [2]

- (ii) Complete Figure 1.1 to show the electronic configuration of an aluminium atom.

**Figure 1.1**

[1]

8. MAR 26/P22/Q1(a)

Context

1 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

(a) Table 1.1 shows some information about the elements Mg to Ba.

Table 1.1

element	proton number	shorthand electronic configuration	sum of first and second ionisation energies / kJ mol^{-1}	third ionisation energy / kJ mol^{-1}
Mg	12	[Ne] $3s^2$	2186	7740
Ca	20	[Ar] $4s^2$	1740	4940
Sr	38	[Kr] $5s^2$	1608	4120
Ba	56	[Xe] $6s^2$	1468	3390

(i) Give the full electronic configuration of a calcium atom.

[1]

(ii) Explain the trend shown in the sum of the first and second ionisation energies, as proton number increases.

[2]

(iii) Write an equation to represent the third ionisation energy of barium.

[1]

(iv) Explain why the third ionisation energy of magnesium is much greater than the sum of its first and second ionisation energies.

[2]

9. M/J 25/P21/Q2(a)

Context

2 A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

(a) Define isotopes.

[1]

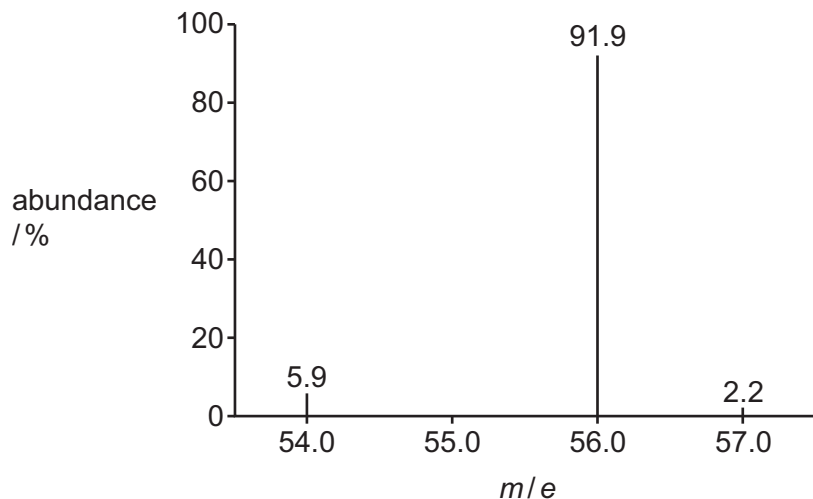
10. M/J 25/P21/Q2(b)(ii)

Context

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

Context: part (b) (introduction)

- (b)** Fig. 2.1 shows the mass spectrum of the sample of iron.

**Fig. 2.1**

- (ii)** Complete Table 2.1 to show the number of protons and nucleons in one atom of ^{56}Fe .

Table 2.1

particle	number of particles in one atom of ^{56}Fe
protons	
nucleons	

[2]

11. M/J 25/P21/Q2(c)

Context

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

- (c)** Deduce the number of pairs of electrons in the shell with principal quantum number $n = 3$ in an Fe atom.

number of pairs of electrons

[1]

12. M/J 25/P21/Q2(d)

Context

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

- (d)** Write an equation to represent the first ionisation energy of iron.

[1]

13. M/J 25/P21/Q2(e)

Context

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

(e) Suggest how the value for the first ionisation energy of ^{54}Fe compares to the first ionisation energy of ^{56}Fe . Explain your answer in terms of the factors that affect ionisation energy.

[4]

14. M/J 25/P22/Q2(b)

Context

- 2** A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(b) Deduce the number of pairs of electrons in the 3d sub-shell in an iron(II) ion.

[1]

15. M/J 25/P22/Q2(c)

Context

- 2** A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(c) Sketch the shape of the lowest energy orbital in the shell with principal quantum number $n = 2$.

[1]

16. M/J 25/P22/Q2(d)

Context

- 2** A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(d) Complete Table 2.2 to show information about particles in one ion of $^{56}\text{Fe}^{3+}$.

Table 2.2

particle	number of particles present in one ion of $^{56}\text{Fe}^{3+}$
electrons	
	30

[2]

17. M/J 25/P22/Q2(e)

Context

- 2** A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(e) The atomic radius of iron is $1.26 \times 10^{-10}\text{ m}$.

Suggest the change to the radius, if any, after an iron atom reacts to produce an Fe^{3+} ion. Explain your answer.

[2]

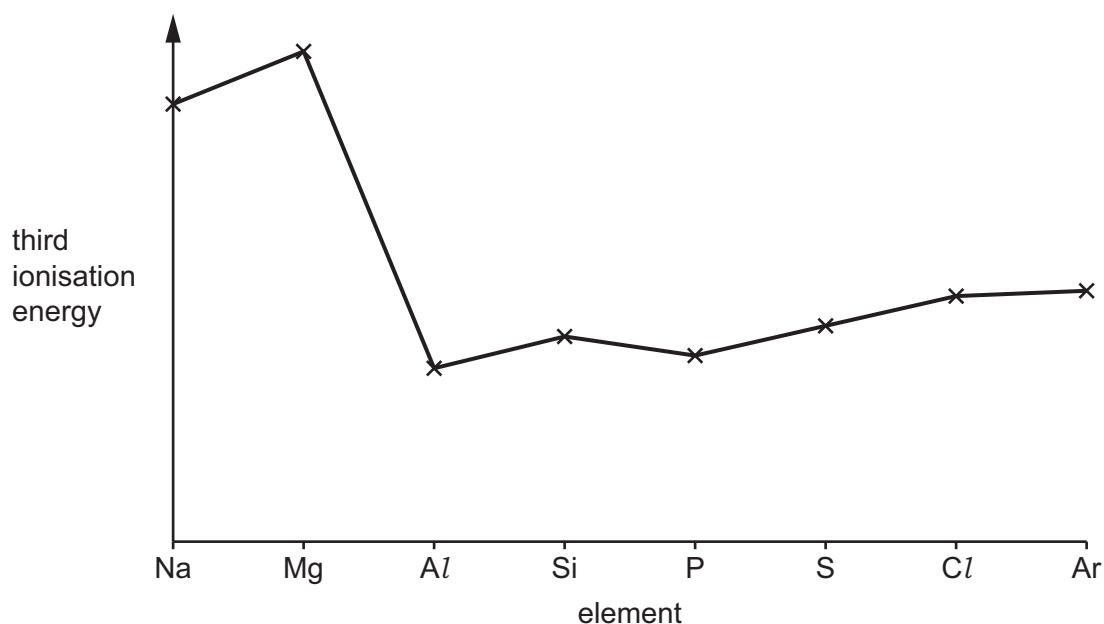


Fig. 1.2

- (i) Write an equation, including state symbols, to represent the **third** ionisation energy of argon. [2]
- (ii) The differences in the values for third ionisation energy shown in Fig. 1.2 are due to differences in the strength of attraction between the nucleus and the outer electron of each ion. State **two** factors that affect the strength of attraction between the nucleus and the outer electron. [2]
- 1
 - 2
- (iii) Use Fig. 1.2 to suggest the most significant factor that determines the size of attraction between the nucleus and the outer electron. Explain your answer. [2]
- (iv) In the third ionisation energy of argon, the ion produced has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^3$. [2]

Complete Fig. 1.3 to show the arrangement of electrons in the orbitals of this ion.

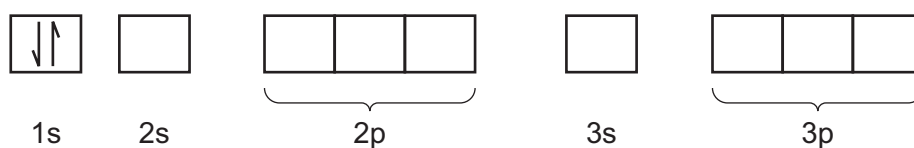


Fig. 1.3

[1]

21. MAR 25/P22/Q2(b)

(b) The Group 2 elements form stable 2+ cations.

(i) State and explain the variation in ionic radius of the Group 2 elements down the group.

[2]

(ii) Table 2.1 shows successive ionisation energy values for beryllium, Be.

Table 2.1

	1st	2nd	3rd	4th
ionisation energy / kJ mol^{-1}	900	1760	14 800	21 000

Use Table 2.1 to state and explain:

- the general trend in these values
- the significance of the large difference between the 2nd and 3rd ionisation energies.

[2]

22. MAR 25/P22/Q2(c)

(c) All the Group 2 elements except beryllium have more than one stable isotope.

(i) Beryllium exists as the single isotope ${}^9_4\text{Be}$.

Describe the distribution of mass within an atom of ${}^9_4\text{Be}$.

[2]

(ii) Complete Table 2.2 to show the numbers of protons and neutrons in the isotopes of magnesium.

Table 2.2

isotope	number of protons	number of neutrons
magnesium-24		
magnesium-25		
magnesium-26		

[2]

(iii) Fig. 2.1 shows the behaviour of a beam of protons in an electric field.

Complete Fig. 2.1 to show the behaviour of separate beams of neutrons and electrons in the same electric field.

Label your diagram clearly. Assume that the beams of each particle are moving at the same velocity.

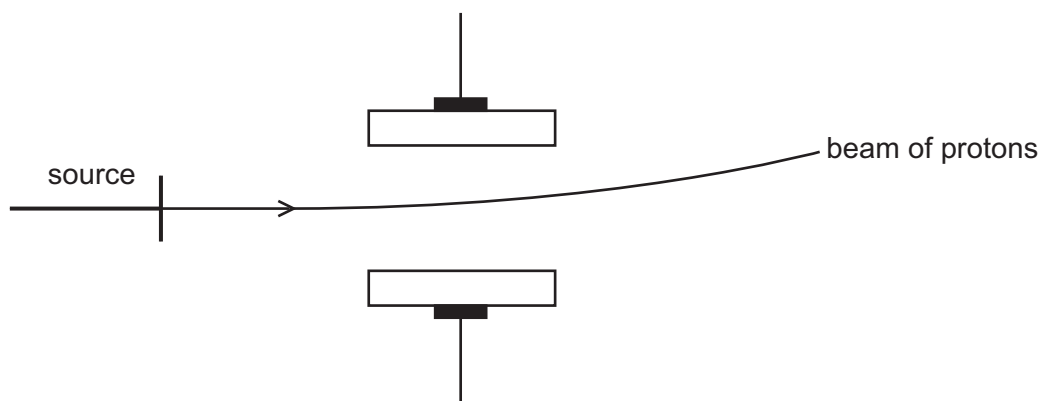


Fig. 2.1

[3]

23. O/N 24/P21/Q1(a)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is $[\text{Ar}]3d^74s^2$.

(i) Identify what is meant by $[\text{Ar}]$ by giving its full electronic configuration.

[1]

(ii) The lowest-energy electrons in cobalt are in the 1s orbital.

Draw the shape of a 1s orbital.

[1]

(iii) Deduce the number of unpaired electrons in a cobalt atom.

[1]

24. O/N 24/P21/Q1(b)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(b) Table 1.1 gives some details of the stable naturally occurring isotopes of rhodium and iridium.

Table 1.1

isotope	number of protons	number of neutrons	total number of electron shells
$^{103}_{45}\text{Rh}$		58	
$^{191}_{77}\text{Ir}$			6
$^{193}_{77}\text{Ir}$			6

Complete Table 1.1.

[3]

25. O/N 24/P21/Q3(c)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(c) Fig. 3.1 shows a sketch of some of the ionisation energies of silicon, Si.

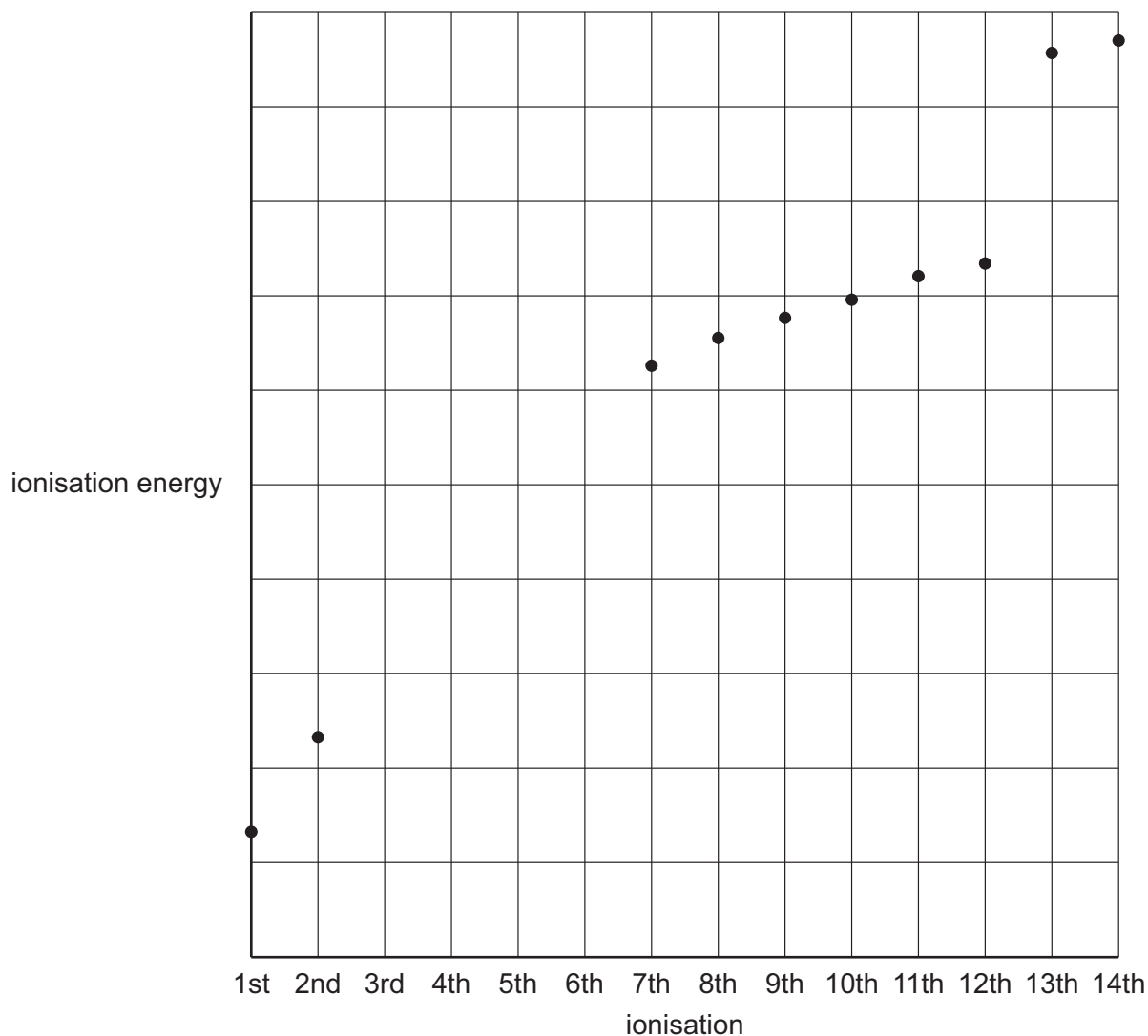


Fig. 3.1

(i) Complete the graph in Fig. 3.1 to show the third to sixth ionisation energies of Si.

[2]

(ii) Construct an equation to represent the second ionisation energy of Si.

[1]

26. O/N 24/P22/Q1(a)

Context

1 Vanadium, niobium and tantalum are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of vanadium in the ground state is $[\text{Ar}]3d^34s^2$.

(i) State what is meant by the term ground state.

[1]

(ii) Show the electronic configuration of vanadium using electrons in boxes notation.



[1]

(iii) Deduce the total number of electrons in the p sub-shells of a vanadium atom.

[1]

27. O/N 24/P22/Q1(b)(i)

Context

1 Vanadium, niobium and tantalum are metals in the same group of the Periodic Table.

Context: part (b) (introduction)

(b) Pelopium was the suggested name for a new element discovered in a mineral.

Pelopium was later found to be a mixture of niobium, Nb, and tantalum, Ta.

Only one naturally occurring isotope exists for each of Nb and Ta.

(i) Complete Table 1.1.

Table 1.1

isotope	relative isotopic mass	number of protons	number of neutrons
${}^{93}_{41}\text{Nb}$	92.91		
${}^{181}_{73}\text{Ta}$	180.95		

[2]

28. O/N 24/P22/Q3(c)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

(c) Fig. 3.1 shows a sketch of some of the ionisation energies of phosphorus, P.

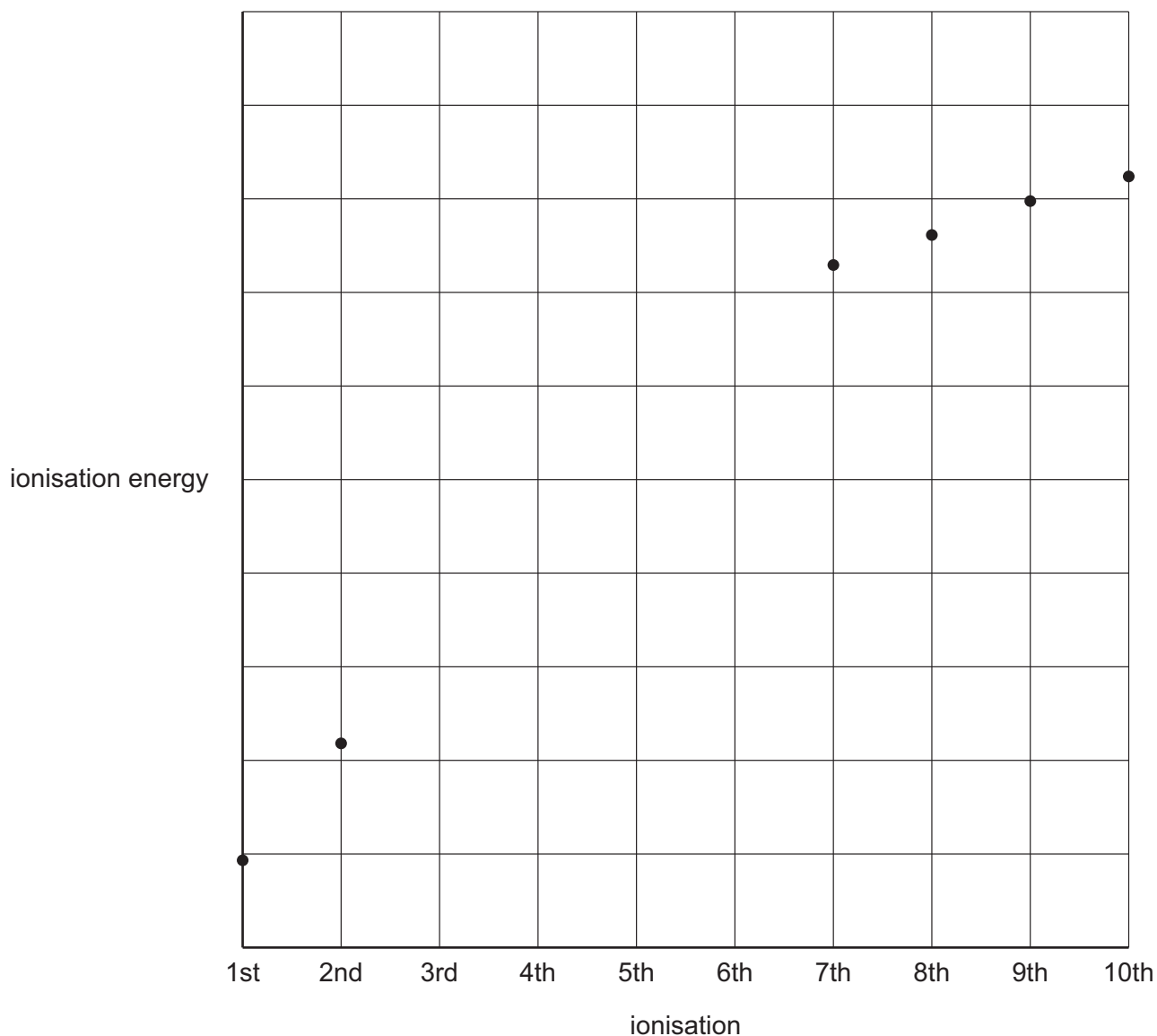


Fig. 3.1

(i) Construct an equation to represent the third ionisation energy of P.

[1]

(ii) Complete the graph in Fig. 3.1 to show the third to sixth ionisation energies of P.

[2]

29. O/N 24/P23/Q1(a)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(a) The shorthand electronic configuration of cobalt is $[\text{Ar}]3d^74s^2$.

(i) Identify what is meant by $[\text{Ar}]$ by giving its full electronic configuration.

[1]

(ii) The lowest-energy electrons in cobalt are in the 1s orbital.

Draw the shape of a 1s orbital.

[1]

(iii) Deduce the number of unpaired electrons in a cobalt atom.

[1]

30. O/N 24/P23/Q1(b)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(b) Table 1.1 gives some details of the stable naturally occurring isotopes of rhodium and iridium.

Table 1.1

isotope	number of protons	number of neutrons	total number of electron shells
$^{103}_{45}\text{Rh}$		58	
$^{191}_{77}\text{Ir}$			6
$^{193}_{77}\text{Ir}$			6

Complete Table 1.1.

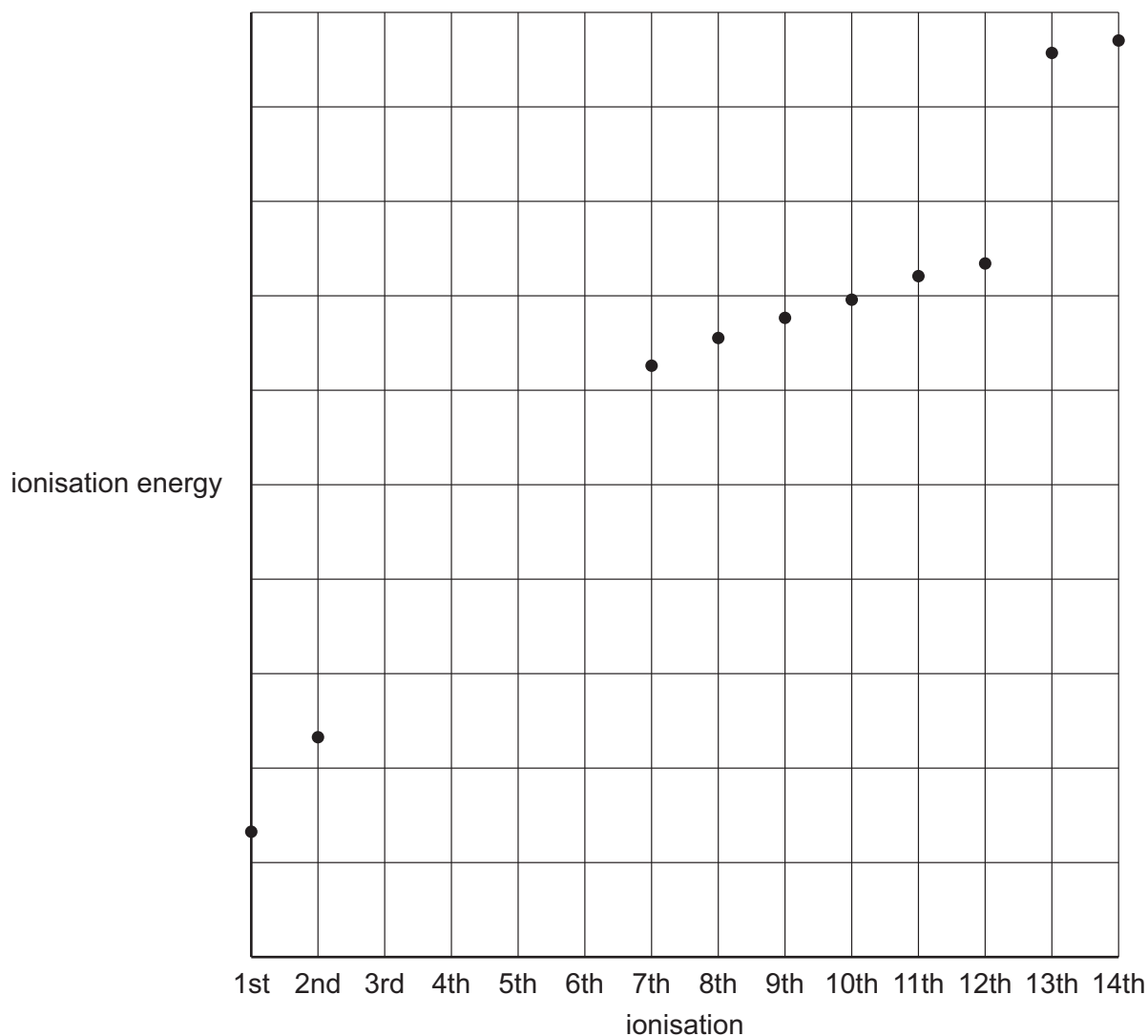
[3]

31. O/N 24/P23/Q3(c)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(c) Fig. 3.1 shows a sketch of some of the ionisation energies of silicon, Si.

**Fig. 3.1**

(i) Complete the graph in Fig. 3.1 to show the third to sixth ionisation energies of Si.

[2]

(ii) Construct an equation to represent the second ionisation energy of Si.

[1]

32. M/J 24/P22/Q1(a)

1 (a) Complete Table 1.1 using relevant information from the Periodic Table.

Table 1.1

	nucleon number	proton number	number of electrons
Mg^{2+}	24		
Al^{3+}	27		

[2]

33. M/J 24/P22/Q1(b)

(b) State and explain the difference in the ionic radius of Al^{3+} compared to Mg^{2+} .

[3]

34. M/J 24/P22/Q3(c)

(c) Determine the full electronic configuration of Fe^{3+} .

[1]

35. O/N 23/P21/Q1(a)

Context

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy / kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) (i) Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and Cl. [2]

(ii) Construct an equation to represent the first ionisation energy of P.

[1]

(iii) Three possible values for the first ionisation energy of S are given.

 1000 kJ mol^{-1} **1160 kJ mol^{-1}** **1320 kJ mol^{-1}**

Circle the correct value.

Explain your choice by comparing your chosen value to those of P and Cl.

[4]

36. O/N 23/P21/Q1(b)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (b)** P³⁻, S²⁻ and Cl⁻ have the same number of electrons.

- (i)** Give the full electronic configuration of P³⁻.

[1]

- (ii)** State the trend in ionic radius shown by P³⁻, S²⁻ and Cl⁻.

Explain your answer.

[2]

37. O/N 23/P22/Q1(b)

also Unit 9 (1 mark), Unit 3 (1 mark); this unit 7 marks

Context

- 1** The elements silicon, phosphorus and sulfur are in Period 3 of the Periodic Table.

- (b)** Table 1.1 shows some properties of the elements Si to S.

The first ionisation energy of P is **not** shown.

Table 1.1

property	Si	P	S
total number of electrons in s subshells			
total number of electrons in p subshells			
first ionisation energy/kJ mol ⁻¹	786		1000
formula of most common chloride	SiCl ₄	PCl ₅	SCl ₂

- (i)** Complete Table 1.1 to show the total number of s and p electrons in an atom of Si, P and S.

[2]

- (ii)** Construct an equation to represent the first ionisation energy of Si.

[1]

- (iii)** Three possible values for the first ionisation energy of P are given.

619 kJ mol⁻¹

893 kJ mol⁻¹

1060 kJ mol⁻¹

Circle the correct value.

Explain your choice, including a comparison of your chosen value to those of Si and S.

[4]

- (iv)** SiCl₄ and PCl₅ each react with water, forming misty fumes.

Identify the chemical responsible for the misty fumes.

[1]

- (v)** Predict the shape of the SCl₂ molecule.

[1]

38. O/N 23/P23/Q1(a)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (a) (i) Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and Cl. [2]
- (ii) Construct an equation to represent the first ionisation energy of P. [1]
- (iii) Three possible values for the first ionisation energy of S are given.

1000 kJ mol⁻¹

1160 kJ mol⁻¹

1320 kJ mol⁻¹

Circle the correct value.

Explain your choice by comparing your chosen value to those of P and Cl.

[4]

39. O/N 23/P23/Q1(b)

Context

- 1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (b) P³⁻, S²⁻ and Cl⁻ have the same number of electrons.

- (i) Give the full electronic configuration of P³⁻.

[1]

- (ii) State the trend in ionic radius shown by P³⁻, S²⁻ and Cl⁻.

Explain your answer.

[2]

40. M/J 23/P21/Q1(a)

Context

- 1 Tellurium is an element in Group 16. The most common isotope of tellurium is ¹³⁰Te. Its electronic configuration is [Kr] 4d¹⁰ 5s² 5p⁴.

- (a) Complete Table 1.1.

Table 1.1

	nucleon number	number of neutrons	number of electrons
¹³⁰ Te			

[3]

41. M/J 23/P21/Q1(b)

Context

- 1 Tellurium is an element in Group 16. The most common isotope of tellurium is ¹³⁰Te. Its electronic configuration is [Kr] 4d¹⁰ 5s² 5p⁴.

- (b) Identify the sub-shell in an atom of Te that contains electrons with the lowest energy.

[1]

42. M/J 23/P21/Q1(c)

Context

- 1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.

(c) Construct an equation to represent the first ionisation energy of Te.

[1]

43. M/J 23/P21/Q1(d)

Context

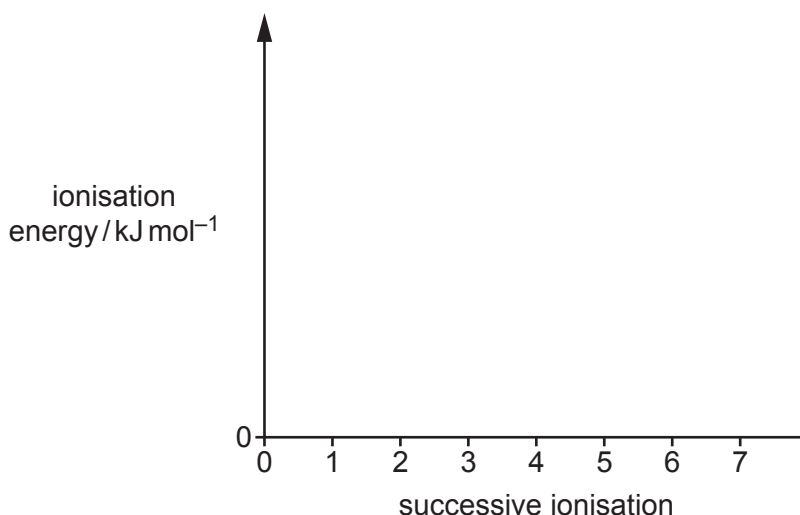
- 1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.

(d) (i) The radius of Te ions decreases after each successive ionisation.

State **two** factors that are responsible for the increase in the first six ionisation energies of Te.

[2]

(ii) Sketch a graph in Fig. 1.1 to show the trend in the first **seven** ionisation energies of Te.

**Fig. 1.1**

[2]

44. M/J 23/P23/Q1(d)(iii)

Context

- 1 Copper is used in electrical equipment. It has a melting point of 1085°C .

Context: part (d) (introduction)

(d) When $\text{KI}(\text{aq})$ is added to $\text{CuSO}_4(\text{aq})$ the blue-coloured solution turns brown and a white precipitate of $\text{CuI}(\text{s})$ is seen.

The reaction between copper ions and iodide forms only two products.

(iii) State the full electronic configuration of Cu^{2+} .

[1]

45. MAR 23/P22/Q1(c)

Context

- 1** The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

- (c) (i)** Write an equation that represents the first ionisation energy of H.

[1]

- (ii)** Explain why there is no information given in Table 1.1 for the second ionisation energy of H.

[1]

- (iii)** Give the full electronic configuration of S²⁺(g).

[1]

46. O/N 22/P21/Q1(a)(i)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

Context: part (a) (introduction)

- (a)** Gallium has two stable isotopes, ⁶⁹Ga and ⁷¹Ga.

- (i)** Complete Table 1.1 to show the numbers of protons, neutrons and electrons in the two stable isotopes of gallium.

Table 1.1

isotope	number of protons	number of neutrons	number of electrons
⁶⁹ Ga			
⁷¹ Ga			

[2]

47. O/N 22/P21/Q1(b)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

(b) Potassium also has two stable isotopes. Both isotopes have the same chemical properties.

- (i)** Explain why both isotopes of potassium have the same chemical properties.

[1]

- (ii)** State the full electronic configuration of an atom of potassium.

[1]

- (iii)** The first, second and third ionisation energies of potassium are 418, 3070 and 4600 kJ mol⁻¹, respectively.

Use this information to explain why potassium is in Group 1.

[2]

48. O/N 22/P22/Q1(b)

Context

- 1** Species such as NH₄⁺, CO₃²⁻ and PO₄³⁻ are examples of molecular ions.

- (b)** Complete Table 1.1 to show the total numbers of protons and electrons in the molecular ions NH₄⁺, CO₃²⁻ and PO₄³⁻.

Table 1.1

molecular ion	total number of protons	total number of electrons
NH ₄ ⁺		
CO ₃ ²⁻		
PO ₄ ³⁻		

[3]

49. O/N 22/P23/Q1(a)(i)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

Context: part (a) (introduction)

(a) Gallium has two stable isotopes, ^{69}Ga and ^{71}Ga .

- (i)** Complete Table 1.1 to show the numbers of protons, neutrons and electrons in the two stable isotopes of gallium.

Table 1.1

isotope	number of protons	number of neutrons	number of electrons
^{69}Ga			
^{71}Ga			

[2]

50. O/N 22/P23/Q1(b)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

(b) Potassium also has two stable isotopes. Both isotopes have the same chemical properties.

- (i)** Explain why both isotopes of potassium have the same chemical properties.

[1]

- (ii)** State the full electronic configuration of an atom of potassium.

[1]

- (iii)** The first, second and third ionisation energies of potassium are 418, 3070 and 4600 kJ mol⁻¹, respectively.

Use this information to explain why potassium is in Group 1.

[2]

51. M/J 22/P21/Q1(a)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

- (a)** Identify the highest energy orbital which contains electrons in a calcium atom. Sketch the shape of this orbital.

identity of highest energy orbital in Ca

shape

[1]

52. M/J 22/P21/Q1(d)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

- (d) (i)** $^{25}_{12}\text{Mg}$ is an isotope of magnesium.

Determine the number of protons and neutrons in an atom of $^{25}_{12}\text{Mg}$.

number of protons

number of neutrons

[1]

- (ii)** State the full electronic configuration of an atom of $^{25}_{12}\text{Mg}$.

[1]

53. M/J 22/P21/Q1(e)(iii)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

Context: part (e) (introduction)

- (e)** A sample of magnesium contains three isotopes, ^{25}Mg , ^{26}Mg and **X**.

The percentage abundance of the three isotopes is shown in Table 1.1.

Table 1.1

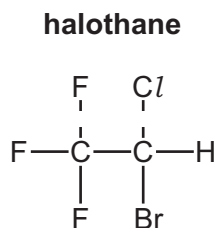
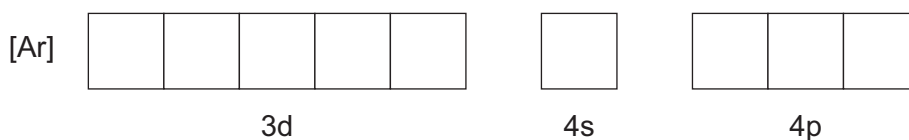
isotope of Mg	mass/a.m.u.	percentage abundance/%
X		78.99
^{25}Mg	24.99	10.00
^{26}Mg	25.98	11.01

- (iii)** State **one** similarity and **one** difference in the properties of these isotopes of magnesium. Explain your answer.

[2]

54. M/J 22/P22/Q4(b)(iii)

Context: part (b) (introduction)

(b) Halothane is an anaesthetic.**Fig. 4.1****(iii)** Complete Fig. 4.2 to show the arrangement of electrons in a bromine atom using the electrons in boxes notation.**Fig. 4.2**

[1]

55. M/J 22/P23/Q1(a)**1 (a)** Define first ionisation energy.

[2]

56. M/J 22/P23/Q1(b)**(b)** Successive ionisation energies for element **A** are shown in Table 1.1.**Table 1.1**

ionisation	1st	2nd	3rd	4th	5th	6th	7th	8th
ionisation energy/kJ mol ⁻¹	1310	3390	5320	7450	11 000	13 300	71 000	84 100

Use Table 1.1 to deduce the group of the Periodic Table that **A** belongs to. Explain your answer.

Group

[1]

57. M/J 22/P23/Q1(c)**(c)** Across Period 3 there is a general trend for first ionisation energies to increase due to the increase in attraction between the nucleus and the outer electron.

Explain why the first ionisation energy of sulfur is less than the first ionisation energy of phosphorus.

[2]

58. M/J 22/P23/Q1(d)

(d) In an Al^{2+} ion the nuclear attraction for the outer electron is stronger than in an atom of Na.

Compare the electronic structures of Al^{2+} and an atom of Na and explain why the third ionisation energy of aluminium is greater than the first ionisation energy of sodium.

[2]

59. M/J 22/P23/Q1(e)

(e) An isotope of copper has a relative isotopic mass of 65.

Complete Table 1.2 for an atom of copper-65.

Table 1.2

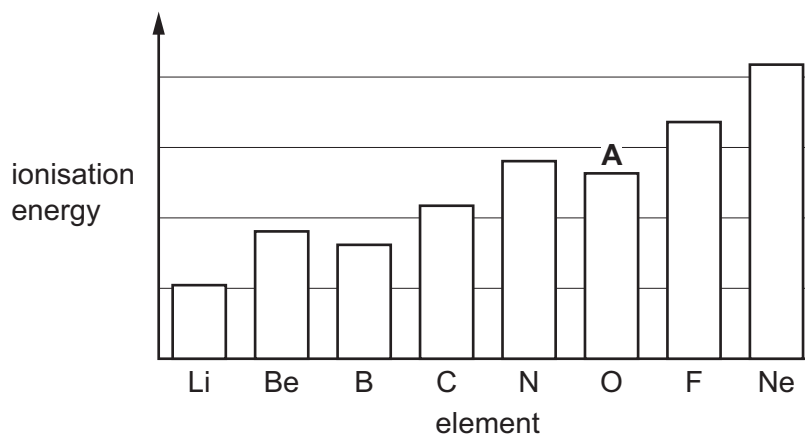
	atomic number	nucleon number	number of neutrons	electronic arrangement
copper-65				

[3]

60. MAR 22/P22/Q1(a)

Context

1 Fig. 1.1 shows how **first** ionisation energies vary across Period 2.

**Fig. 1.1**

(a) Construct an equation to represent the **first** ionisation energy of oxygen. Include state symbols.

[1]

61. MAR 22/P22/Q1(b)

Context

- 1** Fig. 1.1 shows how **first** ionisation energies vary across Period 2.

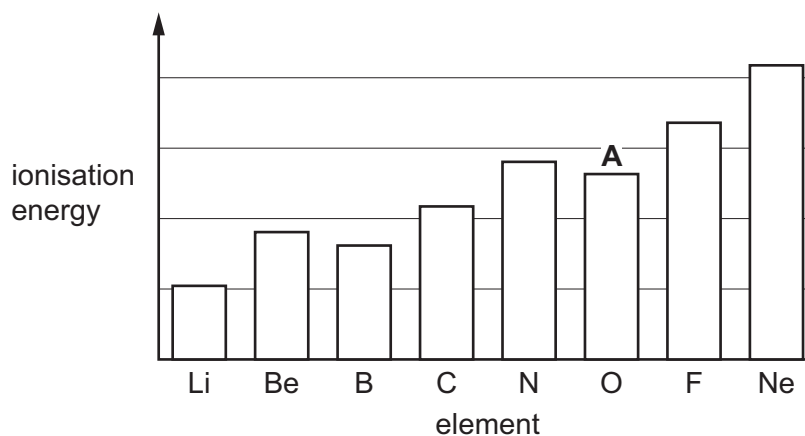


Fig. 1.1

- (b) (i)** State and explain the general trend in first ionisation energies across Period 2.

[3]

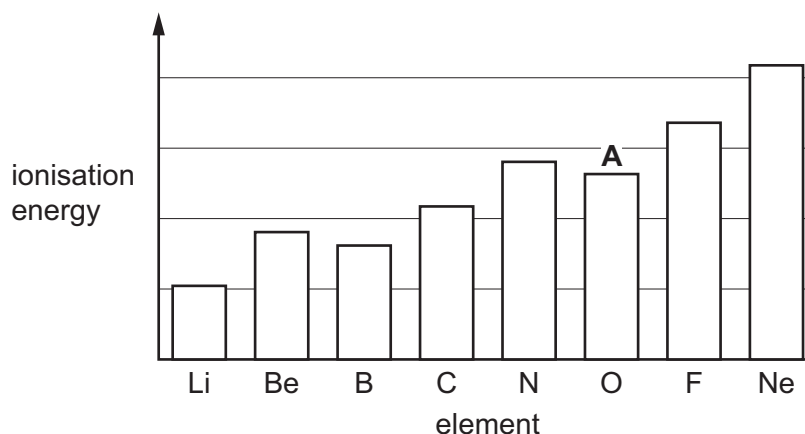
- (ii)** Explain why ionisation energy **A** in Fig. 1.1 does **not** follow the general trend in first ionisation energies across Period 2.

[2]

62. MAR 22/P22/Q1(c)

Context

- 1 Fig. 1.1 shows how **first** ionisation energies vary across Period 2.

**Fig. 1.1**

- (c) Element **E** is in Period 3 of the Periodic Table.
The first eight ionisation energy values of **E** are shown in Table 1.1.

Table 1.1

ionisation	1st	2nd	3rd	4th	5th	6th	7th	8th
ionisation energy/kJ mol ⁻¹	577	1820	2740	11 600	14 800	18 400	23 400	27 500

Deduce the full electronic configuration of **E**.
Explain your answer.

full electronic configuration of **E** =

explanation

[3]

Answers Section

1. M/J 26/P21/Q1(b)

1(b)(i)	 AND label 1s	1	I n in $1s^n$ I spherical I axes I dot-and-cross diagrams
1(b)(ii)	$\text{Cl(g)} \rightarrow \text{Cl}^+(g) + e^{(-)}$	1	A $\text{Cl(g)} - e^{(-)} \rightarrow \text{Cl}^+(g)$ I $\text{Cl(g)} + e^{(-)} \rightarrow \text{Cl}^+(g) + 2 e^{(-)}$
1(b)(iii)	M1 (small) increase in shielding in Cl is outweighed by M2 increased nuclear charge in Cl M3 Cl has greater nuclear attraction OR M1 similar shielding M2 Cl has greater nuclear charge M3 Cl has greater nuclear attraction	ora ora	3 For all 3 marks to be awarded: 'chlorine' needs only to be identified once in M1 or M2 or M3 chlorine can be identified by name / symbol or trend across a period e.g. number of protons increases across the period M1 great(er) = increase = high(er) = more = large(r) etc for shielding: similar = constant = equal = same shielding = screening I electron removed from same (sub) shell I same number of shells I same electron cloud I references to arrangement of electrons in the orbitals I atomic radius references M2 A number of protons for nuclear charge A greater atomic number for greater nuclear charge A Cl has one more proton than S I 'effective' of effective nuclear charge I nucleon number M3 (nuclear) attraction = pull = draws (into nucleus) For M2 and M3 'nuclear' needs only to be mentioned once e.g. greater nuclear attraction may be implied by reference to 'increased attraction due to increased nuclear charge' owtte If M2 and M3 not awarded: Max 1 for greater nuclear attraction linked to greater nuclear charge if species not identified or incorrectly identified

2. M/J 26/P22/Q1(a)

1(a)(i)	4 / four	1	BOD ($1s^2 2s^2$) $2p^4$
1(a)(ii)	2p AND 	1	A $2p^4$ for 2p I p alone I shading but it must have a node I axes I orientation I ignore subscripts $p_x p_y p_z$

3. M/J 26/P22/Q2(a)

2(a)(i)	curve x = electron AND curve y = proton	1	
---------	--	----------	--

2(a)(ii)	particles in y are deflected less because they have larger mass	1	ecf name given for particles in curve y have the heavier mass. Assume 'they' = y = particles of y unless explicitly refer to 'x' in answer A they are heavier A higher mass / charge ratio A greater weight A they have greater momentum A (particles in) x are lighter I 'electrons have negligible mass' (comparison required) I lower charge density I density I protons are larger I kinetic energy R protons have greater charge density
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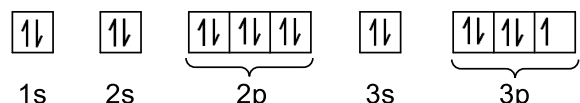
4. M/J 26/P22/Q2(b)

2(b)(i)	(atom) A and (atom) B OR (atom) A and (atom) C	1	A 6 and 17 OR 6 and 14 A carbon and chlorine OR carbon and silicon
2(b)(ii)	(atom) B and (atom) C	1	A 17 and 14 A chlorine and silicon
2(b)(iii)	M1 atom A has (one) fewer (electron) shell(s) (so decreased shielding effect) M2 (so) more attraction of nucleus for outer/valence electrons / outer or valent shell OR In atom C the effect of more protons / increased nuclear charge is less significant than / outweighed by the (increased shielding) effect of the additional shell ora = ✓✓	2	Subject of question is A, so 'it' = A A reverse argument for atom C I subshells / orbitals I incorrect identities of atoms A, B and C M1 A more shells down group shell = energy level = quantum level If number of shells is given, it must be correct (atom A=2 and atom C=3) M1 A atom A has outer electrons in a lower energy shell M1 I distance from the nucleus M1 I reference to one fewer outer/valence shell M2 A more attraction of protons to the outer/valence electrons M2 BOD pull for attraction

5. M/J 26/P23/Q2(a)

2(a)	(down the group) M1 (atomic radius) increases OR (atom) volume increases OR Ne < Ar < Kr < Xe M2 (down the group) the number of shells (of electrons) increases	2	shells = energy levels I number of sub-shells increases I nuclear attraction of outer electron decreases I increased shielding effect outweighs the increase in nuclear charge I outer electron further way
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6. M/J 26/P23/Q5(b)(i)

5(b)(i)	 1s 2s 2p 3s 3p	1	A any arrangement of 2 pairs and one single electron in 3p A double headed arrows.
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7. M/J 26/P24/Q1(a)

1(a)(i)	atom or ion	number of protons	number of neutrons	2	●✓✓✓
	^{27}Al	13	14		
	$^{28}\text{Al}^{3+}$	13	15		

1(a)(ii)		1	A single or double headed arrows A arrows in any direction R lines instead of arrows
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8. MAR 26/P22/Q1(a)

1(a)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$	1
1(a)(ii)	(ionisation energy decreases because of) - more shells / larger (atomic/ionic) radius - increased shielding - less attraction (for the nucleus/protons) / lower energy required (to remove electrons)	2
1(a)(iii)	$\text{Ba}^{2+}(\text{g}) \rightarrow \text{Ba}^{3+}(\text{g}) + \text{e}^{-}$	1
1(a)(iv)	M1 electron removed from closer / inner / lower-energy (p sub)shell/orbital M2 (much) less shielding OR greater nuclear attraction	2

9. M/J 25/P21/Q2(a)

2(a)	atoms (of an element) with same number of protons and different number of neutrons	1
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10. M/J 25/P21/Q2(b)(ii)

2(b)(ii)	particle	number of particles in one atom of ^{56}Fe	2
	protons	26	
	nucleons	56	

11. M/J 25/P21/Q2(c)

2(c)	5	1
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12. M/J 25/P21/Q2(d)

2(d)	$\text{Fe}(\text{g}) \rightarrow \text{Fe}^{+}(\text{g}) + \text{e}^{-}$	1
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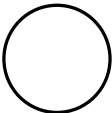
13. M/J 25/P21/Q2(e)

2(e)	M1 same electronic configuration OR same shielding M2 same number of protons OR same nuclear charge M3 so same nuclear attraction to outer shell electron M4 no change / will not be affected	4
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14. M/J 25/P22/Q2(b)

2(b)	1 (pair)	1
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15. M/J 25/P22/Q2(c)

2(c)		1
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16. M/J 25/P22/Q2(d)

2(d)	species present in an ion of 56-Fe^{3+}	number	2
	electrons	M1 23	
	M2 neutrons	30	

17. M/J 25/P22/Q2(e)

2(e)	M1 smaller / decreases M2 loss of outer / valence shell (of electrons) OR loss of electrons so increase in (force of) attraction from nucleus OR loss of electrons so increase nuclear attraction	2
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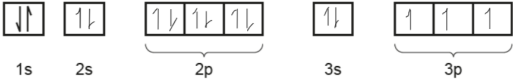
18. M/J 25/P23/Q1(a)

1(a)(i)	$1s^2 2s^2 2p^5$	1
1(a)(ii)	2 (pairs)	1
1(a)(iii)		1

19. M/J 25/P23/Q1(b)

1(b)(i)	$S(g) \rightarrow S^+(g) + e^{(-)}$ OR $S(g) - e^{(-)} \rightarrow S^+(g)$	1
1(b)(ii)	M1 pair of electrons in a 3p orbital (in S) repel each other M2 outweighs the effect of increase in nuclear charge	2
1(b)(iii)	M1 $Na^+ < Ne < F^-$ M2 (all have) same number of electrons / isoelectronic M3 larger species has smaller nuclear charge / fewer protons M4 (so) smaller nuclear attraction (to outer electrons)	4

20. M/J 25/P24/Q1(c)

1(c)(i)	$Ar^{2+}(g) \rightarrow Ar^{3+}(g) + e^{(-)}$ M1 a correct equation which represents an ionisation process e.g. $X^{n+} \rightarrow X^{(n+1)+} + e^{(-)}$ M2 correct equation to show third ionisation energy of Ar including (g) state symbols	2
1(c)(ii)	M1 nuclear charge M2 distance of outer electron(s) from nucleus OR ionic radius OR shielding (by inner shells and / or sub-shells)	2
1(c)(iii)	M1 (most significant factor) = distance of outer electron(s) from nucleus OR ionic radius OR shielding OR shells M2 (3rd IE) value for (Na and) Mg is (much) greater than Al (to Ar) OR (Na and) Mg has electron(s) removed from a shell that is closer to the nucleus / less shielding OR (Na and) Mg has electron(s) removed from a shell that is lower in energy	2
1(c)(iv)	 1s 2s 2p 3s 3p	1

21. MAR 25/P22/Q2(b)

2(b)(i)	ionic radius increases / ions larger (down the group) extra shell (of electrons)	2
2(b)(ii)	ionisation energies increase because less shielding / successive ionic radii are smaller / greater attraction (of electron) to nucleus (large difference between 2nd and 3rd IE) because 3rd e ⁻ coming from an inner shell / lower energy level	2

22. MAR 25/P22/Q2(c)

2(c)(i)	most of the mass is in the nucleus because protons and neutrons / nucleons are found in the nucleus	2												
2(c)(ii)	<table border="1"> <thead> <tr> <th>isotope</th><th>number of protons</th><th>number of neutrons</th></tr> </thead> <tbody> <tr> <td>magnesium-24</td><td>12</td><td>12</td></tr> <tr> <td>magnesium-25</td><td>12</td><td>13</td></tr> <tr> <td>magnesium-26</td><td>12</td><td>14</td></tr> </tbody> </table>	isotope	number of protons	number of neutrons	magnesium-24	12	12	magnesium-25	12	13	magnesium-26	12	14	2
isotope	number of protons	number of neutrons												
magnesium-24	12	12												
magnesium-25	12	13												
magnesium-26	12	14												
2(c)(iii)	neutron beam horizontal straight line from same source electron beam downwards curving line from same source line deflected more than proton beam	3												

23. O/N 24/P21/Q1(a)

1(a)(i)	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶	1
1(a)(ii)		1
1(a)(iii)	3 (three)	1

24. O/N 24/P21/Q1(b)

1(b)					3
	isotope	number of protons	number of neutrons	total number of electron shells	
	$^{103}_{45}\text{Rh}$	● 45	58	● 5	
	$^{191}_{77}\text{Ir}$	● 77	● 114	6	
	$^{193}_{77}\text{Ir}$	● 77	● 116	6	

25. O/N 24/P21/Q3(c)

3(c)(i)	M1 general increase	1
	M2 increase in IE between IE4 and IE5 is noticeably the largest increase between IE's shown	1
3(c)(ii)	Si ⁺ (g) → Si ²⁺ (g) + e ⁻	1

26. O/N 24/P22/Q1(a)

1(a)(i)	lowest energy state owtte examples • (with) lowest energy • (with) no external energy • (with) no added energy • (where) electrons are not promoted to a higher energy level • most energetically stable	1						
1(a)(ii)	[Ar] <table><tr><td>↑</td><td>↑</td><td>↑</td><td></td><td></td><td>↑↓</td></tr></table> <i>The 3 × d electrons may go in any of the d orbitals</i>	↑	↑	↑			↑↓	1
↑	↑	↑			↑↓			
1(a)(iii)	12	1						

27. O/N 24/P22/Q1(b)(i)

1(b)(i)					2
	isotope	RIM	# protons	# neutrons	
	$^{93}_{41}\text{Nb}$	92.91	● 41	● 52	
	$^{181}_{73}\text{Ta}$	180.95	● 73	● 108	

28. O/N 24/P22/Q3(c)

3(c)(i)	$\text{P}^{2+}(\text{g}) \rightarrow \text{P}^{3+}(\text{g}) + \text{e}^{-}$	1
3(c)(ii)	general increase	1
	increase in IE between IE5 and IE6 is noticeably the largest increase between IEs shown	1

29. O/N 24/P23/Q1(a)

1(a)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6$	1
1(a)(ii)		1
1(a)(iii)	3 (three)	1

30. O/N 24/P23/Q1(b)

1(b)					3
	isotope	number of protons	number of neutrons	total number of electron shells	
	$^{103}_{45}\text{Rh}$	● 45	58	● 5	
	$^{191}_{77}\text{Ir}$	● 77	● 114	6	
	$^{193}_{77}\text{Ir}$	● 77	● 116	6	

31. O/N 24/P23/Q3(c)

3(c)(i)	M1 general increase	1
	M2 increase in IE between IE4 and IE5 is noticeably the largest increase between IE's shown	1
3(c)(ii)	$\text{Si}^+(\text{g}) \rightarrow \text{Si}^{2+}(\text{g}) + \text{e}^-$	1

32. M/J 24/P22/Q1(a)

1(a)					2
	Mg ²⁺	24	12	10	
	Al ³⁺	27	13	10	
4 correct = 2 marks 2 or 3 correct = 1 mark 0 or 1 correct = 0 marks					

33. M/J 24/P22/Q1(b)

1(b)	M1 it / Al ³⁺ is smaller (compared to Mg ²⁺) M2 greater nuclear attraction for remaining electrons M3 same shielding effect AND greater nuclear charge	3
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34. M/J 24/P22/Q3(c)

3(c)	full electronic configuration of Fe ³⁺ 1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ⁵	1
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35. O/N 23/P21/Q1(a)

1(a)(i)	total # 3p e ⁻	3	4	5	1
	total # unpaired e ⁻	3	2	1	1
1(a)(ii)	P(g) → P ⁺ (g) + e ⁽⁻⁾				1
1(a)(iii)	1000 (kJmol ⁻¹)				1
	(S) less / least / lowest nuclear attraction (to outer electrons) (than P OR Cl) OR nuclear attraction for Cl is stronger (compared to S (and or P))				1
	<u>S less nuclear charge than Cl</u> OR <u>Cl has a greater nuclear charge than S</u>				1
	(S has a greater nuclear charge (of outer electrons) than P BUT) S has two electrons in a (3) p orbital AND resulting in spin-pair repulsion				1

36. O/N 23/P21/Q1(b)

1(b)(i)	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶				1
1(b)(ii)	- decreases (from P³⁻ to Cl⁻) - increased attraction (of outer e⁻) for nucleus owtte - increased nuclear charge - same shielding •√•√				2

37. O/N 23/P22/Q1(b)

1(b)(i)	total e ⁻ in s subshell	6	6	6	1
	total e ⁻ in p subshell	8	9	10	1
1(b)(ii)	Si(g) → Si ⁺ (g) + e ⁻				1

1(b)(iii)	1060 (is circled)	1
	P / it has greater attraction of nucleus for outer electrons compared to Si ORA	1
	P / it has greater nuclear charge than Si ORA	1
	S has two electrons in a (3)p orbital AND resulting in spin-pair repulsion	1
1(b)(i)(v)	hydrogen chloride / HCl	1
1(b)(v)	non-linear	1

38. O/N 23/P23/Q1(a)

1(a)(i)	total # 3p e ⁻	3	4	5	1
	total # unpaired e ⁻	3	2	1	1
1(a)(ii)	$\text{P(g)} \rightarrow \text{P}^+(\text{g}) + \text{e}^{(-)}$				1
1(a)(iii)	1000 (kJ mol^{-1})				1
	(S) less / least / lowest nuclear attraction (to outer electrons) (than P OR Cl) OR nuclear attraction for Cl is stronger (compared to S (and or P))				1
	<u>S less nuclear charge than Cl</u> OR <u>Cl has a greater nuclear charge than S</u>				1
	(S has a greater nuclear charge (of outer electrons) than P BUT) S has two electrons in a (3)p orbital AND resulting in spin-pair repulsion				1

39. O/N 23/P23/Q1(b)

1(b)(i)	$1s^2 2s^2 2p^6 3s^2 3p^6$	1
1(b)(ii)	- decreases (from P^{3-} to Cl^-) - increased attraction (of outer e⁻) for nucleus owtte - increased nuclear charge - same shielding •√•√	2

40. M/J 23/P21/Q1(a)

1(a)			nucleon number	number of neutrons	number of electrons	3
		Tellurium-130	130	78	52	

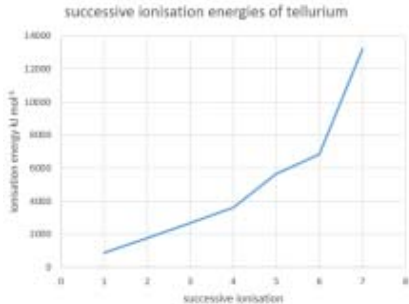
41. M/J 23/P21/Q1(b)

1(b)	1 s	1
------	-----	----------

42. M/J 23/P21/Q1(c)

1(c)	$\text{Te(g)} \rightarrow \text{Te}^+(\text{g}) + \text{e}^{(-)}$	1
------	---	----------

43. M/J 23/P21/Q1(d)

1(d)(i)	any two factors responsible for increase in IE in first 6 IE from: • (increased) nuclear attraction / attraction for nucleus • (decreased) shielding (by sub-shells) (IE 4 to 5) • spin-pair repulsion (IE 1 to 2)	2
1(d)(ii)	 M1 general increase M2 jumps from 4 to 5 and (largest) 6 to 7	2

44. M/J 23/P23/Q1(d)(iii)

1(d)(iii)	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^9 (4s^0)$	1
-----------	--	----------

45. MAR 23/P22/Q1(c)

1(c)(i)	$H(g) \rightarrow H^+(g) + e^-$	1
1(c)(ii)	H (cannot undergo second ionisation because it only) has one electron / H^+ has no electron	1
1(c)(iii)	$1s^2 2s^2 2p^6 3s^2 3p^2$	1

46. O/N 22/P21/Q1(a)(i)

1(a)(i)	columns 1 & 3 identical				1
	isotope	No of p's	No of n's	No of e's	
	⁶⁹ Ga	31	38	31	
	⁷¹ Ga	31	40	31	
	● √ √				1

47. O/N 22/P21/Q1(b)

1(b)(i)	they have the same electron arrangement / electronic configuration	1
1(b)(ii)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$	1
1(b)(iii)	M1 big increase in IE between first and second M2 second (and third) electron(s) is removed from inner shell OR second (and third) electron(s) is removed from a shell closer to the nucleus OR second (and third) electron(s) has a stronger nuclear attraction ora	2

48. O/N 22/P22/Q1(b)

1(b)	NH_4^+ 11 p ⁺ AND 10 e ⁻	1
	CO_3^{2-} 30 p ⁺ AND 32 e ⁻	1
	PO_4^{3-} 47 p ⁺ AND 50 e ⁻	1

49. O/N 22/P23/Q1(a)(i)

1(a)(i)	columns 1 & 3 identical				1
	isotope	No of p's	No of n's	No of e's	
	⁶⁹ Ga	31	38	31	
	⁷¹ Ga	31	40	31	
	● √ √				1

50. O/N 22/P23/Q1(b)

1(b)(i)	they have the same electron arrangement / electronic configuration	1
1(b)(ii)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$	1
1(b)(iii)	M1 big increase in IE between first and second M2 second (and third) electron(s) is removed from inner shell OR second (and third) electron(s) is removed from a shell closer to the nucleus OR second (and third) electron(s) has a stronger nuclear attraction ora	2

51. M/J 22/P21/Q1(a)

1(a)	Identify and draw the shape of highest energy orbital of Ca 4s AND 	1
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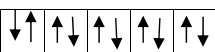
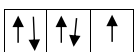
52. M/J 22/P21/Q1(d)

1(d)(i)	number of protons: 12 number of neutrons: 13	1
1(d)(ii)	$1s^2 2s^2 2p^6 3s^2$	1

53. M/J 22/P21/Q1(e)(iii)

1(e)(iii)	M1 (magnesium isotopes have) identical chemical properties AND same electron(ic) arrangement / configuration M2 different physical properties AND different number of neutrons	2
-----------	--	----------

54. M/J 22/P22/Q4(b)(iii)

4(b)(iii)	[Ar]   	1
-----------	--	----------

55. M/J 22/P23/Q1(a)

1(a)	- energy required - when one electron is removed - from each atom in one mole of - gaseous atoms two or three points for one mark, four points for two marks	2
------	---	----------

56. M/J 22/P23/Q1(b)

1(b)	Group VI / 16 AND large increase (in IE) after 6th	1
------	---	----------

57. M/J 22/P23/Q1(c)

1(c)	M1 reference to spin pair repulsion in (3)p orbital (in S) OR due to repulsion of two electrons in a (3)p orbital (in S) M2 outweighs increased nuclear charge (in S)	2
------	--	----------

58. M/J 22/P23/Q1(d)

1(d)	M1 <i>similarity in electronic structure / shielding of Al^{2+} and Na</i> both remove electron from (3)s ¹ / single electron in (3)s (sub-level / orbital) OR Al^{2+} and Na have same electronic configuration OR shielding (of outer electron) is the same M2 <i>greater nuclear charge / number of protons</i> Al^{2+} has greater nuclear charge OR 13p compared to 11p	2
------	---	----------

59. M/J 22/P23/Q1(e)

1(e)		atomic no.	nucleon no.	no.of neutrons	electronic arrangement	3
	copper –65	29	65	65 – 29 = 36	1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ¹⁰ 4s ¹	
	M1 29 AND 65 M2 nucleon no – atomic no ALLOW ecf from M1 M3 electronic arrangement					

60. MAR 22/P22/Q1(a)

1(a)	$O(g) \rightarrow O^+(g) + e^-$	1
------	---------------------------------	----------

61. MAR 22/P22/Q1(b)

1(b)(i)	increase across period AND increased nuclear attraction for (valence / outer) electrons [1] increase in (positive) nuclear charge / number of protons (in the nucleus) [1] similar shielding (of outer electrons) [1]	3
1(b)(ii)	spin-pair repulsion (of electrons) in (2)p <u>orbital</u> [1] outweighs increased nuclear charge [1]	2

62. MAR 22/P22/Q1(c)

1(c)	$1s^2 2s^2 2p^6 3s^2 3p^1$ [1] greatest jump between 3rd and 4th ionisations [1] indicates three electrons in outer shell [1]	3
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Unit 2

ATOMS, MOLECULES AND STOICHIOMETRY

39 items · 86 marks

Syllabus sections in this unit

- 2.1 Relative masses of atoms and molecules
- 2.2 The mole and the Avogadro constant
- 2.3 Formulas
- 2.4 Reacting masses and volumes (of solutions and gases)

Unit 2: Atoms, molecules and stoichiometry

1. M/J 26/P21/Q1(e)

(e) Compound **X** contains sulfur and chlorine only.

100.0 g of **X** contains 47.49 g of sulfur. The relative molecular mass, M_r , of **X** is 135.2.

Determine the molecular formula of **X**. Show your working.

molecular formula of **X** = [3]

2. M/J 26/P22/Q4(b)(ii)-(iii)

Context: part (b) (introduction)

(b) The structure of compound **X** is shown in Figure 4.2.

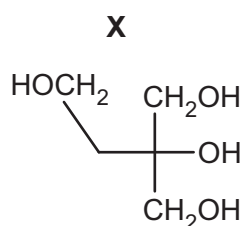


Figure 4.2

The molecular formula of **X** is $C_5H_{12}O_4$. Complete oxidation of 0.255 g of **X** occurs when it reacts with exactly 10.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ acidified $K_2Cr_2O_7(aq)$.

(ii) Calculate the amount, in mol, of $K_2Cr_2O_7(aq)$ used in the reaction.

amount = mol [1]

(iii) Calculate the amount, in mol, of **X** present in the 0.255 g sample.

amount = mol [1]

3. M/J 26/P24/Q1(b)

Context

1 Aluminium is a Group 13 element that forms a number of compounds.

Table 1.1

atom or ion	number of protons	number of neutrons
^{27}Al		
$^{28}\text{Al}^{3+}$		

(b) Deduce the formula of aluminium nitrate(V).

[1]

4. O/N 25/P21/Q3(a)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

[1]

5. O/N 25/P23/Q3(a)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(a) Define empirical formula.

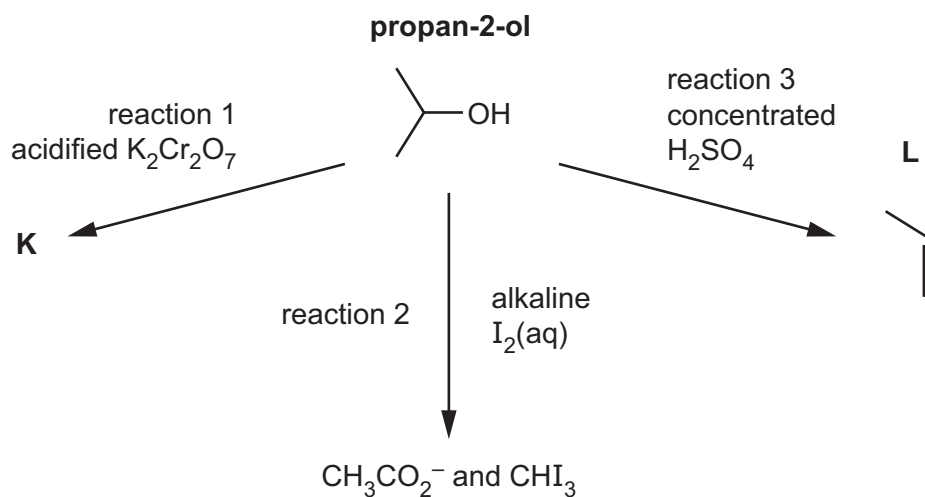
[1]

6. O/N 25/P24/Q3(b)(iv)

Context

- 3** Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

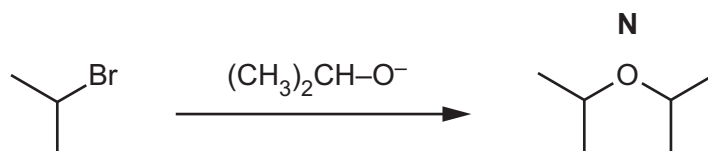
Fig. 3.1 shows some reactions of propan-2-ol.



Context: part (b) (introduction)

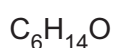
- (b) Propan-2-ol reacts with sodium to produce $(\text{CH}_3)_2\text{CH}-\text{O}^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

**Fig. 3.3**

- (iv) **N** is also added to petrol to make it burn more smoothly.

Construct an equation for the complete combustion of **N**, $\text{C}_6\text{H}_{14}\text{O}$.



[1]

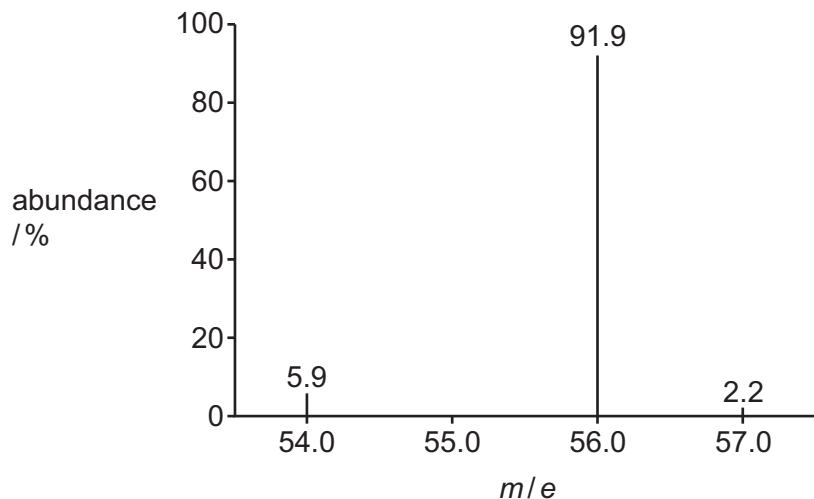
7. M/J 25/P21/Q2(b)(i)

Context

- 2** A sample of iron is analysed using a mass spectrometer. The mass spectrum shows three isotopes of iron are present in the sample.

Context: part (b) (introduction)

- (b)** Fig. 2.1 shows the mass spectrum of the sample of iron.

**Fig. 2.1**

- (i)** Use Fig. 2.1 to calculate the relative atomic mass, A_r , of iron to **one** decimal place.

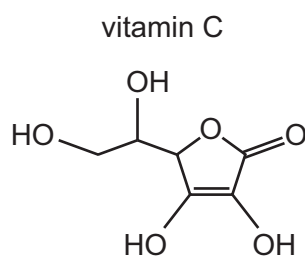
Show your working.

A_r of iron = [2]

8. M/J 25/P21/Q4(a)

Context

- 4** The structure of vitamin C is shown in Fig. 4.1.

**Fig. 4.1**

- (a)** Deduce the empirical formula of vitamin C.

[1]

9. M/J 25/P21/Q4(b)(i)-(ii)

Context

4 The structure of vitamin C is shown in Fig. 4.1.

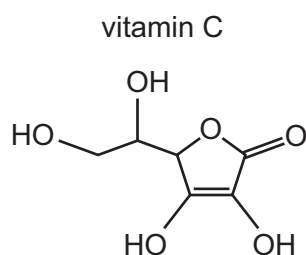


Fig. 4.1

Context: part (b) (introduction)

(b) The concentration of vitamin C is found by titration with $\text{I}_2(\text{aq})$.

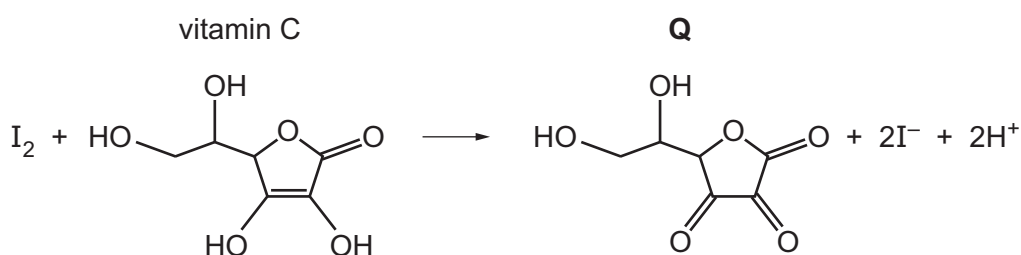


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm^3 of vitamin C solution. 5.00 cm^3 of this vitamin C solution is added to a flask with approximately 150 cm^3 of water and an indicator. Exactly 28.40 cm^3 of $5.00 \times 10^{-4}\text{ mol dm}^{-3}\text{ I}_2(\text{aq})$ reacts with the sample of vitamin C solution in the flask.

[M_r : vitamin C, 176]

(i) Calculate the amount, in mol, of $\text{I}_2(\text{aq})$ added to the flask in this titration.

amount of $\text{I}_2(\text{aq}) =$ mol [1]

(ii) Use your answer to (b)(i) to calculate the mass, in g, of vitamin C in the tablet. Show your working.

(If you were unable to calculate a value for the amount of $\text{I}_2(\text{aq})$ in **(b)(i)**, use the value $2.64 \times 10^{-4} \text{ mol}$. This is **not** the correct value.)

mass of vitamin C in tablet = _____ g [2]

10. M/J 25/P22/Q2(a)

Context

2 A sample of iron contains three different isotopes and has a relative atomic mass, A_r , of 55.8.

(a) (i) Define relative atomic mass.

[2]

(ii) Table 2.1 shows the abundances of two of the isotopes present in the sample of iron.

Table 2.1

isotope	relative isotopic mass	abundance / %
^{54}Fe	53.9	6.0
^{56}Fe	55.9	91.9

Use Table 2.1 to calculate the relative isotopic mass of the third isotope of iron in the sample. Show your working.

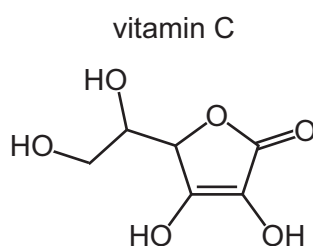
relative isotopic mass =

[2]

11. M/J 25/P22/Q5(a)

Context

5 The structure of vitamin C is shown in Fig. 5.1.

**Fig. 5.1**

(a) Deduce the empirical formula of vitamin C.

[1]

12. M/J 25/P22/Q5(b)(i)-(ii)

Context

5 The structure of vitamin C is shown in Fig. 5.1.

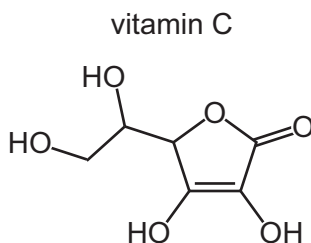


Fig. 5.1

Context: part (b) (introduction)

(b) The mass of vitamin C present in 150.0g of lemon is found in an experiment.

stage 1 All the vitamin C in 150.0g of lemon is extracted and dissolved in water to make 100.0cm³ of solution L.

stage 2 A 25.0 cm^3 sample of solution **L** is titrated with $5.00 \times 10^{-3}\text{ mol dm}^{-3}\text{ I}_2(\text{aq})$.

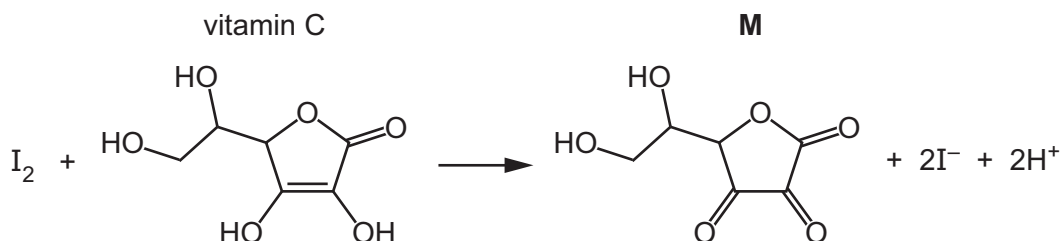


Fig. 5.2

Exactly 36.65 cm^3 of $\text{I}_2(\text{aq})$ reacts with the 25.0 cm^3 sample of solution L.

[M_r : vitamin C, 176]

(i) Calculate the amount, in mol, of $\text{I}_2(\text{aq})$ added in the titration.

amount of $\text{I}_2(\text{aq}) =$ mol [1]

(ii) Use your answer to (b)(i) to calculate the percentage by mass of vitamin C present in 150.0 g of lemon.

(If you were unable to calculate a value for the amount of $\text{I}_2(\text{aq})$ in **(b)(i)**, use the value $7.65 \times 10^{-4} \text{ mol}$. This is **not** the correct value.)

percentage by mass = % [2]

13. M/J 25/P23/Q5(a)(i)

Context: part (a) (introduction)

5 (a) W is a colourless liquid.

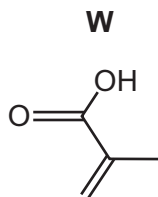


Fig. 5.1

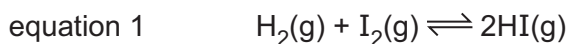
- (i) Deduce the empirical formula of **W**.

[1]

14. M/J 25/P24/Q3(a)

Context

3 $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ react to form $\text{HI}(\text{g})$ in a reversible reaction, as shown in equation 1.



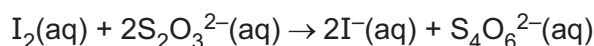
In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

(a) In experiment 1, $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ are combined.

Water is then added to the equilibrium mixture to produce 1.00 dm^3 of solution **A**.

The amount of $\text{I}_2(\text{aq})$ present in 1.00 dm^3 of solution **A** is found by titration with $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$.

Exactly 32.90 cm³ of 0.200 mol dm⁻³ Na₂S₂O₃ reacts with all of the I₂ in a 25.0 cm³ sample of solution **A**.



- (i) Calculate the amount, in mol, of $\text{S}_2\text{O}_3^{2-}$ that reacts in the titration.

amount of $\text{S}_2\text{O}_3^{2-} =$ mol [1]

- (ii) Use your answer to (a)(i) to calculate the amount, in mol, of I_2 present in 1.00 dm^3 of solution **A**.

(If you were unable to obtain an answer in **(a)(i)**, then use 9.42×10^{-3} mol. This is **not** the correct answer.)

amount of I_2 in 1.00 dm^3 of solution **A** = mol [2]

15. M/J 25/P24/Q6(a)(i)-(ii)

Context

6 **P**, **Q** and **R** are three different hydrocarbon molecules.**(a) (i)** Define empirical formula.

[1]

(ii) Hydrocarbon **P** contains 85.7% by mass of carbon.Calculate the empirical formula of **P**.

Show your working.

empirical formula of **P** =

[2]

16. O/N 24/P21/Q1(c)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.**(c)** Table 1.2 shows the relative abundances of isotopes in a sample of an alloy containing rhodium and iridium only.**Table 1.2**

isotope	relative isotopic mass	relative abundance in alloy
$^{103}_{45}\text{Rh}$	102.91	50.00
$^{191}_{77}\text{Ir}$	190.96	15.18
$^{193}_{77}\text{Ir}$	192.96	34.82

(i) Define relative isotopic mass.

[2]

(ii) Use Table 1.2 to calculate the relative atomic mass, A_r , of iridium in the alloy.Give your answer to **two** decimal places.

relative atomic mass of iridium =

[2]

17. O/N 24/P21/Q1(d)(i)

Context

- 1** Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

Context: part (d) (introduction)

- (d)** Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

- (i)** Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$.

Show your working.

$$x =$$

[2]

18. O/N 24/P22/Q1(b)(ii)-(iii)

Context

- 1** Vanadium, niobium and tantalum are metals in the same group of the Periodic Table.

Context: Table 1.1

Table 1.1

isotope	relative isotopic mass	number of protons	number of neutrons
${}^{93}_{41}\text{Nb}$	92.91		
${}^{181}_{73}\text{Ta}$	180.95		

Context: part (b) (introduction)

- (b)** Pelopium was the suggested name for a new element discovered in a mineral.

Pelopium was later found to be a mixture of niobium, Nb, and tantalum, Ta.

Only one naturally occurring isotope exists for each of Nb and Ta.

- (ii)** Define relative isotopic mass.

[2]

- (iii)** A sample of pelopium contains 90.9% by mass ${}^{93}_{41}\text{Nb}$ and 9.1% by mass ${}^{181}_{73}\text{Ta}$.

Calculate the theoretical relative atomic mass of pelopium based on these data and Table 1.1.

Give your answer to two decimal places.

Show your working.

theoretical relative atomic mass of pelopium =

[2]

19. O/N 24/P23/Q1(c)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

(c) Table 1.2 shows the relative abundances of isotopes in a sample of an alloy containing rhodium and iridium only.

Table 1.2

isotope	relative isotopic mass	relative abundance in alloy
$^{103}_{45}\text{Rh}$	102.91	50.00
$^{191}_{77}\text{Ir}$	190.96	15.18
$^{193}_{77}\text{Ir}$	192.96	34.82

(i) Define relative isotopic mass.

[2]

(ii) Use Table 1.2 to calculate the relative atomic mass, A_r , of iridium in the alloy.

Give your answer to **two** decimal places.

relative atomic mass of iridium =

[2]

20. O/N 24/P23/Q1(d)(i)

Context

1 Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

Context: part (d) (introduction)

(d) Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

(i) Hydrated rhodium(III) chloride contains 20.5% by mass of water of crystallisation.

Deduce the integer value of x in $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$.

Show your working.

$x =$

[2]

21. M/J 24/P21/Q2(d)(i)

Context: part (d) (introduction)

(d) Boron nitride is a white solid that melts above 2900 °C.

Fig. 2.2 shows part of the lattice structure of a crystal of boron nitride.

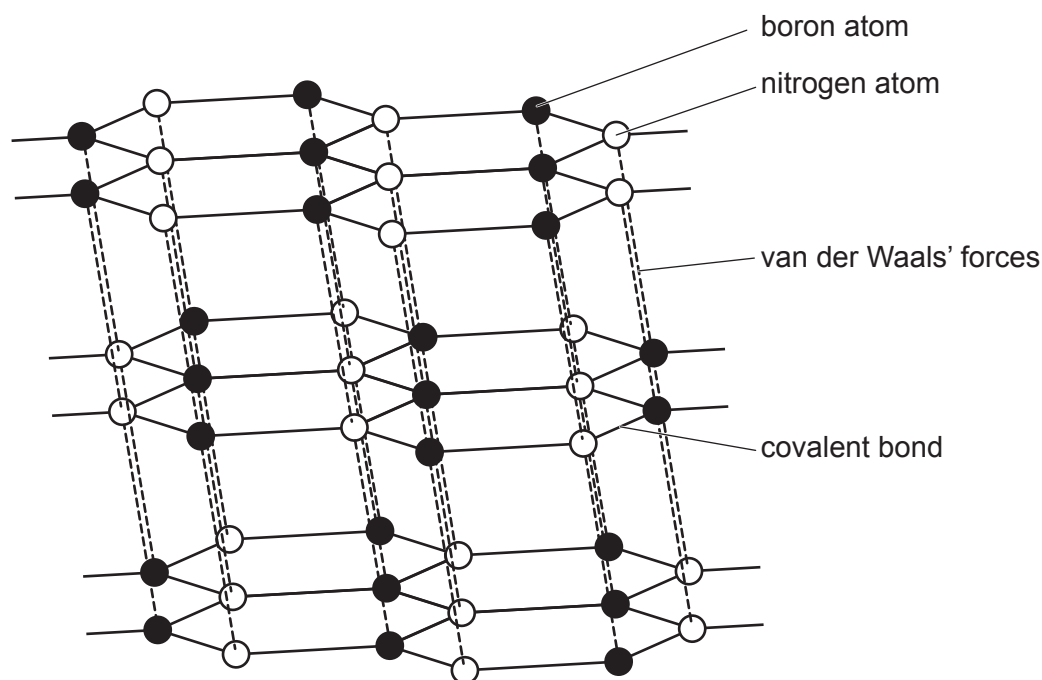


Fig. 2.2

(i) Use Fig. 2.2 to deduce the empirical formula of boron nitride.

[1]

22. M/J 24/P23/Q3(c)(ii)

Context: part (c) (introduction)

(c) Boron carbide is a hard crystalline solid that has a melting point greater than 2000 °C.

(ii) 100 g of pure boron carbide contains 78.26 g of boron.

Calculate the empirical formula of boron carbide.

Show your working.

empirical formula of boron carbide

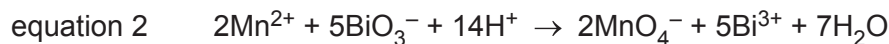
[2]

23. MAR 24/P22/Q1(e)

Context

1 Bismuth is an element in Group 15 of the Periodic Table.

(e) NaBiO_3 can be used to determine the concentration of $\text{Mn}^{2+}(\text{aq})$. The ionic equation for the reaction is shown in equation 2.



A student uses the following procedure in an experiment.

- Add 100.0 cm^3 of a saturated solution of $\text{Mn}^{2+}(\text{aq})$ to a volumetric flask.
- Add distilled water to the flask to make a 1.00 dm^3 diluted solution.
- Titrate a 25.00 cm^3 sample of the diluted solution with 0.100 mol dm^{-3} $\text{NaBiO}_3(\text{aq})$.

The 25.00 cm^3 sample of the diluted solution of $\text{Mn}^{2+}(\text{aq})$ reacts completely with exactly 21.50 cm^3 of 0.100 mol dm^{-3} $\text{NaBiO}_3(\text{aq})$.

Calculate the concentration, in mol dm^{-3} , of $\text{Mn}^{2+}(\text{aq})$ in the saturated solution.

Show your working.

concentration of $\text{Mn}^{2+}(\text{aq})$ in the saturated solution = mol dm^{-3} [3]

24. M/J 23/P21/Q2(b)(i)

Context

2 A neutralisation reaction occurs when $\text{NaOH}(\text{aq})$ is added to $\text{H}_2\text{SO}_4(\text{aq})$.



Context: part (b) (introduction)

(b) An experiment is carried out to calculate ΔH_{neut} for the reaction between $\text{NaOH}(\text{aq})$ and $\text{H}_2\text{SO}_4(\text{aq})$.

100 cm^3 of 1.00 mol dm^{-3} $\text{NaOH}(\text{aq})$ is added to 75 cm^3 of 1.00 mol dm^{-3} $\text{H}_2\text{SO}_4(\text{aq})$ in a polystyrene cup and stirred. Results from the experiment are shown in Table 2.1.

Table 2.1

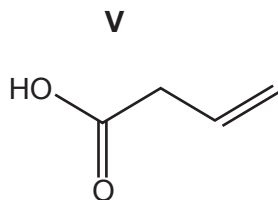
initial temperature of $\text{NaOH}(\text{aq})/^\circ\text{C}$	20.0
initial temperature of $\text{H}_2\text{SO}_4(\text{aq})/^\circ\text{C}$	20.0
maximum temperature of mixture/ $^\circ\text{C}$	27.8

(i) Use equation 1 to calculate the amount, in mol, of $\text{H}_2\text{SO}_4(\text{aq})$ that is neutralised in the experiment.

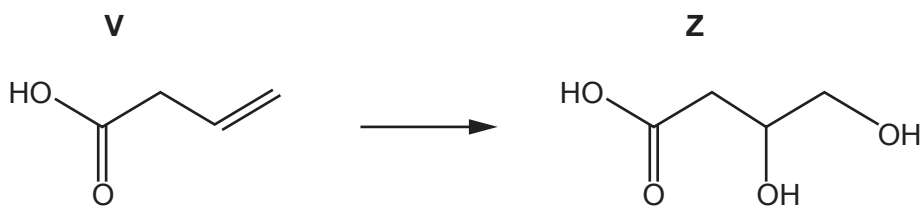
amount of $\text{H}_2\text{SO}_4(\text{aq})$ neutralised = mol [1]

25. M/J 23/P21/Q4(b)(ii)

Context

4 **V** is a colourless liquid.**Fig. 4.1**

Context: part (b) (introduction)

(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.**Fig. 4.2****(ii)** Deduce the empirical formula of **Z**.

[1]

26. M/J 23/P22/Q2(e)(iii)

Context

2 Chlorine is a reactive element. It forms many compounds.

Context: part (e) (introduction)

(e) Bleach is used as a cleaning product to kill bacteria. It is made by adding compounds like sodium chlorate(I), NaClO , to water.**(iii)** A sample of bleach **W** contains an unknown concentration of sodium chlorate(I).

10.0 cm^3 of **W** is diluted with distilled water to make a total volume of 100 cm^3 of bleach solution. 25.0 cm^3 of this diluted bleach solution is added to an excess of hydrogen peroxide and the volume of gas produced measured under room conditions. The experiment is repeated and on average 25.0 cm^3 of diluted bleach solution produces 42.0 cm^3 of gas.

Calculate the concentration, in g dm^{-3} , of sodium chlorate(I) in **W**.

concentration of NaClO in **W** = g dm^{-3} [3]

27. M/J 23/P23/Q1(b)

Context

1 Copper is used in electrical equipment. It has a melting point of 1085 °C.

(b) The relative isotopic masses and natural abundances of the two isotopes in a sample of copper are shown in Table 1.1.

Table 1.1

isotope	relative isotopic mass	% abundance
^{63}Cu	62.930	69.15
^{65}Cu	64.928	30.85

(i) Define the unified atomic mass unit.

[1]

(ii) Define relative atomic mass, A_r , in terms of the unified atomic mass unit.

[1]

(iii) Calculate the relative atomic mass, A_r , of copper in this sample using the data in Table 1.1.

Show your working.

$A_r =$ [1]

28. M/J 23/P23/Q3(b)(i)

Context

3 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).

equation 1 $2\text{NaOH(aq)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{Na}_2\text{SO}_4\text{(aq)} + 2\text{H}_2\text{O(l)}$

Context: part (b) (introduction)

(b) In an experiment, 50.0 cm³ of 2.00 mol dm⁻³ NaOH(aq) is added to 60.0 cm³ of 1.00 mol dm⁻³ H₂SO₄(aq) in a polystyrene cup and stirred. Both solutions have a temperature of 21.4 °C before mixing. The maximum temperature of the mixture is measured.

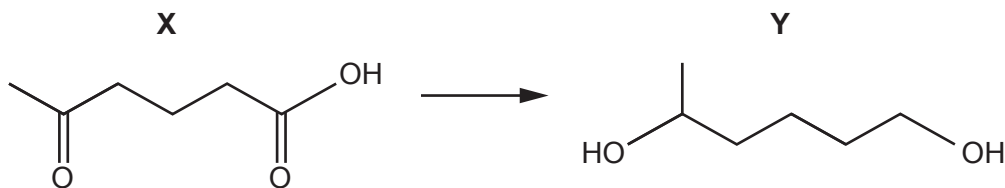
(i) Use equation 1 to calculate the amount, in mol, of H₂SO₄(aq) that is neutralised in the experiment.

amount of H₂SO₄ neutralised = mol [1]

29. M/J 23/P23/Q5(a)

Context

5 Y is formed from X in a single-step reaction, as shown in Fig. 5.1.

**Fig. 5.1**

(a) Deduce the empirical formula of Y.

[1]

30. MAR 23/P22/Q2(d)(ii)

Context

2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(ii) Compound X, an anhydrous Group 2 bromide, is dissolved in water and titrated against aqueous silver nitrate.

A solution containing 0.250 g of X requires 33.65 cm³ of 0.0500 mol dm⁻³ AgNO₃(aq) for complete reaction.

Identify X.

Show your working.

X =

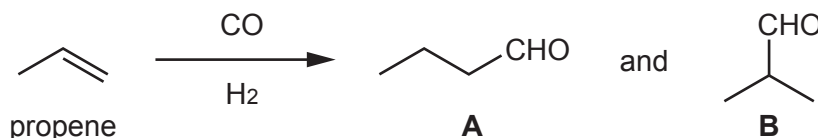
[3]

31. MAR 23/P22/Q3(a)(iv)

Context

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

**Fig. 3.1**

(iv) The complete reaction of propene with a 1:1 mixture of CO and H₂ produces A and B only. The product mixture contains 96% A and 4% B.

Calculate the mass of A produced in this reaction when 5.00 × 10³ kg of propene is used.

mass of A =

kg [1]

32. O/N 22/P21/Q1(a)(ii)-(iii)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

Context: part (a) (introduction)

(a) Gallium has two stable isotopes, ^{69}Ga and ^{71}Ga .

(ii) Define relative atomic mass.

[2]

(iii) The relative atomic mass of gallium, A_r , is 69.723.
The relative isotopic masses of ^{69}Ga and ^{71}Ga are:

^{69}Ga , 68.926; ^{71}Ga , 70.925.

Use this information to calculate the percentage abundance of ^{69}Ga in elemental gallium.
Show your working.

Assume that the element contains only the ^{69}Ga and ^{71}Ga isotopes.
Give your answer to **four** significant figures.

percentage abundance of ^{69}Ga = %
[2]

33. O/N 22/P22/Q1(d)

Context

- 1** Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

(d) There are many naturally occurring hydrated compounds that contain the anion PO_4^{3-} .

(i) Name the anion PO_4^{3-} .

[1]

(ii) Struvite is a soft hydrated mineral with $M_r = 245.3$. The anhydrous form of the mineral has the formula NH_4MgPO_4 .

Calculate the number of molecules of water of crystallisation in struvite.

Give your answer to the nearest integer. Show your working.

number of molecules of water of crystallisation = [2]

34. O/N 22/P23/Q1(a)(ii)-(iii)

Context

- 1** Atoms with nuclei containing an odd number of protons tend to have fewer isotopes than those with an even number of protons.

Context: part (a) (introduction)

(a) Gallium has two stable isotopes, ^{69}Ga and ^{71}Ga .

(ii) Define relative atomic mass.

[2]

(iii) The relative atomic mass of gallium, A_r , is 69.723.
The relative isotopic masses of ^{69}Ga and ^{71}Ga are:

^{69}Ga , 68.926; ^{71}Ga , 70.925.

Use this information to calculate the percentage abundance of ^{69}Ga in elemental gallium.
Show your working.

Assume that the element contains only the ^{69}Ga and ^{71}Ga isotopes.
Give your answer to **four** significant figures.

percentage abundance of ^{69}Ga = %
[2]

35. M/J 22/P21/Q1(e)(i)-(ii)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

Context: part (e) (introduction)

(e) A sample of magnesium contains three isotopes, ^{25}Mg , ^{26}Mg and **X**.

The percentage abundance of the three isotopes is shown in Table 1.1.

Table 1.1

isotope of Mg	mass/a.m.u.	percentage abundance/%
X		78.99
^{25}Mg	24.99	10.00
^{26}Mg	25.98	11.01

- (i)** The relative atomic mass, A_r , is calculated by comparing the average mass of the isotopes of an element to the unified atomic mass unit.

Define the unified atomic mass unit.

[1]

- (ii)** Calculate the mass of **X**. Use data from Table 1.1 and A_r (magnesium) = 24.31 in your calculation. Show your working.

mass of **X** =

[2]

36. M/J 22/P21/Q4(a)

Context

4 Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

(a) Calculate the molecular formula of **V**. Show your working.

molecular formula of **V** = [3]

37. M/J 22/P21/Q4(b)(ii)

Context

4 Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

Context: part (b) (introduction)

(b) **V** contains two types of functional group: a carboxylic acid and an alkene.

(ii) A 3.196 g sample of Br_2 reacts completely with 2.800 g of **V**.

Calculate how many alkene functional groups are present in one molecule of **V**. Show your working.

number of alkene functional groups in **V** = [1]

38. M/J 22/P22/Q6(b)

Context

6 **Z** is a molecule which contains the elements carbon, hydrogen and oxygen only.

Z contains only alkene and carboxyl functional groups.

(b) Table 6.2 shows the percentage by mass of each element present in **Z**.

Table 6.2

element	percentage by mass/%
carbon	41.38
hydrogen	3.45
oxygen	55.17

Using the data in Table 6.2, demonstrate that the empirical formula of **Z** is CHO . Show your working.

[1]

39. M/J 22/P23/Q1(f)(i)

(f) (i) The element copper has a relative atomic mass of 63.5.

Calculate how many atoms are present in 1.05 g of copper.

atoms of copper present = [1]

Answers Section

1. M/J 26/P21/Q1(e)

1(e)	M1 mass $Cl = 52.51$ M2 $47.49 / 32.1 : 52.51 / 35.5 (= 1.48 : 1.48)$ (empirical formula =) SCl M3 ($M_r(X) = 135.2$) molecular formula = S_2Cl_2	3	check working M1 allow $100 - 47.49$ without being evaluated A owtte $135.2 - \times(32.1) s = 2$ so $M_r = S_2Cl_2$ ✓✓✓ A alternative methods M2 A ecf from an incorrect M1 M3 A only S_2Cl_2 (with no working)
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2. M/J 26/P22/Q4(b)(ii)-(iii)

4(b)(ii)	0.005 OR 5×10^{-3}	1	1 / 200
4(b)(iii)	1.875×10^{-3}	1	A 1.88×10^{-3} OR 1.9×10^{-3} A 2×10^{-3} with appropriate working R 1.880×10^{-3} OR 1.90×10^{-3} OR 2.0×10^{-3} OR 1.87×10^{-3}

3. M/J 26/P24/Q1(b)

1(b)	Al(NO ₃) ₃	1	A (NO ₃) ₃ Al
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4. O/N 25/P21/Q3(a)

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	1
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5. O/N 25/P23/Q3(a)

3(a)	simplest /lowest (whole number / integer) ratio of atoms (of each element) in a compound / molecule	1
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6. O/N 25/P24/Q3(b)(iv)

3(b)(iv)	C ₆ H ₁₄ O + 9O ₂ → 6CO ₂ + 7H ₂ O	1
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7. M/J 25/P21/Q2(b)(i)

2(b)(i)	M1 method showing each abundance multiplied by m/e value (5.9 × 54) and (91.9 × 56) and (2.2 × 57) OR (0.59 × 54) and (9.19 × 56) and (0.22 × 57) M2 the three values from M1 expressed as a percentage and added together correctly with final answer to 1 d.p $\frac{5.9}{100} \times 54 + \frac{91.9}{100} \times 56 + \frac{2.2}{100} \times 57 = 55.9$	2
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8. M/J 25/P21/Q4(a)

4(a)	C ₃ H ₄ O ₃	1
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9. M/J 25/P21/Q4(b)(i)-(ii)

4(b)(i)	(28.40 / 1000 × 0.0005) = 1.42×10^{-5} mol	1
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4(b)(ii)	$(1.42 \times 10^{-5} \times 176 \times 200 / 5) = 0.099968$ M1 correct method to deduce mass of vit C in 5 cm ³ sample $(1.42 \times 10^{-5} \times 176) = 2.49 \times 10^{-3}$ M2 correct calculation for mass of vit C in original solution $(2.49 \times 10^{-3} \times 200 / 5) = 0.099968$ OR M1 correct method to deduce amount, mol, vit C in 5 cm ³ sample $(1.42 \times 10^{-5} \times 200 / 5) = 5.68 \times 10^{-4}$ M2 correct calculation for mass of vit C in original solution = M1 $\times$ 176 $(5.68 \times 10^{-4}) \times 176 = 0.099968$	2
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10. M/J 25/P22/Q2(a)

2(a)(i)	route 1 M1 average (weighted) mass of (an) atom(s) (of an element) OR average (weighted) mass of the isotope(s) (of an element) M2 compared to the mass of the unified atomic mass unit OR on a scale in which a carbon-12 atom / isotope has a mass of exactly 12 (units) OR on a scale in which the mass of a carbon-12 atom/isotope is exactly 12 (units) OR compared to $\frac{1}{12}$ mass of one atom of C-12 route 2 M1 average mass of one mol of isotopes / atoms (of an element) M2 relative / compared to $\frac{1}{12}$ of the mass of one mol of C-12 atoms/isotopes OR when the mass of one mol C-12 atoms / isotopes is exactly 12 (000) g	2
2(a)(ii)	M1 relative abundance of unknown isotope = $(100 - 91.9 - 6.0) = 2.1\%$ M2 $55.8 = (53.9 \times 6.0 + 55.9 \times 91.9 + ? \times M1) / 100 = 56.9$	2

11. M/J 25/P22/Q5(a)

5(a)	C ₃ H ₄ O ₃	1
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12. M/J 25/P22/Q5(b)(i)-(ii)

5(b)(i)	$5.00 \times 10^{-3} \times 36.65 / 1000$ $= 1.833 \times 10^{-4} / 1.8325 \times 10^{-4}$	1
5(b)(ii)	M1 expression and / or correct calculation to show how the amount of vitamin C in 150 g of lemon peel is calculated $1.8325 \times 10^{-4} \text{ mol} \times 176 \times 4$ M2 correct calculation based on $(M1 / 150) \times 100 = 0.086(\%)$ Using 1.8325×10^{-4} for M1 , M2 = 0.086 OR 0.087(%) Alternative answer using 7.65×10^{-4} : M1 $7.65 \times 10^{-4} \times 176 \times 4 = 0.5386 / 0.539$ M2 correct calculation based on $(M1 / 150) \times 100 = 0.359 / 0.36 (\%)$	2

13. M/J 25/P23/Q5(a)(i)

5(a)(i)	C ₂ H ₃ O	1
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14. M/J 25/P24/Q3(a)

3(a)(i)	0.00658(0) mol	1
3(a)(ii)	M1 I ₂ in 25.0 cm ³ $0.00658 / 2 = 0.00329$ (mol) M2 $0.00329 \times 1000 / 25.0 = 0.132$ (mol) alternative answer using 9.42×10^{-3} : M1 0.00471 M2 0.1884	2

15. M/J 25/P24/Q6(a)(i)-(ii)

6(a)(i)	simplest (whole number) ratio of atoms (of the elements) in a compound	1
6(a)(ii)	C : H M1 $\frac{85.7}{12}$ or 7.14 : $\frac{14.3}{1}$ or 14.3 1 : 2 M2 CH₂	2

16. O/N 24/P21/Q1(c)

1(c)	mass of an atom of an isotope	1
	compared to the mass of the unified atomic mass unit (amu) OR on a scale on which a carbon-12 atom has a mass of exactly 12 units OR divided by $\frac{1}{12}$ mass of a carbon-12 atom	1
1(c)(ii)	$\frac{190.96 \times 15.18 + 192.96 \times 34.82}{50.00}$	1
	= 192.35	1

17. O/N 24/P21/Q1(d)(i)

1(d)(i)	M1 $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$ OR $18x / (102.9 + 3(35.5) + 18x) = 0.205$ $x = 3$ OR $(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$ $x = 3$ M2 $x = 3$	1
		1

18. O/N 24/P22/Q1(b)(ii)-(iii)

1(b)(ii)	M1 mass of an (atom of an) isotope	1
	M2 relative/compared to (the mass of) the unified atomic mass unit OR on a scale in which a carbon-12 atom / isotope has a mass of exactly 12 units OR divided by / compared to $\frac{1}{12}$ of the mass of a carbon-12 atom / isotope Alternative route using mass of 1 mol throughout M1 mass of one mol of an (atom of an) isotope M2 relative/compared to $\frac{1}{12}$ of the mass of 1 mol of C-12 atom / isotope OR when the mass of one mol C-12 atom / isotope is exactly 12(.000) g	1
1(b)(iii)	$92.91 \times 0.909 + 180.95 \times 0.091$	1
	= 100.92	1

19. O/N 24/P23/Q1(c)

1(c)	mass of an atom of an isotope	1
	compared to the mass of the unified atomic mass unit (amu) OR on a scale on which a carbon-12 atom has a mass of exactly 12 units OR divided by $\frac{1}{12}$ mass of a carbon-12 atom	1
1(c)(ii)	$\frac{190.96 \times 15.18 + 192.96 \times 34.82}{50.00}$	1
	= 192.35	1

20. O/N 24/P23/Q1(d)(i)

1(d)(i)	M1 $\frac{79.5}{209.4} : \frac{20.5}{18.0} = 0.380 : 1.14 = 1:3$ OR $18x / (102.9 + 3(35.5) + 18x) = 0.205$ $x = 3$ OR $(102.9 + 3(35.5)) / (209.4 + 18x) = 0.795$ $x = 3$	1
	M2 $x = 3$	1

21. M/J 24/P21/Q2(d)(i)

2(d)(i)	BN	1
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22. M/J 24/P23/Q3(c)(ii)

3(c)(ii)	M1 give correct expression or calculate mol B compared to mol C $78.26 / 10.8 (= 7.246 / 7.25 \text{ mol B})$ AND $21.74 / 12 (= 1.81(2) \text{ mol C})$ M2 Find B : C ratio AND use this to state empirical formula $7.246 \text{ to } 7.25 / 1.817 \text{ to } 1.82 = 3.99 \text{ to } 3.98 \text{ mol B : 1 mol C}$ (so) B ₄ C OR CB ₄	2
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23. MAR 24/P22/Q1(e)

1(e)	moles of NaBiO ₃ = $21.50 / 1000 \times 0.100$ (= $2.15 \times 10^{-3} \text{ mol}$)	1
	moles of Mn ²⁺ in dil. = $2.15 \times 10^{-3} \times 2 / 5 \times 40$ (= 0.0344 mol)	1
	conc. saturated Mn ²⁺ = $0.0344 \times 1000 / 100$ = 0.344 mol dm⁻³	1

24. M/J 23/P21/Q2(b)(i)

2(b)(i)	0.05 mol	1
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25. M/J 23/P21/Q4(b)(ii)

4(b)(ii)	CH ₂ O	1
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26. M/J 23/P22/Q2(e)(iii)

2(e)(iii)	M1 (show how to calculate) mol O ₂ gas 42 cm ³ gas = 42 / 24000 mol oxygen gas (= 1.75 × 10 ⁻³ mol) M2 (process to find) amount NaClO in 10 cm ³ sample = 100 cm ³ (diluted) bleach M1 × 4 (= 4 × 1.75 × 10 ⁻³ mol in 10 cm ³ = 0.007) M3 correct calculation using M2 to find concentration = (M2 / 10) × 1000 × 74.5 = 52.2 (g / dm ³)	3
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27. M/J 23/P23/Q1(b)

1(b)(i)	(the mass of) one twelfth of the mass of a C-12 / ¹² C atom (or isotope)	1
1(b)(ii)	average mass of the isotopes of an element compared to the unified atomic mass unit	1
1(b)(iii)	(62.93 × 69.15) ÷ 100 + (64.928 × 30.85) ÷ 100 = 63.55 / 63.546	1

28. M/J 23/P23/Q3(b)(i)

3(b)(i)	0.05 mol (H ₂ SO ₄ (aq))	1
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29. M/J 23/P23/Q5(a)

5(a)	C ₃ H ₇ O	1
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30. MAR 23/P22/Q2(d)(ii)

2(d)(ii)	moles of AgNO ₃ = $\frac{33.65}{1000} \times 0.0500$ (= 1.68(25) × 10 ⁻³)	1
	moles of X = ½ × moles of AgNO ₃ = 0.250 ÷ M _r (X) ∴ M _r (X) = 297.2	1
	A _r of Group 2 element is 297.2 – 2(79.9) = 137.4 AND X is BaBr ₂	1

31. MAR 23/P22/Q3(a)(iv)

3(a)(iv)	96% × (5000 / 42.0) × 72.0 = 8230 or 8229 (kg)	1
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32. O/N 22/P21/Q1(a)(ii)-(iii)

1(a)(ii)	M1 (weighted) average / mean mass of the isotopes / average mass of the atom(s) (of an element) M2 compared to (the mass of) the unified atomic mass unit	1
1(a)(iii)	69.723 = 68.926x + 70.925(1 – x) ∴ x = 0.6013 / 69.723 = $\frac{68.926x + 70.925(100 - x)}{100}$	1
	60.13%	1

33. O/N 22/P22/Q1(d)

1(d)(i)	phosphate(V)	1
1(d)(ii)	(molecular mass of NH ₄ MgPO ₄) = 137.3	1
	(245.3 – M _r (NH ₄ MgPO ₄) ÷ 18.0) = 6	1

34. O/N 22/P23/Q1(a)(ii)-(iii)

1(a)(ii)	M1 (weighted) average / mean mass of the isotopes / average mass of the atom(s) (of an element) M2 compared to (the mass of) the unified atomic mass unit	1
		1

1(a)(iii)	$69.723 = 68.926x + 70.925(1-x) \therefore x = 0.6013$ $/ 69.723 = \frac{68.926x + 70.925(100-x)}{100}$	1
	60.13%	1

35. M/J 22/P21/Q1(e)(i)-(ii)

1(e)(i)	1 / 12 (one twelfth) the mass of a carbon-12 / ^{12}C atom	1
1(e)(ii)	M1 correct expression relating A_r to the mass / % abundance of the three isotopes $24.31 = x \times 0.7899 + 24.99 \times 0.1000 + 25.98 \times 0.1101$ M2 correct answer to 4 sig figs atomic mass of X = 23.99	2

36. M/J 22/P21/Q4(a)

4(a)	M1 % / A_r for C H O M2 each % / A_r for C H O divided by the smallest value for % / A_r to give simplest whole number ratio / empirical formula M3 compare M_r from M2 ratio with 280 to deduce the actual molecular formula <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="text-align: center;">C</td><td style="text-align: center;">H</td><td style="text-align: center;">O</td></tr> <tr> <td style="text-align: center;">$77.2 / 12 = 6.433$</td><td style="text-align: center;">$11.4 / 1 = 11.4$</td><td style="text-align: center;">$11.4 / 16 = 0.7125$</td></tr> <tr> <td style="text-align: center;">9(.03)</td><td style="text-align: center;">16</td><td style="text-align: center;">1</td></tr> </table> $M_r(\text{C}_9\text{H}_{16}\text{O}) = 140$ so molecular formula of V = $\text{C}_{18}\text{H}_{32}\text{O}_2$	C	H	O	$77.2 / 12 = 6.433$	$11.4 / 1 = 11.4$	$11.4 / 16 = 0.7125$	9(.03)	16	1	3
C	H	O									
$77.2 / 12 = 6.433$	$11.4 / 1 = 11.4$	$11.4 / 16 = 0.7125$									
9(.03)	16	1									

37. M/J 22/P21/Q4(b)(ii)

4(b)(ii)	$3.196 \text{ g Br}_2 = 3.196 / 159.8 (= 0.02 \text{ mol Br}_2)$ AND $2.8(00) / 280 (= 0.01 \text{ mol V})$ 2 alkene / 2 C=C (groups)	1
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38. M/J 22/P22/Q6(b)

6(b)	<table><tr><td>C</td><td>:</td><td>H</td><td>:</td><td>O</td></tr><tr><td>41.38 / 12</td><td></td><td>3.45 / 1</td><td></td><td>55.17 / 16</td></tr><tr><td>3.45</td><td></td><td>3.45</td><td></td><td>3.45</td></tr></table> (so C ₍₁₎ H ₍₁₎ O ₍₁₎)	C	:	H	:	O	41.38 / 12		3.45 / 1		55.17 / 16	3.45		3.45		3.45	1
C	:	H	:	O													
41.38 / 12		3.45 / 1		55.17 / 16													
3.45		3.45		3.45													

39. M/J 22/P23/Q1(f)(i)

1(f)(i)	$M_r = 63.5$ $(1.05 / 63.5) \times 6.022 \times 10^{23} = 9.958 \times 10^{21}$ OR 9.96×10^{21}	1
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Unit 3

CHEMICAL BONDING

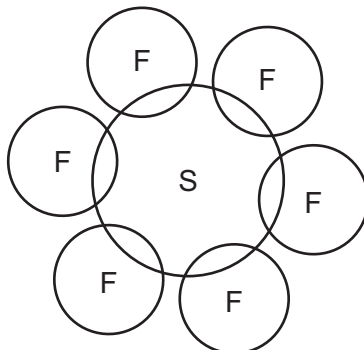
57 items · 145 marks

Syllabus sections in this unit

- 3.1 Electronegativity and bonding
- 3.2 Ionic bonding
- 3.3 Metallic bonding
- 3.4 Covalent bonding and coordinate (dative covalent) bonding
- 3.5 Shapes of molecules
- 3.6 Intermolecular forces, electronegativity and bond properties
- 3.7 Dot-and-cross diagrams

Unit 3: Chemical bonding**1. M/J 26/P21/Q1(a)**

- 1 (a) (i) Complete the dot-and-cross diagram in Figure 1.1 to show the arrangement of outer shell electrons in a molecule of SF_6 .

**Figure 1.1**

- (ii) Name the shape of a molecule of SF_6 .

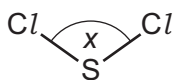
[1]

[1]

2. M/J 26/P21/Q1(c)

- (c) A molecule of SCl_2 is polar.

- (i) Predict bond angle x in Figure 1.2 using VSEPR theory. Explain your answer.

**Figure 1.2**

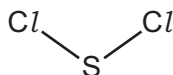
bond angle $x =$

°

explanation

[3]

- (ii) Show on Figure 1.3 all the dipoles present in the molecule of SCl_2 .

**Figure 1.3**

[1]

3. M/J 26/P21/Q1(d)

(d) SCl_2 and H_2O both contain instantaneous dipole–induced dipole attractions and permanent dipole attractions.

(i) Complete Table 1.2.

Table 1.2

	SCl_2	H_2O
total number of electrons per molecule		10

[1]

(ii) Identify which molecule, SCl_2 or H_2O , has the stronger instantaneous dipole–induced dipole attractions. Explain your answer.

[1]

(iii) H_2O has a greater dipole moment than SCl_2 . Explain why.

[1]

4. M/J 26/P22/Q1(b)

Context

1 Oxygen reacts with many elements.

(b) Lithium reacts with oxygen to produce lithium oxide, Li_2O .

Li_2O is a white solid with a melting point over 1400°C .

Draw a dot-and-cross diagram to show the arrangement of **all** the electrons in Li_2O .

[2]

5. M/J 26/P24/Q1(c)

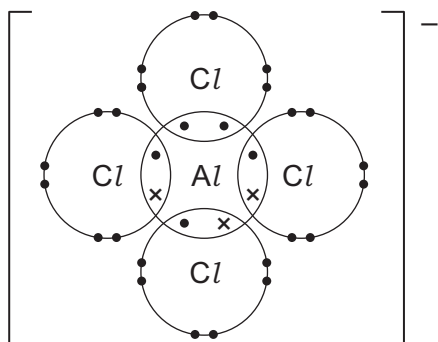
Context

- 1** Aluminium is a Group 13 element that forms a number of compounds.

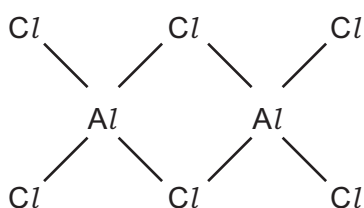
Table 1.1

atom or ion	number of protons	number of neutrons
^{27}Al		
$^{28}\text{Al}^{3+}$		

Context: Fig. 1.3

**Figure 1.3**

- (c) AlCl_3 dimerises to form Al_2Cl_6 . Al_2Cl_6 is shown in Figure 1.2.

**Figure 1.2**

Describe how coordinate (dative covalent) bonds form when two AlCl_3 molecules combine to form Al_2Cl_6 .

[1]

6. M/J 26/P24/Q1(d)

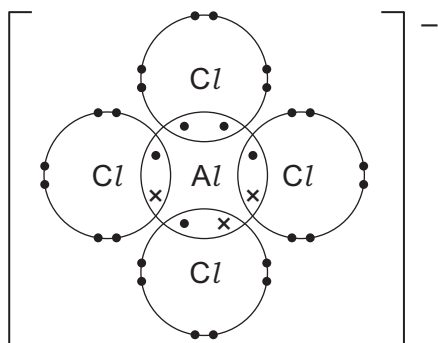
Context

- 1** Aluminium is a Group 13 element that forms a number of compounds.

Table 1.1

atom or ion	number of protons	number of neutrons
^{27}Al		
$^{28}\text{Al}^{3+}$		

- (d)** Aluminium forms the AlCl_4^- ion. The dot-and-cross diagram for AlCl_4^- in Figure 1.3 shows the arrangement of outer electrons.

**Figure 1.3**

- (i)** Use Figure 1.3 and VSEPR theory to name the shape of the AlCl_4^- ion and to predict the $\text{Cl}-\text{Al}-\text{Cl}$ bond angle.

name of shape

bond angle

[2]

- (ii)** Draw the three-dimensional shape of the AlCl_4^- ion.

[1]

7. M/J 26/P24/Q2(a)(i)

- 2 (a) (i)** Nitrogen, N_2 , contains sigma (σ) and pi (π) bonds.

Describe how a pi (π) bond is formed.

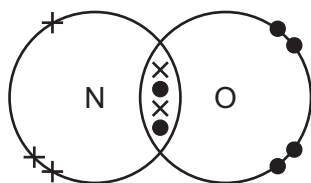
[2]

8. MAR 26/P22/Q2(a)(ii)

Context

2 Nitrogen monoxide, NO, is an atmospheric pollutant.

Context: part (a) (introduction)

(a) Figure 2.1 shows the dot-and-cross diagram of NO.**Figure 2.1****(ii)** In NO, the N and O atoms are bonded by a sigma (σ) and a pi (π) bond.

Complete Figure 2.2 to show the pi bond in NO that is formed from orbital overlap.

**Figure 2.2**

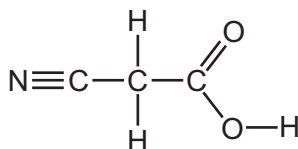
[1]

9. MAR 26/P22/Q4(b)

Context

4 Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.**(b)** $\text{CH}_2(\text{CN})\text{COOH}$ is very soluble in water because it can form hydrogen bonds with water.Complete Figure 4.2 to show **one** hydrogen bond between a molecule of $\text{CH}_2(\text{CN})\text{COOH}$ and a molecule of H_2O .

Label the hydrogen bond clearly, and show any relevant partial charges and lone pairs of electrons.

**Figure 4.2**

[3]

10. O/N 25/P22/Q3(d)(ii)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

Context: part (d) (introduction)

- (d)** Hexane, C_6H_{14} , has four structural isomers.

Fig. 3.1 shows hexane and two of its structural isomers.

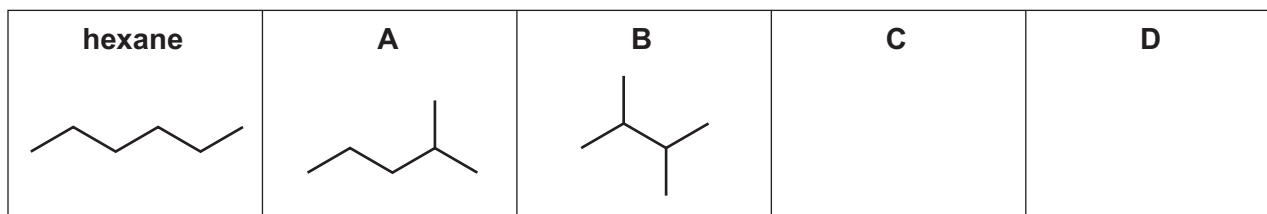


Fig. 3.1

- (ii)** **A**, **B** and hexane have different boiling points.

Arrange **A**, **B** and hexane in order of increasing boiling point.

Explain your answer.

lowest < < highest
[3]

11. O/N 25/P24/Q1(b)(v)

Context

- 1** The Group 17 elements are oxidising agents.

Context: part (b) (introduction)

- (b)** Cl_2 and Br_2 can each react with NH_3 to give N_2 and a hydrogen halide, HX .



The relative bond strengths of $X-X$ and $H-X$ determine the difference in enthalpy change of the two reactions.

- (v)** I_2 reacts with NH_3 to form NI_3 .

Predict the shape of a molecule of NI_3 . Explain your answer.

shape

explanation

[2]

12. M/J 25/P21/Q1(a)(ii)

Context: part (a) (introduction)

- 1** **(a)** Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.
- (ii)** Explain how solid sodium conducts electricity.

[1]

13. M/J 25/P23/Q4(b)

Context

- 4** The reaction between $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and $\text{HCl}(\text{aq})$ is monitored at constant temperature.

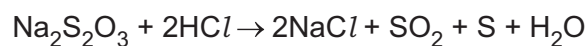
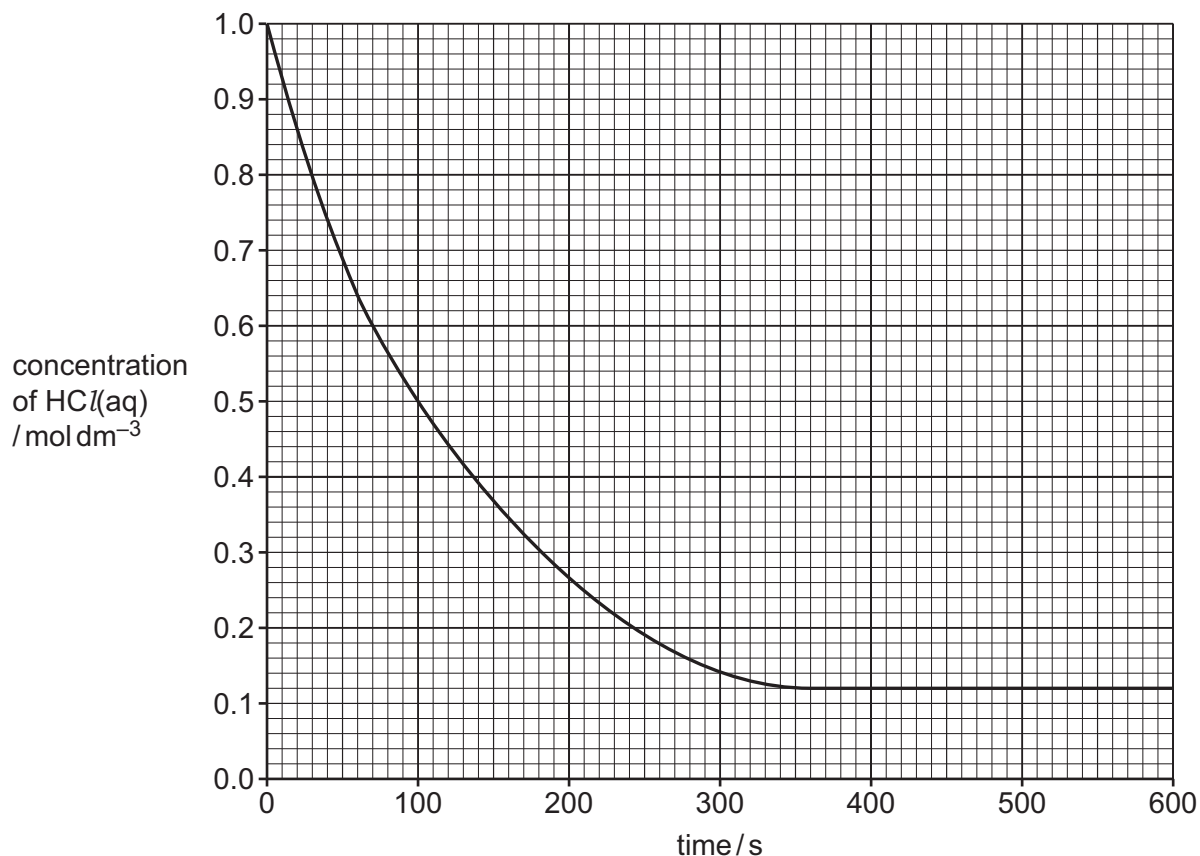
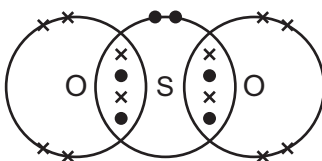


Fig. 4.1 shows how the concentration of $\text{HCl}(\text{aq})$ varies with time.

**Fig. 4.1**

- (b)** Fig. 4.2 shows a possible arrangement of outer-shell electrons in one SO_2 molecule.

**Fig. 4.2**

- (i)** Use Fig. 4.2 to predict the shape and bond angle of a molecule of SO_2 .

shape

bond angle

°

[2]

- (ii)** Use Table 4.1 to predict the strength of the dipole moment of SO_2 , if any, compared to that of H_2O . Explain your answer.

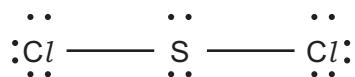
Table 4.1

	H	O	S
electronegativity	2.1	3.5	2.6

[2]

14. M/J 25/P24/Q2(a)

- 2 (a) Fig. 2.1 shows the covalent bonds and lone pairs of electrons in a molecule of SCl_2 .

**Fig. 2.1**

- (i) Use VSEPR theory to predict the shape of a molecule of SCl_2 .

[1]

- (ii) Predict the bond angle in a molecule of SCl_2 .

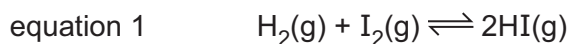
°

[1]

15. M/J 25/P24/Q3(d)(i)

Context

- 3 $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ react to form $\text{HI}(\text{g})$ in a reversible reaction, as shown in equation 1.



In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

Context: part (d) (introduction)

- (d) The value of K_c for the dissociation of HCl at 298 K is 5.50×10^{-34} .



- (i) Define covalent bond.

[1]

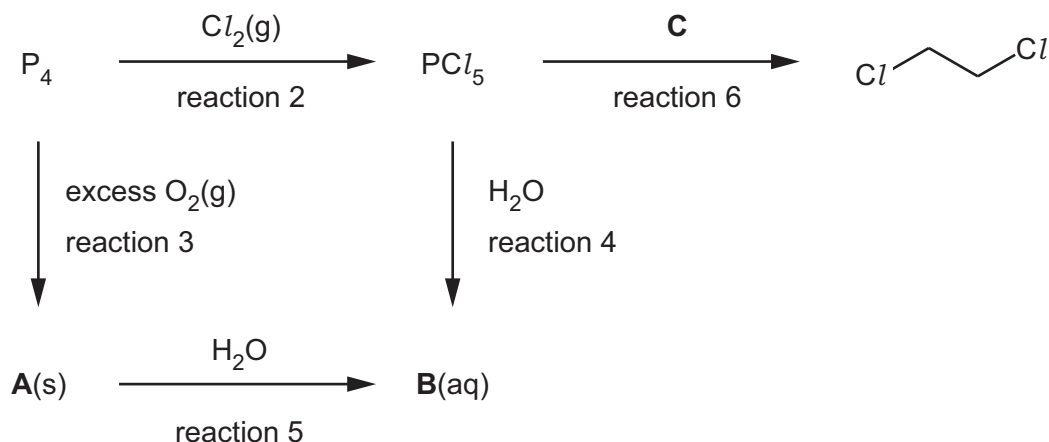
16. MAR 25/P22/Q1(b)(ii)

Context

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

Context: part (b) (introduction)

(b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2 .

**Fig. 1.1**

(ii) Predict the shape of a molecule of PCl_5 .

[1]

17. MAR 25/P22/Q2(a)

2 **(a)** The Group 2 elements show metallic bonding.

Define metallic bonding.

[2]

18. O/N 24/P21/Q3(a)(iii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (a) (introduction)

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

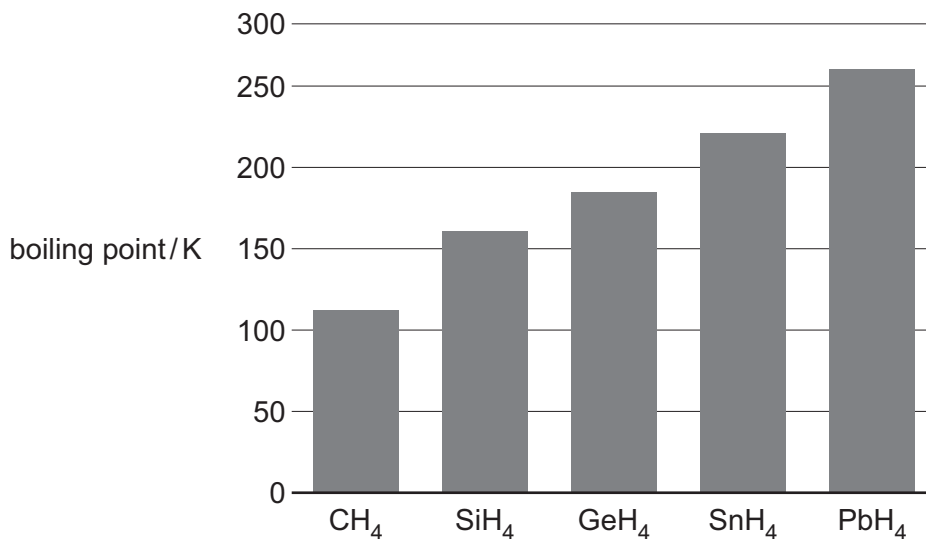
(iii) Explain why Sn has good electrical conductivity.

[1]

19. O/N 24/P21/Q3(d)

Context

- 3** The Group 14 elements show a change from non-metallic to metallic character down the group.
- (d)** Fig. 3.2 shows the boiling points of the simplest hydrides of the Group 14 elements, C to Pb.

**Fig. 3.2**

- (i)** Explain the trend in the boiling points of the Group 14 hydrides shown in Fig. 3.2.

[2]

- (ii)** Deduce the shape of a molecule of SiH₄.

[1]

20. O/N 24/P22/Q2(a)(ii)

Context

- 2** Oxygen is a Group 16 element.

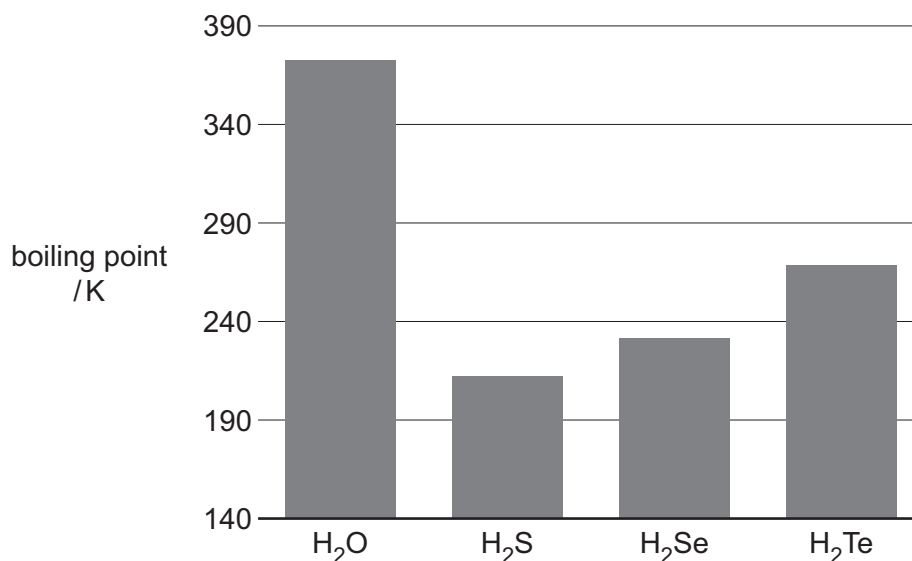
- (ii)** Draw a dot-and-cross diagram to show the species present in Al₂O₃.

Draw outer electrons only.

[1]

21. O/N 24/P22/Q2(c)

Context

2 Oxygen is a Group 16 element.**(c)** Fig. 2.1 shows the boiling points of H_2O and other Group 16 hydrides.**Fig. 2.1****(i)** Explain the trend in the boiling points of the Group 16 hydrides H_2S to H_2Te .

[2]

(ii) Explain why the boiling point of H_2O is much higher than that of H_2S .

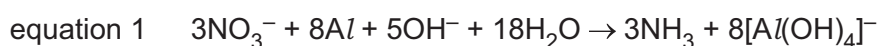
[1]

22. O/N 24/P22/Q3(b)(iv)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (b) (introduction)

(b) A common test for nitrates is the reaction with NaOH and Al . Equation 1 shows the reaction.**(iv)** Suggest the shape of the $[\text{Al}(\text{OH})_4]^-$ ion.

[1]

23. O/N 24/P22/Q3(f)(i)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (f) (introduction)

(f) At very high temperatures, phosphorus can form P_2 molecules. P_2 contains a triple bond, $\text{P}\equiv\text{P}$.**(i)** Describe the formation of the $\text{P}\equiv\text{P}$ bond in terms of orbital overlap.

[2]

24. O/N 24/P23/Q3(a)(iii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (a) (introduction)

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(iii) Explain why Sn has good electrical conductivity.

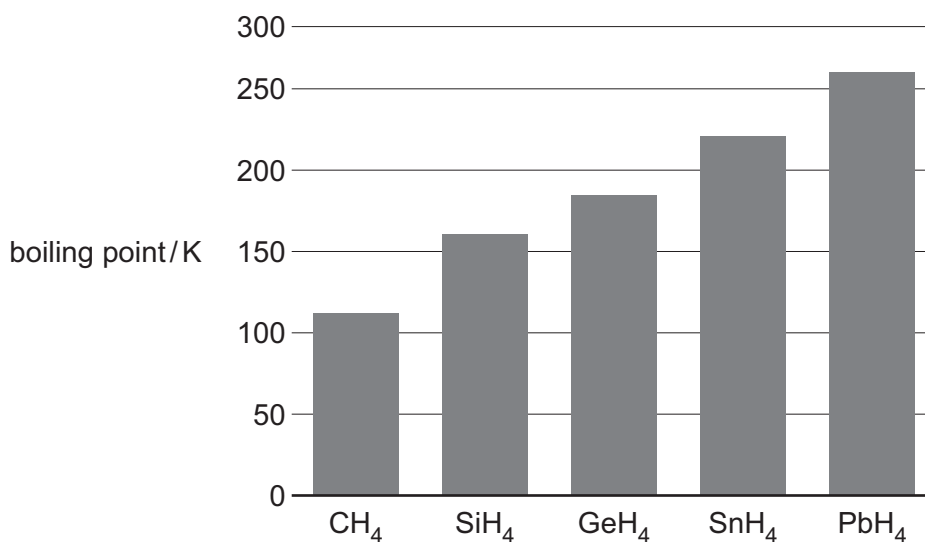
[1]

25. O/N 24/P23/Q3(d)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(d) Fig. 3.2 shows the boiling points of the simplest hydrides of the Group 14 elements, C to Pb.

**Fig. 3.2**

(i) Explain the trend in the boiling points of the Group 14 hydrides shown in Fig. 3.2.

[2]

(ii) Deduce the shape of a molecule of SiH₄.

[1]

26. M/J 24/P21/Q2(b)

(b) A molecule of SCl_2 contains two S–Cl covalent bonds.

- (i) Complete the dot-and-cross diagram in Fig. 2.1 to show the arrangement of the outer electrons in a molecule of SCl_2 .

Use × to show electrons from the chlorine atoms.

Use • to show electrons from the sulfur atom.

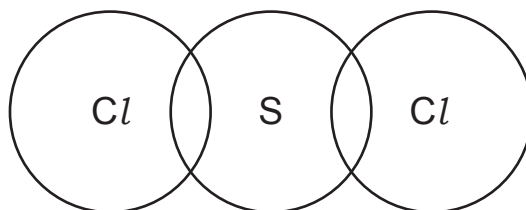


Fig. 2.1

[2]

- (ii) Predict the shape of, and bond angle in, a molecule of SCl_2 by using VSEPR theory.

shape

bond angle

[2]

27. M/J 24/P21/Q5(a)

Context

- 5 Hydrocarbon molecules contain covalent bonds.

(a) Define covalent bond.

[1]

28. M/J 24/P22/Q1(c)

(c) Draw a labelled diagram to show the structure and bonding in sodium.

[1]

29. M/J 24/P22/Q3(d)

(d) SCN^- (aq) is colourless.

Complete the dot-and-cross diagram in Fig. 3.1 to show the arrangement of outer electrons in an SCN^- ion.

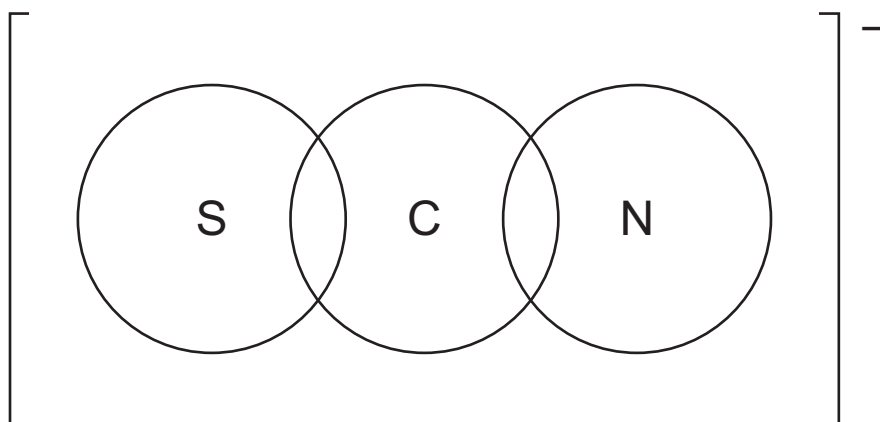


Fig. 3.1

[2]

30. M/J 24/P23/Q1(a)(ii)

(ii) Covalent bonds can be σ bonds or π bonds.

Complete Table 1.1 to show the number of σ and π bonds in a molecule of N_2 and to describe how the orbitals overlap to form σ and π bonds.

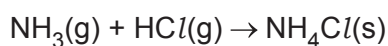
Table 1.1

	σ bond	π bond
number of bonds in N_2		
how the orbitals overlap		

[4]

31. M/J 24/P23/Q4(a)

4 (a) NH_3 (g) reacts with HCl (g) to produce NH_4Cl (s), as shown.



Draw a diagram to show the ionic, covalent and coordinate bonding present in a formula unit of NH_4Cl .

[2]

32. MAR 24/P22/Q1(a)

Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

(a) Bismuth has metallic bonding.

Draw a labelled diagram to show the metallic bonding in bismuth.

[1]

33. MAR 24/P22/Q2(c)(i)

Context

- 2** Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

Context: part (c) (introduction)

- (c) An excess of Cl_2 reacts with phosphorus to form PCl_5 .

(i) PCl_5 is a simple molecule in the gas phase.

It also exists in a solid form as two ions, PCl_4^+ and PCl_6^- .

Complete Table 2.2 to identify the shapes of each of these species.

Table 2.2

species	PCl_5	PCl_4^+	PCl_6^-
shape		tetrahedral	

[2]

34. MAR 24/P22/Q3(d)

Context

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

(d) The N_2F_2 molecule has a double covalent bond between its nitrogen atoms. This consists of a σ and a π bond.

(i) Complete Fig. 3.2 to show the dot-and-cross diagram for N_2F_2 .

Show outer electrons only.

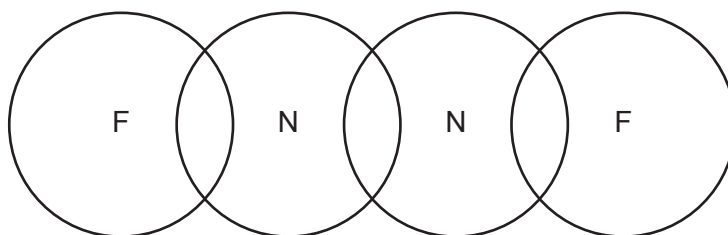


Fig. 3.2

[2]

(ii) Deduce the hybridisation of the N atoms in N_2F_2 .

[1]

(iii) Draw a diagram of the π bond between the N atoms in N_2F_2 and describe how it forms.

[2]

35. O/N 23/P21/Q1(d)(iii)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

Context: part (d) (introduction)

- (d)** POC_l₃ shows similar chemical properties to PCl₅.

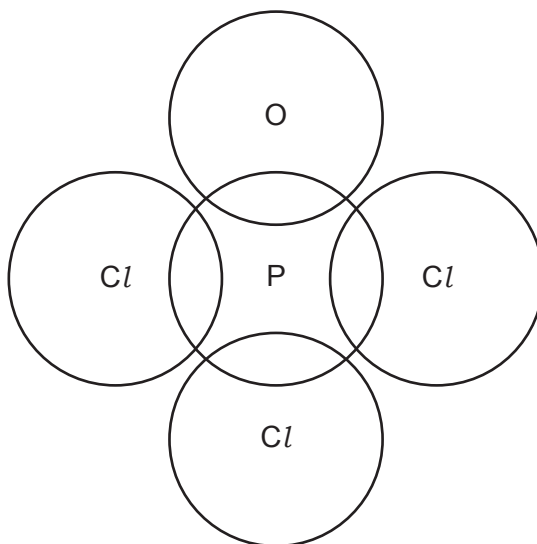
POC_l₃ has a melting point of 1°C and a boiling point of 106°C.

POC_l₃ reacts vigorously with water, forming misty fumes and an acidic solution.

- (iii)** POC_l₃ contains a double covalent bond between P and O.

Complete the dot-and-cross diagram, in Fig. 1.1, to show the bonding in POC_l₃.

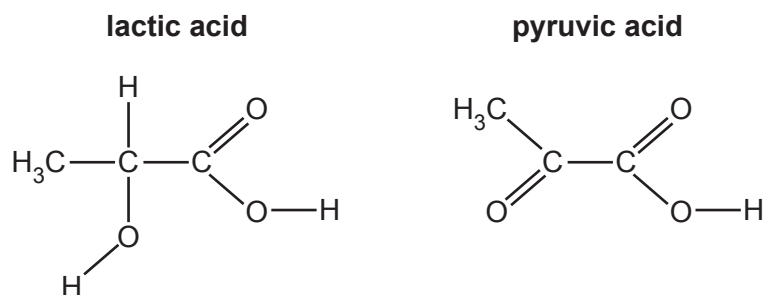
Show outer shell electrons only.

**Fig. 1.1**

36. O/N 23/P22/Q4(a)(iii)

Context

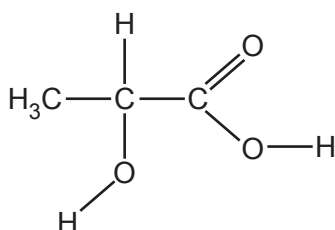
- 4** Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, CH_3COCOOH , both contain two functional groups.

**Fig. 4.1**

- (iii)** Lactic acid forms hydrogen bonds with water.

Complete Fig. 4.2 to show the formation of a hydrogen bond between one molecule of lactic acid and one molecule of water.

Label the hydrogen bond. Show any relevant dipoles and lone pairs of electrons.

**Fig. 4.2**

[3]

37. O/N 23/P23/Q1(d)(iii)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

Context: part (d) (introduction)

- (d)** POC_l₃ shows similar chemical properties to PCl₅.

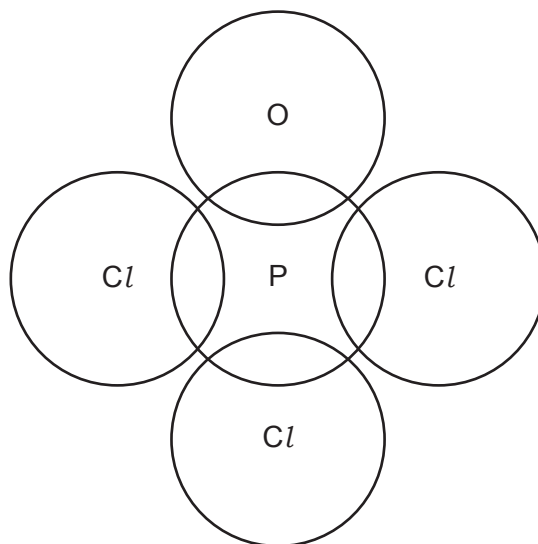
POC_l₃ has a melting point of 1°C and a boiling point of 106°C.

POC_l₃ reacts vigorously with water, forming misty fumes and an acidic solution.

- (iii)** POC_l₃ contains a double covalent bond between P and O.

Complete the dot-and-cross diagram, in Fig. 1.1, to show the bonding in POC_l₃.

Show outer shell electrons only.

**Fig. 1.1**

38. M/J 23/P21/Q1(e)

Context

- 1** Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.

(e) Te reacts with F_2 at 150°C to form TeF_x . Molecules of TeF_x are octahedral with bond angles of 90° .

Explain why TeF_x is octahedral with bond angles of 90° .

[2]

39. M/J 23/P22/Q1(b)

Context

- 1** The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point / K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

- (b) (i) Define electronegativity.

[1]

- (ii) Explain why electronegativity increases across a period.

[2]

- (iii) Name the strongest intermolecular force that exists between $\text{NH}_3(\text{l})$ molecules.

[1]

- (iv) Draw a diagram to show the formation of the strongest intermolecular force between **two** molecules of $\text{NH}_3(\text{l})$.

Include any relevant lone pairs of electrons and dipoles.

[2]

- (v) The melting points of ice and ammonia are shown in Table 1.2.

Table 1.2

solid	melting point / K
ice	273
ammonia	195

Suggest **two** reasons for the difference in the melting points of ice and ammonia.

[2]

40. M/J 23/P23/Q1(a)(ii)

Context

- 1** Copper is used in electrical equipment. It has a melting point of 1085°C .

- (ii) Draw a labelled diagram to show the bonding present in copper.

[1]

41. MAR 23/P22/Q1(a)

Context

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/ kJ mol^{-1}	1310	519	1090	1310	1000
second ionisation energy/ kJ mol^{-1}	—	7300	2350	3390	2260

- (a) (i) Define electronegativity.

[1]

- (ii) O and S are in Group 16.

Explain the difference in the Pauling electronegativity values of O and S.

[2]

42. MAR 23/P22/Q1(b)

Context

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/ kJ mol^{-1}	1310	519	1090	1310	1000
second ionisation energy/ kJ mol^{-1}	—	7300	2350	3390	2260

- (b) (i) LiH is an ionic compound.

Draw a dot-and-cross diagram of LiH.

Include **all** electrons.

[2]

- (ii) Suggest the shape of a molecule of H_2S .

[1]

43. MAR 23/P22/Q1(e)(i)

Context

- 1** The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy / kJ mol^{-1}	1310	519	1090	1310	1000
second ionisation energy / kJ mol^{-1}	—	7300	2350	3390	2260

- (e) (i)** Complete Table 1.2 by placing a tick (✓) to show which of the compounds have molecules with an overall dipole moment.

Table 1.2

compound	O=C=O	O=S=O	S=C=S	S=C=O
overall dipole moment				

[2]

44. MAR 23/P22/Q2(a)

Context

- 2** The Group 2 elements Mg to Ba are all silvery-white reactive metals.

- (a) (i)** Draw a labelled diagram to show the bonding and structure of the Group 2 metals at room temperature.

[2]

- (ii)** Explain why Mg has a higher electrical conductivity than Na.

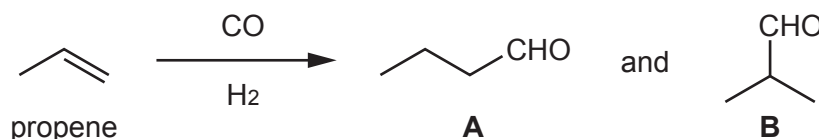
[1]

45. MAR 23/P22/Q3(d)(iv)

Context

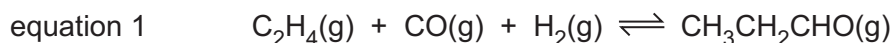
- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

**Fig. 3.1**

Context: part (d) (introduction)

- (d)** The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



At atmospheric pressure a cobalt-based catalyst is used in this reaction.

- (iv)** The reaction mixture is cooled to collect CH₃CH₂CHO as a liquid.

Identify all types of van der Waals' forces that are present between molecules of CH₃CH₂CHO.

[1]

46. O/N 22/P21/Q2(a)

Context

- 2** Magnesium shows reactions typical of a Group 2 metal.

- (a)** Draw a labelled diagram to show the bonding in magnesium metal.

[2]

47. O/N 22/P21/Q3(d)

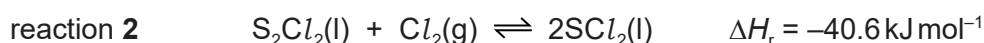
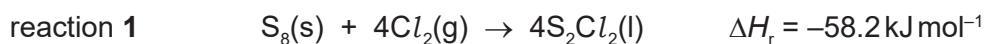
also Unit 4 (2 marks), Unit 5 (2 marks), Unit 13 (2 marks), Unit 7 (1 mark); this unit 4 marks

Context

- 3** Some of the common chlorides of Period 3 elements are shown in the list.



- (d)** Sulfur, S₈, reacts with chlorine to form several different chlorides. The most common are S₂Cl₂ and SCl₂. SCl₂ forms when sulfur reacts with an excess of chlorine.



- (i)** SCl₂ is a cherry-red liquid that reacts vigorously with water to form an acidic solution.

Use this information to deduce the bonding and structure shown by SCl₂.
Explain your answer.

[2]

- (ii) Calculate the enthalpy change of formation, ΔH_f , of $\text{SCl}_2(\text{l})$. You may find it useful to use Hess's Law to construct an energy cycle.

enthalpy change of formation of $\text{SCl}_2(\text{l})$, $\Delta H_f =$ kJ mol^{-1}
[2]

- (iii) State the effect of a decrease in pressure on the position of equilibrium in reaction 2. Explain your answer.

[1]

Fig. 3.1 shows the two structural isomers of S_2Cl_2 .

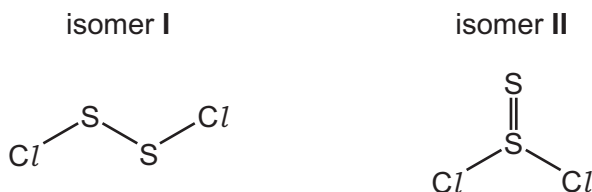


Fig. 3.1

- (iv) Define the term structural isomer.

[2]

- (v) Suggest a value for the Cl-S-S bond angle in isomer I. Explain your answer.

bond angle = °

explanation

[2]

- (vi) Draw a dot-and-cross diagram to show the bonding in isomer II. Show outer shell electrons only.

[2]

48. O/N 22/P22/Q1(a)

Context

- 1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

- (a) Ionic and covalent bonds both involve an electrostatic attraction between different species.

Identify the species that are electrostatically attracted to one another in:

- an ionic bond
- a covalent bond.

[2]

49. O/N 22/P22/Q1(c)(iii)

Context

1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

Context: part (c) (introduction)

(c) NH_4^+ is a Brønsted–Lowry acid.

(iii) The nitrogen atom in NH_4^+ is sp^3 hybridised. sp^3 orbitals form from the mixing of one 2s and three 2p orbitals.

Sketch the shapes of a 2s and a $2p_x$ orbital on the axes in Fig. 1.1.

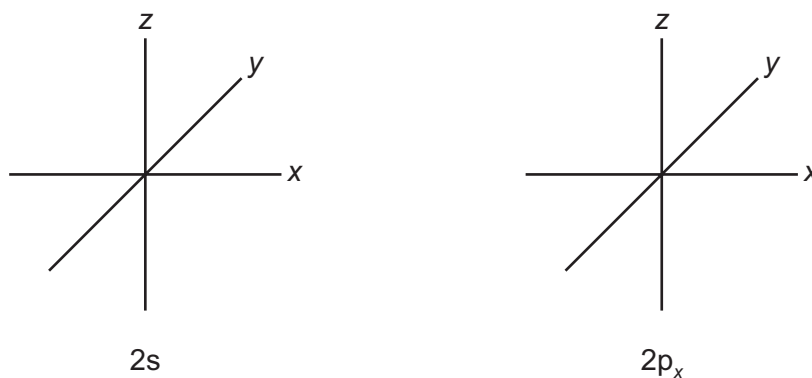


Fig. 1.1

[2]

50. O/N 22/P22/Q2(c)(i)

Context

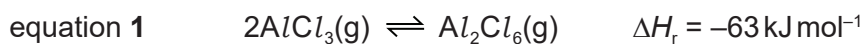
2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

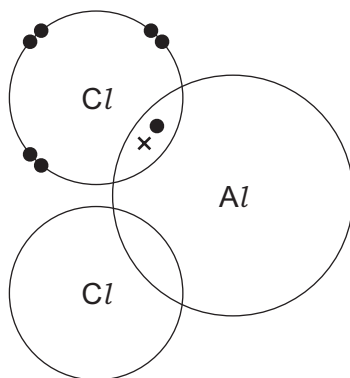
Period 3 chloride	NaCl	AlCl ₃	SiCl ₄	PCl ₅	PCl ₃	SCl ₂
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

Context: part (c) (introduction)

(c) In the gas phase, AlCl₃(g) exists at equilibrium with Al₂Cl₆(g) as shown.



(i) Complete the dot-and-cross diagram to show the bonding in Al₂Cl₆.



[2]

51. O/N 22/P22/Q2(e)(ii)

Context

2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl ₃	SiCl ₄	PCl ₅	PCl ₃	SCl ₂
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

(ii) SCl₂ has $M_r = 103.1$ and is a liquid at room temperature. SBr₂ has $M_r = 191.9$ and is a gas at room temperature.

Explain the difference in the physical state of SCl₂ and SBr₂. Give your answer in terms of intermolecular forces.

[2]

52. O/N 22/P22/Q2(f)(i)

Context

- 2** The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl ₃	SiCl ₄	PCl ₅	PCl ₃	SCl ₂
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

Context: part (f) (introduction)

- (f)** Bismuth is a dense metal in the same group as phosphorus.

- (i)** Draw a labelled diagram to show the bonding in bismuth metal.

[2]

53. O/N 22/P23/Q2(a)

Context

- 2** Magnesium shows reactions typical of a Group 2 metal.

- (a)** Draw a labelled diagram to show the bonding in magnesium metal.

[2]

54. O/N 22/P23/Q3(d)

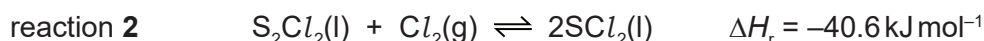
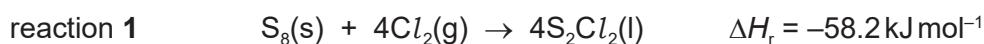
also Unit 4 (2 marks), Unit 5 (2 marks), Unit 13 (2 marks), Unit 7 (1 mark); this unit 4 marks

Context

- 3** Some of the common chlorides of Period 3 elements are shown in the list.



- (d)** Sulfur, S₈, reacts with chlorine to form several different chlorides. The most common are S₂Cl₂ and SCl₂. SCl₂ forms when sulfur reacts with an excess of chlorine.



- (i)** SCl₂ is a cherry-red liquid that reacts vigorously with water to form an acidic solution.

Use this information to deduce the bonding and structure shown by SCl₂.
Explain your answer.

[2]

- (ii)** Calculate the enthalpy change of formation, ΔH_f , of SCl₂(l). You may find it useful to use Hess's Law to construct an energy cycle.

enthalpy change of formation of SCl₂(l), $\Delta H_f =$ kJ mol⁻¹
[2]

- (iii) State the effect of a decrease in pressure on the position of equilibrium in reaction 2. Explain your answer.

[1]

Fig. 3.1 shows the two structural isomers of S_2Cl_2 .

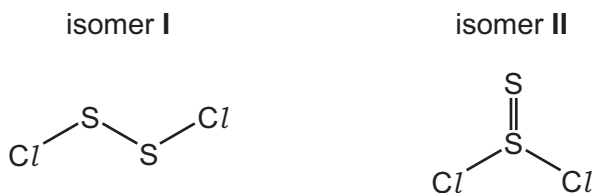


Fig. 3.1

- (iv) Define the term structural isomer.

[2]

- (v) Suggest a value for the $Cl-S-S$ bond angle in isomer I. Explain your answer.

bond angle = °

explanation

[2]

- (vi) Draw a dot-and-cross diagram to show the bonding in isomer II. Show outer shell electrons only.

[2]

55. M/J 22/P21/Q2(a)

Context

- 2 Nitrogen molecules, $N_2(g)$, contain two atoms attracted to each other by a triple covalent bond.

- (a) Describe how the triple covalent bond forms in a $N_2(g)$ molecule. Refer to orbital overlap and hybridisation in your answer.

[3]

56. M/J 22/P22/Q2(b)

Context

- 2 Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

Radium chloride, $RaCl_2$, has a melting point of $900^\circ C$ and is soluble in water.

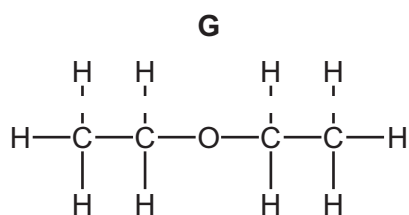
- (b) Draw a dot-and-cross diagram to show the arrangement of outer electrons in $RaCl_2$.

[1]

57. M/J 22/P23/Q3(a)(i)

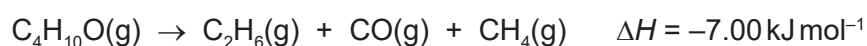
Context

3 **G** belongs to a group of compounds called ethers.

**Fig. 3.1**

Context: part (a) (introduction)

(a) When **G**, $\text{C}_4\text{H}_{10}\text{O}$, is heated, thermal decomposition occurs.



- (i)** The atoms in a molecule of CO are held together by a triple covalent bond. One of these bonds is a coordinate (dative covalent) bond.

Draw a dot-and-cross diagram to show the arrangement of outer electrons in a CO molecule.

Use $\bullet$ to represent electrons from an oxygen atom.

Use $\times$ to represent electrons from a carbon atom.

[1]

Answers Section

1. M/J 26/P21/Q1(a)

1(a)(i)		1	A any symbol for electrons – symbols can all be the same or different A electrons can be on the lines or in the spaces A non-bonding electrons need not drawn in pairs I inner electrons
1(a)(ii)	octahedral	1	A square bipyramidal A octahedron A near miss in spelling e.g. octohedral I octagonal / octagon

2. M/J 26/P21/Q1(c)

1(c)(i)	M1 104.5(°) M2 2 bonding pairs (of electrons) and 2 lone pairs (of electrons) M3 lone pairs (of electrons) repel more than bonding pairs (of electrons)	3	M1 A range 103–105° inclusive I reference to stated shape even if incorrect I reference to tetrahedral geometry and 109.5 in text A 'bonding pairs' and 'lone pairs' assuming they are describing electrons unless stated otherwise. A 2 shared pairs for 2 bonding pairs For 2 lone pairs: A 2 pair of lone pairs A 2 non-bonding pairs I S—Cl bond / 2 bonds for bonding pair I any electrons on outer shell of chlorine R one unpaired electron If M2 is not awarded from text, award M2 for a suitably annotated Figure 1.2 or dot-and-cross diagram The electrons in both bond pairs and lone pairs must be shown on Figure 1.2 M3 A repulsion from lone pair greater than from bond pair M3 A repulsion lone pair – lone pair > lone pair – bonding pair > bonding pair – bonding pair M3 A repulsion lone pair – bonding pair > bonding pair – bonding pair Max 1 for 2 bonding pairs and 1 lone pair AND between 107–109° inclusive
1(c)(ii)	S as δ+ AND Both Cl as δ-	1	A δ+ δ+ / δ2+ / 2+δ on S A +δ and -δ I Δ I any electrons drawn on the atoms R ± without δ BOD TWO overall dipole direction arrows parallel to bonds I ONE overall dipole direction arrow / arrow

3. M/J 26/P21/Q1(d)

1(d)(i)		SCl ₂	H ₂ O	1	
	total number of electrons per molecule	50	10		

1(d)(ii)	SCl_2 AND greater number of electrons (per molecule) OR H_2O has weaker id (id) as it has smaller number of electrons (per molecule)	1 greater = higher = larger = bigger = more M1 awarded if incorrect number of electrons are given in 1(d)(i) AND still greater than 10 A ecf from 1(d)(i) for $\text{SCl}_2 < \text{H}_2\text{O}$ AND fewer (than 10) electrons per molecule if less than 10 shown in 1(d)(i) A ecf from 1(d)(i) $\text{SCl}_2 = \text{H}_2\text{O}$ AND same number of electrons per molecule if 10 shown in 1(d)(i) I greater electron density / electron cloud I SCl_2 has more valence / outer electrons
1(d)(iii)	electronegativity difference (between H and O) is greater (than S and Cl)	1 A larger for greater / bigger owtte A H–O has large electronegativity difference and S–Cl has small electronegativity difference I H_2O is more polar than SCl_2 I oxygen/O is more electronegative than chlorine I any comments on hydrogen bonding R H_2O is more electronegative than SCl_2 R O₂ is more electronegative than Cl₂

4. M/J 26/P22/Q1(b)

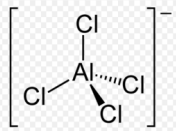
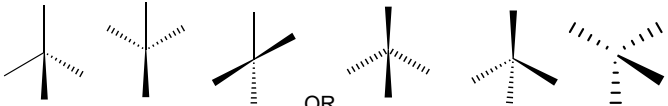
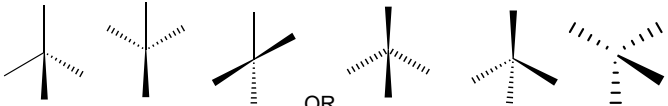
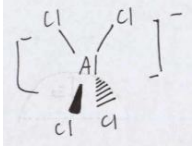
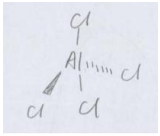
1(b)		2 M1 correct charges M1 A –2 for O ion and +1 / 1+ for Li ion M2 all else fully correct I symbols for Li and O A any symbol for electrons Max 1 for incorrect identification of type of bonding as covalent as long as all else correct Max 1 for any ionic compound with a positively charged species and a negatively charged species
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5. M/J 26/P24/Q1(c)

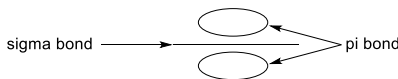
1(c)	Lone pair (of electrons) on a Cl (could be seen as : on a diagram) AND donated / shared with Al (could be seen by the arrow on a diagram) OR A fully annotated Figure 1.2:	1 A full annotations on Figure 1.2 i.e. lone pair AND dative arrow on <u>both</u> Cl to Al bonds R dative bond shown from Al to Cl on diagram R chloride / chlorine ion BOD lone pair A chlorine A donates / gives / shares / provides a lone pair / electron pair / both electrons <u>to Al</u>
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6. M/J 26/P24/Q1(d)

1(d)(i)	M1 (name of shape) tetrahedral M2 (bond angle) 109–109.5°	2 M1 A tetrahedra M1 A phonetically spelt name of shape
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1(d)(ii)	3D shape required:  Acceptable 3D descriptions of different bond types behind the plane and out of the plane are shown below  are all interchangeable  are all interchangeable Although the 3D structure shown in the syllabus is drawn as the example shown in the mark scheme it is acceptable to allow a combination of bond types shown below (3D structure has two types of wedge adjacent)  OR  A BOD any structures which show unambiguous use of bond shapes representing bonds behind the plane and out of the plane as long as they are different.	1 I negative charge (not required) I bond angles I dative bonds A BOD  R 
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7. M/J 26/P24/Q2(a)(i)

2(a)(i)	M1 sideways overlap of (adjacent) p orbitals M2 above and below the (covalent/σ) bond	2 M1 A unambiguous labelled diagram A 'sideways overlap of p' A sideways = lateral = side on = side by side = side to side = parallel I indirect I horizontal / vertical I shoulder-to-shoulder M2 A above and below the (covalent) bond A labelled diagram: 
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8. MAR 26/P22/Q2(a)(ii)

2(a)(ii)		1
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9. MAR 26/P22/Q4(b)

4(b)	M1 clearly distinct / labelled H bond M2 from lone pair on any oxygen to H atom attached to O M3 with three correct successive partial charges across two molecules	3
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10. O/N 25/P22/Q3(d)(ii)

3(d)(ii)	M1 B < A < hexane. M2 hexane / unbranched has stronger / more IMFs M3 hexane / unbranched requires most / more energy	3
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11. O/N 25/P24/Q1(b)(v)

1(b)(v)	M1 (trigonal) pyramidal M2 three bonding pairs AND one lone pair (of electrons)	2
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12. M/J 25/P21/Q1(a)(ii)

1(a)(ii)	delocalised electrons move (through the structure)	1
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13. M/J 25/P23/Q4(b)

4(b)(i)	M1 non-linear M2 bond angle – $120 > x \geq 116^\circ$	2
4(b)(ii)	M1 weaker dipole moment (in SO_2) M2 electronegativity (difference) is larger (/ 1.4) for H and O than for S and O (/ 0.9) OR S=O bond is less polar than H—O OR stronger dipoles in S=O / SO_2	2

14. M/J 25/P24/Q2(a)

2(a)(i)	non-linear	1
2(a)(ii)	104.5°	1

15. M/J 25/P24/Q3(d)(i)

3(d)(i)	(electrostatic) attraction between the nuclei (of two atoms) and a shared pair of electrons	1
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16. MAR 25/P22/Q1(b)(ii)

1(b)(ii)	trigonal bipyramid(al)	1
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17. MAR 25/P22/Q2(a)

2(a)	electrostatic attraction between positive metal ions / cations and delocalised electrons	2
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18. O/N 24/P21/Q3(a)(iii)

3(a)(iii)	its delocalised electrons are free to move	1
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19. O/N 24/P21/Q3(d)

3(d)(i)	M1 molecules CH_4 to PbH_4 have greater number of electrons / number of electrons increase	1
	M2 greater / stronger instantaneous dipole–induced forces / London forces / dispersion forces OR more energy required to overcome the instantaneous dipole–induced forces / London forces / dispersion forces	1
3(d)(ii)	tetrahedral	1

20. O/N 24/P22/Q2(a)(ii)

2(a)(ii)	Al^{3+} 	1
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21. O/N 24/P22/Q2(c)

2(c)(i)	(molecules) H_2S to H_2Te / they have greater number of electrons	1
	stronger instantaneous dipole–induced dipole / London dispersion forces OR more energy required to overcome the instantaneous dipole–induced dipole / London dispersion forces	1
2(c)(ii)	(only) H_2O has hydrogen bonding AND hydrogen bonding (much) stronger than the other intermolecular forces OR H_2O has hydrogen bonding AND because of higher electronegativity of O compared to S OR stronger van der Waals' forces because (only) H_2O has hydrogen bonding OR hydrogen bond AND increased strength of permanent dipoles in H_2O outweighs the increase in strength of id-id in the others	1

22. O/N 24/P22/Q3(b)(iv)

3(b)(iv)	tetrahedral	1
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23. O/N 24/P22/Q3(f)(i)

3(f)(i)	M1 one sigma / σ bond AND head-on (overlap of) p / sp (orbitals)	1
	M2 two pi / π bond(s) AND side-on (overlap of / involving) p (orbitals)	1

24. O/N 24/P23/Q3(a)(iii)

3(a)(iii)	its delocalised electrons are free to move	1
-----------	--	----------

25. O/N 24/P23/Q3(d)

3(d)(i)	M1 molecules CH_4 to PbH_4 have greater number of electrons / number of electrons increase	1
	M2 greater / stronger instantaneous dipole–induced forces / London forces / dispersion forces OR more energy required to overcome the instantaneous dipole–induced forces / London forces / dispersion forces	1
3(d)(ii)	tetrahedral	1

26. M/J 24/P21/Q2(b)

2(b)(i)	M1 1 shared pair of electrons between each S–Cl shown as $\bullet\bullet$ M2 6 non-bonding electrons around each Cl shown as $\times$ AND 4 non-bonding electrons around S shown as $\bullet$	2
2(b)(ii)	M1 shape non-linear M2 bond angle $103\text{--}105^\circ$	2

27. M/J 24/P21/Q5(a)

5(a)	(electrostatic) attraction between nuclei of two atoms and shared pair of electrons	1
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28. M/J 24/P22/Q1(c)

1(c)	positively charged ions delocalised electrons labelled diagram shows: - regular arrangement of minimum of 2×2 circles containing + for positive ions AND surrounded by (sea of) delocalised electrons	1
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29. M/J 24/P22/Q3(d)

3(d)	M1 6 electrons between C and N AND 2 electrons between C and S M2 3 lone pairs around S AND none around C AND 1 lone pair around N	2
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30. M/J 24/P23/Q1(a)(ii)

1(a)(ii)	<table border="1"> <thead> <tr> <th></th><th>σ bond</th><th>π bond</th></tr> </thead> <tbody> <tr> <td>no of bonds</td><td>1</td><td>2</td></tr> <tr> <td>how orbitals overlap</td><td>direct (orbital overlap)</td><td>sideways (overlap of adjacent p orbitals)</td></tr> </tbody> </table>		σ bond	π bond	no of bonds	1	2	how orbitals overlap	direct (orbital overlap)	sideways (overlap of adjacent p orbitals)	4
	σ bond	π bond									
no of bonds	1	2									
how orbitals overlap	direct (orbital overlap)	sideways (overlap of adjacent p orbitals)									

31. M/J 24/P23/Q4(a)

4(a)	OR Correct representation or label showing M1 (ionic bond between) NH_4^+ and Cl^- AND 4 N—H covalent bonds OR 4 shared pairs of electrons in NH_4 M2 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons	2
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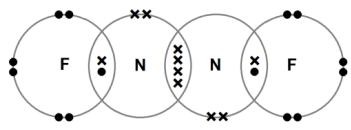
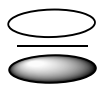
32. MAR 24/P22/Q1(a)

1(a)	labelled diagram shows: - regular arrangement of positive ions AND surrounded by / (sea of) delocalised electrons	1
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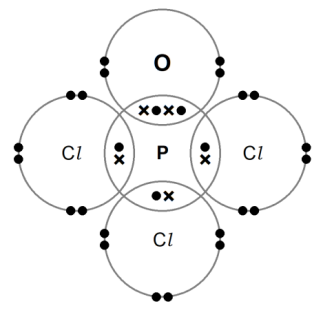
33. MAR 24/P22/Q2(c)(i)

2(c)(i)	• trigonal bipyramidal • octahedral	2
---------	-------------------------------------	----------

34. MAR 24/P22/Q3(d)

3(d)(i)	 correct bonding pairs	1
	lone pairs	1
3(d)(ii)	sp^2	1
3(d)(iii)	 M1 shows the pi (π) orbital	1
	M2 refers to overlap of (two) (2)p orbitals side-on / above and below the plane	1

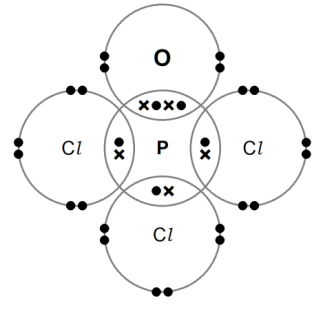
35. O/N 23/P21/Q1(d)(iii)

1(d)(iii)	 Correct bonding e^-	1
	rest of electrons correct (i.e. to make 32 electrons in total)	1

36. O/N 23/P22/Q4(a)(iii)

4(a)(iii)	dashed (or distinctly labelled) bond between O and H in different molecules of H-O-H and propanoic acid	1
	correct sequence of three partial charges over the H-bond $H^{\delta+} \quad O^{\delta-} \cdots H^{\delta+}$ OR $O^{\delta-} \quad H^{\delta+} \cdots O^{\delta-}$	1
	lone pair on O in line with H-bond	1

37. O/N 23/P23/Q1(d)(iii)

1(d)(iii)	 Correct bonding e^-	1
	rest of electrons correct (i.e. to make 32 electrons in total)	1

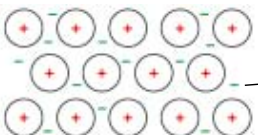
38. M/J 23/P21/Q1(e)

1(e)	M1 6 bonding pairs (of electrons and 0 lone pairs) OR coordination number = 6 OR x = 6 M2 bonding pairs repel (equally)	2
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39. M/J 23/P22/Q1(b)

1(b)(i)	power of an atom to attract electrons to itself	1
1(b)(ii)	(across a period) - increase in nuclear charge - similar shielding (so) increase in nuclear attraction for bonding / outer / valence electrons OR bonding / outer / valence electron(s) are more strongly attracted to nucleus Two correct for one mark, three correct for two marks	2
1(b)(iii)	hydrogen bond	1
1(b)(iv)	M1 link shown as a dashed line between the lone pair of electrons from N of one NH ₃ to one H on other NH ₃ M2 minimum 3 correct partial charges (on adjacent atoms) over two NH ₃ molecules EITHER $\delta^- \text{N} - \delta^+ \text{H} \cdots \delta^- \text{N}$ OR $\delta^+ \text{H} \cdots \delta^- \text{N} - \delta^+ \text{H}$	2
1(b)(v)	M1 O is more electronegative than N M2 two H-bonds per water molecule : 1 per ammonia molecule.	2

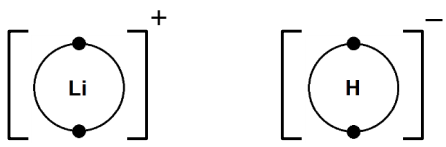
40. M/J 23/P23/Q1(a)(ii)

1(a)(ii)	 delocalised electrons M1 diagram showing minimum of 4 particles - circles containing just a + do not have to be labelled - empty circles / circles with Cu must be labelled + ion / positive ion / cation / Cuⁿ⁺ AND with the circles surrounded by electrons shown as e⁻ / – OR little circles labelled delocalised electrons	1
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41. MAR 23/P22/Q1(a)

1(a)(i)	power of an atom to attract electrons to itself	1
1(a)(ii)	- O lower nuclear charge / lower proton number - O has (one) fewer shell than S / less shielding - greater attraction (for nucleus) in O	2

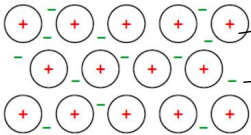
42. MAR 23/P22/Q1(b)

1(b)(i)		2
1(b)(ii)	non-linear	1

43. MAR 23/P22/Q1(e)(i)

1(e)(i)						2
	compound	O=C=O	O=S=O	S=C=S	S=C=O	
	overall dipole moment		✓		✓	

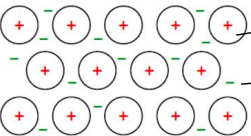
44. MAR 23/P22/Q2(a)

2(a)(i)	 M1 diagram showing minimum of 4 particles (in total in 2 rows) - circles containing X^{n+} do not have to be labelled - must be labelled as 'ion' OR empty circles / circles with X must be labelled + ion / positive ion / cation / X^{n+} AND with the circles surrounded by electrons shown as e^- / – OR little circles labelled electrons	1
	M2 label / legend showing delocalised electrons	1
2(a)(ii)	Mg has more delocalised e^- (than Na)	1

45. MAR 23/P22/Q3(d)(iv)

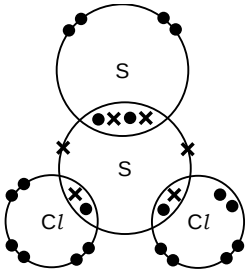
3(d)(iv)	instantaneous dipole—induced dipole / id—id AND permanent dipole—permanent dipole / pd—pd	1
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46. O/N 22/P21/Q2(a)

2(a)	 M1 diagram showing minimum of 4 particles (in total in two rows) (circles) - circles containing Mg^{n+} do not have to be labelled - must be labelled as 'ion' OR empty circles / circles with Mg must be labelled + ion / positive ion / cation / Mgⁿ⁺ AND - circles surrounded by electrons shown as e^- / – OR in an area around the circles labelled as 'electrons' OR little circles labelled electrons OR electrons drawn only on perimeter of structure	1
	M2 label / legend showing <u>delocalised</u> electrons	1

47. O/N 22/P21/Q3(d)

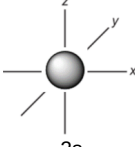
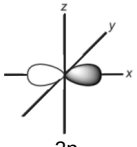
3(d)(i)	M1 (structure $\Rightarrow$) simple / molecular, because it has a low melting / boiling point	1
	M2 (bonding $\Rightarrow$) covalent, because it is hydrolysed	1
3(d)(ii)	Correct use of enthalpy values / correct direction / use of arrows (with or without an energy cycle) M1 $(-58.2) + (-40.6)$ OR (-98.8) M2 $[(-58.2) + 4(-40.6) = 8\Delta H_f]$ Correct calculation and correct stoichiometry $\Delta H_f = -27.6 \text{ (kJ mol}^{-1}\text{)}$	2
3(d)(iii)	equilibrium moves to left AND more moles / molecules of gas on LHS	1
3(d)(iv)	(molecules / isomers with) the same molecular formula	1
	but different structural formulae	1
3(d)(v)	M1 $90^\circ < Cl-S-S \leq 108^\circ$	1
	M2 sulfur has two lone pairs (of e^-) (and two bonding pairs) AND repulsion from lone pairs (greater)	1

3(d)(vi)	 bonding electrons	1
	all other electrons correct	1

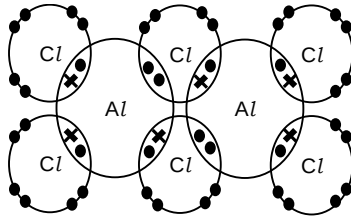
48. O/N 22/P22/Q1(a)

1(a)	M1 ionic = oppositely charged ions / cations AND anions	1
	M2 covalent = shared pair(s) of electrons AND two nuclei	1

49. O/N 22/P22/Q1(c)(iii)

1(c)(iii)	 	2
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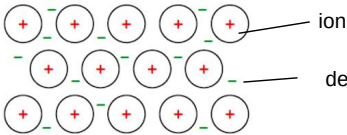
50. O/N 22/P22/Q2(c)(i)

2(c)(i)		1
	<i>'dot-and-cross' diagram to show</i> M1 correct number and connectivity of Cl and Al atoms M2 2 correct dative bonds (••) between Cl and Al of AlCl₃ neighbour AND correct number of non-bonding electrons around each atom of Al₂Cl₆	1

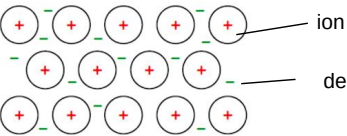
51. O/N 22/P22/Q2(e)(ii)

2(e)(ii)	M1 sum of instantaneous dipole – dipole attractions and permanent dipole attractions in SCl₂ is greater than those in SBr₂ OR permanent dipole-permanent dipole (forces) in SCl₂ are stronger OR intermolecular forces are stronger in SCl₂ ora	1
	M2 SCl₂ (molecule / bond) is more polar than SBr₂ OR Cl is more electronegative than Br so SCl₂ has stronger (permanent) dipoles ora	1

52. O/N 22/P22/Q2(f)(i)

2(f)(i)	 M1 diagram showing minimum of 4 particles (in total in 2 rows) - circles containing Bi^{n+} do not have to be labelled - must be labelled as 'ion' OR empty circles / circles with Bi must be labelled + ion / positive ion / cation / Bi^{n+} AND - with the circles surrounded by electrons shown as e^- / – OR little circles labelled electrons	1
	M2 label / legend showing <u>delocalised</u> electrons	1

53. O/N 22/P23/Q2(a)

2(a)	 M1 diagram showing minimum of 4 particles (in total in two rows) (circles) - circles containing Mg^{n+} do not have to be labelled - must be labelled as 'ion' OR empty circles / circles with Mg must be labelled + ion / positive ion / cation / Mg^{n+} AND - circles surrounded by electrons shown as e^- / – OR in an area around the circles labelled as 'electrons' OR little circles labelled electrons OR electrons drawn only on perimeter of structure	1
	M2 label / legend showing <u>delocalised</u> electrons	1

54. O/N 22/P23/Q3(d)

3(d)(i)	M1 (structure =) simple / molecular, because it has a low melting / boiling point	1
	M2 (bonding =) covalent, because it is hydrolysed	1
3(d)(ii)	Correct use of enthalpy values / correct direction / use of arrows (with or without an energy cycle) M1 $(-58.2) + (-40.6)$ OR (-98.8) M2 $[(-58.2 + 4(-40.6)) = 8\Delta H_f]$ Correct calculation and correct stoichiometry $\Delta H_f = -27.6 \text{ (kJ mol}^{-1}\text{)}$	2
3(d)(iii)	equilibrium moves to left AND more moles / molecules of gas on LHS	1
3(d)(iv)	(molecules / isomers with) the same molecular formula	1
	but different structural formulae	1
3(d)(v)	M1 $90^\circ < \text{Cl-S-S} \leq 108^\circ$	1
	M2 sulfur has two lone pairs (of e^-) (and two bonding pairs) AND repulsion from lone pairs (greater)	1

3(d)(vi)	bonding electrons	1
	all other electrons correct	1

55. M/J 22/P21/Q2(a)

2(a)	M1 one sigma / σ bond and two pi / π bonds M2 sp hybridisation (in each N atom) M3 sigma / σ forms from direct / head-on / end-on overlap of orbitals AND pi / π forms sideways / lateral overlap of (p) orbitals	3
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56. M/J 22/P22/Q2(b)

2(b)	Ra^{2+} and 2 x Cl^- 0 electrons surrounding 2Ra AND 8 electrons surrounding Cl	1
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57. M/J 22/P23/Q3(a)(i)

3(a)(i)		1
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Unit 4

STATES OF MATTER

35 items · 84 marks

Syllabus sections in this unit

4.1 The gaseous state: ideal and real gases and $pV = nRT$

4.2 Bonding and structure

Unit 4: States of matter**1. M/J 26/P21/Q2(c)(ii)**

Context

2 The Contact process uses sulfur to produce sulfuric acid.**stage 1** Sulfur is burned in excess air.**stage 2** $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$ **stage 3** SO_3 reacts to produce H_2SO_4 .

Context: part (c) (introduction)

(c) CaCO_3 and O_2 are added to the waste gases from the Contact process before they escape. This prevents the release of SO_2 into the atmosphere.**(ii)** Use the ideal gas equation to calculate the volume of $\text{O}_2(\text{g})$ that reacts with 0.500 mol of SO_2 at 400°C and 101 kPa in equation 1.

Include units in your answer.

volume of $\text{O}_2(\text{g}) =$

units

[4]

2. M/J 26/P23/Q2(b)

Context

2 Group 18 elements exist as colourless gases at room temperature and pressure.**Table 2.1**

atom	relative volume of atom
neon	172
argon	287
krypton	355
xenon	446

(b) Identify **one** Group 18 element from Table 2.1 that behaves most like an ideal gas under standard conditions. Explain your answer.

[3]

3. M/J 26/P23/Q2(c)

Context

- 2** Group 18 elements exist as colourless gases at room temperature and pressure.

Table 2.1

atom	relative volume of atom
neon	172
argon	287
krypton	355
xenon	446

- (c) A sample of 1.775 g of gas **X** occupies 617.0 cm³ at 300 K and 101 kPa.

Calculate the relative molecular mass, M_r , of **X** assuming that **X** behaves as an ideal gas.

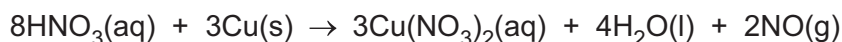
$$M_r \text{ of } \mathbf{X} = \quad [4]$$

4. MAR 26/P22/Q2(b)

Context

- 2** Nitrogen monoxide, NO, is an atmospheric pollutant.

- (b) In the laboratory, NO is made in the reaction of copper metal with dilute nitric acid. The reaction is kept cold to prevent the oxidation of NO.



A student reacts 0.00393 mol Cu with excess dilute HNO₃(aq) at 5 °C and 101 kPa.

Calculate the maximum volume, in cm³, of NO(g) that is produced under these conditions.

$$\text{volume of NO(g) produced} = \quad \text{cm}^3 [4]$$

5. O/N 25/P24/Q2(a)

Context

- 2** Aluminium oxide, Al₂O₃, and phosphorus(V) oxide, P₄O₁₀, are both used as reagents and catalysts.

- (a) The melting point of Al₂O₃ is 2072 °C. The melting point of P₄O₁₀ is 340 °C.

Explain the difference in the melting points of these two compounds.

[3]

6. M/J 25/P21/Q1(a)(i)

Context: part (a) (introduction)

- 1** (a) Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.

- (i) Name the type of bonding present in sodium and in sodium oxide.

bonding in sodium

bonding in sodium oxide

[1]

7. M/J 25/P21/Q1(a)(iii)

Context: part (a) (introduction)

- 1 (a) Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.
- (iii) Explain why sodium oxide conducts electricity when molten.

[1]

8. M/J 25/P21/Q1(d)

- (d) Table 1.2 shows melting points of some oxides.

Table 1.2

oxide	melting point/°C
SO ₂	-73
H ₂ O	0
SO ₃	17
SiO ₂	1610
MgO	2852
Al ₂ O ₃	2072

A student suggests the following hypothesis.

*The higher the oxidation number of the element combined with oxygen,
the higher the melting point of the oxide.*

Use Table 1.2 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

[2]

9. M/J 25/P22/Q1(a)

- 1 (a) Diamond and graphite are both crystalline solids made from carbon atoms. Graphite conducts electricity. Diamond does **not** conduct electricity.
- (i) Name the type of lattice structure present in the crystalline solids diamond and graphite.
- (ii) Explain how graphite conducts electricity.
- (iii) Explain why diamond does **not** conduct electricity.

[1]

[1]

[1]

10. M/J 25/P22/Q1(d)

(d) The melting points of different oxides are shown in Table 1.1.

Table 1.1

oxide	melting point / °C	force of attraction broken during melting
SO ₂	-73	
H ₂ O	0	
SiO ₂	1610	
Na ₂ O	1132	
MgO	2852	

- (i) Complete Table 1.1 by identifying the strongest force of attraction in each oxide that is broken during melting. Use the abbreviations below.

i.d. = instantaneous dipole–induced dipole

p.d. = permanent dipole–permanent dipole

H = hydrogen bond

C = covalent bond

I = ionic bond

[3]

- (ii) A student suggests the following hypothesis.

*The stronger the covalent bond between atoms in non-metal oxides,
the higher the melting point.*

Use Table 1.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

[2]

11. M/J 25/P23/Q2(c)(ii)

Context

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

Context: part (c) (introduction)

(c) Table 2.1 shows the melting points of some oxides.

Table 2.1

oxide	melting point/°C
SO ₂	−73
H ₂ O	0
SO ₃	17
SiO ₂	1610
Na ₂ O	1132
MgO	2852
Al ₂ O ₃	2072

(ii) A student suggests the following hypothesis.

The melting point of an ionically bonded oxide is only determined by the charge on the cation.

Use Table 2.1 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

[2]

12. M/J 25/P24/Q1(a)

Context

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(a) Complete Table 1.1 by identifying the lattice structures of the crystalline solids of Mg, Si and P.

Table 1.1

element	Mg	Si	P
lattice structure in crystalline solid			

[3]

13. M/J 25/P24/Q6(b)

Context

6 P, Q and R are three different hydrocarbon molecules.

(b) Q is a volatile hydrocarbon.

0.194 g of gaseous Q occupies a volume of 71.5 cm³ at 100 °C and 100 kPa.

Use the ideal gas equation to calculate the M_r of Q.

M_r of Q =

[2]

14. O/N 24/P21/Q3(a)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (a) (introduction)

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.

Table 3.1

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1. [3]

(ii) Identify the lattice structure shown by graphite.

[1]

15. O/N 24/P21/Q3(e)(iii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (e) (introduction)

(e) Silicon readily reacts with elements of high electronegativity.

(iii) SiO_2 is a white solid that melts above 1700°C .

SiCl_4 is a colourless liquid at room temperature.

Explain the difference in the melting points of these two compounds with reference to their structure and bonding.

[2]

16. O/N 24/P22/Q3(d)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.**(d)** Complete Table 3.1 to show the properties of nitrogen and phosphorus in their standard states.**Table 3.1**

	nitrogen	phosphorus
state and appearance of standard state	colourless gas	white solid
electrical conductivity		poor
type of bonding		
type of structure	simple	

[2]

17. O/N 24/P22/Q3(e)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.**(e)** A form of solid nitrogen has a lattice structure similar to solid iodine.

Identify the type of lattice structure of solid nitrogen.

[1]

18. O/N 24/P23/Q3(a)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (a) (introduction)

(a) Table 3.1 shows some properties of two Group 14 elements, C and Sn, in their standard states. The table is incomplete.**Table 3.1**

	C (graphite)	Sn
state and appearance in standard state	grey shiny solid	silvery solid
electrical conductivity		good
type of bonding		metallic
type of structure	giant	

(i) Complete Table 3.1.

[3]

(ii) Identify the lattice structure shown by graphite.

[1]

19. O/N 24/P23/Q3(e)(iii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (e) (introduction)

(e) Silicon readily reacts with elements of high electronegativity.

(iii) SiO_2 is a white solid that melts above 1700°C .

SiCl_4 is a colourless liquid at room temperature.

Explain the difference in the melting points of these two compounds with reference to their structure and bonding.

[2]

20. M/J 24/P21/Q2(d)(ii)

Context: part (d) (introduction)

(d) Boron nitride is a white solid that melts above 2900°C .

Fig. 2.2 shows part of the lattice structure of a crystal of boron nitride.

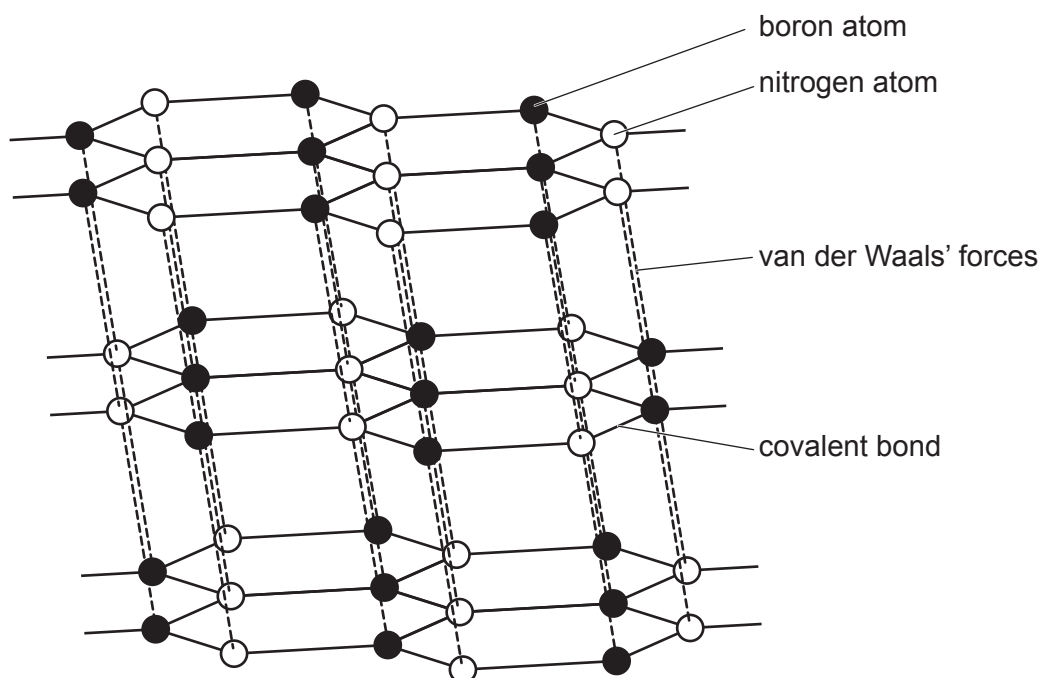


Fig. 2.2

(ii) Suggest the identity of another crystalline solid that has atoms arranged in layers similar to that of solid boron nitride.

[1]

21. M/J 24/P23/Q3(c)(i)

Context: part (c) (introduction)

(c) Boron carbide is a hard crystalline solid that has a melting point greater than 2000°C .

(i) Suggest the structure and bonding in boron carbide.

[1]

22. MAR 24/P22/Q1(b)(ii)

Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

Context: part (b) (introduction)

- (b)** Bismuth reduces water to form bismuth oxide, Bi_2O_3 . A colourless gas that ignites with a squeaky pop also forms.
- (ii)** Bi_2O_3 is a yellow insoluble solid that melts at 1090 K. The molten compound conducts electricity.

Deduce the structure and bonding of Bi_2O_3 . Explain your answer.

[2]

23. O/N 23/P21/Q1(d)(i)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy / kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^-

Context: part (d) (introduction)

- (d)** POCl_3 shows similar chemical properties to PCl_5 .

POCl_3 has a melting point of 1°C and a boiling point of 106°C .

POCl_3 reacts vigorously with water, forming misty fumes and an acidic solution.

- (i)** Explain how the information in **(d)** suggests the structure and bonding of POCl_3 is simple covalent.

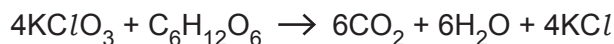
[2]

24. O/N 23/P21/Q3(d)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(d) Molten KClO_3 reacts with glucose, $\text{C}_6\text{H}_{12}\text{O}_6$.



KClO_3 melts at 630 K. At this temperature, both CO_2 and H_2O are gases.

- (i)** Use the ideal gas equation to calculate the volume, in m^3 , of one mole of gas at 630 K and $1.00 \times 10^5 \text{ Pa}$.

Show your working. Give your answer to 3 significant figures.

volume of 1 mol of gas = m^3
[1]

- (ii)** 5.00 g of $\text{C}_6\text{H}_{12}\text{O}_6$ reacts completely with molten KClO_3 .

Use your answer to **(d)(i)** to calculate the total volume of gas released at 630 K and $1.00 \times 10^5 \text{ Pa}$ in this reaction.

(If you were unable to answer **(d)(i)**, use 0.0463 m^3 in this question. This is **not** the correct answer to **(d)(i)**.)

total volume of gas released = m^3
[2]

25. O/N 23/P23/Q1(d)(i)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

Context: part (d) (introduction)

- (d)** POC_l₃ shows similar chemical properties to PC_l₅.

POC_l₃ has a melting point of 1°C and a boiling point of 106°C.

POC_l₃ reacts vigorously with water, forming misty fumes and an acidic solution.

- (i)** Explain how the information in **(d)** suggests the structure and bonding of POC_l₃ is simple covalent.

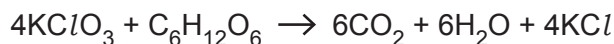
[2]

26. O/N 23/P23/Q3(d)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(d) Molten KClO_3 reacts with glucose, $\text{C}_6\text{H}_{12}\text{O}_6$.



KClO_3 melts at 630 K. At this temperature, both CO_2 and H_2O are gases.

- (i) Use the ideal gas equation to calculate the volume, in m^3 , of one mole of gas at 630 K and $1.00 \times 10^5 \text{ Pa}$.

Show your working. Give your answer to 3 significant figures.

volume of 1 mol of gas = m^3
[1]

- (ii) 5.00 g of $\text{C}_6\text{H}_{12}\text{O}_6$ reacts completely with molten KClO_3 .

Use your answer to (d)(i) to calculate the total volume of gas released at 630 K and $1.00 \times 10^5 \text{ Pa}$ in this reaction.

(If you were unable to answer (d)(i), use 0.0463 m^3 in this question. This is **not** the correct answer to (d)(i).)

total volume of gas released = m^3
[2]

27. M/J 23/P22/Q1(a)(i)-(ii)

Context

- 1** The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point / K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

- (a) (i) State the type of bonding present in magnesium and in sodium chloride.

bonding in magnesium

bonding in sodium chloride

[1]

- (ii) Explain the difference in the melting points of magnesium and sodium chloride.

[1]

28. M/J 23/P23/Q1(a)(i)

Context

- 1** Copper is used in electrical equipment. It has a melting point of 1085 °C.

(a) (i) Identify the lattice structure of copper.

[1]

29. MAR 23/P22/Q1(e)(ii)

Context

- 1** The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

Context: Table 1.2

Table 1.2

compound	O=C=O	O=S=O	S=C=S	S=C=O
overall dipole moment				

- (ii) At 150 °C and 103 kPa, all of the compounds listed in Table 1.2 are gases.

Under these conditions, 0.284 g of one of the compounds occupies a volume of 127 cm³.

Use this information to calculate the M_r of the compound. Hence, identify the compound from those given in Table 1.2.

Show your working.

 $M_r =$

identity of compound =

[3]

30. O/N 22/P22/Q2(d)

Context

- 2** The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

- (d)** A 3.30 g sample of a Period 3 chloride is heated to 500 K in a sealed flask. At this temperature, the chloride is a gas of volume 250 cm^3 and the pressure in the flask is 323 kPa.

Use the ideal gas equation $pV = nRT$ to calculate the M_r of the Period 3 chloride. Deduce its formula.

$$M_r =$$

formula of Period 3 chloride =

[3]

31. O/N 22/P22/Q2(f)(ii)

Context

- 2** The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

Context: part (f) (introduction)

- (f)** Bismuth is a dense metal in the same group as phosphorus.

- (ii)** Bismuth reacts with chlorine to form BiCl_3 .
 BiCl_3 is a solid at room temperature. It melts when heated gently.
 BiCl_3 reacts vigorously with water at room temperature to form an acidic solution.

Suggest the type of bonding and structure shown by BiCl_3 . Explain your answer.

[2]

32. M/J 22/P22/Q1(a)

- 1 (a) Magnesium has a melting point of 650°C and high electrical conductivity.

Explain these properties of magnesium by referring to its structure and bonding.

[2]

33. M/J 22/P22/Q2(a)

Context

- 2 Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

Radium chloride, RaCl_2 , has a melting point of 900°C and is soluble in water.

- (a) Predict the lattice structure of $\text{RaCl}_2(\text{s})$ based on the properties described.

[1]

34. M/J 22/P23/Q1(f)(ii)

- (ii) Copper has a melting point of 1085°C and a high electrical conductivity.

Explain these properties of copper by referring to its structure and bonding.

[2]

35. MAR 22/P22/Q2(c)(i)

Context

- 2 Some oxides of elements in Period 3 are shown.



Context: part (c) (introduction)

- (c) P_4O_6 is a white solid that has a melting point of 24°C . Solid P_4O_6 reacts with water to form H_3PO_3 .

- (i) Deduce the type of structure and bonding shown by P_4O_6 . Explain your answer.

[2]

Answers Section

1. M/J 26/P21/Q2(c)(ii)

2(c)(ii)	M1 $PV=nRT$ M2 number of mol of O_2 required $n = 0.5 \div 2 = 0.25$ M3 temperature conversion to kelvin $= 400 + 273 = 673$ M4 $vol = (M2 \times 8.31 \times M3) \div 101\,000$ $= 0.0138431$ AND m^3	4	check the answer line first. M1 BOD upper / lower case discrepancies 0.0138(431) AND m^3 ✓✓✓✓ 13.8 and dm^3 OR 13 800 and cm^3 ✓✓✓✓ 0.014 and m^3 OR 14 and dm^3 or 14 000 and cm^3 ✓✓✓✓ 0.0276(86287) AND m^3 ✓✓✓ 0.00822(7722772) AND m^3 ✓✓✓✓ 0.0164(55445) AND m^3 ✓✓ A 2 sig figs up to calculator value final answers must be correctly rounded. M1–M3 may be seen in working if not described explicitly M1 A $V = nRT / P$ or A from working in M4 M3 A 273.15 OR 273.2 for 273 M3 A $400 + 273 / 673.15 / 673.2$ M4 volume calculated correctly using nRT / P AND correct / appropriate unit M4 A ecf from M2 and / or M3
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2. M/J 26/P23/Q2(b)

2(b)	M1 neon / Ne (most like an ideal gas) M2 it has the smallest volume M3 weakest vdw / temporary dipole – induced dipole attractions / London (dispersion) forces / London forces	3	M2 and M3 comparative term required once only e.g. Ne has smaller volume and weak vdw M2 BOD 'it is the smallest (atom)' M2 I mass BOD molecules for atoms intermolecular forces / IMF's = vdw M3 BOD least for weakest M3 A 'forces of attraction between atoms' I smallest number of electrons
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3. M/J 26/P23/Q2(c)

2(c)	M1 $PV=nRT$ M2 use of appropriate pressure value with volume e.g. convert volume to m^3 and kPa to Pa OR $101 \times 617.0 \times 10^{-3}$ etc. M3 calculate value for n $n = (101\,000 \times 617.0 \times 10^{-6}) \div 8.31 \times 300$ $= 0.024997 / 0.025$ M4 $(1.775 / M_r(X) = 0.02499 / 0.025)$ $M_r = 71(.0)$	4	Final answer 71(.0) ✓✓✓✓ M1 I subsequent incorrect rearrangement M2 A from calculations which produce consistent conversion for volume and pressure ie 101 and $617(.0) \times 10^{-3}$ (I ref to units for volume & pressure even if incorrect) M3 & M4 A ecf from incorrect conversion of volume and / or pressure units M3 A ecf from transcription errors / incorrect T using correct manipulation of $PV=nRT$ M4 A ecf from number of moles calculated M4 A ecf from M3 I units
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4. MAR 26/P22/Q2(b)

2(b)	M1 moles of NO = $0.00393 \times 2 / 3 = 0.00262$ M2 where $p = 101\,000$ (Pa) $R = 8.31$ $T = 278$ $n = 0.00262$ OR $p = 101$ (kPa) M3 use of $pV = nRT$ OR $V = nRT / p$ M4 volume = $5.99 \times 10^{-5} \times 10^6 = 59.9 \text{ cm}^3$ OR $5.99 \times 10^{-2} \times 10^3$	4
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5. O/N 25/P24/Q2(a)

2(a)	M1 Al_2O_3 is ionic M2 P_4O_{10} is a (simple) molecule M3 ionic bonds are stronger than intermolecular forces	3
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6. M/J 25/P21/Q1(a)(i)

1(a)(i)	sodium – metallic AND sodium oxide – ionic	1
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7. M/J 25/P21/Q1(a)(iii)

1(a)(iii)	(sodium and oxide) ions move (through the liquid)	1
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8. M/J 25/P21/Q1(d)

1(d)	<i>All three for two marks:</i> • false • comparison or statement of oxidation number • correct link to melting point from their comparison or statement	2
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9. M/J 25/P22/Q1(a)

1(a)(i)	(both) giant molecular	1
1(a)(ii)	delocalised electrons move through the structure OR delocalised electrons move through / between the layers	1
1(a)(iii)	no delocalised electrons (to carry charge through the structure) OR <u>all</u> (4) outer electrons are used in bonding	1

10. M/J 25/P22/Q1(d)

1(d)(i)	<table border="1"> <tbody> <tr> <td>SO_2</td><td>–73</td><td>pd</td></tr> <tr> <td>H_2O</td><td>0</td><td>H</td></tr> <tr> <td>SiO_2</td><td>1600</td><td>C</td></tr> <tr> <td>Na_2O</td><td>1132</td><td>I</td></tr> <tr> <td>MgO</td><td>2852</td><td>I</td></tr> </tbody> </table>	SO_2	–73	pd	H_2O	0	H	SiO_2	1600	C	Na_2O	1132	I	MgO	2852	I	3
SO_2	–73	pd															
H_2O	0	H															
SiO_2	1600	C															
Na_2O	1132	I															
MgO	2852	I															
1(d)(ii)	<i>Evaluate hypotheses based on the information candidates have given in the table</i> M1 not enough information M2 SO_2 and / or H_2O break intermolecular forces OR SO_2 and / or H_2O does not break covalent bonds OR only SiO_2 breaks covalent bonds	2															

11. M/J 25/P23/Q2(c)(ii)

2(c)(ii)	• false • Al_2O_3 melting point is lower than MgO • (but) charge on Al ion is greater than that on Mg Any 2 points = 1 mark , all 3 points = 2 marks	2
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12. M/J 25/P24/Q1(a)

1(a)	<table border="1"> <tr> <td><i>Mg</i></td><td><i>Si</i></td><td><i>P</i></td></tr> <tr> <td>M1 Giant metallic</td><td>M2 Giant molecular</td><td>M3 Simple molecular</td></tr> </table>	<i>Mg</i>	<i>Si</i>	<i>P</i>	M1 Giant metallic	M2 Giant molecular	M3 Simple molecular	3
<i>Mg</i>	<i>Si</i>	<i>P</i>						
M1 Giant metallic	M2 Giant molecular	M3 Simple molecular						

13. M/J 25/P24/Q6(b)

6(b)	M1 convert temp to 373 K (and use) M2 correct rearrangement and use $7.15 = n \cdot 8.31(4) \times T$ to find n AND $Mr = 0.194 / n$ $Mr = 84.1$	2
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14. O/N 24/P21/Q3(a)(i)-(ii)

3(a)(i)	<table border="1"> <tr> <td>element</td><td>C (graphite)</td><td>Sn</td></tr> <tr> <td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr> <tr> <td>electrical conductivity</td><td>• good/ conductor</td><td>good</td></tr> <tr> <td>type of bonding</td><td>• covalent</td><td>metallic</td></tr> <tr> <td>type of structure</td><td>giant</td><td>• giant</td></tr> </table>	element	C (graphite)	Sn	state and appearance	grey shiny solid	silvery solid	electrical conductivity	• good/ conductor	good	type of bonding	• covalent	metallic	type of structure	giant	• giant	3
element	C (graphite)	Sn															
state and appearance	grey shiny solid	silvery solid															
electrical conductivity	• good/ conductor	good															
type of bonding	• covalent	metallic															
type of structure	giant	• giant															
3(a)(ii)	giant molecular	1															

15. O/N 24/P21/Q3(e)(iii)

3(e)(iii)	M1 SiO_2 is giant covalent AND $SiCl_4$ is simple covalent / simple molecular / molecular M2 so need less energy to overcome IMF's in $SiCl_4$, compared to the bonds in SiO_2 ora	2
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16. O/N 24/P22/Q3(d)

3(d)	<table border="1"> <tr> <td>element</td><td>nitrogen</td><td>phosphorus</td></tr> <tr> <td>state and appearance</td><td>colourless gas</td><td>white solid</td></tr> <tr> <td>electrical conductivity</td><td>• poor</td><td>poor</td></tr> <tr> <td>type of bonding</td><td>• covalent</td><td>• covalent</td></tr> <tr> <td>type of structure</td><td>simple</td><td>• simple</td></tr> </table>	element	nitrogen	phosphorus	state and appearance	colourless gas	white solid	electrical conductivity	• poor	poor	type of bonding	• covalent	• covalent	type of structure	simple	• simple	2
element	nitrogen	phosphorus															
state and appearance	colourless gas	white solid															
electrical conductivity	• poor	poor															
type of bonding	• covalent	• covalent															
type of structure	simple	• simple															

17. O/N 24/P22/Q3(e)

3(e)	simple molecular (lattice structure)	1
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18. O/N 24/P23/Q3(a)(i)-(ii)

3(a)(i)	element	C (graphite)	Sn	3
	state and appearance	grey shiny solid	silvery solid	
	electrical conductivity	• good/ conductor	good	
	type of bonding	• covalent	metallic	
	type of structure	giant	• giant	
3(a)(ii)	giant molecular			1

19. O/N 24/P23/Q3(e)(iii)

3(e)(iii)	M1 SiO₂ is giant covalent AND SiCl₄ is simple covalent / simple molecular / molecular M2 so need less energy to overcome IMF's in SiCl₄, compared to the bonds in SiO₂ ora	2
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20. M/J 24/P21/Q2(d)(ii)

2(d)(ii)	graphite	1
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21. M/J 24/P23/Q3(c)(i)

3(c)(i)	giant (molecular and) covalent	1
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22. MAR 24/P22/Q1(b)(ii)

1(b)(ii)	giant AND has a high melting point	1
	ionic AND conducts electricity (only) when molten	1

23. O/N 23/P21/Q1(d)(i)

1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	1
	(vigorous reaction with water suggests) hydrolysis	1

24. O/N 23/P21/Q3(d)

3(d)(i)	$pV = nRT \therefore V = 8.31 \times 630 \div 100\,000 = \underline{\underline{0.0524}}$ (m ³)	1
3(d)(ii)	moles of gas = $12 \times 5.00 / 180$	1
	volume of gas = $M1 \times \text{(d)(i)} = \underline{\underline{0.0175}}$ (m ³)	1

25. O/N 23/P23/Q1(d)(i)

1(d)(i)	(relatively low melting and boiling points suggests) weak intermolecular / VdW forces	1
	(vigorous reaction with water suggests) hydrolysis	1

26. O/N 23/P23/Q3(d)

3(d)(i)	$pV = nRT \therefore V = 8.31 \times 630 \div 100\,000 = \underline{\underline{0.0524}}$ (m ³)	1
3(d)(ii)	moles of gas = $12 \times 5.00 / 180$	1
	volume of gas = $M1 \times \text{(d)(i)} = \underline{\underline{0.0175}}$ (m ³)	1

27. M/J 23/P22/Q1(a)(i)-(ii)

1(a)(i)	bonding in magnesium – metallic AND bonding in sodium chloride – ionic	1
1(a)(ii)	bonds in NaCl are stronger than bonds in Mg	1

28. M/J 23/P23/Q1(a)(i)

1(a)(i)	giant metallic	1
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29. MAR 23/P22/Q1(e)(ii)

1(e)(ii)	conversion of units 103000 Pa $127 \times 10^{-6} \text{ m}^3$ 423 K	1
	Use of $pV = (m/M_r)RT$ $M_r = \frac{0.284 \times 8.31 \times 423}{103000 \times 127 \times 10^{-6}}$	1
	$M_r = 76.3$ AND compound = CS ₂	1

30. O/N 22/P22/Q2(d)

2(d)	M1 value of n based on re-arranged equation for 'n' and substitution of correct values $n = \frac{3.23 \times 10^5 \cdot 250 \times 10^{-6}}{8.31 \cdot 500} = 0.0194(34416)$	1
	M2 calculated value of M_r based on M1 $M_r = 3.30 \div n = 170$	1
	M3 period 3 chloride which matches M_r in M2 (formula) SiCl ₄	1

31. O/N 22/P22/Q2(f)(ii)

2(f)(ii)	M1 (structure =) simple/molecular, because it has a low melting/boiling point <i>owtte</i>	1
	M2 (bonding =) covalent, because it is hydrolysed	1

32. M/J 22/P22/Q1(a)

1(a)	M1 comment explaining high melting point of Mg in terms of many strong metallic bonds OR many strong (electrostatic) attractions between cations and delocalised electrons OR strong bonds in giant metallic structure. M2 comment explaining electrical conductivity of Mg in terms of movement of delocalised electrons delocalised electrons can move through the structure	2
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33. M/J 22/P22/Q2(a)

2(a)	giant ionic	1
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34. M/J 22/P23/Q1(f)(ii)

1(f)(ii)	M1 comment explaining high melting point of Cu many strong metallic bonds OR many strong (electrostatic) attractions between cations and delocalised electrons OR strong bonds in giant metallic structure. M2 comment explaining electrical conductivity of Cu delocalised electrons are free to move through the structure (<i>owtte</i>)	2
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35. MAR 22/P22/Q2(c)(i)

2(c)(i)	(structure =) simple/molecular, because it has a low melting/boiling point [1] (bonding =) covalent, because it is hydrolysed [1]	2
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Unit 5

CHEMICAL ENERGETICS

30 items · 77 marks

Syllabus sections in this unit

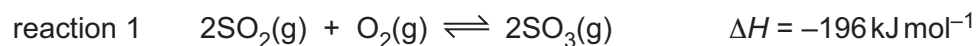
5.1 Enthalpy change, ΔH

5.2 Hess's law

Unit 5: Chemical energetics**1. M/J 26/P23/Q3(a)**

Context

- 3** The conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process are carefully controlled.



- (a)** Complete Figure 3.1 to show the reaction pathway diagram for reaction 1.

Draw arrows to represent the enthalpy change, ΔH , and activation energy, E_{A} .

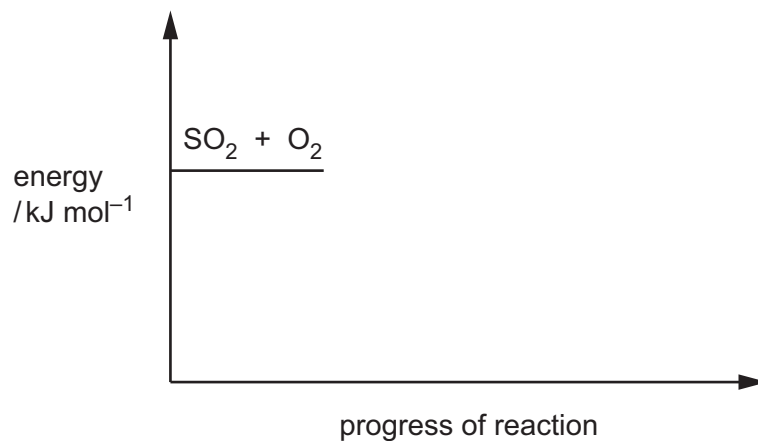


Figure 3.1

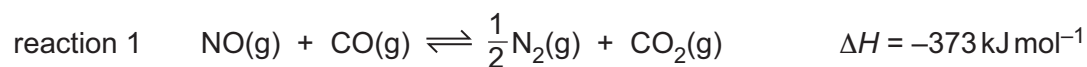
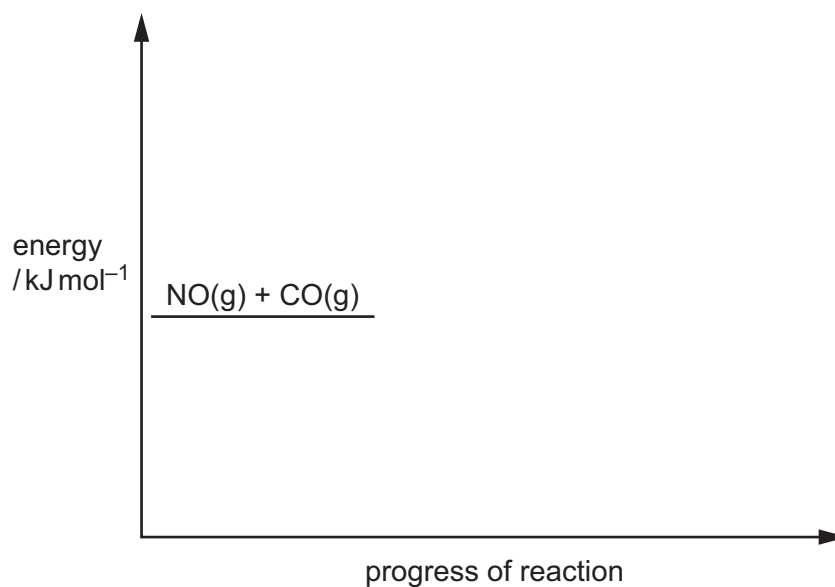
[2]

2. MAR 26/P22/Q2(d)(i)

Context

2 Nitrogen monoxide, NO, is an atmospheric pollutant.

Context: part (d) (introduction)

(d) Reaction 1 occurs during the catalytic removal of NO from car exhaust gases.The activation energy, E_a , of this reaction is 570 kJ mol^{-1} .**(i)** Complete Figure 2.3 to sketch the energy profile for reaction 1.Show the enthalpy change, ΔH , and the activation energy, E_a .**Figure 2.3**

[2]

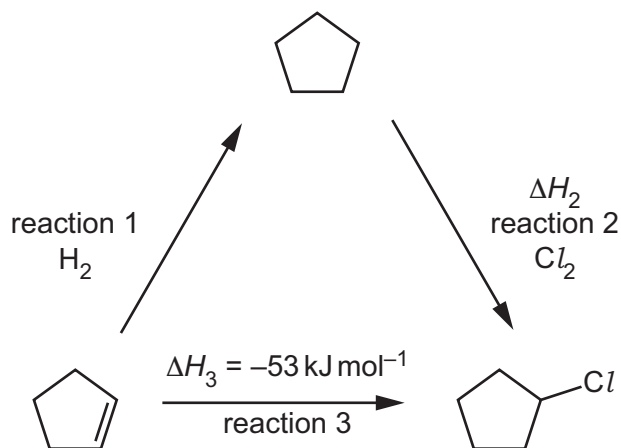
3. O/N 25/P21/Q3(d)(ii)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (d) (introduction)

- (d)** Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and $\text{C}_5\text{H}_9\text{Cl}$.

**Fig. 3.2**

- (ii)** Use the data in Fig. 3.2 and in Table 3.1 to calculate the enthalpy change of reaction 2, ΔH_2 .

Table 3.1

compound	enthalpy change of combustion, $\Delta H_c / \text{kJ mol}^{-1}$
	-3292
	-3115
H_2	-286

$\Delta H_2 =$ kJ mol⁻¹ [2]

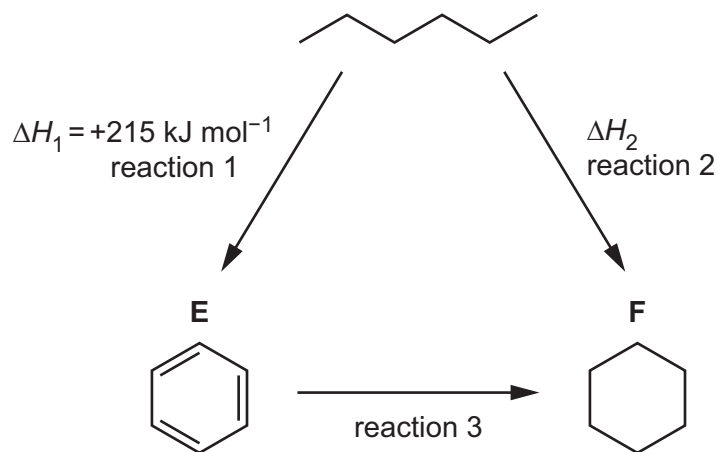
4. O/N 25/P22/Q3(e)(ii)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

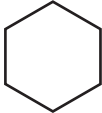
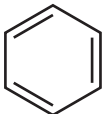
Context: part (e) (introduction)

- (e)** Hexane can be converted into compounds **E** and **F** at high temperature and pressure. Fig. 3.2 shows the reaction scheme involving hexane, **E** and **F**.

**Fig. 3.2**

- (ii)** Use the data in Fig. 3.2 and in Table 3.1 to calculate the enthalpy change of reaction 2, ΔH_2 .

Table 3.1

compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
	-156
	+48

 $\Delta H_2 =$ kJ mol^{-1} [2]

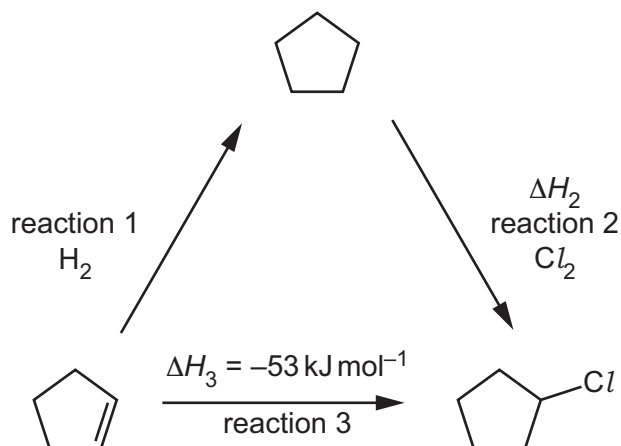
5. O/N 25/P23/Q3(d)(ii)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (d) (introduction)

- (d)** Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and $\text{C}_5\text{H}_9\text{Cl}$.

**Fig. 3.2**

- (ii)** Use the data in Fig. 3.2 and in Table 3.1 to calculate the enthalpy change of reaction 2, ΔH_2 .

Table 3.1

compound	enthalpy change of combustion, $\Delta H_c / \text{kJ mol}^{-1}$
	-3292
	-3115
H_2	-286

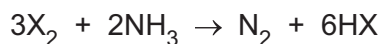
$\Delta H_2 =$ kJ mol⁻¹ [2]

6. O/N 25/P24/Q1(b)(iii)-(iv)

Context

1 The Group 17 elements are oxidising agents.

Context: part (b) (introduction)

(b) Cl_2 and Br_2 can each react with NH_3 to give N_2 and a hydrogen halide, HX . $\text{X} = \text{Cl or Br}$

The relative bond strengths of X-X and H-X determine the difference in enthalpy change of the two reactions.

(iii) Define enthalpy change of formation, ΔH_f .

[2]

(iv) Table 1.1 gives data relevant to the reaction of $\text{Cl}_2(\text{g})$ with $\text{NH}_3(\text{g})$.**Table 1.1**

compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
$\text{NH}_3(\text{g})$	-46
$\text{HCl}(\text{g})$	-92

Use the data in Table 1.1 to calculate the enthalpy change of the reaction of $\text{Cl}_2(\text{g})$ with $\text{NH}_3(\text{g})$.

enthalpy change of reaction = kJ mol^{-1} [2]

7. M/J 25/P22/Q3(c)**(c)** Hydrocarbon **Y** contains two $\text{C}=\text{C}$ groups. There are no other functional groups present.**Y** reacts with an excess of $\text{H}_2(\text{g})$ to produce hexane, C_6H_{14} .**Table 3.1**

bond	energy / kJ mol^{-1}
C-C	350
$\text{C}=\text{C}$	610
$\text{C}\equiv\text{C}$	840
H-H	436
C-H	410

Use Table 3.1 to calculate the enthalpy change per mole of C_6H_{14} produced in this reaction.

enthalpy change = kJ mol^{-1} [2]

8. M/J 24/P21/Q4(a)**4** **(a)** Define enthalpy change of formation.

[2]

9. M/J 24/P21/Q4(b)

(b) Iron is made when iron(III) oxide is heated with carbon monoxide, as shown by reaction 2.

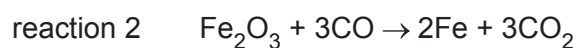


Table 4.1 shows enthalpy change of formation data measured at 298 K and 101 kPa.

Table 4.1

substance	equation	value for $\Delta H_f^\ominus / \text{kJ mol}^{-1}$
Fe_2O_3		–824.2
CO		–110.5
CO_2	$\text{C(s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$	–393.5

(i) Complete Table 4.1 by adding equations with relevant state symbols to represent:

- standard enthalpy change of formation for Fe_2O_3
- standard enthalpy change of formation for CO.

[2]

(ii) Use the data in Table 4.1 to calculate the enthalpy change of reaction, ΔH_r , in kJ mol^{-1} , for reaction 2.

Show your working.

$\Delta H_r =$ kJ mol^{-1} [2]

10. M/J 24/P22/Q2(a)

Context

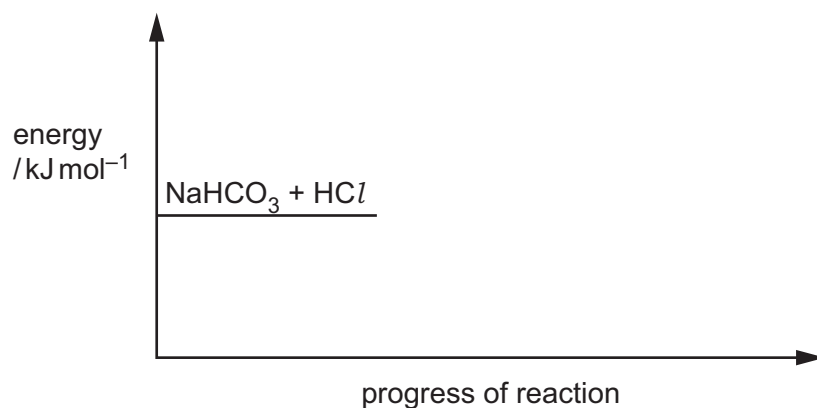
- 2** Separate samples of Na_2CO_3 and NaHCO_3 react with $\text{HCl}(\text{aq})$ to produce the same products, as shown in Table 2.1.

Table 2.1

reaction	equation	$\Delta H/\text{kJ mol}^{-1}$
1	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	ΔH_1
2	$\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	$\Delta H_2 = +27.2$

- (a)** Complete the reaction pathway diagram in Fig. 2.1 for reaction 2.

Label the diagram to show the enthalpy change, ΔH_2 , and the activation energy, E_A .

**Fig. 2.1**

[2]

11. M/J 24/P22/Q2(b)

Context

- 2** Separate samples of Na_2CO_3 and NaHCO_3 react with $\text{HCl}(\text{aq})$ to produce the same products, as shown in Table 2.1.

Table 2.1

reaction	equation	$\Delta H/\text{kJ mol}^{-1}$
1	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	ΔH_1
2	$\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	$\Delta H_2 = +27.2$

- (b)** The value for ΔH_1 is determined by experiment using the following method.

- 50.0 cm^3 of 2.00 mol dm^{-3} $\text{HCl}(\text{aq})$ is added to a polystyrene cup.
- The initial temperature of the acid is recorded as 19.6°C .
- 0.0400 mol of Na_2CO_3 is added and the mixture is stirred.
- All the solid Na_2CO_3 disappears and a colourless solution is produced.

The maximum temperature recorded during the reaction is 26.2°C .

- (i)** Describe **one** other observation that shows the reaction is complete.

[1]



5

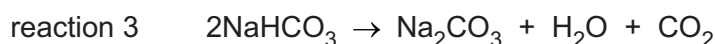
- (ii)** Calculate the value of ΔH_1 in kJ mol^{-1} .

Assume the specific heat capacity of the reaction mixture is the same as for water and no heat is lost to the surroundings.

Show your working.

$$\Delta H_1 = \quad \text{kJ mol}^{-1} \quad [3]$$

- (iii)** Thermal decomposition occurs when NaHCO_3 is heated.



Calculate the enthalpy change for reaction 3, ΔH_r , using the data in Table 2.1 and the value of ΔH_1 calculated in **(b)(ii)**.

(If you were unable to calculate a value for ΔH_1 in **(b)(ii)**, assume the enthalpy change is -38.4 kJ mol^{-1} . This is **not** the correct value.)

$$\Delta H_r = \quad \text{kJ mol}^{-1} \quad [2]$$

12. M/J 24/P23/Q3(a)

- 3** **(a)** Write an equation to show the reaction for the standard enthalpy change of formation of H_2O . Include state symbols.

[2]

13. M/J 24/P23/Q3(b)

(b) Water is one of the products in the reaction of B_2O_3 and NH_3 , as shown in reaction 2.

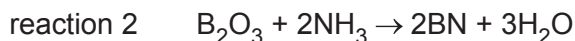


Table 3.1 shows information about the standard enthalpy change of formation, $\Delta H_f^\ominus$, of some substances.

Table 3.1

substance	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
B_2O_3	-1264
NH_3	-46
BN	-134
H_2O	-286

Calculate the enthalpy change, ΔH , for reaction 2 using the data from Table 3.1.

$$\Delta H = \quad \quad \quad \text{kJ mol}^{-1} \quad [2]$$

14. MAR 24/P22/Q3(c)(i)

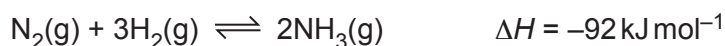
Context

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

Context: part (c) (introduction)

(c) The Haber process involves the reaction of N_2 and H_2 to form ammonia, NH_3 .

A catalyst is used, which allows the process to be carried out at a lower temperature and pressure.



(i) Use the information in (c) to complete Table 3.1.

Table 3.1

compound	enthalpy change of formation, $\Delta H_f^\ominus / \text{kJ mol}^{-1}$
N_2	
H_2	
NH_3	

[2]

15. O/N 23/P21/Q1(e)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (e) POC_l₃(g) forms when PC_l₃(g) reacts with O₂(g).

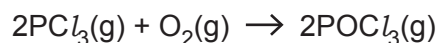


Table 1.3 gives some relevant data.

Table 1.3

process	value /kJ mol ⁻¹
enthalpy change of formation of PC _l ₃ (g)	-289
enthalpy change of formation of POC _l ₃ (g)	-592
O ₂ (g) → 2O(g)	+496

- (i) Define enthalpy change of formation, ΔH_f.

[2]

- (ii) Calculate the bond energy of P=O in POC_l₃ using the data in Table 1.3.

Show your working.

bond energy of P=O =

kJ mol⁻¹

[2]

16. O/N 23/P22/Q3(b)(i)-(ii)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (b) (introduction)

(b) H_3PO_4 is also formed in the process shown in reaction 1.

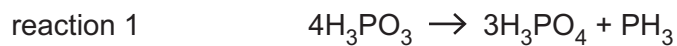


Table 3.1 shows some relevant thermodynamic data.

Table 3.1

compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
H_3PO_3	-972
H_3PO_4	-1281
PH_3	+9

(i) Define enthalpy change of formation.

[2]

(ii) Use the data in Table 3.1 to calculate the enthalpy change, ΔH_r , of reaction 1.

$\Delta H_r =$ kJ mol^{-1}
[2]

17. O/N 23/P23/Q1(e)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy / kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (e) POC_l₃(g) forms when PC_l₃(g) reacts with O₂(g).

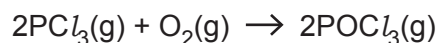


Table 1.3 gives some relevant data.

Table 1.3

process	value / kJ mol ⁻¹
enthalpy change of formation of PC _l ₃ (g)	-289
enthalpy change of formation of POC _l ₃ (g)	-592
O ₂ (g) → 2O(g)	+496

- (i) Define enthalpy change of formation, ΔH_f.

[2]

- (ii) Calculate the bond energy of P=O in POC_l₃ using the data in Table 1.3.

Show your working.

bond energy of P=O =

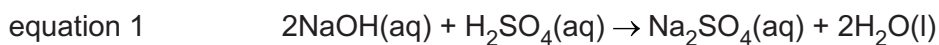
kJ mol⁻¹

[2]

18. M/J 23/P21/Q2(a)

Context

- 2** A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



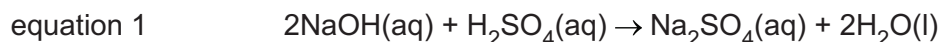
- (a) Define enthalpy change of neutralisation, ΔH_{neut}.

[2]

19. M/J 23/P21/Q2(b)(ii)

Context

2 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



Context: part (b) (introduction)

(b) An experiment is carried out to calculate ΔH_{neut} for the reaction between NaOH(aq) and H₂SO₄(aq).

100 cm³ of 1.00 mol dm⁻³ NaOH(aq) is added to 75 cm³ of 1.00 mol dm⁻³ H₂SO₄(aq) in a polystyrene cup and stirred. Results from the experiment are shown in Table 2.1.

Table 2.1

initial temperature of NaOH(aq)/°C	20.0
initial temperature of H ₂ SO ₄ (aq)/°C	20.0
maximum temperature of mixture/°C	27.8

(ii) Calculate ΔH_{neut} using the results in Table 2.1. Include units in your answer.

Assume that:

- the specific heat capacity of the final solution is 4.18 J g⁻¹ K⁻¹
- 1.00 cm³ of the final solution has a mass of 1.00 g
- there is no heat loss to the surroundings
- full dissociation of H₂SO₄(aq) occurs
- the experiment takes place at constant pressure.

Show your working.

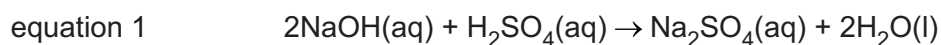
$$\Delta H_{\text{neut}} = \quad \text{units}$$

[3]

20. M/J 23/P21/Q2(c)(ii)

Context

2 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



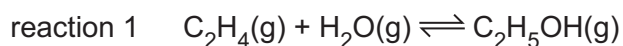
(ii) Suggest why the enthalpy change of neutralisation cannot be determined using the addition of dilute sulfuric acid to aqueous barium hydroxide.

[1]

21. M/J 23/P22/Q4(a)

Context

- 4** In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



- (a) Use the bond energy values in Table 4.1 to calculate the enthalpy change, ΔH_r , for reaction 1.

Table 4.1

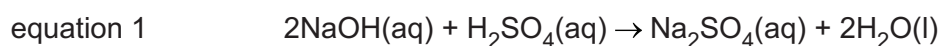
bond	bond energy / kJ mol^{-1}
C–C	350
C=C	610
C≡C	840
C–H	410
C–O	360
C=O	740
O–H	460

$$\Delta H_r = \quad \quad \quad \text{kJ mol}^{-1} \quad [2]$$

22. M/J 23/P23/Q3(a)

Context

- 3** A neutralisation reaction occurs when $\text{NaOH}(\text{aq})$ is added to $\text{H}_2\text{SO}_4(\text{aq})$.



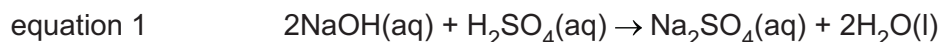
- (a) Define enthalpy change of neutralisation, ΔH_{neut} .

[2]

23. M/J 23/P23/Q3(b)(ii)

Context

3 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



Context: part (b) (introduction)

(b) In an experiment, 50.0 cm³ of 2.00 mol dm⁻³ NaOH(aq) is added to 60.0 cm³ of 1.00 mol dm⁻³ H₂SO₄(aq) in a polystyrene cup and stirred. Both solutions have a temperature of 21.4 °C before mixing. The maximum temperature of the mixture is measured.

(ii) Calculate the theoretical maximum temperature of the mixture in this experiment.

Assume that:

- enthalpy change of neutralisation, ΔH_{neut} , of NaOH(aq) and H₂SO₄(aq) is -57.1 kJ mol⁻¹
- full dissociation of H₂SO₄(aq) occurs
- the specific heat capacity of the final solution is 4.18 J g⁻¹ K⁻¹
- 1.00 cm³ of the final solution has a mass of 1.00 g
- there is no heat loss to the surroundings
- the experiment takes place at constant pressure.

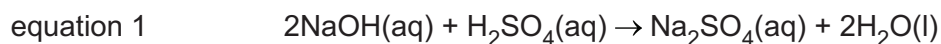
Show your working.

theoretical maximum temperature = °C [3]

24. M/J 23/P23/Q3(c)(ii)

Context

3 A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



Context: part (c) (introduction)

(c) The enthalpy change of neutralisation of CH₃COOH(aq) and NaOH(aq) is -55.2 kJ mol⁻¹.

(ii) Values for the enthalpy change of neutralisation, ΔH_{neut} , are shown in Table 3.1.

Table 3.1

reagents	$\Delta H_{\text{neut}}/\text{kJ mol}^{-1}$
NaOH + HCl	-57.1
NaOH + CH ₃ COOH	-55.2

Suggest why the value for ΔH_{neut} of the weak acid, CH₃COOH, reacting with NaOH is different to the value obtained using the strong acid, HCl. Assume that the values are determined under the same conditions.

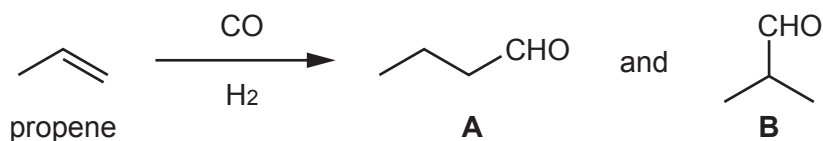
[1]

25. MAR 23/P22/Q3(d)(iii)

Context

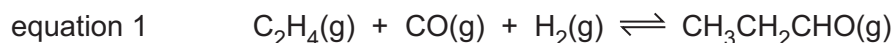
- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

**Fig. 3.1**

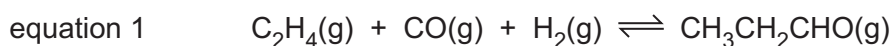
Context: part (d) (introduction)

- (d)** The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



At atmospheric pressure a cobalt-based catalyst is used in this reaction.

- (iii)** Use the information in Table 3.2 to calculate the enthalpy change, ΔH_r , of the reaction in equation 1.

**Table 3.2**

compound	enthalpy change of formation, $\Delta H_f^\circ/\text{kJ mol}^{-1}$
C ₂ H ₄ (g)	+52
CO(g)	−111
CH ₃ CH ₂ CHO(g)	−187

$\Delta H_r =$ kJ mol^{−1} [2]

26. M/J 22/P21/Q1(f)(ii)

Context

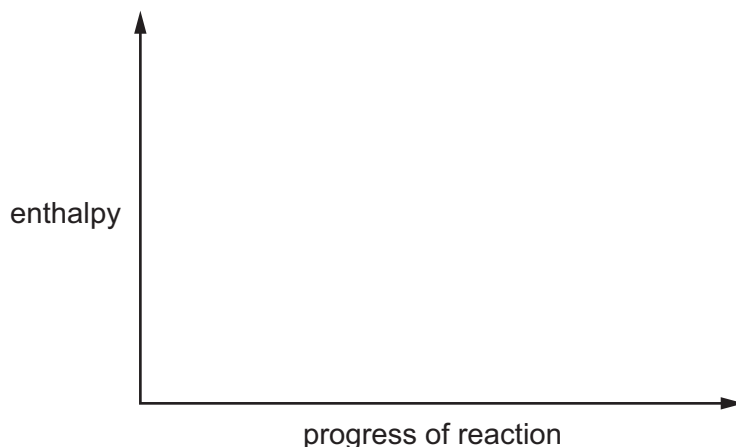
- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

Context: part (f) (introduction)

- (f)** Magnesium, Mg, burns in oxygen, O₂.
The activation energy, E_a , for this reaction is +148 kJ mol⁻¹.

- (ii)** On Fig. 1.1:

- sketch a reaction pathway diagram for the reaction that occurs when Mg burns in O₂
- label the diagram to show the enthalpy change, ΔH , and the activation energy, E_a , for the reaction.

**Fig. 1.1**

[3]

27. M/J 22/P22/Q1(b)(iii)-(iv)

Context: part (b) (introduction)

- (b)** When magnesium is heated in air, magnesium oxide, MgO, is the major product. Smaller amounts of magnesium nitride, Mg₃N₂, are also made.

- (iii)** Define enthalpy change of formation.

[2]

- (iv)** When 3.645 g of Mg(s) burns in excess N₂(g) to form Mg₃N₂(s), 23.05 kJ of energy is released.

Calculate the enthalpy change of formation, ΔH_f , of Mg₃N₂. Show your working.

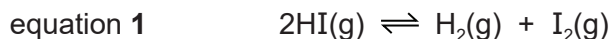
$$\Delta H_f (\text{Mg}_3\text{N}_2) =$$

[3]

28. M/J 22/P22/Q3(b)

Context: part (a) (introduction)

- 3 (a)** 0.025 mol of HI(g) is added to a closed vessel and left to reach dynamic equilibrium. The total pressure of the vessel is 100 kPa.



- (b)** Use equation 1 and the bond energy values in Table 3.1 to calculate the change in enthalpy, ΔH , for the thermal decomposition of 1 mole of HI(g). Show your working.

Table 3.1

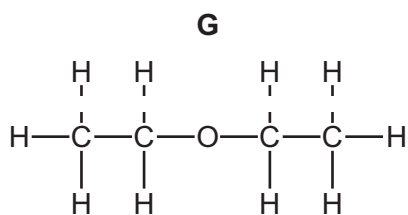
bond	bond energy / kJ mol ⁻¹
H–H	436
I–I	151
H–I	299

$$\Delta H = \quad \text{kJ mol}^{-1} \quad [2]$$

29. M/J 22/P23/Q3(a)(ii)

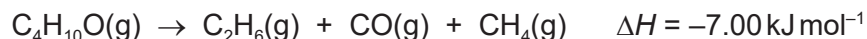
Context

- 3 G** belongs to a group of compounds called ethers.

**Fig. 3.1**

Context: part (a) (introduction)

- (a)** When **G**, C₄H₁₀O, is heated, thermal decomposition occurs.



- (ii)** Calculate the bond energy of C≡O using the bond energy values in Table 3.1 and the enthalpy change, ΔH , for the thermal decomposition of **G**. Show your working.

Table 3.1

bond	bond energy / kJ mol ⁻¹
C–C	350
C–O (in G)	360
C–H	410

$$\text{bond energy (C}\equiv\text{O)} = \quad \text{kJ mol}^{-1} \quad [2]$$

30. MAR 22/P22/Q2(c)(iii)

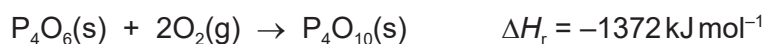
Context

2 Some oxides of elements in Period 3 are shown.

Context: part (c) (introduction)

(c) P_4O_6 is a white solid that has a melting point of 24°C . Solid P_4O_6 reacts with water to form H_3PO_3 .

(iii) When $\text{P}_4\text{O}_6(\text{s})$ is heated with oxygen it forms $\text{P}_4\text{O}_{10}(\text{s})$.



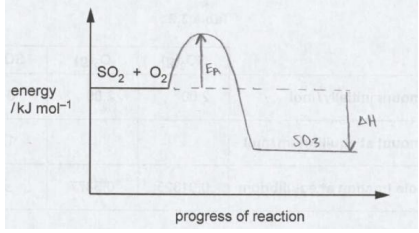
The enthalpy change of formation, ΔH_f , of $\text{P}_4\text{O}_{10}(\text{s})$ is $-3012 \text{ kJ mol}^{-1}$.

Calculate the enthalpy change of formation, ΔH_f , of $\text{P}_4\text{O}_6(\text{s})$.

ΔH_f of $\text{P}_4\text{O}_6(\text{s}) =$ kJ mol^{-1} [1]

Answers Section

1. M/J 26/P23/Q3(a)

3(a)	 • correct shape profile • arrow label ΔH from reactant line to product line • arrow label E_a (from reactant line to maximum on hump)	2 max 1 endothermic profile with correct labels & arrows for E_a and / or ΔH ΔH single headed arrows used in correct direction only. A E_a shown as a double headed arrow BOD vertical line A half-headed arrows A 'square' humps I missing products / incorrect labels on product line I '-' sign on $-\Delta H$ BOD poorly represented lines / arrows if approximate position and length (and direction) is correct
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2. MAR 26/P22/Q2(d)(i)

2(d)(i)	• reactants higher than labelled products AND hump in the middle • reactants to top of hump = E_a OR (+)570 (kJ mol⁻¹) • reactants to products = ΔH OR (-)373 (kJ mol⁻¹)	2
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3. O/N 25/P21/Q3(d)(ii)

3(d)(ii)	M1 (reaction 1): $-3115 - 286 + 3292 = -109$ (kJ mol⁻¹) M2 $\Delta H_2 = -53 - (-109) = (+)56$ (kJ mol⁻¹)	2
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4. O/N 25/P22/Q3(e)(ii)

3(e)(ii)	M1 $+215 + (-156) - (+48) =$ M2 $+11$ kJ mol⁻¹	2
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5. O/N 25/P23/Q3(d)(ii)

3(d)(ii)	M1 (reaction 1): $-3115 - 286 + 3292 = -109$ (kJ mol⁻¹) M2 $\Delta H_2 = -53 - (-109) = (+)56$ (kJ mol⁻¹)	2
----------	---	----------

6. O/N 25/P24/Q1(b)(iii)-(iv)

1(b)(iii)	M1 (enthalpy change / energy change) when one mole of a compound / substance is formed M2 from its (constituent) elements in their standard state(s)	2
1(b)(iv)	M1 $(6 \times -92) - (2 \times -46) =$ M2 -460 (kJ mol⁻¹)	2

7. M/J 25/P22/Q3(c)

3(c)	method 1 – breaking all bonds in molecules $(2 \times 610) + (2 \times 436) + (10 \times 410) + (3 \times 350) - (14 \times 410) - (5 \times 350)$ - energy involved when all bonds are broken = (7242) - energy involved when all bonds are made = (7490) - energy to break bonds – energy to make bonds $= (7242 - 7490) = -248$ method 2 – ignore any bonds that do not change from reactants to products $(2 \times 610) + (2 \times 436) - (4 \times 410) - (2 \times 350)$ - energy involved when all bonds are broken = (2092) - energy involved when all bonds are made = (2340) - energy to break bonds – energy to make bonds $(2092 - 2340) = -248$	2
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8. M/J 24/P21/Q4(a)

4(a)	M1 (enthalpy change / energy change) when one mole of a compound / substance is formed M2 from its elements in their standard states	2
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9. M/J 24/P21/Q4(b)

4(b)(i)	<table border="1"> <thead> <tr> <th></th><th>equation for reaction</th><th>$\Delta H_{\text{formation}}$</th></tr> </thead> <tbody> <tr> <td>Fe₂O₃</td><td>$2\text{Fe (s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)}$</td><td>-824.2</td></tr> <tr> <td>CO</td><td>$\text{C (s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO (g)}$</td><td>-110.5</td></tr> <tr> <td>CO₂</td><td>$\text{C (s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$</td><td>-393.5</td></tr> </tbody> </table>		equation for reaction	$\Delta H_{\text{formation}}$	Fe ₂ O ₃	$2\text{Fe (s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)}$	-824.2	CO	$\text{C (s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO (g)}$	-110.5	CO ₂	$\text{C (s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$	-393.5	2
	equation for reaction	$\Delta H_{\text{formation}}$												
Fe ₂ O ₃	$2\text{Fe (s)} + \frac{3}{2}\text{O}_2\text{(g)} \rightarrow \text{Fe}_2\text{O}_3\text{(s)}$	-824.2												
CO	$\text{C (s)} + \frac{1}{2}\text{O}_2\text{(g)} \rightarrow \text{CO (g)}$	-110.5												
CO ₂	$\text{C (s)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)}$	-393.5												
4(b)(ii)	$\Delta H_{\text{reaction}} = -\Delta H_{\text{Fe}_2\text{O}_3} - (3 \times -\Delta H_{\text{CO}}) + (3 \times -\Delta H_{\text{CO}_2})$ $= -(-824.2) - (3 \times -110.5) + (3 \times -393.5)$ $= -24.8 \text{ kJ mol}^{-1}$ - evidence of use of correct expression and / or any rearrangement of only these bond energy values - use correct stoichiometry for these bond energy values only - calculate their expression correctly provided the expression uses only addition / subtraction and all 3 values	2												

10. M/J 24/P22/Q2(a)

2(a)	<i>All three correct (2 marks) or two correct (1 mark):</i> - profile for endothermic reaction (include 'hump') - arrow representing E_a - arrow representing ΔH OR (+) 27.2	2
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11. M/J 24/P22/Q2(b)

2(b)(i)	no more fizzing / effervescence stops	1
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2(b)(ii)	M1 calculate Q from $= 50 \times 4.18 \times (26.2 - 19.6) = 1379.4$ M2 Use M1 to find energy change when 1 mol Na_2CO_3 / 2 mol HCl reacts M1 $\div 0.04 = 34485$ M3 express Q as $\Delta H_{(1)}$ in $\text{kJ mol}^{-1} = -(\text{value for M2}) \div 1000$ $= -34.5 \text{ kJ mol}^{-1}$ <i>Alternative method</i> M1 evidence of use $Q = mc\Delta T = 50 \times 4.18 / 1000 \times (26.2 - 19.6) = 1.3794$ M2 Use M1 to find energy change when 1 mol Na_2CO_3 / 2 mol HCl reacts M1 $\div 0.04 = 34.485$ M3 express Q as $\Delta H_{(1)}$ in $\text{kJ mol}^{-1} = -(\text{value for M2})$ (expressed in kJ) $= -34.5 \text{ kJ mol}^{-1}$	3
2(b)(iii)	Using answer from 2(b)(ii) $(=) 2\Delta H_2 - (\Delta H_1)$ $(=) 2 \times 27.2 - (2b(ii))$ $2 \times 27.2 - (-34.50) = (+) 88.9 \text{ kJ mol}^{-1}$ All three correct (2 marks) or two correct (1 mark): $(27.2) - (-34.5)$ OR $(27.2) - (2b(ii))$ 2×27.2 and $1 \times 2b(ii)$ - calculate correctly their expression Using alternative value of $\Delta H_1 = -38.4 \text{ kJ mol}^{-1}$ $\Delta H_r = (+)92.8 \text{ kJ mol}^{-1} = 2 \text{ marks}$	2

12. M/J 24/P23/Q3(a)

3(a)	$\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$ M1 correct equation only M2 correct state symbols based on this equation	2
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13. M/J 24/P23/Q3(b)

3(b)	$\Delta H = -(-1264) - (2 \times -46) + (2 \times -134) + (3 \times -286) = (+)230 \text{ kJ mol}^{-1}$ M1 correct expression in terms of all 4 enthalpy changes of formation AND correct stoichiometry (as described in given equation) M2 correct calculation based on figures given	2
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14. MAR 24/P22/Q3(c)(i)

3(c)(i)	0 for both ΔH_f of N_2 and H_2	1
	-46 for ΔH_f of NH_3	1

15. O/N 23/P21/Q1(e)

1(e)(i)	enthalpy change when one mole of a compound / substance is formed	1
	from its constituent elements in their standard states	1
1(e)(ii)	$2(-592) = 2(-289) + (496) - 2(\text{P=O})$ OR $(2) \text{P=O} = 2(-289) + (496) + 2(592)$	1
	$E_{\text{P=O}} = \frac{1}{2}(1102) = +551 \text{ (kJ mol}^{-1}\text{)}$	1

16. O/N 23/P22/Q3(b)(i)-(ii)

3(b)(i)	enthalpy change when one mole of a compound / substance is formed	1
	from its constituent elements in their standard states	1
3(b)(ii)	$\Delta H_f = +9 + 3(-1281) - 4(-972) = (+)54 \text{ (kJ mol}^{-1}\text{)} (1)$ Any two from (1): - use correct stoichiometry for all three values - use correct expression regardless of stoichiometry for all three values - calculated correctly	2

17. O/N 23/P23/Q1(e)

1(e)(i)	enthalpy change when one mole of a compound / substance is formed	1
	from its constituent elements in their standard states	1
1(e)(ii)	$2(-592) = 2(-289) + (496) - 2(P=O)$ OR $(2) P=O = 2(-289) + (496) + 2(592)$	1
	$E_{P=O} = \frac{1}{2}(1102) = +551 \text{ (kJ mol}^{-1}\text{)}$	1

18. M/J 23/P21/Q2(a)

2(a)	M1 (enthalpy change) when 1 mole of water is formed M2 from an (aqueous) acid and an alkali / (aqueous) base	2
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19. M/J 23/P21/Q2(b)(ii)

2(b)(ii)	M1 correct volume and correct temperature change M2 calculation to calculate Q based on M1 volume and temp and $Q = mc\Delta T$ $Q = mc\Delta T = 175 \times 4.18 \times (27.8 - 20) = (-)5705.7$ M3 energy change when 0.1 mol H_2O is made with appropriate unit and sign $\Delta H = M2 / 0.1 = -57.1 \text{ kJ mol}^{-1}$ OR $-57100 \text{ J mol}^{-1}$	3
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20. M/J 23/P21/Q2(c)(ii)

2(c)(ii)	precipitation (of $BaSO_4$) (also) involves an energy change	1
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21. M/J 23/P22/Q4(a)

4(a)	Method 1 - identification of number and type of bonds when all are broken $2O-H + 4C-H + C=C$ OR $(2 \times 460) + (4 \times 410) + 610$ - Identification of number and type of bonds when all are made $O-H + 5C-H + C-C + C-O$ OR $(460 + (5 \times 410) + 350 + 360)$ - calculation of energy to break bonds – energy to make bonds $3170 - 3220 = -50 \text{ (kJ mol}^{-1}\text{)}$ Two correct for one mark, three correct for two marks. OR Method 2 - identification of number and type of relevant bonds broken (including the new C-H only) $2O-H + C=C$ OR $(2 \times 460) + 610$ - identification of number and type of relevant bonds made (including the new C-H only) $O-H + C-H + C-C + C-O$ OR $(460 + 410 + 350 + 360)$ - calculation using energy to break bonds – energy to make bonds $1530 - 1580 = -50 \text{ (kJ mol}^{-1}\text{)}$ Two correct for one mark, three correct for two marks.	2
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22. M/J 23/P23/Q3(a)

3(a)	M1 (energy change) when 1 mole of water is formed M2 from an (aqueous) acid and an alkali / aqueous base	2
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23. M/J 23/P23/Q3(b)(ii)

3(b)(ii)	M1 correct value for Q (using 0.1 mol H_2O / OR $2 \times (b)(i)$) (and Q in Joules) M2 calculate ΔT (using their Q AND 110) $Q/mc = \Delta T = (Q) \div (4.18 \times 110) = 12.42 \text{ (}^\circ\text{C)}$ M3 use ΔT and initial temperature to find temperature of final solution $21.4 + M2 = 33.8 \text{ (}^\circ\text{C)}$ [M3 $\Delta T + 21.4 = T_{\text{max}}$]	3
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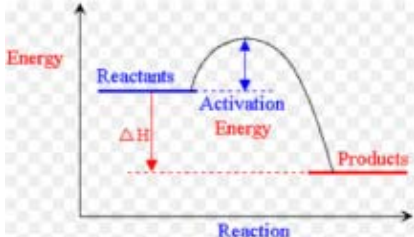
24. M/J 23/P23/Q3(c)(ii)

3(c)(ii)	(additional) energy is involved / required to fully ionise / dissociate (weak / ethanoic acid) OR (additional) energy is involved / required to donate/ release H^+ ions	1
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25. MAR 23/P22/Q3(d)(iii)

3(d)(iii)	$-187 - (-111) - (+52)$	1
	$= -128 \text{ (kJ mol}^{-1}\text{)}$	1

26. M/J 22/P21/Q1(f)(ii)

1(f)(ii)	M1 sketch shows exothermic reaction with a 'hump' AND labelled reactants ($Mg + O_2$) and products (MgO) M2 arrow from reactants / $Mg + O_2$ to products / MgO shown as ΔH M3 arrow showing activation energy / E_a / (+)148 	3
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27. M/J 22/P22/Q1(b)(iii)-(iv)

1(b)(iii)	M1 (enthalpy change / energy change) when one mole of a compound / substance is formed M2 from its elements in their standard states	2
1(b)(iv)	M1 correct method to find amount Mg_3N_2 formed in reaction $3.645 \div 24.3 \text{ mol Mg produces } 0.15 \div 3 \text{ mol of } Mg_3N_2$ OR $0.05 \text{ mol of } Mg_3N_2$ M2 use M1 in correct expression to find energy released by 1 mol Mg_3N_2 $23.05 \div \text{M1}$ OR $23.05 \div \text{mol } Mg_3N_2$ OR $23\,050 \div \text{M1}$ OR $23\,050 \div Mg_3N_2$ M3 calculate value from M2 expression correctly $\Delta H = -(M2) = -461(.0) \text{ kJ mol}^{-1}$ OR $-461\,000 \text{ J mol}^{-1}$	3

28. M/J 22/P22/Q3(b)

3(b)	evaluation, to give a value, based on calculation using: - all three correct bond energies [$HI = 299$, $H_2 = 436$, $I_2 = 151$] - correct use of stoichiometry to calculate ΔH for 1 mol HI $x = 1$ and $y = \frac{1}{2}$ - calculate a value for ΔH correctly i.e. $x(299) - y(436 + 151)$ $\Delta H = (+)5.5 \text{ kJ mol}^{-1}$	2
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29. M/J 22/P23/Q3(a)(ii)

3(a)(ii)	M1 expression OR value derived from using all bond energy values and ΔH correctly, ignoring stoichiometry $-7 = (n350 + m360 + p410) - (q410 + r350 + sC\equiv O)$. M2 correct calculation based on correct expression AND stoichiometry $[-7 = 2(350) + 2(360) + 10(410) - 10(410) - 350 - x] = (+)1077 \text{ kJ mol}^{-1}$	2
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30. MAR 22/P22/Q2(c)(iii)

2(c)(iii)	$-1640 \text{ (kJ mol}^{-1}\text{)}$	1
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Unit 6

ELECTROCHEMISTRY

24 items · 43 marks

Syllabus sections in this unit

6.1 Redox processes: electron transfer and changes in oxidation number (oxidation state)

Unit 6: Electrochemistry

1. M/J 26/P21/Q3(d)

Context

- 3 Separate samples of NaCl, NaBr and NaI each react with concentrated H_2SO_4 . Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smell of SO_2
3	NaI	- effervescence - steamy fumes initially that turn blue litmus paper red - purple vapour - pungent smell of SO_2 and then a rotten egg smell of H_2S

(d) Complete Table 3.2.

Table 3.2

species	oxidation number of sulfur
H_2SO_4	
SO_2	
H_2S	

[2]

2. M/J 26/P21/Q4(a)(ii)

Context: part (a) (introduction)

- 4 (a) Compound **Z** is oxidised to produce organic compound **W**, $\text{C}_5\text{H}_6\text{O}_7$, by heating under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$.

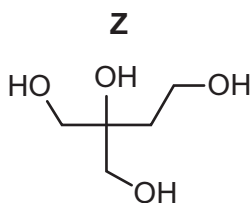
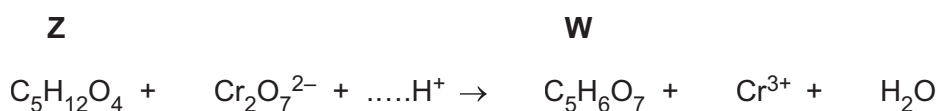


Figure 4.1

- (ii) The complete oxidation of 0.025 mol of **Z** requires 0.050 mol of $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$. **W** is produced in this reaction.

Balance the equation for this reaction.

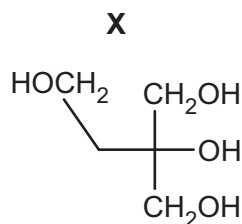


[1]

3. M/J 26/P22/Q4(c)

also Unit 17 (2 marks); this unit 2 marks

Context: part (b) (introduction)

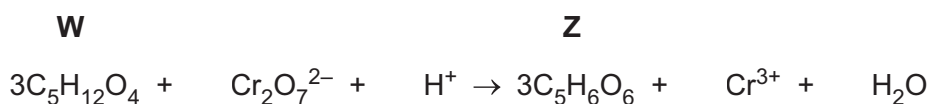
(b) The structure of compound **X** is shown in Figure 4.2.**Figure 4.2**

The molecular formula of **X** is $\text{C}_5\text{H}_{12}\text{O}_4$. Complete oxidation of 0.255 g of **X** occurs when it reacts with exactly 10.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$.

(c) (i) Compounds **X** and **W** are a pair of structural isomers.

0.0012 mol of **W** is completely oxidised when it is heated under reflux with exactly 0.0020 mol of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce compound **Z**.

Complete the equation for this reaction.



[2]

(ii) Each molecule of **W** contains four hydroxyl groups. Only two of these are primary hydroxyl groups.

Table 4.1 shows the observations when separate samples of **Z** are added to each reagent.

Table 4.1

reagent	observations with Z
2,4-dinitrophenylhydrazine (2,4-DNPH)	orange precipitate
Fehling's reagent	none

Use the information in **4(c)** to suggest the structure of **W**.

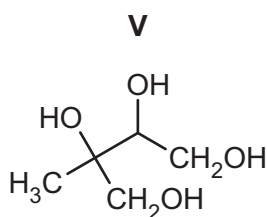


[2]

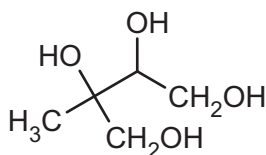
4. M/J 26/P23/Q4(d)(ii)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

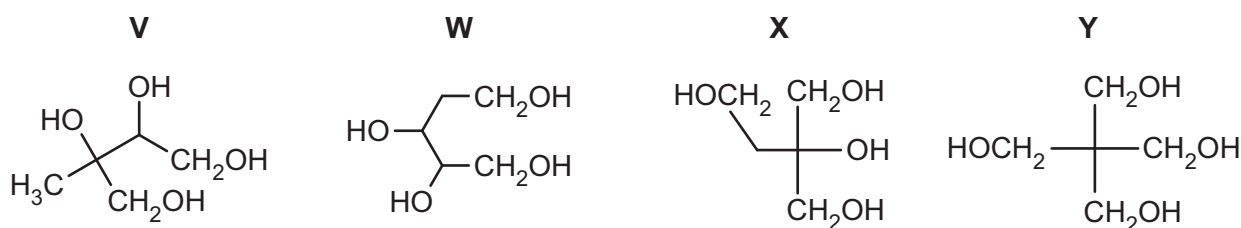
**Figure 4.1**

Context: Fig. 4.1

**Figure 4.1**

Context: part (c) (introduction)

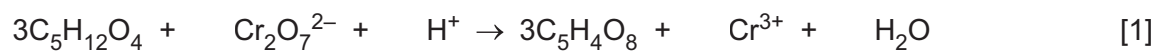
- (c)** Compounds **W**, **X** and **Y** are structural isomers of **V**.

**Figure 4.2**

Context: part (d) (introduction)

- (d)** Exactly 0.03 mol of one of the isomers, **V**, **W**, **X** or **Y**, is completely oxidised by 0.08 mol of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce only **one** organic compound with molecular formula $\text{C}_5\text{H}_4\text{O}_8$.

- (ii)** Complete the equation for this reaction.



5. M/J 25/P21/Q4(b)(iii)

Context

4 The structure of vitamin C is shown in Fig. 4.1.

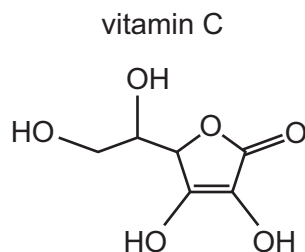


Fig. 4.1

Context: part (b) (introduction)

(b) The concentration of vitamin C is found by titration with $\text{I}_2(\text{aq})$.

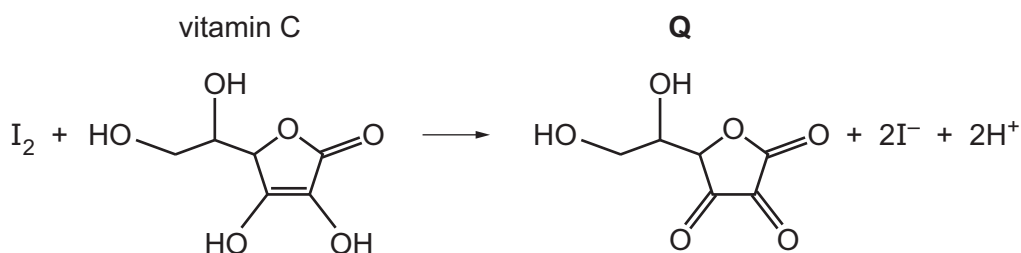


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm³ of vitamin C solution. 5.00 cm³ of this vitamin C solution is added to a flask with approximately 150 cm³ of water and an indicator. Exactly 28.40 cm³ of 5.00 × 10⁻⁴ mol dm⁻³ I₂(aq) reacts with the sample of vitamin C solution in the flask.

[M_r : vitamin C, 176]

(iii) Deduce the role of $\text{I}_2(\text{aq})$ in the reaction in Fig. 4.2.

[1]

6. M/J 25/P24/Q2(b)(ii)

Context: part (b) (introduction)

(b) SCl_2 is a red liquid that reacts with water.



(ii) When SCl_2 reacts with water, the SCl_2 is broken down and a disproportionation reaction occurs.

State the oxidation numbers of sulfur in SCl_2 and H_2SO_3 .

oxidation number of S in SCl_2

oxidation number of S in H_2SO_3

[2]

7. MAR 25/P22/Q1(a)

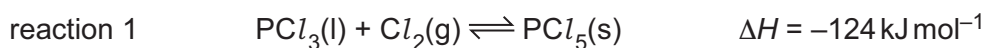
also Unit 7 (2 marks), Unit 2 (2 marks), Unit 4 (2 marks); this unit 2 marks

Context

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.**(a)** Chlorine forms three different compounds with phosphorus.The most common compounds are PCl_3 and PCl_5 .**(i)** Complete Table 1.1.**Table 1.1**

compound	oxidation number of P	oxidation number of Cl
PCl_3		
PCl_5		

[2]

(ii) In a closed system, PCl_3 and PCl_5 exist in an equilibrium mixture as shown in reaction 1.Deduce **two** conditions that favour the production of PCl_5 in reaction 1.

1

2

[2]

(iii) The third compound of phosphorus and chlorine, **W**, has a relative molecular mass, M_r , between that of PCl_3 and PCl_5 . The compound contains 69.6% by mass of chlorine.Determine the molecular formula of **W**.molecular formula of **W** = [2]**(iv)** **W** is a liquid at room temperature and pressure. It reacts vigorously with water to form an acidic solution.Suggest the structure and bonding in **W**. Explain your answer.

[2]

8. O/N 24/P21/Q2(c)(ii)-(iii)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

(c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO . NaClO is made from chlorine in a disproportionation reaction.**(ii)** Define disproportionation.

[1]

(iii) Write numbers in the boxes to balance the equation showing the reaction of propanone with NaClO .

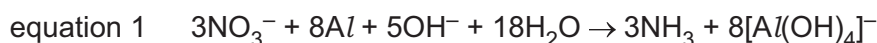
[1]

9. O/N 24/P22/Q3(b)(i)-(ii)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (b) (introduction)

(b) A common test for nitrates is the reaction with NaOH and Al . Equation 1 shows the reaction.**(i)** Deduce the oxidation state of nitrogen in NO_3^- .

[1]

(ii) Identify the species that is oxidised in equation 1.

[1]

10. O/N 24/P23/Q2(c)(ii)-(iii)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

(c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO . NaClO is made from chlorine in a disproportionation reaction.**(ii)** Define disproportionation.

[1]

(iii) Write numbers in the boxes to balance the equation showing the reaction of propanone with NaClO .

[1]

11. M/J 24/P23/Q1(d)(i)

Context: part (d) (introduction)

- (d) The oxide of silicon reacts with calcium oxide in an addition reaction to produce calcium silicate, CaSiO_3 . The oxidation number of calcium in CaSiO_3 is +II.

- (i) Deduce the oxidation number of silicon in calcium silicate.

oxidation number of silicon

[1]

12. MAR 24/P22/Q1(b)(i)

Context

- 1 Bismuth is an element in Group 15 of the Periodic Table.

Context: part (b) (introduction)

- (b) Bismuth reduces water to form bismuth oxide, Bi_2O_3 . A colourless gas that ignites with a squeaky pop also forms.

- (i) Construct an equation for the reduction of water by bismuth.

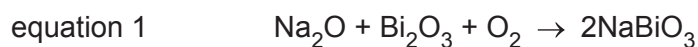
[1]

13. MAR 24/P22/Q1(c)

Context

- 1 Bismuth is an element in Group 15 of the Periodic Table.

- (c) Bi_2O_3 can be used to form NaBiO_3 , as shown in equation 1.



- (i) Deduce the oxidation number of Bi in Bi_2O_3 and in NaBiO_3 .

oxidation number of Bi:

in Bi_2O_3 in NaBiO_3

[1]

- (ii) Identify the reducing agent in equation 1.

[1]

14. MAR 24/P22/Q1(d)(i)

Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

Context: part (d) (introduction)

- (d)** NaBiO_3 is an oxidising agent with similar properties to KMnO_4 .

Fig. 1.1 shows an example of the use of NaBiO_3 as an oxidising agent.

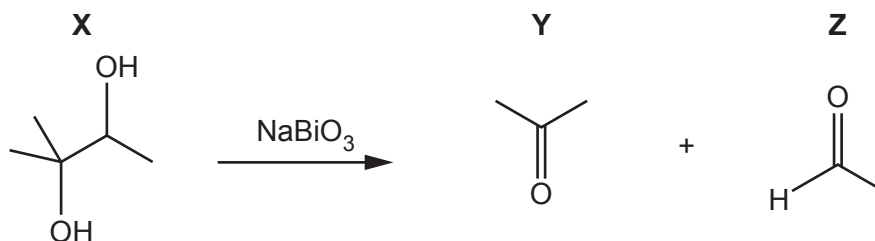


Fig. 1.1

- (i)** Explain the term oxidising agent.

[1]

15. O/N 23/P21/Q3(a)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

- (a)** Define oxidising agent.

[1]

16. O/N 23/P21/Q3(c)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

- (c)** When KClO_3 is heated without a catalyst, KClO_4 and KCl form.



Explain why this reaction is described as a disproportionation reaction.

[1]

17. O/N 23/P22/Q3(b)(iii)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (b) (introduction)

(b) H_3PO_4 is also formed in the process shown in reaction 1.

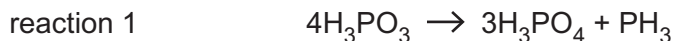


Table 3.1 shows some relevant thermodynamic data.

Table 3.1

compound	enthalpy change of formation, $\Delta H_f / \text{kJ mol}^{-1}$
H_3PO_3	-972
H_3PO_4	-1281
PH_3	+9

(iii) Explain why reaction 1 is a disproportionation reaction.

Explain your reasoning with reference to relevant oxidation numbers.

[2]

18. O/N 23/P23/Q3(a)

Context

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(a) Define oxidising agent.

[1]

19. O/N 23/P23/Q3(c)

Context

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(c) When KClO_3 is heated without a catalyst, KClO_4 and KCl form.



Explain why this reaction is described as a disproportionation reaction.

[1]

20. M/J 23/P22/Q2(e)(ii)

Context

2 Chlorine is a reactive element. It forms many compounds.

Context: part (e) (introduction)

- (e) Bleach is used as a cleaning product to kill bacteria. It is made by adding compounds like sodium chlorate(I), NaClO , to water.
- (ii) Sodium chlorate(I), NaClO , reacts with hydrogen peroxide to produce sodium chloride, water and oxygen gas.

Construct an equation for this reaction.

[1]

21. M/J 23/P23/Q1(d)(i)-(ii)

Context

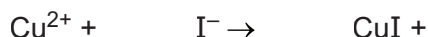
1 Copper is used in electrical equipment. It has a melting point of 1085°C .

Context: part (d) (introduction)

- (d) When KI(aq) is added to $\text{CuSO}_4\text{(aq)}$ the blue-coloured solution turns brown and a white precipitate of CuI(s) is seen.

The reaction between copper ions and iodide forms only two products.

- (i) Complete the equation for this reaction.



[1]

- (ii) Identify the oxidising agent in this reaction. Explain your answer in terms of electron transfer.

[1]

22. M/J 22/P22/Q1(b)(i)-(ii)

Context: part (b) (introduction)

- (b) When magnesium is heated in air, magnesium oxide, MgO , is the major product. Smaller amounts of magnesium nitride, Mg_3N_2 , are also made.

- (i) Calculate the oxidation number for magnesium and for the nitrogen species in Mg_3N_2 to complete Table 1.1.

Table 1.1

species	magnesium in Mg_3N_2	nitrogen in Mg_3N_2
oxidation number		

[1]

- (ii) Identify the type of reaction which takes place between magnesium and nitrogen. Explain your answer.

[1]

23. MAR 22/P22/Q2(a)(i)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (a) (introduction)

(a) Na reacts with O_2 to form Na_2O . Na is the reducing agent in this reaction.

(i) Define reducing agent.

[1]

24. MAR 22/P22/Q2(c)(ii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (c) (introduction)

(c) P_4O_6 is a white solid that has a melting point of 24°C . Solid P_4O_6 reacts with water to form H_3PO_3 .

(ii) Determine the oxidation number of P in H_3PO_3 .

[1]

Answers Section

1. M/J 26/P21/Q3(d)

3(d)	species	oxidation number of sulfur	2 •✓✓ A 6 (+) 4 (+) and 2– A VI (+) AND IV (+) AND II – R S ⁺⁴ S ⁺⁶ or S ^{–2} R S ⁴⁺ S ⁶⁺ or S ^{2–}
	H ₂ SO ₄	(+)VI OR (+)6	
	SO ₂	(+)IV OR (+)4	
	H ₂ S	–II OR –2	

2. M/J 26/P21/Q4(a)(ii)

4(a)(ii)	$\text{C}_5\text{H}_{12}\text{O}_4 + 2\text{Cr}_2\text{O}_7^{2-} + 16\text{H}^+ \rightarrow \text{C}_5\text{H}_6\text{O}_7 + 4\text{Cr}^{3+} + 11\text{H}_2\text{O}$	1	
----------	--	----------	--

3. M/J 26/P22/Q4(c)

4(c)(i)	$3\text{C}_5\text{H}_{12}\text{O}_4 + 5\text{Cr}_2\text{O}_7^{2-} + 40\text{H}^+ \rightarrow 3\text{C}_5\text{H}_6\text{O}_6 + 10\text{Cr}^{3+} + 29\text{H}_2\text{O}$ M1 $5\text{Cr}_2\text{O}_7^{2-}$ M2 fully correct equation	2	
4(c)(ii)	$\text{HOCH}_2\text{CH}(\text{OH})\text{C}(\text{OH})(\text{CH}_3)\text{CH}_2\text{OH}$ OR $(\text{HOCH}_2)_2\text{C}(\text{OH})(\text{CHOHCH}_3)$  • molecular formula C ₅ H ₁₂ O ₄ • 2 primary alcohol groups • 1 secondary alcohol group • 1 tertiary alcohol group •✓•✓	2	A unambiguous structure I any name attempted Max 1 all correct but 1 or more connectivity issue e.g. C–H–O

4. M/J 26/P23/Q4(d)(ii)

4(d)(ii)	$3\text{C}_5\text{H}_{12}\text{O}_4 + 8\text{Cr}_2\text{O}_7^{2-} + 64\text{H}^+ \rightarrow 3\text{C}_5\text{H}_4\text{O}_8 + 16\text{Cr}^{3+} + 44\text{H}_2\text{O}$	1	
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5. M/J 25/P21/Q4(b)(iii)

4(b)(iii)	oxidising agent	1	
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6. M/J 25/P24/Q2(b)(ii)

2(b)(ii)	oxidation number of S in $\text{SCl}_2 = (+)2 / (+)\text{II}$ oxidation number of S in $\text{H}_2\text{SO}_3 = (+)4 / (+)\text{IV}$	2	
----------	---	----------	--

7. MAR 25/P22/Q1(a)

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	2	
1(a)(ii)	high pressure low temperature	2	
1(a)(iii)	ratio of moles of P : Cl = $\frac{30.4}{31.0} : \frac{69.6}{35.5} = 0.98 : 1.96 (= 1 : 2)$ molecular formula = P_2Cl_4	2	
1(a)(iv)	(structure =) simple / molecular because low melting point / weak intermolecular forces (bonding =) covalent because hydrolysed	2	

8. O/N 24/P21/Q2(c)(ii)-(iii)

2(c)(ii)	a reaction where a species / same species is both oxidised and reduced simultaneously / in the same reaction	1
2(c)(iii)	$\text{CH}_3\text{COCH}_3 + 3\text{NaClO} \rightarrow 1\text{CHCl}_3 + 1\text{CH}_3\text{COONa} + 2\text{NaOH}$	1

9. O/N 24/P22/Q3(b)(i)-(ii)

3(b)(i)	(+)5 / V	1
3(b)(ii)	aluminium / Al	1

10. O/N 24/P23/Q2(c)(ii)-(iii)

2(c)(ii)	a reaction where a species / same species is both oxidised and reduced simultaneously / in the same reaction	1
2(c)(iii)	$\text{CH}_3\text{COCH}_3 + 3\text{NaClO} \rightarrow 1\text{CHCl}_3 + 1\text{CH}_3\text{COONa} + 2\text{NaOH}$	1

11. M/J 24/P23/Q1(d)(i)

1(d)(i)	(+)IV	1
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12. MAR 24/P22/Q1(b)(i)

1(b)(i)	$2\text{Bi} + 3\text{H}_2\text{O} \rightarrow \text{Bi}_2\text{O}_3 + 3\text{H}_2$	1
---------	--	----------

13. MAR 24/P22/Q1(c)

1(c)(i)	Bi_2O_3 (+)3 / III AND NaBiO_3 (+)5 / V	1
1(c)(ii)	Bi_2O_3	1

14. MAR 24/P22/Q1(d)(i)

1(d)(i)	a substance that removes electrons from another / causes electron loss / gains electrons	1
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15. O/N 23/P21/Q3(a)

3(a)	compound / molecule / substance that oxidises another / is reduced	1
------	--	----------

16. O/N 23/P21/Q3(c)

3(c)	Cl / chlorine is both oxidised and reduced	1
------	--	----------

17. O/N 23/P22/Q3(b)(iii)

3(b)(iii)	P (in H_3PO_3) is (both) oxidised and reduced (simultaneously)	1
	P (is oxidised) from (+)3 / (+)III $\rightarrow$ (+)5 / (+)V AND (reduced to) -3 / -III	1

18. O/N 23/P23/Q3(a)

3(a)	compound / molecule / substance that oxidises another / is reduced	1
------	--	----------

19. O/N 23/P23/Q3(c)

3(c)	Cl / chlorine is both oxidised and reduced	1
------	--	----------

20. M/J 23/P22/Q2(e)(ii)

2(e)(ii)	$\text{NaClO} + \text{H}_2\text{O}_2 \rightarrow \text{O}_2 + \text{NaCl} + \text{H}_2\text{O}$	1
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21. M/J 23/P23/Q1(d)(i)-(ii)

1(d)(i)	$2\text{Cu}^{2+} + 4\text{I}^{-} \rightarrow 2\text{CuI} + \text{I}_2$	1
1(d)(ii)	Cu^{2+} OR CuSO_4 AND copper (species / ion) has gained / taken electron(s) from iodide (ion)	1

22. M/J 22/P22/Q1(b)(i)-(ii)

1(b)(i)	species	magnesium in Mg_3N_2	nitrogen in Mg_3N_2	1
	oxidation number	2(+) / (+)II	3- / -III	
1(b)(ii)	- redox - Mg loses electrons / Mg increases ox no / Mg is oxidised - N gains electrons / N reduces ox no / N is reduced			1

23. MAR 22/P22/Q2(a)(i)

2(a)(i)	species that donates electrons	1
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24. MAR 22/P22/Q2(c)(ii)

2(c)(ii)	(+)3 / III	1
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Unit 7

EQUILIBRIA

44 items · 129 marks

Syllabus sections in this unit

7.1 Chemical equilibria: reversible reactions, dynamic equilibrium

7.2 Brønsted–Lowry theory of acids and bases

Unit 7: Equilibria

1. M/J 26/P21/Q2(b)(i)

Context

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

Context: part (b) (introduction)

(b) The reaction conditions for stage 2 include:

- using a ratio of 1 mol SO_2 to 1 mol O_2
- heat in the presence of V_2O_5 .

(i) Stage 2 is repeated using a ratio of more than 1 mol SO_2 to 1 mol O_2 .

Predict the effect on the percentage yield of SO_3 at equilibrium. Use Le Chatelier's Principle to explain your answer.

[1]

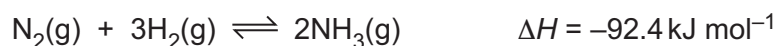
2. M/J 26/P22/Q1(e)(i)

Context

1 Oxygen reacts with many elements.

Context: part (e) (introduction)

(e) Ammonia is made in the Haber process.



A temperature of 400–450 °C is used in the Haber process.

(i) State **two** other conditions used in the Haber process.

1

2

[2]

3. M/J 26/P22/Q3(a)

also Unit 11 (1 mark); this unit 2 marks

Context

- 3** Separate samples of NaCl and NaBr are reacted with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smelling gas

- (a) (i) Write an equation for reaction 1.

[1]

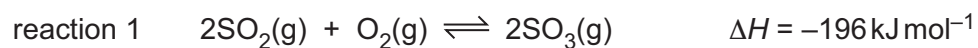
- (ii) Explain why reaction 1 is described as a Brønsted–Lowry acid–base reaction.

[2]

4. M/J 26/P23/Q3(b)

Context

- 3** The conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process are carefully controlled.



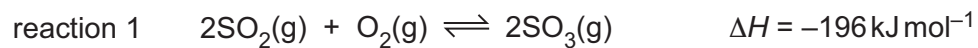
- (b) Define Le Chatelier's principle.

[2]

5. M/J 26/P23/Q3(c)

Context

- 3** The conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process are carefully controlled.



- (c) In industry, reaction 1 occurs at 400–450 °C and 110–200 kPa with a catalyst.

Complete Table 3.1 by using the terms 'increase', 'decrease' or 'no change'.

Table 3.1

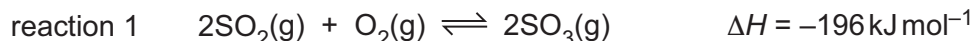
	use temperature greater than 450 °C	without catalyst
effect on value of equilibrium constant, K_p		
effect on yield of product at equilibrium		
effect on rate of reaction		

[3]

6. M/J 26/P23/Q3(d)

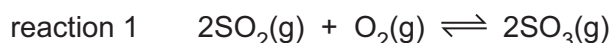
Context

- 3** The conditions used for the conversion of sulfur dioxide to sulfur trioxide in the Contact process are carefully controlled.



- (d)** A sealed tube that initially contains only 2.00 mol of $\text{SO}_2(\text{g})$ and 2.00 mol of $\text{O}_2(\text{g})$ is left to reach equilibrium.

At equilibrium, 1.96 mol of $\text{SO}_3(\text{g})$ is present and the total pressure is 150 kPa.



- (i)** Complete Table 3.2.

Table 3.2

	$\text{SO}_2(\text{g})$	$\text{O}_2(\text{g})$	$\text{SO}_3(\text{g})$
amount initially / mol	2.00	2.00	0
amount at equilibrium / mol			1.96
mole fraction at equilibrium	0.01325	0.3377	

[3]

- (ii)** Table 3.3 shows the mole fractions at equilibrium of $\text{SO}_2(\text{g})$, $\text{O}_2(\text{g})$ and $\text{SO}_3(\text{g})$ under different conditions.

Table 3.3

	$\text{SO}_2(\text{g})$	$\text{O}_2(\text{g})$	$\text{SO}_3(\text{g})$
mole fraction at equilibrium	0.360	0.019	0.621

Calculate K_p at 200 kPa using the data from Table 3.3.

Include units in your answer.

$$K_p =$$

units

[3]

7. M/J 26/P24/Q2(b)(ii)

Context: part (b) (introduction)

- (b)** Nitrogen is used to manufacture ammonia, NH_3 , in the Haber process. An iron catalyst is used.

- (ii)** State the temperature and pressure used in the Haber process.

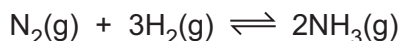
temperature =

pressure =

[2]

8. M/J 26/P24/Q2(c)

(c) The equation for the Haber process is shown.



Three separate experiments are carried out using mixtures of $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$ to find the amount of $\text{NH}_3(\text{g})$ produced at equilibria under different conditions.

- (i) In experiment 1, 3.55 mol of $\text{N}_2(\text{g})$ and 8.60 mol of $\text{H}_2(\text{g})$ are mixed and left to reach equilibrium.

At equilibrium, 3.24 mol of NH_3 is present.

Complete Table 2.1.

Table 2.1

	$\text{N}_2(\text{g})$	$\text{H}_2(\text{g})$	$\text{NH}_3(\text{g})$
initial amount/mol	3.55	8.60	0
change in amount/mol			3.24
amount at equilibrium/mol			3.24

[3]

- (ii) In experiment 2, under different conditions $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$ are mixed and left to reach equilibrium.

Table 2.2 shows the amount, in mol, at equilibrium of $\text{N}_2(\text{g})$, $\text{H}_2(\text{g})$ and $\text{NH}_3(\text{g})$.

Table 2.2

	$\text{N}_2(\text{g})$	$\text{H}_2(\text{g})$	$\text{NH}_3(\text{g})$
amount at equilibrium/mol	0.874	1.340	0.347

Calculate the mole fraction of $\text{NH}_3(\text{g})$ in the equilibrium mixture in experiment 2.

$$\text{mole fraction of } \text{NH}_3(\text{g}) =$$

[2]

- (iii) The expression for the equilibrium constant, K_p , for the Haber process is shown.

$$K_p = \frac{p(\text{NH}_3)^2}{p(\text{N}_2) \times p(\text{H}_2)^3}$$

Describe what is meant by partial pressure.

[1]

- (iv) In experiment 3, the mixture at equilibrium contains the following partial pressures of $\text{N}_2(\text{g})$ and $\text{H}_2(\text{g})$.

$$p(\text{N}_2) = 1590 \text{ kPa}$$

$$p(\text{H}_2) = 1750 \text{ kPa}$$

For experiment 3, the equilibrium constant, K_p , is $8.49 \times 10^{-7} \text{ kPa}^{-2}$.

Calculate the partial pressure, in kPa, of $\text{NH}_3(\text{g})$ at equilibrium.

$$p(\text{NH}_3) = \text{ kPa [3]}$$

9. MAR 26/P22/Q4(a)(i)-(iii)

Context

4 Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.

Context: part (a) (introduction)

(a) At room temperature, $\text{CH}_2(\text{CN})\text{COOH}$ undergoes an acid–base reaction with $\text{NaOH}(\text{aq})$.

(i) Define weak acid.

[1]

(ii) Sketch the pH titration curve on Figure 4.1 for the reaction of:

- 25.00 cm^3 of 0.10 mol dm^{-3} $\text{CH}_2(\text{CN})\text{COOH}(\text{aq})$ and
- 0.10 mol dm^{-3} $\text{NaOH}(\text{aq})$, added to excess.

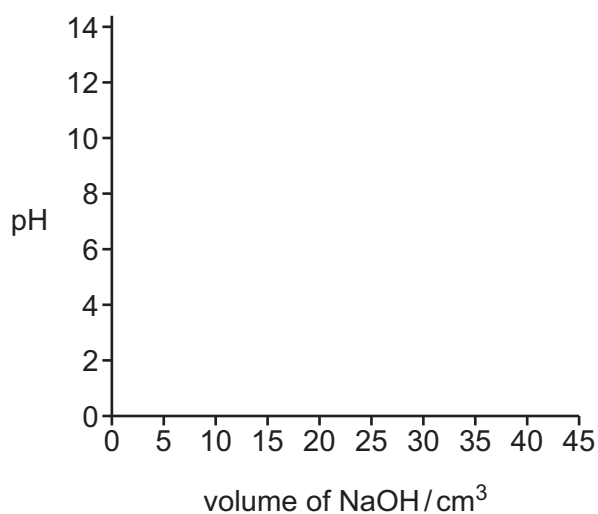


Figure 4.1

(iii) Write the overall ionic equation for the neutralisation of $\text{CH}_2(\text{CN})\text{COOH}(\text{aq})$ by $\text{NaOH}(\text{aq})$ in **4(a)(ii)**.

[3]

Include state symbols.

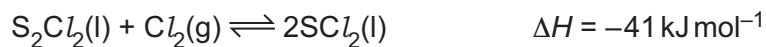
[1]

10. O/N 25/P22/Q2(e)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

- (e) $\text{S}_2\text{Cl}_2(\text{l})$ reacts with $\text{Cl}_2(\text{g})$ in a reversible reaction to form $\text{SCl}_2(\text{l})$. Under certain conditions, a dynamic equilibrium is established.



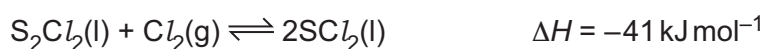
- (i) State what is meant by dynamic equilibrium.

[1]

- (ii) Identify the condition necessary to establish dynamic equilibrium.

[1]

- (iii) $\text{S}_2\text{Cl}_2(\text{l})$ is yellow and $\text{SCl}_2(\text{l})$ is red.



State what is observed when the following changes are made to an equilibrium mixture of $\text{S}_2\text{Cl}_2(\text{l})$ and $\text{SCl}_2(\text{l})$.

Explain your answers.

- The equilibrium mixture is warmed gently.
observation
explanation
- The overall pressure of the equilibrium mixture is increased.
observation
explanation

[4]

11. O/N 25/P24/Q2(b)

Context

- 2** Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

(b) A 5.00 dm^3 sealed flask contains 0.400 mol of $CO(g)$ and 0.800 mol of $H_2(g)$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium.

Equation 1 shows the reaction.



The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[CH_3OH]}{[CO][H_2]^2}$$

- (i) State the units of K_c .

[1]

- (ii) The equilibrium mixture contains 0.280 mol of $CH_3OH(g)$.

Calculate the value of K_c .

Give your answer to **three** significant figures.

value of $K_c =$ [3]

- (iii) State and explain the effect, if any, on the value of K_c when the overall pressure in the sealed flask is increased.

[1]

12. M/J 25/P21/Q3(b)

- (b) Define Le Chatelier's principle.

[2]

13. M/J 25/P21/Q3(c)

- (c) Reaction 2 shows the equilibrium reaction between X(g) and Y(g) to produce Z(g) in a sealed container.

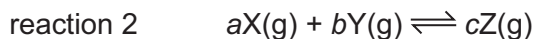


Fig. 3.3 shows the effect of changing pressure on the percentage yield of Z(g) at two different temperatures, 300 K and 350 K.

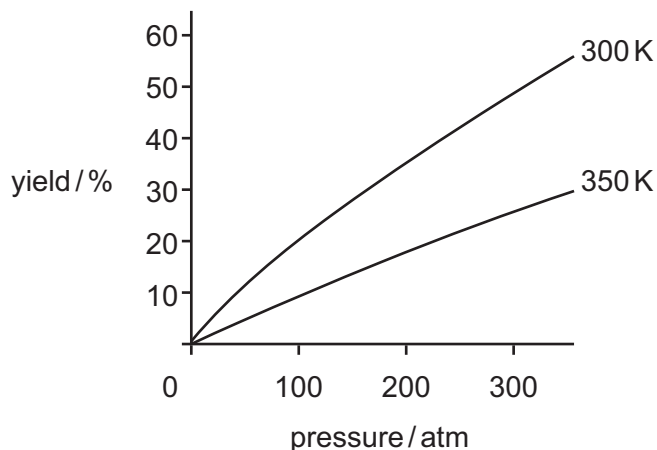


Fig. 3.3

Deduce **two** conclusions about reaction 2 using Fig 3.3.

[2]

14. M/J 25/P23/Q2(d)

Context

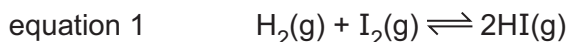
- 2** The chemical properties of oxides are related to the chemical bonding present in these compounds.
- (d) State why ZnO is described as a Brønsted–Lowry base when it is added to H₂SO₄(aq).

[1]

15. M/J 25/P24/Q3(b)

Context

- 3** H₂(g) and I₂(g) react to form HI(g) in a reversible reaction, as shown in equation 1.



In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

- (b) In experiment 2, 0.100 mol of H₂(g), 0.200 mol of I₂(g) and 0.300 mol of HI(g) are combined.

At equilibrium 0.154 mol of I₂ is present.

Calculate the amounts, in mol, of H₂ and HI in the equilibrium mixture produced in experiment 2.

amount of H₂ in equilibrium mixture = _____ mol

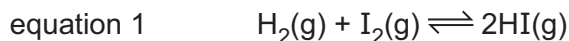
amount of HI in equilibrium mixture = _____ mol

[3]

16. M/J 25/P24/Q3(c)

Context

- 3** $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ react to form $\text{HI}(\text{g})$ in a reversible reaction, as shown in equation 1.



In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

- (c)** In experiment 3, 0.0772 mol of H_2 and 0.0986 mol of I_2 are present in an equilibrium mixture at 298 K.



Use the equilibrium constant, K_{c} , to calculate the amount, in mol, of HI present in the equilibrium mixture in experiment 3.

amount of HI in equilibrium mixture = _____ mol [1]

17. O/N 24/P21/Q3(b)(iii)

Context

- 3** The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (b) (introduction)

- (b)** Carbon is found in inorganic compounds such as carbonates.

- (iii)** Ammonium carbonate undergoes an acid–base reaction with $\text{NaOH}(\text{aq})$.

Explain this statement.

[2]

18. O/N 24/P23/Q3(b)(iii)

Context

- 3** The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (b) (introduction)

- (b)** Carbon is found in inorganic compounds such as carbonates.

- (iii)** Ammonium carbonate undergoes an acid–base reaction with $\text{NaOH}(\text{aq})$.

Explain this statement.

[2]

19. M/J 24/P21/Q2(c)

also Unit 6 (1 mark), Unit 10 (1 mark); this unit 2 marks

(c) Solid magnesium nitride, Mg_3N_2 , is a crystalline solid.

- (i) Deduce the oxidation numbers of magnesium and nitrogen in magnesium nitride to complete Table 2.1.

Table 2.1

	oxidation number in Mg_3N_2
magnesium	
nitrogen	

[1]

- (ii) Magnesium nitride reacts with an excess of water to produce ammonia and magnesium hydroxide only. Construct an equation to describe this reaction.

[1]

- (iii) Explain why the solution produced in the reaction in (c)(ii) has a pH greater than 7. Refer to the products of the reaction in your answer.

[2]

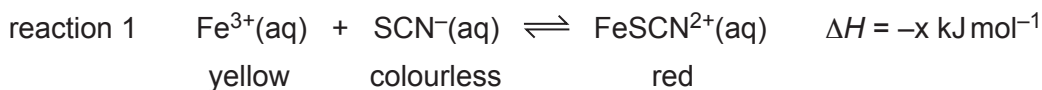
20. M/J 24/P21/Q3(a)

- 3 (a) Define Le Chatelier's principle.

[2]

21. M/J 24/P21/Q3(b)-(c)

- (b) Reaction 1 describes the reversible reaction between yellow $\text{Fe}^{3+}(\text{aq})$ and colourless $\text{SCN}^{-}(\text{aq})$ to produce red $\text{FeSCN}^{2+}(\text{aq})$.



A mixture of $\text{Fe}^{3+}(\text{aq})$, $\text{SCN}^{-}(\text{aq})$ and $\text{FeSCN}^{2+}(\text{aq})$ is at equilibrium at 20°C .

The temperature of this mixture is then increased to 50°C and allowed to reach equilibrium.

Deduce the changes that occur, if any, in the equilibrium mixture at 50°C compared to the equilibrium mixture at 20°C .

- change in appearance
- change in relative concentration of $\text{FeSCN}^{2+}(\text{aq})$
- change in value of the equilibrium constant, K_c

[3]

- (c) In another experiment, equimolar amounts of $\text{Fe}^{3+}(\text{aq})$ and $\text{SCN}^{-}(\text{aq})$ are mixed together and allowed to reach equilibrium. The total volume of the mixture is 25.0 cm^3 .



At equilibrium the mixture contains:

- $[\text{SCN}^{-}] = 1.30 \times 10^{-3} \text{ mol dm}^{-3}$
- $[\text{FeSCN}^{2+}] = 0.300 \times 10^{-3} \text{ mol dm}^{-3}$.

- (i) Calculate the initial amount, in mol, of $\text{Fe}^{3+}(\text{aq})$ added to $\text{SCN}^{-}(\text{aq})$ to produce this mixture.

initial amount of $\text{Fe}^{3+}(\text{aq}) =$ mol [2]

- (ii) Calculate K_c for reaction 1 and state its units.

Show your working.

 $K_c =$

units

[2]

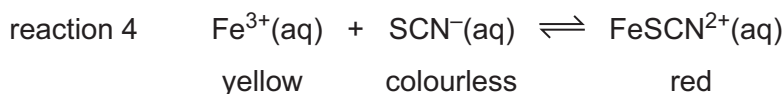
22. M/J 24/P22/Q3(a)

- 3 (a) Describe what is meant by dynamic equilibrium.

[2]

23. M/J 24/P22/Q3(b)

- (b) Reaction 4 describes the reversible reaction between yellow $\text{Fe}^{3+}(\text{aq})$ and colourless $\text{SCN}^{-}(\text{aq})$ to produce red $\text{FeSCN}^{2+}(\text{aq})$.



An equilibrium mixture contains $\text{Fe}^{3+}(\text{aq})$, $\text{SCN}^{-}(\text{aq})$ and $\text{FeSCN}^{2+}(\text{aq})$. A few colourless crystals of soluble $\text{KSCN}(\text{s})$ are added. The mixture is then left until it reaches equilibrium again. The temperature of both equilibrium mixtures is the same.

- (i) Deduce the changes that occur, if any, in the equilibrium mixture after $\text{KSCN}(\text{s})$ is added compared to the original equilibrium mixture.
- change in appearance
 - change in relative concentration of $\text{Fe}^{3+}(\text{aq})$
 - change in value of the equilibrium constant, K_c

[3]



7

- (ii) The expression for the equilibrium constant, K_c , for reaction 4 is shown.

$$K_c = \frac{[\text{FeSCN}^{2+}(\text{aq})]}{[\text{Fe}^{3+}(\text{aq})] \times [\text{SCN}^{-}(\text{aq})]}$$

$5.00 \times 10^{-5} \text{ mol}$ of $\text{Fe}^{3+}(\text{aq})$ and $5.00 \times 10^{-5} \text{ mol}$ of $\text{SCN}^{-}(\text{aq})$ are added together and allowed to reach equilibrium. The total volume of the mixture is 25.0 cm^3 .

At equilibrium the concentration of $\text{FeSCN}^{2+}(\text{aq})$ is $4.23 \times 10^{-4} \text{ mol dm}^{-3}$.

Calculate the equilibrium constant, K_c , for reaction 4.

Include the units in your answer.

 $K_c =$

units

[4]



8

24. M/J 24/P23/Q2(a)

Context

- 2** $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.

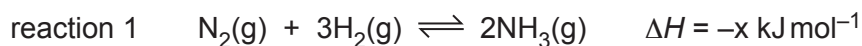


Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

- (a) Describe and explain the change, if any, to the percentage yield of $\text{NH}_3(\text{g})$ produced in **B** compared to **A**.

[1]

25. M/J 24/P23/Q2(b)(i)

Context

- 2** $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.

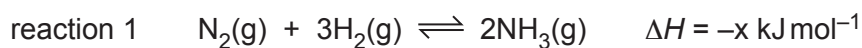


Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

- (b) (i) Describe and explain the change, if any, to the percentage yield of $\text{NH}_3(\text{g})$ produced in **C** compared to **A**.

[1]

26. M/J 24/P23/Q2(c)

Context

- 2** $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.

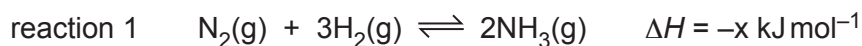


Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

- (c) (i)** Write an expression for the equilibrium constant, K_p , for reaction 1. State the units.

$$K_p =$$

units

[2]

- (ii)** Equilibrium mixture **D** is made when 1.0 mol of $\text{N}_2(\text{g})$ and 3.0 mol of $\text{H}_2(\text{g})$ are added to a sealed container at 750 °C and 1000 atm and left to reach equilibrium. This mixture contains 1.16 mol of $\text{NH}_3(\text{g})$.

Calculate the mole fraction of $\text{NH}_3(\text{g})$ in **D**.

$$\text{mole fraction of } \text{NH}_3(\text{g}) =$$

[2]

- (iii)** The mole fraction of $\text{N}_2(\text{g})$ is 0.625 in a new equilibrium mixture, **E**.

Calculate the partial pressure of $\text{N}_2(\text{g})$ in **E** when the total pressure is 1000 atm.

$$\text{partial pressure of } \text{N}_2(\text{g}) =$$

atm [1]

27. M/J 24/P23/Q4(b)(i)

Context: part (b) (introduction)

- (b)** An exothermic reaction occurs when $\text{NH}_4^+(\text{aq})$ is added to $\text{OH}^-(\text{aq})$.

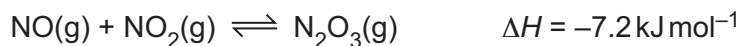
- (i)** Identify the type of reaction.

[1]

28. O/N 23/P22/Q2(a)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

- (a)** Fig. 2.1 shows how the rate of the forward reaction changes with time.

Initially, the rate of the reverse reaction is zero.

Complete Fig. 2.1 to sketch how the rate of the **reverse** reaction changes with time.

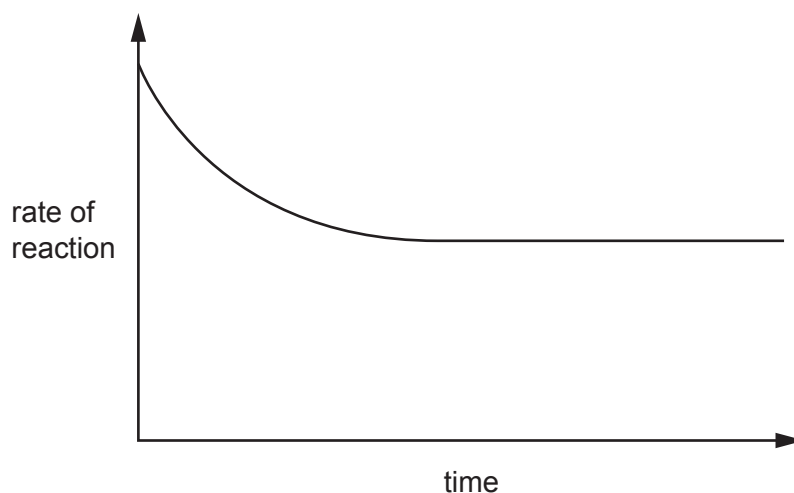


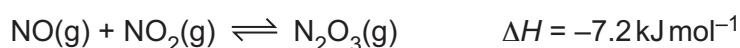
Fig. 2.1

[1]

29. O/N 23/P22/Q2(b)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

- (b)** State how the position of equilibrium changes, if at all, when the reaction takes place at 100 °C.

Explain your answer.

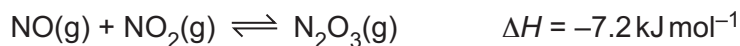
Assume the pressure remains constant.

[2]

30. O/N 23/P22/Q2(c)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

- (c)** Table 2.1 shows the composition of an equilibrium mixture of NO(g), NO₂(g) and N₂O₃(g) at 101 kPa.

Table 2.1

gas	number of moles at equilibrium / mol
NO	0.605
NO ₂	0.605
N ₂ O ₃	0.390

Calculate K_p , the equilibrium constant with respect to partial pressures.

Deduce the units of K_p .

$K_p =$ units

[3]

31. O/N 23/P22/Q3(a)(ii)

Context

- 3** Phosphoric(V) acid, H₃PO₄, is used in both inorganic and organic reactions.

Context: part (a) (introduction)

- (a)** H₃PO₄ is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H₃PO₄.

- (ii)** H₃PO₄ is a weak Brønsted–Lowry acid.

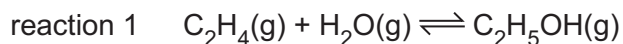
Define weak Brønsted–Lowry acid.

[2]

32. M/J 23/P22/Q4(b)

Context

- 4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



- (b) Reaction 1 reaches equilibrium at constant temperature and pressure.

Deduce what effect increasing the pressure will have on the amount of ethanol in the new equilibrium mixture. Use Le Chatelier's principle to explain your answer.

effect of increasing pressure

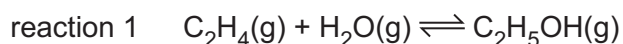
explanation

[2]

33. M/J 23/P22/Q4(c)(i)

Context

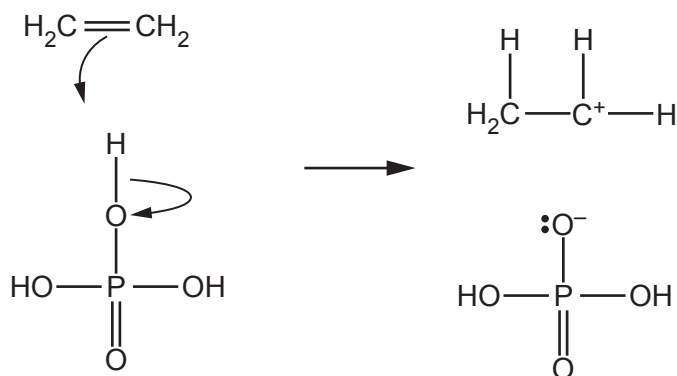
- 4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



Context: part (c) (introduction)

- (c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

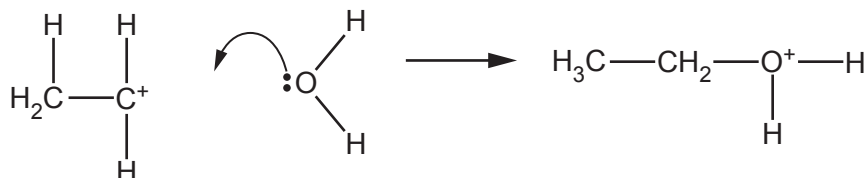


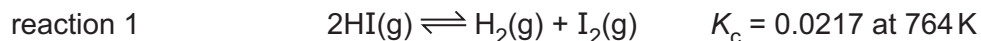
Fig. 4.1

- (i) Describe the behaviour of H_3PO_4 in step 1 in Fig. 4.1. Explain your answer.

[1]

34. M/J 23/P23/Q4(b)

- (b) A sample of HI(g) is added to a 2.00 dm³ sealed vessel at 764 K and allowed to reach equilibrium.



At equilibrium the mixture contains 1.70 mol of HI(g).

- (i) State **one** difference in the appearance of the initial reaction mixture compared to the mixture at equilibrium.

[1]

- (ii) Deduce the expression for equilibrium constant K_c for reaction 1.

$$K_c =$$

[1]

- (iii) Calculate the concentration of I₂ present in the reaction mixture at equilibrium.
Show your working.

concentration of I₂ = mol dm⁻³ [3]

35. M/J 23/P23/Q4(c)

- (c) The experiment is repeated at 500 K. The value of K_c under these conditions is 0.00625.

- (i) Describe the difference in the composition of the equilibrium mixture at 500 K compared to 764 K.

[1]

- (ii) Use Le Chatelier's principle to deduce whether the decomposition of HI(g) is endothermic or exothermic. Explain your answer.

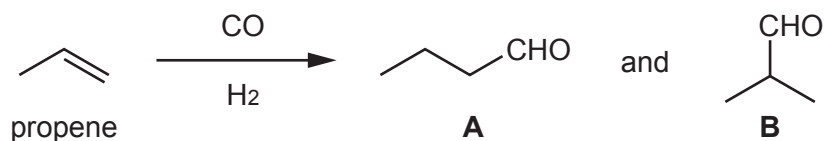
[1]

36. MAR 23/P22/Q3(d)(ii)

Context

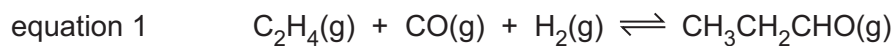
- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

**Fig. 3.1**

Context: part (d) (introduction)

- (d)** The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



At atmospheric pressure a cobalt-based catalyst is used in this reaction.

- (ii)** Explain why the yield of CH₃CH₂CHO(g) increases when the overall pressure of the reaction mixture is increased.

[1]

37. O/N 22/P22/Q1(c)(i)

Context

- 1** Species such as NH₄⁺, CO₃²⁻ and PO₄³⁻ are examples of molecular ions.

Context: part (c) (introduction)

- (c)** NH₄⁺ is a Brønsted–Lowry acid.

- (i)** Define Brønsted–Lowry acid.

[1]

38. O/N 22/P22/Q2(c)(ii)

Context

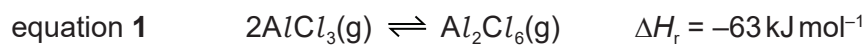
2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl ₃	SiCl ₄	PCl ₅	PCl ₃	SCl ₂
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

Context: part (c) (introduction)

(c) In the gas phase, AlCl₃(g) exists at equilibrium with Al₂Cl₆(g) as shown.



(ii) State the effect of an increase in temperature on the equilibrium mixture in equation 1. Explain your answer.

[1]

39. M/J 22/P21/Q2(d)

Context

2 Nitrogen molecules, $\text{N}_2(\text{g})$, contain two atoms attracted to each other by a triple covalent bond.

(d) 25 cm^3 of $0.10\text{ mol dm}^{-3}\text{ HCl}(\text{aq})$ is added to a beaker and its pH is recorded.

50 cm^3 of $0.10\text{ mol dm}^{-3}\text{ NH}_3(\text{aq})$ is added to the $\text{HCl}(\text{aq})$ in 5 cm^3 portions.

The pH of the mixture is monitored until all the $\text{NH}_3(\text{aq})$ is added.

HCl is a strong Brønsted-Lowry acid.

(i) Describe what is meant by a strong Brønsted-Lowry acid.

[2]

(ii) NH_3 is a weak base.

Construct an equation that shows the behaviour of NH_3 as a weak Brønsted-Lowry base when dissolved in water.

[1]

(iii) On Fig. 2.1 sketch a graph to show the change in pH which occurs when $\text{HCl}(\text{aq})$ is titrated with $\text{NH}_3(\text{aq})$ as described in **(d)**.

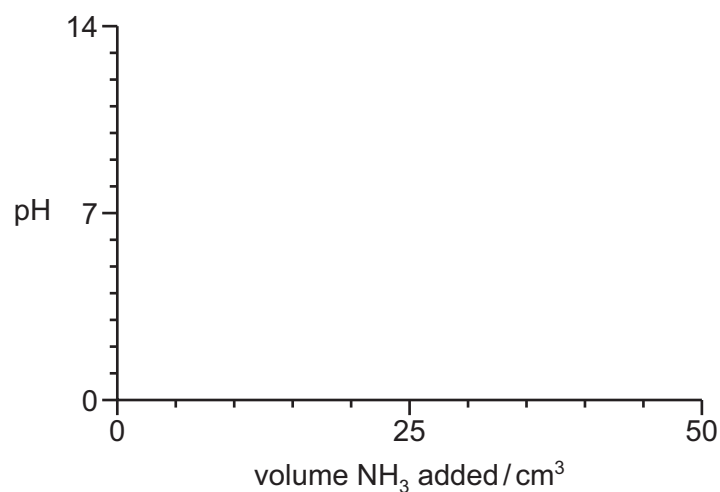
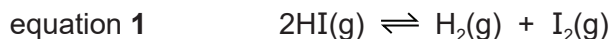


Fig. 2.1

[2]

40. M/J 22/P22/Q3(a)

- 3 (a)** 0.025 mol of HI(g) is added to a closed vessel and left to reach dynamic equilibrium. The total pressure of the vessel is 100 kPa.



- (i)** Explain what is meant by dynamic equilibrium.

[2]

- (ii)** Describe **one** difference in the initial appearance of the reaction mixture compared to the mixture at equilibrium.

[1]

- (iii)** Write an expression for K_p for the reaction described in equation 1.

$$K_p =$$

[1]

- (iv)** At equilibrium the partial pressure of HI(g) is 86.4 kPa.

Calculate the amount of HI(g) present in the mixture at equilibrium. Show your working.

amount of HI(g) = mol [2]

41. M/J 22/P22/Q3(c)

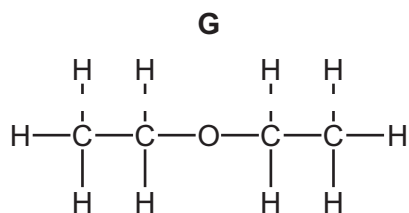
- (c)** Describe the effect of increasing pressure on the value of K_p for the decomposition of HI(g).

[1]

42. M/J 22/P23/Q3(a)(iii)

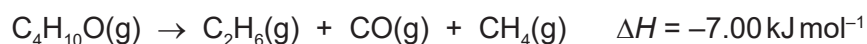
Context

3 **G** belongs to a group of compounds called ethers.

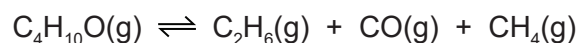
**Fig. 3.1**

Context: part (a) (introduction)

(a) When **G**, $\text{C}_4\text{H}_{10}\text{O}$, is heated, thermal decomposition occurs.



(iii) When **G**, $\text{C}_4\text{H}_{10}\text{O}$, is heated in a sealed container, an equilibrium mixture is produced.



Complete the expression for the equilibrium constant, K_c , for this reaction.
State the units of K_c .

 $K_c =$

units =

[2]

43. MAR 22/P22/Q2(d)(iii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (d) (introduction)

(d) SO_2 and SO_3 are found in the atmosphere.

The oxidation of SO_2 to SO_3 in the atmosphere is catalysed by NO_2 .

The first step of the catalytic oxidation is shown in equation 1.



(iii) Fig. 2.2 shows how the temperature of the atmosphere varies with height from the ground.

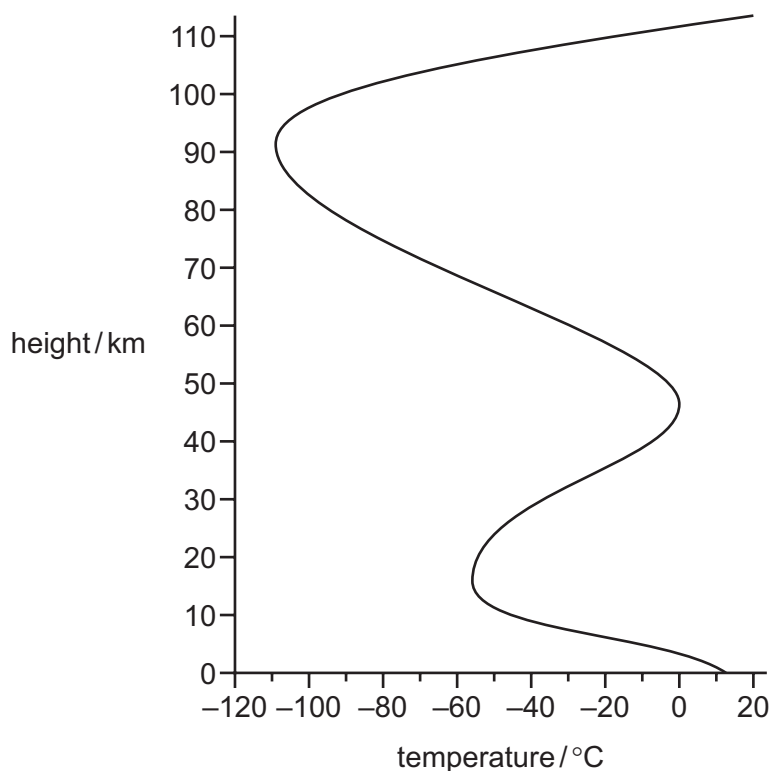


Fig. 2.2

The equilibrium reaction in equation 1 has $\Delta H_r = -168 \text{ kJ mol}^{-1}$.

Suggest how the position of this equilibrium differs at a height of 20 km compared with a height of 50 km from the ground.

Explain your answer.

[2]

44. MAR 22/P22/Q3(d)

Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

(d) The hydrogen halides dissolve in water to form strong Brønsted–Lowry acids.

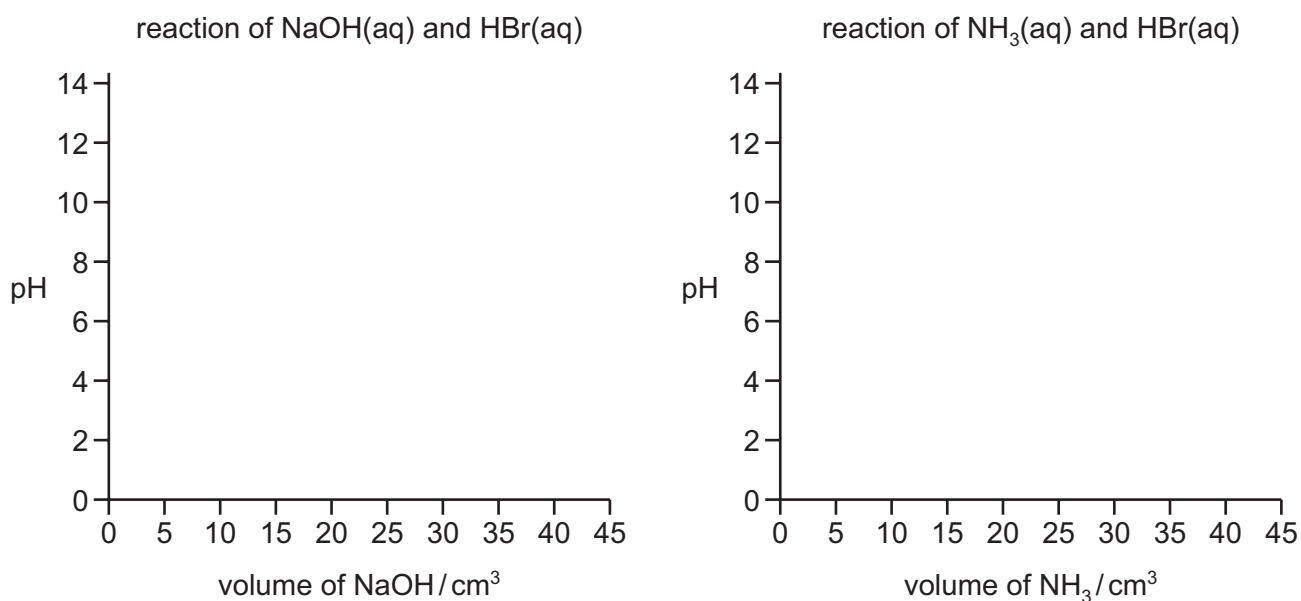
The concentration of a strong acid can be determined by titration.

(i) State what is meant by strong Brønsted–Lowry acid.

[2]

(ii) On Fig. 3.2, sketch the pH titration curves produced when:

- $0.1 \text{ mol dm}^{-3} \text{ NaOH(aq)}$ is added to 25 cm^3 of $0.1 \text{ mol dm}^{-3} \text{ HBr(aq)}$, to excess
- $0.1 \text{ mol dm}^{-3} \text{ NH}_3(\text{aq})$ is added to 25 cm^3 of $0.1 \text{ mol dm}^{-3} \text{ HBr(aq)}$, to excess.

**Fig. 3.2**

[3]

Answers Section

1. M/J 26/P21/Q2(b)(i)

2(b)(i)	increases yield AND <u>equilibrium</u> shifts to right (hand side)	1 A alternatives to increased yield = increased amount / more product / SO_3 A <u>equilibrium</u> shifts to the side with fewer gaseous molecules / to side with fewer gaseous moles A <u>equilibrium</u> shifts to the right to reduce the concentration of SO_2 BOD <u>equilibrium</u> shifts to (favour) forwards reaction / direction I equilibrium shifts to side with lower volume I limiting reagent I any comments on rate R equilibrium shifts to side with more gaseous moles
---------	--	---

2. M/J 26/P22/Q1(e)(i)

1(e)(i)	M1 high pressure OR ≥ 200 atm M2 iron catalyst / Fe catalyst	2 A answers on either line M1 BOD equivalent pressure of ≥ 150 atm or equivalent in alternative units with appropriate conversion M1 A $\geq 15\,000$ kPa / ≥ 150 bar M1 I ATP M1 I numbers with no units M2 A alternative correct named catalysts e.g. Co (consult TL if necessary) M2 BOD Fe(II) catalyst / Fe^{2+} catalyst / Fe^{3+} catalyst / iron oxide catalyst M2 BOD magnetite catalyst M2 I Ni M2 I promoters M2 R Pt / platinum / vanadium oxide
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3. M/J 26/P22/Q3(a)

3(a)(i)	$\text{H}_2\text{SO}_4 + \text{NaCl} \rightarrow \text{NaHSO}_4 + \text{HCl}$ OR $\text{H}_2\text{SO}_4 + 2\text{NaCl} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{HCl}$	1 A $\text{Cl}^- + \text{H}_2\text{SO}_4 \rightarrow \text{HCl} + \text{HSO}_4^-$ A $2\text{Cl}^- + \text{H}_2\text{SO}_4 \rightarrow 2\text{HCl} + \text{SO}_4^{2-}$
3(a)(ii)	M1 H_2SO_4 acid AND Cl^- base M2 H_2SO_4 donates proton(s) AND Cl^- accepts protons OR H_2SO_4 gives / transfers protons <u>to Cl^-</u>	2 proton = hydrogen ion I general definition of BL acids and bases. $\text{Cl}^- = \text{Cl}$ in NaCl I HCl is base/protons given to HCl Alternative if neither M1 nor M2 can be awarded: • H_2SO_4 acid • Cl^- base • H_2SO_4 donates proton(s) • Cl^- accepts proton(s) •✓•✓

4. M/J 26/P23/Q3(b)

3(b)	M1 (if a change in conditions occurs) the equilibrium shifts M2 to minimise the change (in conditions)	2 M1 A equilibrium moves / equilibrium changes M1 I there will be a 'change in the system' M2 minimise = oppose = counteract = resists = reverse
------	--	--

5. M/J 26/P23/Q3(c)

3(c)		use temp. greater than 450°C	without a catalyst	3	All independent including K_p and yield marks
	effect on value of equilibrium constant, K_p	decreases	no change		
	effect on yield of product at equilibrium	decreases	no change		
	effect on rate of reaction	increases	decreases		

•✓•✓•✓

6. M/J 26/P23/Q3(d)

3(d)(i)		SO ₂ (g)	O ₂ (g)	SO ₃ (g)	3	Independent marks If M3 not 0.649 look to see if ecf from M1 and / or M2 M3 A 0.649 / 0.65 M3 A ecf from M1 and / or M2 M3 A 1 – (0.01325 + 0.3377) = 0.649(05)
	amount initially / mol	2	2	0		
	amount at equilibrium / mol	M1 0.04(0)	M2 (2-0.98) = 1.02	1.96		
	mol fraction at equilibrium	0.01325	0.3377	M3 (1.96 / 3.02) = 0.649(007)		

3(d)(ii)	$\frac{(0.621 \times 200)(0.621 \times 200)}{(0.360 \times 200)(0.360 \times 200)(0.019 \times 200)} = 0.783(1) \text{ kPa}^{-1}$ OR $7.83(1) \times 10^{-4} \text{ Pa}^{-1}$ ✓✓✓ M1 correct expression for K_p using numerical values M2 correct value for K_p. M3 kPa^{-1}	3	0.783(1) kPa^{-1} ✓✓✓ $7.83(1) \times 10^{-4} \text{ Pa}^{-1}$ ✓✓✓ M1 I use of [] for () M2 A ecf from M1 for transcription errors of mol fraction values M2 A ecf from M1 for inverted expression i.e. $\frac{0.360 \times 200)(0.360 \times 200)(0.019 \times 200)}{(0.621 \times 200)(0.621 \times 200)} = 1.27(7)$ AND M3 kPa M3 A kPa^{-1} if no expression / incorrect K_p expression M3 A ecf for appropriate units based on incorrect expression for K_p M3 A 'no unit' if expression used involved no reference to 200kPa and only mol fraction (= 156.6) M3 A Pa^{-1} if calculation is based on conversion of kPa to Pa i.e. $7.83(1) \times 10^{-4} \text{ Pa}^{-1}$ ✓✓✓ I 1 / kPa I $K_p = (\text{ppSO}_3)^2 \div (\text{ppSO}_2)^2 \times (\text{ppO}_2)$
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7. M/J 26/P24/Q2(b)(ii)

2(b)(ii)	M1 (temperature:) 400–450 °C	2	Units required M1 A alternative temperatures with appropriate units A 400–500 °C A 673–773 K M2 A 15 200–25 300 kPa A 15 200 000–25 300 000 Pa A 150–250 bar
	M2 (pressure:) 150–250 atm		

8. M/J 26/P24/Q2(c)

2(c)(i)	<table><tr><td></td><td>N₂(g)</td><td>H₂(g)</td><td>NH₃(g)</td></tr><tr><td>initial amount / mol</td><td>3.55</td><td>8.60</td><td>0</td></tr><tr><td>change in amount / mol</td><td>(-) 1.62</td><td>(-) 4.86</td><td>3.24</td></tr><tr><td>amount at equilibrium / mol</td><td>(+) 1.93</td><td>(+) 3.74</td><td>3.24</td></tr></table>		N ₂ (g)	H ₂ (g)	NH ₃ (g)	initial amount / mol	3.55	8.60	0	change in amount / mol	(-) 1.62	(-) 4.86	3.24	amount at equilibrium / mol	(+) 1.93	(+) 3.74	3.24	3 M1 (-) 1.62 M2 (-) 4.86 M3 N₂ and H₂ amount / moles at equilibrium (initial amount / moles minus change in amount / moles) M3 A ecf from incorrect changes in amount / moles in M1 / M2 M3 R negative amounts / moles at equilibrium Max 2: <table><tr><td></td><td>N₂(g)</td><td>H₂(g)</td><td>NH₃(g)</td></tr><tr><td>initial amount / mol</td><td>3.55</td><td>8.60</td><td>0</td></tr><tr><td>change in amount / mol</td><td>(-)1.93</td><td>(-)3.74</td><td>3.24</td></tr><tr><td>amount at equilibrium /mol</td><td>(+)1.62</td><td>(+)4.86</td><td>3.24</td></tr></table>		N ₂ (g)	H ₂ (g)	NH ₃ (g)	initial amount / mol	3.55	8.60	0	change in amount / mol	(-)1.93	(-)3.74	3.24	amount at equilibrium /mol	(+)1.62	(+)4.86	3.24
	N ₂ (g)	H ₂ (g)	NH ₃ (g)																															
initial amount / mol	3.55	8.60	0																															
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change in amount / mol	(-)1.93	(-)3.74	3.24																															
amount at equilibrium /mol	(+)1.62	(+)4.86	3.24																															
2(c)(ii)	M1 Total amount / mol = 2.561 OR 0.874 + 1.340 + 0.347 M2 Mole fraction of NH₃ = 0.347 ÷ 2.561 = 0.135	2 R M1 and M2 if 0.874 + 1.340 + 0.347 is not used to calculate total amount / moles e.g. total moles = 0.874 + 1.340 M2 A ecf if 0.874 + 1.340 + 0.347 are used to calculate total amount / moles, but incorrectly e.g. 0.874 + 1.340 + 0.347² A correctly rounded to minimum of 2 significant figures A as integer fraction e.g. 347 / 2561 I 13.5% A calculator answer 0.135494																																
2(c)(iii)	(partial pressure) = mole fraction (of a gas) × total pressure OR the pressure that would be exerted by (only) one of the gases in a mixture (of gases) (if it occupied the same volume)	1 A product of mole fraction and total pressure A the pressure that is exerted by one of the gases to the total pressure A The pressure that a gas in a mixture of gases exerts on the surroundings A ratio / proportion / fraction of the pressure of a gas to the total pressure (in a closed system) A gas = molecules = particles R compound / element I amount of pressure that a particular gas occupies compared to total pressure																																
2(c)(iv)	M1 p(NH₃)² = K_p × p(N₂) × p(H₂)³ OR p(NH₃)² = 8.49 × 10⁻⁷ × 1590 × (1750)³ OR p(NH₃) = √ 8.49 × 10⁻⁷ × 1590 × (1750)³ M2 p(NH₃)² = 7.23(4673906) × 10⁶ M3 √ M2 = 2690 / 2689.73	3 M2 ecf from.... M1 transcription error or M1 incorrect rearrangement for p(NH₃)² using all the terms K_p AND p(N₂) AND p(H₂)³ M2 A calculator answer 723 4673.906 M3 ecf √ M2 correctly calculated to minimum 3 significant figures M3 A calculator answer 2689.73																																

9. MAR 26/P22/Q4(a)(i)-(iii)

4(a)(i)	(proton donor that) partially dissociates (in aqueous solution)	1
4(a)(ii)	M1 weak acid curve starts pH 1.7 (calculation not necessary) M2 vertical section at 25 cm ³ M3 strong alkali curve ends at pH 13	3
4(a)(iii)	$\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$	1

10. O/N 25/P22/Q2(e)

2(e)(i)	rate of forward and backward (reaction) are the same / equal OR (ratio of) concentrations of reactants and products are constant	1
2(e)(ii)	closed system	1
2(e)(iii)	M1 (mixture / solution becomes more) yellow M2 (forward) reaction is exothermic (equilibrium shifts left) OR equilibrium / reaction shifts (left) to absorb (additional) heat / energy M3 (mixture / solution becomes more) red M4 fewer moles of gas on right-hand side OR (gas on left and) no gas on the right	4

11. O/N 25/P24/Q2(b)

2(b)(i)	mol ⁻² dm ⁶	1
2(b)(ii)	M1 at equilibrium: moles of CO = 0.120 AND moles of H ₂ = 0.240 M2 [CH ₃ OH] = 0.056 AND [CO] = 0.024 AND [H ₂] = 0.048 M3 use of K_c with correct evaluation = 1.01×10^3	3
2(b)(iii)	no effect AND K_c is not affected by pressure	1

12. M/J 25/P21/Q3(b)

3(b)	M1 (if a change is made at a system is dynamic equilibrium) the equilibrium moves M2 to minimise the change in conditions	2
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13. M/J 25/P21/Q3(c)

3(c)	(forward) reaction is exothermic (a + b) > c	2
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14. M/J 25/P23/Q2(d)

2(d)	it / ZnO accepts protons	1
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15. M/J 25/P24/Q3(b)

3(b)	M1 $0.2 - 0.154 = 0.046$ (mol H ₂ and / or I ₂ reacts) M2 mol H ₂ at equilibrium $0.1 - 0.046 = 0.054$ M3 mol HI at equilibrium $0.3 + (0.046 \times 2) = 0.392$	3
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16. M/J 25/P24/Q3(c)

3(c)	2.46	1
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17. O/N 24/P21/Q3(b)(iii)

3(b)(iii)	M1 NH_4^+ / ammonium ion / ammonium cation is a proton / H^+ donor M2 OH^- is a proton / H^+ acceptor OR the OH^- / NaOH is a stronger base than NH_3	2
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18. O/N 24/P23/Q3(b)(iii)

3(b)(iii)	M1 NH_4^+ / ammonium ion / ammonium cation is a proton / H^+ donor M2 OH^- is a proton / H^+ acceptor OR the OH^- / NaOH is a stronger base than NH_3	2
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19. M/J 24/P21/Q2(c)

2(c)(i)	<table><tr><td></td><td>oxidation number in Mg₃N₂</td></tr><tr><td>magnesium</td><td>(+)II / 2(+)</td></tr><tr><td>nitrogen</td><td>–III/ 3 –</td></tr></table> Both correct for one mark		oxidation number in Mg ₃ N ₂	magnesium	(+)II / 2(+)	nitrogen	–III/ 3 –	1
	oxidation number in Mg ₃ N ₂							
magnesium	(+)II / 2(+)							
nitrogen	–III/ 3 –							
2(c)(ii)	Mg ₃ N ₂ + 6H ₂ O → 3Mg(OH) ₂ + 2NH ₃	1						
2(c)(iii)	M1 due to OH [–] (aq) M2 EITHER acid-base reaction of water by ammonia OR dissociation of (some) Mg(OH) ₂	2						

20. M/J 24/P21/Q3(a)

3(a)	M1 (if a change in conditions occurs) the equilibrium shifts M2 to minimise the change in conditions	2
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21. M/J 24/P21/Q3(b)-(c)

3(b)	M1 change in appearance paler red / more yellow / less orange M2 change in relative concentration of $\text{FeSCN}^{2+}(\text{aq})$ lower M3 change in value of the equilibrium constant, K_c lower	3
3(c)(i)	M1 deduce the concentration of SCN^- present in initial mixture $[\text{SCN}^- \text{ eq'm}] = [\text{Fe}^{3+} \text{ eq'm}] = [\text{original SCN}^-] - [\text{FeSCN}^{2+} \text{ produced}]$ $1.30 \times 10^{-3} + 0.300 \times 10^{-3} = 1.60 \times 10^{-3} \text{ mol dm}^{-3}$ M2 find initial amount of Fe^{3+} initially $\text{M1} \times 25 / 1000 = \text{amount of Fe}^{3+} \text{ added initially} = 4.00 \times 10^{-5} \text{ (mol)}$	2
3(c)(ii)	M1 value for $K_c = 178$ M2 units = $\text{mol}^{-1} \text{ dm}^3$	2

22. M/J 24/P22/Q3(a)

3(a)	M1 rates of forward and reverse / backward reactions are equal M2 no change in measurable properties OR concentration of reactants AND products remain constant	2
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23. M/J 24/P22/Q3(b)

3(b)(i)	M1 change to appearance of the mixture: (goes) darker red / more red M2 change to relative concentration of $\text{Fe}^{3+}(\text{aq})$: decreases M3 change to the value of equilibrium constant, K_c : constant / none	3
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3(b)(ii)	Method 1 M1 conc of initial conc of Fe^{3+} and $\text{SCN}^- = 0.0020 \text{ (mol dm}^{-3}\text{)}$ M2 at equilibrium the conc of Fe^{3+} and SCN^- are ($\text{M1} - 0.000423$) M3 use $0.000423 \div (\text{M1} - 0.000423)^2$ to calculate $K_c = 170(.089)$ M4 units = $\text{mol}^{-1} \text{ dm}^3$ OR $\text{dm}^3 \text{ mol}^{-1}$ Method 2 M1 no mol $\text{FeSCN}^{2+} = 4.23 \times 10^{-4} \times 25 / 1000 = 1.0575 \times 10^{-5}$ M2 no mol Fe^{3+} and SCN^- are ($5 \times 10^{-5} - \text{M1}$) = 3.9425×10^{-5} M3 $0.000423 \div (\text{M2} \times 1000 / 25)^2 = 170(.089)$ M4 units = $\text{mol}^{-1} \text{ dm}^3$ OR $\text{dm}^3 \text{ mol}^{-1}$	4
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24. M/J 24/P23/Q2(a)

2(a)	58% / no change / stays the same AND catalysts don't affect yield / iron is a catalyst	1
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25. M/J 24/P23/Q2(b)(i)

2(b)(i)	less yield (in C than A) AND (as) exothermic (forward reaction)	1
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26. M/J 24/P23/Q2(c)

2(c)(i)	M1 $K_p = (\text{pNH}_3)^2 / (\text{pN}_2)(\text{pH}_2)^3$ M2 atm^{-2}	2
2(c)(ii)	M1 correct amount N_2 at equilibrium = $1 - 1.16 / 2 = 0.42$ AND correct amount H_2 at equilibrium = $3.0 - (1.16 \times 3 / 2) = 1.26$ M2 express amount of ammonia as a mol fraction in terms of total amount of $\text{N}_2 + \text{H}_2 + \text{NH}_3$ at equilibrium = $1.16 / (\text{N}_2 + \text{H}_2 + 1.16) = 0.408$ OR 0.41	2
2(c)(iii)	$0.625 \times 1000 = 625 \text{ (atm)}$	1

27. M/J 24/P23/Q4(b)(i)

4(b)(i)	acid–base reaction	1
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28. O/N 23/P22/Q2(a)

2(a)	curved line from (0,0) to reach same horizontal line	1
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29. O/N 23/P22/Q2(b)

2(b)	equilibrium position would / it moves to left / reactants	1
	forward reaction is exothermic (so disfavoured by increase in T)	1

30. O/N 23/P22/Q2(c)

2(c)	NO mole fraction $\frac{0.605}{1.60}$ $(= 0.378125)$ NO_2 $\frac{0.605}{1.60}$ $(= 0.378125)$ N_2O_3 $\frac{0.390}{1.60}$ $(= 0.24375)$	1
	(partial pressure / kPa 38.19 38.19 24.62)	1
	$K_p = 24.62 \div 38.19^2 = 0.0169$	
	kPa^{-1}	1

31. O/N 23/P22/Q3(a)(ii)

3(a)(ii)	proton / H^+ donor	1
	partially dissociates (in solution / in water / when aqueous)	1

32. M/J 23/P22/Q4(b)

4(b)	M1 amount ethanol increases M2 equilibrium moves to right AND to produce fewer moles gas OR smaller amount of gas (to reduce the pressure)	2
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33. M/J 23/P22/Q4(c)(i)

4(c)(i)	acid AND it donates a proton / H^+	1
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34. M/J 23/P23/Q4(b)

4(b)(i)	(from colourless) to purple	1
4(b)(ii)	$K_c = [H_2][I_2] / [HI]^2$ OR $0.0217 = [H_2][I_2] / [HI]^2$	1
4(b)(iii)	M1 $[HI]$ at equilibrium = $1.7 / 2 = 0.85$ M2 idea that $[H_2] = [I_2]$ in equilibrium mixture M3 correct calculation based on correct rearrangement of equation in (b)(ii), 0.0217 and their M1 = 0.125	3

35. M/J 23/P23/Q4(c)

4(c)(i)	new mixture will have a lower / smaller conc / amount of H_2 and I_2	1
4(c)(ii)	endothermic AND decrease in temperature causes position of equilibrium to shift in exothermic direction	1

36. MAR 23/P22/Q3(d)(ii)

3(d)(ii)	equilibrium moves to right/products where fewer moles / smaller amount of gas	1
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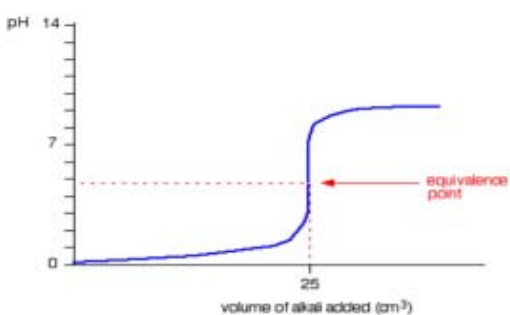
37. O/N 22/P22/Q1(c)(i)

1(c)(i)	proton / H^+ donor	1
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38. O/N 22/P22/Q2(c)(ii)

2(c)(ii)	reaction is exothermic so equilibrium shifts to the LHS OR equilibrium shifts to LHS because forward reaction is exothermic OR equilibrium shifts to LHS because backward reaction is endothermic OR endothermic reaction is favoured so more reactants / are $AlCl_3$ is made	1
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39. M/J 22/P21/Q2(d)

2(d)(i)	M1 proton / H^+ donor M2 fully dissociates (in aqueous solution / water / solvent)	2
2(d)(ii)	$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$	1
2(d)(iii)	M1 correct basic shape extending to $\sim 50\text{ cm}^3$ with vertical portion of curve at 25 cm^3 M2 initial pH at 0–2 (based on idea that HCl is a strong acid) AND final pH at between 8–12 (based on idea that NH_3 is a weak alkali) 	2

40. M/J 22/P22/Q3(a)

3(a)(i)	M1 rate(s) of forward and reverse / backward reactions are equal / are the same M2 no change in measurable properties OR concentration of reactants and products remain constant	2
3(a)(ii)	(colourless gas) becomes purple (gas)	1
3(a)(iii)	$K_p = \frac{(pH^2)(pI^2)}{(pHI)^2}$	1
3(a)(iv)	M1 use correct expression for finding amount of HI used up in reaction $86.4 = 100 \times (0.025 - x) / 0.025 - x + x / 2 + x / 2$ where $x = HI \text{ mol used up}$ OR $86.4 = 100 \times (0.025 - 2x) / (0.025 - 2x + x + x)$ where $2x = HI \text{ mol used up}$ M2 correct calculation to find amount of HI(mol) at equilibrium based on $HI(mol) \text{ at eq'm} = 0.025 - HI(mol) \text{ used} = 0.025 - 0.0034 = 0.0216$ OR 0.022 (mol) alternative method M1 use the pp H_2 and pp I_2 at equilibrium (= 13.6) to calculate no mol H_2 & I_2 at equilibrium (either 0.0034 in total OR 0.0017 each) M2 use value for no mol $H_2 + I_2$ in expression to find no mol HI $0.025 - (\text{no mol } H_2 + I_2) = 0.0216$	2

41. M/J 22/P22/Q3(c)

3(c)	none / no change	1
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42. M/J 22/P23/Q3(a)(iii)

3(a)(iii)	M1 $K_c = \frac{[C_2H_6][CO][CH_4]}{[C_4H_{10}O]}$ M2 $\text{mol}^2 \text{ dm}^{-6}$	2
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43. MAR 22/P22/Q2(d)(iii)

2(d)(iii)	position of equilibrium moves / farther to right (at 20 km) [1] (forward) reaction is exothermic AND temperature colder at 20 km (cf. 50 km) [1]	2
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44. MAR 22/P22/Q3(d)

3(d)(i)	proton / H^+ donor [1] fully dissociates (in aqueous solution / water / solvent) [1]	2
3(d)(ii)	M1: correct sigmoid shape with vertical section at 25 cm^3 for both M2: both curves show initial $\text{pH} \leq 2$ M3: (with NaOH) heading to $\text{pH} \geq 12$ (with NH_3) heading to $\text{pH} 8-12$ 	3

Unit 8

REACTION KINETICS

21 items · 57 marks

Syllabus sections in this unit

8.1 Rate of reaction

8.2 Effect of temperature on reaction rates and the concept of activation energy

8.3 Homogeneous and heterogeneous catalysts

Unit 8: Reaction kinetics

1. M/J 26/P21/Q2(b)(ii)

Context

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

Context: part (b) (introduction)

(b) The reaction conditions for stage 2 include:

- using a ratio of 1 mol SO_2 to 1 mol O_2
- heat in the presence of V_2O_5 .

(ii) Explain how V_2O_5 affects the rate of reaction.

[2]

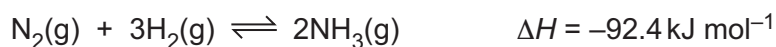
2. M/J 26/P22/Q1(e)(ii)

Context

1 Oxygen reacts with many elements.

Context: part (e) (introduction)

(e) Ammonia is made in the Haber process.



A temperature of 400–450 °C is used in the Haber process.

(ii) Describe how the initial rate of the forward reaction changes, if at all, when a temperature greater than 450 °C is used. Explain your answer.

Assume all other variables remain constant.

change in initial rate

explanation

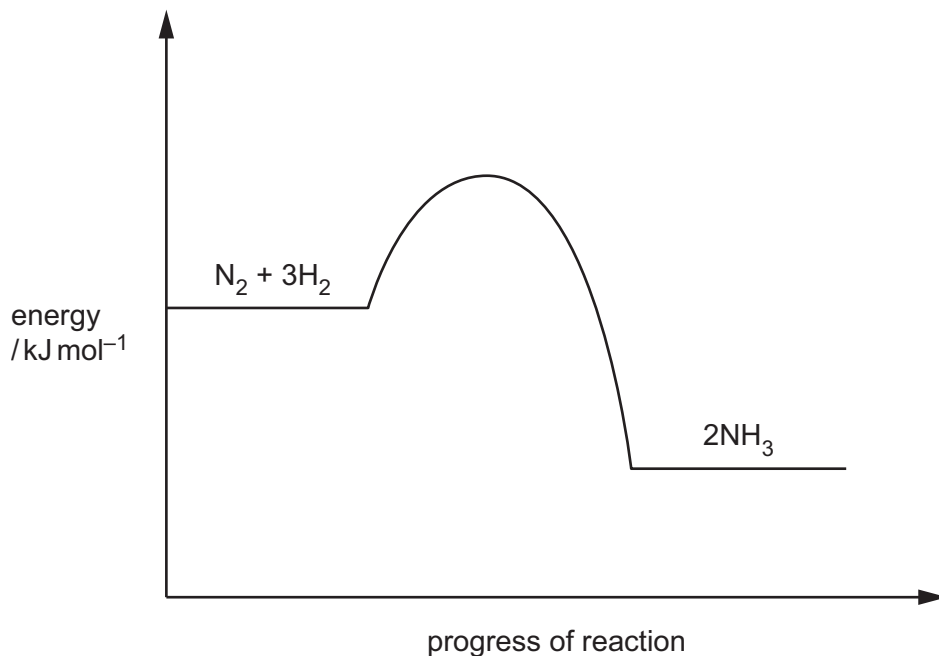
[3]

3. M/J 26/P24/Q2(b)(i)

Context: part (b) (introduction)

(b) Nitrogen is used to manufacture ammonia, NH_3 , in the Haber process. An iron catalyst is used.

(i) Complete the reaction pathway diagram in Figure 2.1 to show the effect of adding an iron catalyst to the reaction of nitrogen and hydrogen.

**Figure 2.1**

[1]

4. M/J 26/P24/Q2(d)

(d) Explain the effect of increasing temperature on rate of reaction.

[3]

5. M/J 26/P24/Q2(e)

(e) Figure 2.2 shows the Boltzmann distribution for a sample of X(g) at temperature T_1 .

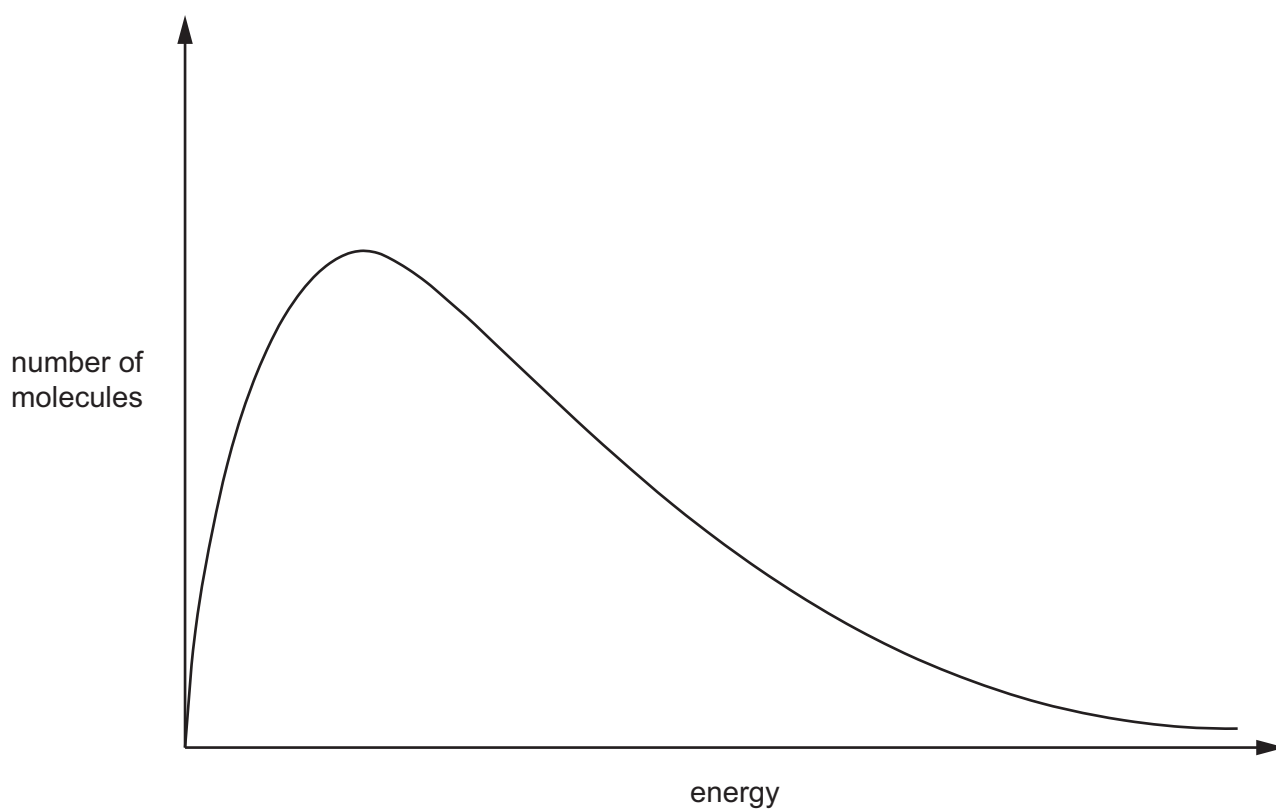


Figure 2.2

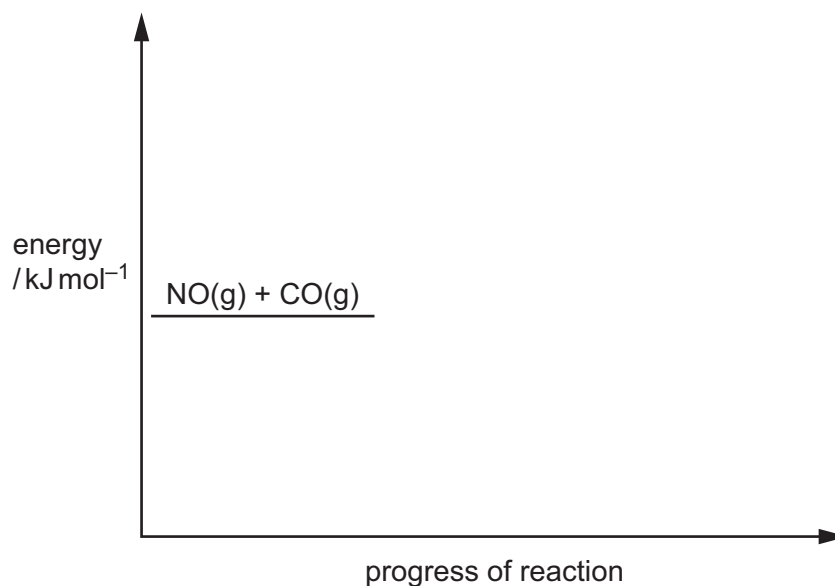
Sketch a curve on Figure 2.2 to show the Boltzmann distribution for the sample of X(g) at a temperature higher than T_1 . [2]

6. MAR 26/P22/Q2(d)(ii)

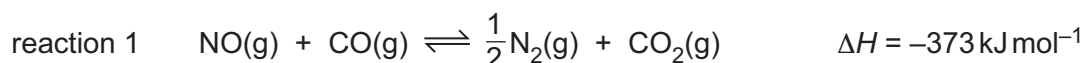
Context

2 Nitrogen monoxide, NO, is an atmospheric pollutant.

Context: Fig. 2.3

**Figure 2.3**

Context: part (d) (introduction)

(d) Reaction 1 occurs during the catalytic removal of NO from car exhaust gases.The activation energy, E_a , of this reaction is 570 kJ mol^{-1} .**(ii)** On your diagram in Figure 2.3, show how the addition of a catalyst affects reaction 1. [1]**7. O/N 25/P24/Q2(c)(i)**

Context

2 Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

Context: part (b) (introduction)

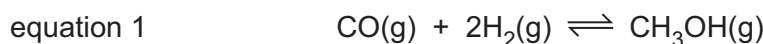
(b) A 5.00 dm^3 sealed flask contains 0.400 mol of CO(g) and 0.800 mol of $\text{H}_2\text{(g)}$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium.

Equation 1 shows the reaction.

The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2}$$

Context: part (c) (introduction)

(c) P_4O_{10} catalyses the reversible reaction of CO with H_2 to form CH_3OH . P_4O_{10} then acts as a dehydrating agent, causing CH_3OH to form CH_3OCH_3 .**(i)** Explain how the presence of a catalyst affects a chemical reaction.

[1]

8. M/J 25/P21/Q3(a)

- 3 (a) Hexene reacts with hydrogen gas to produce hexane.

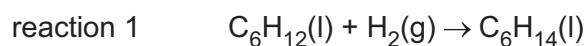


Fig. 3.1 shows the distribution of energies of $\text{H}_2(\text{g})$ molecules at temperature T .

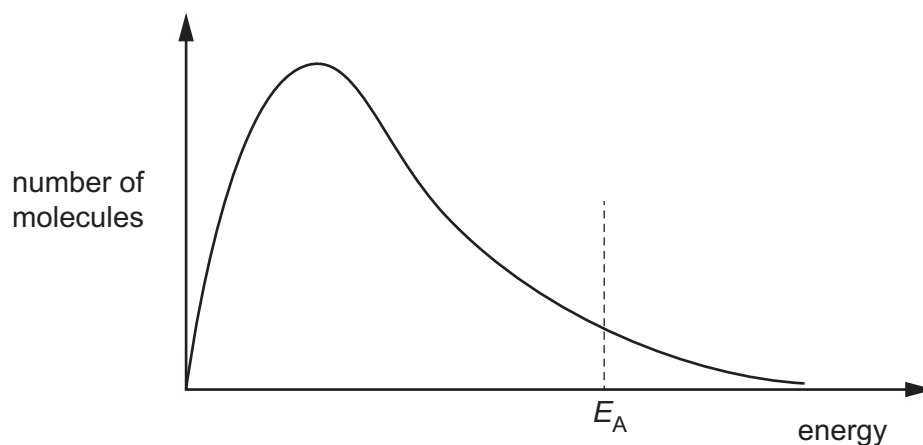


Fig. 3.1

- (i) Sketch on Fig. 3.1 the shape of the curve for the same sample of $\text{H}_2(\text{g})$ molecules when the temperature is increased. [2]
- (ii) Explain why increasing the temperature increases the rate of reaction 1. [2]
- (iii) State the role of nickel when it is added to reaction 1. [1]
- (iv) Annotate Fig. 3.2 to show the effect of adding nickel to reaction 1 at temperature T .

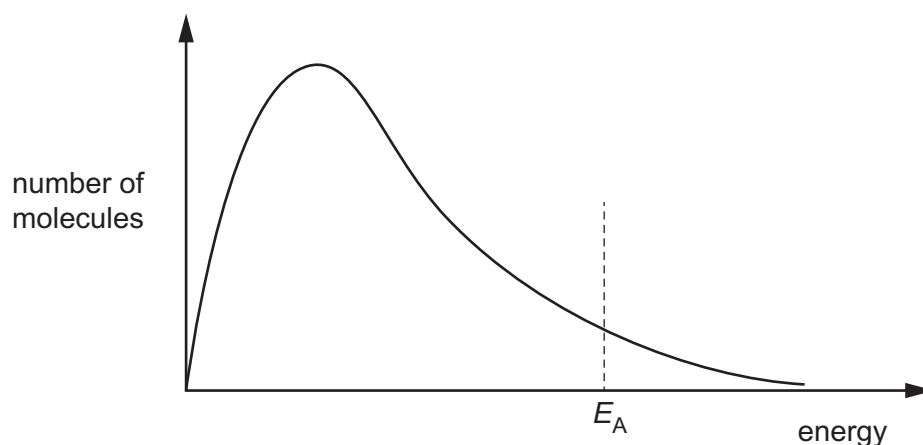


Fig. 3.2

[1]

9. M/J 25/P22/Q3(b)

- (b) $\text{H}_2(\text{g})$ and nickel are added to alkene **X**. Fig. 3.2 shows how the concentration of **X** changes with time.

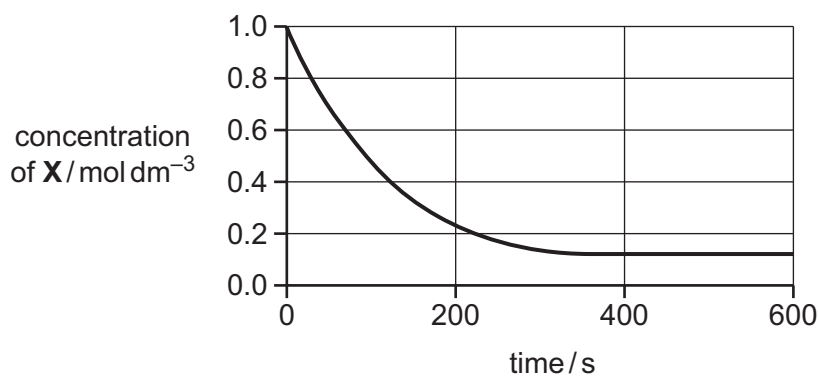


Fig. 3.2

- (i) Use Fig. 3.2 to identify the limiting reagent in this reaction. Explain your answer. [1]
- (ii) Use Fig. 3.2 to describe how the gradient changes as the reaction proceeds. State what this shows about the rate during this reaction. [1]

10. M/J 25/P22/Q3(d)

Context: part (c)

(c) Hydrocarbon **Y** contains two C=C groups. There are no other functional groups present.

Y reacts with an excess of $\text{H}_2(\text{g})$ to produce hexane, C_6H_{14} .

Table 3.1

bond	energy / kJ mol^{-1}
C–C	350
C=C	610
$\text{C}\equiv\text{C}$	840
H–H	436
C–H	410

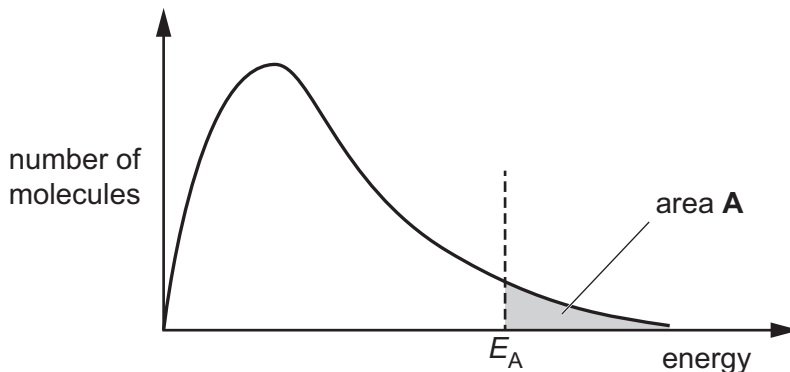
Use Table 3.1 to calculate the enthalpy change per mole of C_6H_{14} produced in this reaction.

enthalpy change = kJ mol^{-1} [2]

(d) Hydrocarbon **Y** reacts with $\text{H}_2(\text{g})$.

Fig. 3.3 shows the distribution of energies of $\text{H}_2(\text{g})$ at temperature T .

Area **A** represents the number of molecules with energy greater than or equal to the activation energy, E_{A} , at temperature T .

**Fig. 3.3**

- (i) Annotate Fig. 3.3 to show the effect of adding nickel to **Y** and $\text{H}_2(\text{g})$ at temperature T . [1]
- (ii) Area **B** (not labelled on Fig. 3.3) represents the number of molecules with energy greater than or equal to the activation energy when nickel is added at temperature T .

State the difference, if any, between areas **A** and **B**. Explain the significance of your answer on the rate of hydrogenation of **Y**. Give your answer in terms of collisions.

[2]

11. M/J 25/P23/Q3(b)(ii)-(iii)

Context: part (b) (introduction)

(b) C_4H_8 reacts with an excess of H_2 to produce C_4H_{10} .



(ii) Define activation energy, E_{A} .

[1]

(iii) The Boltzmann distribution for the reaction mixture in reaction 1 is shown in Fig. 3.1. Use the Boltzmann distribution to explain the effect of adding a catalyst on the rate of reaction.

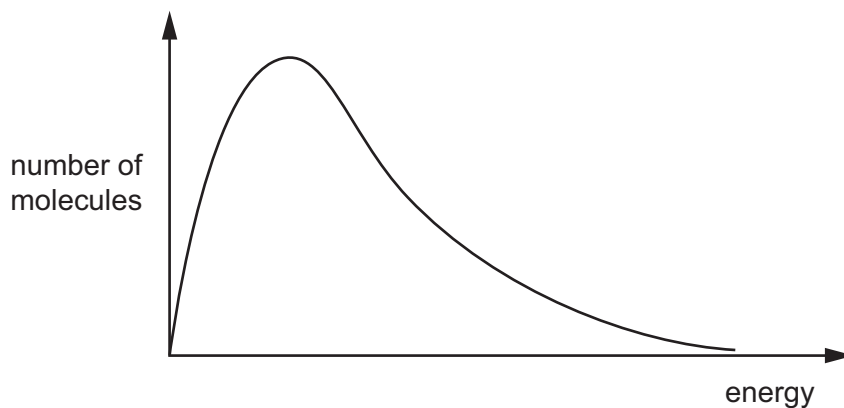


Fig. 3.1

[2]

12. M/J 25/P23/Q4(a)

Context

4 The reaction between $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$ and $\text{HCl}(\text{aq})$ is monitored at constant temperature.

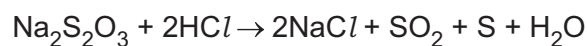


Fig. 4.1 shows how the concentration of $\text{HCl}(\text{aq})$ varies with time.

13. M/J 24/P23/Q2(b)(ii)

Context

- 2** $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.



Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

- (ii) Describe and explain the change to the rate of the forward reaction that occurs to establish the equilibrium in **C** compared to **A**.

You do **not** need to refer to the Boltzmann distribution in your answer.

[2]

14. MAR 24/P22/Q3(c)(ii)-(iii)

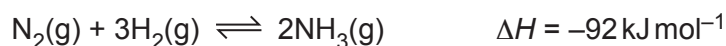
Context

- 3** Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

Context: part (c) (introduction)

- (c) The Haber process involves the reaction of N_2 and H_2 to form ammonia, NH_3 .

A catalyst is used, which allows the process to be carried out at a lower temperature and pressure.



- (ii) Explain how the presence of a catalyst affects the reaction.

[1]

- (iii) State and explain the effect, if any, on the **rate** of the Haber process as the pressure is lowered.

[2]

15. O/N 23/P21/Q3(b)

Context

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(b) $\text{KClO}_3(\text{s})$ decomposes when heated.

$\text{MnO}_2(\text{s})$ catalyses the exothermic decomposition reaction.

Complete and label the diagram in Fig. 3.1 to show the effect of $\text{MnO}_2(\text{s})$ on the decomposition of $\text{KClO}_3(\text{s})$.

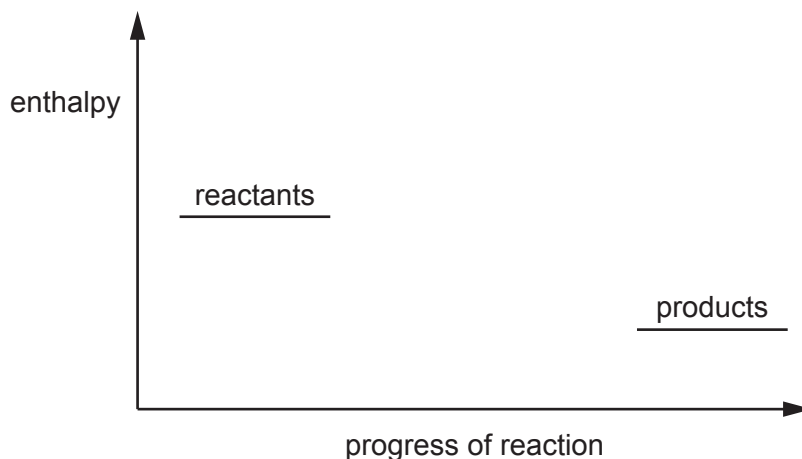


Fig. 3.1

[2]

16. O/N 23/P23/Q3(b)

Context

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

(b) $\text{KClO}_3(\text{s})$ decomposes when heated.

$\text{MnO}_2(\text{s})$ catalyses the exothermic decomposition reaction.

Complete and label the diagram in Fig. 3.1 to show the effect of $\text{MnO}_2(\text{s})$ on the decomposition of $\text{KClO}_3(\text{s})$.

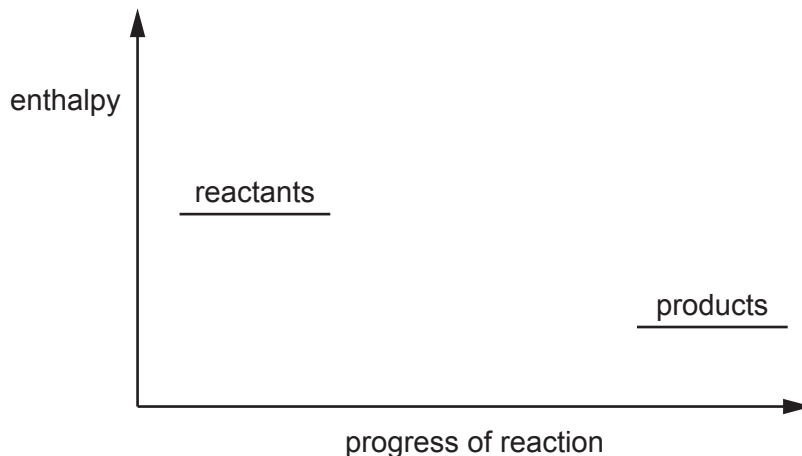


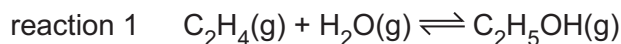
Fig. 3.1

[2]

17. M/J 23/P22/Q4(c)(iv)-(v)

Context

4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



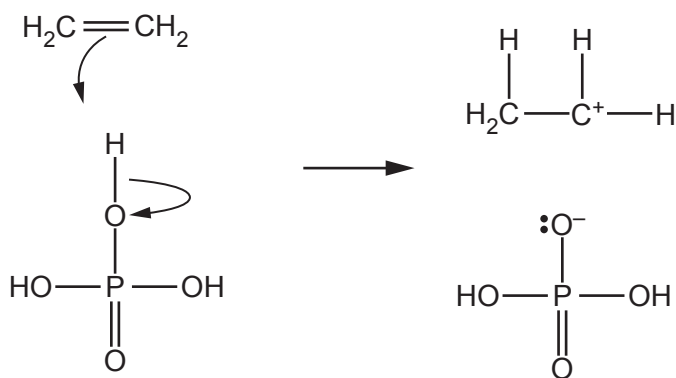
Context: Fig. 4.2

**Fig. 4.2**

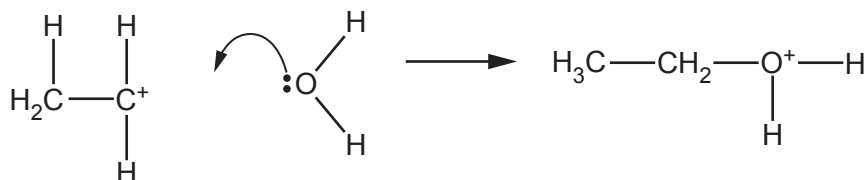
Context: part (c) (introduction)

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

**Fig. 4.1**

(iv) Describe how a catalyst affects a reaction. Explain your answer.

[2]

(v) Use Fig. 4.1 and Fig. 4.2 to justify why H_3PO_4 is described as a catalyst in reaction 1.

[1]

18. MAR 23/P22/Q3(d)(i)

Context

- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

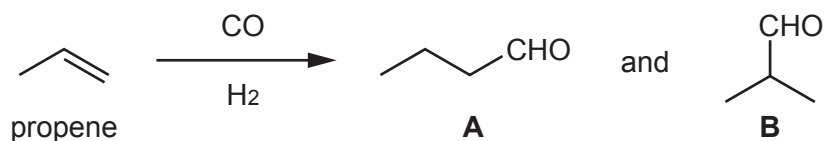


Fig. 3.1

Context: part (d) (introduction)

- (d)** The reaction of ethene, C₂H₄, with a 1:1 mixture of CO and H₂ is shown in equation 1.



At atmospheric pressure a cobalt-based catalyst is used in this reaction.

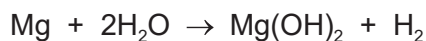
- (i)** State and explain the effect of using a catalyst on this reaction.

[2]

19. M/J 22/P21/Q1(g)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.
- (g)** Cold water reacts slowly with a piece of Mg to produce bubbles of H₂(g).
Cold water reacts rapidly with burning Mg to produce H₂(g) in an explosive mixture.

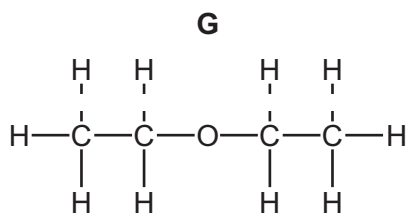


Explain why the rate of reaction of cold water with burning magnesium is greater.

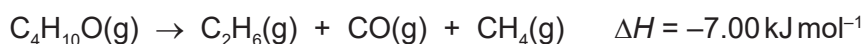
[2]

20. M/J 22/P23/Q3(a)(iv)-(v)

Context

3 G belongs to a group of compounds called ethers.**Fig. 3.1**

Context: part (a) (introduction)

(a) When **G**, $C_4H_{10}O$, is heated, thermal decomposition occurs.**(iv)** Thermal decomposition of **G** in the presence of I_2 affects the activation energy, E_a , for the reaction. Table 3.2 shows E_a for the thermal decomposition of **G** with and without I_2 .**Table 3.2**

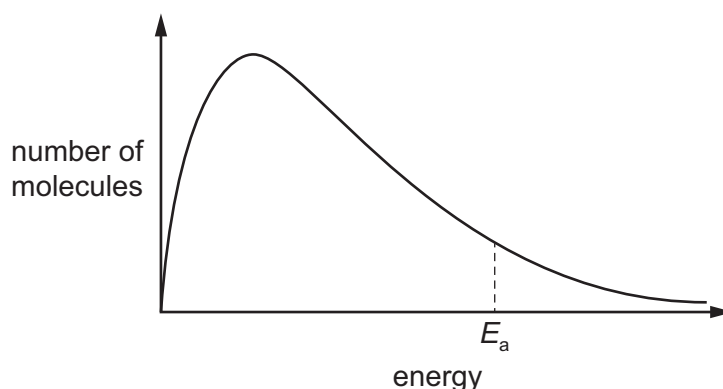
reaction	E_a (with I_2)/kJ mol ⁻¹	E_a /kJ mol ⁻¹
$C_4H_{10}O(g) \rightarrow C_2H_6(g) + CO(g) + CH_4(g)$	143	224

State what effect adding I_2 to the reaction mixture has on the value of K_c .
Explain your answer.

[2]

(v) Fig. 3.2 shows the Boltzmann distribution of energies for molecules of **G** at constant temperature, $T^\circ\text{C}$.

Sketch, on Fig. 3.2, the Boltzmann distribution of energies for molecules of **G** at a higher temperature, $(T+100)^\circ\text{C}$.

**Fig. 3.2**

[2]

21. MAR 22/P22/Q2(b)(iii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (b) (introduction)

(b) Al_2O_3 is an amphoteric oxide found in bauxite.

(iii) Al_2O_3 is used as a catalyst in the dehydration of alcohols.

State the effect of using Al_2O_3 as a catalyst in the dehydration of alcohols. Use the Boltzmann distribution in Fig. 2.1 to help explain your answer.

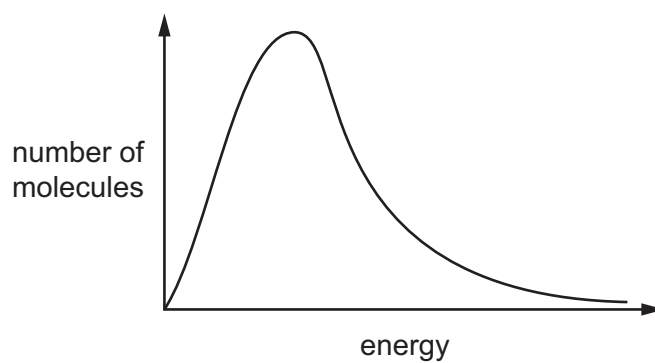


Fig. 2.1

[3]

Answers Section

1. M/J 26/P21/Q2(b)(ii)

2(b)(ii)	M1 it / rate (of reaction) increases M2 (it) lowers EA OR (used) to lower EA	2 rate = speed successful = effective particles = molecules M1 A greater frequency of successful collisions / more successful collisions per unit time / higher successful collision rate / more frequent successful collisions / more successful collisions per second A for per = / M2 A more O₂ and/or SO₂ (molecules) OR more particles have sufficient / enough / minimum energy to react M2 A more O₂ and / or SO₂ (molecules) OR more particles have energy $\geq$ EA A Ea / EA / AE for EA I more of the collisions have energy greater than or equal to activation energy I pathway (even if wrong) I redox I it is a catalyst
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2. M/J 26/P22/Q1(e)(ii)

1(e)(ii)	change in rate explanation M1 increases M2 more particles have $E \geq E_A$ M3 so greater frequency of effective collisions.	3 I yield / Le Chatelier arguments M1 might be seen in explanation if not on first answer line M1 A it takes less time to reach equilibrium M2 A more particles have energy greater than / equal to the activation energy / E_A M2 R particles have more activation energy M2 I particles have more energy M2 I atoms / reactants M3 A more effective collisions per unit time / increased rate of effective collisions M3 I higher chance / probability of more successful collisions M3 I more collisions / more effective collisions alone effective = successful particles = molecules activation energy = sufficient / enough energy to react
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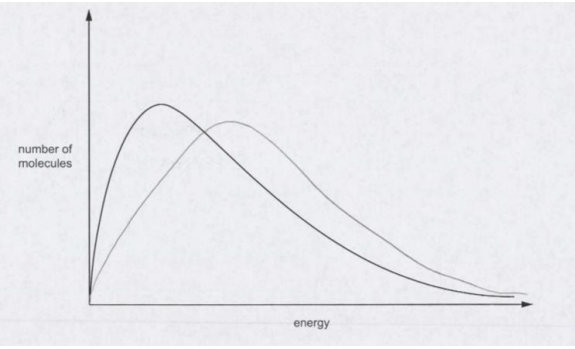
3. M/J 26/P24/Q2(b)(i)

2(b)(i)		1. A BOD double hump for cat complex intermediate A line that joins original line before products A BOD
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4. M/J 26/P24/Q2(d)

2(d)	M1 (increasing temperature) increases rate M2 increased number of particles/collisions with E_A OR larger proportion of particles/collisions have energy greater than or equal to E_A / $E \geq E_A$ M3 increased / higher frequency of effective / successful collisions	3 M1 A BOD increases speed of reaction M2 A more particles with enough / sufficient energy to react M3 A increased rate of effective / successful collisions A increase in effective / successful collisions per unit time A 'increased frequency of collisions' AND 'more effective / successful collisions' I increased chance / probability of effective / successful collision I more collisions (only) I references to Le Chatelier / yield A particles = molecules = atoms
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5. M/J 26/P24/Q2(e)

2(e)	 • start at origin and not crossing original until beyond the peak • peak to the RHS of original peak AND lower than original • line does not cross the original after the peak reached AND crosses the original once only ✓✓	2	A BOD slight upwards 'kick' at the end. Max 1 completely correct graphs for temperature < T_1
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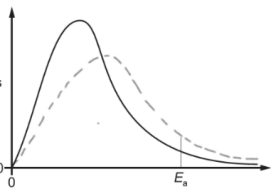
6. MAR 26/P22/Q2(d)(ii)

2(d)(ii)	diagram has lower-energy transition state(s)	1
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7. O/N 25/P24/Q2(c)(i)

2(c)(i)	increases rate by providing a different mechanism with a lower activation energy	1
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8. M/J 25/P21/Q3(a)

3(a)(i)	 Two correct for one mark, all four for two marks: • start at origin • peak sits below the original • peak sits to the right-hand side of the original • new line crosses the original once only AND to the right of the original peak	2
3(a)(ii)	M1 greater proportion of (H_2) molecules have energy greater than or equal to the activation energy $/ \geq E_A$ M2 increase in frequency of successful collisions (between H_2 and C_6H_{12})	2
3(a)(iii)	catalyst	1
3(a)(iv)	annotation to show new E_A (nickel / catalyst) further to the left than original E_A	1

9. M/J 25/P22/Q3(b)

3(b)(i)	H_2 / hydrogen AND some X / alkene left(over) / remains (at the end of this reaction)	1
3(b)(ii)	link the significance of gradient values to rate AND describe either the decrease in rate or the decrease in gradient as the gradient decreases the rate decreases OR rate decreases until the reaction stops when gradient = 0	1

10. M/J 25/P22/Q3(d)

3(d)(i)	E_A shown on Fig. 3.3 to the left of original (on the x axis)	1
3(d)(ii)	M1 it / (area) B larger AND rate (reaction / hydrogenation) increases OR (area) B identified by shading or labelling on Fig 3.3 AND rate (reaction / hydrogenation) increases in prose M2 increase in the frequency of effective / successful collisions OR increase in rate of effective / successful collisions	2

Answer for context part (c)

3(c)	method 1 – breaking all bonds in molecules $(2 \times 610) + (2 \times 436) + (10 \times 410) + (3 \times 350) - (14 \times 410) - (5 \times 350)$ - energy involved when all bonds are broken = (7242) - energy involved when all bonds are made = (7490) - energy to break bonds – energy to make bonds $= (7242 - 7490) = -248$ method 2 – ignore any bonds that do not change from reactants to products $(2 \times 610) + (2 \times 436) - (4 \times 410) - (2 \times 350)$ - energy involved when all bonds are broken = (2092) - energy involved when all bonds are made = (2340) - energy to break bonds – energy to make bonds $(2092 - 2340) = -248$	2
------	--	---

11. M/J 25/P23/Q3(b)(ii)-(iii)

3(b)(ii)	minimum energy required for a collision to be effective	1
3(b)(iii)	M1 • area under the curve with energy $\geq E_A$ using a catalyst is larger so • greater proportion of particles with sufficient energy to react M2 rate faster because higher frequency of successful collisions	2

12. M/J 25/P23/Q4(a)

4(a)(i)	Check final graph for range of acceptable answers $\pm(1-0.5)/100 = \pm 0.005$ 0 $\text{mol dm}^{-3} \text{ s}^{-1}$ for both Any 2 points = 1 mark, all 3 points = 2 marks	2
4(a)(ii)	$\text{Na}_2\text{S}_2\text{O}_3$ because reaction stops before all the HCl is used up. OR $\text{Na}_2\text{S}_2\text{O}_3$ because there is still HCl remaining (when the reaction has stopped)	1
4(a)(iii)	the frequency of effective collisions decreases as concentration of HCl(aq) and / or $\text{Na}_2\text{S}_2\text{O}_3$ decreases	1
4(a)(iv)	M1 0.001(00) mol sulfur produced M2 $\text{M1} \times 6.02 \times 10^{23}$ $= 6.02 \times 10^{20}$	2
4(a)(v)	It reacts with water (to produce H_2SO_3)	1

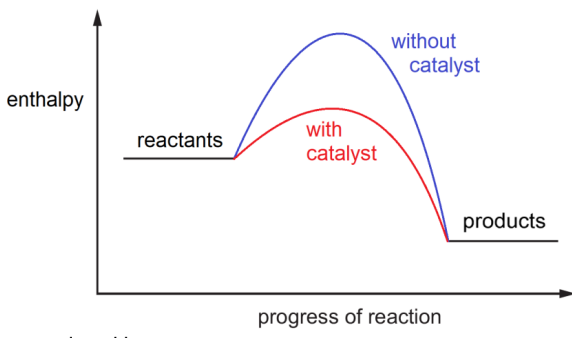
13. M/J 24/P23/Q2(b)(ii)

2(b)(ii)	(increase in temperature / energy from A to C (so)) M1 more collisions with particles that have $E > E_a$ AND increase frequency of effective collisions M2 (initial) rate (of the forward reaction) increases	2
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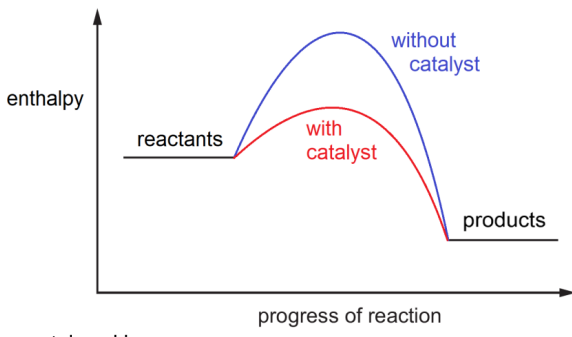
14. MAR 24/P22/Q3(c)(ii)-(iii)

3(c)(ii)	increases rate by providing a different mechanism with a lower activation energy	1
3(c)(iii)	rate is lowered	1
	lower frequency of successful collisions / number of successful collisions per unit time	1

15. O/N 23/P21/Q3(b)

3(b)	 enthalpy reactants without catalyst with catalyst products progress of reaction uncatalysed hump catalysed hump lower than uncatalysed	1
		1

16. O/N 23/P23/Q3(b)

3(b)	 enthalpy reactants without catalyst with catalyst products progress of reaction uncatalysed hump catalysed hump lower than uncatalysed	1
		1

17. M/J 23/P22/Q4(c)(iv)-(v)

4(c)(iv)	M1 a catalyst increases rate of reaction M2 by producing an alternative reaction pathway with lower activation energy	2
4(c)(v)	it is regenerated OR it is reformed	1

18. MAR 23/P22/Q3(d)(i)

3(d)(i)	increases the rate of reaction	1
	by providing an alternative reaction pathway of lower E_a	1

19. M/J 22/P21/Q1(g)

1(g)	M1 (heat / energy released from burning Mg) provides more particles with energy $\geq E_a$ M2 frequency of successful / effective collisions is greater	2
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20. M/J 22/P23/Q3(a)(iv)-(v)

3(a)(iv)	M1 no change / none M2 I_2 is a catalyst AND does not affect position of equilibrium OR I_2 is a catalyst AND reduces time taken to reach (same) equilibrium	2
3(a)(v)	• start at origin and not crossing original until beyond the peak • peak to the right-hand side of original peak AND lower than original • line does not cross the original after the peak reached	2

21. MAR 22/P22/Q2(b)(iii)

2(b)(iii)	two lines shown on diagram, e.g. E_A and $E_{A,cat}$ [1] greater proportion of molecules with $E \geq E_A$ [1] frequency of effective collisions increases [1]	3
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Unit 9

THE PERIODIC TABLE: CHEMICAL PERIODICITY

69 items · 177 marks

Syllabus sections in this unit

- 9.1 Periodicity of physical properties of the elements in Period 3
- 9.2 Periodicity of chemical properties of the elements in Period 3
- 9.3 Chemical periodicity of other elements

Unit 9: The Periodic Table: chemical periodicity**1. M/J 26/P22/Q1(c)**

Context

1 Oxygen reacts with many elements.**(c)** Li_2O reacts with water to produce a solution with pH greater than 10.**(i)** State the name of a Period 3 oxide that behaves in the same way as Li_2O when added to water.

[1]

(ii) Construct an equation for the reaction of Li_2O with water.

[1]

(iii) Identify **all** the Period 3 chlorides that are solid at room temperature and that react with water to produce an acidic solution.

[2]

2. M/J 26/P22/Q1(d)

Context

1 Oxygen reacts with many elements.**(d)** LiNO_3 decomposes on heating to produce Li_2O , O_2 and a brown gas.

Identify the brown gas.

[1]

3. M/J 26/P23/Q1(a)

Context

1 The chlorides of Period 3 elements show different physical and chemical properties.**(a) (i)** Complete Table 1.1.**Table 1.1**

element	formula of chloride	type of bonding in the chloride	type of reaction, if any, of the chloride with cold water
sodium			
silicon			

[3]

(ii) Write an equation for the reaction of PCl_5 with an excess of cold water.

[1]

4. M/J 26/P24/Q3(a)

3 (a) (i) State the trend in atomic radius of the elements across Period 3.

[1]

(ii) State the formula of the oxide formed when sodium reacts with oxygen.

[1]

(iii) Write an equation for the reaction of MgO with water.

[1]

5. M/J 26/P24/Q3(b)

(b) Table 3.1 shows some of the physical properties of Period 3 chlorides.

Table 3.1

	melting point /°C	boiling point /°C	type of bonding	type of structure
NaCl	801	1413		
SiCl₄	−69	58		

(i) Complete Table 3.1 using the abbreviations shown.

M = metallic

C = covalent

I = ionic

S = simple

G = giant

[2]

(ii) State the formula of the phosphorus-containing product formed when PCl_5 reacts with an excess of water. Suggest the pH of the resulting solution.

formula =

pH of solution =

[2]

6. O/N 25/P21/Q2(a)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) Fig. 2.1 shows the variation in atomic and ionic radii of the Period 3 elements Na to Cl.

The ionic radius of Si is **not** shown.

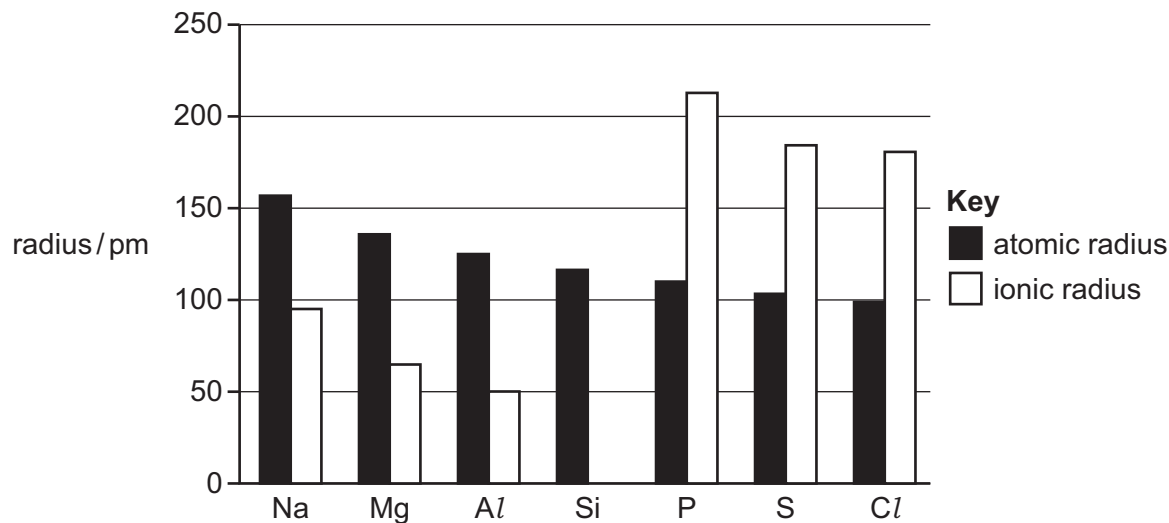


Fig. 2.1

(i) Explain the trend shown in the atomic radii of the Period 3 elements Na to Cl.

[2]

(ii) Explain why there is a large difference in the ionic radii of Al and P.

[2]

7. O/N 25/P21/Q2(b)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(b) Table 2.1 gives some information about some of the Period 3 oxides.

Row B gives the pH of the solution that forms when the Period 3 oxide is added to water.

Table 2.1

	formula of Period 3 oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₃
A	oxidation number of Period 3 element			+3			
B	pH of solution			—	—		

(i) Complete Table 2.1.

[2]

(ii) State why there is no data given in row B for Al₂O₃ and SiO₂.

[1]

8. O/N 25/P21/Q2(c)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(c) (i) Write an equation for the reaction of Na_2O with dilute hydrochloric acid.

[1]

(ii) Construct an equation for the reaction of Al_2O_3 with a base to form NaAlO_2 .

[1]

9. O/N 25/P22/Q2(a)-(c)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) (i) Explain why the elements Na to Al are good electrical conductors.

[1]

(ii) Explain why the elements P, S and Cl do not conduct electricity.

[1]

(b) Fig. 2.1 shows the variation in melting point of the Period 3 elements Si to Cl.

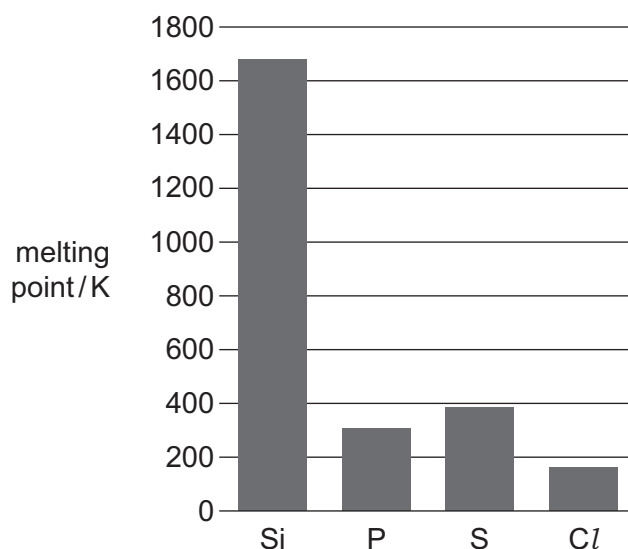


Fig. 2.1

The Period 3 elements Si to Cl are all non-metals.

Explain why there is a large difference between the melting point of Si and the melting points of P, S and Cl.

[2]

(c) Table 2.1 gives some information about some Period 3 chlorides.

Row B refers to the pH of the solution that forms when the Period 3 chloride is added to water.

Table 2.1

	formula of Period 3 chloride	NaCl	MgCl_2	AlCl_3	SiCl_4	PCl_5
A	oxidation number of element bonded to Cl					
B	pH of solution		6.5			
C	bonding			ionic		
D	structure			giant		

Complete Table 2.1.

You may use the following abbreviations.

I = ionic, C = covalent, M = metallic

G = giant, S = simple

[4]

10. O/N 25/P22/Q2(d)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(d) (i) Write an equation for the formation of AlCl_3 from its elements.

[1]

(ii) Write an equation for the formation of H_3PO_4 from PCl_5 .

[1]

11. O/N 25/P23/Q2(a)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(a) Fig. 2.1 shows the variation in atomic and ionic radii of the Period 3 elements Na to Cl.

The ionic radius of Si is **not** shown.

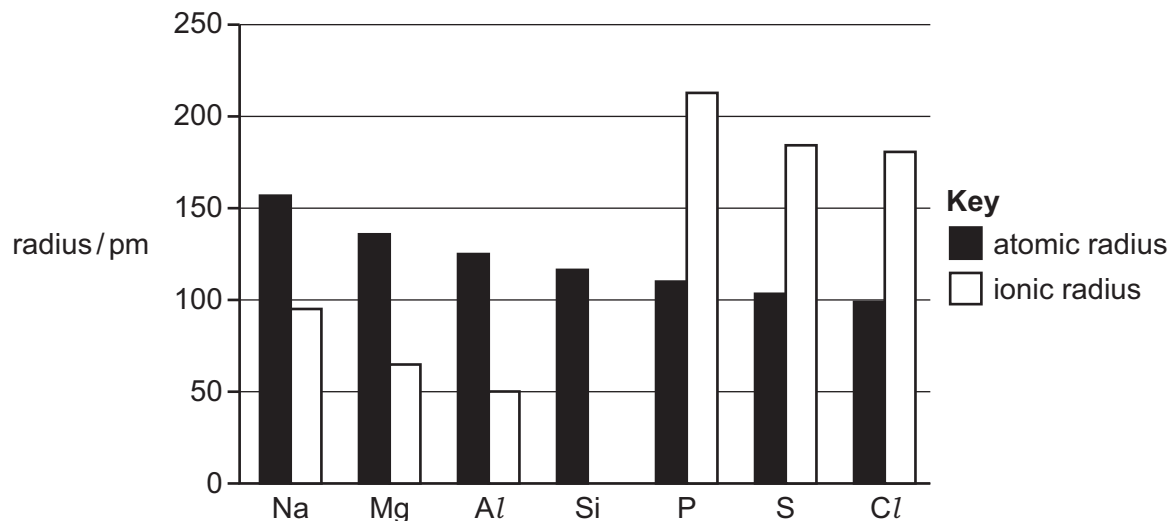


Fig. 2.1

(i) Explain the trend shown in the atomic radii of the Period 3 elements Na to Cl.

[2]

(ii) Explain why there is a large difference in the ionic radii of Al and P.

[2]

12. O/N 25/P23/Q2(b)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(b) Table 2.1 gives some information about some of the Period 3 oxides.

Row B gives the pH of the solution that forms when the Period 3 oxide is added to water.

Table 2.1

	formula of Period 3 oxide	Na ₂ O	MgO	Al ₂ O ₃	SiO ₂	P ₄ O ₁₀	SO ₃
A	oxidation number of Period 3 element			+3			
B	pH of solution			—	—		

(i) Complete Table 2.1.

[2]

(ii) State why there is no data given in row B for Al₂O₃ and SiO₂.

[1]

13. O/N 25/P23/Q2(c)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(c) (i) Write an equation for the reaction of Na_2O with dilute hydrochloric acid.

[1]

(ii) Construct an equation for the reaction of Al_2O_3 with a base to form NaAlO_2 .

[1]

14. O/N 25/P24/Q1(a)(ii)

Context

1 The Group 17 elements are oxidising agents.

(ii) Write an equation to show the reaction in which Cl_2 oxidises aluminium metal.

[1]

15. O/N 25/P24/Q1(d)

Context

1 The Group 17 elements are oxidising agents.

(d) Table 1.3 gives some information about MgCl_2 and SiCl_4 .

Table 1.3

	MgCl_2	SiCl_4
electrical conductivity when liquid	conducts	does not conduct
observation when added to water	dissolves	vigorous reaction

(i) Explain the difference between the electrical conductivity of liquid MgCl_2 and of liquid SiCl_4 . Refer to bonding and relevant particles in your answer.

[2]

(ii) Suggest the pH of the solutions that form when each chloride is added to water.

 MgCl_2 SiCl_4

[2]

16. O/N 25/P24/Q2(c)(iii)

Context

- 2** Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

Context: part (b) (introduction)

- (b)** A 5.00 dm^3 sealed flask contains 0.400 mol of $CO(g)$ and 0.800 mol of $H_2(g)$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium. Equation 1 shows the reaction.

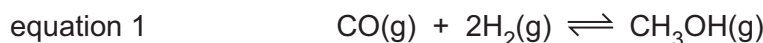


The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[CH_3OH]}{[CO][H_2]^2}$$

Context: part (c) (introduction)

- (c)** P_4O_{10} catalyses the reversible reaction of CO with H_2 to form CH_3OH .



P_4O_{10} then acts as a dehydrating agent, causing CH_3OH to form CH_3OCH_3 .

- (iii)** Write an equation to show the reaction of P_4O_{10} with an excess of water.

[1]

17. M/J 25/P21/Q1(b)

- (b)** Separate samples of sodium and sodium oxide are each added to an excess of cold water.

- (i)** Write an equation for the reaction of sodium with cold water.

[1]

- (ii)** Write an equation for the reaction of sodium oxide with cold water.

[1]

- (iii)** Complete Table 1.1.

Do **not** refer to temperature changes when considering observations for these reactions.

Table 1.1

	sodium	sodium oxide
one similarity in observation on addition to cold water		
one difference in observation on addition to cold water		

[2]

18. M/J 25/P21/Q1(c)

- (c) State the oxidation number of the Period 3 elements bonded to Cl in NaCl and PCl_5 .
Explain the difference in the oxidation number.

[2]

19. M/J 25/P22/Q1(b)

- (b) Separate samples of phosphorus(V) chloride and silicon(IV) chloride are each added to an excess of cold water.

- (i) Write an equation for each reaction.

[2]

- (ii) Describe the appearance of phosphorus(V) chloride and silicon(IV) chloride at room temperature.

phosphorus(V) chloride

silicon(IV) chloride

[2]

- (iii) Compare the appearance of the mixtures produced when each reaction is complete.

[2]

20. M/J 25/P22/Q1(c)

- (c) Some oxides are amphoteric.

- (i) Describe what is meant by amphoteric.

[1]

- (ii) Identify the formula of a Period 3 oxide that is amphoteric.

[1]

21. M/J 25/P23/Q2(a)

Context

- 2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

- (a) A Period 3 oxide produces a solution with a pH greater than 10 when it is added to water.
State the formula of the oxide.

[1]

22. M/J 25/P23/Q2(b)

Context

- 2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

- (b) P_4O_{10} is added to an excess of aqueous NaOH.
Write an equation to describe the reaction.

[1]

23. M/J 25/P23/Q2(c)(i)

Context

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

Context: part (c) (introduction)

(c) Table 2.1 shows the melting points of some oxides.

Table 2.1

oxide	melting point/°C
SO ₂	−73
H ₂ O	0
SO ₃	17
SiO ₂	1610
Na ₂ O	1132
MgO	2852
Al ₂ O ₃	2072

- (i)** Identify the oxide from Table 2.1 that contains the element with the highest oxidation number.

[1]

24. M/J 25/P23/Q2(e)

Context

2 The chemical properties of oxides are related to the chemical bonding present in these compounds.

(e) Al₂O₃ is a white amphoteric compound.

- (i)** State the formula of the aluminium-containing species produced when Al₂O₃ reacts with NaOH(aq).

[1]

- (ii)** State the formula of the aluminium-containing salt produced when Al₂O₃ reacts with H₂SO₄(aq).

[1]

25. M/J 25/P24/Q1(b)

Context

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(b) The relative electrical conductivities of the Period 3 elements are shown in Fig. 1.1.

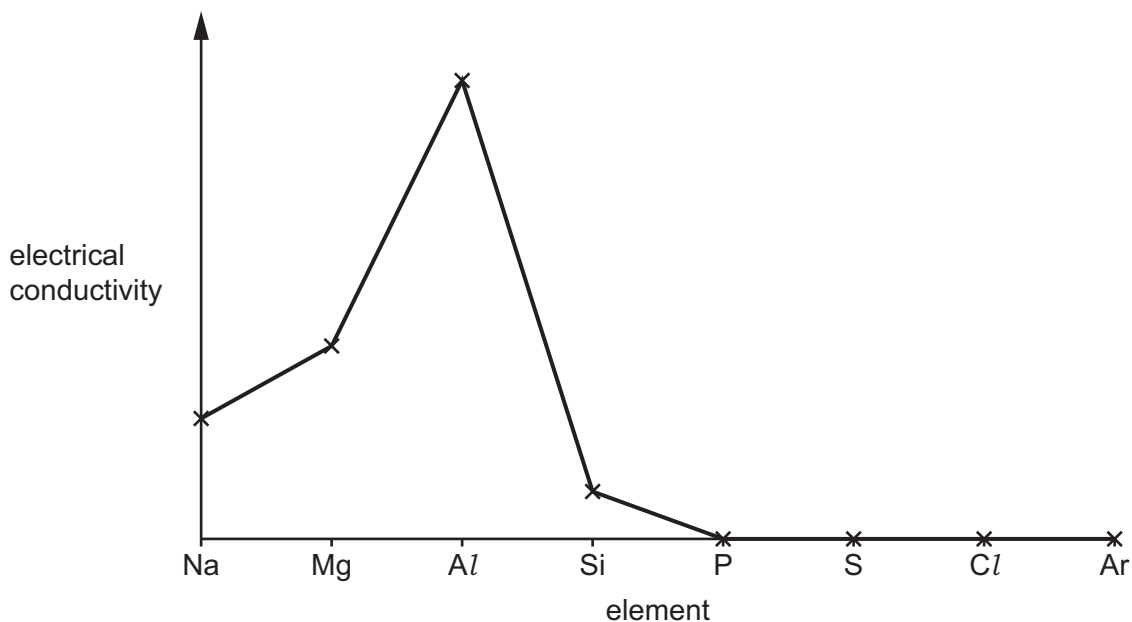


Fig. 1.1

Explain why there is an increase in conductivity from Na to Al and why P, S and Cl are non-conductors of electricity.

[2]

26. M/J 25/P24/Q1(d)

Context

1 Elements in Period 3 of the Periodic Table show trends in their properties.

(d) Most Period 3 elements react with oxygen to form oxides.

(i) Write an equation for the reaction of phosphorus with oxygen.

[1]

(ii) Write an equation for the reaction of aluminium oxide with an excess of aqueous sodium hydroxide.

[1]

27. M/J 25/P24/Q2(b)(i)

Context: part (b) (introduction)

(b) SCl_2 is a red liquid that reacts with water.



(i) Describe **two** observations, other than temperature change, that are made when an excess of water is added to SCl_2 .

[2]

28. M/J 25/P24/Q2(b)(iii)

Context: part (b) (introduction)

(b) SCl_2 is a red liquid that reacts with water.



(iii) State the general name to describe the type of reaction that occurs when a substance is broken down by water.

[1]

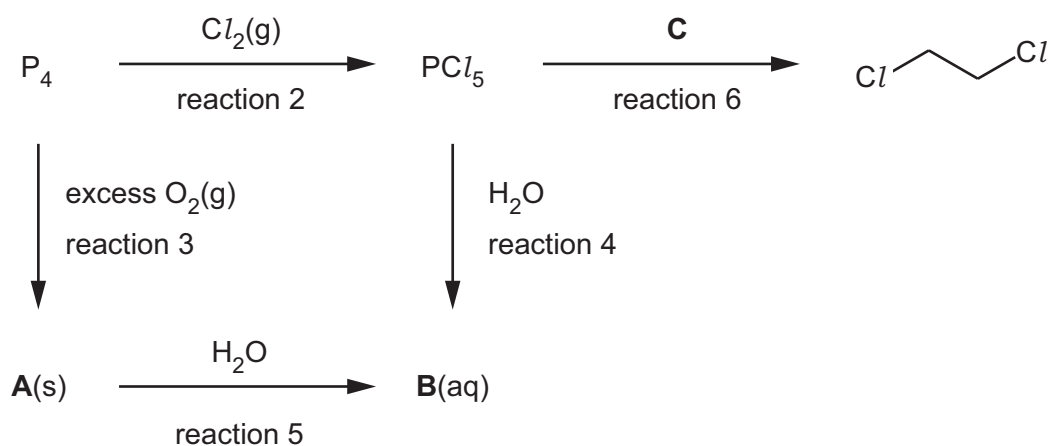
29. MAR 25/P22/Q1(b)(i)

Context

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

Context: part (b) (introduction)

(b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2 .

**Fig. 1.1**

(i) Suggest what is observed in reaction 2.

[1]

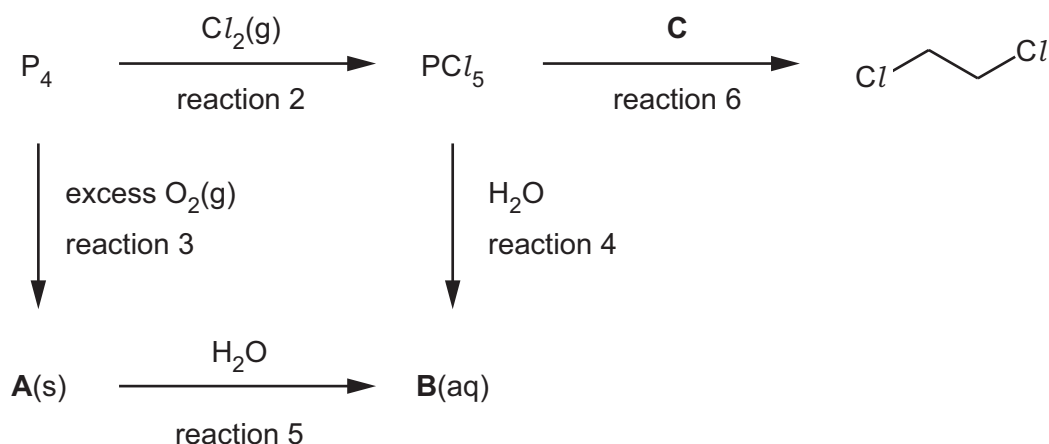
30. MAR 25/P22/Q1(b)(iii)-(v)

Context

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

Context: part (b) (introduction)

(b) Fig. 1.1 shows a reaction scheme involving P_4 and Cl_2 .

**Fig. 1.1**

(iii) Write an equation for the formation of **A** in reaction 3.

[1]

(iv) Name **B**, which is formed in both reactions 4 and 5.

[1]

(v) Draw the structure of **C**, used in reaction 6.

[1]

31. O/N 24/P21/Q3(e)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (e) (introduction)

(e) Silicon readily reacts with elements of high electronegativity.

(i) Write an equation for the formation of $SiCl_4$ from its constituent elements.

[1]

(ii) Describe what is observed when a small sample of $SiCl_4$ is added to water.

[1]

32. O/N 24/P21/Q3(f)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(f) Tin forms an amphoteric oxide, SnO_2 .

Suggest the formula of the tin compound that forms when SnO_2 reacts with H_2SO_4 in an acid–base reaction.

[1]

33. O/N 24/P22/Q2(a)(i)

Context

2 Oxygen is a Group 16 element.**(a) (i)** Write equations for the following reactions.

- sodium and oxygen
- sulfur and oxygen

[2]

34. O/N 24/P22/Q2(a)(iii)

Context

2 Oxygen is a Group 16 element.**(iii)** The maximum oxidation state of the Period 3 elements in their oxides varies across the period.

State and explain the variation.

[2]

35. O/N 24/P22/Q2(b)(i)

Context

2 Oxygen is a Group 16 element.

Context: part (b) (introduction)

(b) H_2O reacts with both inorganic and organic compounds.**(i)** Complete Table 2.1 to give details of the reactions of some Period 3 oxides with H_2O .**Table 2.1**

Period 3 oxide	product of reaction with H_2O	pH of solution formed
	$\text{Mg}(\text{OH})_2$	
P_4O_{10}		

[2]

36. O/N 24/P23/Q3(e)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (e) (introduction)

(e) Silicon readily reacts with elements of high electronegativity.**(i)** Write an equation for the formation of SiCl_4 from its constituent elements.

[1]

(ii) Describe what is observed when a small sample of SiCl_4 is added to water.

[1]

37. O/N 24/P23/Q3(f)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

(f) Tin forms an amphoteric oxide, SnO_2 .

Suggest the formula of the tin compound that forms when SnO_2 reacts with H_2SO_4 in an acid–base reaction.

[1]

38. M/J 24/P21/Q2(a)

2 (a) Sulfur chloride, SCl_2 , is a liquid at room temperature. When SCl_2 is added to water, misty fumes are seen and a solution is made that turns universal indicator red.

(i) Identify the type of reaction that occurs when SCl_2 is added to water.

[1]

(ii) Name a chloride of a different Period 3 element that is also a liquid at room temperature and produces misty fumes when added to water.

[1]

39. M/J 24/P22/Q1(d)-(e)

- (d) Fig. 1.1 shows the variation in melting point of some Period 3 elements in their standard states at room temperature and pressure.

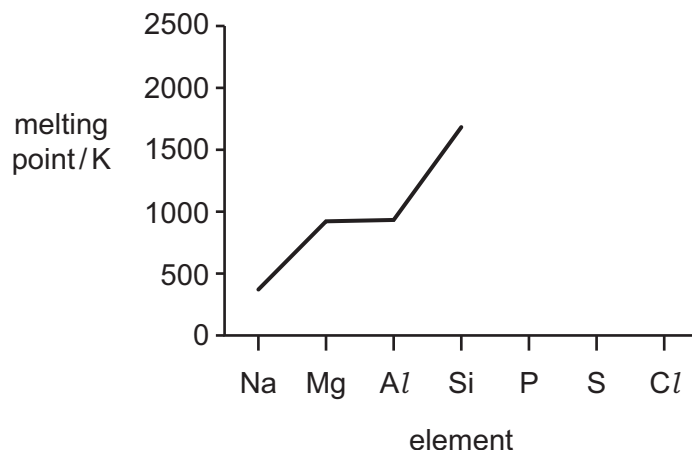


Fig. 1.1
3

- (i) Explain why Si has a high melting point.

[1]

- (ii) Complete Fig. 1.1 to show the variation in the melting points of the elements P, S and Cl.

[2]

- (e) Two Period 3 elements react with an excess of oxygen at room pressure.

- (i) Complete Table 1.2.

Table 1.2

1	2	3
Period 3 element	state of oxide at room temperature and pressure	approximate pH of solution made when oxide is added to water
Na		
S		

[2]

- (ii) The solutions made in column 3 of Table 1.2 are mixed together.
Name the type of reaction that occurs.

[1]

- (iii) Write an equation to describe the reaction between P_4O_{10} and an excess of water.

[1]

40. M/J 24/P22/Q1(f)

(f) Aluminium hydroxide is amphoteric.

(i) Explain what is meant by amphoteric.

[1]

(ii) Write an equation to describe the reaction that occurs when aluminium hydroxide, $\text{Al}(\text{OH})_3$, reacts with $\text{NaOH}(\text{aq})$.

[1]

41. M/J 24/P23/Q1(b)

(b) (i) A sample of Al reacts with an excess of Cl_2 .

State the oxidation number of Al in the product of the reaction.

oxidation number of Al

[1]

(ii) State what determines the maximum oxidation number of the Period 3 elements in their oxides.

[1]

42. M/J 24/P23/Q1(c)

(c) Separate samples of aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , react with an excess of $\text{NaOH}(\text{aq})$ at room temperature.

(i) Give the state of Al_2O_3 and P_4O_{10} at room temperature.

Al_2O_3

P_4O_{10}

[1]

(ii) Write an equation for the reaction of each oxide with an excess of $\text{NaOH}(\text{aq})$ at room temperature.

Al_2O_3 +

P_4O_{10} +

[4]

43. MAR 24/P22/Q2(a)

Context

- 2** Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

- (a)** Complete Table 2.1.

[3]

44. MAR 24/P22/Q2(c)(ii)

Context

- 2** Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

Context: part (c) (introduction)

- (c)** An excess of Cl_2 reacts with phosphorus to form PCl_5 .

- (ii)** PCl_5 reacts with **J** to form H_3PO_4 .

Identify **J** and state the type of reaction.

J type of reaction

[2]

45. O/N 23/P21/Q1(d)(ii)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

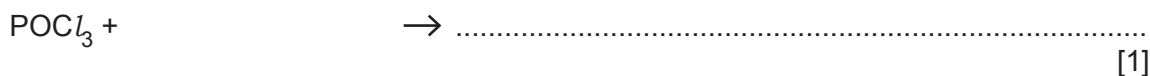
Context: part (d) (introduction)

- (d)** POCl_3 shows similar chemical properties to PCl_5 .

POCl_3 has a melting point of 1°C and a boiling point of 106°C .

POCl_3 reacts vigorously with water, forming misty fumes and an acidic solution.

- (ii) Construct an equation for the reaction of POCl_3 with water.



46. O/N 23/P22/Q1(a)

Context

- 1** The elements silicon, phosphorus and sulfur are in Period 3 of the Periodic Table.

- (a) (i)** Describe the variation in atomic radius from silicon to sulfur.

[1]

- (ii) The melting point of silicon is 1410°C . The melting point of sulfur is 113°C .

Explain this difference.

[3]

47. O/N 23/P22/Q3(a)(i)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (a) (introduction)

(a) H_3PO_4 is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form H_3PO_4 .

(i) Write an equation for each step.

step 1

step 2

[2]

48. O/N 23/P23/Q1(d)(ii)

Context

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

Context: part (d) (introduction)

(d) POCl_3 shows similar chemical properties to PCl_5 .

POCl_3 has a melting point of 1°C and a boiling point of 106°C .

POCl_3 reacts vigorously with water, forming misty fumes and an acidic solution.

(ii) Construct an equation for the reaction of POCl_3 with water.

$$\text{POCl}_3 + \dots \rightarrow \dots$$

[1]

49. M/J 23/P21/Q1(f)

Context

1 Tellurium is an element in Group 16. The most common isotope of tellurium is ^{130}Te . Its electronic configuration is $[\text{Kr}] 4d^{10} 5s^2 5p^4$.

(f) TeF_x reacts with water to form tellurium hydroxide and HF. The oxidation number of tellurium does **not** change during this reaction.

(i) Construct an equation for the reaction of TeF_x with water.

[1]

(ii) Name the type of reaction that occurs when TeF_x reacts with water.

[1]

50. M/J 23/P21/Q3(a)

Context

3 Chlorine is a very reactive element.

(a) Chlorine reacts with silicon to form silicon(IV) chloride.
Describe the appearance of silicon(IV) chloride at room temperature and pressure. State its structure and bonding.

appearance

structure and bonding

[2]

51. M/J 23/P21/Q3(b)

Context

3 Chlorine is a very reactive element.

(b) Samples of magnesium chloride and phosphorus(V) chloride are added to separate beakers of cold water.

Complete Table 3.1. Ignore temperature changes when considering observations for these reactions.

Table 3.1

	magnesium chloride	phosphorus(V) chloride
appearance at room temperature		
one similarity in observation on addition to cold water		
one difference in observation on addition to cold water		
pH of final solution		

[4]

52. M/J 23/P22/Q1(a)(iii)

Context

- 1** The melting points of some solids are shown in Table 1.1.

Table 1.1

solid	melting point / K
magnesium	923
phosphorus	317
sodium chloride	1074
sulfur	392

- (iii) Explain the difference in the melting points of phosphorus and sulfur in terms of structure and bonding.

[2]

53. M/J 23/P22/Q2(a)

Context

- 2** Chlorine is a reactive element. It forms many compounds.

- (a) (i) Complete Table 2.1 to show the maximum oxidation number of the elements Na to P in their chlorides.

Table 2.1

element	Na	Mg	Al	Si	P
maximum oxidation number					

[1]

- (ii) State what determines the maximum oxidation number of elements in Period 3.

[1]

54. M/J 23/P22/Q2(b)

Context

- 2** Chlorine is a reactive element. It forms many compounds.

- (b) An excess of cold water is added to the chloride of silicon.

- (i) Write an equation for the reaction between an excess of cold water and the chloride of silicon.

[1]

- (ii) Suggest the pH of the solution produced in (b)(i).

[1]

55. M/J 23/P22/Q2(c)

Context

2 Chlorine is a reactive element. It forms many compounds.

(c) An excess of cold water is added to the chloride of phosphorus.

(i) Write an equation for the reaction between an excess of cold water and the chloride of phosphorus.

[1]

(ii) Suggest the pH of the solution produced in **(c)(i)**.

[1]

56. M/J 23/P23/Q2(a)

2 (a) The reaction of pure aluminium is only observed if the aluminium oxide layer is removed first. When pure aluminium is added to cold water, bubbles of gas are seen.

(i) State **one** property of aluminium oxide that explains why an aluminium object does **not** react with cold water until the aluminium oxide layer is removed.

[1]

(ii) Write an equation, with state symbols, for the reaction of aluminium oxide with an excess of NaOH(aq).

[2]

(iii) Name **one** other Period 3 element that also produces bubbles of gas when added to cold water.

[1]

57. M/J 23/P23/Q2(c)

(c) Aluminium chloride and phosphorus chloride are both white solids.

- (i) State the maximum oxidation number of aluminium and of phosphorus in these solid chloride salts.

maximum oxidation number of aluminium

maximum oxidation number of phosphorus

[1]

- (ii) State why the maximum oxidation number of aluminium is different from that of phosphorus.

[1]

- (iii) Write an equation for the reaction of solid phosphorus chloride and excess water.

[1]

- (iv) Name the type of reaction that occurs when aluminium chloride is added to water.

[1]

- (v) Explain why the solution produced after aluminium chloride is added to water has a pH of 1–2.

[1]

58. MAR 23/P22/Q1(d)(i)-(ii)

Context

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

Context: part (d) (introduction)

- (d) CO₂ and SO₂ are acidic gases.

- (i) Write an equation for the reaction of SO₂ with H₂O.

[1]

- (ii) Write an equation for the reaction of SO₂ with NaOH.

[1]

59. O/N 22/P21/Q3(a)

Context

- 3** Some of the common chlorides of Period 3 elements are shown in the list.



(a) From this list, identify:

(i) all the chlorides that have giant ionic structures in the solid state

[1]

(ii) all the chlorides that react vigorously with water to form strongly acidic solutions

[1]

(iii) the chloride that dissolves in water to form a neutral solution

[1]

(iv) the chloride formed from the **element** with the highest melting point.

[1]

60. O/N 22/P22/Q2(a)

Context

- 2** The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

(a) Complete Table 2.1.

- Identify the bonding shown by each chloride under standard conditions.
Use C = covalent, I = ionic, M = metallic.
- Identify the structure shown by each chloride under standard conditions.
Use G = giant, S = simple.
- Deduce the oxidation state of the Period 3 element in each chloride.

[4]

61. O/N 22/P22/Q2(b)

Context

- 2** The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

- (b)** Write equations for the reactions of NaCl and PCl_5 with water. Include state symbols in both equations.

NaCl

PCl_5

[3]

62. O/N 22/P23/Q3(a)

Context

- 3** Some of the common chlorides of Period 3 elements are shown in the list.

NaCl MgCl_2 AlCl_3 SiCl_4 PCl_5

- (a)** From this list, identify:

(i) all the chlorides that have giant ionic structures in the solid state

[1]

(ii) all the chlorides that react vigorously with water to form strongly acidic solutions

[1]

(iii) the chloride that dissolves in water to form a neutral solution

[1]

(iv) the chloride formed from the **element** with the highest melting point.

[1]

63. M/J 22/P21/Q1(f)(i)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

Context: part (f) (introduction)

- (f)** Magnesium, Mg, burns in oxygen, O₂.
The activation energy, E_a , for this reaction is +148 kJ mol⁻¹.
- (i)** State **one** observation when magnesium burns in oxygen.
Do **not** refer to temperature changes in your answer.

[1]

65. M/J 22/P23/Q2(a)

2 (a) Period 3 elements and their compounds show trends in their physical properties.

(i) On Fig. 2.1 sketch a graph to show the melting points of the first five elements in Period 3.

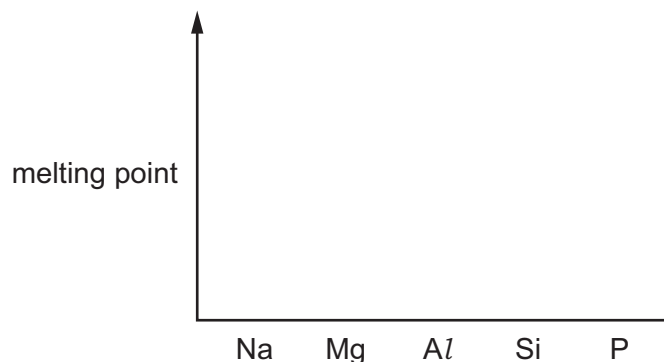


Fig. 2.1

[3]

(ii) Complete Table 2.1 with information for sodium chloride and phosphorus(V) chloride.

Table 2.1

	sodium chloride	phosphorus(V) chloride
state at room temperature		
name of change which occurs on addition of water		
pH of final solution		

[3]

66. MAR 22/P22/Q2(a)(ii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (a) (introduction)

(a) Na reacts with O_2 to form Na_2O . Na is the reducing agent in this reaction.

(ii) Write an equation for the reaction of Na_2O with water.

[1]

67. MAR 22/P22/Q2(b)(i)-(ii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (b) (introduction)

(b) Al_2O_3 is an amphoteric oxide found in bauxite.

(i) State what is meant by amphoteric.

[1]

(ii) Al_2O_3 is purified from bauxite in several steps. The first step involves heating Al_2O_3 with an excess of NaOH(aq) . A colourless solution forms.

Write an equation for this reaction.

[1]

68. MAR 22/P22/Q2(c)(iv)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (c) (introduction)

(c) P_4O_6 is a white solid that has a melting point of 24°C . Solid P_4O_6 reacts with water to form H_3PO_3 .

(iv) Write an equation for the reaction of P_4O_{10} with water.

[1]

69. MAR 22/P22/Q3(b)(i)

Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

Context: part (b) (introduction)

(b) HCl is a product of several different reactions. Some of these are shown in Fig. 3.1.

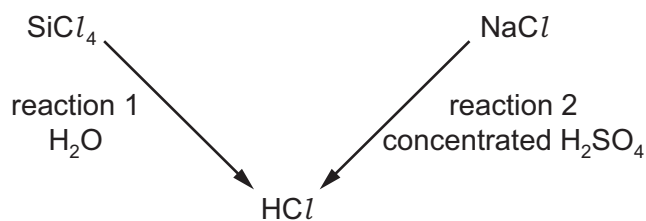


Fig. 3.1

(i) Write an equation for reaction 1.

[1]

Answers Section

1. M/J 26/P22/Q1(c)

1(c)(i)	sodium oxide	1	I formulas BOD sodium
1(c)(ii)	$\text{Li}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{LiOH}$	1	A $2\text{Li}^+ + 2\text{OH}^-$ for 2LiOH BOD LiHO I equations with water as both a reactant and a product
1(c)(iii)	M1 phosphorus (V) chloride / PCl_5 M2 aluminium chloride / AlCl_3	2	M1 A phosphorus pentachloride M1 I phosphorus chloride / phosphorus trichloride M1 I phosphorus alone M2 A Al_2Cl_6 M2 A aluminium alone / aluminium trichloride / aluminium(III) chloride M2 I incorrect aluminium chloride formula I MgCl_2 / magnesium chloride / magnesium I sulfur chlorides / sulfur R silicon chlorides / silicon R NaCl / Na R any period 3 chloride with additional elements R any chloride not from period 3

2. M/J 26/P22/Q1(d)

1(d)	nitrogen dioxide OR NO_2	1	A nitrogen(IV) oxide I nitrogen oxide alone
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3. M/J 26/P23/Q1(a)

1(a)(i)	element	formula of chloride	type of bonding	type of reaction, if any of the chloride with cold water	3	•✓•✓•✓ I giant / molecular in 'bonding'
	sodium	NaCl	ionic	no reaction / none A dissolves / dissociates / dissolution		
	silicon	SiCl_4	covalent	hydrolysis R hydration		
1(a)(ii)	$\text{PCl}_5 + 4 \text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5 \text{HCl}$				1	I $\text{PCl}_5 + \text{H}_2\text{O} \rightarrow \text{POCl}_3 + 2 \text{HCl}$

4. M/J 26/P24/Q3(a)

3(a)(i)	decreases (across the period)	1	A smaller / goes down A decreases down the period R decreases down the group
3(a)(ii)	Na_2O	1	A Na_2O_2 and NaO_2 A equation that gives Na_2O as the only product I balancing
3(a)(iii)	$\text{MgO} + \text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2$	1	A $\text{MgO} + \text{H}_2\text{O} \rightarrow \text{Mg}^{2+} + 2\text{OH}^-$ A in equation =

5. M/J 26/P24/Q3(b)

3(b)(i)		type of bonding	type of structure	2	●✓●✓ A			
	NaCl	I (ionic)	G (giant)				type of bonding	type of structure
	SiCl₄	C (covalent)	S (simple)			NaCl	GI	GI
						SiCl₄	SC	SC
3(b)(ii)	M1 (formula:) H ₃ PO ₄ M2 (pH of solution:) 0–4			2	M1 A PO(OH) ₃ M1 R H ₃ PO ₃ (as excess water) M1 R HCl on formula answer line M1 R PCl ₅ + 4H ₂ O → H ₃ PO ₄ + 5HCl on formula answer line			

6. O/N 25/P21/Q2(a)

2(a)(i)	M1 nuclear charge increases (across period) AND similar shielding M2 greater / stronger attraction to nucleus	2
2(a)(ii)	M1 Al loses (3) electrons / Al is 3+, and P gains (3) electrons / P is 3– M2 Al ⁽³⁺⁾ has (one) fewer shell (than P ^(3–))	2

7. O/N 25/P21/Q2(b)

2(b)(i)	M1	(+)1	(+)2	(+)3	(+)4	(+)5	(+)6	2
	M2	12-14	8-10	—	—	0-4	0-4	
2(b)(ii)	they are insoluble							1

8. O/N 25/P21/Q2(c)

2(c)(i)	Na ₂ O + 2HCl → 2NaCl + H ₂ O	1
2(c)(ii)	Al ₂ O ₃ + 2NaOH → 2NaAlO ₂ + H ₂ O	1

9. O/N 25/P22/Q2(a)-(c)

2(a)(i)	delocalised electrons move (through the structure / between layers)						1
2(a)(ii)	P / S / Cl have no charge carriers OR (no ions and) no delocalised electrons OR no mobile / moving ions						1
2(b)	M1 Si has (lots of) strong covalent bonds broken M2 P / S / Cl (only) has weak Van der Waals' (forces) broken						2
2(c)	M1 M2 M3 M4	+1 7 I G	+2 6.5 I G	+3 0–4 I G	+4 0–4 C S	+5 0–4 C S	4

10. O/N 25/P22/Q2(d)

2(d)(i)	2Al + 3Cl ₂ → 2AlCl ₃	1
2(d)(ii)	PCl ₅ + 4H ₂ O → H ₃ PO ₄ + 5HCl	1

11. O/N 25/P23/Q2(a)

2(a)(i)	M1 nuclear charge increases (across period) AND similar shielding M2 greater / stronger attraction to nucleus	2
2(a)(ii)	M1 Al loses (3) electrons / Al is 3+, and P gains (3) electrons / P is 3– M2 Al ⁽³⁺⁾ has (one) fewer shell (than P ^(3–))	2

12. O/N 25/P23/Q2(b)

2(b)(i)	<table><tr><td>M1</td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td><td>(+)6</td></tr><tr><td>M2</td><td>12-14</td><td>8-10</td><td>—</td><td>—</td><td>0-4</td><td>0-4</td></tr></table>	M1	(+)1	(+)2	(+)3	(+)4	(+)5	(+)6	M2	12-14	8-10	—	—	0-4	0-4	2
M1	(+)1	(+)2	(+)3	(+)4	(+)5	(+)6										
M2	12-14	8-10	—	—	0-4	0-4										
2(b)(ii)	they are insoluble	1														

13. O/N 25/P23/Q2(c)

2(c)(i)	$\text{Na}_2\text{O} + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O}$	1
2(c)(ii)	$\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + \text{H}_2\text{O}$	1

14. O/N 25/P24/Q1(a)(ii)

1(a)(ii)	$2\text{Al} + 3\text{Cl}_2 \rightarrow 2\text{AlCl}_3$	1
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15. O/N 25/P24/Q1(d)

1(d)(i)	M1 MgCl ₂ (is ionic and) has mobile ions M2 SiCl ₄ is covalent AND no charge carriers OR (no ions and) no delocalised electrons OR no mobile / moving ions	2
1(d)(ii)	M1 MgCl ₂ pH = 7 M2 SiCl ₄ pH 0 – 4	2

16. O/N 25/P24/Q2(c)(iii)

2(c)(iii)	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1
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17. M/J 25/P21/Q1(b)

1(b)(i)	$2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$	1									
1(b)(ii)	$\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$	1									
1(b)(iii)	<table border="1"> <tr> <td></td><td>sodium</td><td>sodium oxide</td></tr> <tr> <td>one similarity</td><td colspan="2">(both) solids disappear OR (both produce) colourless solutions</td></tr> <tr> <td>one difference</td><td>effervescence</td><td>no effervescence</td></tr> </table>		sodium	sodium oxide	one similarity	(both) solids disappear OR (both produce) colourless solutions		one difference	effervescence	no effervescence	2
	sodium	sodium oxide									
one similarity	(both) solids disappear OR (both produce) colourless solutions										
one difference	effervescence	no effervescence									

18. M/J 25/P21/Q1(c)

1(c)	M1 NaCl + 1 AND PCl ₅ + 5 M2 Na has only 1 valence (shell) electrons AND P has 5	2
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19. M/J 25/P22/Q1(b)

1(b)(i)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$	1
	$\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	1
1(b)(ii)	M1 phosphorus(V) chloride white / yellow solid M2 silicon(IV) chloride colourless liquid	2
1(b)(iii)	M1 (phosphorus(V) chloride) produces colourless solution M2 silicon(IV) chloride produces white suspension / white precipitate (in a colorless liquid / mixture / solution)	2

20. M/J 25/P22/Q1(c)

1(c)(i)	species behaves as acid and base OR species reacts with (both) acid and base	1
1(c)(ii)	Al_2O_3	1

21. M/J 25/P23/Q2(a)

2(a)	Na_2O	1
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22. M/J 25/P23/Q2(b)

2(b)	$\text{P}_4\text{O}_{10} + 12\text{NaOH} \rightarrow 4\text{Na}_3\text{PO}_4 + 6\text{H}_2\text{O}$	1
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23. M/J 25/P23/Q2(c)(i)

2(c)(i)	SO_3	1
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24. M/J 25/P23/Q2(e)

2(e)(i)	$\text{NaAl}(\text{OH})_4$	1
2(e)(ii)	$\text{Al}_2(\text{SO}_4)_3$	1

25. M/J 25/P24/Q1(b)

1(b)	M1 Na to Al have (delocalised) electrons that can move (through the structure / lattice) AND (P / S / Cl) do not (have delocalised electrons / mobile charge carriers) M2 Na to Al increasing number of delocalised electrons	2
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26. M/J 25/P24/Q1(d)

1(d)(i)	$\text{P}_4 + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$	1
1(d)(ii)	$\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$	1

27. M/J 25/P24/Q2(b)(i)

2(b)(i)	<i>In any order:</i> M1 (solution / liquid) (red) to colourless M2 yellow solid appears	2
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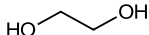
28. M/J 25/P24/Q2(b)(iii)

2(b)(iii)	hydrolysis	1
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29. MAR 25/P22/Q1(b)(i)

1(b)(i)	green colour (of gas) disappears	1
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30. MAR 25/P22/Q1(b)(iii)-(v)

1(b)(iii)	$P_4 + 5O_2 \rightarrow P_4O_{10}$	1
1(b)(iv)	phosphoric(V) acid	1
1(b)(v)		1

31. O/N 24/P21/Q3(e)(i)-(ii)

3(e)(i)	$Si + 2Cl_2 \rightarrow SiCl_4$	1
3(e)(ii)	effervescence / misty / steamy fumes	1

32. O/N 24/P21/Q3(f)

3(f)	$Sn(SO_4)_2$	1
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33. O/N 24/P22/Q2(a)(i)

2(a)(i)	M1 $4Na + O_2 \rightarrow 2Na_2O$	1
	M2 $S + O_2 \rightarrow SO_2$	1

34. O/N 24/P22/Q2(a)(iii)

2(a)(iii)	it increases	1
	as number of valence electrons (which can be used in bonding / lost / shared / donated) increases	1

35. O/N 24/P22/Q2(b)(i)

2(b)(i)	Period 3 oxide	product of rxn with water	pH of solution formed	●✓●✓
	● MgO	Mg(OH) ₂	● 8 ≤ pH ≤ 12	
	P ₄ O ₁₀	● H ₃ PO ₄	● 1 ≤ pH ≤ 4	

36. O/N 24/P23/Q3(e)(i)-(ii)

3(e)(i)	$Si + 2Cl_2 \rightarrow SiCl_4$	1
3(e)(ii)	effervescence / misty / steamy fumes	1

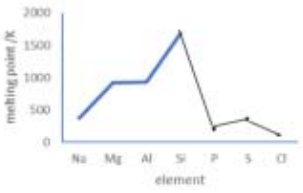
37. O/N 24/P23/Q3(f)

3(f)	$Sn(SO_4)_2$	1
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38. M/J 24/P21/Q2(a)

2(a)(i)	hydrolysis	1
2(a)(ii)	silicon(IV) chloride / silicon tetrachloride ALLOW phosphorus(III) chloride / phosphorus trichloride	1

39. M/J 24/P22/Q1(d)-(e)

1(d)(i)	large amount of energy required to break bonds in giant covalent structure	1						
1(d)(ii)	 M1 all 3 melting points lower than Mg M2 show melting point Cl < P < S	2						
1(e)(i)	<table border="1" data-bbox="236 560 933 654"> <tr> <td>Na</td><td>solid</td><td>10–14</td></tr> <tr> <td>S</td><td>gas</td><td>0–4</td></tr> </table> 4 correct = 2 marks 2 or 3 correct = 1 mark 0 or 1 correct = 0 marks	Na	solid	10–14	S	gas	0–4	2
Na	solid	10–14						
S	gas	0–4						
1(e)(ii)	acid–base OR neutralisation	1						
1(e)(iii)	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1						

40. M/J 24/P22/Q1(f)

1(f)(i)	reacts with both acids and bases OR shows both acidic and basic behaviour	1
1(f)(ii)	$\text{Al}(\text{OH})_3 + \text{NaOH} \rightarrow \text{NaAl}(\text{OH})_4$	1

41. M/J 24/P23/Q1(b)

1(b)(i)	(+)III	1
1(b)(ii)	number of outer / valence electrons	1

42. M/J 24/P23/Q1(c)

1(c)(i)	both states shown as solid	1
1(c)(ii)	M1 correctly balanced equation for aluminium oxide $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$ M2 correctly balanced equation for P_4O_{10} and an excess of NaOH $\text{P}_4\text{O}_{10} + 12\text{NaOH} \rightarrow 4\text{Na}_3\text{PO}_4 + 6\text{H}_2\text{O}$	2

43. MAR 24/P22/Q2(a)

2(a)	formula of chloride	• NaCl	• MgCl ₂	• SiCl ₄
	structure	giant	• giant	• simple
	bonding of chloride	• ionic	• ionic	covalent
	pH of solution formed	• 7	6.2	• 1–4

3

44. MAR 24/P22/Q2(c)(ii)

2(c)(ii)	H ₂ O / water	1
	hydrolysis	1

45. O/N 23/P21/Q1(d)(ii)

1(d)(ii)	$\text{POCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 3\text{HCl}$	1
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46. O/N 23/P22/Q1(a)

1(a)(i)	decreases (from Si to S)	1
1(a)(ii)	M1 Si giant covalent AND S is simple molecular OR Si giant covalent AND S has IMFs	1
	M2 Si involves breaking covalent bonds AND S involves breaking IMFs	1
	M3 more energy needed to break covalent bonds than IMFs OR covalent bonds are strong AND IMFs are weak	1

47. O/N 23/P22/Q3(a)(i)

3(a)(i)	$4\text{P} / \text{P}_4 + 5\text{O}_2 \rightarrow \text{P}_4\text{O}_{10}$	1
	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1

48. O/N 23/P23/Q1(d)(ii)

1(d)(ii)	$\text{POCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 3\text{HCl}$	1
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49. M/J 23/P21/Q1(f)

1(f)(i)	$\text{TeF}_x + x\text{H}_2\text{O} \rightarrow \text{Te}(\text{OH})_x + x\text{HF}$ OR $\text{TeF}_6 + 6\text{H}_2\text{O} \rightarrow \text{Te}(\text{OH})_6 + 6\text{HF}$	1
1f(ii)	hydrolysis	1

50. M/J 23/P21/Q3(a)

3(a)	M1 <i>appearance</i> (colourless / yellow) liquid M2 <i>structure and bonding</i> simple / molecular AND covalent	2
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51. M/J 23/P21/Q3(b)

3(b)			MgCl_2	PCl_5		4
	appearance		• (white) crystals / solid	• (white OR pale yellow) crystals / solid	[1]	
	one similarity in observation		(colourless) solution made		[1]	
	one difference in observation		no (misty / steamy) fumes	misty / steamy fumes	[1]	
	pH		• 6.5–7	• 0–4	[1]	

52. M/J 23/P22/Q1(a)(iii)

1(a)(iii)	M1 S_8 / molecules of sulfur have more electrons (than P_4 / molecules of phosphorus) M2 S has stronger instantaneous dipole–induced dipole forces (than phosphorus / P)	2
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53. M/J 23/P22/Q2(a)

2(a)(i)		<table><tr><td>element</td><td>Na</td><td>Mg</td><td>Al</td><td>Si</td><td>P</td></tr><tr><td>maximum oxidation number of elements in chlorides</td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td></tr></table>	element	Na	Mg	Al	Si	P	maximum oxidation number of elements in chlorides	(+)1	(+)2	(+)3	(+)4	(+)5	1
		element	Na	Mg	Al	Si	P								
		maximum oxidation number of elements in chlorides	(+)1	(+)2	(+)3	(+)4	(+)5								
All correct for one mark															
2(a)(ii)	number of outer/valence electrons					1									

54. M/J 23/P22/Q2(b)

2(b)(i)	$\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	1
2(b)(ii)	1–4	1

55. M/J 23/P22/Q2(c)

2(c)(i)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow 5\text{HCl} + \text{H}_3\text{PO}_4$	1
2(c)(ii)	1–4	1

56. M/J 23/P23/Q2(a)

2(a)(i)	it / aluminium oxide is insoluble (in water) OR it / aluminium oxide does not react (with water)	1
2(a)(ii)	M1 correct state of correct aluminium product M2 balanced equation for reaction of aluminium oxide with NaOH $\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4(\text{aq})$ OR $\text{Al}_2\text{O}_3 + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2(\text{aq}) + \text{H}_2\text{O}$	2
2(a)(iii)	sodium (only)	1

57. M/J 23/P23/Q2(c)

2(c)(i)	Al (+)3 / III AND P (+)5 / V	1
2(c)(ii)	different number of valence/outer shell electrons	1
2(c)(iii)	$\text{PCl}_5 + 4\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$	1
2(c)(iv)	hydrolysis / substitution	1
2(c)(v)	$\text{HCl}(\text{g})$ dissolves in water to make (hydrochloric) acid	1

58. MAR 23/P22/Q1(d)(i)-(ii)

1(d)(i)	$\text{SO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3$	1
1(d)(ii)	$\text{SO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$	1

59. O/N 22/P21/Q3(a)

3(a)(i)	NaCl AND MgCl_2	1
3(a)(ii)	AlCl_3 AND SiCl_4 AND PCl_5	1
3(a)(iii)	NaCl	1
3(a)(iv)	SiCl_4	1

60. O/N 22/P22/Q2(a)

2(a)		NaCl	AlCl ₃	SiCl ₄	PCl ₅	1
		I	I OR C	C	C	
		G	G	S	S	1
		+1	+3	+4	+5	1
				+3	+2	1

61. O/N 22/P22/Q2(b)

2(b)	M1 $\text{NaCl(s)} \rightarrow \text{Na}^{\text{+}}(\text{aq}) + \text{Cl}^{-}(\text{aq})$	1
	$\text{PCl}_5(\text{s}) + 4\text{H}_2\text{O(l)} \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + 5\text{HCl(aq)}$ M2 balanced equation	1
	M3 all state symbols for PCl_5 equation	1

62. O/N 22/P23/Q3(a)

3(a)(i)	NaCl AND MgCl ₂	1
3(a)(ii)	AlCl ₃ AND SiCl ₄ AND PCl ₅	1
3(a)(iii)	NaCl	1
3(a)(iv)	SiCl ₄	1

63. M/J 22/P21/Q1(f)(i)

1(f)(i)	white flame / light OR white solid / smoke	1
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64. M/J 22/P21/Q3(d)(iii)

3(d)(iii)	M1 water reacts with / hydrolyses PCl_5	2
	M2 $\text{H}_2\text{O} + \text{PCl}_5 \rightarrow \text{POCl}_3 + 2\text{HCl}$ OR $4\text{H}_2\text{O} + \text{PCl}_5 \rightarrow \text{H}_3\text{PO}_4 + 5\text{HCl}$	

65. M/J 22/P23/Q2(a)

2(a)(i)				3
2(a)(ii)		sodium chloride	phosphorus (V) chloride	3
	state at room temperature	solid	solid	
	name of change which occurs on addition of water	dissolves	hydrolysis	
	pH of final solution	7	1–2	

66. MAR 22/P22/Q2(a)(ii)

2(a)(ii)	$\text{Na}_2\text{O} + \text{H}_2\text{O} \rightarrow 2\text{NaOH}$	1
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67. MAR 22/P22/Q2(b)(i)-(ii)

2(b)(i)	reacts with both acids and bases / shows both acidic and basic behaviour	1
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2(b)(ii)	$\text{Al}_2\text{O}_3 + 2\text{NaOH} + 3\text{H}_2\text{O} \rightarrow 2\text{NaAl}(\text{OH})_4$	1
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68. MAR 22/P22/Q2(c)(iv)

2(c)(iv)	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$	1
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69. MAR 22/P22/Q3(b)(i)

3(b)(i)	$\text{SiCl}_4 + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 4\text{HCl}$	1
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Unit 10

GROUP 2

27 items · 70 marks

Syllabus sections in this unit

10.1 Similarities and trends in the properties of the Group 2 metals, magnesium to barium, and their compounds

Unit 10: Group 2**1. M/J 26/P24/Q3(c)**

(c) (i) Describe what is observed when white solid strontium nitrate decomposes on heating.

[1]

(ii) State the trend in solubility down the Group 2 hydroxides.

[1]

2. MAR 26/P22/Q1(b)

Context

1 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

(b) When strontium reacts with water, an alkaline solution is produced.

(i) Write an equation for this reaction. Include state symbols.

[2]

(ii) Samples of the alkaline solution produced in 1(b)(i) are tested with dilute HCl(aq) and separately with dilute $\text{H}_2\text{SO}_4\text{(aq)}$.

State what you would observe in each test.

observation with dilute HCl(aq)

observation with dilute $\text{H}_2\text{SO}_4\text{(aq)}$

[2]

3. MAR 26/P22/Q1(c)

Context

1 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

(c) The nitrates and carbonates of Group 2 elements decompose on heating.

(i) State the trend in the thermal stability of the Group 2 carbonates down the group.

[1]

(ii) Write an equation for the thermal decomposition of $\text{Ca(NO}_3)_2$.

[1]

4. O/N 25/P21/Q2(d)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(d) Group 2 nitrates decompose on heating to form oxides.

(i) State the trend in thermal stability of the Group 2 nitrates down the group.

[1]

(ii) Identify the other products of the thermal decomposition of Group 2 nitrates.

[1]

5. O/N 25/P22/Q2(f)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(f) Aqueous MgCl_2 reacts with aqueous Na_2CO_3 to form a white precipitate. Upon heating, the white precipitate undergoes thermal decomposition.

(i) Construct an equation to show the reaction of aqueous MgCl_2 with aqueous Na_2CO_3 . Use state symbols in your equation.

[2]

(ii) Identify the products of the thermal decomposition of the white precipitate.

[1]

(iii) State the trend in thermal stability of the Group 2 carbonates down the group.

[1]

6. O/N 25/P23/Q2(d)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(d) Group 2 nitrates decompose on heating to form oxides.

(i) State the trend in thermal stability of the Group 2 nitrates down the group.

[1]

(ii) Identify the other products of the thermal decomposition of Group 2 nitrates.

[1]

7. MAR 25/P22/Q2(d)

(d) (i) State what is observed when dilute hydrochloric acid is added to separate samples of barium oxide and barium carbonate.

barium oxide

barium carbonate

[2]

(ii) Write an equation for the reaction of strontium, Sr, with an excess of cold water.

[1]

(iii) State the variation in solubility of the Group 2 sulfates down the group.

[1]

8. O/N 24/P21/Q3(b)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (b) (introduction)

(b) Carbon is found in inorganic compounds such as carbonates.

(i) Write an equation for the reaction of magnesium carbonate with dilute HCl(aq) .

[1]

(ii) Describe the thermal stability of the carbonates down Group 2.

[1]

9. O/N 24/P22/Q3(a)(iv)-(v)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (a) (introduction)

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.

(iv) Nitric acid reacts with basic oxides to form nitrates.

Write an equation for the reaction of nitric acid with calcium oxide.

[1]

(v) Describe what is seen when solid calcium nitrate is heated strongly.

[1]

10. O/N 24/P23/Q3(b)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (b) (introduction)

(b) Carbon is found in inorganic compounds such as carbonates.

(i) Write an equation for the reaction of magnesium carbonate with dilute HCl(aq) .

[1]

(ii) Describe the thermal stability of the carbonates down Group 2.

[1]

11. M/J 24/P22/Q2(c)

Context

2 Separate samples of Na_2CO_3 and NaHCO_3 react with HCl(aq) to produce the same products, as shown in Table 2.1.

Table 2.1

reaction	equation	$\Delta H/\text{kJ mol}^{-1}$
1	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	ΔH_1
2	$\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	$\Delta H_2 = +27.2$

(c) **Z** is a salt that contains a Period 4 element from Group 2. When **Z** is heated brown gas forms.

Identify the formula of **Z** and use it to write an equation for the reaction.

[2]

12. M/J 24/P23/Q1(d)(ii)

Context: part (d) (introduction)

(d) The oxide of silicon reacts with calcium oxide in an addition reaction to produce calcium silicate, CaSiO_3 . The oxidation number of calcium in CaSiO_3 is +II.

(ii) Calcium oxide can be made from calcium carbonate in a single-step reaction.

Identify the type of reaction that occurs.

[1]

13. O/N 23/P21/Q2(a)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

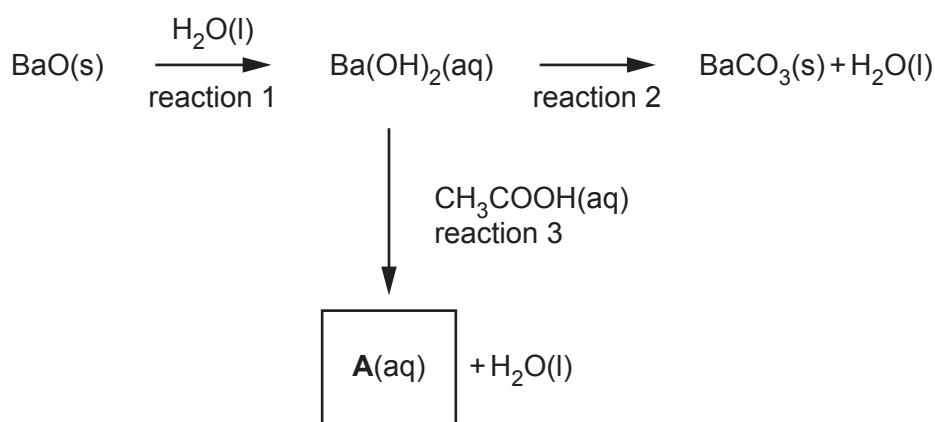


Fig. 2.1

(a) (i) State the variation in solubilities of group 2 hydroxides.

[1]

(ii) State what is observed in reaction 1.

[1]

(iii) Suggest a reactant for reaction 2.

[1]

(iv) Identify **A**.

[1]

(v) $\text{Ba}(\text{OH})_2$ is made by the reaction of Ba with water.

Write an equation for this reaction.

[1]

14. O/N 23/P21/Q2(b)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

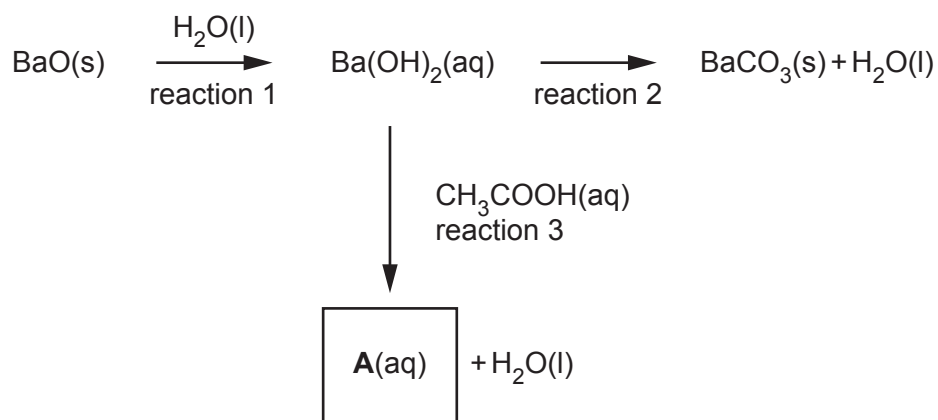


Fig. 2.1

(b) The mineral barytocalcite contains both BaCO_3 and CaCO_3 . Both compounds decompose on heating.

(i) State which compound decomposes first when barytocalcite is heated.

Explain your answer.

[1]

(ii) Construct an equation for the complete thermal decomposition of barytocalcite.

The formula of barytocalcite is $\text{BaCa}(\text{CO}_3)_2$.

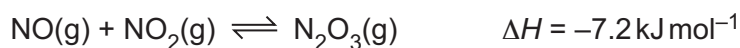
$\text{BaCa}(\text{CO}_3)_2$

[1]

15. O/N 23/P22/Q2(f)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

- (f)** A student titrates nitric acid with a base to form a solution containing aqueous magnesium nitrate.

- (i)** Identify a base that the student could use.

[1]

- (ii)** The student evaporates the water to obtain magnesium nitrate solid. When this solid is heated it decomposes.

Write an equation for the decomposition of magnesium nitrate.

[1]

- (iii)** State how the thermal stability of Group 2 nitrates changes down the group.

[1]

16. O/N 23/P23/Q2(a)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

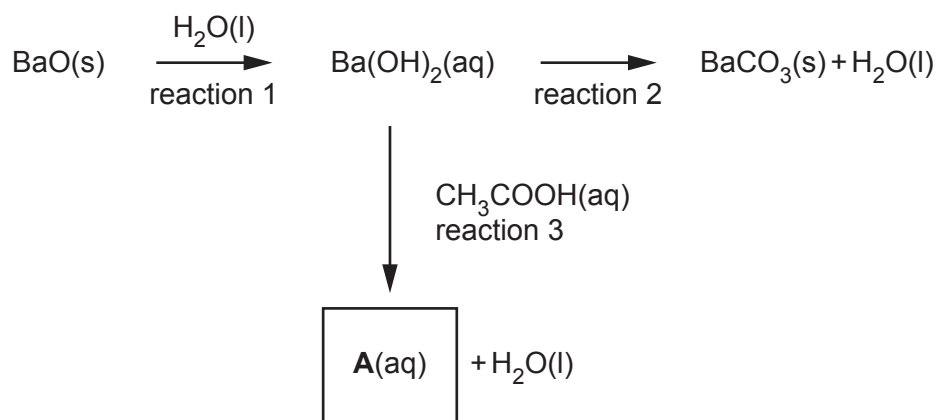


Fig. 2.1

(a) (i) State the variation in solubilities of group 2 hydroxides.

[1]

(ii) State what is observed in reaction 1.

[1]

(iii) Suggest a reactant for reaction 2.

[1]

(iv) Identify **A**.

[1]

(v) $\text{Ba}(\text{OH})_2$ is made by the reaction of Ba with water.

Write an equation for this reaction.

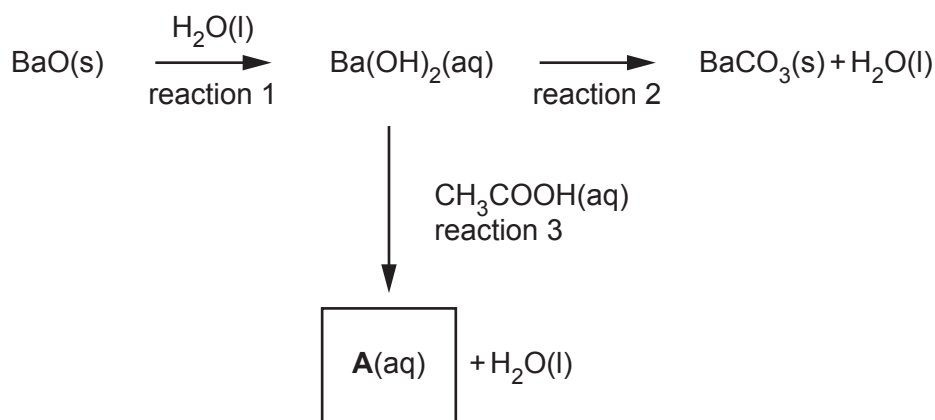
[1]

17. O/N 23/P23/Q2(b)

Context

- 2** Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

**Fig. 2.1**

- (b)** The mineral barytocalcite contains both BaCO_3 and CaCO_3 . Both compounds decompose on heating.

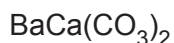
- (i)** State which compound decomposes first when barytocalcite is heated.

Explain your answer.

[1]

- (ii)** Construct an equation for the complete thermal decomposition of barytocalcite.

The formula of barytocalcite is $\text{BaCa}(\text{CO}_3)_2$.

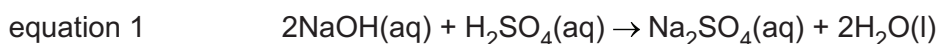


[1]

18. M/J 23/P21/Q2(c)(i)

Context

- 2** A neutralisation reaction occurs when NaOH(aq) is added to $\text{H}_2\text{SO}_4\text{(aq)}$.



- (c) (i)** Complete the equation for the reaction that occurs when a solution of $\text{Ba}(\text{OH})_2$ is added to aqueous sulfuric acid. Include state symbols.



[4]

19. M/J 23/P23/Q2(b)

- (b) Aluminium nitrate is a white soluble salt. On heating aluminium nitrate, thermal decomposition occurs and a brown gas is seen.

State the formula of the salt of another element in Period 3 which also decomposes on heating to produce a brown gas.

[1]

20. MAR 23/P22/Q1(d)(iii)

Context

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

Context: part (d) (introduction)

- (d) CO₂ and SO₂ are acidic gases.

(iii) Construct an equation for the reaction of CO₂ with Mg(OH)₂.

[1]

21. MAR 23/P22/Q2(b)

Context

- 2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(b) Write an equation for the reaction of magnesium with cold water.

[1]

22. MAR 23/P22/Q2(c)

Context

- 2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(c) Identify a single reagent that can be used to distinguish separate samples of dilute Mg(NO₃)₂(aq) and dilute Ba(NO₃)₂(aq).

Explain your answer.

reagent

explanation

[2]

23. O/N 22/P21/Q2(b)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(b) Fig. 2.1 shows some reactions of magnesium and its compounds.

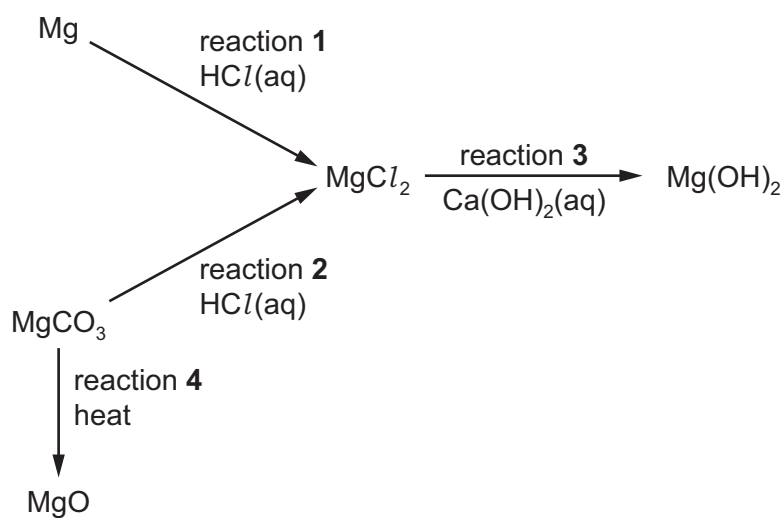


Fig. 2.1

(i) Identify the other products of reactions 1 and 2.

reaction 1

reaction 2

[2]

(ii) Reaction 3 is used to form a precipitate of Mg(OH)_2 from $\text{MgCl}_2(\text{aq})$.

State why $\text{Ca(OH)}_2(\text{aq})$ would **not** form a precipitate of Ba(OH)_2 from $\text{BaCl}_2(\text{aq})$.

[1]

(iii) State the type of reaction that occurs in reaction 4.

[1]

24. O/N 22/P23/Q2(b)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(b) Fig. 2.1 shows some reactions of magnesium and its compounds.

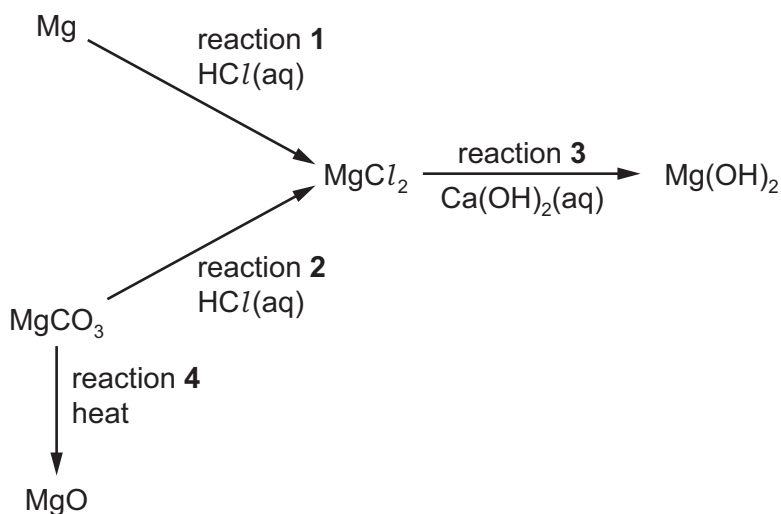


Fig. 2.1

(i) Identify the other products of reactions 1 and 2.

reaction 1

reaction 2

[2]

(ii) Reaction 3 is used to form a precipitate of $Mg(OH)_2$ from $MgCl_2(aq)$.

State why $Ca(OH)_2(aq)$ would **not** form a precipitate of $Ba(OH)_2$ from $BaCl_2(aq)$.

[1]

(iii) State the type of reaction that occurs in reaction 4.

[1]

25. M/J 22/P21/Q1(b)

Context

1 Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

(b) (i) Write the equation for the thermal decomposition of calcium nitrate.

[1]

(ii) Suggest which of the Group 2 nitrates, calcium, magnesium or radium, requires the highest temperature to decompose. Explain your answer.

[1]

26. M/J 22/P21/Q1(c)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

(c) Predict what you would observe when aqueous radium chloride is added to aqueous sodium sulfate.

Do **not** refer to temperature changes in your answer.

[1]

27. M/J 22/P22/Q2(c)

Context

- 2** Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

Radium chloride, RaCl_2 , has a melting point of 900°C and is soluble in water.

(c) Solid Ra and Ca show similar reactions with H_2O , but the reactions occur at different rates.

Separate samples, each containing a single piece of solid Ra or Ca, are added to equal volumes of cold water.

Each sample contains equal numbers of moles of solid and the H_2O is in excess.

(i) Construct an equation for the reaction of Ra with H_2O .

[1]

(ii) Identify which element, Ra or Ca, reacts with H_2O at a faster rate. Suggest how the observations of each reaction would differ.

[1]

(iii) Suggest why these reactions occur at different rates.

[2]

(iv) One of the solutions is cloudy when the reaction has finished.

At the end of each reaction, universal indicator is added to each reaction mixture.

Suggest pH values of the solutions made in both reactions. Explain your answer.

[2]

Answers Section

1. M/J 26/P24/Q3(c)

3(c)(i)	brown gas	1	gas = fumes = vapour A red-brown gas, orange-brown gas A shades of brown R red gas (alone) I white solid (formed) I any other observations (e.g. melting, effervescence, solid disappears)
3(c)(ii)	increases	1	A decrease <u>up</u> the group. A more soluble

2. MAR 26/P22/Q1(b)

1(b)(i)	M1 $\text{Sr(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Sr(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$ M2 state symbols for <u>all</u> species	2
1(b)(ii)	M1 with HCl no visible reaction / no change (observed) M2 with H_2SO_4 white precipitate/solid forms	2

3. MAR 26/P22/Q1(c)

1(c)(i)	(thermal stability) increases (down the group)	1
1(c)(ii)	$2\text{Ca(NO}_3)_2 \rightarrow 2\text{CaO} + 4\text{NO}_2 + \text{O}_2$	1

4. O/N 25/P21/Q2(d)

2(d)(i)	(thermal stability) increases (down the group)	1
2(d)(ii)	NO_2 AND O_2	1

5. O/N 25/P22/Q2(f)

2(f)(i)	$\text{MgCl}_2\text{(aq)} + \text{Na}_2\text{CO}_3\text{(aq)} \rightarrow \text{MgCO}_3\text{(s)} + 2\text{NaCl(aq)}$ M1 formulae in balanced equation M2 state symbols	2
2(f)(ii)	MgO / magnesium oxide AND CO_2 / carbon dioxide	1
2(f)(iii)	(thermal stability) increases (down the group)	1

6. O/N 25/P23/Q2(d)

2(d)(i)	(thermal stability) increases (down the group)	1
2(d)(ii)	NO_2 AND O_2	1

7. MAR 25/P22/Q2(d)

2(d)(i)	<i>barium oxide</i> (white) solid disappears / dissolves <i>barium carbonate</i> effervescence AND (white) solid disappears / dissolves	2
2(d)(ii)	$\text{Sr} + 2\text{H}_2\text{O} \rightarrow \text{Sr(OH)}_2 + \text{H}_2$	1
2(d)(iii)	decreases (down the group)	1

8. O/N 24/P21/Q3(b)(i)-(ii)

3(b)(i)	$\text{MgCO}_3 + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$	1
3(b)(ii)	thermal stability increases down the group	1

9. O/N 24/P22/Q3(a)(iv)-(v)

3(a)(iv)	$2\text{HNO}_3 + \text{CaO} \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_2\text{O}$	1
3(a)(v)	brown fumes given off	1

10. O/N 24/P23/Q3(b)(i)-(ii)

3(b)(i)	$\text{MgCO}_3 + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$	1
3(b)(ii)	thermal stability increases down the group	1

11. M/J 24/P22/Q2(c)

2(c)	$\text{Ca}(\text{NO}_3)_2 \rightarrow \text{CaO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$ M1 correct formula for calcium nitrate: $\text{Ca}(\text{NO}_3)_2$ M2 correct balanced equation	2
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12. M/J 24/P23/Q1(d)(ii)

1(d)(ii)	thermal decomposition	1
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13. O/N 23/P21/Q2(a)

2(a)(i)	increases down the group	1
2(a)(ii)	white solid disappears	1
2(a)(iii)	CO_2 / carbon dioxide	1
2(a)(iv)	$\text{Ba}(\text{CH}_3\text{COO})_2$ / barium ethanoate	1
2(a)(v)	$\text{Ba} + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$	1

14. O/N 23/P21/Q2(b)

2(b)(i)	CaCO_3 AND thermal stability of carbonates increases down the group	1
2(b)(ii)	$\text{BaCa}(\text{CO}_3)_2 \rightarrow \text{BaO} + \text{CaO} + 2\text{CO}_2$	1

15. O/N 23/P22/Q2(f)

2(f)(i)	magnesium hydroxide / $\text{Mg}(\text{OH})_2$	1
2(f)(ii)	$\text{Mg}(\text{NO}_3)_2 \rightarrow \text{MgO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$	1
2(f)(iii)	increases	1

16. O/N 23/P23/Q2(a)

2(a)(i)	increases down the group	1
2(a)(ii)	white solid disappears	1
2(a)(iii)	CO_2 / carbon dioxide	1
2(a)(iv)	$\text{Ba}(\text{CH}_3\text{COO})_2$ / barium ethanoate	1

2(a)(v)	$\text{Ba} + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$	1
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17. O/N 23/P23/Q2(b)

2(b)(i)	CaCO_3 AND thermal stability of carbonates increases down the group	1
2(b)(ii)	$\text{BaCa}(\text{CO}_3)_2 \rightarrow \text{BaO} + \text{CaO} + 2\text{CO}_2$	1

18. M/J 23/P21/Q2(c)(i)

2(c)(i)	..(1)..$\text{H}_2\text{SO}_4(\text{aq}) + \text{..(1)..Ba}(\text{OH})_2(\text{aq}) \rightarrow \text{...(1)..BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$..... M1 correct state symbols for both products M2 correctly balanced equation	2
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19. M/J 23/P23/Q2(b)

2(b)	$\text{Mg}(\text{NO}_3)_2$	1
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20. MAR 23/P22/Q1(d)(iii)

1(d)(iii)	$\text{CO}_2 + \text{Mg}(\text{OH})_2 \rightarrow \text{MgCO}_3 + \text{H}_2\text{O}$	1
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21. MAR 23/P22/Q2(b)

2(b)	$\text{Mg} + 2\text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + \text{H}_2$	1
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22. MAR 23/P22/Q2(c)

2(c)	reagent = any named/formula of soluble sulfate OR H_2SO_4 OR any named/formula of soluble hydroxide	1
	BaSO₄ insoluble (& MgSO₄ soluble) OR (Ba(OH)₂ soluble &) Mg(OH)₂ insoluble	1

23. O/N 22/P21/Q2(b)

2(b)(i)	reaction 1 = hydrogen / H_2	1
	reaction 2 = carbon dioxide / CO_2 AND water / H_2O	1
2(b)(ii)	$\text{Ba}(\text{OH})_2$ is soluble (in aqueous solution) / solubility of Group 2 hydroxides increases down group	1
2(b)(iii)	thermal decomposition	1

24. O/N 22/P23/Q2(b)

2(b)(i)	reaction 1 = hydrogen / H_2	1
	reaction 2 = carbon dioxide / CO_2 AND water / H_2O	1
2(b)(ii)	$\text{Ba}(\text{OH})_2$ is soluble (in aqueous solution) / solubility of Group 2 hydroxides increases down group	1
2(b)(iii)	thermal decomposition	1

25. M/J 22/P21/Q1(b)

1(b)(i)	$\text{Ca}(\text{NO}_3)_2 \rightarrow \text{CaO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$	1
1(b)(ii)	radium (nitrate) as thermal stability increases down group / has the greatest thermal stability	1

26. M/J 22/P21/Q1(c)

1(c)	white precipitate / solid (of radium sulfate)	1
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27. M/J 22/P22/Q2(c)

2(c)(i)	$\text{Ra} + 2\text{H}_2\text{O} \rightarrow \text{Ra}(\text{OH})_2 + \text{H}_2$	1
2(c)(ii)	Ra – more bubbles per unit time OR With Ra solid disappears more quickly OR Ra is the first to stop fizzing ora	1
2(c)(iii)	option 1 <i>suggest why these reactions occur at different rates (collision theory)</i> M1 difference in activation energy / ionisation energy M2 affects the frequency of effective collisions (between particles / molecules) OR affects the proportion of particles with energy greater than activation energy option 2 <i>suggest why the elements have different reactivity in terms of atomic structure (which results in a different rate)</i> M1 ionisation energies are different M2 a decrease in nuclear attraction due to EITHER increase in shielding OR increase in distance of outer electron from nucleus OR increase in number of shells of electrons ora	2
2(c)(iv)	M1 - pH value (or values) for each solution must be greater > 7 AND - pH value (or range values) identified for solution made from Ra must be greater than pH values (or range of values) stated for Ca M2 any one from: - solubility of group 2 hydroxides increases down the group $\text{Ra}(\text{OH})_2$ is more soluble - greater concentration of $\text{OH}^-(\text{aq})$ in the solution involving Ra - more $\text{OH}^-(\text{aq})$ in the solution involving Ra	2

Unit 11

GROUP 17

47 items · 114 marks

Syllabus sections in this unit

- 11.1 Physical properties of the Group 17 elements
- 11.2 The chemical properties of the halogen elements and the hydrogen halides
- 11.3 Some reactions of the halide ions
- 11.4 The reactions of chlorine

Unit 11: Group 17**1. M/J 26/P21/Q3(a)**

Context

- 3** Separate samples of NaCl, NaBr and NaI each react with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smell of SO₂
3	NaI	- effervescence - steamy fumes initially that turn blue litmus paper red - purple vapour - pungent smell of SO₂ and then a rotten egg smell of H₂S

- (a) (i) Write an equation for reaction 1.

[1]

- (ii) Identify the reactant that behaves as a Brønsted–Lowry base in reaction 1.

[1]

2. M/J 26/P21/Q3(b)-(c)

Context

- 3** Separate samples of NaCl, NaBr and NaI each react with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	• effervescence • steamy fumes that turn blue litmus paper red
2	NaBr	• effervescence • steamy fumes initially that turn blue litmus paper red • brown vapour • pungent smell of SO₂
3	NaI	• effervescence • steamy fumes initially that turn blue litmus paper red • purple vapour • pungent smell of SO₂ and then a rotten egg smell of H₂S

- (b)** State the formulae of the species responsible for the brown vapour in reaction 2 and for the purple vapour in reaction 3.

brown vapour

purple vapour

[1]

- (c)** Explain why reaction 3 produces purple vapour.

[1]

3. M/J 26/P21/Q3(e)

Context

- 3** Separate samples of NaCl, NaBr and NaI each react with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smell of SO₂
3	NaI	- effervescence - steamy fumes initially that turn blue litmus paper red - purple vapour - pungent smell of SO₂ and then a rotten egg smell of H₂S

Context: part (b)

- (b)** State the formulae of the species responsible for the brown vapour in reaction 2 and for the purple vapour in reaction 3.

brown vapour

purple vapour

[1]

- (e)** Explain why reaction 3 produces H₂S but reaction 2 does **not** produce H₂S.

[1]

4. M/J 26/P21/Q3(f)

Context

- 3** Separate samples of NaCl, NaBr and NaI each react with concentrated H₂SO₄. Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steamy fumes that turn blue litmus paper red
2	NaBr	- effervescence - steamy fumes initially that turn blue litmus paper red - brown vapour - pungent smell of SO₂
3	NaI	- effervescence - steamy fumes initially that turn blue litmus paper red - purple vapour - pungent smell of SO₂ and then a rotten egg smell of H₂S

- (f) (i)** Describe what is observed when Ag⁺(aq), followed by NH₃(aq), are added to a sample of NaI(aq).

observation after addition of Ag⁺(aq)

observation after addition of NH₃(aq)

[2]

- (ii)** Write an ionic equation for the reaction of Ag⁺(aq) with NaI(aq).

[1]

5. M/J 26/P22/Q3(b)-(d)

Context

- 3** Separate samples of NaCl and NaBr are reacted with concentrated H_2SO_4 . Table 3.1 shows the observations for each reaction.

Table 3.1

reaction	compound	observations
1	NaCl	- effervescence - steaminess that turn blue litmus paper red
2	NaBr	- effervescence - steaminess initially that turn blue litmus paper red - brown vapour - pungent smelling gas

Context: part (a)(i)

- (a) (i)** Write an equation for reaction 1.

[1]

- (b)** Identify the formula of each species responsible for the observations in reaction 2.

steaminess

brown vapour

pungent smelling gas

[3]

- (c)** Explain why reaction 2 produces brown vapour.

[1]

- (d)** Explain why a gas with a pungent smell is **not** produced in reaction 1 but is produced in reaction 2.

[1]

6. M/J 26/P23/Q1(b)

Context

- 1** The chlorides of Period 3 elements show different physical and chemical properties.

- (b)** The reaction of chlorine with cold water is used in water purification.

- (i)** Write an equation for the reaction of chlorine with cold water.

[1]

- (ii)** Explain the use of chlorine in water purification.

[1]

7. M/J 26/P23/Q1(c)

Context

1 The chlorides of Period 3 elements show different physical and chemical properties.

(c) The relative strength of Cl_2 , Br_2 and I_2 as oxidising agents can be compared by adding NaBr(aq) to separate samples of Cl_2 and I_2 .

(i) Describe the relative reactivity of Cl_2 , Br_2 and I_2 as oxidising agents.

[1]

(ii) Complete Table 1.2 to describe the reactions of NaBr(aq) with separate samples of $\text{Cl}_2(\text{aq})$ and $\text{I}_2(\text{aq})$.

Table 1.2

aqueous halogen	colour of aqueous halogen	appearance after NaBr(aq) is added	halide ions present in the mixture after the reaction
$\text{Cl}_2(\text{aq})$	pale green		
$\text{I}_2(\text{aq})$	red–brown		

[3]

8. M/J 26/P23/Q1(d)

Context

1 The chlorides of Period 3 elements show different physical and chemical properties.

(d) Describe what is observed when $\text{Ag}^+(\text{aq})$, followed by $\text{NH}_3(\text{aq})$, are added to a sample of NaI(aq) .

observation after addition of $\text{Ag}^+(\text{aq})$

observation after addition of $\text{NH}_3(\text{aq})$

[2]

9. MAR 26/P22/Q1(d)

Context

1 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

(d) Calcium chlorate(I), Ca(ClO)_2 , is used for water purification in the same way as sodium chlorate(I), NaClO .

Ca(ClO)_2 can be prepared by the reaction of aqueous Ca(OH)_2 with Cl_2 .

(i) Write an ionic equation for the reaction of cold dilute OH^- ions with Cl_2 .

[1]

(ii) State why NaClO and Ca(ClO)_2 are used in water purification.

[1]

10. O/N 25/P24/Q1(a)(i)

Context

- 1** The Group 17 elements are oxidising agents.

(a) (i) Explain how the Group 17 elements act as oxidising agents.

[1]

11. O/N 25/P24/Q1(a)(iii)

Context

- 1** The Group 17 elements are oxidising agents.

(iii) A student heats equal amounts of $I_2(g)$ and $H_2(g)$ in a sealed flask. The student leaves the contents to cool.

State what you would observe during the reaction.

[1]

12. O/N 25/P24/Q1(b)(i)-(ii)

Context

- 1** The Group 17 elements are oxidising agents.

Context: part (b) (introduction)

(b) Cl_2 and Br_2 can each react with NH_3 to give N_2 and a hydrogen halide, HX .



The relative bond strengths of $X-X$ and $H-X$ determine the difference in enthalpy change of the two reactions.

(i) Describe and explain the difference in the $X-X$ bond strengths of Cl_2 and Br_2 .

[2]

(ii) Describe the relative thermal stabilities of HCl and HBr .

[1]

13. O/N 25/P24/Q1(c)

Context

1 The Group 17 elements are oxidising agents.

(c) Table 1.2 shows some information about reactions of NaCl, NaBr and NaI.

Table 1.2

	NaCl	NaBr	NaI
observation with $\text{Ag}^+(\text{aq})$	white precipitate		
type of reaction with concentrated H_2SO_4	acid–base	acid–base, then redox	acid–base, then redox
observations with concentrated H_2SO_4			• black solid • yellow solid • effervescence

(i) Complete Table 1.2. [4]

(ii) Suggest an identity for the species that produces each observation in the reaction of NaI with concentrated H_2SO_4 .

black solid

yellow solid

effervescence

[2]

14. M/J 25/P23/Q6(a)(ii)-(iii)

Context: part (a) (introduction)

6 (a) HOCl reacts with $\text{CH}_2=\text{CH}_2$ to produce $\text{HOCH}_2\text{CH}_2\text{Cl}$ in an electrophilic addition reaction.(ii) Describe how Cl_2 is used to produce HOCl .

[1]

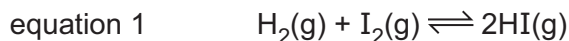
(iii) State **one** use for HOCl .

[1]

15. M/J 25/P24/Q3(d)(ii)-(iii)

Context

- 3** $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ react to form $\text{HI}(\text{g})$ in a reversible reaction, as shown in equation 1.



In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

Context: part (c)

- (c)** In experiment 3, 0.0772 mol of H_2 and 0.0986 mol of I_2 are present in an equilibrium mixture at 298 K.



Use the equilibrium constant, K_c , to calculate the amount, in mol, of HI present in the equilibrium mixture in experiment 3.

amount of HI in equilibrium mixture = _____ mol [1]

Context: part (d) (introduction)

- (d)** The value of K_c for the dissociation of HCl at 298 K is 5.50×10^{-34} .



- (ii)** Describe and explain the relative thermal stabilities of the hydrogen halides HCl and HI . [2]

- (iii)** Use the data given in **(c)** and **(d)** to suggest a value for the equilibrium constant for the **dissociation** of $\text{HBr}(\text{g})$ at 298 K.

[1]

16. MAR 25/P22/Q3(a)

Context

- 3** The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol^{-1}	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

- (a) (i)** Complete Table 3.1. [1]

- (ii)** The bond energy values in Table 3.1 refer to the $\text{X}-\text{X}$ bond where X is the halogen.

Explain the trend in the bond strength of the $\text{X}-\text{X}$ bond in the halogens.

[2]

17. MAR 25/P22/Q3(b)

Context

- 3** The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol ⁻¹	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

- (b)** Explain, with the use of an equation, how chlorine, Cl₂, is used in water purification. State the role of the active species produced.

[2]

18. MAR 25/P22/Q3(c)

Context

- 3** The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol ⁻¹	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

- (c)** The sodium halides in Table 3.1 also show trends in chemical properties.

- (i)** Identify the sodium halide that reacts with concentrated H₂SO₄ to form H₂S.

[1]

- (ii)** Identify the sodium halide that **only** undergoes a Brønsted–Lowry acid–base reaction with concentrated H₂SO₄.

[1]

- (iii)** A student adds a few drops of AgNO₃(aq) to a solution of NaBr(aq). State what is observed.

[1]

19. O/N 24/P21/Q2(a)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

- (a) (i) Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature. [1]
- (ii) Describe the relative reactivity of the elements chlorine to iodine as oxidising agents. [1]
- (iii) State what is observed when chlorine reacts with hydrogen. [1]
- (iv) Explain why the thermal stability of the hydrogen halides decreases down the group. [1]

20. O/N 24/P21/Q2(c)(i)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

(c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO .

NaClO is made from chlorine in a disproportionation reaction.

- (i) Identify a reagent and conditions that can be used to convert chlorine to NaClO . [1]

21. O/N 24/P23/Q2(a)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

- (a) (i) Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature. [1]
- (ii) Describe the relative reactivity of the elements chlorine to iodine as oxidising agents. [1]
- (iii) State what is observed when chlorine reacts with hydrogen. [1]
- (iv) Explain why the thermal stability of the hydrogen halides decreases down the group. [1]

22. O/N 24/P23/Q2(c)(i)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

(c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO .

NaClO is made from chlorine in a disproportionation reaction.

- (i) Identify a reagent and conditions that can be used to convert chlorine to NaClO . [1]

23. M/J 24/P21/Q1(a)

- 1 (a) The elements of Group 17 are called halogens.

Complete Table 1.1.

Table 1.1

halogen	colour at 293 K
chlorine	
bromine	
iodine	

[1]

24. M/J 24/P21/Q1(b)

- (b) State the trend in volatility of the halogens chlorine, bromine and iodine. Explain your answer.

[3]

25. M/J 24/P21/Q1(c)

- (c) Iodine is made by reacting bromine with sodium iodide.

- (i) Construct an ionic equation for the reaction of bromine with sodium iodide.

[1]

- (ii) State the role of bromine in the reaction. Explain your answer.

[1]

26. M/J 24/P21/Q1(d)

- (d) Concentrated sulfuric acid is added to separate samples containing equal amounts of NaCl, NaBr and NaI. All three samples initially react to produce the hydrogen halide.

- (i) Write an equation to describe the acid–base reaction that occurs when concentrated sulfuric acid reacts with NaBr.

[1]

- (ii) Deduce which sodium halide, NaCl, NaBr or NaI, produces the largest percentage yield of hydrogen halide when concentrated sulfuric acid is added. Explain your answer by considering the relative reactivity of the halide ions as reducing agents.

identity of sodium halide

explanation

[3]

27. MAR 24/P22/Q2(b)

Context

- 2** Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

- (b)** When Cl_2 reacts with **cold** NaOH(aq) , Cl_2 is both oxidised and reduced. The products are NaCl , water and **G**.

- (i)** State the type of redox reaction in which the same species is both oxidised and reduced.

[1]

- (ii)** Identify **G**.

[1]

- (iii)** Write an equation for the reaction between Cl_2 and **hot** NaOH(aq) .

[1]

- (iv)** Describe fully what is observed when $\text{AgNO}_3(\text{aq})$ is added to the aqueous solution of the chloride of sodium, followed by dilute $\text{NH}_3(\text{aq})$.

[2]

28. O/N 23/P21/Q1(c)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (c) A student does three tests on separate samples of NaCl(aq).

Complete Table 1.2 with the observations the student makes in each test.

Table 1.2

test	test	observations
1	addition of a few drops of Br ₂ (aq)	
2	addition of a few drops of concentrated H ₂ SO ₄	
3	addition of a few drops of dilute AgNO ₃ (aq)	

[3]

29. O/N 23/P23/Q1(c)

Context

- 1** The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to Cl.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	Cl
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol ⁻¹	1060		1260
formula of most common anion	P ³⁻	S ²⁻	Cl ⁻

- (c)** A student does three tests on separate samples of NaCl(aq).

Complete Table 1.2 with the observations the student makes in each test.

Table 1.2

test	test	observations
1	addition of a few drops of Br ₂ (aq)	
2	addition of a few drops of concentrated H ₂ SO ₄	
3	addition of a few drops of dilute AgNO ₃ (aq)	

[3]

30. M/J 23/P21/Q3(c)

Context

- 3** Chlorine is a very reactive element.

- (c) (i)** State the reagent and conditions required for the formation of sodium chlorate(V) from Cl₂(g).

[1]

- (ii)** Explain why the reaction in **(c)(i)** is described as a disproportionation reaction. Your answer should refer to relevant species and their oxidation numbers.

[1]

31. M/J 23/P22/Q2(d)

Context

2 Chlorine is a reactive element. It forms many compounds.

(d) (i) Write an equation for the reaction of chlorine with water.

[1]

(ii) Write an equation for the reaction of chlorine with hot NaOH(aq).

[1]

32. M/J 23/P22/Q2(e)(i)

Context

2 Chlorine is a reactive element. It forms many compounds.

Context: part (e) (introduction)

(e) Bleach is used as a cleaning product to kill bacteria. It is made by adding compounds like sodium chlorate(I), NaClO, to water.

(i) Identify the formula of the ion present in bleach that kills bacteria.

[1]

33. M/J 23/P23/Q4(a)

4 (a) Hydrogen chloride gas is made in the laboratory by adding concentrated sulfuric acid to potassium chloride.

(i) Construct an equation for this reaction.

[1]

(ii) Explain why hydrogen iodide is **not** prepared by adding concentrated sulfuric acid to sodium iodide.

[2]

34. MAR 23/P22/Q2(d)(i)

Context

2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(d) (i) Describe what is observed when SrI₂(aq) reacts with concentrated sulfuric acid.

[2]

35. O/N 22/P21/Q2(c)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(c) 1 cm³ of MgCl₂(aq) is placed in a test-tube. A few drops of AgNO₃(aq) are added, followed by 1 cm³ of dilute NH₃(aq).

State in full what is observed in this experiment.

[2]

36. O/N 22/P21/Q2(d)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(d) When 1 cm^3 of $\text{MgCl}_2(\text{aq})$ is added to 1 cm^3 of $\text{Br}_2(\text{aq})$ in a test-tube, the solution remains orange.

Explain this observation.

[1]

37. O/N 22/P21/Q3(b)

Context

3 Some of the common chlorides of Period 3 elements are shown in the list.



(b) NaCl is one product of the reaction of chlorine gas and cold aqueous sodium hydroxide.

Identify the other products.

[1]

38. O/N 22/P22/Q2(e)(i)

Context

2 The chlorides of some of the Period 3 elements are shown in Table 2.1.

Table 2.1

Period 3 chloride	NaCl	AlCl_3	SiCl_4	PCl_5	PCl_3	SCl_2
bonding					C	C
structure					S	S
oxidation state of Period 3 element						

(e) (i) An excess of $\text{Cl}^-(\text{aq})$ is added to 1 cm^3 of $\text{Br}_2(\text{aq})$.

Describe what is observed. Explain your answer.

[2]

39. O/N 22/P23/Q2(c)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(c) 1 cm^3 of $\text{MgCl}_2(\text{aq})$ is placed in a test-tube. A few drops of $\text{AgNO}_3(\text{aq})$ are added, followed by 1 cm^3 of dilute $\text{NH}_3(\text{aq})$.

State in full what is observed in this experiment.

[2]

40. O/N 22/P23/Q2(d)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(d) When 1 cm^3 of $\text{MgCl}_2(\text{aq})$ is added to 1 cm^3 of $\text{Br}_2(\text{aq})$ in a test-tube, the solution remains orange.

Explain this observation.

[1]

41. O/N 22/P23/Q3(b)

Context

3 Some of the common chlorides of Period 3 elements are shown in the list.



(b) NaCl is one product of the reaction of chlorine gas and cold aqueous sodium hydroxide.

Identify the other products.

[1]

42. M/J 22/P22/Q2(d)

Context

2 Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

Radium chloride, RaCl_2 , has a melting point of 900°C and is soluble in water.

(d) A sample of aqueous calcium halide, $\text{CaX}_2(\text{aq})$, contains either chloride, bromide or iodide ions.

Complete Table 2.1 to describe a two-step process that could be used to identify the halide ion present.

Table 2.1

step	method	observation with CaCl_2	observation with CaBr_2	observation with CaI_2
step 1				
step 2				

[3]

43. M/J 22/P22/Q3(d)

(d) HCl(g) is prepared by adding NaCl(s) to concentrated H_2SO_4 .

HI(g) is **not** prepared by adding NaI(s) to concentrated H_2SO_4 because the HI(g) produced also reacts with concentrated H_2SO_4 .

(i) Identify the type of reaction that occurs when NaI(s) reacts with concentrated H_2SO_4 to form HI(g) .

[1]

(ii) Write an equation for the reaction of HI(g) and concentrated H_2SO_4 .

[1]

(iii) Explain why HI(g) reacts with concentrated H_2SO_4 whereas HCl does not.

[1]

44. M/J 22/P23/Q2(b)

(b) Tennessine, Ts, is an unstable man-made element. It is found below astatine, At, in Group 17. The chemical properties of Ts and its compounds have only been predicted.

(i) Suggest an equation for the reaction of NaTs and Br_2 . Assume that Ts follows the same trends as the other elements in Group 17. Explain your answer.

equation

explanation

[2]

(ii) Some scientists predict that Ts has properties typical of metals like copper.

Complete Table 2.2 with:

- the predicted melting point of tennessine
- the lattice structure of solid chlorine, bromine and tennessine.

Assume that Ts has properties typical of metals like copper.

Table 2.2

element	chlorine	bromine	tennessine
melting point/ $^{\circ}\text{C}$	-101	-7.2	
lattice structure of crystalline solid			

[2]

45. MAR 22/P22/Q3(a)

Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

(a) The hydrogen halides can be formed by reacting the halogens with hydrogen.

Describe and explain the relative reactivity of the halogens down the group when they react with hydrogen to form HCl , HBr and HI .

[2]

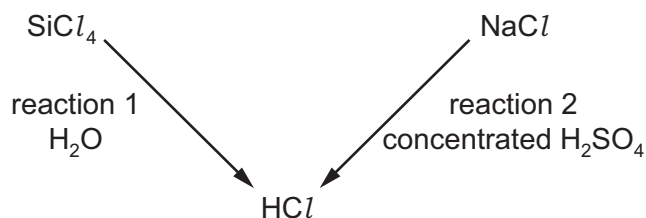
46. MAR 22/P22/Q3(b)(ii)

Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

Context: part (b) (introduction)

(b) HCl is a product of several different reactions. Some of these are shown in Fig. 3.1.

**Fig. 3.1**

(ii) In reaction 2, NaCl reacts with concentrated H_2SO_4 to form HCl and NaHSO_4 . When NaBr reacts with concentrated H_2SO_4 , the products include Br_2 and SO_2 .

Identify the type(s) of reaction that occur in each case by completing Table 3.1. Explain the difference in these reactions.

Table 3.1

reactants	type(s) of reaction
NaCl and concentrated H_2SO_4	
NaBr and concentrated H_2SO_4	

explanation

.....

[3]

47. MAR 22/P22/Q3(c)

Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

(c) When heated with a Bunsen burner, HCl does not decompose, whereas HI forms H_2 and I_2 .

Explain the difference in the effect of heating on HCl and HI .

[1]

Answers Section

1. M/J 26/P21/Q3(a)

3(a)(i)	$\text{H}_2\text{SO}_4 + \text{NaCl} \rightarrow \text{NaHSO}_4 + \text{HCl}$	1	A $\text{H}_2\text{SO}_4 + 2\text{NaCl} \rightarrow \text{Na}_2\text{SO}_4 + 2\text{HCl}$ A $\text{Cl}^- + \text{H}_2\text{SO}_4 \rightarrow \text{HCl} + \text{HSO}_4^-$ A $2\text{Cl}^- + \text{H}_2\text{SO}_4 \rightarrow 2\text{HCl} + \text{SO}_4^{2-}$
3(a)(ii)	$\text{Cl}^- / \text{NaCl}$	1	A sodium chloride / chloride (ion) A chlorine anion I chlorine ion I chlorine R Cl_2

2. M/J 26/P21/Q3(b)-(c)

3(b)	Brown vapour Br_2 Purple vapour I_2	1	I bromine I Br I iodine I I R iodide / bromide
3(c)	(purple vapour is produced when) HI / iodide (anions) are oxidised (by H_2SO_4) OR (sulfuric) acid / H_2SO_4 oxidises I^- (to iodine)	1	A iodide / I^- loses electrons A H_2SO_4 / S (in H_2SO_4) is reduced by HI / I^- / I in NaI / I in HI A I^- oxidised by S ((VI))(in H_2SO_4) For (VI) A 6+ / +6 / 6 I displacement I oxidation products I oxidation number unless they are specifying S(VI) in H_2SO_4 I The reaction is exothermic so the iodine / I_2 produced is vaporised R iodine is oxidised by sulfuric acid

3. M/J 26/P21/Q3(e)

3(e)	iodide / I^- is a stronger reducing agent (than bromide / Br^-) OR HBr / bromide / Br^- is a weaker reducing agent (than iodide / I^-) OR HBr / Bromide / Br^- can't reduce S(IV) / SO_2	1	A the reducing ability of the halides increases down the group A for I^- HI / NaI / I in NaI / I in HI / iodide A bromide is not strong enough to fully reduce sulfuric acid strong enough = good enough I bond energy arguments I reactivity arguments if not linked to redox I electronegativity arguments I references to H_2SO_4 being an oxidising agent I explanations
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Answer for context part (b)

3(b)	Brown vapour Br_2 Purple vapour I_2	1	I bromine I Br I iodine I I R iodide / bromide
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4. M/J 26/P21/Q3(f)

3(f)(i)	(observation after addition of $\text{Ag}^+(\text{aq})$)(pale) yellow precipitate (observation after addition of $\text{NH}_3(\text{aq})$)precipitate remains / precipitate does not dissolve / precipitate does not disappear	2	precipitate = ppt(e) = crystals = powder = (insoluble) solid M1 ignore shades R any other colour M2 A no change/no reaction/no visible change if ppt seen in M1 M2 A precipitate is insoluble M2 yellow ppt if yellow ppt seen in M1 M2 A ecf ppt remains on incorrect colour ppt in M1
3(f)(ii)	$\text{Ag}^+ + \text{I}^- \rightarrow \text{AgI}$	1	I $\text{Ag}^+ + \text{NaI} \rightarrow \text{AgI} + \text{Na}^+$ I $\text{Ag}^+ + \text{Na}^+ + \text{I}^- \rightarrow \text{AgI} + \text{I}^- + \text{Na}^+$

5. M/J 26/P22/Q3(b)-(d)

3(b)	M1 (steamy fumes) HBr M2 (brown vapour) Br ₂ M3 (pungent gas) SO ₂	3	If formula given, I name. If no formula given, I name <u>once only</u> and then A correct name. I state symbols M1 I HCl M2 I Br M3 I Br ₂ and / or HBr
3(c)	(brown vapour is made when) HBr / bromide (anions) are oxidised (by H ₂ SO ₄)	1	A H ₂ SO ₄ / S (in H ₂ SO ₄) is reduced by HBr / Br ⁻ / Br in NaBr / Br in HBr A bromide / Br ⁻ loses electrons A Br ⁻ oxidised by S (VI) (in H ₂ SO ₄) A Br ⁻ / HBr / Br in NaBr reduces S(VI) (to S(IV)) I displacement I oxidation products I oxidation numbers unless they are specifying the S(VI) in H ₂ SO ₄ R bromine is oxidised by sulfuric acid For VI allow 6+ / 6 / +6
3(d)	H ₂ SO ₄ is not (a) strong enough (oxidising agent) to oxidise Cl ⁻ / NaCl / HCl OR Cl ⁻ is not (a) strong enough (reducing agent) to reduce H ₂ SO ₄ / S (in sulfuric acid)	1	A HBr / Br ⁻ is a stronger reducing agent than HCl / Cl ⁻ A reducing ability of halides increases down the group Cl ⁻ = HCl / NaCl / Cl in NaCl / Cl in HCl / chloride Strong enough = good enough Not strong enough oxidising agent = can't oxidise I bond energy arguments I reactivity arguments if not linked to redox I electronegativity arguments I reduction product e.g. H ₂ S A H ₂ SO ₄ is (a) strong enough (oxidising agent) to oxidise Br ⁻ A Br ⁻ is (a) strong enough (reducing agent) to reduce H ₂ SO ₄ / S (in sulfuric acid) / sulfate ions Br ⁻ = HBr / NaBr / Br in NaBr / Br in HBr / bromide R HCl is a stronger oxidising agent than H ₂ SO ₄ R Cl ⁻ is a stronger oxidising agent than H ₂ SO ₄ R NaCl is a stronger oxidising agent than H ₂ SO ₄

Answer for context part (a)(i)

3(a)(i)	H ₂ SO ₄ + NaCl → NaHSO ₄ + HCl OR H ₂ SO ₄ + 2NaCl → Na ₂ SO ₄ + 2HCl	1	A Cl ⁻ + H ₂ SO ₄ → HCl + HSO ₄ ⁻ A 2Cl ⁻ + H ₂ SO ₄ → 2HCl + SO ₄ ²⁻
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6. M/J 26/P23/Q1(b)

1(b)(i)	Cl ₂ + H ₂ O → HCl + HOCl	1	A Cl ₂ + H ₂ O → 2H ⁺ + Cl ⁻ + OCl ⁻
1(b)(ii)	(HOCl AND / OR ClO ⁻) kill bacteria	1	chloric(I) acid / hypochlorous acid = HOCl ClO ⁻ = chlorate(I) I chlorine / Cl ₂ A ecf product(s) of the reaction kill bacteria if 1b(i) correct A microbes / microorganisms / microbes BOD pathogens I viruses A 'destroys' for kills I sterilises / harms / removes

7. M/J 26/P23/Q1(c)

1(c)(i)	$\text{Cl}_2 > \text{Br}_2 > \text{I}_2$ OR oxidising power decreases down the group.				1	A relative reactivity for relative oxidising power I attempted explanations												
1(c)(ii)	<table><tr><th>aqueous halogen</th><th>colour of aqueous halogen</th><th>appearance after NaBr(aq) is added</th><th>identity of halide ions present in mixture after the reaction</th></tr><tr><td>$\text{Cl}_2(\text{aq})$</td><td>pale green</td><td>orange / brown A yellow alone R red-brown</td><td>Cl^- (and Br^-)</td></tr><tr><td>$\text{I}_2(\text{aq})$</td><td>red-brown</td><td>pale(r) (red-brown) A (paler) red-brown A no change</td><td>Br^-</td></tr></table>				aqueous halogen	colour of aqueous halogen	appearance after NaBr(aq) is added	identity of halide ions present in mixture after the reaction	$\text{Cl}_2(\text{aq})$	pale green	orange / brown A yellow alone R red-brown	Cl^- (and Br^-)	$\text{I}_2(\text{aq})$	red-brown	pale(r) (red-brown) A (paler) red-brown A no change	Br^-	3	•✓✓✓ M1 shade of brown on addition NaBr to Cl_2 must be different to that when it is added to I_2 e.g. orange in test-tube with Cl_2 AND brown in test-tube with I_2 then BOD 'brown'
aqueous halogen	colour of aqueous halogen	appearance after NaBr(aq) is added	identity of halide ions present in mixture after the reaction															
$\text{Cl}_2(\text{aq})$	pale green	orange / brown A yellow alone R red-brown	Cl^- (and Br^-)															
$\text{I}_2(\text{aq})$	red-brown	pale(r) (red-brown) A (paler) red-brown A no change	Br^-															

8. M/J 26/P23/Q1(d)

1(d)	M1 (after addition of $\text{Ag}^+(\text{aq})$).....(pale) yellow precipitate M2 (after addition of $\text{NH}_3(\text{aq})$).....precipitate remains / precipitate does not dissolve / precipitate does not disappear	2	precipitate = ppt(e) = crystals = powder = (insoluble) solid M1 I shades R any other colour M2 A no change / no reaction / no visible change if ppt seen in M1 M2 A insoluble (if ppt given in M1) M2 A yellow ppt if yellow ppt seen in M1 M2 A precipitate is insoluble M2 A ecf ppt remains on incorrect colour ppt in M1 M2 BOD 'no observation' M2 I colour of M1 precipitate M2 I 'no reaction'
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9. MAR 26/P22/Q1(d)

1(d)(i)	$2\text{OH}^- + \text{Cl}_2 \rightarrow \text{ClO}^- + \text{Cl}^- + \text{H}_2\text{O}$	1
1(d)(ii)	(ClO^-) kills bacteria	1

10. O/N 25/P24/Q1(a)(i)

1(a)(i)	(halogens can) remove electron(s) (from a reactant)	1
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11. O/N 25/P24/Q1(a)(iii)

1(a)(iii)	purple (gas) disappears	1
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12. O/N 25/P24/Q1(b)(i)-(ii)

1(b)(i)	M1 Br—Br is weaker (than Cl—Cl) AND Br atoms are larger M2 less (significant) overlap of orbitals (in Br-Br)	2
1(b)(ii)	HCl is more thermally stable	1

13. O/N 25/P24/Q1(c)

1(c)(i)		NaCl	NaBr	NaI	4
	with Ag ⁺ (aq)	white precipitate	M1 cream / off-white ppt	M2 yellow ppt	
	with conc. H ₂ SO ₄	acid – base	acid – base, then redox	acid – base, then redox	
	observations with conc. H ₂ SO ₄	M3 steamy fumes OR fizzing	M4 brown liquid OR fizzing	- black solid - yellow solid - effervescence	
1(c)(ii)	M1 black solid = iodine / I ₂ AND yellow solid = sulfur / S _(s) M2 effervescence = hydrogen sulfide / H ₂ S(g)				2

14. M/J 25/P23/Q6(a)(ii)-(iii)

6(a)(ii)	mix / add Cl ₂ into water	1
6(a)(iii)	water purification / killing bacteria or microbes	1

15. M/J 25/P24/Q3(d)(ii)-(iii)

3(d)(ii)	M1 HCl is more stable AND (because) Cl atom is smaller / less distance between electron pair and nucleus / shorter (H-X) bond length M2 (In HCl) stronger attraction between nucleus and shared pair of electrons OR (In HCl) (fewer shells of electrons so) less shielding OR (In HCl) greater orbital overlap so stronger interaction	2
3(d)(iii)	any value or range of values for K _c between $5.50 \times 10^{-34} < K_c < 1.26 \times 10^{-3}$	1

Answer for context part (c)

3(c)	2.46	1
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16. MAR 25/P22/Q3(a)

3(a)(i)	bromine: (dark / red-)brown liquid iodine: grey / black solid	1
3(a)(ii)	- atoms larger (down the group) - distance between electron pair and nucleus greater / longer bond length / lesser orbital overlap - weaker attraction	2

17. MAR 25/P22/Q3(b)

3(b)	$\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HOCl}$ ClO ⁻ / HOCl kills bacteria	2
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18. MAR 25/P22/Q3(c)

3(c)(i)	sodium iodide / NaI	1
3(c)(ii)	sodium chloride / NaCl	1
3(c)(iii)	cream(-coloured) / off-white precipitate / solid forms	1

19. O/N 24/P21/Q2(a)

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	1
2(a)(ii)	oxidising strength decreases from chlorine to iodine	1
2(a)(iii)	green colour disappears	1
2(a)(iv)	H—Hal covalent bond strength decreases down the group	1

20. O/N 24/P21/Q2(c)(i)

2(c)(i)	cold NaOH(aq)	1
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21. O/N 24/P23/Q2(a)

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	1
2(a)(ii)	oxidising strength decreases from chlorine to iodine	1
2(a)(iii)	green colour disappears	1
2(a)(iv)	H—Hal covalent bond strength decreases down the group	1

22. O/N 24/P23/Q2(c)(i)

2(c)(i)	cold NaOH(aq)	1
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23. M/J 24/P21/Q1(a)

1(a)	halogen	colour at 293 K
	chlorine	green
	bromine	brown
	iodine	(dark) grey
All three correct for one mark		

24. M/J 24/P21/Q1(b)

1(b)	M1 (it / volatility) decreases M2 more electrons in molecules M3 (so) increased strength of instantaneous dipole – induced dipole forces (of attraction)	3
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25. M/J 24/P21/Q1(c)

1(c)(i)	$\text{Br}_2 + 2\text{I}^- \rightarrow 2\text{Br}^- + \text{I}_2$	1
1(c)(ii)	(Br ₂ / bromine is an) oxidising agent AND removes electron(s) from iodide (ions) I ⁻ OR increases the oxidation number of I ⁻ / iodide (ion)	1

26. M/J 24/P21/Q1(d)

1(d)(i)	$\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow \text{HBr} + \text{NaHSO}_4$ OR $2\text{NaBr} + \text{H}_2\text{SO}_4 \rightarrow 2\text{HBr} + \text{Na}_2\text{SO}_4$	1
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1(d)(ii)	M1 sodium chloride M2 Cl^- is not a strong enough reducing agent to reduce H_2SO_4 M3 Br^- & I^- are strong enough reducing agents to react (further) with (the S in) H_2SO_4 <i>Alternative M2 and M3</i> M2 ability to behave as reducing agents increases down the group M3 EITHER Cl^- is not strong enough to reduce H_2SO_4 OR Br^- and I^- react (further) with (the S in) H_2SO_4	3
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27. MAR 24/P22/Q2(b)

2(b)(i)	disproportionation	1
2(b)(ii)	NaClO / sodium chlorate(I)	1
2(b)(iii)	$3\text{Cl}_2 + 6\text{NaOH} \rightarrow \text{NaClO}_3 + 5\text{NaCl} + 3\text{H}_2\text{O}$	1
2(b)(iv)	white ppt / solid	1
	dissolves (in $\text{NH}_3(\text{aq})$)	1

28. O/N 23/P21/Q1(c)

1(c)	(Br_2) no visible reaction OR colourless to orange / brown / yellow solution	1
	(conc. H_2SO_4) no visible reaction OR Solution remains colourless	1
	(dil. AgNO_3) white precipitate forms	1

29. O/N 23/P23/Q1(c)

1(c)	(Br_2) no visible reaction OR colourless to orange / brown / yellow solution	1
	(conc. H_2SO_4) no visible reaction OR Solution remains colourless	1
	(dil. AgNO_3) white precipitate forms	1

30. M/J 23/P21/Q3(c)

3(c)(i)	hot aqueous sodium hydroxide / hot $\text{NaOH}(\text{aq})$	1
3(c)(ii)	oxidation number / state of <u>chlorine / Cl</u> increases / goes (from 0) to +5 AND decreases / goes (from 0) to -1	1

31. M/J 23/P22/Q2(d)

2(d)(i)	$\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HOCl} + \text{HCl}$	1
2(d)(ii)	$3\text{Cl}_2 + 6\text{NaOH} \rightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$	1

32. M/J 23/P22/Q2(e)(i)

2(e)(i)	ClO^-	1
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33. M/J 23/P23/Q4(a)

4(a)(i)	$2\text{KCl} + \text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{SO}_4 + 2\text{HCl}$ OR $\text{KCl} + \text{H}_2\text{SO}_4 \rightarrow \text{KHSO}_4 + \text{HCl}$	1
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4(a)(ii)	M1 HI(g) is not the only gaseous product OR SO ₂ and / or H ₂ S and / or I ₂ (g) is (also) made M2 HI / iodide (ion) is oxidised OR H ₂ SO ₄ oxidises HI OR HI reduces sulfuric acid	2
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34. MAR 23/P22/Q2(d)(i)

2(d)(i)	white precipitate yellow solid (dark) grey solid / purple gas	effervescence / misty fumes rotten egg smell	2
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35. O/N 22/P21/Q2(c)

2(c)	white ppt / solid /	1
	solid (re)dissolves (on addition of NH ₃)	1

36. O/N 22/P21/Q2(d)

2(d)	bromine / Br ₂ is not a strong enough / (too) weak as a oxidising agent (to oxidise chloride / Cl ⁻) owtte / Bromine / Br ₂ cannot oxidise chloride (ion) / Cl ⁻	1
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37. O/N 22/P21/Q3(b)

3(b)	NaClO / sodium hypochlorite / chlorate(I) AND H ₂ O / water	1
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38. O/N 22/P22/Q2(e)(i)

2(e)(i)	M1 orange colour (of solution) remains / no visible reaction	1
	M2 bromine/Br ₂ cannot oxidise chloride/Cl ⁻ OR bromine/Br ₂ is not a strong enough oxidising agent (to oxidise chloride/Cl ⁻)	1

39. O/N 22/P23/Q2(c)

2(c)	white ppt / solid /	1
	solid (re)dissolves (on addition of NH ₃)	1

40. O/N 22/P23/Q2(d)

2(d)	bromine / Br ₂ is not a strong enough / (too) weak as a oxidising agent (to oxidise chloride / Cl ⁻) owtte / Bromine / Br ₂ cannot oxidise chloride (ion) / Cl ⁻	1
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41. O/N 22/P23/Q3(b)

3(b)	NaClO / sodium hypochlorite / chlorate(I) AND H ₂ O / water	1
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42. M/J 22/P22/Q2(d)

2(d)						3
	step	method	observation with CaCl_2	observation with CaBr_2	observation with CaI_2	
	step 1	$\text{AgNO}_3(\text{aq})$ (+ $\text{HNO}_3(\text{aq})$)	white ppt	cream OR off-white ppt	(pale) yellow ppt	
	step 2	$\text{NH}_3(\text{aq})$	(ppt) dissolves / (completely) soluble (in dilute or conc) OR (forms) colourless solution	(ppt) partly soluble / slightly soluble (in dilute or conc) ALLOW (ppt) dissolves in excess (in dilute or conc) ALLOW (ppt) soluble in conc. NH_3	(ppt) insoluble / solid remains (in dilute or concentrated)	
M1 step 1 $\text{AgNO}_3(\text{aq})$ / silver nitrate solution AND step 2 $\text{NH}_3(\text{aq})$ / ammonia solution / dilute NH_3 / concentrated NH_3						
M2 ALL solid / precipitate AND correct colours described in row 1						
M3 correct observations on addition of ammonia						

43. M/J 22/P22/Q3(d)

3(d)(i)	acid-base	1
3(d)(ii)	$8\text{HI} + \text{H}_2\text{SO}_4 \rightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$ OR $8\text{H}^+ + 8\text{I}^- + \text{H}_2\text{SO}_4 \rightarrow 4\text{I}_2 + \text{H}_2\text{S} + 4\text{H}_2\text{O}$	1
3(d)(iii)	<i>explanation in terms of comparison of reducing nature of HI to HCl</i> i.e. HI / I^- is stronger reducing agent compared to Cl^- / HCl OR <i>explanation in terms of additional behaviour of HI (compared to HCl)</i> i.e. HI / I^- reduces S (in sulfuric acid) / (concentrated) sulfuric acid	1

44. M/J 22/P23/Q2(b)

2(b)(i)	M1 $2\text{NaTs} + \text{Br}_2 \rightarrow 2\text{NaBr} + \text{Ts}_2$ OR $2\text{Ts}^- + \text{Br}_2 \rightarrow 2\text{Br}^- + \text{Ts}_2$ M2 Ts^- reduces Br / Br_2 OR reducing strength of halide (ion) increases down the group				2												
2(b)(ii)	<table><tr><td>element</td><td>chlorine</td><td>bromine</td><td>tennessine</td></tr><tr><td>melting point / °C</td><td>−101</td><td>−7.2</td><td>> 250</td></tr><tr><td>lattice structure of crystalline solid</td><td>simple molecular</td><td>simple molecular</td><td>giant metallic</td></tr></table>				element	chlorine	bromine	tennessine	melting point / °C	−101	−7.2	> 250	lattice structure of crystalline solid	simple molecular	simple molecular	giant metallic	2
element	chlorine	bromine	tennessine														
melting point / °C	−101	−7.2	> 250														
lattice structure of crystalline solid	simple molecular	simple molecular	giant metallic														

45. MAR 22/P22/Q3(a)

3(a)	M1: reaction less vigorous (down the group) M2: <i>Any two of the following for one mark:</i> • electronegativity decreases • less attractive to e^- addition • weaker oxidising agent • greater nuclear charge outweighing increased shielding (ENC argument)	2
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46. MAR 22/P22/Q3(b)(ii)

3(b)(ii)	M1: <i>All three correct for two marks:</i> row 1 • acid–base row 2 • acid–base • redox M2: <i>explanation</i> H_2SO_4 is strong enough to oxidise / is an oxidising agent with $\text{NaBr} / \text{HBr} / \text{bromide}$	3
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47. MAR 22/P22/Q3(c)

3(c)	H—Cl bond is stronger than H—I / BDE decreases down the group	1
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Unit 12

NITROGEN AND SULFUR

20 items · 47 marks

Syllabus sections in this unit

12.1 Nitrogen and sulfur

Unit 12: Nitrogen and sulfur

1. M/J 26/P21/Q2(a)

Context

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

(a) Write an overall equation for stage 3.

[1]

2. M/J 26/P21/Q2(c)(i)

Context

2 The Contact process uses sulfur to produce sulfuric acid.

stage 1 Sulfur is burned in excess air.

stage 2 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g}) \quad \Delta H = -196 \text{ kJ mol}^{-1}$

stage 3 SO_3 reacts to produce H_2SO_4 .

Context: part (c) (introduction)

(c) CaCO_3 and O_2 are added to the waste gases from the Contact process before they escape. This prevents the release of SO_2 into the atmosphere.



(i) State **one** environmental consequence of the release of $\text{SO}_2(\text{g})$ into the atmosphere.

[1]

3. M/J 26/P24/Q2(a)(ii)

(ii) Explain the lack of reactivity of nitrogen.

[2]

4. MAR 26/P22/Q2(c)

Context

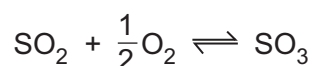
2 Nitrogen monoxide, NO, is an atmospheric pollutant.**(c) (i)** State a natural process that creates atmospheric NO.

[1]

(ii) Photochemical smog forms when NO reacts with unburned hydrocarbons.

Identify the major component of photochemical smog.

[1]

(iii) NO has a catalytic role in the oxidation of atmospheric SO₂.

Show, using relevant equations, the catalytic role that NO has in this oxidation.

[2]

5. M/J 25/P23/Q3(a)(ii)

Context: part (a) (introduction)

3 (a) Different hydrocarbon mixtures produced from fractional distillation of crude oil have different uses.**(ii)** Describe how photochemical smog is produced during the combustion of petrol in an internal combustion engine.

[2]

6. O/N 24/P22/Q3(a)(i)-(iii)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (a) (introduction)

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.**(i)** Identify **one** natural and **one** man-made occurrence of nitrogen oxides in the atmosphere.

natural

man-made

[2]

(ii) Write an equation to describe the role of NO₂ in the direct formation of acid rain.

[1]

(iii) Peroxyacetyl nitrate, PAN, is a component of photochemical smog.Describe how PAN forms from NO₂.

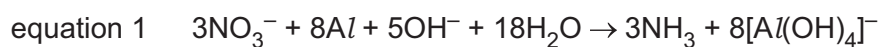
[1]

7. O/N 24/P22/Q3(b)(iii)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (b) (introduction)

(b) A common test for nitrates is the reaction with NaOH and Al. Equation 1 shows the reaction.**(iii)** NH_3 is a basic gas.Describe how NH_3 is able to act as a base.

[1]

8. O/N 24/P22/Q3(f)(ii)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (f) (introduction)

(f) At very high temperatures, phosphorus can form P_2 molecules. P_2 contains a triple bond, $\text{P}\equiv\text{P}$.**(ii)** The bond energy of $\text{P}\equiv\text{P}$ is 485kJ mol^{-1} . The bond energy of $\text{N}\equiv\text{N}$ is 944kJ mol^{-1} . Compare the reactivity of P_2 and N_2 . Explain your answer.

[1]

9. M/J 24/P23/Q1(a)(i)**1 (a) (i)** Explain the lack of reactivity of nitrogen gas, $\text{N}_2(\text{g})$.

[2]

10. M/J 24/P23/Q2(d)

Context

2 $\text{N}_2(\text{g})$ reacts with $\text{H}_2(\text{g})$ in the Haber process, as shown in reaction 1.

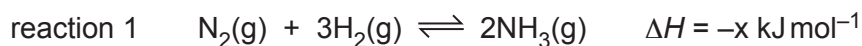


Table 2.1 shows the different conditions used to produce three equilibrium mixtures, **A**, **B** and **C**.

Table 2.1

	A	B	C
initial molar ratio of $\text{N}_2 : \text{H}_2$ added	1 : 3	1 : 3	1 : 3
temperature / °C	500	500	1000
pressure / atm	1000	1000	1000
iron present in mixture	no	yes	no
percentage yield of $\text{NH}_3(\text{g})$ at equilibrium	58	x	y

(d) When oxides of nitrogen escape into the atmosphere they may be involved in:

- formation of acid rain from sulfur dioxide
- formation of photochemical smog.

(i) Identify the role of NO and NO_2 in the formation of H_2SO_4 from SO_2 in the atmosphere to produce acid rain.

Use relevant equations to support your answer.

[3]

(ii) Outline how NO and NO_2 may contribute to the formation of photochemical smog.

[2]

11. M/J 24/P23/Q4(b)(ii)

Context: part (b) (introduction)

(b) An exothermic reaction occurs when $\text{NH}_4^+(\text{aq})$ is added to $\text{OH}^-(\text{aq})$.

(ii) Construct an ionic equation for the reaction of NH_4^+ and OH^- .

[1]

12. MAR 24/P22/Q3(a)

Context

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

(a) Give **two** reasons to explain the lack of reactivity of nitrogen.

1

2

[2]

13. MAR 24/P22/Q3(b)

Context

3 Nitrogen, N_2 , is generally an unreactive molecule but it does react under certain conditions.

(b) N_2 can react with oxygen in an internal combustion engine to form a mixture of NO and NO_2 . Fig. 3.1 shows a reaction scheme involving N_2 .

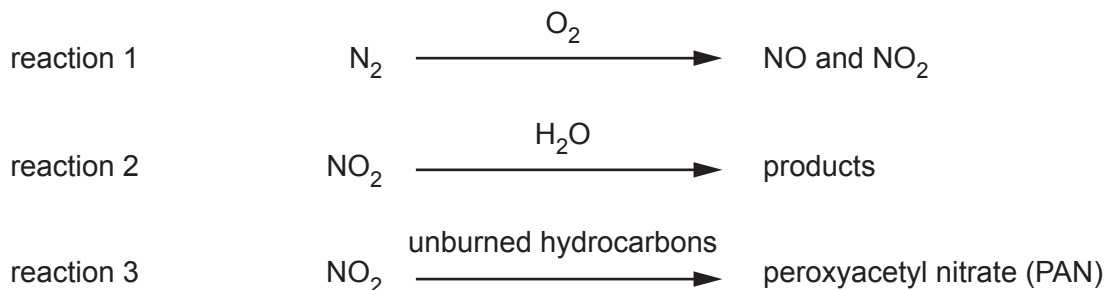


Fig. 3.1

(i) Write an equation to show the formation of a mixture of NO and NO_2 in reaction 1.

[1]

(ii) Give the formulae of the products of reaction 2.

[1]

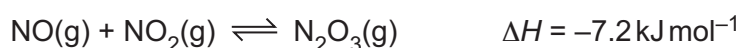
(iii) State **one** environmental consequence of reaction 3.

[1]

14. O/N 23/P22/Q2(d)

Context

2 NO and NO_2 react at 25°C to give N_2O_3 as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

(d) Identify one natural process and one man-made process that cause the formation of atmospheric NO and NO_2 .

natural process

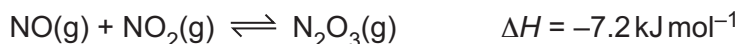
man-made process

[2]

15. O/N 23/P22/Q2(e)(ii)-(iii)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

Context: part (e) (introduction)

- (e)** NO₂ is a brown gas that can be used to form nitric acid.

- (ii)** NO₂ has a catalytic role in the oxidation of atmospheric sulfur dioxide.

Write equations to show the catalytic role of NO₂ in this oxidation.

[2]

- (iii)** State **one** environmental consequence of the oxidation of atmospheric sulfur dioxide.

[1]

16. M/J 23/P21/Q5(b)

Context: part (a) (introduction)

- 5** **(a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.

- (b)** When petrol is burned in an internal combustion engine, oxides of nitrogen are released into the atmosphere. Oxides of nitrogen are responsible for the formation of acid rain.

- (i)** Suggest the conditions required for the production of oxides of nitrogen during combustion of **M** in an internal combustion engine. Use an appropriate equation in your answer.

[2]

- (ii)** Describe how acid rain is formed in the atmosphere in the presence of oxides of nitrogen and SO₂. Identify the role of the oxides of nitrogen in this process. Include **all** relevant equations.

[3]

- (iii)** State **one** other type of air pollution that is caused by the production of oxides of nitrogen in an internal combustion engine.

[1]

17. O/N 22/P22/Q1(c)(ii)

Context

1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

Context: part (c) (introduction)

(c) NH_4^+ is a Brønsted–Lowry acid.

(ii) When $\text{NH}_4^+(\text{aq})$ is heated with $\text{NaOH}(\text{aq})$, a pungent gas is produced.

Write an ionic equation for this reaction.

[1]

18. M/J 22/P21/Q2(b)

Context

2 Nitrogen molecules, $\text{N}_2(\text{g})$, contain two atoms attracted to each other by a triple covalent bond.

(b) Nitrogen oxides, NO_2 and NO , are produced in internal combustion engines. Release of these gases into the atmosphere leads to the formation of photochemical smog.

(i) Outline how nitrogen oxides are involved in the formation of photochemical smog.

[2]

(ii) Construct an equation to demonstrate how a catalytic converter reduces the amount of nitrogen oxide gases released into the atmosphere.

[1]

19. M/J 22/P21/Q2(c)

Context

2 Nitrogen molecules, $\text{N}_2(\text{g})$, contain two atoms attracted to each other by a triple covalent bond.

(c) $\text{N}_2(\text{g})$ is very unreactive. It is difficult to make ammonia, $\text{NH}_3(\text{g})$, directly from its elements but it can be made from $\text{NH}_4\text{Cl}(\text{s})$.

Identify a reagent and the conditions required to make $\text{NH}_3(\text{g})$ from $\text{NH}_4\text{Cl}(\text{s})$.

[1]

20. MAR 22/P22/Q2(d)(i)-(ii)

Context

2 Some oxides of elements in Period 3 are shown.



Context: part (d) (introduction)

(d) SO_2 and SO_3 are found in the atmosphere.

The oxidation of SO_2 to SO_3 in the atmosphere is catalysed by NO_2 .

The first step of the catalytic oxidation is shown in equation 1.



(i) Construct an equation to show how NO_2 is regenerated in the catalytic oxidation of SO_2 .

[1]

(ii) NO_2 can also react with unburned hydrocarbons to form photochemical smog.

State the product of this reaction that contributes to photochemical smog.

[1]

Answers Section

1. M/J 26/P21/Q2(a)

2(a)	$\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$	1	BOD $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$ BOD $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2 \text{H}_2\text{SO}_4$
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2. M/J 26/P21/Q2(c)(i)

2(c)(i)	acid rain	1	A breathing / respiratory problems / asthma A harmful to fish / plants / animals A damages buildings / statues / trees BOD (photochemical) smog I ozone I greenhouse effect / global warming / climate change I harmful to health
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3. M/J 26/P24/Q2(a)(ii)

2(a)(ii)	M1 (N_2 molecules / N_2) have a strong triple (covalent) bond OR $\text{N} \equiv \text{N}$ (bond) is strong M2 (N_2 molecules / N_2) are non-polar	2	M1 A strong = a lot of energy to break / high activation energy / high bond energy BOD hard/difficult to break I three strong (covalent) bonds (alone) A three strong bonds between N atoms R three strong bonds AND triple bond ($\therefore = 9$) M2 A same electronegativity / no difference in electronegativity
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4. MAR 26/P22/Q2(c)

2(c)(i)	lightning	1
2(c)(ii)	PAN / peroxyac(et)lynitrate	1
2(c)(iii)	M1 first equation $\text{NO} + \frac{1}{2}\text{O}_2 \rightarrow \text{NO}_2$ M2 second equation $\text{SO}_2 + \text{NO}_2 \rightarrow \text{SO}_3 + \text{NO}$	2

5. M/J 25/P23/Q3(a)(ii)

3(a)(ii)	- unburned hydrocarbons - react with NO / NO_2 (to) produce PAN (– a component of photochemical smog) Any 2 points = 1 mark, all 3 points = 2 marks	2
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6. O/N 24/P22/Q3(a)(i)-(iii)

3(a)(i)	(natural =)lightning	1
	(man-made =)internal combustion engines	1
3(a)(ii)	$2\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HNO}_2 + \text{HNO}_3$ OR $4\text{NO}_2 + \text{O}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HNO}_3$	1
3(a)(iii)	It / NO_2 reacts with (unburned) hydrocarbons / VOCs ALLOW reaction of unburned hydrocarbons / VOCs in presence of NO_2	1

7. O/N 24/P22/Q3(b)(iii)

3(b)(iii)	(NH_3 is an) H^+ acceptor	1
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8. O/N 24/P22/Q3(f)(ii)

3(f)(ii)	$P \equiv P$ is much weaker so P_2 is more reactive (than N_2)	1
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9. M/J 24/P23/Q1(a)(i)

1(a)(i)	M1 (N_2 molecules /molecule(s) of N_2) have a strong triple (covalent) bond M2 (N_2 molecules /molecule(s) of N_2) are non-polar	2
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10. M/J 24/P23/Q2(d)

2(d)(i)	M1 catalyst M2 2 equations to show the catalytic activity of NO_2 in formation of SO_3 from SO_2 $SO_2 + NO_2 \rightarrow NO + SO_3$ $2NO + O_2 \rightarrow 2NO_2$ M3 describe production of H_2SO_4 from SO_3 and water (can be shown in an equation). SO_3 then reacts with (rain)water / moisture / water vapour to make sulfuric acid / H_2SO_4 OR $SO_3 + H_2O \rightarrow H_2SO_4$	3
2(d)(ii)	M1 unburned hydrocarbons combine with NO and / NO_2 M2 to form PAN / peroxyac(et)yl nitrate	2

11. M/J 24/P23/Q4(b)(ii)

4(b)(ii)	$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$	1
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12. MAR 24/P22/Q3(a)

3(a)	strong triple bond / high triple bond enthalpy	1
	non-polar (molecule)	1

13. MAR 24/P22/Q3(b)

3(b)(i)	e.g. $N_2 + 1\frac{1}{2}O_2 \rightarrow NO + NO_2$	1
3(b)(ii)	HNO_2 and HNO_3	1
3(b)(iii)	photochemical smog	1

14. O/N 23/P22/Q2(d)

2(d)	natural process: lightning	1
	man-made process: internal combustion engines	1

15. O/N 23/P22/Q2(e)(ii)-(iii)

2(e)(ii)	$NO_2 + SO_2 \rightarrow SO_3 + NO$	1
	$NO + \frac{1}{2}O_2 \rightarrow NO_2$	1
2(e)(iii)	(formation of) acid rain	1

16. M/J 23/P21/Q5(b)

5(b)(i)	M1 • high temperature • high pressure • N_2 Any two correct for M1 M2 $N_2 + O_2 \rightarrow 2NO$ OR $N_2 + 2O_2 \rightarrow 2NO_2$ OR $N_2 + xO_2 \rightarrow 2NO_x$	2
5(b)(ii)	M1 NO_2 is the catalyst (for $SO_2 \rightarrow SO_3$) OR catalytic oxidation (of SO_2) by NO_2 M2 $NO_2 + SO_2 \rightarrow NO + SO_3$ AND $2NO + O_2 \rightarrow 2NO_2$ M3 SO_3 reacts with rain OR moisture/water (in atmosphere) to form sulfuric acid / H_2SO_4 OR $SO_3 + H_2O \rightarrow H_2SO_4$	3
5(b)(iii)	photochemical smog / PAN	1

17. O/N 22/P22/Q1(c)(ii)

1(c)(ii)	<i>ionic equation</i> $\text{NH}_4^+ + \text{OH}^- \rightarrow \text{NH}_3 + \text{H}_2\text{O}$	1
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18. M/J 22/P21/Q2(b)

2(b)(i)	M1 react with (unburnt) hydrocarbons M2 (form) PAN / peroxyac(et)yl nitrate	2
2(b)(ii)	$2\text{NO} + 2\text{CO} \rightarrow 2\text{CO}_2 + \text{N}_2$ OR $\text{NO}_2 + 2\text{CO} \rightarrow \frac{1}{2}\text{N}_2 + 2\text{CO}_2$	1

19. M/J 22/P21/Q2(c)

2(c)	any Group 1 hydroxide or $\text{Ca}(\text{OH})_2$ / $\text{Sr}(\text{OH})_2$ / $\text{Ba}(\text{OH})_2$	1
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20. MAR 22/P22/Q2(d)(i)-(ii)

2(d)(i)	$\text{NO} + \frac{1}{2}\text{O}_2 \rightarrow \text{NO}_2$	1
2(d)(ii)	peroxyac(et)ynitrate / PAN	1

Unit 13

AN INTRODUCTION TO AS LEVEL ORGANIC CHEMISTRY

69 items · 145 marks

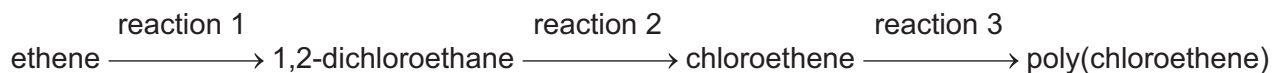
Syllabus sections in this unit

- 13.1 Formulas, functional groups and the naming of organic compounds
- 13.2 Characteristic organic reactions
- 13.3 Shapes of organic molecules; σ and π bonds
- 13.4 Isomerism: structural isomerism and stereoisomerism

Unit 13: An introduction to AS Level organic chemistry**1. M/J 26/P21/Q5(a)**

Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

- (a) Complete Table 5.1.

Table 5.1

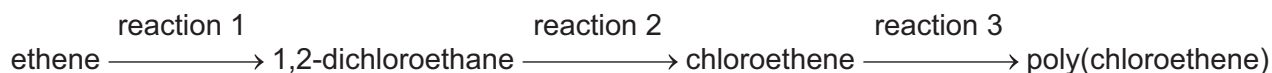
compound	type of hybridisation of carbon atoms
1,2-dichloroethane	
chloroethene	

[2]

2. M/J 26/P21/Q5(c)

Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

- (c) Describe what is meant by heterolytic fission of a covalent bond.

[2]

3. M/J 26/P22/Q5(a)

Context

- 5** Hexane, $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_5\text{Cl}$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_2\text{Cl}$	1,2-dichlorohexane
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$	dodecane
$\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$	5,6-dimethyldecane

(a) Complete Table 5.1.

[2]

4. M/J 26/P22/Q5(b)(i)-(ii)

Context

- 5** Hexane, $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_5\text{Cl}$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_2\text{Cl}$	1,2-dichlorohexane
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$	dodecane
$\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$	5,6-dimethyldecane

Context: part (b) (introduction)

- (b)** The mechanism for the reaction of hexane with $\text{Cl}_2(\text{g})$ involves three stages.

stage 1 initiation

stage 2 propagation

stage 3 termination

In stage 1 and 2, free radicals are produced.

- (i)** Describe what is meant by free radical.

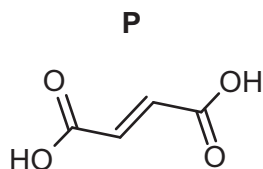
[1]

- (ii)** Name the type of bond breaking that occurs to produce the free radical $\bullet\text{Cl}$ from Cl_2 .

[1]

5. M/J 26/P22/Q6(a)

- 6** **(a)** Figure 6.1 shows the structure of compound **P**.

**Figure 6.1**

- (i)** Draw the structure of the stereoisomer of **P**.

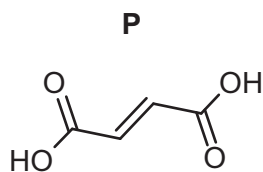
[1]

- (ii)** Explain why **P** shows stereoisomerism.

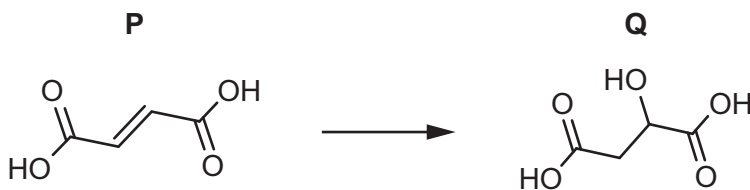
[2]

6. M/J 26/P22/Q6(b)(ii)

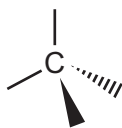
Context: part (a) (introduction)

6 (a) Figure 6.1 shows the structure of compound **P**.**Figure 6.1**

Context: part (b) (introduction)

(b) **P** reacts to produce compound **Q**.**Figure 6.2****(ii)** Complete Figure 6.3 to show the pair of stereoisomers of **Q**.

isomer 1



isomer 2

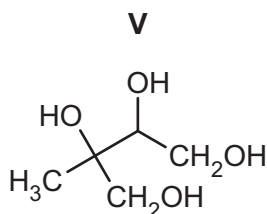
Figure 6.3

[2]

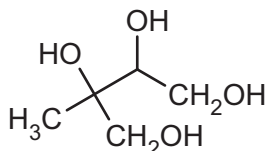
7. M/J 26/P23/Q4(c)(i)-(ii)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

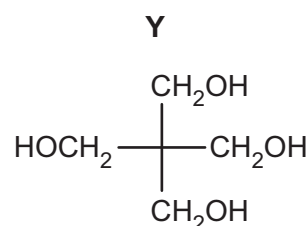
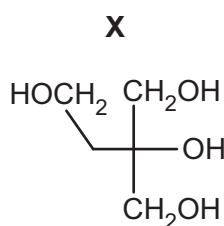
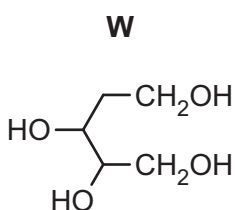
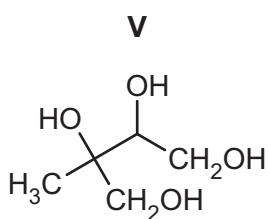
**Figure 4.1**

Context: Fig. 4.1

**Figure 4.1**

Context: part (c) (introduction)

- (c)** Compounds **W**, **X** and **Y** are structural isomers of **V**.

**Figure 4.2**

- (i)** Describe structural isomerism.

[2]

- (ii)** Identify which of structures **V**, **W**, **X** and **Y** show stereoisomerism.

Explain your answer by referring to the relevant parts of the molecule that cause stereoisomerism.

[2]

8. M/J 26/P23/Q5(a)(i)

Context

- 5** Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

- (a) (i) Complete Table 5.1.

[2]

9. M/J 26/P24/Q5(c)

Context

- 5** Long-chain alkanes from crude oil can be cracked to make alkenes.

- (c) But-1-ene and but-2-ene are structural isomers.

State the type of structural isomerism shown by but-1-ene and but-2-ene.

[1]

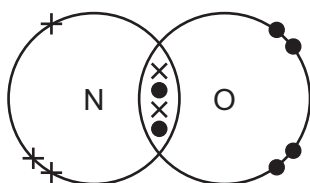
10. MAR 26/P22/Q2(a)(i)

Context

- 2** Nitrogen monoxide, NO, is an atmospheric pollutant.

Context: part (a) (introduction)

- (a) Figure 2.1 shows the dot-and-cross diagram of NO.

**Figure 2.1**

- (i) NO has an unpaired electron.

State the term for a molecule with an unpaired electron.

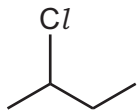
[1]

11. MAR 26/P22/Q3(b)

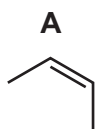
Context

- 3** Secondary halogenoalkanes are important building blocks in organic chemistry.

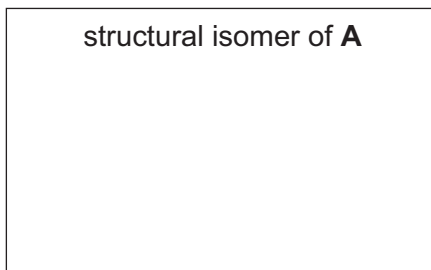
2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane**Figure 3.1**

- (b) 2-chlorobutane can be formed by the electrophilic addition of HCl to different isomers of C_4H_8 . Compound **A** is one of these isomers.

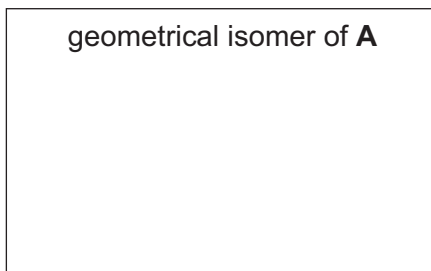
**Figure 3.2**

- (i) Draw a structural isomer of **A** that can also form 2-chlorobutane when it reacts with HCl .



[1]

- (ii) Draw and give the systematic name of the geometrical isomer of **A**.



systematic name

[2]

- (iii) State and explain the origin of geometrical isomerism in alkenes.

[2]

12. MAR 26/P22/Q4(d)

also Unit 22 (2 marks), Unit 18 (1 mark), Unit 20 (1 mark); this unit 2 marks

Context

- 4** Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.

- (d) Figure 4.4 describes a sequence of reactions, starting with $\text{CH}_2(\text{CN})\text{COOH}$, to form **E**.

E

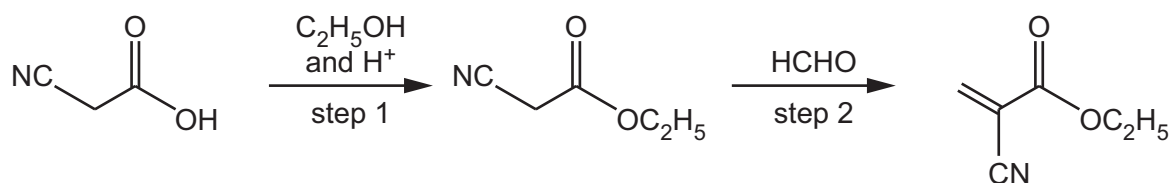


Figure 4.4

- (i) Identify the type of reaction that occurs in step 1. [1]
- (ii) Name HCHO , the reactant used in step 2. [1]
- (iii) Deduce the molecular formula of **E**. [1]
- (iv) **E** can be polymerised to form **F**.
Draw the structure of **one** repeat unit of polymer **F**. [1]
- (v) Figure 4.5 shows the infrared spectrum of **E**.

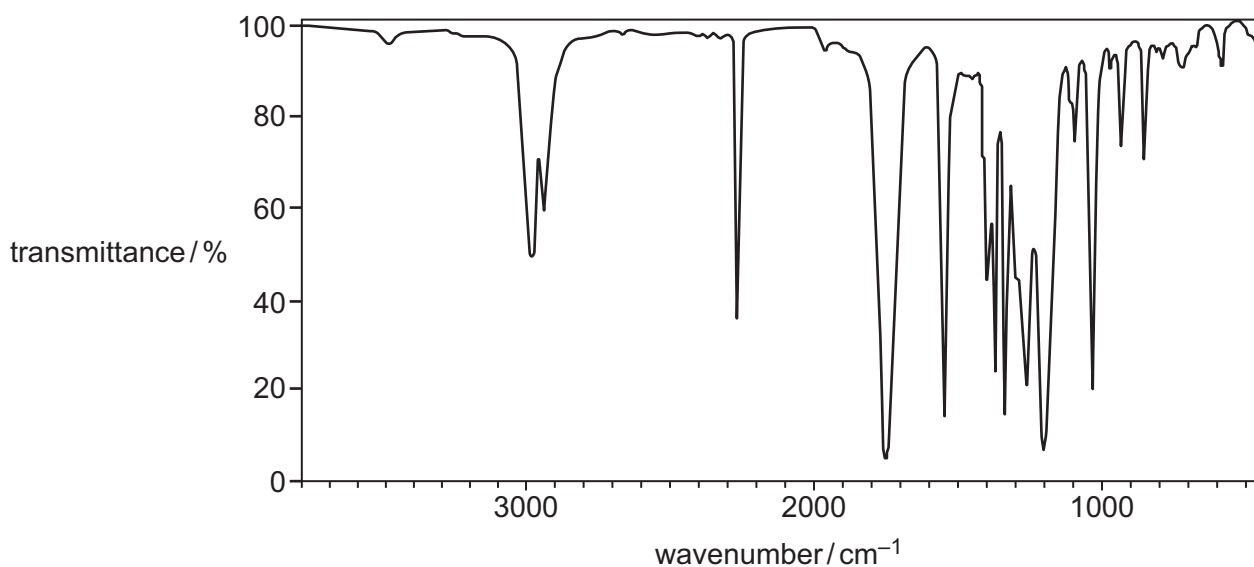


Figure 4.5

Table 4.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Using the information given in Table 4.1, identify with a cross (X) on Figure 4.5 an absorption that can be used to monitor the progress of the polymerisation of **E** to form **F**.

Explain your answer with reference to the relevant bond.

[2]

13. O/N 25/P21/Q3(b)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(b) Cyclopentane, C₅H₁₀, has four cyclic structural isomers. One of these isomers is **C**, shown in Fig. 3.1.

Complete Fig. 3.1 to show two other cyclic structural isomers of C₅H₁₀.

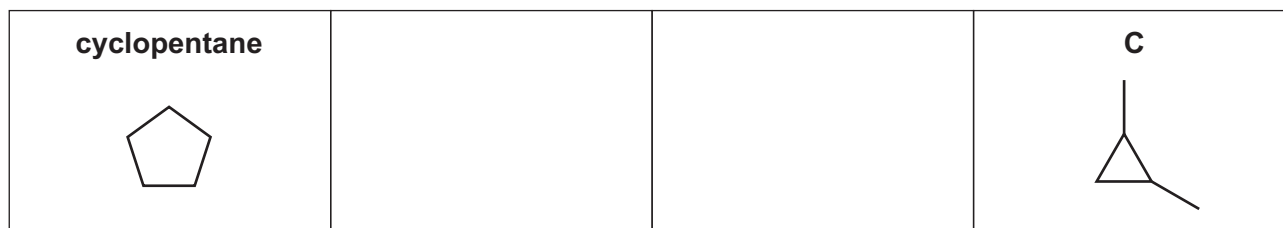


Fig. 3.1

[2]

14. O/N 25/P21/Q3(c)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (c) (introduction)

(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form $\text{C}_5\text{H}_9\text{Cl}$.

- (i)** The reaction is initiated by the bond fission of Cl_2 .

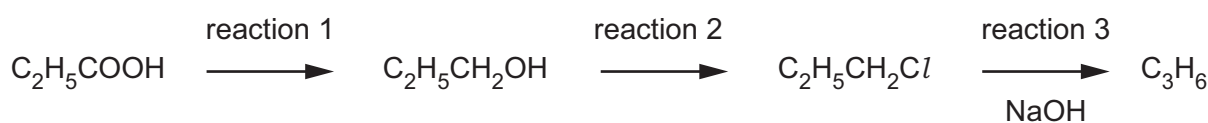
State the type of bond fission shown in the initiation step.

[1]

15. O/N 25/P21/Q4(b)(ii)

Context

- 4** Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

Context: part (b) (introduction)

- (b)** Under suitable conditions, propene polymerises to form poly(propene).

Poly(propene) exhibits stereoisomerism.

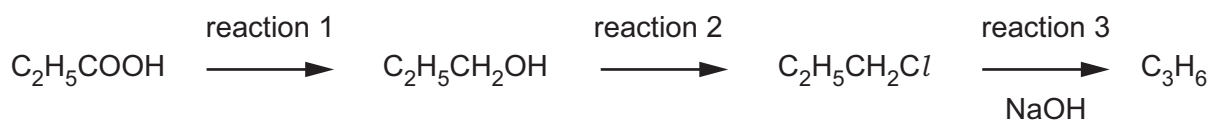
- (ii)** Define stereoisomerism.

[2]

16. O/N 25/P21/Q4(c)(i)

Context

- 4** Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

Context: part (c) (introduction)

- (c)** Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.



- (i)** Name **X**.

[1]

17. O/N 25/P22/Q3(a)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(a) Define homologous series.

[2]

18. O/N 25/P22/Q3(d)(i)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

Context: part (d) (introduction)

(d) Hexane, C_6H_{14} , has four structural isomers.

Fig. 3.1 shows hexane and two of its structural isomers.

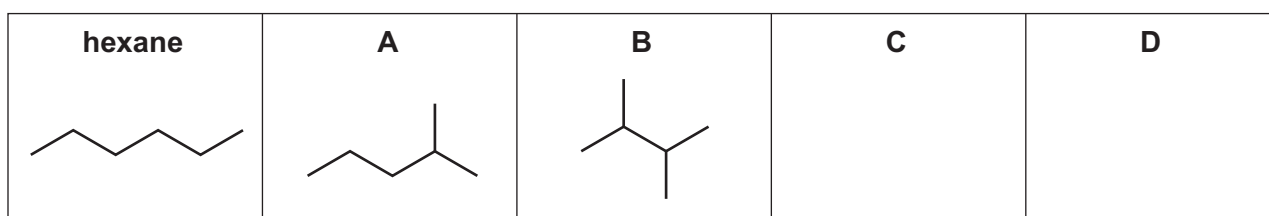


Fig. 3.1

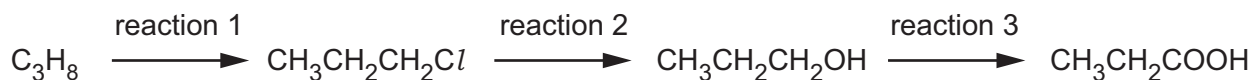
- (i) Complete Fig. 3.1 by drawing structures for **C** and **D**, the other two structural isomers of hexane.

[2]

19. O/N 25/P22/Q4(a)(iii)-(v)

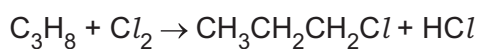
Context

- 4** Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, CH_3CH_2COOH .

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 in Fig. 4.1 takes place in the presence of sunlight.



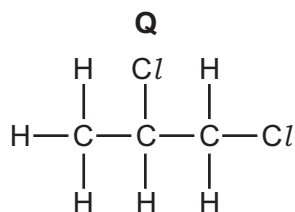
The reaction takes place via initiation, propagation and termination steps.

- (iii)** Reaction 1 is initiated by the bond fission of Cl_2 .

State the type of bond fission shown in the initiation step.

[1]

- (iv)** Compound **Q** is a by-product of reaction 1.



Name **Q**.

[1]

- (v)** The molecular formula of **Q** is $C_3H_6Cl_2$.

Identify the types of structural isomerism and stereoisomerism that a molecule with molecular formula $C_3H_6Cl_2$ can show.

type of structural isomerism

type of stereoisomerism

[2]

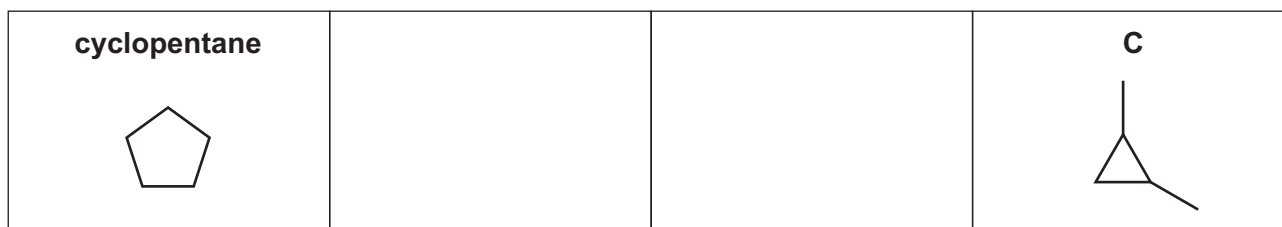
20. O/N 25/P23/Q3(b)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

(b) Cyclopentane, C_5H_{10} , has four cyclic structural isomers. One of these isomers is **C**, shown in Fig. 3.1.

Complete Fig. 3.1 to show two other cyclic structural isomers of C_5H_{10} .

**Fig. 3.1**

[2]

21. O/N 25/P23/Q3(c)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (c) (introduction)

(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form C_5H_9Cl .

- (i) The reaction is initiated by the bond fission of Cl_2 .

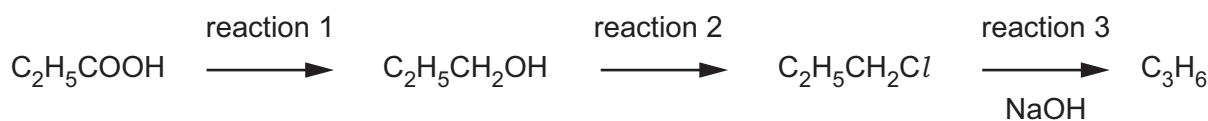
State the type of bond fission shown in the initiation step.

[1]

22. O/N 25/P23/Q4(b)(ii)

Context

- 4** Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

Context: part (b) (introduction)

- (b) Under suitable conditions, propene polymerises to form poly(propene).

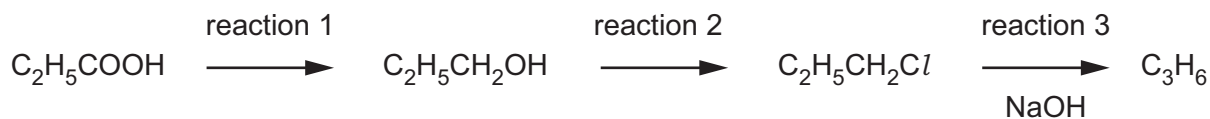
Poly(propene) exhibits stereoisomerism.

- (ii) Define stereoisomerism.

[2]

23. O/N 25/P23/Q4(c)(i)

Context

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .**Fig. 4.1**

Context: part (c) (introduction)

(c) Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.**(i)** Name **X**.

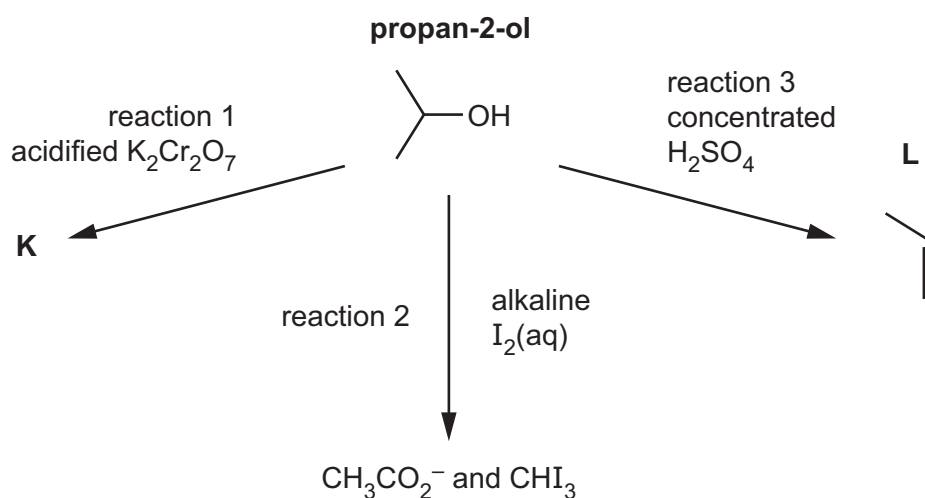
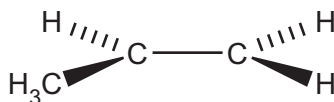
[1]

24. O/N 25/P24/Q3(a)(iv)

Context

3 Propan-2-ol, $(CH_3)_2CHOH$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

**Fig. 3.1****(iv)** Complete Fig. 3.2 to show the pi (π) bond in **L** that is formed from orbital overlap.**Fig. 3.2**

[1]

25. M/J 25/P22/Q3(a)(i)

Context: part (a) (introduction)

3 (a) C_3H_6 reacts with $HBr(g)$ in an addition reaction.**(i)** Define addition reaction.

[1]

26. M/J 25/P22/Q4(c)(i)

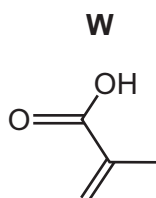
Context: part (c) (introduction)

(c) Compounds **A**, **B** and **C** belong to the alcohol homologous series. Each molecule of **A**, **B** and **C** contains four saturated carbon atoms.**(i)** Identify the type of hybridisation shown in the saturated carbon atoms of all alcohols.

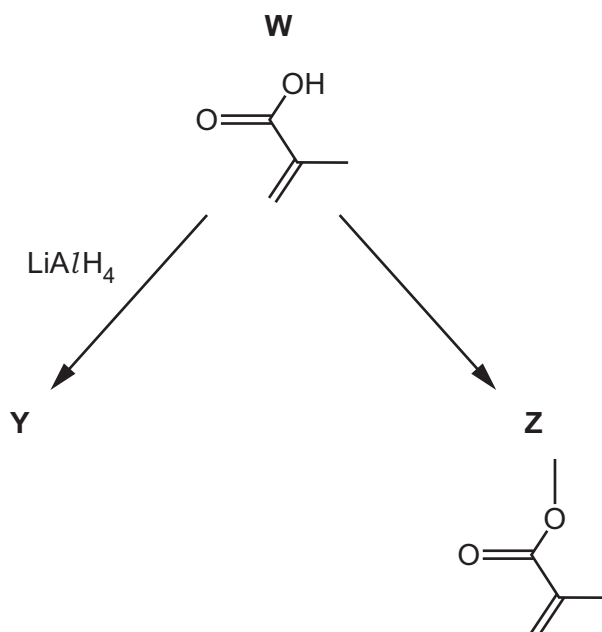
[1]

27. M/J 25/P23/Q5(b)(i)

Context: part (a) (introduction)

5 (a) **W** is a colourless liquid.**Fig. 5.1**

Context: part (b) (introduction)

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.**Fig. 5.2****(i)** Deduce the number of sigma (σ) bonds and pi (π) bonds present in **Z**.number of σ bondsnumber of π bonds

[2]

28. M/J 25/P23/Q6(a)(i)

Context: part (a) (introduction)

6 (a) HOCl reacts with $\text{CH}_2=\text{CH}_2$ to produce $\text{HOCH}_2\text{CH}_2\text{Cl}$ in an electrophilic addition reaction.

(i) Define addition reaction.

[1]

29. M/J 25/P24/Q4(a)

Context

4 The skeletal formula of **E** is shown in Fig. 4.1.

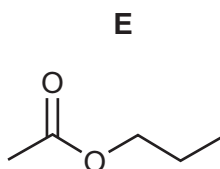


Fig. 4.1

(a) State the molecular formula of **E**.

[1]

30. M/J 25/P24/Q5(a)-(b)

Context

5 The skeletal formula of **W** is shown in Fig. 5.1.

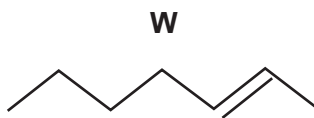
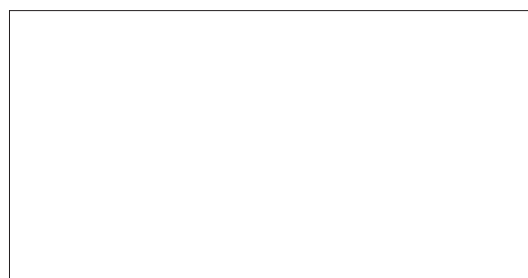
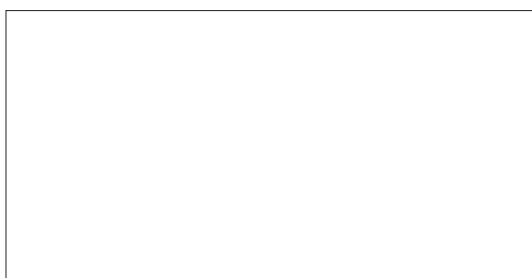


Fig. 5.1

(a) **W** has two positional isomers. Only one shows stereoisomerism.

Draw the structures of the **two** positional isomers of **W** in the boxes.



[2]

(b) (i) Describe the origin of stereoisomerism in **W**.

[1]

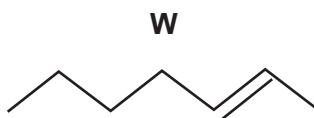
(ii) State why **W** shows stereoisomerism but one of the positional isomers in (a) does **not**.

[1]

31. M/J 25/P24/Q5(c)

Context

- 5** The skeletal formula of **W** is shown in Fig. 5.1.

**Fig. 5.1**

- (c) Draw the skeletal structure of the stereoisomer of **W**.

[1]

32. M/J 25/P24/Q6(a)(iii)

Context

- 6** **P**, **Q** and **R** are three different hydrocarbon molecules.

- (iii) Molecules of **P** have straight chains.

Name the homologous series to which **P** belongs.

[1]

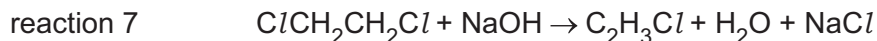
33. MAR 25/P22/Q1(c)(i)

Context

- 1** Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

Context: part (c) (introduction)

- (c) 1,2-dichloroethane, $\text{ClCH}_2\text{CH}_2\text{Cl}$, reacts with NaOH to produce an unsaturated compound, $\text{C}_2\text{H}_3\text{Cl}$, as shown in reaction 7.



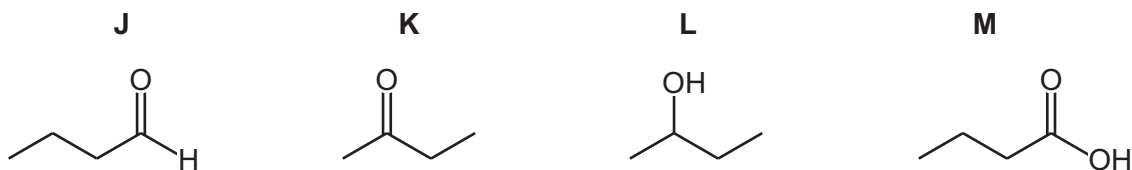
- (i) State what is meant by unsaturated.

[1]

34. MAR 25/P22/Q4(b)(iii)

Context

- 4** Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

**Fig. 4.1**

Context: part (b) (introduction)

- (b) **K** reacts with HCN in the presence of a KCN catalyst, forming **N**.

- (iii) **N** has a chiral centre.

Explain what is meant by a chiral centre.

[1]

35. O/N 24/P21/Q1(d)(ii)-(iv)

Context

- 1** Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

Context: part (d) (introduction)

- (d)** Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

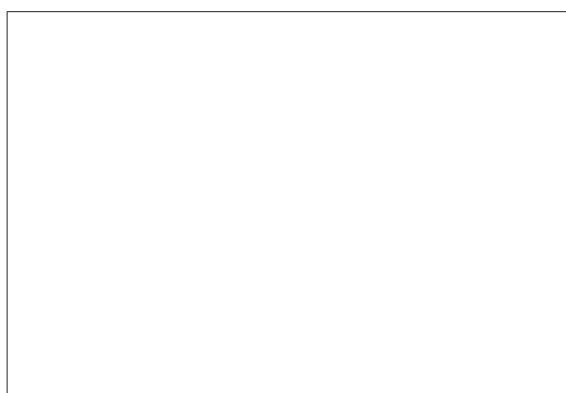
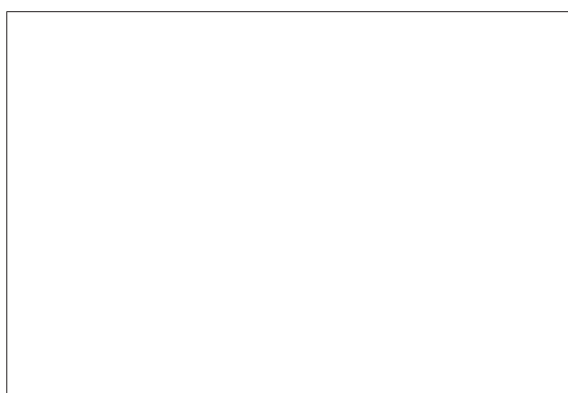
- (ii)** Define stereoisomers.

[1]

- (iii)** Explain how the conversion of ethene to but-2-ene can be described as an addition reaction.

[1]

- (iv)** Draw the two stereoisomers of but-2-ene.



[2]

36. O/N 24/P21/Q2(b)(i)

Context

- 2** Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (b) (introduction)

- (b)** The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.

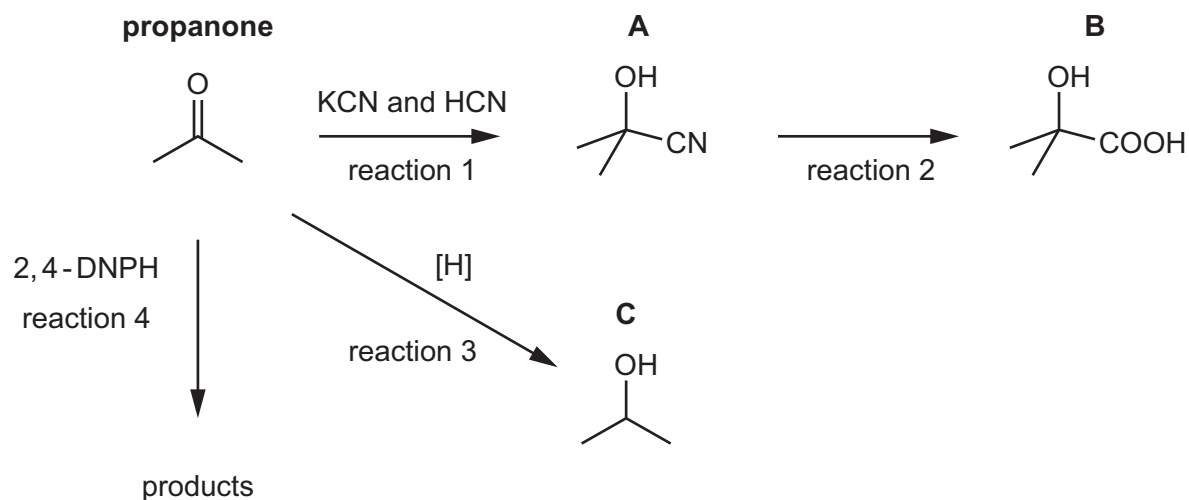
- (i)** Define free radical.

[1]

37. O/N 24/P21/Q4(a)(ii)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 is a nucleophilic addition reaction.
- (ii)** Explain why **A** does **not** show optical isomerism.

[1]

38. O/N 24/P23/Q1(d)(ii)-(iv)

Context

- 1** Cobalt, rhodium and iridium are metals in the same group of the Periodic Table.

Context: part (d) (introduction)

- (d)** Hydrated rhodium(III) chloride, $\text{RhCl}_3 \cdot x\text{H}_2\text{O}$, catalyses the conversion of ethene to but-2-ene.

Both stereoisomers of but-2-ene are formed in the reaction.

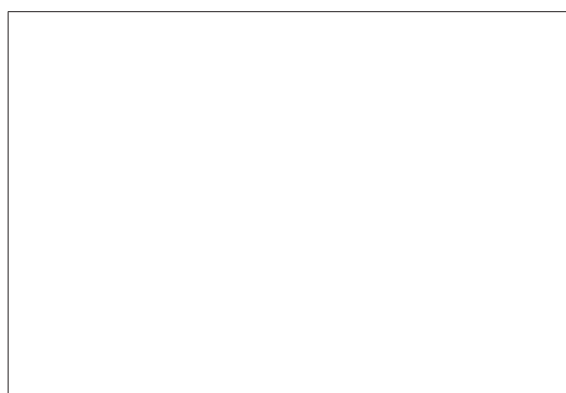
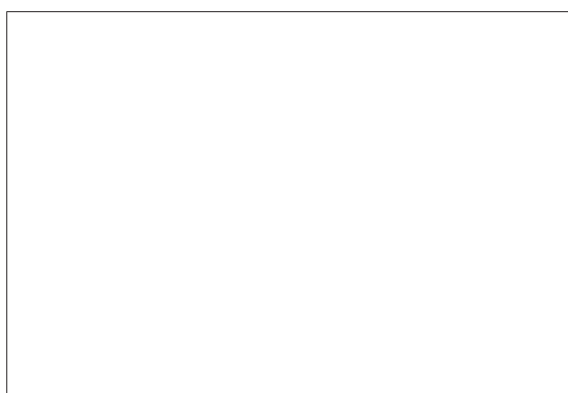
- (ii)** Define stereoisomers.

[1]

- (iii)** Explain how the conversion of ethene to but-2-ene can be described as an addition reaction.

[1]

- (iv)** Draw the two stereoisomers of but-2-ene.



[2]

39. O/N 24/P23/Q2(b)(i)

Context

- 2** Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (b) (introduction)

- (b)** The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.

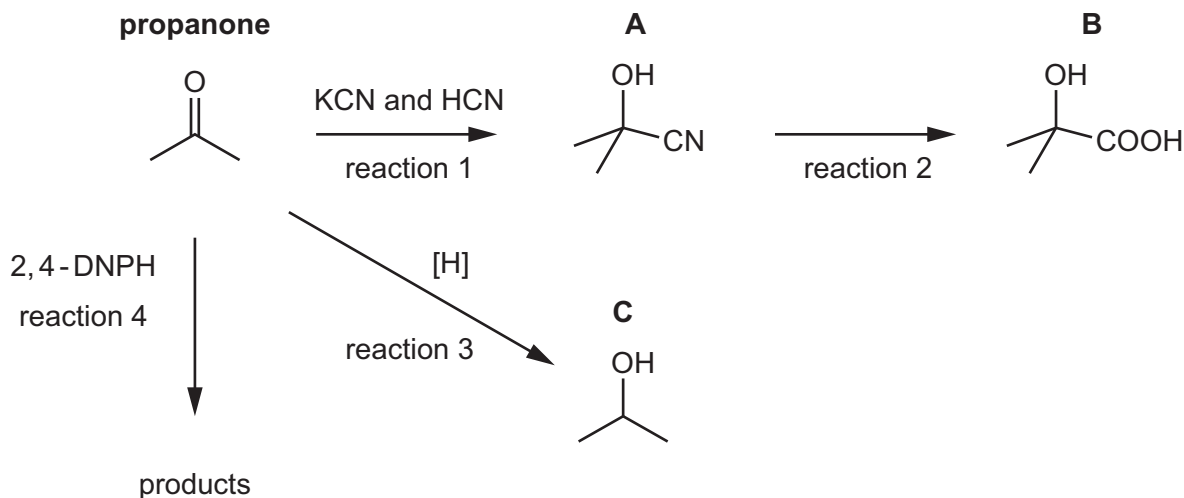
- (i)** Define free radical.

[1]

40. O/N 24/P23/Q4(a)(ii)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 is a nucleophilic addition reaction.
- (ii)** Explain why **A** does **not** show optical isomerism.

[1]

41. M/J 24/P21/Q5(b)

Context

- 5** Hydrocarbon molecules contain covalent bonds.

- (b)** A $\text{C}=\text{C}$ bond in an alkene is made from a σ bond and a π bond.

- (i)** Identify the hybridisation of the carbon atoms in a $\text{C}=\text{C}$ bond in an alkene.

[1]

- (ii)** Draw labelled diagrams to show, in terms of orbital overlap, how the σ and π bonds are made in a $\text{C}=\text{C}$ bond.

 σ bond π bond

[2]

42. M/J 24/P21/Q5(c)(i)

Context

- 5** Hydrocarbon molecules contain covalent bonds.

Context: part (c) (introduction)

- (c)** In electrophilic reactions involving alkenes the π bond of $\text{C}=\text{C}$ is broken.

- (i)** Suggest **one** difference between σ and π bonds that explains why the π bond is broken in electrophilic addition reactions involving alkenes.

[1]

43. M/J 24/P21/Q6(a)

6 (a) **V** shows stereoisomerism.

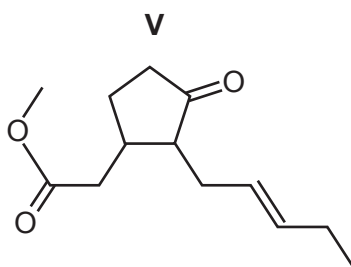


Fig. 6.1

(i) Explain what is meant by stereoisomerism.

[1]

(ii) Deduce the number of stereoisomers of **V**. Explain your reasoning.

[2]

(iii) Deduce the molecular formula of **V**.

[1]

(iv) Name **all** the functional groups present in **V**.

[1]

44. M/J 24/P22/Q4(a)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

(a) Draw the three-dimensional structures of the **two** stereoisomers of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$.
R can be used to represent $\text{CH}_3(\text{CH}_2)_5$.

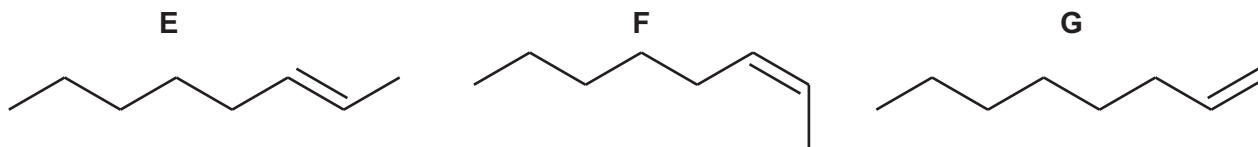
[2]

45. M/J 24/P22/Q4(e)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

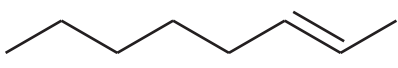
Context: part (d) (introduction)

(d) $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ is heated with **D** to produce three different molecules, **E**, **F** and **G**.**Fig. 4.2**

(e) (i) Both σ and π bonds are present in a molecule of **E** as a result of different types of hybridisation in the carbon atoms.

Complete Table 4.2 to show the number of carbon atoms with each type of hybridisation in a molecule of **E**.

Table 4.2

	number of carbon atoms		
	sp hybridised	sp ² hybridised	sp ³ hybridised
E 			

[2]

(ii) Describe the essential feature of an unbranched hydrocarbon that causes its molecules to show stereoisomerism. Explain how this feature leads to stereoisomerism.

[3]

46. M/J 24/P23/Q4(e)(iii)

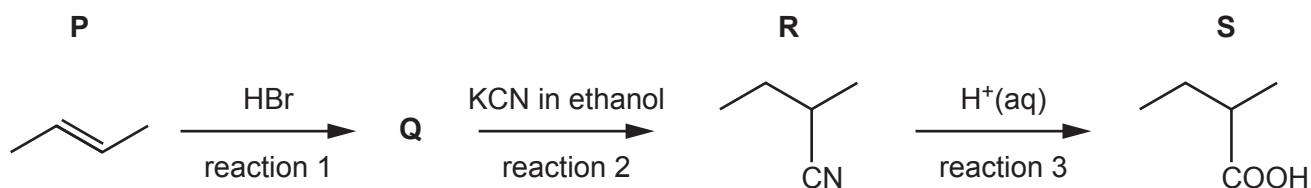
(iii) Give the systematic name for the organic product $\text{CH}_3\text{CH}(\text{NH}_2)\text{C}_2\text{H}_5$.

[1]

47. MAR 24/P22/Q4(a)

Context

4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

**Fig. 4.1**

(a) **P**, **Q**, **R** and **S** show stereoisomerism.

Complete Table 4.1 by identifying with a tick (✓) the type of stereoisomerism that each molecule shows.

The type of stereoisomerism shown by **Q** is given.

Table 4.1

	P	Q	R	S
geometrical isomerism				
optical isomerism		✓		

[2]

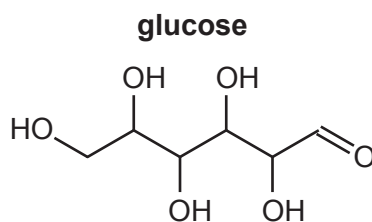
48. O/N 23/P21/Q3(e)(iii)

Context

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

Context: part (e) (introduction)

(e) The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.

**Fig. 3.2**

(iii) There are many structural isomers of $\text{C}_6\text{H}_{12}\text{O}_6$.

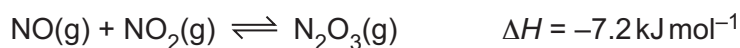
Define structural isomers.

[1]

49. O/N 23/P22/Q2(e)(i)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

Context: part (e) (introduction)

- (e)** NO₂ is a brown gas that can be used to form nitric acid.

- (i)** NO₂ is a free radical.

Define free radical.

[1]

50. O/N 23/P22/Q4(a)(i)-(ii)

Context

- 4** Lactic acid, CH₃CH(OH)COOH, and pyruvic acid, CH₃COCOOH, both contain two functional groups.

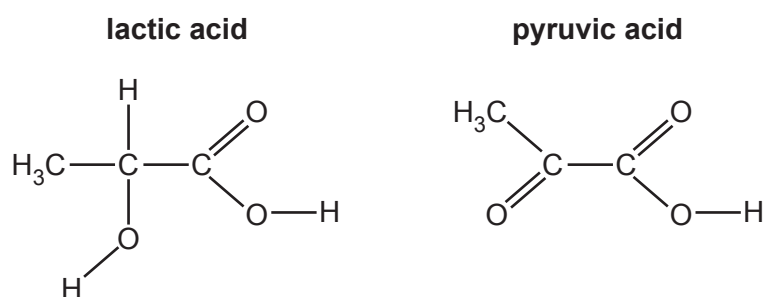


Fig. 4.1

- (a) (i)** Explain why lactic acid exists as optical isomers.

[1]

- (ii)** Give the systematic name of lactic acid.

[1]

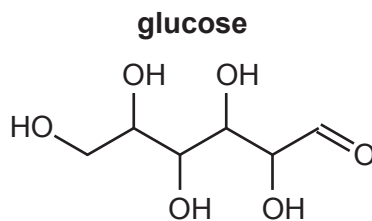
51. O/N 23/P23/Q3(e)(iii)

Context

3 Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

Context: part (e) (introduction)

(e) The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.

**Fig. 3.2**

(iii) There are many structural isomers of $\text{C}_6\text{H}_{12}\text{O}_6$.

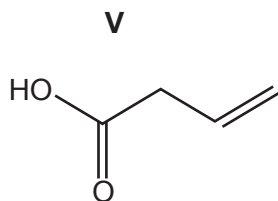
Define structural isomers.

[1]

52. M/J 23/P21/Q4(b)(iii)

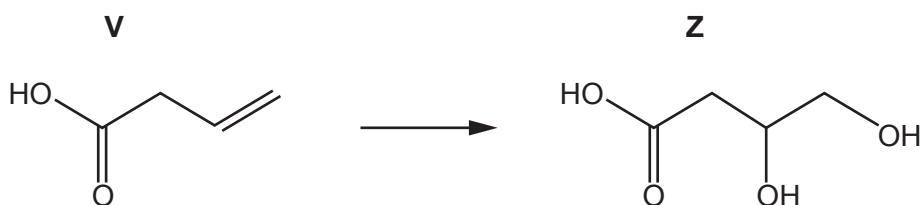
Context

4 **V** is a colourless liquid.

**Fig. 4.1**

Context: part (b) (introduction)

(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.

**Fig. 4.2**

(iii) Complete Table 4.1 to show the number of sp^2 and sp^3 hybridised carbon atoms that are present in a molecule of **V**.

Table 4.1

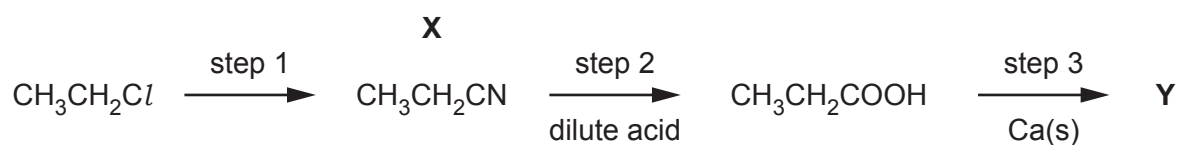
type of hybridisation	sp^2	sp^3
number of carbon atoms in V		

[2]

53. M/J 23/P22/Q3(a)(ii)

Context

- 3 Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

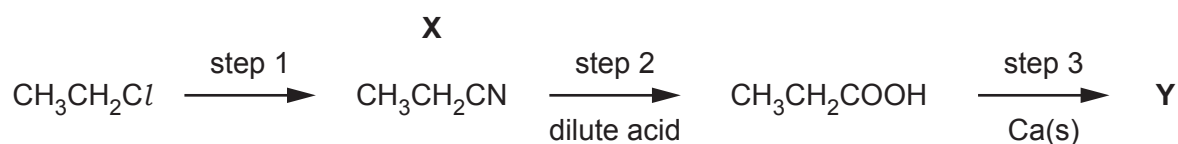
- (ii) Give the systematic name of **X**.

[1]

54. M/J 23/P22/Q3(c)

Context

- 3 Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

- (c) Complete Table 3.1 to show the number of sigma bonds (σ) and pi bonds (π) present in a molecule of **X**.

Table 3.1

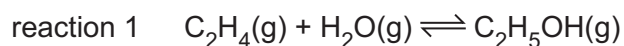
type of bond	number of bonds in X
sigma (σ)	
pi (π)	

[2]

55. M/J 23/P22/Q4(d)

Context

- 4** In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



- (d)** Describe the covalent bonds present between the carbon atoms in an ethene molecule by completing Table 4.2.

Table 4.2

	sigma (σ)	pi (π)
type of orbitals involved in bond		
how the orbitals overlap		

[2]

56. M/J 23/P22/Q5(a)

- 5** **(a)** Describe structural isomerism.

[1]

57. M/J 23/P22/Q5(c)

(c) **C** is a structural isomer of **A** and **B**.

C is straight chained and has two functional groups.

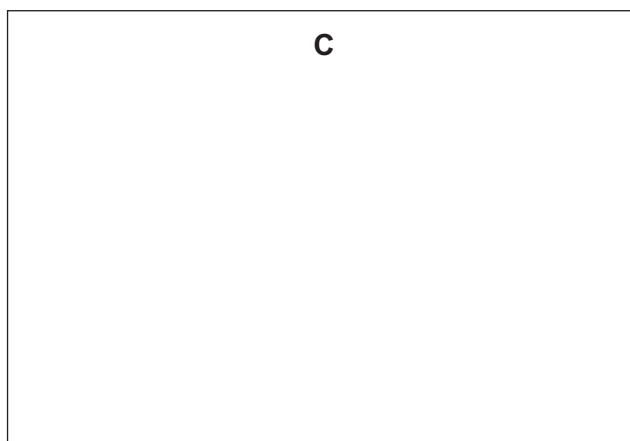
C shows only one type of stereoisomerism.

Table 5.2 describes observations when separate samples of **C** are added to different reagents.

Table 5.2

reagent	C
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	no reaction
Br ₂ (aq)	orange to colourless
alkaline I ₂ (aq)	yellow precipitate appears

(i) Draw the structure of **C** in the box.



[2]

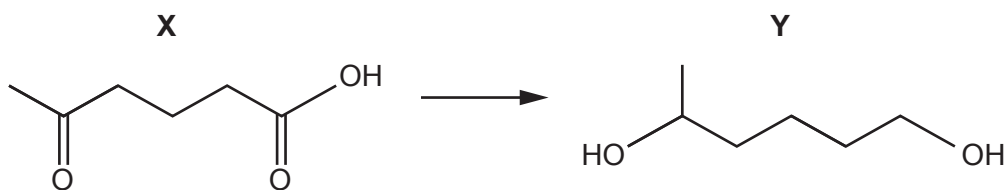
(ii) Name the type of stereoisomerism shown by molecules of **C**.

[1]

58. M/J 23/P23/Q5(c)

Context

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.

**Fig. 5.1**

(c) Complete Table 5.1 to show the number of sp^2 and sp^3 hybridised carbon atoms in a molecule of **X**.

Table 5.1

type of hybridisation	sp^2	sp^3
number of carbon atoms in X		

[2]

59. M/J 23/P23/Q6(c)

Context

6 Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

(c) Molecules of **W**, $\text{CH}_2=\text{CHCN}$, do **not** show stereoisomerism.

(i) Describe stereoisomerism.

[1]

(ii) Describe the **two** essential features of an alkene molecule that cause it to show geometrical stereoisomerism.

[2]

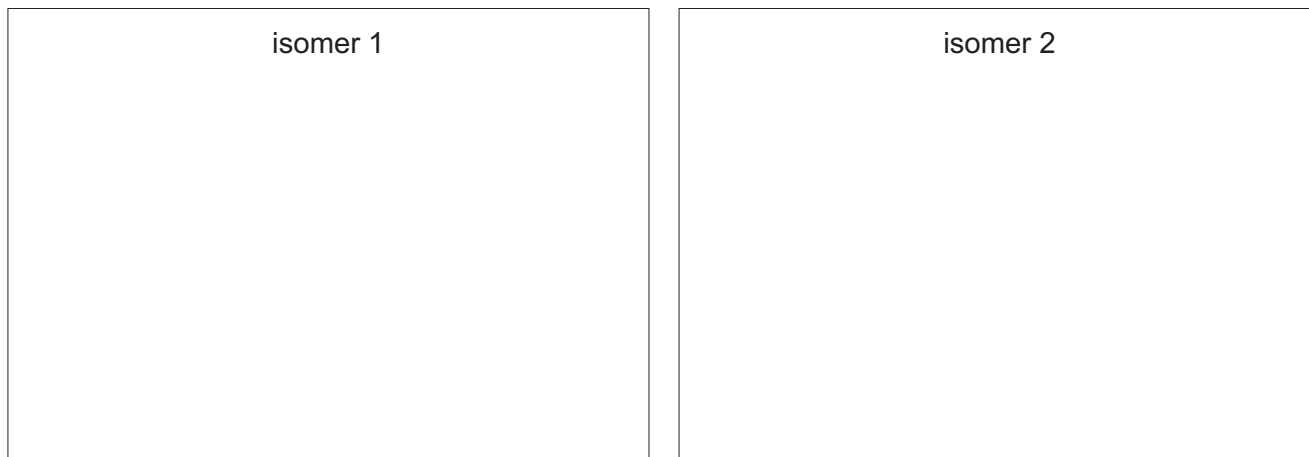
60. M/J 23/P23/Q6(d)

Context

6 Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

(d) Molecules of $\text{CH}_3\text{CH}(\text{OH})\text{CN}$ exist as a pair of optical isomers.

Draw three-dimensional diagrams in the boxes to show the optical isomers of $\text{CH}_3\text{CH}(\text{OH})\text{CN}$.



[1]

61. MAR 23/P22/Q3(a)(i)-(iii)

Context

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2 .

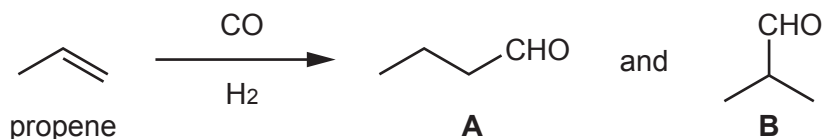


Fig. 3.1

(a) (i) Define addition reaction.

[1]

(ii) Aldehydes **A** and **B** are structural isomers.

State the type of structural isomerism shown by **A** and **B**.

[1]

(iii) Name **A**.

[1]

62. MAR 23/P22/Q3(c)

Context

- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H₂ to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H₂.

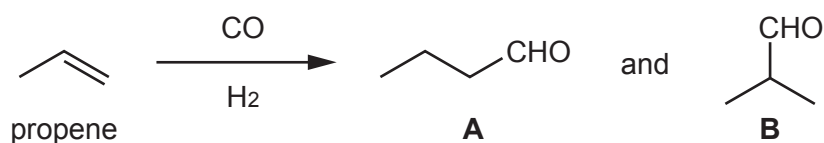


Fig. 3.1

- (c)** When propene reacts with CO and an excess of H₂, an alkane and a mixture of alcohols are formed instead. The alcohols are isomers of each other.

Suggest the molecular formulae of the alkane and the alcohols that are formed under these conditions.

molecular formula of alkane

molecular formula of alcohols

[2]

63. O/N 22/P22/Q3(a)(ii)-(iv)

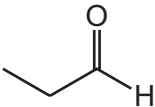
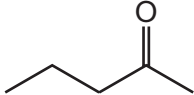
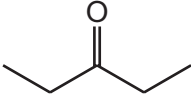
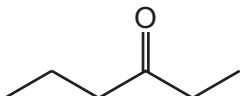
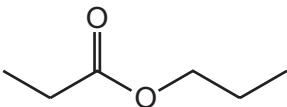
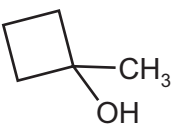
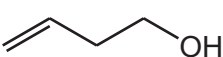
Context

3 Organic compounds can be distinguished using chemical tests and analytical techniques.

Context: part (a) (introduction)

(a) Table 3.1 shows four pairs of organic compounds.

Table 3.1

organic compounds		reagent	positive result of chemical test on identified compound
A1 	A2 		
B1 	B2 		
C1 	C2 		
D1 	D2 		

(ii) **A1** and **A2** are structural isomers.

Define structural isomers.

[1]

(iii) Give the systematic name of **B2**.

[1]

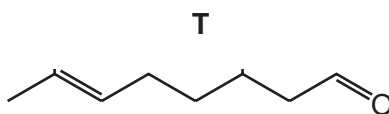
(iv) Deduce the molecular formula of **D1**.

[1]

64. M/J 22/P21/Q3(a)

Context

- 3** Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

- (a)** Molecules of **T** exist as a pair of stereoisomers.

Name the type of stereoisomerism shown by molecules of **T**. Explain your answer.

[2]

65. M/J 22/P21/Q4(c)(iii)

Context

4 Compound **V** is a liquid.**V** contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.**V** has a relative molecular mass of 280.

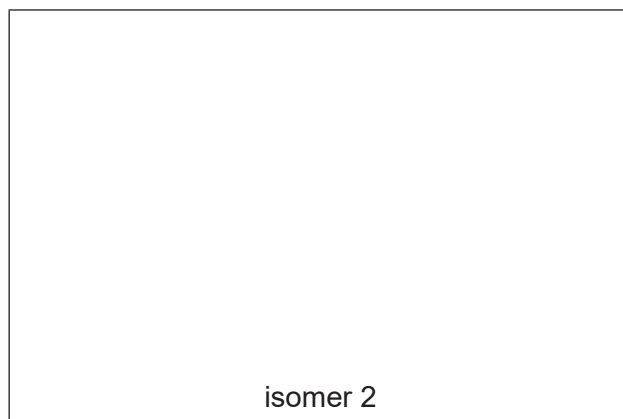
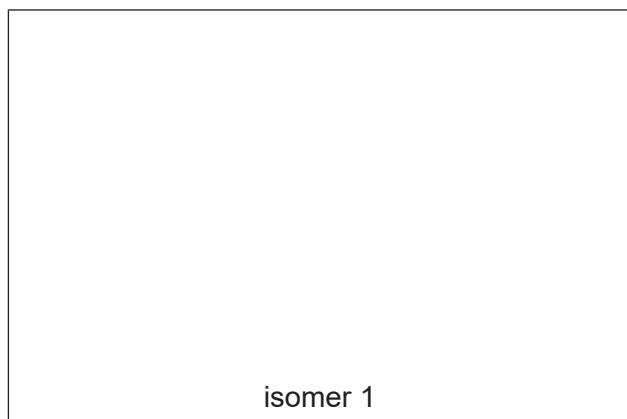
Context: part (c) (introduction)

(c) **W**, **X** and **Y** have the same molecular formula, $C_5H_{10}O$.**W**, **X** and **Y** are added separately to different reagents. Observations for these reactions are described in Table 4.1.**Table 4.1**

	+ 2,4-dinitrophenylhydrazine	+ alkaline $I_2(aq)$	+ Fehling's reagent and warm
W	orange precipitate seen	no change	orange-red precipitate seen
X	orange precipitate seen	yellow precipitate seen	no change
Y	orange precipitate seen		

(iii) **W** could be one of four structural isomers.

- Draw the skeletal formulae for two possible structural isomers of **W**.
- Describe the type of structural isomerism shown.



type of structural isomerism

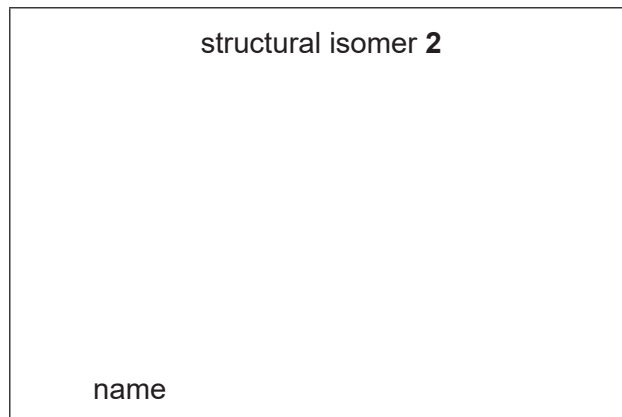
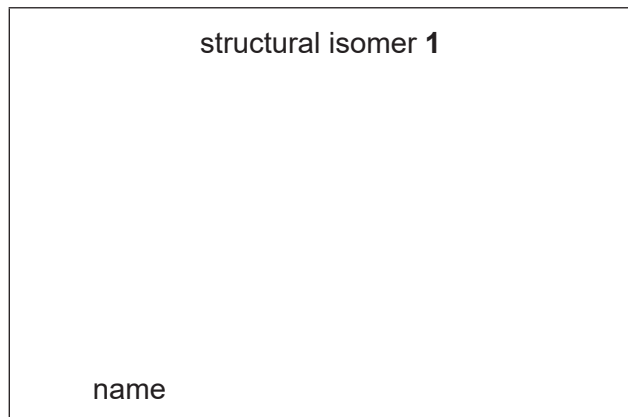
[3]

66. M/J 22/P22/Q4(a)

- 4 (a) Bromine reacts with butane in the presence of ultraviolet light to form bromobutane.

Two structural isomers with the molecular formula C_4H_9Br are produced during this reaction.

- (i) Draw the two structural isomers and state the systematic name of each isomer.



[2]

- (ii) Identify the type of structural isomerism shown in (a)(i).

[1]

67. M/J 22/P22/Q4(b)(i)-(ii)

Context: part (b) (introduction)

- (b) Halothane is an anaesthetic.

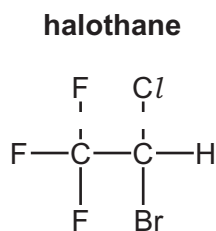


Fig. 4.1

- (i) Identify the chiral centre in halothane and mark it with an asterisk (*). [1]

When halothane reacts in ultraviolet light, homolytic fission occurs and the C–Br bond is broken.

- (ii) Construct an equation to show the homolytic fission of halothane, $CF_3CHBrCl$.

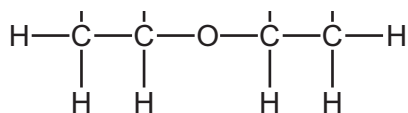
[1]

68. M/J 22/P23/Q3(b)

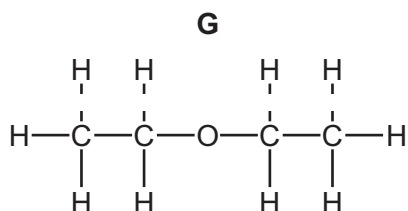
Context

- 3 G belongs to a group of compounds called ethers.



**Fig. 3.1**

- (b) The functional group in **G** is an oxygen atom bonded to two carbon atoms.

**Fig. 3.3**

G, **H** and **J** are structural isomers with molecular formula $\text{C}_4\text{H}_{10}\text{O}$.

H and **J** are straight chain molecules.

Table 3.3 shows the boiling points and reactions of **G**, **H** and **J** when heated under reflux with excess acidified $\text{K}_2\text{Cr}_2\text{O}_7$.

Table 3.3

	G	H	J
boiling point/ $^{\circ}\text{C}$	35	117	98
heat under reflux with excess acidified $\text{K}_2\text{Cr}_2\text{O}_7$	remains orange	orange to green	orange to green

- (i) Identify the type of structural isomerism shown between **G** and **H** using the information in Table 3.3.

[1]

- (ii) Identify the type of structural isomerism shown between **H** and **J** using the information in Table 3.3.

[1]

- (iii) Draw a possible structure for **H** and for **J**. State the systematic name for each structure.

H

name

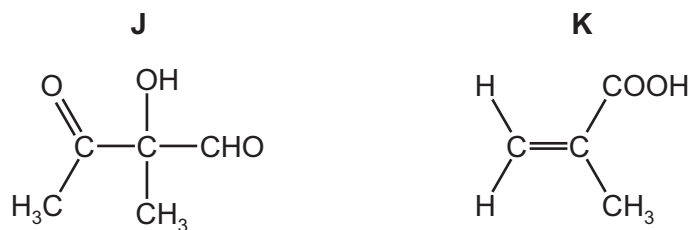
J

name

[3]

69. MAR 22/P22/Q4(a)(ii)

Context

4 Compounds **J** and **K** are found in plant oils.**Fig. 4.1****(ii)** **J** has **two** optical isomers.Draw the three-dimensional structures of the **two** optical isomers of **J**.

[2]

Answers Section

1. M/J 26/P21/Q5(a)

5(a)	compound	type of hybridisation of carbon atoms	2	M1 A sp ³ sp ₃ SP ³ etc. M2 A sp ² sp ₂ SP ² etc.
	1,2 dichloroethane	sp ³		
	chloroethene	sp ²		

2. M/J 26/P21/Q5(c)

5(c)	One atom takes both of the electrons when the covalent bond is broken ✓✓ M1 when a (covalent) bond / it is broken M2 one atom takes both of the electrons from the shared pair	2	A breaking = splitting = separation A breaking of a specific covalent bond e.g. when a CH bond is broken shared pair of electrons = covalent bond M2 A 'one atom takes both electrons' if reference to process involving breaking a (covalent) bond is described A takes = receives A electrons moves to / electrons transferred to I attract / pull towards M2 A species = atom R element, molecule, compound I radical I uneven number of electrons I formation of (positive and negative) ions I both electrons unequally distributed to one atom I formation of nucleophile and electrophile
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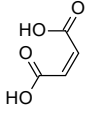
3. M/J 26/P22/Q5(a)

5(a)	structure of product	systematic name of product	2	Max 1 for consistently incorrect alkane stem that has correct idea of 1-chloro and 2-chloro. Max 1 6-chlorohexane and 5-chlorohexane. Max 1 for consistent errors e.g. 1-chlorohexane and 2-chlorohexane OR 1-chlorohexane and 2-chlorohexane OR chloro-1-hexane and chloro-2-hexane A missing hyphen
	CH ₃ (CH ₂) ₅ Cl	1-chlorohexane		
	CH ₃ (CH ₂) ₃ CHClCH ₃	2-chlorohexane		
	CH ₃ (CH ₂) ₃ CH ₂ ClCH ₂ Cl	1,2 dichlorohexane		
	CH ₃ (CH ₂) ₁₀ CH ₃	dodecane		
	CH ₃ (CH ₂) ₃ CH(CH ₃)CH(CH ₃)(CH ₂) ₃ CH ₃	5,6-dimethyldecane		

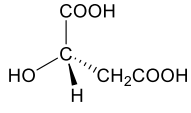
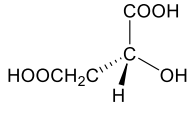
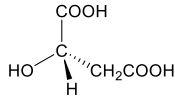
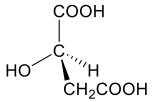
4. M/J 26/P22/Q5(b)(i)-(ii)

5(b)(i)	(species) with one or more unpaired electrons	1	BOD species with (an) unpaired electron(s) I lone / single / free / odd number of electrons (in valence shell) I species with specified number (except 1) unpaired electrons BOD species with one or two unpaired electrons R it is an unpaired electron BOD species = atom, ion, particle, molecule, substance, element, compound
5(b)(ii)	homolytic (fission)	1	BOD homolytic A homolysis I covalent I photolytic

5. M/J 26/P22/Q6(a)

6(a)(i)		1	A any unambiguous cis / Z structure of P I connectivity I trans structure
6(a)(ii)	M1 limited / restricted / lack of / no rotation about the pi bond / π bond of C=C M2 (two) different groups attached to each C of the C=C / each end of the C=C	2	M1 A limited / restricted / lack of / no rotation about the C=C reference to C=C needs to be made once only BOD C–C double bond I double bond (there is a C=O also in the molecule) I movement rotate = twist = spin R (compound) P has a chiral carbon M2 each = both A different atoms / functional groups / H and COOH / species I each side (of the C=C) R different compounds / molecules

6. M/J 26/P22/Q6(b)(ii)

6(b)(ii)	M1  M2 	2	COOH = CO ₂ H A C ₂ H ₃ O ₂ M1 4 correct groups around central C in any orientation (might be seen on 'isomer 2') M2 enantiomer of their M1 structure (does not have to be shown as its mirror image).   M2 R bonds in the plane that are opposite to each other I bond shapes that are not associated with the chiral carbon. I bond angles I connectivity
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7. M/J 26/P23/Q4(c)(i)-(ii)

4(c)(i)	M1 (molecules / isomers with) the same molecular formula / same number of atoms of each element M2 different arrangement of atoms	2	A similar = same M1 A same molecular formula I same chemical formulas I same / similar molecular number M2 A different structures M2 A different structural formulas / different displayed formulas / different skeletal formulas I arrangement of bonds / different arrangements in space I different carbon skeleton / functional groups / chain
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4(c)(ii)	M1 compound W AND compound V M2 both have (2) chiral centres / (2) asymmetric carbon atoms	2 M1 compound W AND compound V ✓ M2 one or two chiral centre on V & W ✓ M1 compound V OR compound W M2 show one or two chiral centre on the isomer identified ^✓ R molecules / compounds for atoms / groups R C attached to 4 different alkyl groups M2 A they have a chiral centre / an asymmetric carbon (atom) chiral = 4 different groups / atoms bonded to a carbon M2 may be indicated by annotation on Figure 4.2 e.g. CH₂OHCH₂ *CH(OH) and / or *CH(OH)CH₂OH
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8. M/J 26/P23/Q5(a)(i)

5(a)(i)	structure of product	systematic name of product	2	A any unambiguous structures
	$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1 chloropropane		
	$\text{CH}_3\text{CHCl} / \text{CH}_3$	2-chloropropane		
	$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,4-dimethylbutane		
	$\text{CH}_3(\text{CH}_2)_4\text{CH}_3$	hexane		
	$\text{CH}_3(\text{CH}_2)_2\text{CH}(\text{CH}_3)_2$	2-methylpentane		

9. M/J 26/P24/Q5(c)

5(c)	positional	1 A regoisomerism A position (isomer)
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10. MAR 26/P22/Q2(a)(i)

2(a)(i)	(free) radical	1
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11. MAR 26/P22/Q3(b)

3(b)(i)		1
3(b)(ii)	M1 M2 <i>trans</i>-but-2-ene	2
3(b)(iii)	M1 (two) different groups attached to each carbon of C=C M2 restricted / lack of / no rotation about double / C=C / pi / π bond	2

12. MAR 26/P22/Q4(d)

4(d)(i)	condensation	1
4(d)(ii)	methanal	1
4(d)(iii)	C ₆ H ₇ NO ₂	1
4(d)(iv)		1

4(d)(v)	• $\times$ on absorption at $\sim 1550\text{ cm}^{-1}$ • $\text{C}=\text{C}$ • (peak / $\text{C}=\text{C}$ / bond) disappears (during / after polymerisation)	2
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13. O/N 25/P21/Q3(b)

3(b)	two of: 	2
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14. O/N 25/P21/Q3(c)(i)

3(c)(i)	homolytic	1
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15. O/N 25/P21/Q4(b)(ii)

4(b)(ii)	M1 (molecules that have the) same structural formula (and same molecular formula) M2 but different 3D / spatial arrangement of atoms / groups	2
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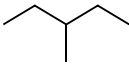
16. O/N 25/P21/Q4(c)(i)

4(c)(i)	2,3-dimethylbut-1-ene	1
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17. O/N 25/P22/Q3(a)

3(a)	M1 (family of molecules / compounds / series with) same general formula OR (family of molecules where) molecular formula differs (only) by (units of) $\text{CH}_{2(n)}$. M2 similar chemical properties / reactions	2
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18. O/N 25/P22/Q3(d)(i)

3(d)(i)	M1  M2 	2
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19. O/N 25/P22/Q4(a)(iii)-(v)

4(a)(iii)	homolytic	1
4(a)(iv)	1,2-dichloropropane	1
4(a)(v)	M1 (type of structural isomerism) positional M2 (type of stereoisomerism) optical	2

20. O/N 25/P23/Q3(b)

3(b)	two of: 	2
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21. O/N 25/P23/Q3(c)(i)

3(c)(i)	homolytic	1
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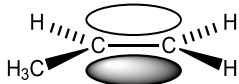
22. O/N 25/P23/Q4(b)(ii)

4(b)(ii)	M1 (molecules that have the) same structural formula (and same molecular formula) M2 but different 3D / spatial arrangement of atoms / groups	2
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23. O/N 25/P23/Q4(c)(i)

4(c)(i)	2,3-dimethylbut-1-ene	1
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24. O/N 25/P24/Q3(a)(iv)

3(a)(iv)		1
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25. M/J 25/P22/Q3(a)(i)

3(a)(i)	two (or more) reactants / molecules react to make (only) one product	1
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26. M/J 25/P22/Q4(c)(i)

4(c)(i)	sp ³	1
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27. M/J 25/P23/Q5(b)(i)

5(b)(i)	number of σ bonds = 14 number of π bonds = 2	2
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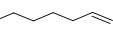
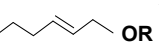
28. M/J 25/P23/Q6(a)(i)

6(a)(i)	two or more reactants / molecules react to make (only) one product	1
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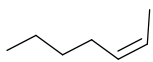
29. M/J 25/P24/Q4(a)

4(a)	C ₅ H ₁₀ O ₂	1
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30. M/J 25/P24/Q5(a)-(b)

5(a)	M1  M2  OR 	2
5(b)(i)	no rotation / restricted rotation of C=C / pi bond	1
5(b)(ii)	(W / it has two) different groups bonded to each C of the double bond	1

31. M/J 25/P24/Q5(c)

5(c)		1
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32. M/J 25/P24/Q6(a)(iii)

6(a)(iii)	alkene(s)	1
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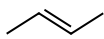
33. MAR 25/P22/Q1(c)(i)

1(c)(i)	contains a(/ one or more) C=C / double / multiple bonds	1
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34. MAR 25/P22/Q4(b)(iii)

4(b)(iii)	atom / carbon with four different atoms / (functional) groups attached / bonded	1
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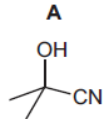
35. O/N 24/P21/Q1(d)(ii)-(iv)

1(d)(ii)	molecules with the same structural formula and same molecular formula with different 3D / different spatial arrangement of atoms / groups	1
1(d)(iii)	a single product is made	1
1(d)(iv)		1
		1

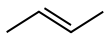
36. O/N 24/P21/Q2(b)(i)

2(b)(i)	a species with one or more unpaired electrons	1
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37. O/N 24/P21/Q4(a)(ii)

4(a)(ii)	A's central carbon atom is not bonded to four different atoms/groups of atoms 	1
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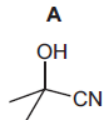
38. O/N 24/P23/Q1(d)(ii)-(iv)

1(d)(ii)	molecules with the same structural formula and same molecular formula with different 3D / different spatial arrangement of atoms / groups	1
1(d)(iii)	a single product is made	1
1(d)(iv)		1
		1

39. O/N 24/P23/Q2(b)(i)

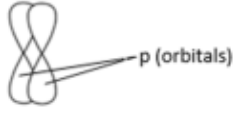
2(b)(i)	a species with one or more unpaired electrons	1
---------	---	----------

40. O/N 24/P23/Q4(a)(ii)

4(a)(ii)	A's central carbon atom is not bonded to four different atoms/groups of atoms 	1
----------	--	----------

41. M/J 24/P21/Q5(b)

5(b)(i)	sp^2	1
---------	--------	----------

5(b)(ii)	M1 σ bond – diagram showing how σ bond is made from direct overlap of orbitals i.e.  OR  M2 π bond - diagram showing how π bond is made from sideways overlap of 2$\times$ p-orbitals i.e.  OR  OR  sideways overlap	2
----------	--	---

42. M/J 24/P21/Q5(c)(i)

5(c)(i)	EITHER (pair of) electrons in π bond are further away from the nuclei so weaker attraction OR (pair of) electrons in σ bond are closer to the two nuclei so stronger attraction	1
---------	--	---

43. M/J 24/P21/Q6(a)

6(a)(i)	(molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	1
6(a)(ii)	M1 number of stereoisomers = 8 M2 reasoning: 2 chiral centres AND 1 C=C (producing a cis-and trans or geometrical pair)	2
6(a)(iii)	C ₁₃ H ₂₀ O ₃	1
6(a)(iv)	ester AND carbonyl / ketone AND C=C (bond) / alkene	1

44. M/J 24/P22/Q4(a)

4(a)	Three-dimensional stereoisomers of CH₃(CH₂)₅CHBrCH₃ M1 correct representation of left-hand answer as 3d structure of one of the enantiomers of CH₃(CH₂)₅CHBrCH₃ M2 correct representation of 3d structure of the second enantiomer of CH₃(CH₂)₅CHBrCH₃	2
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45. M/J 24/P22/Q4(e)

4(e)(i)	 <table border="1" data-bbox="411 1626 660 1704"> <tr> <td>0</td> <td>2</td> <td>6</td> </tr> </table> 3 correct answers = 2 marks 2 correct answers = 1 mark 0 or 1 correct answers = 0 marks	0	2	6	2
0	2	6			
4(e)(ii)	M1 presence of a π bond M2 limited rotation of C=C bond M3 EITHER two different groups on each / both double bonded carbon atoms OR two different atoms on each / both double bonded carbon atom OR one H / atom and one alkyl group on each / both double bonded carbon atom	3			

46. M/J 24/P23/Q4(e)(iii)

4(e)(iii)	butan-2-amine OR butyl-2-amine	1
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47. MAR 24/P22/Q4(a)

4(a)	<table><tr><td>molecule</td><td>P</td><td>Q</td><td>R</td><td>S</td></tr><tr><td>geometrical</td><td>✓</td><td>(✗)</td><td>(✗)</td><td>(✗)</td></tr><tr><td>optical</td><td>(✗)</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	molecule	P	Q	R	S	geometrical	✓	(✗)	(✗)	(✗)	optical	(✗)	✓	✓	✓	1
	molecule	P	Q	R	S												
	geometrical	✓	(✗)	(✗)	(✗)												
	optical	(✗)	✓	✓	✓												
<table><tr><td>molecule</td><td>P</td><td>Q</td><td>R</td><td>S</td></tr><tr><td>geometrical</td><td>✓</td><td>(✗)</td><td>(✗)</td><td>(✗)</td></tr><tr><td>optical</td><td>(✗)</td><td>✓</td><td>✓</td><td>✓</td></tr></table> CH ₃ CHBrCH ₂ CH ₃	molecule	P	Q	R	S	geometrical	✓	(✗)	(✗)	(✗)	optical	(✗)	✓	✓	✓	1	
molecule	P	Q	R	S													
geometrical	✓	(✗)	(✗)	(✗)													
optical	(✗)	✓	✓	✓													

48. O/N 23/P21/Q3(e)(iii)

3(e)(iii)	molecules with the same molecular formula but different structural formulae	1
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49. O/N 23/P22/Q2(e)(i)

2(e)(i)	species with 1 or more unpaired electrons	1
---------	---	----------

50. O/N 23/P22/Q4(a)(i)-(ii)

4(a)(i)	(it has molecules that) contain a chiral carbon / centre OR are non-superimposable mirror images	1
4(a)(ii)	2-hydroxypropanoic acid	1

51. O/N 23/P23/Q3(e)(iii)

3(e)(iii)	molecules with the same molecular formula but different structural formulae	1
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52. M/J 23/P21/Q4(b)(iii)

4(b)(iii)	type of hybridisation	sp ²	sp ³	2
	number of carbon atoms in V	3	1	

53. M/J 23/P22/Q3(a)(ii)

3(a)(ii)	propanenitrile	1
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54. M/J 23/P22/Q3(c)

3(c)	type of bond in X	number of bonds in X	2
	sigma 	8	
	pi 	2	

55. M/J 23/P22/Q4(d)

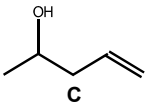
4(d)		sigma	pi	2
	type of orbitals involved in bond	$sp^2(-sp^2)$	$(2)p(-(2)p)$	
	how the orbitals overlap	direct	sideways	

Two correct for one mark, four correct for two marks.

56. M/J 23/P22/Q5(a)

5(a)	molecules / isomers with the same molecular formula / same number of atoms of each element AND different structural formulae / different structures	1
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57. M/J 23/P22/Q5(c)

5(c)(i)	 M1 $CH_3CH(OH)$ present M2 five carbons in a straight chain AND molecular formula $C_5H_{10}O$ AND 1 $C=C$ structure which shows only one type of stereoisomerism	2
5(c)(ii)	optical	1

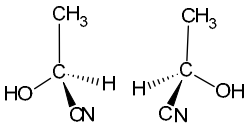
58. M/J 23/P23/Q5(c)

5(c)	type of hybridisation	sp^2	sp^3	2
	number of carbon atoms in X	2	4	

59. M/J 23/P23/Q6(c)

6(c)(i)	(molecules / compounds with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	1
6(c)(ii)	M1 restricted rotation of $C=C$ (bond) M2 each C of the double bond has two different groups / two different atoms (attached)	2

60. M/J 23/P23/Q6(d)

6(d)	 A pair of molecules of 2-hydroxypropanenitrile with bonds represented in 3-d AND the pair of molecules are non-superimposable images	1
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61. MAR 23/P22/Q3(a)(i)-(iii)

3(a)(i)	(a reaction where) two (or more) compounds / reagents / molecules form (only) one product	1
3(a)(ii)	position(al isomerism)	1
3(a)(iii)	butanal	1

62. MAR 23/P22/Q3(c)

3(c)	alkane	C_3H_8	1
	alcohols	$C_4H_{10}O$	1

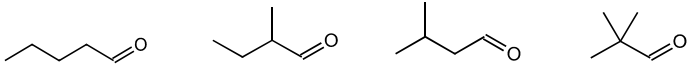
63. O/N 22/P22/Q3(a)(ii)-(iv)

3(a)(ii)	molecules with the same molecular formula AND different structural formulae <i>owtte</i>	1
3(a)(iii)	pentan-3-one	1
3(a)(iv)	<u>molecular</u> formula $C_5H_{10}O$	1

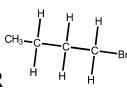
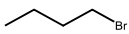
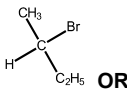
64. M/J 22/P21/Q3(a)

3(a)	M1 optical M2 one of the C atoms has 4 different groups / atoms attached	2
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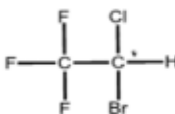
65. M/J 22/P21/Q4(c)(iii)

4(c)(iii)	M1 / M2  M3 chain (isomerism)	3
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66. M/J 22/P22/Q4(a)

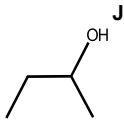
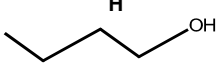
4(a)(i)	$CH_3(CH_2)_2CH_2Br$  OR  name: 1-bromobutane $CH_3CH_2CH(Br)CH_3$  OR  name: 2-bromobutane M1 structure of isomer 1 AND name M2 structure of isomer 2 AND name (in any order)	2
4(a)(ii)	positional	1

67. M/J 22/P22/Q4(b)(i)-(ii)

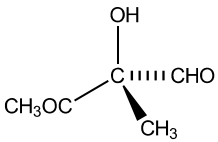
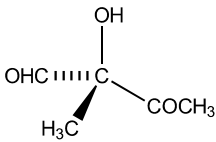
4(b)(i)		1
4(b)(ii)	$CF_3CBrClH \rightarrow CF_3CHCl(\bullet) + Br(\bullet)$	1

68. M/J 22/P23/Q3(b)

3(b)(i)	functional group	1
3(b)(ii)	positional	1

3(b)(iii)	 name: butan-2-ol  name: butan-1-ol	3
M1 correct structures M2 correct structures in correct boxes H and J M3 correct corresponding name of both H and J		

69. MAR 22/P22/Q4(a)(ii)

4(a)(ii)	 	2
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Unit 14

HYDROCARBONS

54 items · 141 marks

Syllabus sections in this unit

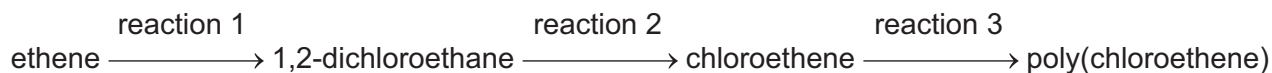
14.1 Alkanes

14.2 Alkenes

Unit 14: Hydrocarbons**1. M/J 26/P21/Q5(b)**

Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

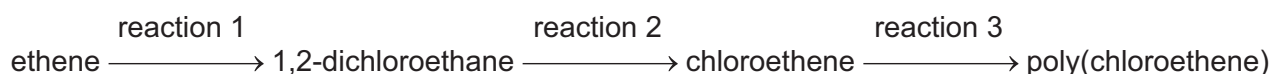
- (b)** In reaction 1, ethene reacts with $\text{Cl}_2(\text{g})$ at room temperature.
Name the mechanism for reaction 1.

[1]

2. M/J 26/P21/Q5(d)

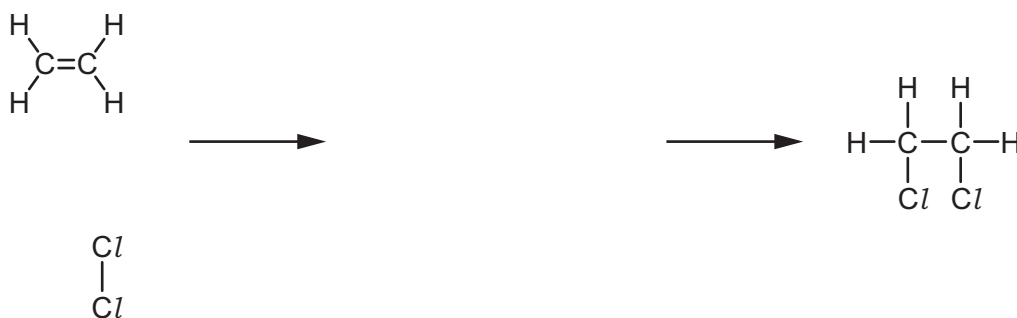
Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

- (d)** Complete Figure 5.2 to show the mechanism for the reaction of ethene with Cl_2 in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Figure 5.2**

[4]

3. M/J 26/P22/Q5(b)(iii)-(iv)

Context

- 5** Hexane, $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_5\text{Cl}$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_2\text{Cl}$	1,2-dichlorohexane
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$	dodecane
$\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$	5,6-dimethyldecane

Context: part (b) (introduction)

- (b)** The mechanism for the reaction of hexane with $\text{Cl}_2(\text{g})$ involves three stages.

stage 1 initiation

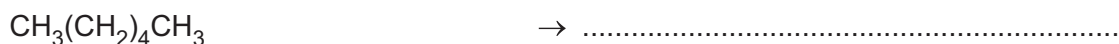
stage 2 propagation

stage 3 termination

In stage 1 and 2, free radicals are produced.

- (iii)** Suggest a reaction sequence of **two** propagation reactions that result in $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$ forming $\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$.

propagation reaction 1



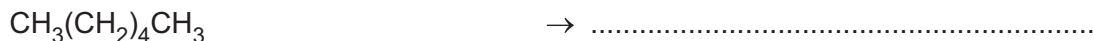
propagation reaction 2



[2]

- (iv)** Suggest a reaction sequence of **one** propagation reaction followed by a termination reaction that results in $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$ forming $\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$.

propagation reaction



termination reaction



[2]

4. M/J 26/P22/Q5(c)

Context

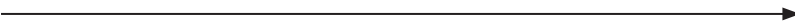
- 5** Hexane, $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_5\text{Cl}$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_3$	
$\text{CH}_3(\text{CH}_2)_3\text{CHClCH}_2\text{Cl}$	1,2-dichlorohexane
$\text{CH}_3(\text{CH}_2)_{10}\text{CH}_3$	dodecane
$\text{CH}_3(\text{CH}_2)_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)(\text{CH}_2)_3\text{CH}_3$	5,6-dimethyldecane

- (c) The proportion of each product varies depending on the type of free radical intermediate that is produced.

Table 5.2

primary free radical	secondary free radical	tertiary free radical
$\bullet\text{C}(\text{CH}_3)\text{H}_2$	$\bullet\text{C}(\text{CH}_3)_2\text{H}$	$\bullet\text{C}(\text{CH}_3)_3$
 increased stability of radical		

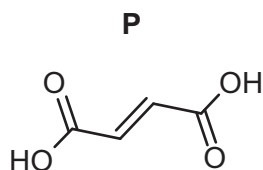
Use Table 5.2 to suggest which factor determines the increased stability of free radicals from primary to secondary to tertiary.

[1]

5. M/J 26/P22/Q6(b)(i)

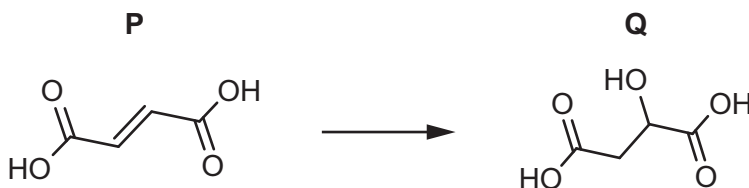
Context: part (a) (introduction)

- 6 (a)** Figure 6.1 shows the structure of compound **P**.

**Figure 6.1**

Context: part (b) (introduction)

- (b)** **P** reacts to produce compound **Q**.

**Figure 6.2**

- (i)** Identify the reagent and conditions required to convert **P** to **Q**.

reagent

conditions

[2]

6. M/J 26/P23/Q5(a)(ii)

Context

- 5** Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

- (ii)** Name the mechanism for the reaction of propane and $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light.

[1]

7. M/J 26/P23/Q5(b)(ii)-(iii)

Context

- 5** Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

Context: part (b) (introduction)

- (b)** The mechanism for this reaction involves three stages.

stage 1 initiation

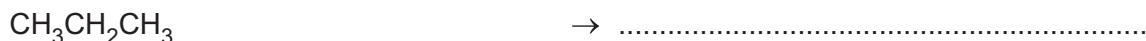
stage 2 propagation

stage 3 termination

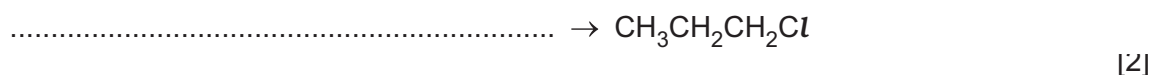
During the initiation reaction, a chlorine free radical is produced.

- (ii)** Suggest a reaction sequence of **two** propagation reactions that result in $\text{CH}_3\text{CH}_2\text{CH}_3$ forming $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$.

propagation reaction 1

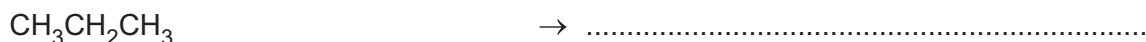


propagation reaction 2

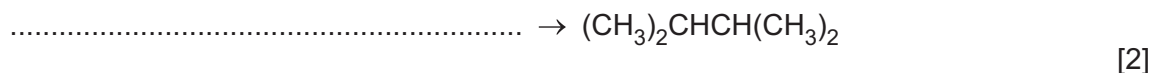


- (iii)** Suggest a reaction sequence of **one** propagation reaction followed by a termination reaction that results in $\text{CH}_3\text{CH}_2\text{CH}_3$ forming $(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$.

propagation reaction 3



termination reaction

**8. M/J 26/P24/Q5(a)**

Context

- 5** Long-chain alkanes from crude oil can be cracked to make alkenes.

- (a)** State the conditions used for the cracking of alkanes.

[1]

9. M/J 26/P24/Q5(b)(iii)

Context

5 Long-chain alkanes from crude oil can be cracked to make alkenes.

Context: part (b) (introduction)

(b) Propene is an alkene that is made by cracking. Alkenes are used in addition polymerisation.

(iii) Aqueous bromine is used to test for the presence of the C=C bond in propene.

State what is observed when aqueous bromine is mixed with propene.

[1]

10. M/J 26/P24/Q5(d)

Context

5 Long-chain alkanes from crude oil can be cracked to make alkenes.

(d) But-2-ene reacts with HBr(g). The HBr acts as an electrophile in the reaction.

(i) State how HBr acts as an electrophile in terms of electron transfer.

[1]

(ii) Complete Figure 5.1 to show the mechanism for the reaction of but-2-ene with HBr.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Figure 5.1

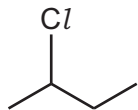
[4]

11. MAR 26/P22/Q3(c)

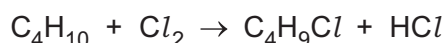
Context

3 Secondary halogenoalkanes are important building blocks in organic chemistry.

2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane**Figure 3.1**

(c) Butane, C_4H_{10} , reacts with Cl_2 in the presence of ultraviolet light to form 2-chlorobutane, which has molecular formula C_4H_9Cl .



(i) Name the mechanism of this reaction.

[1]

(ii) Complete the reaction scheme given to show the formation of 2-chlorobutane.

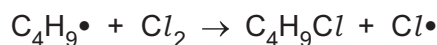
You may use molecular formulas in your answer.

initiation



propagation

propagation



termination

[2]

12. O/N 25/P21/Q3(c)(ii)-(iii)

Context

3 Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (c) (introduction)

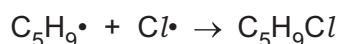
(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form C_5H_9Cl .

(ii) Complete the equations to show the two propagation steps that follow the initiation step.



[2]

(iii) The final step is shown.



Give the name for this step in the reaction.

[1]

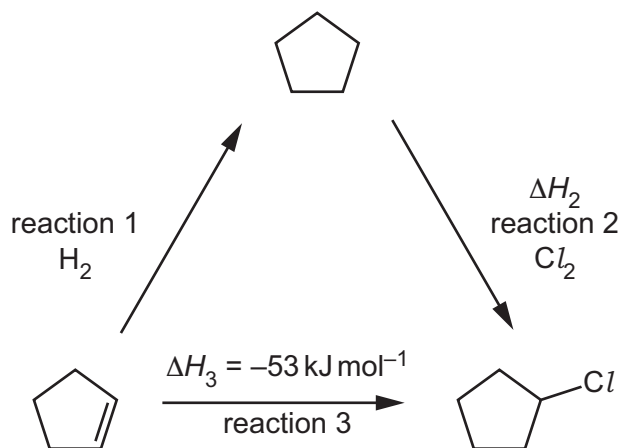
13. O/N 25/P21/Q3(d)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (d) (introduction)

- (d)** Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and C_5H_9Cl .

**Fig. 3.2**

- (i)** Identify a suitable reagent for reaction 3.

[1]

14. O/N 25/P21/Q3(e)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (e) (introduction)

- (e)** Cyclopentene, C_5H_8 , reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

- (i)** Draw the structure of **W**.

[1]

15. O/N 25/P22/Q3(b)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

- (b)** Give two reasons to explain the general unreactivity of alkanes.

1

2

[2]

16. O/N 25/P22/Q3(c)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

(c) Alkanes with low relative molecular mass, M_r , are more useful than those found in heavier crude oil fractions.

Name the process that is used to obtain alkanes with low M_r from heavier crude oil fractions.

[1]

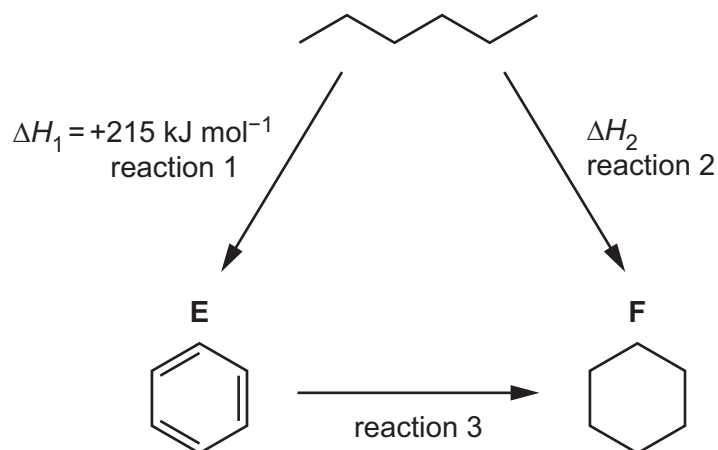
17. O/N 25/P22/Q3(e)(i)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

Context: part (e) (introduction)

(e) Hexane can be converted into compounds **E** and **F** at high temperature and pressure. Fig. 3.2 shows the reaction scheme involving hexane, **E** and **F**.

**Fig. 3.2**

- (i) Identify a suitable reagent for reaction 3.

[1]

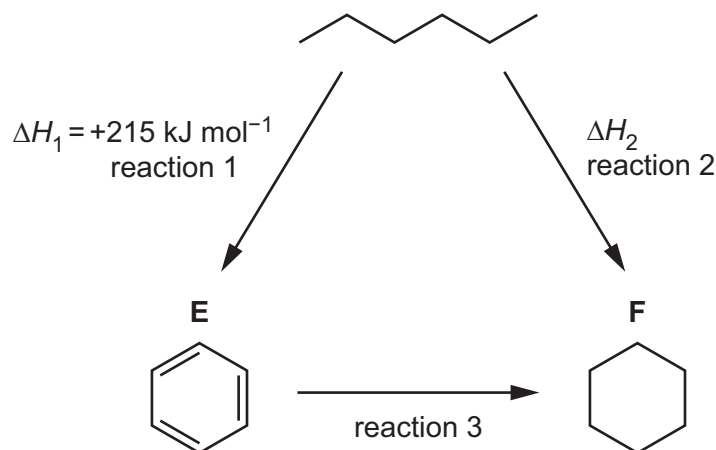
18. O/N 25/P22/Q3(e)(iii)

Context

- 3** The alkanes are a homologous series of organic molecules. Alkanes are generally unreactive and are commonly used as fuels.

Context: part (e) (introduction)

- (e)** Hexane can be converted into compounds **E** and **F** at high temperature and pressure. Fig. 3.2 shows the reaction scheme involving hexane, **E** and **F**.

**Fig. 3.2**

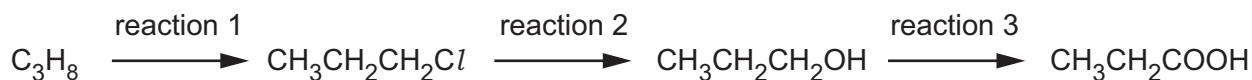
- (iii)** Write an equation for the complete combustion of hexane.

[1]

19. O/N 25/P22/Q4(a)(i)-(ii)

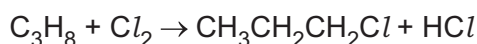
Context

- 4** Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, CH_3CH_2COOH .

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 in Fig. 4.1 takes place in the presence of sunlight.



The reaction takes place via initiation, propagation and termination steps.

- (i)** Name the mechanism shown by reaction 1.

[1]

- (ii)** Complete the mechanism for reaction 1.

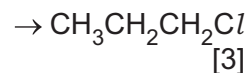
Construct equations to describe the steps of the mechanism.



propagation 1

propagation 2

termination



[3]

20. O/N 25/P23/Q3(c)(ii)-(iii)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (c) (introduction)

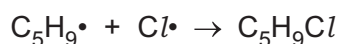
(c) Cyclopentane reacts with Cl_2 in the presence of ultraviolet light to form $\text{C}_5\text{H}_9\text{Cl}$.

- (ii) Complete the equations to show the two propagation steps that follow the initiation step.



[2]

- (iii) The final step is shown.



Give the name for this step in the reaction.

[1]

21. O/N 25/P23/Q3(d)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (d) (introduction)

(d) Fig. 3.2 shows a reaction cycle involving cyclopentane, cyclopentene and $\text{C}_5\text{H}_9\text{Cl}$.

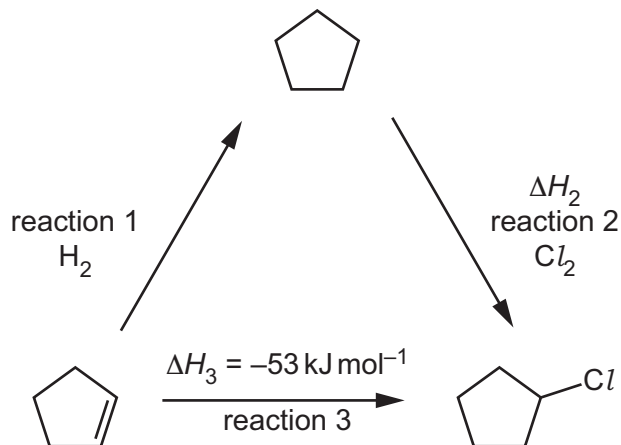


Fig. 3.2

- (i) Identify a suitable reagent for reaction 3.

[1]

22. O/N 25/P23/Q3(e)(i)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (e) (introduction)

- (e)** Cyclopentene, C_5H_8 , reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

- (i)** Draw the structure of **W**.

[1]

23. O/N 25/P24/Q4(a)

also Unit 13 (4 marks), Unit 22 (2 marks); this unit 5 marks

Context

- 4** Compounds **P** and **Q** are structural isomers.

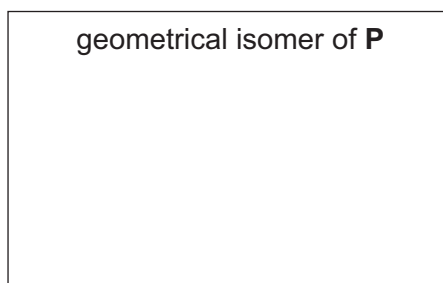
**Fig. 4.1**

- (a) (i)** Define structural isomerism.

[2]

- (ii)** **P** shows geometrical isomerism.

Draw the geometrical isomer of **P**. Explain why the two isomers are **not** identical.



explanation

[2]

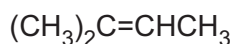
- (iii)** Both **P** and **Q** react with aqueous bromine.

Name the mechanism of this reaction.

[1]

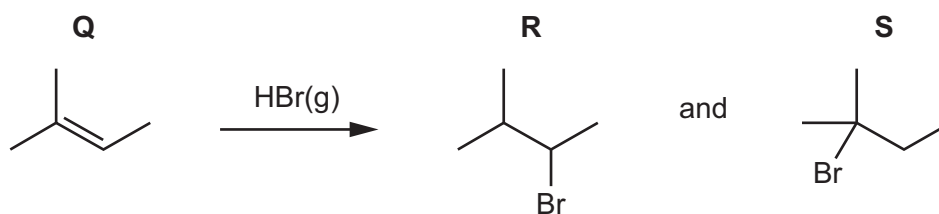
- (iv)** **Q** is oxidised by hot concentrated acidified $KMnO_4(aq)$, forming two different organic products.

Construct an equation for this reaction. Use [O] to represent an atom of oxygen from the oxidising agent.



[2]

- (v)** **Q** reacts with $HBr(g)$ to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.

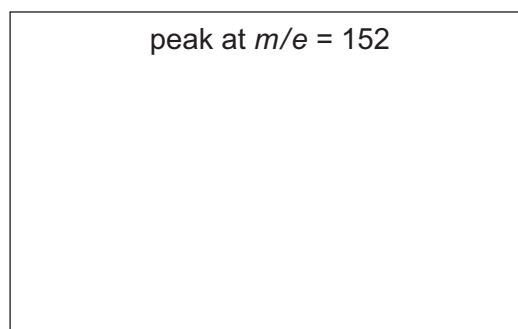
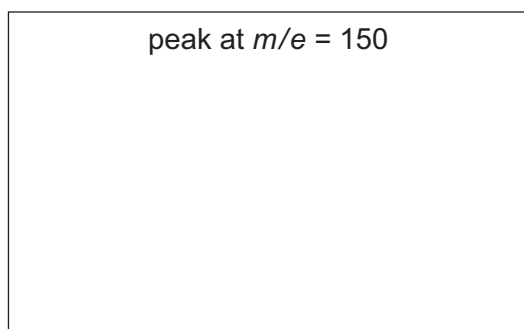
**Fig. 4.2**

State and explain why isomer **S** is the major product of the reaction.

[2]

(vi) The mass spectrum of **R** shows peaks at $m/e = 150$ and $m/e = 152$.

Suggest structures for the ions responsible for these peaks.



[2]

24. M/J 25/P21/Q4(b)(iv)

Context

4 The structure of vitamin C is shown in Fig. 4.1.

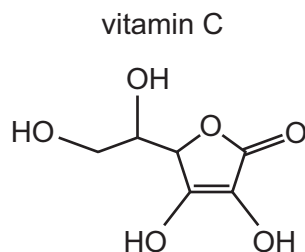


Fig. 4.1

Context: part (b) (introduction)

(b) The concentration of vitamin C is found by titration with $\text{I}_2(\text{aq})$.

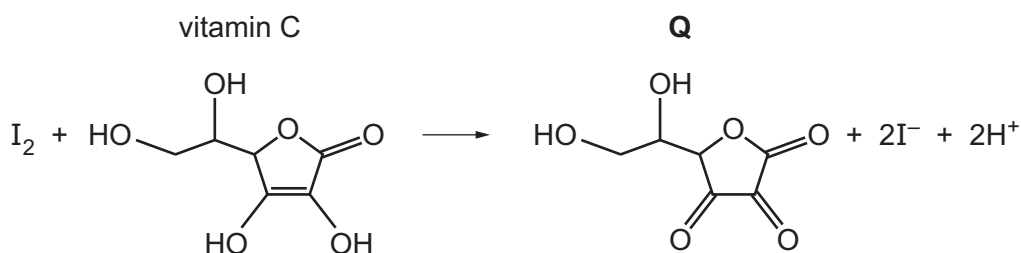


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm^3 of vitamin C solution. 5.00 cm^3 of this vitamin C solution is added to a flask with approximately 150 cm^3 of water and an indicator. Exactly 28.40 cm^3 of $5.00 \times 10^{-4}\text{ mol dm}^{-3}\text{ I}_2(\text{aq})$ reacts with the sample of vitamin C solution in the flask.

[M_r : vitamin C, 176]

(iv) Suggest **two** reasons why hot concentrated acidified potassium manganate(VII) is **not** a suitable reagent for producing **Q** from vitamin C.

1

2

25. M/J 25/P22/Q3(a)(ii)-(iii)

Context: part (a) (introduction)

3 (a) C_3H_6 reacts with HBr(g) in an addition reaction.

- (ii) Complete Fig. 3.1 to show the mechanism for the addition reaction between C_3H_6 and HBr to produce 2-bromopropane. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 3.1**

- (iii) Explain why the major product of this reaction is 2-bromopropane rather than 1-bromopropane.

26. M/J 25/P22/Q3(e)

- (e) Alkene **Z** contains two $\text{C}=\text{C}$ bonds. **Z** reacts with an excess of hot concentrated acidified KMnO_4 to produce **only** CH_3COCH_3 , $\text{HOOCCH}_2\text{COOH}$, CO_2 and H_2O .

Suggest the structure of **Z**.

27. M/J 25/P23/Q3(a)(i)

Context: part (a) (introduction)

3 (a) Different hydrocarbon mixtures produced from fractional distillation of crude oil have different uses.

- (i) State the compound that is heated with long-chain hydrocarbons to produce more useful smaller alkanes and alkenes.

28. M/J 25/P23/Q3(b)(i)

Context: part (b) (introduction)

- (b) C_4H_8 reacts with an excess of H_2 to produce C_4H_{10} .



- (i) Name a catalyst for reaction 1.

29. M/J 25/P23/Q6(b)

- (b) Complete Fig. 6.1 to show the mechanism for the reaction between HOCl and $\text{CH}_2=\text{CH}_2$ to produce $\text{HOCH}_2\text{CH}_2\text{Cl}$.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 6.1

[4]

30. M/J 25/P24/Q5(d)

Context

- 5** The skeletal formula of **W** is shown in Fig. 5.1.

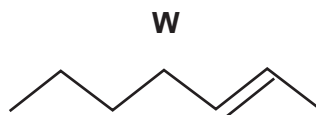


Fig. 5.1

- (d) **W** reacts with reagent **X** to produce ethanoic acid and pentanoic acid.

- (i) State the role of **X** in this reaction.

[1]

- (ii) Identify **X** and state the conditions used for this reaction.

[1]

31. MAR 25/P22/Q3(d)

also Unit 3 (1 mark), Unit 13 (1 mark); this unit 5 marks

Context

- 3** The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol^{-1}	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

- (d)** Iodine monobromide, IBr , is a dark red solid that melts near room temperature.

IBr reacts with propene. The mechanism for this reaction is the same as the mechanism for that of HBr with propene.

- (i)** Identify **all** the intermolecular forces that exist between molecules of IBr .

[1]

- (ii)** Name the mechanism involved in the reaction of IBr with propene.

[1]

- (iii)** The reaction of IBr with propene forms two structural isomers.

Draw the **two** structural isomers shown by these molecules.



[2]

- (iv)** Identify the type of structural isomerism shown by the molecules in **(d)(iii)**.

[1]

- (v)** Explain why the two structural isomers do **not** form in equal amounts.

[2]

32. O/N 24/P21/Q2(b)(ii)-(iii)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (b) (introduction)

(b) The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.**(ii)** State the essential condition for chlorine to react with C_2H_6 at room temperature.

[1]

(iii) Write **two** equations to show the propagation steps in this reaction.

1

2

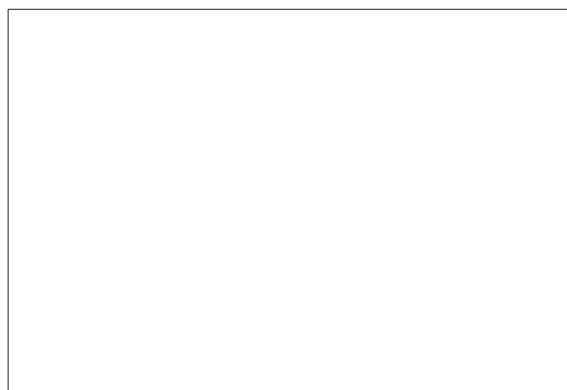
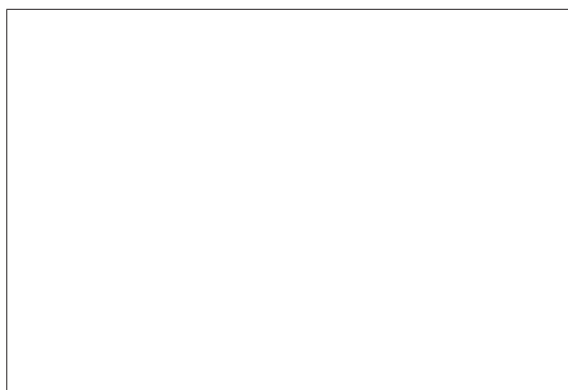
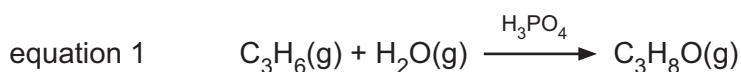
[2]

33. O/N 24/P22/Q2(b)(iii)

Context

2 Oxygen is a Group 16 element.

Context: part (b) (introduction)

(b) H_2O reacts with both inorganic and organic compounds.**(iii)** Draw the structures of the two alcohols formed in the reaction shown in equation 1.

[2]

34. O/N 24/P23/Q2(b)(ii)-(iii)

Context

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (b) (introduction)

(b) The halogenoalkane $\text{CH}_3\text{CH}_2\text{Cl}$ forms when chlorine reacts with C_2H_6 via a free-radical substitution mechanism.**(ii)** State the essential condition for chlorine to react with C_2H_6 at room temperature.

[1]

(iii) Write **two** equations to show the propagation steps in this reaction.

1

2

[2]

35. M/J 24/P21/Q5(c)(ii)

Context

5 Hydrocarbon molecules contain covalent bonds.

Context: part (c) (introduction)

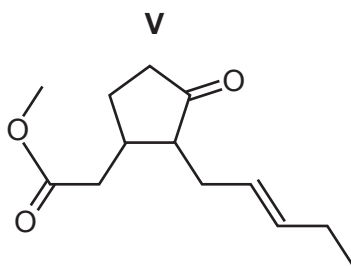
(c) In electrophilic reactions involving alkenes the π bond of $C=C$ is broken.**(ii)** Complete Fig. 5.1 to show the mechanism for the electrophilic addition of hydrogen bromide to 2-methylpropene to produce the major organic product.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

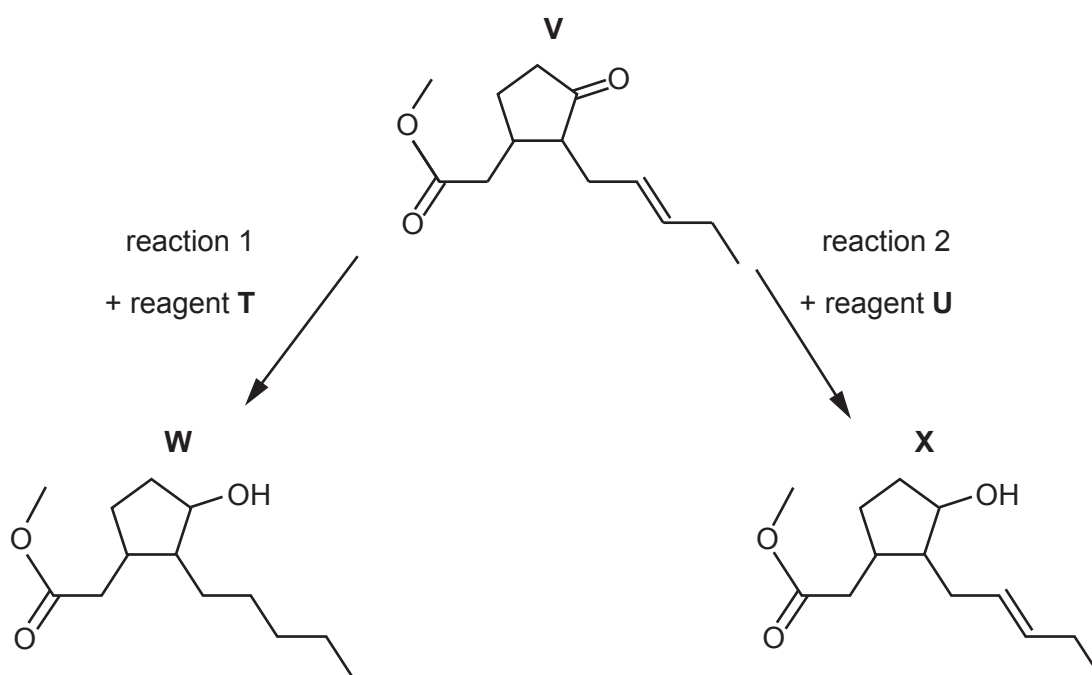
**Fig. 5.1****[4]**

36. M/J 24/P21/Q6(b)(ii)

Context: part (a) (introduction)

6 (a) V shows stereoisomerism.**Fig. 6.1**

Context: part (b) (introduction)

(b) Fig. 6.2 shows two reactions involving **V**.**Fig. 6.2****(ii)** Suggest the identity of reagent **U** in reaction 2.

37. M/J 24/P23/Q5(a)

5 (a) **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

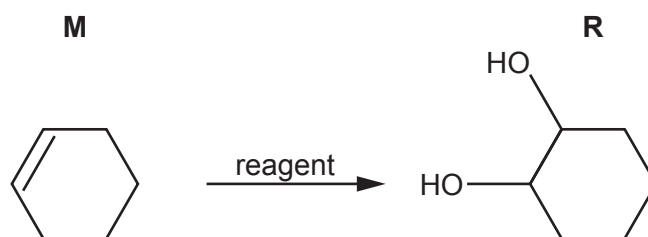


Fig. 5.1

Identify the reagent and conditions for this reaction.

[1]

38. M/J 24/P23/Q5(b)(ii)

Context: part (a)

5 (a) **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

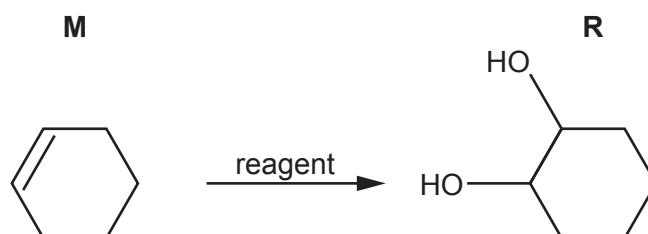


Fig. 5.1

Identify the reagent and conditions for this reaction.

[1]

Context: part (b) (introduction)

(b) **R** is also made from **M** by two steps, as shown in Fig. 5.2.

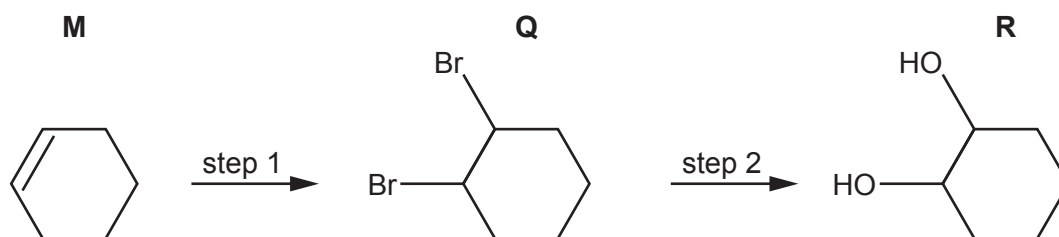


Fig. 5.2

(ii) Name the mechanism for step 1 in Fig. 5.2.

[1]

39. MAR 24/P22/Q1(d)(ii)

Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

Context: part (d) (introduction)

- (d)** NaBiO_3 is an oxidising agent with similar properties to KMnO_4 .

Fig. 1.1 shows an example of the use of NaBiO_3 as an oxidising agent.

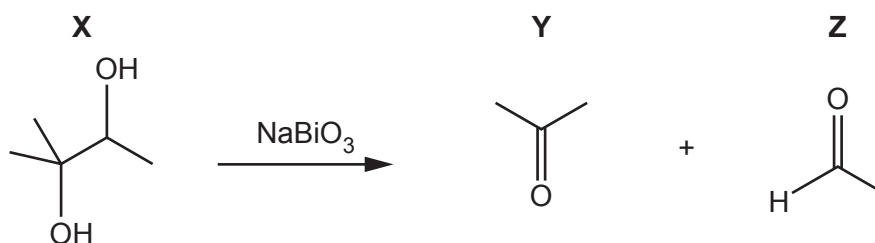


Fig. 1.1

- (ii)** Compound **X** forms when methylbut-2-ene reacts with KMnO_4 .

State the essential conditions for this reaction.

[1]

40. MAR 24/P22/Q2(d)(i)

Context

2 Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

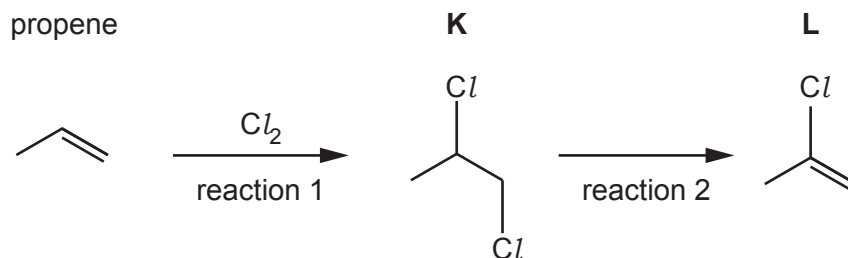
Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

Context: part (d) (introduction)

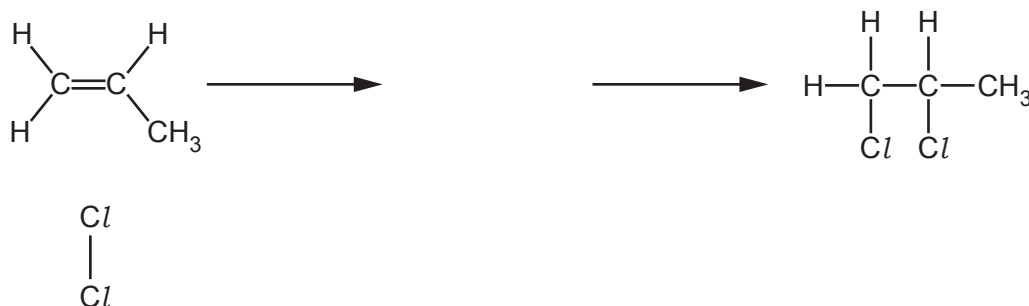
(d) Cl_2 reacts readily with propene to form **K**, 1,2-dichloropropane.

K can be used to form **L**.

**Fig. 2.1**

(i) Complete Fig. 2.2 to show the mechanism for the reaction of Cl_2 with propene in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

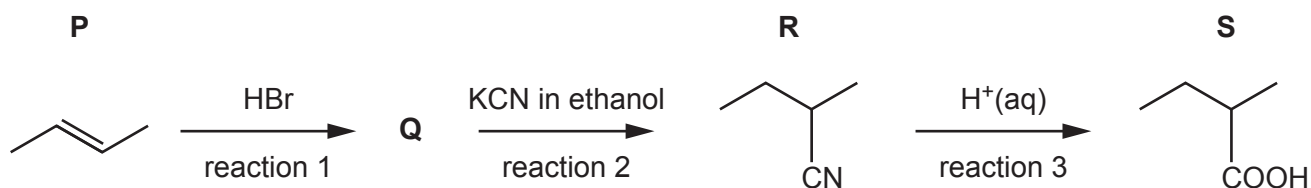
**Fig. 2.2**

[4]

41. MAR 24/P22/Q4(b)(i)

Context

4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

**Fig. 4.1**

(b) (i) Give the structural formula of **Q**.

[1]

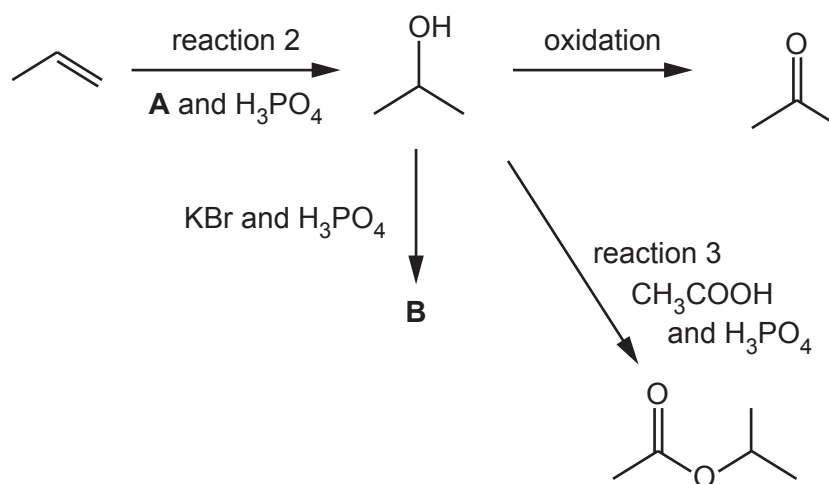
42. O/N 23/P22/Q3(c)(i)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (c) (introduction)

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

**Fig. 3.1**

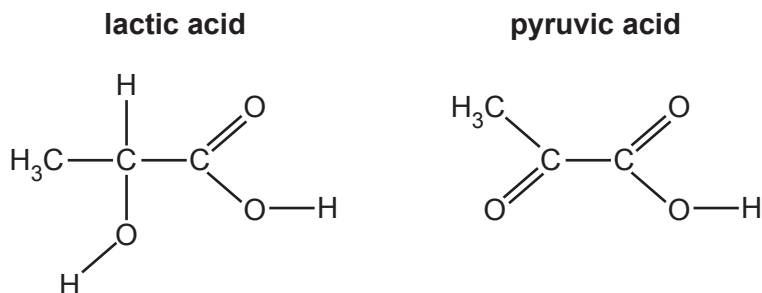
(i) Identify **A**, which reacts with propene in the presence of H_3PO_4 in reaction 2.

[1]

43. O/N 23/P22/Q4(b)(i)

Context

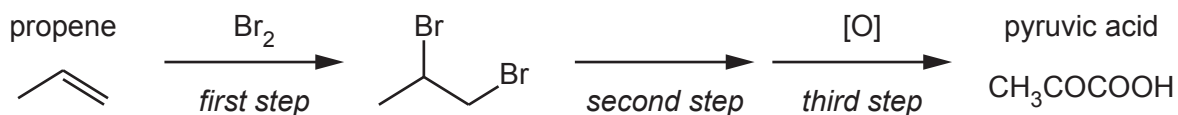
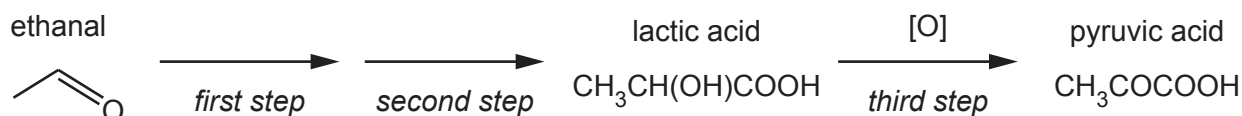
- 4** Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, $\text{CH}_3\text{COCO}\text{OH}$, both contain two functional groups.

**Fig. 4.1**

Context: part (b) (introduction)

- (b)** Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

**Fig. 4.3****Fig. 4.4**

- (i)** Complete the diagram in Fig. 4.5 to show the mechanism for the reaction of propene with Br_2 .

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 4.5**

44. M/J 23/P21/Q3(d)

Context

3 Chlorine is a very reactive element.**(d)** Chlorine reacts with methane in a series of reactions to produce chloroalkanes.**(i)** State the conditions required for chlorine to react with methane.

[1]

(ii) One of the products of the reaction is CH_2Cl_2 which reacts further to produce CHCl_3 .Complete Table 3.2 to show details of the mechanism that forms CHCl_3 from CH_2Cl_2 .**Table 3.2**

name of step	equation
initiation	
propagation	$\text{CH}_2\text{Cl}_2 + \text{Cl}\cdot \rightarrow$
termination	 $\rightarrow \text{CHCl}_3$

[3]

45. M/J 23/P21/Q5(a)**5 (a)** Molecule **M** is present in petrol, a fuel used in cars. **M** is a saturated, non-cyclic hydrocarbon. **M** contains eight carbon atoms.**(i)** Construct an equation for the complete combustion of **M**.

[2]

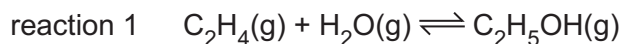
(ii) Describe how the composition of products differs when incomplete combustion of **M** occurs.

[2]

46. M/J 23/P22/Q4(c)(ii)-(iii)

Context

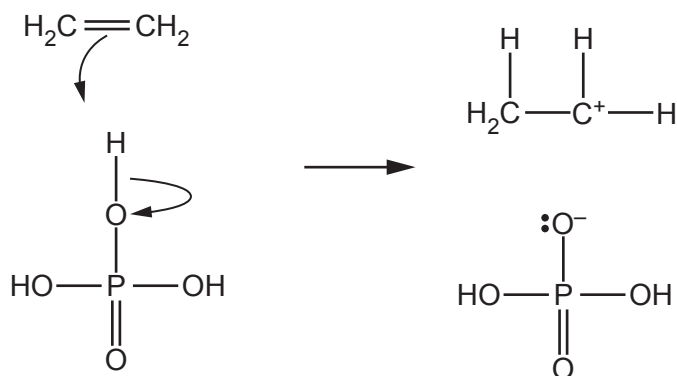
4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



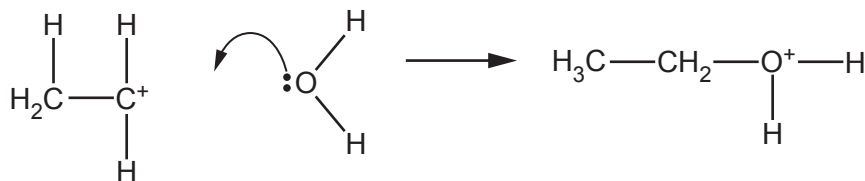
Context: part (c) (introduction)

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



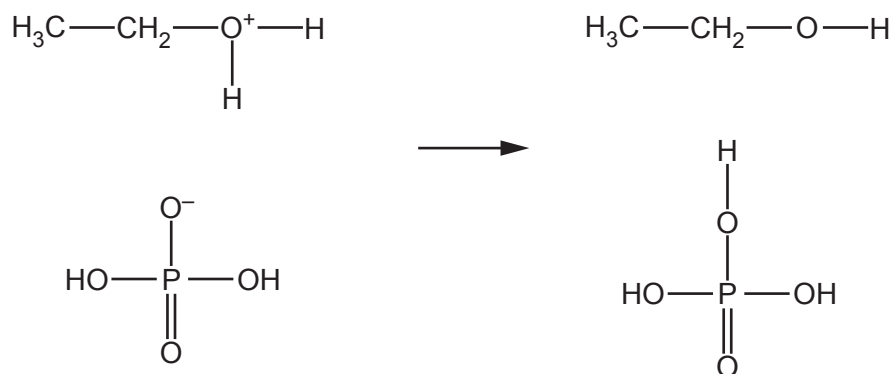
step 2

**Fig. 4.1**

(ii) Identify the species that behaves as an electrophile in step 2 in Fig. 4.1. Explain your answer.

[1]

(iii) Complete Fig. 4.2 to show the mechanism for step 3 of reaction 1. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

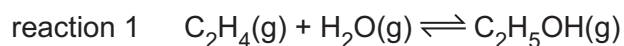
**Fig. 4.2**

[2]

47. M/J 23/P22/Q4(c)(vi)

Context

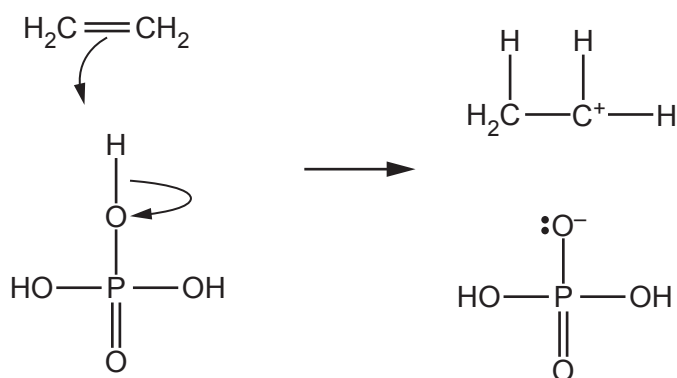
4 In industry, ethanol is made by reacting ethene with steam in the presence of H_3PO_4 .



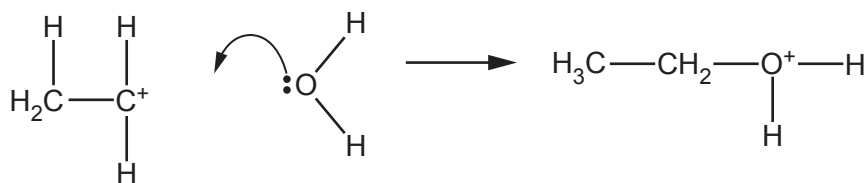
Context: part (c) (introduction)

(c) The mechanism for reaction 1 can be described in three steps. Steps 1 and 2 for reaction 1 are shown in Fig. 4.1.

step 1



step 2

**Fig. 4.1**

(vi) Propene also reacts with steam. A mixture of organic products is produced.

Explain why propan-2-ol is produced in the higher yield.

[2]

48. O/N 22/P21/Q5(a)(i)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

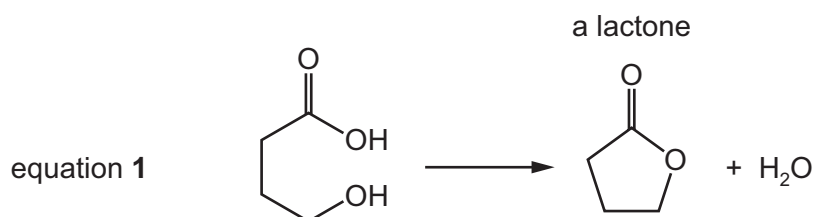


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

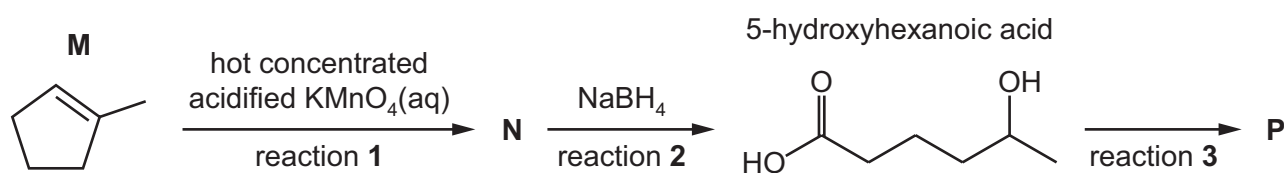


Fig. 5.1

- (a) (i)** **M** reacts with hot concentrated acidified KMnO₄(aq) to form **N**, C₆H₁₀O₃, in reaction 1.

Draw the structure of **N**.

[1]

49. O/N 22/P23/Q5(a)(i)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

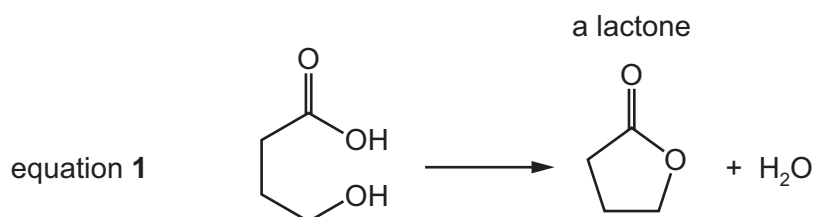


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

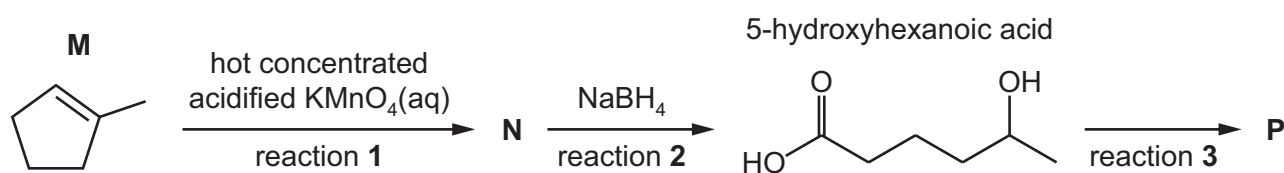


Fig. 5.1

- (a) (i)** **M** reacts with hot concentrated acidified $\text{KMnO}_4(\text{aq})$ to form **N**, $\text{C}_6\text{H}_{10}\text{O}_3$, in reaction 1.

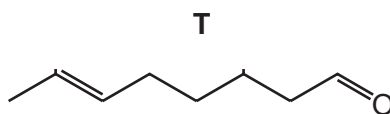
Draw the structure of **N**.

[1]

50. M/J 22/P21/Q3(b)

Context

- 3** Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

- (b)** Two organic products are produced when a sample of **T** is heated under reflux with excess acidified concentrated KMnO_4 .

Draw the structure of the two organic products, from this reaction, in the boxes.

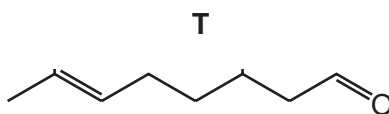


[2]

51. M/J 22/P21/Q3(c)(ii)-(iii)

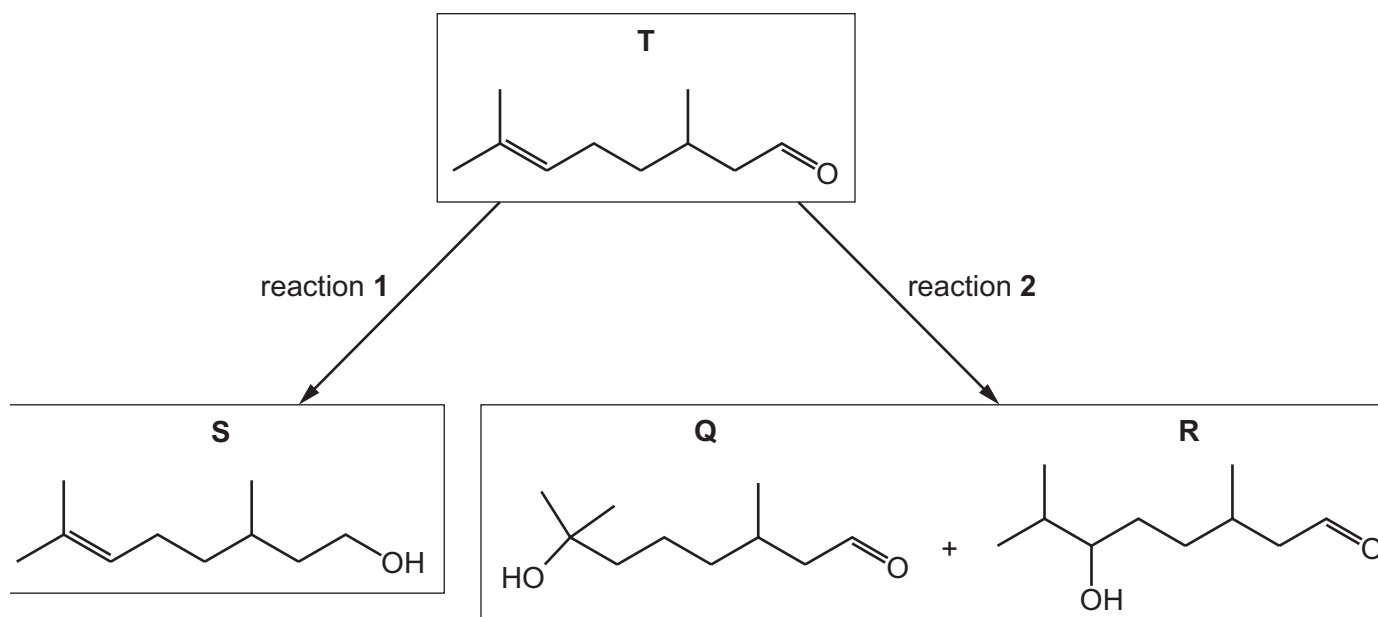
Context

- 3** Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

Context: part (c) (introduction)

- (c)** Fig. 3.2 shows two reactions of **T**.

**Fig. 3.2**

- (ii)** Identify the reagent and conditions needed for reaction 2.

[2]

- (iii)** Suggest which product formed in reaction 2 has a higher yield. Explain your answer.

[3]

52. M/J 22/P23/Q4(a)

- 4 (a) 2-methylpropene reacts with HCl(g) at room temperature. The major organic product is 2-chloro-2-methylpropane.

(i) Complete Fig. 4.1 to show the structure of the intermediate and mechanism for this reaction. Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.1

[3]

(ii) Explain why, in this reaction, 2-chloro-2-methylpropane is produced at a higher yield than 1-chloro-2-methylpropane.

[2]

53. M/J 22/P23/Q5(a)(i)

Context: part (a) (introduction)

- 5 (a) But-2-ene reacts with KMnO_4 to form organic product, **Y**.
Y does not react with Na_2CO_3 .
A gas is produced when an excess of Na is added to **Y**.

(i) Describe the conditions for the KMnO_4 used in the reaction to form **Y** from but-2-ene.

[1]

54. MAR 22/P22/Q3(e)(i)-(ii)

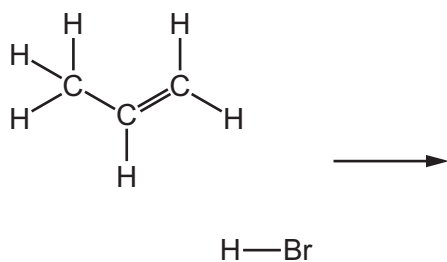
Context

3 The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

Context: part (e) (introduction)

(e) HBr reacts with propene to form two bromoalkanes, $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and $(\text{CH}_3)_2\text{CHBr}$.

- (i)** Complete the diagram to show the mechanism of the reaction of HBr and propene to form the major organic product.
Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.
Draw the structures of the intermediate and the major organic product.



[4]

- (ii)** Explain why the two bromoalkanes are **not** produced in equal amounts by this reaction.

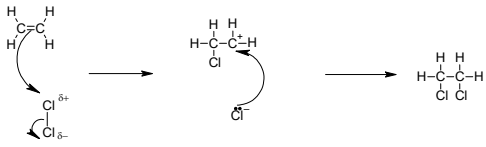
[2]

Answers Section

1. M/J 26/P21/Q5(b)

5(b)	electrophilic addition	1	1 addition alone
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2. M/J 26/P21/Q5(d)

5(d)	 M1 curly arrow from C=C bond to Cl closest to C=C M2 correct dipoles $Cl\delta^+ - Cl\delta^-$ AND curly arrow from bond of $Cl\delta^+ - Cl\delta^-$ to $Cl\delta^-$ M3 correct carbocation intermediate M4 curly arrow from lone pair of : Cl^- to C^+ of intermediate Judgement of arrows: A curly arrows that appear straight NOT single headed arrows (fishhooks) first time seen-and then A M1 judgement of arrows: π bonds M1 starts from anywhere on π bond AND goes to Cl or close to Cl atom of $Cl-Cl$ M1 if arrow does not start at the double bond, but when it is extrapolated back the base of the arrow hits the π bond AND the arrow end goes to Cl or close to Cl atom of $Cl-Cl$ M2 judgement of arrows: $Cl-Cl$ bond M2 A curly arrow from bond of $Cl-Cl$ to just beyond the Cl (if dipole correct).	4	M1 δ^-. I π or δ^- label on the double bond (e.g. $C = C$) I partial charges on Cl R partial charges on C of $C = C$ (e.g. $\delta^+C = \delta^+C / \delta^-C = \delta^-C$) R arrows from any other part of the alkene M2 A dipole moment of Cl_2 shown as an arrow with + convention I electrons (i.e. dots / crosses above and/or below bond) M3 A unambiguous structure R δ^+ charge on C^+ R lone pair on C^+ M4 A ecf from incorrect intermediate structure in M3 when no positive charge on C of intermediate OR $C\delta^+$ instead of C^+ I δ^-Cl^- if all else ok
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3. M/J 26/P22/Q5(b)(iii)-(iv)

5(b)(iii)	M1 $CH_3(CH_2)_4CH_3 + Cl\cdot \rightarrow CH_3(CH_2)_3\cdot CHCH_3 + HCl$ M2 $CH_3(CH_2)_3\cdot CHCH_3 + Cl_2 \rightarrow CH_3(CH_2)_3CHClCH_3 + Cl\cdot$	2	Independent marks A radicals with or without $\cdot$ M1 I alkyl radical as $\cdot C_6H_{13}$ OR $CH_3(CH_2)_5\cdot$ I position of the dot BOD $CH_3(CH_2)_3CH_3CH\cdot$ OR $CH_3(CH_2)_3CCH_3H\cdot$ M2 A ecf on their alkyl structure from M1 (even if M2 does not balance) M2 A alkyl radical as $\cdot C_6H_{13}$ OR $CH_3(CH_2)_5\cdot$ M2 ecf use of anion/cation
5(b)(iv)	M1 $CH_3(CH_2)_4CH_3 + \cdot Cl \rightarrow CH_3(CH_2)_4\cdot CH_2 + HCl$ M2 $CH_3(CH_2)_4\cdot CH_2 + CH_3(CH_2)_4\cdot CH_2 \rightarrow CH_3(CH_2)_{10}CH_3$ OR $2CH_3(CH_2)_4\cdot CH_2 \rightarrow CH_3(CH_2)_{10}CH_3$	2	Independent marks A radicals with or without $\cdot$ M1 A $CH_3(CH_2)_5\cdot$ M1 I alkyl radical as $\cdot C_6H_{13}$ I position of the dot M2 A alkyl radical as $\cdot C_6H_{13}$ OR $CH_3(CH_2)_5\cdot$ M2 ecf use of anion/cation

4. M/J 26/P22/Q5(c)

5(c)	number of alkyl / methyl groups	1	A number of C atoms bonded to the central carbon / branches / R groups BOD number of alkyl groups bonded to the carbocation I number of C atoms bonded to the carbocation I number of C atoms I number of ethyl/propyl groups I number of hydrogens I inductive effect I tertiary>secondary>primary stability I carbocation stability
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5. M/J 26/P22/Q6(b)(i)

6(b)(i)	M1 steam / H ₂ O (g) M2 H ₃ PO ₄ (catalyst) / phosphoric (V) acid (catalyst)	2	M1 A water <u>and</u> then temperature $\geq 100^\circ\text{C}$ (temperature might be seen on conditions line) M2 A H ₃ PO ₄ (catalyst) / phosphoric (V) acid (catalyst) / sulfuric acid (catalyst) / H ₂ SO ₄ (catalyst) on reagent line A orthophosphoric acid / phosphoric acid / sulfuric acid I heat under reflux / heat alone I pressure I state symbols
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6. M/J 26/P23/Q5(a)(ii)

5(a)(ii)	free-radical substitution	1	
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7. M/J 26/P23/Q5(b)(ii)-(iii)

5(b)(ii)	M1 $\text{CH}_3\text{CH}_2\text{CH}_3 + \bullet\text{Cl} \rightarrow \bullet\text{CH}_2\text{CH}_2\text{CH}_3 + \text{HCl}$ M2 $\bullet\text{CH}_2\text{CH}_2\text{CH}_3 + \text{Cl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \bullet\text{Cl}$	2	M1 I $\bullet\text{C}_3\text{H}_7$ M1 R $\text{CH}_3\bullet\text{CHCH}_3$ I position of the dot I absence of the dot M2 A alkyl radical as $\bullet\text{C}_3\text{H}_7$ or $\text{CH}_3\bullet\text{CHCH}_3$
5(b)(iii)	M1 $\text{CH}_3\text{CH}_2\text{CH}_3 + \bullet\text{Cl} \rightarrow \text{CH}_3\bullet\text{CHCH}_3 + \text{HCl}$ M2 $2\text{CH}_3\bullet\text{CHCH}_3 \rightarrow (\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2	A radicals with or without $\bullet$ $(\text{CH}_3)_2\text{CH}\bullet = \text{CH}_3\bullet\text{CHCH}_3$ M1 A C_3H_7 for CH_3CHCH_3 M1 R $\bullet\text{CH}_2\text{CH}_2\text{CH}_3$ I position of the dot M2 I alkyl radical as $\bullet\text{C}_3\text{H}_7$ M2 R $\bullet\text{CH}_2\text{CH}_2\text{CH}_3$

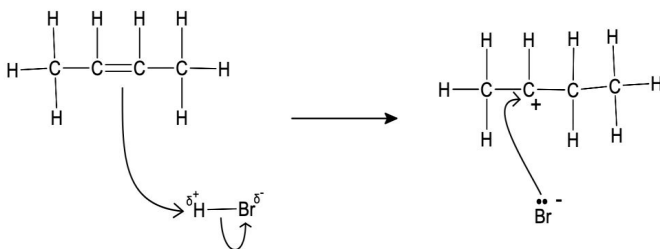
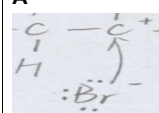
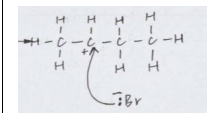
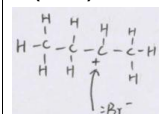
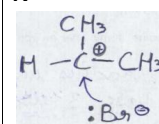
8. M/J 26/P24/Q5(a)

5(a)	Al ₂ O ₃ AND heat / high temperature	1	A zeolite / alumina / porous pot / porcelain / ceramics AND heat A high temperature/high temp = heat A $\geq 100^\circ\text{C}$ I catalyst I steam I pressure
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9. M/J 26/P24/Q5(b)(iii)

5(b)(iii)	orange / brown (solution / liquid) to colourless	1	I turns clear I colour disappears I discolouration / colour discharged A orange / brown disappears A decolourises / turns colourless A brown / yellow / yellow-brown / orange-brown / yellow-orange R red / red-brown I 'liquid'
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10. M/J 26/P24/Q5(d)

5(d)(i)	electron pair acceptor	1	I reference to likes electrons / attracted to electrons / area of low electron density. A lone pair acceptor A gains both electrons I electron acceptor
5(d)(ii)	 M1 curly arrow from $=$ of $C=C$ to H • I π or $\delta-$ label on the double bond • R partial charges shown on C of $C=C$ R arrows from any other part of the alkene M2 correct dipoles $H^{\delta+}-Br^{\delta-}$ AND curly arrow from bond of $H^{\delta+}-Br^{\delta-}$ to $Br^{\delta-}$ • I electrons (i.e. dots / crosses above and / or below bond) M3 correct intermediate – secondary carbocation only • A unambiguous structure • I $\delta+$ charge on C^+ • R lone pair on C^+	4	Judgement of arrows: A curly arrows that appear straight R single headed arrows first time seen and then A M1 judgement of arrows: π bonds M1 starts from anywhere on π bond AND goes to H or close to H atom of $H-Br$ ✓ M1 if arrow does not start at the double bond, but when it is extrapolated back the base of the arrow hits the π bond AND the arrow end goes to H or close to H atom of $H-Br$ ✓ M2 judgement of arrows: $H-Br$ bond M2 A curly arrow from bond of $H-Br$ to just beyond the Br (if dipole correct).
5(d)(ii) ctd.	M4 curly arrow from lone pair of $:Br^-$ to C^+ of intermediate A ecf from incorrect intermediate structure in M3 when: - no positive charge on C of intermediate OR $C^{\delta+}$ instead of C^+ - I $\delta-$ on $:Br^-$		Judgement of arrows: lone pairs M4 A arrow starts anywhere closely associated with lone pair A  A  A (BOD)  R 

11. MAR 26/P22/Q3(c)

3(c)(i)	(free-)radical substitution	1
3(c)(ii)	M1 <i>propagation</i> $\text{Cl}^\bullet + \text{C}_4\text{H}_{10} \rightarrow \text{C}_4\text{H}_9^\bullet + \text{HCl}$ M2 <i>termination</i> $\text{C}_4\text{H}_9^\bullet + \text{Cl}^\bullet \rightarrow \text{C}_4\text{H}_9\text{Cl}$	2

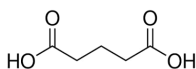
12. O/N 25/P21/Q3(c)(ii)-(iii)

3(c)(ii)	M1 $\text{C}_5\text{H}_{10} + \text{Cl}^\bullet \rightarrow \text{C}_5\text{H}_9^\bullet + \text{HCl}$ M2 $\text{C}_5\text{H}_9^\bullet + \text{Cl}_2 \rightarrow \text{C}_5\text{H}_9\text{Cl} + \text{Cl}^\bullet$	2
3(c)(iii)	termination	1

13. O/N 25/P21/Q3(d)(i)

3(d)(i)	HCl (g)	1
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14. O/N 25/P21/Q3(e)(i)

3(e)(i)		1
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15. O/N 25/P22/Q3(b)

3(b)	M1 strong C—H (bonds) M2 non-polar (C—H / molecule / bonds)	2
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16. O/N 25/P22/Q3(c)

3(c)	cracking	1
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17. O/N 25/P22/Q3(e)(i)

3(e)(i)	H_2 ((g)) OR hydrogen (gas)	1
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18. O/N 25/P22/Q3(e)(iii)

3(e)(iii)	$\text{C}_6\text{H}_{14} + 9\frac{1}{2}\text{O}_2 \rightarrow 6\text{CO}_2 + 7\text{H}_2\text{O}$	1
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19. O/N 25/P22/Q4(a)(i)-(ii)

4(a)(i)	free-radical substitution	1
4(a)(ii)	M1 $\text{C}_3\text{H}_8 + \text{Cl}^\bullet \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2^\bullet + \text{HCl}$ OR $\text{C}_3\text{H}_8 + \text{Cl}^\bullet \rightarrow \text{C}_3\text{H}_7^\bullet + \text{HCl}$ M2 $\text{CH}_3\text{CH}_2\text{CH}_2^\bullet + \text{Cl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{Cl}^\bullet$ OR $\text{C}_3\text{H}_7^\bullet + \text{Cl}_2 \rightarrow \text{C}_3\text{H}_7\text{Cl} + \text{Cl}^\bullet$ etc M3 $\text{CH}_3\text{CH}_2\text{CH}_2^\bullet + \text{Cl}^\bullet \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ OR $\text{C}_3\text{H}_7^\bullet + \text{Cl}^\bullet \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$	3

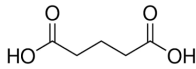
20. O/N 25/P23/Q3(c)(ii)-(iii)

3(c)(ii)	M1 $\text{C}_5\text{H}_{10} + \text{Cl}^\bullet \rightarrow \text{C}_5\text{H}_9^\bullet + \text{HCl}$ M2 $\text{C}_5\text{H}_9^\bullet + \text{Cl}_2 \rightarrow \text{C}_5\text{H}_9\text{Cl} + \text{Cl}^\bullet$	2
3(c)(iii)	termination	1

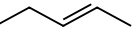
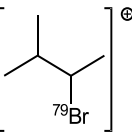
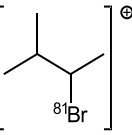
21. O/N 25/P23/Q3(d)(i)

3(d)(i)	HCl (g)	1
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22. O/N 25/P23/Q3(e)(i)

3(e)(i)		1
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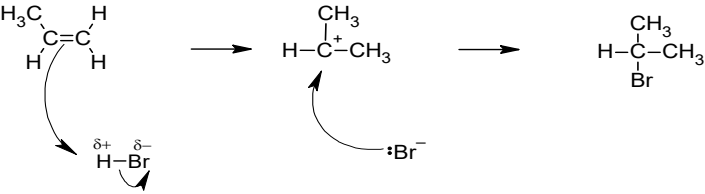
23. O/N 25/P24/Q4(a)

4(a)(i)	M1 (molecules with the) same molecular formula / same number of atoms of each element M2 different structural formulae/structures	2
4(a)(ii)	M1  M2 restricted / lack of / no rotation about C=C	2
4(a)(iii)	electrophilic addition	1
4(a)(iv)	M1 CH ₃ COCH ₃ + CH ₃ COOH as products M2 (CH ₃) ₂ C = CHCH ₃ + 3[O] → CH ₃ COCH ₃ + CH ₃ COOH	2
4(a)(v)	M1 (reaction proceeds via a) more stable intermediate / carbocation M2 because of greater positive inductive effect of more alkyl groups	2
4(a)(vi)	M1  M2 	2

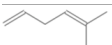
24. M/J 25/P21/Q4(b)(iv)

4(b)(iv)	M1 (also) oxidises / reacts with (primary and / or secondary) alcohol groups M2 (also) breaks the sigma and pi bond in C=C	2
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25. M/J 25/P22/Q3(a)(ii)-(iii)

3(a)(ii)	 M1 curly arrow from = of C=C to H (of HBr) M2 curly arrow from bond of H ^{δ+} —Br ^{δ-} to Br ^{δ-} (dipole is essential) M3 correct intermediate M4 curly arrow from lone pair of Br ⁻ to C ⁺ of intermediate drawn	4
3(a)(iii)	M1 (2-bromopropane is made from the) more stable intermediate / carbocation M2 (due to positive) inductive effect of more alkyl groups (attached to the C ⁺ / central carbon atom) OR (due to) greater (positive) inductive effect of 2 methyl groups (rather than 1 ethyl group)	2

26. M/J 25/P22/Q3(e)

3(e)	$\text{H}_2\text{C}=\text{CHCH}_2\text{CH}=\text{C}(\text{CH}_3)_2$ ✓✓  ✓✓ $\text{H}_2\text{CCHCH}_2\text{CHC}(\text{CH}_3)_2$ ✓✓ M1 $\text{H}_2\text{C}=\text{C}\dots$ on one end of their molecule OR $\dots\text{C}=\text{C}(\text{CH}_3)_2$ on one end of their molecule OR $\dots\text{C}=\text{CHCH}_2\text{CH}=\text{C}\dots$ in the middle of their molecule M2 correct structure	2
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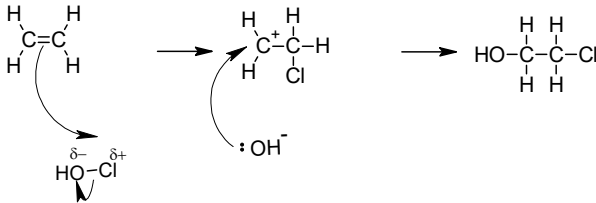
27. M/J 25/P23/Q3(a)(i)

3(a)(i)	Al_2O_3 / aluminium oxide	1
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28. M/J 25/P23/Q3(b)(i)

3(b)(i)	nickel / platinum	1
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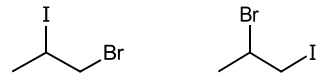
29. M/J 25/P23/Q6(b)

6(b)	 M1 correct arrow for $\text{HO}-\text{Cl}$ M2 arrow from double bond of $\text{C}=\text{C}$ to Cl M3 correct intermediate M4 arrow from lone pair on O of $:\text{OH}^-$ to the appropriate C	4
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30. M/J 25/P24/Q5(d)

5(d)(i)	oxidising agent	1
5(d)(ii)	hot concentrated acidified KMnO_4	1

31. MAR 25/P22/Q3(d)

3(d)(i)	instantaneous dipole—induced dipole / id—id / London / dispersion (forces) AND permanent dipole—permanent dipole / pd—pd (forces)	1
3(d)(ii)	electrophilic addition	1
3(d)(iii)		2
3(d)(iv)	position(al) / regio- (isomerism)	1
3(d)(v)	- secondary (carbo)(cat)ion/intermediate is more stable / $[\text{CH}_3\text{CHCH}_2\text{I}]^+$ is preferred - electrophile approaches with I nearest / $\text{I}^{\delta+}$ / I^+ attacks first I and Br have different electronegativities / IBr is a polar molecule	2

32. O/N 24/P21/Q2(b)(ii)-(iii)

2(b)(ii)	ultraviolet	1
2(b)(iii)	$\text{C}_2\text{H}_6 + \text{Cl}^\bullet \rightarrow \text{C}_2\text{H}_5^\bullet + \text{HCl}$	1
	$\text{C}_2\text{H}_5^\bullet + \text{Cl}_2 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{Cl}^\bullet$	1

33. O/N 24/P22/Q2(b)(iii)

2(b)(iii)		2
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34. O/N 24/P23/Q2(b)(ii)-(iii)

2(b)(ii)	ultraviolet	1
2(b)(iii)	$C_2H_6 + Cl^\bullet \rightarrow C_2H_5^\bullet + HCl$	1
	$C_2H_5^\bullet + Cl_2 \rightarrow C_2H_5Cl + Cl^\bullet$	1

35. M/J 24/P21/Q5(c)(ii)

5(c)(ii)	M1 arrow from = of C=C to H of H-Br M2 correct dipole on H-Br AND arrow from H-Br bond to Br M3 correct intermediate AND 2-bromopropane M4 arrow from lone pair on Br⁻ to C⁺ of their intermediate (with +ve or δ+ charge)	4
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36. M/J 24/P21/Q6(b)(ii)

6(b)(ii)	sodium borohydride / NaBH ₄	1
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37. M/J 24/P23/Q5(a)

5(a)	cold dilute KMnO ₄ OR cold dilute potassium manganate(VII)	1
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38. M/J 24/P23/Q5(b)(ii)

5(b)(ii)	electrophilic addition	1
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Answer for context part (a)

5(a)	cold dilute KMnO ₄ OR cold dilute potassium manganate(VII)	1
------	--	----------

39. MAR 24/P22/Q1(d)(ii)

1(d)(ii)	(KMnO ₄ is) cold and dilute	1
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40. MAR 24/P22/Q2(d)(i)

2(d)(i)	• curly arrow from C=C bond to Cl(δ⁺) • correct dipole on Cl₂ AND curly arrow from Cl-Cl bond to Cl^{δ-} • correct carbocation intermediate • curly arrow from lone pair on Cl⁻ to C⁺	1
	• curly arrow from C=C bond to Cl(δ ⁺)	1
	• correct dipole on Cl ₂ AND curly arrow from Cl-Cl bond to Cl ^{δ-}	1
	• correct carbocation intermediate	1
	• curly arrow from lone pair on Cl ⁻ to C ⁺	1

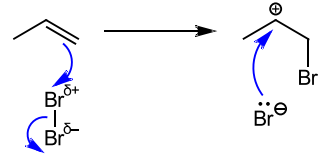
41. MAR 24/P22/Q4(b)(i)

4(b)(i)	molecule	P	Q	R	S	1
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42. O/N 23/P22/Q3(c)(i)

3(c)(i)	H ₂ O / steam	1
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43. O/N 23/P22/Q4(b)(i)

4(b)(i)	 curly arrow from C=C bond to Br^{δ+} AND curly arrow from Br^{δ+}—Br^{δ-} bond to Br^{δ-} AND Br^{δ+} closest to the double bond	1
	correct intermediate	1
	curly arrow from lone pair on Br ⁻ to C ⁽⁺⁾ of the intermediate	1

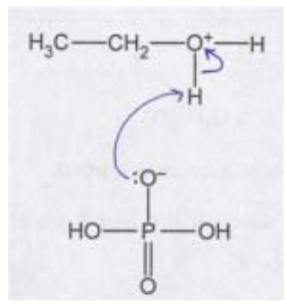
44. M/J 23/P21/Q3(d)

3(d)(i)	ultraviolet light			1	
3(d)(ii)				3	
		name of stage	equation		
		initiation	$\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$		[1]
		propagation	$\text{CH}_2\text{Cl}_2 + \text{Cl}\cdot \rightarrow \text{CHCl}_2\cdot + \text{HCl}$		[1]
		termination	$\text{CHCl}_2\cdot + \text{Cl}\cdot \rightarrow \text{CHCl}_3$		[1]

45. M/J 23/P21/Q5(a)

5(a)(i)	$\text{C}_8\text{H}_{18} + 12\frac{1}{2} \text{O}_2 \rightarrow 8\text{CO}_2 + 9\text{H}_2\text{O}$ M1 correct formula of M C ₈ H ₁₈ M2 balanced equation showing complete combustion of M	2
5(a)(ii)	• C • CO • unburnt hydrocarbons • less/no CO₂ Two correct for one mark, three correct for two marks.	2

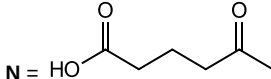
46. M/J 23/P22/Q4(c)(ii)-(iii)

4(c)(ii)	(CH ₃ CH ₂) ⁺ OR CH ₃ C ⁺ H ₂ AND accepts a pair of electrons	1
4(c)(iii)	 M1 arrow from bond of one of the O-H to O of the cation M2 arrow from lone pair on O of the anion to between H of H-O	2

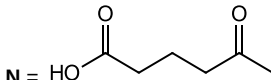
47. M/J 23/P22/Q4(c)(vi)

4(c)(vi)	M1 (propan-2-ol is produced via a) more stable carbocation OR more stable intermediate OR $(\text{CH}_3)_2\text{C}^+\text{H}$ is more stable M2 (positive) inductive effect of more alkyl/methyl groups OR more electron donating alkyl / methyl groups OR greater electron donating effect of two alkyl / two methyl groups	2
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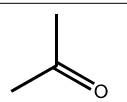
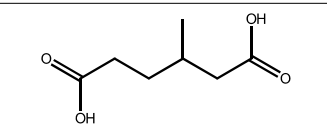
48. O/N 22/P21/Q5(a)(i)

5(a)(i)		1
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49. O/N 22/P23/Q5(a)(i)

5(a)(i)		1
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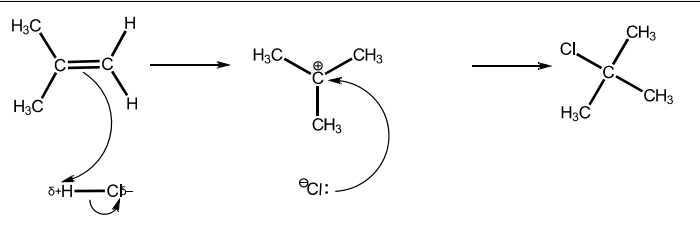
50. M/J 22/P21/Q3(b)

3(b)	 OR $(\text{CH}_3)_2\text{CO}$  OR $\text{HOOC}(\text{CH}_2)_2\text{CH}(\text{CH}_3)\text{CH}_2\text{COOH}$	2
------	--	----------

51. M/J 22/P21/Q3(c)(ii)-(iii)

3(c)(ii)	M1 $\text{H}_2\text{O}(\text{g})$ / steam M2 phosphoric acid (catalyst) / H_3PO_4 (catalyst)	2
3(c)(iii)	M1 Q AND intermediate / cation is (more) stable M2 (as the) C^+ has 3 / more alkyl groups attached M3 (so) greater (positive) inductive effect	3

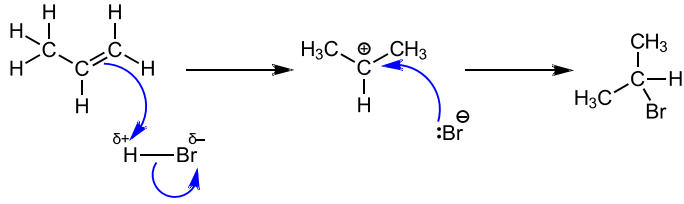
52. M/J 22/P23/Q4(a)

4(a)(i)	 M1 curly arrow from = to H AND correct dipole on H-Cl M2 correct intermediate M3 curly arrow from H-Cl bond to Cl AND curly arrow from lp on: Cl^- to C^+	3
4(a)(ii)	- intermediate OR C^+ OR carbocation (formed) is more stable - due to greater (positive) inductive effect. - caused by 3 methyl groups / 3 alkyl groups / more alkyl groups attached to C^+ / central C	2

53. M/J 22/P23/Q5(a)(i)

5(a)(i)	cold + dilute	1
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54. MAR 22/P22/Q3(e)(i)-(ii)

3(e)(i)	 M1: curly arrow from C=C to H of HBr M2: correct dipole ($\delta^+ \text{H} - \text{Br} \delta^-$) AND curly arrow from H—Br to Br M3: curly arrow from lone pair on :Br⁻ to carbocation M4: correct intermediate AND product (2-bromopropane)	4
3(e)(ii)	2° carbocation more stable (than 1°) (or reverse argument) [1] greater positive inductive effect of two methyl groups (<i>cf.</i> one ethyl) (or reverse argument) [1]	2

Unit 15

HALOGEN COMPOUNDS

33 items · 70 marks

Syllabus sections in this unit

15.1 Halogenoalkanes

Unit 15: Halogen compounds

1. M/J 26/P21/Q5(e)

Context

- 5 Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

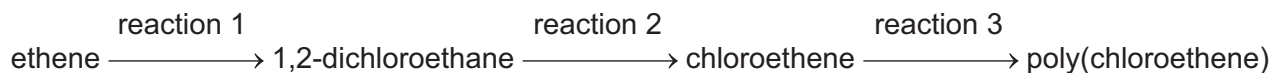


Figure 5.1

- (e) When 1,2-dichloroethane is heated in reaction 2, thermal decomposition occurs. The reaction is endothermic.

State the type of reaction that occurs.

[1]

2. MAR 26/P22/Q3(a)

Context

- 3 Secondary halogenoalkanes are important building blocks in organic chemistry. 2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane

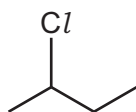


Figure 3.1

- (a) Explain why 2-chlorobutane is classified as a secondary halogenoalkane.

[1]

3. MAR 26/P22/Q3(d)

Context

3 Secondary halogenoalkanes are important building blocks in organic chemistry.

2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane

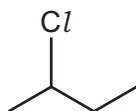


Figure 3.1

(d) The rates of hydrolysis of three secondary halogenoalkanes, **B**, **C** and **D**, are compared.

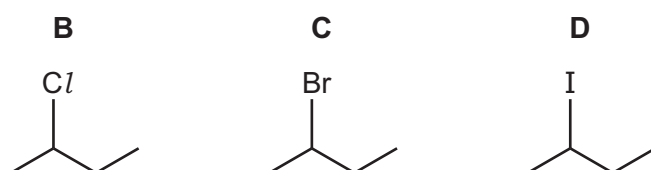


Figure 3.3

Each halogenoalkane is dissolved in ethanol and reacted with $\text{AgNO}_3(\text{aq})$.

Identify the order of the rates of hydrolysis of compounds **B**, **C** and **D**. Explain your answer.

fastest > > slowest

explanation

[2]

4. MAR 26/P22/Q4(c)

Context

4 Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.

(c) Figure 4.3 shows a synthesis of $\text{CH}_2(\text{CN})\text{COOH}$.

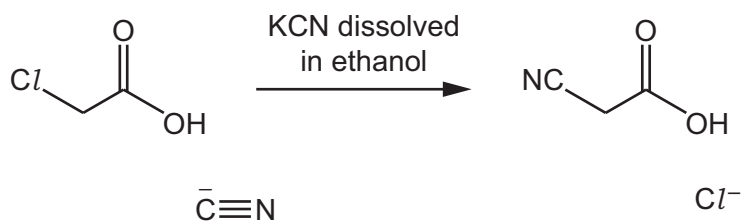


Figure 4.3

(i) The reaction mechanism of the synthesis in Figure 4.3 has a single step.

Complete Figure 4.3 to show this mechanism. Include lone pairs, charges, dipoles and curly arrows as appropriate.

[2]

(ii) Identify the nucleophile in Figure 4.3.

[1]

(iii) Identify the mechanism of the reaction in Figure 4.3.

[1]

5. O/N 25/P22/Q4(b)

Context

4 Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$.

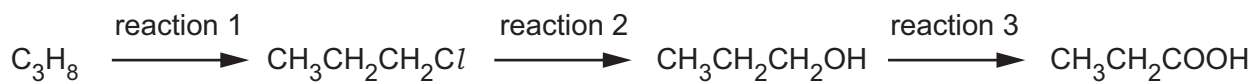


Fig. 4.1

(b) State the reagent and solvent required for reaction 2.

[1]

6. O/N 25/P24/Q3(b)(ii)-(iii)

Context

- 3 Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

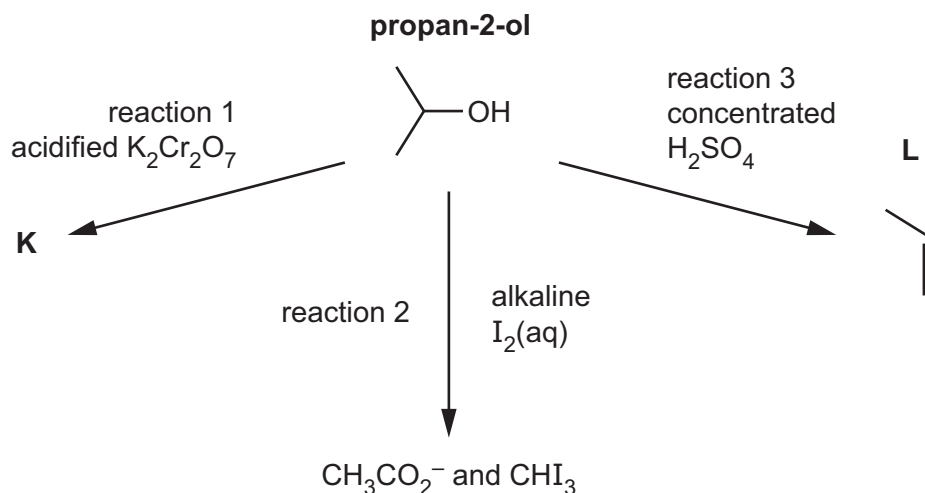


Fig. 3.1

Context: part (b) (introduction)

- (b) Propan-2-ol reacts with sodium to produce $(\text{CH}_3)_2\text{CH-O}^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

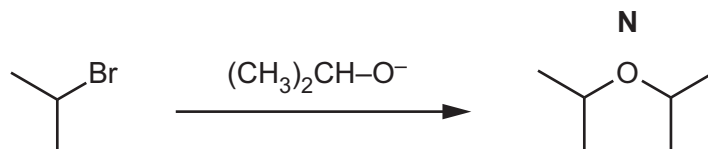


Fig. 3.3

- (ii) The reaction of 2-bromopropane with $(\text{CH}_3)_2\text{CH-O}^-$ anions follows an $\text{S}_{\text{N}}1$ mechanism.

Complete Fig. 3.4 to show this mechanism. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 3.4

- (iii) Suggest how the rate of the $\text{S}_{\text{N}}1$ reaction would change, if at all, if 2-chloropropane were used instead of 2-bromopropane.

Explain your answer.

[3]

[2]

7. O/N 25/P24/Q4(b)(i)

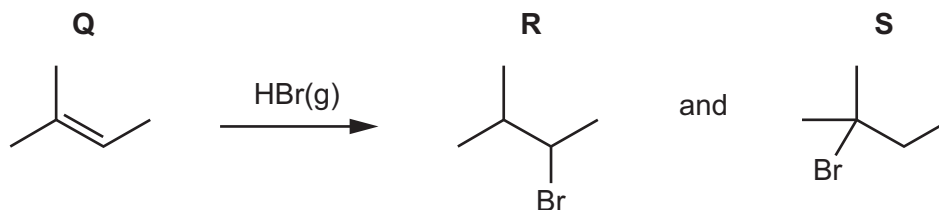
Context

4 Compounds **P** and **Q** are structural isomers.

**Fig. 4.1**

Context: part (a)(v)

(v) **Q** reacts with HBr(g) to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.

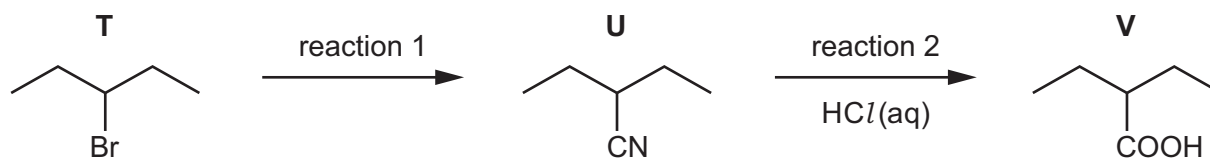
**Fig. 4.2**

State and explain why isomer **S** is the major product of the reaction.

[2]

Context: part (b) (introduction)

(b) Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.

**Fig. 4.3**

(i) Identify the reagent and conditions for reaction 1.

[1]

8. M/J 25/P22/Q4(b)(ii)

Context: part (b) (introduction)

(b) Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, SOCl_2 is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with NaOH(aq) .

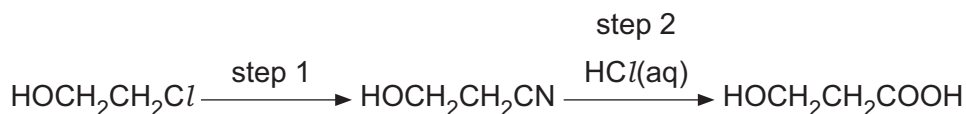
(ii) Identify the reagent and conditions required in step 2.

[1]

9. M/J 25/P23/Q6(c)(i)

Context: part (c) (introduction)

- (c) $\text{HOCH}_2\text{CH}_2\text{Cl}$ reacts in a two-step synthesis to produce $\text{HOCH}_2\text{CH}_2\text{COOH}$, as shown in Fig. 6.2.

**Fig. 6.2**

- (i) State the reagent and conditions for step 1.

[1]

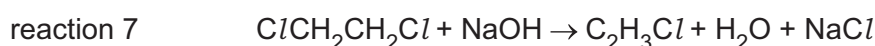
10. MAR 25/P22/Q1(c)(ii)

Context

- 1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

Context: part (c) (introduction)

- (c) 1,2-dichloroethane, $\text{ClCH}_2\text{CH}_2\text{Cl}$, reacts with NaOH to produce an unsaturated compound, $\text{C}_2\text{H}_3\text{Cl}$, as shown in reaction 7.



- (ii) State the conditions for reaction 7.

[1]

11. O/N 24/P21/Q2(c)(iv)

Context

- 2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

- (c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO .

NaClO is made from chlorine in a disproportionation reaction.

- (iv) Aqueous AgNO_3 dissolved in ethanol reacts with an aqueous solution of CHCl_3 .

State what is observed in this reaction. Explain your answer.

[2]

12. O/N 24/P23/Q2(c)(iv)

Context

- 2 Chlorine is one of the elements in Group 17 of the Periodic Table.

Context: part (c) (introduction)

- (c) CHCl_3 is another halogenoalkane. CHCl_3 forms when propanone reacts with NaClO .

NaClO is made from chlorine in a disproportionation reaction.

- (iv) Aqueous AgNO_3 dissolved in ethanol reacts with an aqueous solution of CHCl_3 .

State what is observed in this reaction. Explain your answer.

[2]

13. M/J 24/P22/Q4(b)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

(b) A sample of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ reacts with NaOH to make $\text{CH}_3(\text{CH}_2)_5\text{CH(OH)CH}_3$ in an $\text{S}_{\text{N}}1$ mechanism.

Complete Fig. 4.1 to show the mechanism for the reaction of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ and NaOH .

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 4.1**

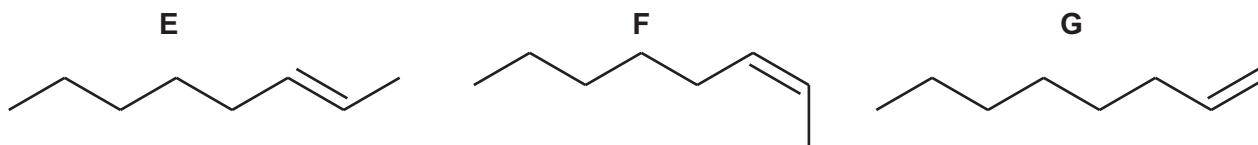
[3]

14. M/J 24/P22/Q4(d)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

(d) $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ is heated with **D** to produce three different molecules, **E**, **F** and **G**.

**Fig. 4.2**

(i) Name the type of reaction.

[1]

(ii) Identify **D** and the conditions used.

[1]

15. M/J 24/P23/Q4(c)

(c) Substitution reactions of NH_3 and OH^- with halogenoalkanes both involve a lone pair of electrons.

(i) Name the role of NH_3 and OH^- in these reactions.

[1]

(ii) Suggest which species, NH_3 or OH^- , is more reactive during these reactions. Explain your answer.

[1]

16. M/J 24/P23/Q4(d)

- (d) When 2-bromo-2-methylpropane reacts with OH^- , two mechanisms, $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$, both occur. The $\text{S}_{\text{N}}2$ mechanism has a slower rate.

Fig. 4.1 shows the reaction pathway diagram for the $\text{S}_{\text{N}}1$ mechanism.

Sketch a graph on Fig. 4.1 to show the reaction pathway for the $\text{S}_{\text{N}}2$ mechanism.

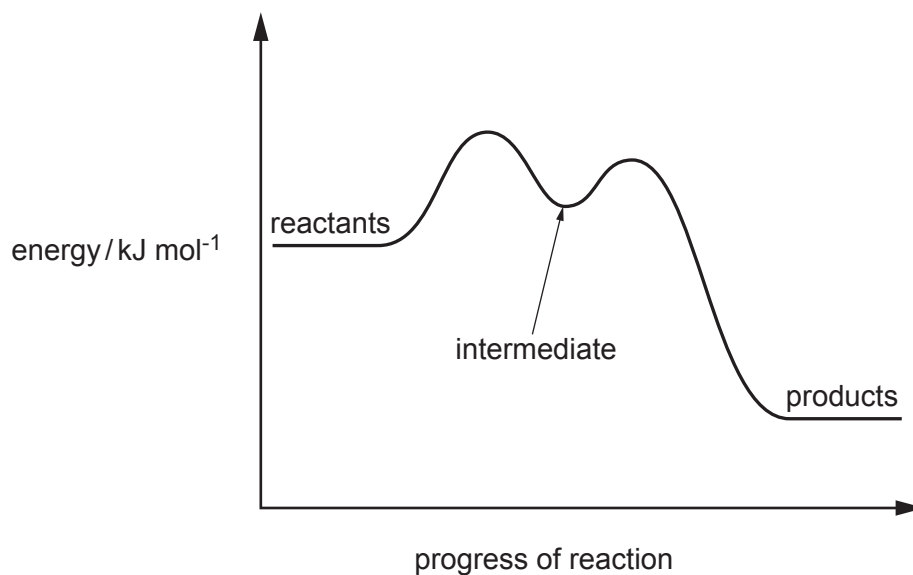
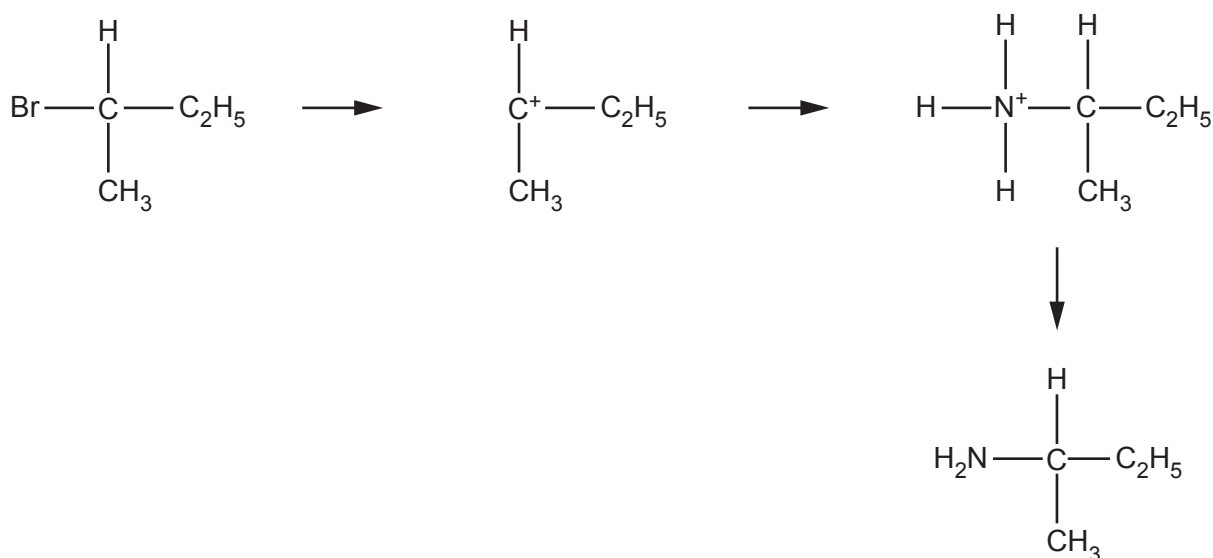


Fig. 4.1

[2]

17. M/J 24/P23/Q4(e)(i)-(ii)

- (e) (i) Complete Fig. 4.2 to show the mechanism for the S_N1 reaction that occurs when $\text{CH}_3\text{CHBrC}_2\text{H}_5$ reacts with NH_3 to produce $\text{CH}_3\text{CH}(\text{NH}_2)\text{C}_2\text{H}_5$. Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 4.2**

[3]

- (ii) Identify the inorganic product that forms in the reaction in Fig. 4.2.

[1]

18. M/J 24/P23/Q4(f)

- (f) (i) Complete Table 4.1 by drawing the structural formula of the intermediate that is formed when 2-bromo-2-methylpropane reacts in an S_N1 reaction.

Table 4.1

	2-bromobutane	2-bromo-2-methylpropane
structural formula of intermediate in S_N1 reaction	$\begin{array}{c} \text{H} \\ \\ \text{C}^+ - \text{C}_2\text{H}_5 \\ \\ \text{CH}_3 \end{array}$	

[1]

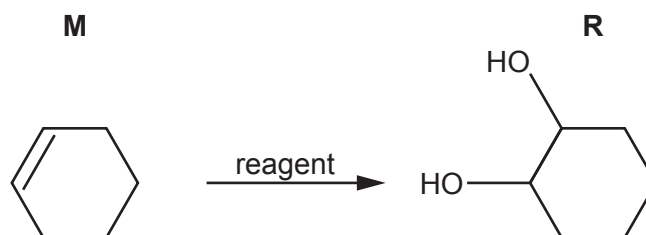
- (ii) Identify the halogenoalkane in Table 4.1 that has the greater tendency to react using the S_N1 mechanism. Explain your answer.

[2]

19. M/J 24/P23/Q5(b)(i)

Context: part (a)

- 5 (a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

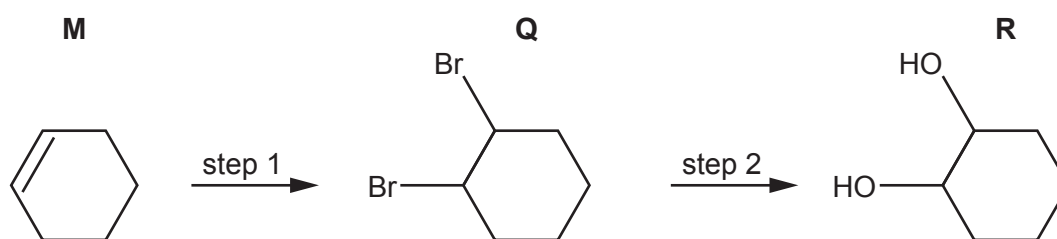
**Fig. 5.1**

Identify the reagent and conditions for this reaction.

[1]

Context: part (b) (introduction)

- (b)** **R** is also made from **M** by two steps, as shown in Fig. 5.2.

**Fig. 5.2**

- (i)** Identify the reagents and conditions for steps 1 and 2 in Fig. 5.2.

step 1

step 2

[2]

20. MAR 24/P22/Q2(d)(ii)

Context

2 Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

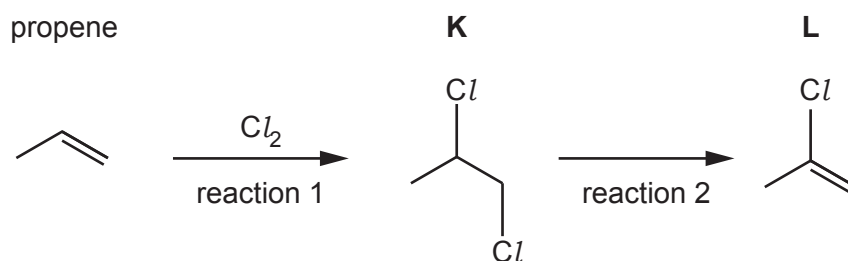
Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

Context: part (d) (introduction)

(d) Cl_2 reacts readily with propene to form **K**, 1,2-dichloropropane.

K can be used to form **L**.

**Fig. 2.1**

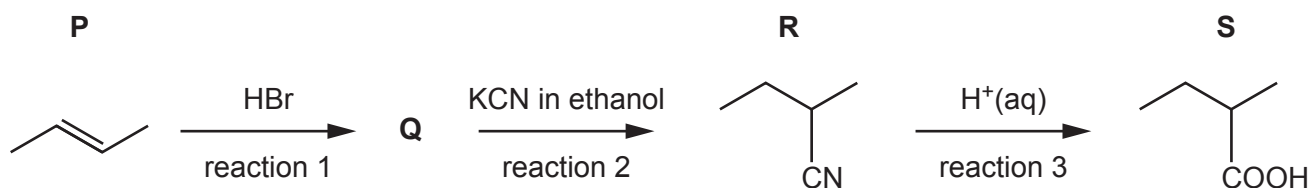
(ii) Identify the reagent and conditions for reaction 2.

[1]

21. MAR 24/P22/Q4(b)(ii)

Context

- 4** Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

**Fig. 4.1**

- (ii) Name the mechanism in reaction 2.

[1]

22. M/J 23/P21/Q3(e)

Context

- 3** Chlorine is a very reactive element.

- (e) CHCl_3 and HF are used to form CHClF_2 in a substitution reaction.

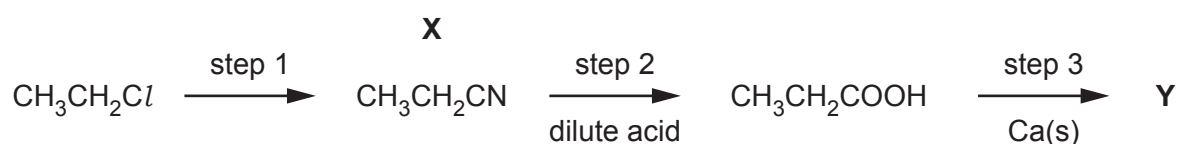
Construct an equation for this reaction.

[1]

23. M/J 23/P22/Q3(a)(i)

Context

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

- (a) (i) State the reagent and conditions for step 1 in Fig. 3.1.

[1]

24. MAR 23/P22/Q4(a)(i)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

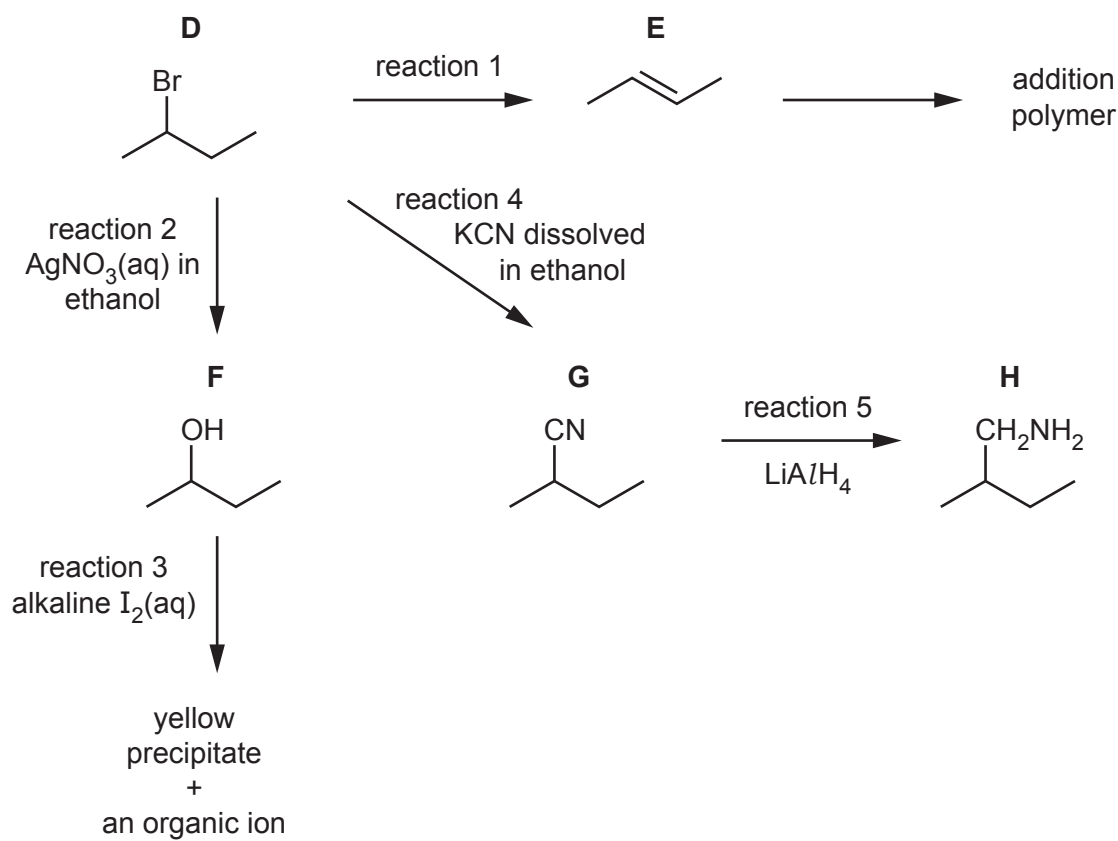


Fig. 4.1

(a) (i) State the reagent and conditions used to form **E** in reaction 1.

[1]

25. MAR 23/P22/Q4(b)(i)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

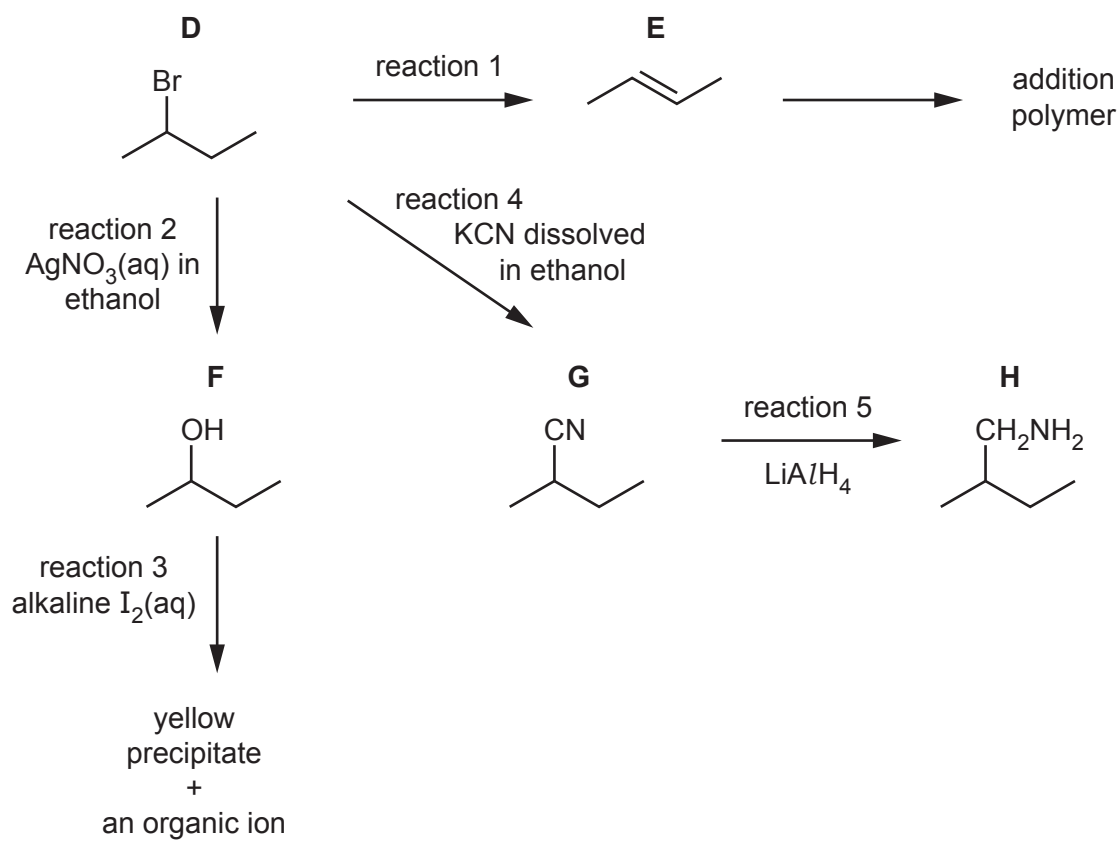


Fig. 4.1

(b) (i) Predict what is observed in reaction 2.

[1]

26. MAR 23/P22/Q4(c)(i)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

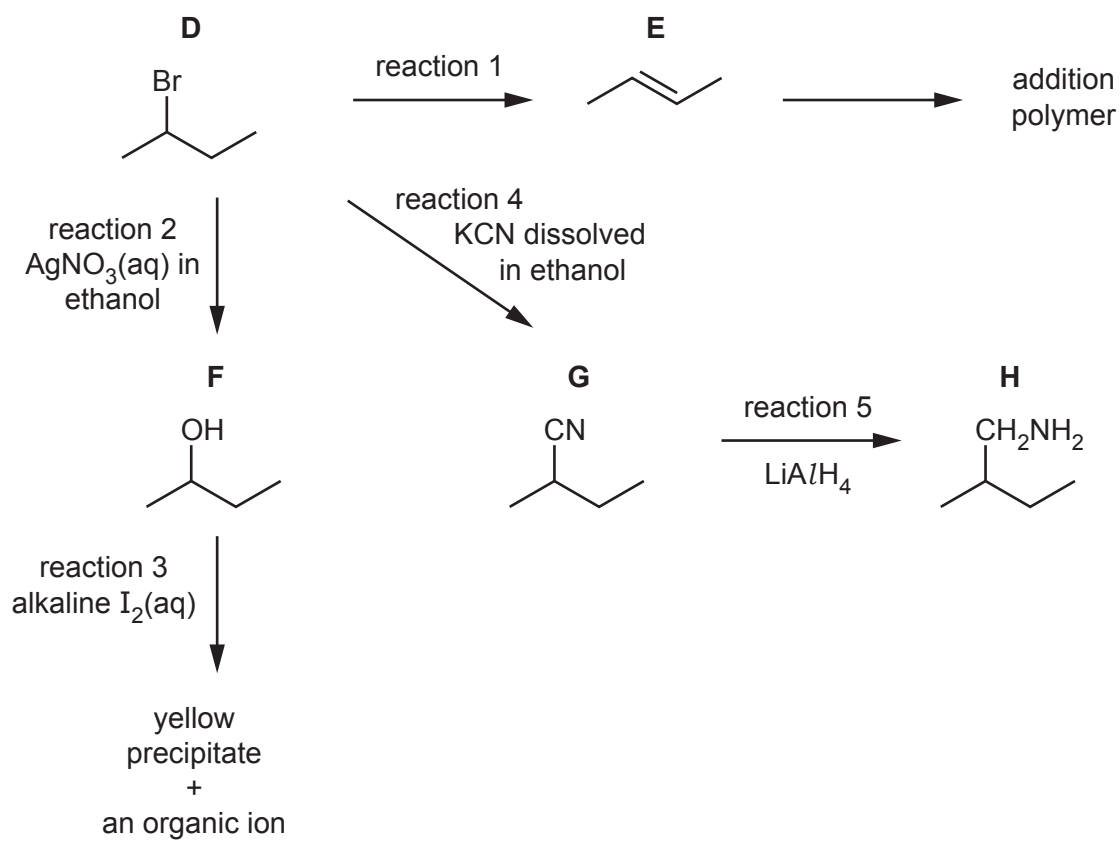


Fig. 4.1

(c) (i) State the type of reaction that occurs in reaction 4.

[1]

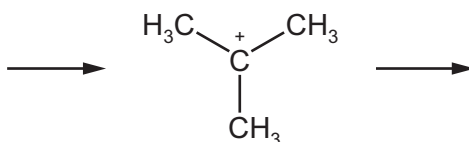
27. O/N 22/P22/Q1(e)

Context

1 Species such as NH_4^+ , CO_3^{2-} and PO_4^{3-} are examples of molecular ions.

(e) OH^- (aq) reacts with 2-bromo-2-methylpropane in an $\text{S}_{\text{N}}1$ reaction.
The molecular ion $(\text{CH}_3)_3\text{C}^+$ forms as the intermediate in this reaction.

(i) Draw the mechanism for the $\text{S}_{\text{N}}1$ reaction of OH^- with 2-bromo-2-methylpropane. Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate. Draw the structures of the organic reactant and organic product.



[3]

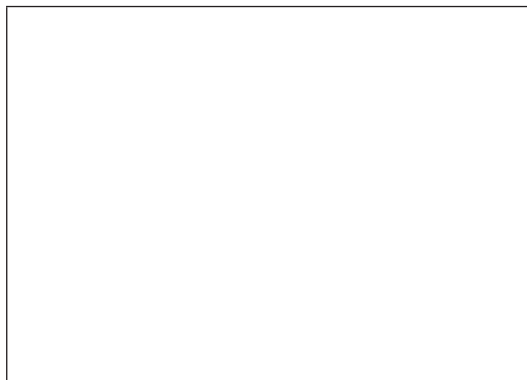
(ii) 2-bromo-2-methylpropane is a tertiary bromoalkane.

Define tertiary bromoalkane.

[1]

(iii) Organic compound **M** forms when 2-bromo-2-methylpropane is heated with **ethanolic** OH^- .

Draw the structure of **M**.

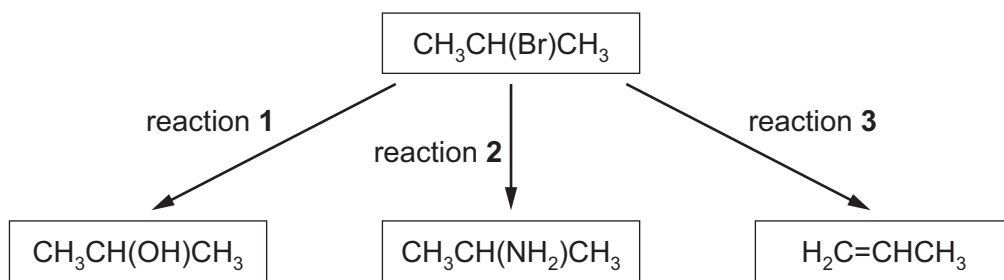


[1]

28. M/J 22/P22/Q5(a)

Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

**Fig. 5.1**

(a) Complete Table 5.1 for each reaction, by:

- stating the reagent and conditions used
- identifying the type of reaction that occurs.

Table 5.1

reaction	reagent and conditions	type of reaction
1		
2		
3		

[6]

29. M/J 22/P22/Q5(b)

Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

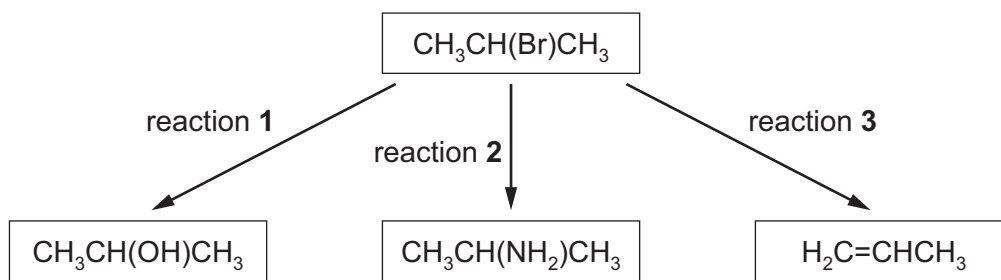


Fig. 5.1

(b) A sample of 2-iodopropane, $\text{CH}_3\text{CH}(\text{I})\text{CH}_3$, reacts under the same conditions as reaction **1** to produce $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$.

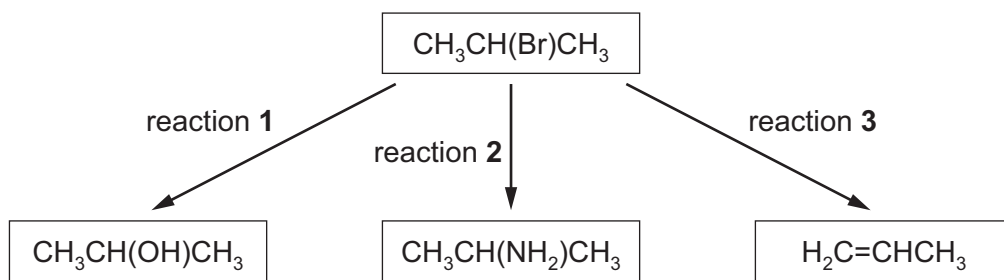
Explain why 2-iodopropane reacts at a faster rate than 2-bromopropane.

[2]

30. M/J 22/P22/Q5(c)(i)

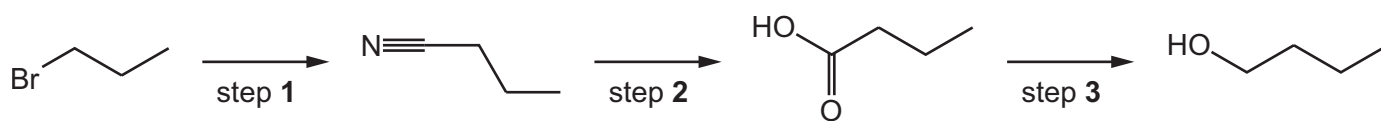
Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

**Fig. 5.1**

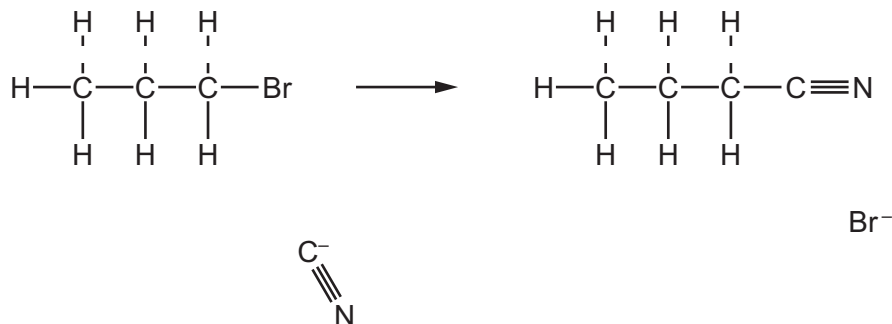
Context: part (c) (introduction)

(c) Fig. 5.2 shows how butan-1-ol can be made from 1-bromopropane in three steps.

**Fig. 5.2**

(i) In step 1, 1-bromopropane reacts with CN^- to form butanenitrile.

Complete Fig. 5.3 to show the mechanism for step 1. Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.

**Fig. 5.3**

[2]

31. M/J 22/P23/Q4(b)(i)-(iii)

Context: part (b) (introduction)

- (b) Two bottles labelled **Q** and **M** each contain a straight-chain halogenoalkane with molecular formula C_4H_9X , where **X** represents Cl, Br or I.
A sample from each bottle is added to separate samples of equal amounts of aqueous silver nitrate in ethanol. In each reaction, the same organic product, **T**, and a precipitate are made, as shown in Fig. 4.2.

**Fig. 4.2**

Table 4.1 describes the colour of each of the precipitates made.

Table 4.1

halogenoalkane added to $\text{AgNO}_3(\text{aq})$ in ethanol	colour of precipitate
Q	white
M	yellow

- (i) Identify the functional group present in **T** and name the type of reaction that occurs using the information in Fig. 4.2 and Table 4.1.

functional group in **T**

type of reaction

[2]

- (ii) Construct an ionic equation to describe the formation of the yellow precipitate produced when **M** reacts with $\text{AgNO}_3(\text{aq})$ in ethanol.

[1]

- (iii) Describe which reagent, **Q** or **M**, will produce a precipitate more quickly when each is added to $\text{AgNO}_3(\text{aq})$ in ethanol. Explain your answer.

reagent

[1]

32. M/J 22/P23/Q4(b)(v)

Context: part (b) (introduction)

- (b) Two bottles labelled **Q** and **M** each contain a straight-chain halogenoalkane with molecular formula C_4H_9X , where **X** represents Cl, Br or I.
A sample from each bottle is added to separate samples of equal amounts of aqueous silver nitrate in ethanol. In each reaction, the same organic product, **T**, and a precipitate are made, as shown in Fig. 4.2.

**Fig. 4.2**

Table 4.1 describes the colour of each of the precipitates made.

Table 4.1

halogenoalkane added to $\text{AgNO}_3(\text{aq})$ in ethanol	colour of precipitate
Q	white
M	yellow

- (v) Deduce the structure of the straight-chain halogenoalkane **M**.

[2]

33. MAR 22/P22/Q3(e)(iii)

Context

- 3** The hydrogen halides HCl , HBr and HI are all colourless gases at room temperature.

Context: part (e) (introduction)

- (e) HBr reacts with propene to form two bromoalkanes, $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and $(\text{CH}_3)_2\text{CHBr}$.

- (iii) The reaction of $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and NaOH is different depending on whether water or ethanol is used as a solvent.

Complete Table 3.2 to identify the organic and inorganic products of the reaction of $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ and NaOH in each solvent.

Table 3.2

solvent	organic product(s)	inorganic product(s)
water		
ethanol		

[2]

Answers Section

1. M/J 26/P21/Q5(e)

5(e)	elimination	1
------	-------------	---

2. MAR 26/P22/Q3(a)

3(a)	(organic molecule where) carbon (of C—Hal) is bonded to two other C atoms	1
------	---	---

3. MAR 26/P22/Q3(d)

3(d)	M1 fastest $D > C > B$ slowest M2 carbon—halogen gets weaker down the group (so breaks faster)	2
------	---	---

4. MAR 26/P22/Q4(c)

4(c)(i)	M1 curly arrow from lone pair on C of CN^- to δ^+C of $C-Cl$ M2 correct dipole of $\delta^+C-Cl^{\delta-}$ AND curly arrow from bond to Cl	2
4(c)(ii)	CN^- / cyanide	1
4(c)(iii)	nucleophilic substitution / $S_N(2)$	1

5. O/N 25/P22/Q4(b)

4(b)	sodium hydroxide AND water	1
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6. O/N 25/P24/Q3(b)(ii)-(iii)

3(b)(ii)	M1 correct dipole on $\delta^+C-Br^{\delta-}$ AND curly arrow from $C-Br$ bond to Br  M2 intermediate M3 curly arrow from lone pair of $(CH_3)_2CHO^-$ to C^+	3
3(b)(iii)	M1 rate would decrease M2 $C-Br$ is weaker (than $C-Cl$, so breaks more easily)	2

7. O/N 25/P24/Q4(b)(i)

4(b)(i)	KCN in ethanol (and heat)	1
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Answer for context part (a)(v)

4(a)(v)	M1 (reaction proceeds via a) more stable intermediate / carbocation M2 because of greater positive inductive effect of more alkyl groups	2
---------	---	---

8. M/J 25/P22/Q4(b)(ii)

4(b)(ii)	KCN / potassium cyanide in ethanol / C_2H_5OH AND heat	1
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9. M/J 25/P23/Q6(c)(i)

6(c)(i)	KCN/potassium cyanide in ethanol AND heat	1
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10. MAR 25/P22/Q1(c)(ii)

1(c)(ii)	ethanol (solvent) AND heat	1
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11. O/N 24/P21/Q2(c)(iv)

2(c)(iv)	M1 white precipitate/white ppt forms	1
	M2 CHCl_3 is hydrolysed / chloride ions are released OR Cl^- / chloride ions which are released react with Ag^+ OR silver chloride / AgCl is formed	1

12. O/N 24/P23/Q2(c)(iv)

2(c)(iv)	M1 white precipitate/white ppt forms	1
	M2 CHCl_3 is hydrolysed / chloride ions are released OR Cl^- / chloride ions which are released react with Ag^+ OR silver chloride / AgCl is formed	1

13. M/J 24/P22/Q4(b)

4(b)	M1 correct dipole ($\delta^+\text{C}-\text{Br}^{\delta-}$) AND curly arrow from bond of $\text{C}-\text{Br}$ to Br (or just beyond) M2 correct intermediate ALLOW R for $-(\text{CH}_2)_5\text{CH}_3$ M3 curly arrow from lone pair on OH^- to the C of intermediate (or suitable position where $\text{C}-\text{OH}$ bond is made) <i>If $\text{S}_{\text{N}}2$ mechanism shown:</i> Apply $\text{S}_{\text{N}}2$ mechanism mark scheme only if mechanism shows OH^- / NaOH species attacking the reactant and no OH M1 correct dipole ($\delta^+\text{C}-\text{Br}^{\delta-}$) AND curly arrow from bond of $\text{C}-\text{Br}$ to Br of reactant (or just beyond) NOT M2 M3 curly arrow from lone pair on $:\text{OH}^-$ to the appropriate C of the reactant	3
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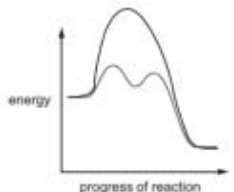
14. M/J 24/P22/Q4(d)

4(d)(i)	elimination	1
4(d)(ii)	NaOH / sodium hydroxide in ethanol	1

15. M/J 24/P23/Q4(c)

4(c)(i)	nucleophile	1
4(c)(ii)	OH^- (more reactive because) negatively charged species more attracted to halogenoalkane and / or $\text{C}^{\delta+}$ and / or intermediate and / or C^+ OR NH_3 (more reactive because) N electrons less tightly held OR N is more able to donate (lp) electrons (to make the covalent bond) OR N is less electronegative ORA	1

16. M/J 24/P23/Q4(d)

4(d)	 M1 same start energy and finishing energy AND single hump M2 transition state / activation energy shown in answer is higher than both transition state energies shown in reaction profile (as the reaction is slower)	2
------	--	---

17. M/J 24/P23/Q4(e)(i)-(ii)

4(e)(i)	M1 dipole and correct arrow on C—Br of 2-bromobutane M2 lone pair on NH₃ attacking C(+) of the intermediate drawn M3 curly arrow showing bond between N—H breaking and pointing towards the N	3
4(e)(ii)	NH ₄ Br / ammonium bromide	1

18. M/J 24/P23/Q4(f)

4(f)(i)		1
4(f)(ii)	M1 2-bromo-2-methylpropane AND more stable carbocation / C⁺ M2 alkyl groups have an inductive effect (of alkyl groups) AND 2-bromo-2-methylpropane has more / most alkyl groups attached (to C⁺) / carbocation contains more alkyl groups	2

19. M/J 24/P23/Q5(b)(i)

5(b)(i)	step 1Br₂ (in the dark) step 2NaOH(aq) + heat	2
---------	--	---

Answer for context part (a)

5(a)	cold dilute KMnO ₄ OR cold dilute potassium manganate(VII)	1
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20. MAR 24/P22/Q2(d)(ii)

2(d)(ii)	NaOH in ethanol AND heat	1
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21. MAR 24/P22/Q4(b)(ii)

4(b)(ii)						1
	geometrical	✓	(x)	(x)	(x)	
	optical	(x)	✓	✓	✓	
CH ₃ CHBrCH ₂ CH ₃ nucleophilic substitution						

22. M/J 23/P21/Q3(e)

3(e)	CHCl ₃ + 2HF → CHClF ₂ + 2HCl	1
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23. M/J 23/P22/Q3(a)(i)

3(a)(i)	KCN in ethanol + heat	1
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24. MAR 23/P22/Q4(a)(i)

4(a)(i)	NaOH in alcohol / ethanol AND heat (under reflux)	1
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25. MAR 23/P22/Q4(b)(i)

4(b)(i)	cream(-coloured) / off-white precipitate (forms)	1
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26. MAR 23/P22/Q4(c)(i)

4(c)(i)	substitution	1
---------	--------------	----------

27. O/N 22/P22/Q1(e)

1(e)(i)	M1 correct dipole in on $\delta^+C-Br^{\delta-}$ (in tBuBr) AND curly arrow from $C-Br$ to Br M2 curly arrow from electron pair on O of $^-OH / :OH^-$ to C^+ in carbocation M3 tBuOH product	1
1(e)(ii)	The $C-Br$ carbon is bonded to three other carbon atoms <i>owtte</i>	1
1(e)(iii)		1

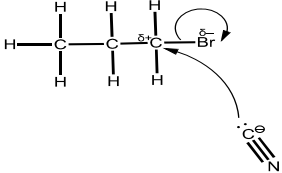
28. M/J 22/P22/Q5(a)

5(a)	reaction	reagent and conditions	type of reaction	6
	1	NaOH(aq) [M1]	substitution	
	2	NH ₃ [M2] in ethanol AND heat under pressure [M3]	substitution	
	3	NaOH in ethanol AND heat [M4]	elimination	
	M5 and M6 types of reaction [3 correct types = 2 marks and 2 correct types = 1 mark] All reagents can be identified as (correct) formula or in words			

29. M/J 22/P22/Q5(b)

5(b)	M1 C–I (covalent bond) is weaker / or a M2 lower activation energy / lower E_a (with 2-iodopropane) OR M2 explain in terms of the SN_1 mechanism (that is dominant / preferred / occurring with 2-iodopropane is fast(er)) (identical) carbocation is made (more) quickly (with 2-iodopropane) OR low / less energy is required to make the (same) carbocation / intermediate (from 2-iodopropane) OR M2 iodine / I (of C–I) has weaker attraction of nucleus to bonding / shared pair (of electrons) due to any one of: • more / high shielding (of electrons from inner shells) • more / 5 electron shells in iodine • C–I greater distance from nuclei (to bonding pair of electrons)	2
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30. M/J 22/P22/Q5(c)(i)

5(c)(i)	 M1 dipole $C^{\delta+}-Br^{\delta-}$ AND curly arrow from C–Br bond to $Br^{\delta-}$ M2 lone pair on C of ^-CN AND curly arrow from lone pair to C (on C-1 of bromopropane)	2
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31. M/J 22/P23/Q4(b)(i)-(iii)

4(b)(i)	M1 hydroxy(l) M2 substitution	2
4(b)(ii)	$I^- + Ag^+ \rightarrow AgI$	1
4(b)(iii)	M AND C–I (bond) is weak(er)	1

32. M/J 22/P23/Q4(b)(v)

4(b)(v)	$CH_3CH_2CHICH_3$ M1 correct identification of I on 4C molecule M2 unambiguous structure of M – unbranched with 4C and I on 2C	2
---------	---	----------

33. MAR 22/P22/Q3(e)(iii)

3(e)(iii)	M1: (water solvent) $CH_3CH_2CH_2OH$ AND NaBr / sodium bromide / propan-1-ol M2: (ethanol solvent) CH_3CHCH_2 AND H_2O / water / propene AND NaBr / sodium bromide	2
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Unit 16

HYDROXY COMPOUNDS

43 items · 87 marks

Syllabus sections in this unit

16.1 Alcohols

Unit 16: Hydroxy compounds

1. M/J 26/P21/Q4(a)(i)

Context: part (a) (introduction)

- 4 (a) Compound **Z** is oxidised to produce organic compound **W**, $\text{C}_5\text{H}_6\text{O}_7$, by heating under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$.

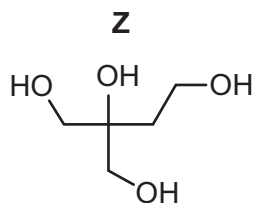


Figure 4.1

- (i) Describe the colour change in this reaction.

colour of mixture at start of the reaction

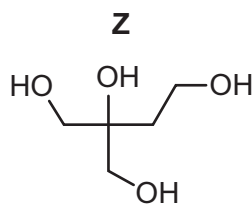
colour of mixture after the reaction

[1]

2. M/J 26/P21/Q4(b)(i)

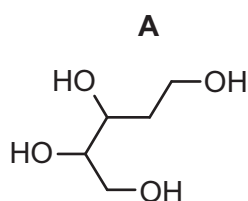
Context: part (a) (introduction)

- 4 (a)** Compound **Z** is oxidised to produce organic compound **W**, $\text{C}_5\text{H}_6\text{O}_7$, by heating under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$.

**Figure 4.1**

Context: part (b) (introduction)

- (b)** Compound **A** is a structural isomer of **Z**.

**Figure 4.2**

- (i)** Complete Table 4.1 to show what is observed when separate samples of **A** are added to each reagent. State 'none' if no reaction occurs.

Table 4.1

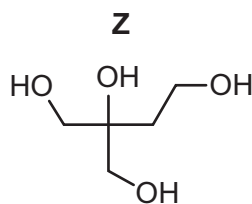
reagent	observation with A
Na(s)	
2,4-dinitrophenylhydrazine (2,4-DNPH)	
Fehling's reagent	none

[2]

3. M/J 26/P21/Q4(b)(iii)

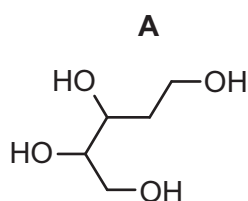
Context: part (a) (introduction)

- 4 (a)** Compound **Z** is oxidised to produce organic compound **W**, $\text{C}_5\text{H}_6\text{O}_7$, by heating under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$.

**Figure 4.1**

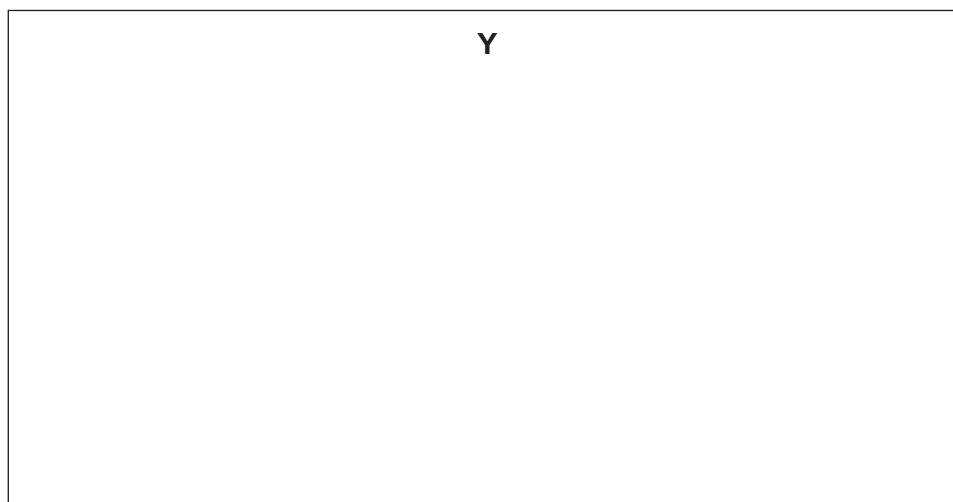
Context: part (b) (introduction)

- (b)** Compound **A** is a structural isomer of **Z**.

**Figure 4.2**

- (iii)** Compound **A** is heated under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$ to produce organic compound **Y**.

Draw the structure of **Y**.



[2]

4. M/J 26/P22/Q4(a)

4 (a) Figure 4.1 shows two reactions of compound **D**.

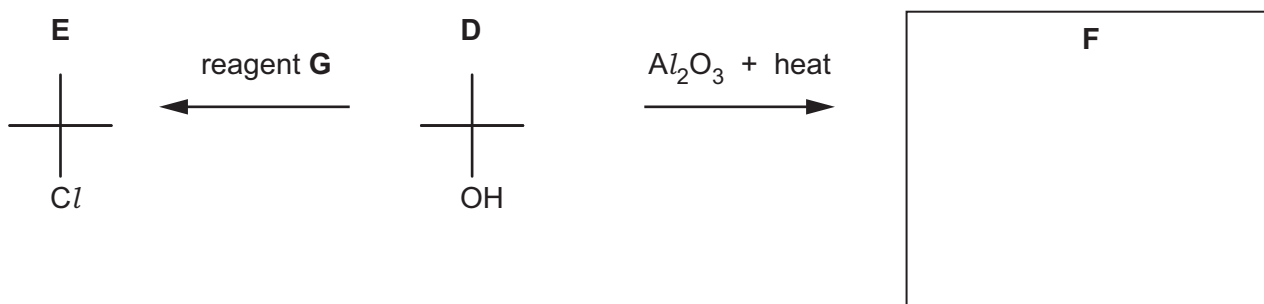


Figure 4.1

(i) **D** reacts with a single reagent, **G**, to produce compound **E**. Identify reagent **G**.

[1]

(ii) **D** is heated with Al_2O_3 to produce organic compound **F**. Identify the role of Al_2O_3 in this reaction.

[1]

(iii) Draw the structure of **F** in Figure 4.1.

[1]

5. M/J 26/P22/Q4(b)(i)

Context: part (b) (introduction)

(b) The structure of compound **X** is shown in Figure 4.2.

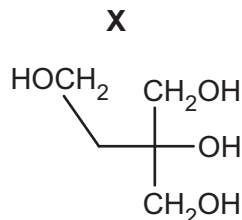


Figure 4.2

The molecular formula of **X** is $\text{C}_5\text{H}_{12}\text{O}_4$. Complete oxidation of 0.255 g of **X** occurs when it reacts with exactly 10.0 cm^3 of $0.500 \text{ mol dm}^{-3}$ acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$.

(i) Describe the colour change in this reaction.

colour of mixture at start of the reaction

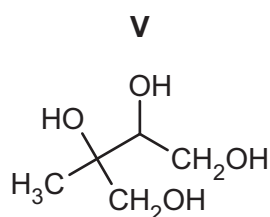
colour of mixture after the reaction

[1]

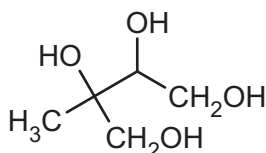
6. M/J 26/P23/Q4(a)(i)-(ii)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

**Figure 4.1**

Context: Fig. 4.1

**Figure 4.1**

- (a) (i) Complete Table 4.1.

Table 4.1

	primary hydroxyl groups	secondary hydroxyl groups	tertiary hydroxyl groups
number of groups in one molecule of V			

[2]

- (ii) Complete Table 4.2 to show what is observed when separate samples of **V** are added to each reagent. State 'none' if no reaction occurs.

Table 4.2

reagent	observation with V
Na(s)	
2,4-dinitrophenylhydrazine (2,4-DNPH)	
Fehling's reagent	none

[2]

7. M/J 26/P23/Q4(b)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

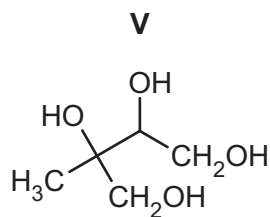


Figure 4.1

- (b)** Figure 4.1 is repeated for use in **4(b)**.

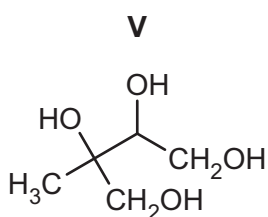


Figure 4.1

V is completely oxidised by heating under reflux with an excess of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce organic compound **Z**.

- (i)** Describe the colour change in this reaction.

colour of mixture at start of the reaction

colour of mixture after the reaction

[1]

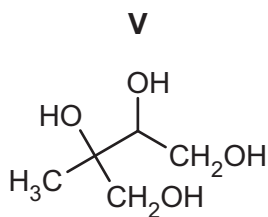
- (ii)** Draw the structure of **Z**.

[1]

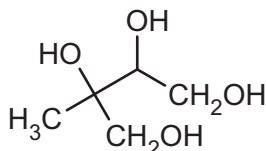
8. M/J 26/P23/Q4(c)(iii)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

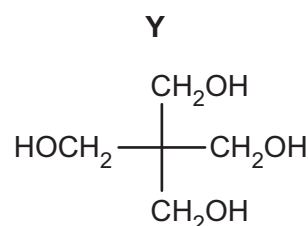
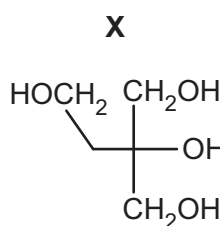
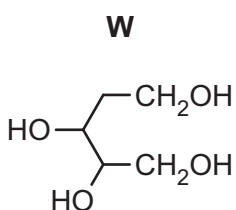
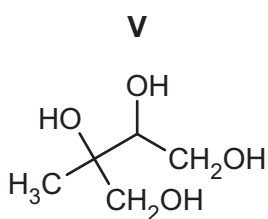
**Figure 4.1**

Context: Fig. 4.1

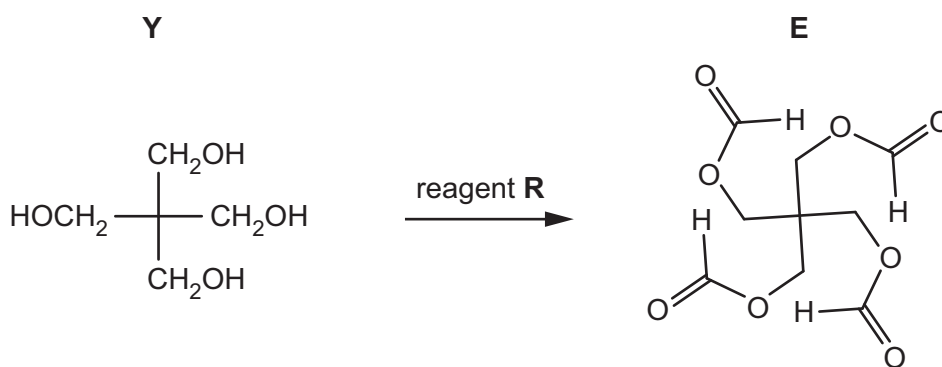
**Figure 4.1**

Context: part (c) (introduction)

- (c)** Compounds **W**, **X** and **Y** are structural isomers of **V**.

**Figure 4.2**

- (iii)** **Y** reacts with reagent **R** to produce compound **E**.

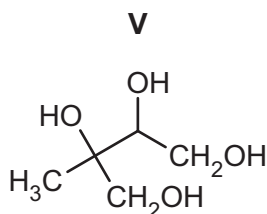
**Figure 4.3**

Name reagent **R**.

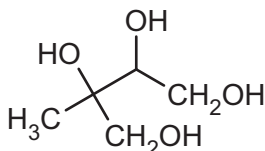
9. M/J 26/P23/Q4(d)(i)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

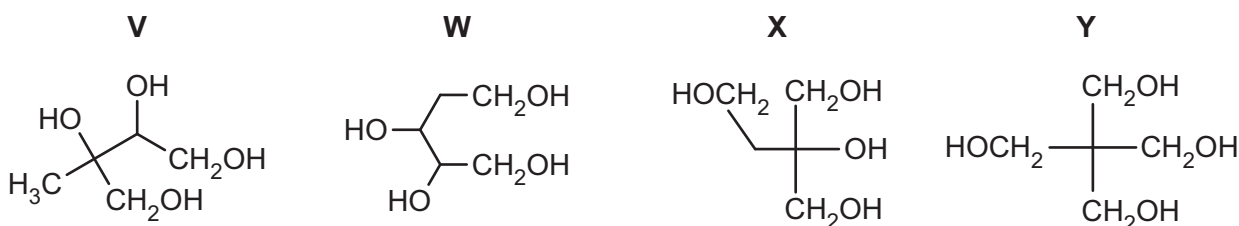
**Figure 4.1**

Context: Fig. 4.1

**Figure 4.1**

Context: part (c) (introduction)

- (c)** Compounds **W**, **X** and **Y** are structural isomers of **V**.

**Figure 4.2**

Context: part (d) (introduction)

- (d)** Exactly 0.03 mol of one of the isomers, **V**, **W**, **X** or **Y**, is completely oxidised by 0.08 mol of acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ to produce only **one** organic compound with molecular formula $\text{C}_5\text{H}_4\text{O}_8$.

- (i)** Identify which isomer, **V**, **W**, **X** or **Y**, is used in this reaction.

[1]

10. M/J 26/P24/Q4(a)

- 4** **(a)** Alcohol **S** has the structural formula $\text{CH}_3\text{C}(\text{OH})(\text{CH}_3)\text{CH}_2\text{CH}_3$.

Classify alcohol **S**.

[1]

11. M/J 26/P24/Q4(b)

(b) Figure 4.1 shows compound **T**.

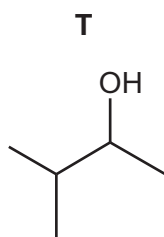


Figure 4.1

Tests are carried out on **T** using different reagents.

Complete Table 4.1 to show the observations of each test on **T**. If no reaction occurs, write 'no reaction' in the relevant box.

Table 4.1

reagent	observations
Na	
acidified $\text{K}_2\text{Cr}_2\text{O}_7$	
alkaline $\text{I}_2(\text{aq})$	

[3]

12. MAR 26/P22/Q3(e)

Context

3 Secondary halogenoalkanes are important building blocks in organic chemistry.

2-chlorobutane is an example of a secondary halogenoalkane.

2-chlorobutane

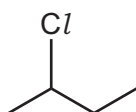


Figure 3.1

Context: part (d)

(d) The rates of hydrolysis of three secondary halogenoalkanes, **B**, **C** and **D**, are compared.

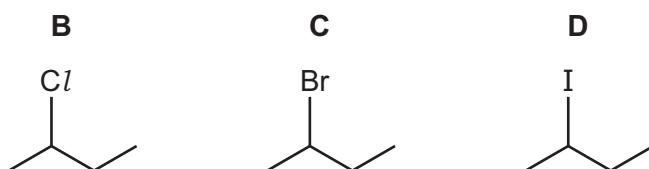


Figure 3.3

Each halogenoalkane is dissolved in ethanol and reacted with $\text{AgNO}_3(\text{aq})$.

Identify the order of the rates of hydrolysis of compounds **B**, **C** and **D**. Explain your answer.

>

>

fastest
explanation

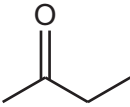
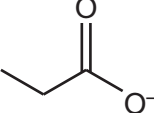
slowest

[2]

(e) The organic product of hydrolysis of each of **B**, **C** and **D** is butan-2-ol.

Complete Table 3.1 to give details of some typical chemical reactions of butan-2-ol.

Table 3.1

reagent and conditions	observation with butan-2-ol	organic product(s)
acidified $\text{K}_2\text{Cr}_2\text{O}_7$		
Na(s)	effervescence	
	yellow precipitate	 and CHI_3

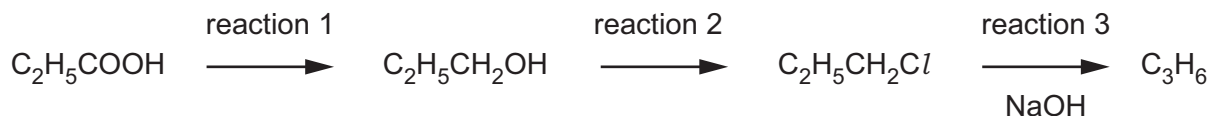
[3]

13. O/N 25/P21/Q4(a)

also Unit 15 (2 marks), Unit 18 (1 mark); this unit 2 marks

Context

- 4** Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

- (a) (i) Identify the type of reaction that occurs in reaction 1.

[1]

- (ii) Suggest a suitable reagent for reaction 2.

[1]

- (iii) Reaction 3 is an elimination reaction.

Write an equation for this reaction and identify the solvent and conditions used.

equation

solvent and conditions

[2]

- (iv) $C_2H_5CH_2OH$ can be directly converted to C_3H_6 .

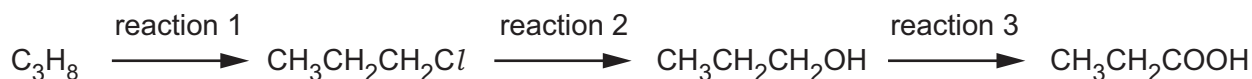
Suggest the reagent and conditions for this conversion.

[1]

14. O/N 25/P22/Q4(c)

Context

- 4** Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, CH_3CH_2COOH .

**Fig. 4.1**

- (c) Reaction 3 takes place when $CH_3CH_2CH_2OH$ is heated under reflux with acidified potassium dichromate(VI) solution.

- (i) State the colour change that takes place in the reaction mixture.

[1]

- (ii) Construct an equation to represent reaction 3. Use [O] to represent an atom of oxygen from the oxidising agent.

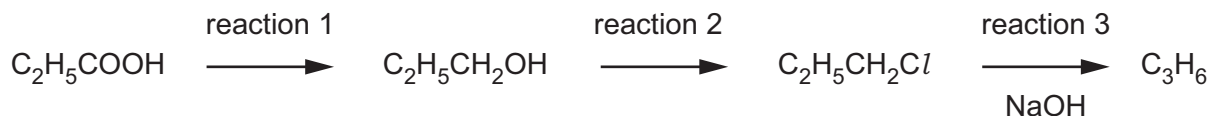
[1]

15. O/N 25/P23/Q4(a)

also Unit 15 (2 marks), Unit 18 (1 mark); this unit 2 marks

Context

- 4** Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

- (a) (i)** Identify the type of reaction that occurs in reaction 1.

[1]

- (ii)** Suggest a suitable reagent for reaction 2.

[1]

- (iii)** Reaction 3 is an elimination reaction.

Write an equation for this reaction and identify the solvent and conditions used.

equation

solvent and conditions

[2]

- (iv)** $C_2H_5CH_2OH$ can be directly converted to C_3H_6 .

Suggest the reagent and conditions for this conversion.

[1]

16. O/N 25/P24/Q2(c)(ii)

Context

- 2** Aluminium oxide, Al_2O_3 , and phosphorus(V) oxide, P_4O_{10} , are both used as reagents and catalysts.

Context: part (b) (introduction)

- (b)** A 5.00 dm^3 sealed flask contains 0.400 mol of $CO(g)$ and 0.800 mol of $H_2(g)$ and an Al_2O_3 catalyst. The flask is heated to a temperature of 290°C and allowed to reach equilibrium. Equation 1 shows the reaction.



The equilibrium constant, K_c , of equation 1 is given.

$$K_c = \frac{[CH_3OH]}{[CO][H_2]^2}$$

Context: part (c) (introduction)

- (c)** P_4O_{10} catalyses the reversible reaction of CO with H_2 to form CH_3OH .



P_4O_{10} then acts as a dehydrating agent, causing CH_3OH to form CH_3OCH_3 .

- (ii)** Construct an equation for the dehydration reaction of CH_3OH to form CH_3OCH_3 .

[1]

17. O/N 25/P24/Q3(a)(i)-(iii)

Context

3 Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

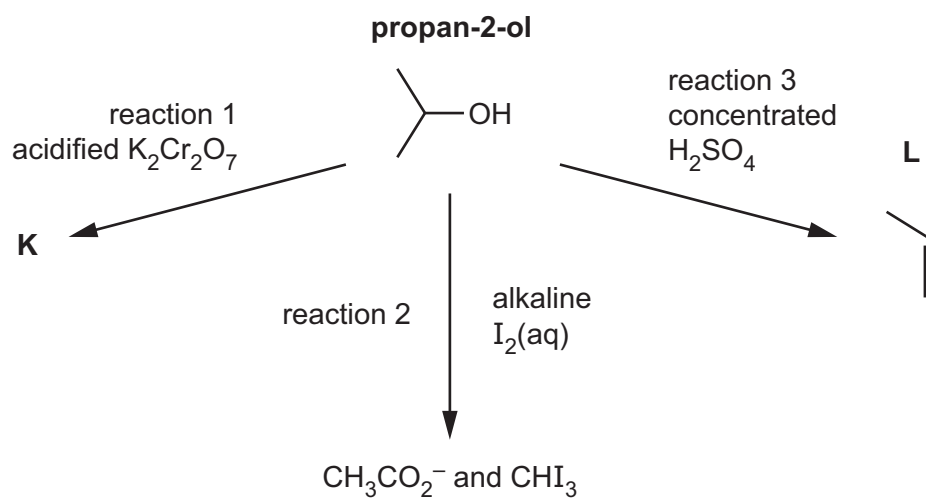


Fig. 3.1

(a) (i) Draw the structure of organic compound **K**.

[1]

(ii) State an observation you would make in reaction 2.

[1]

(iii) State the type of reaction that is shown in reaction 3.

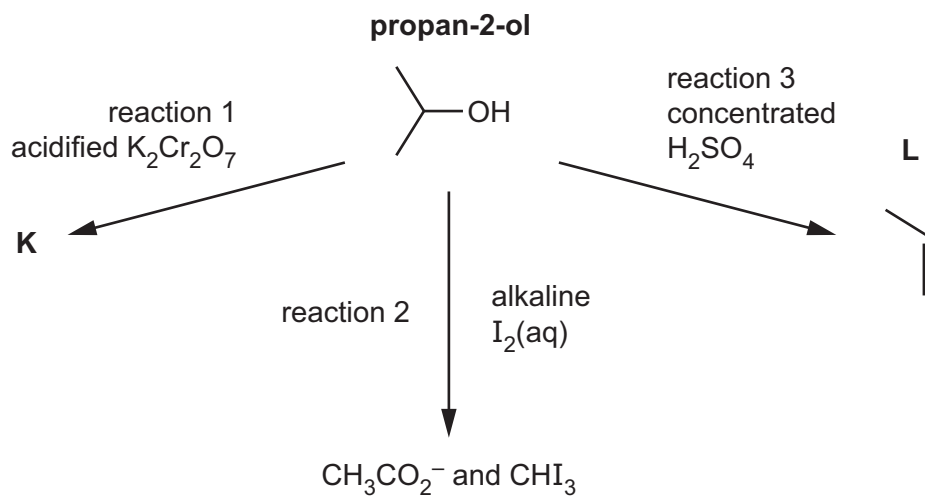
[1]

18. O/N 25/P24/Q3(b)(i)

Context

- 3** Propan-2-ol, $(\text{CH}_3)_2\text{CHOH}$, is sometimes added to fuel to help it burn.

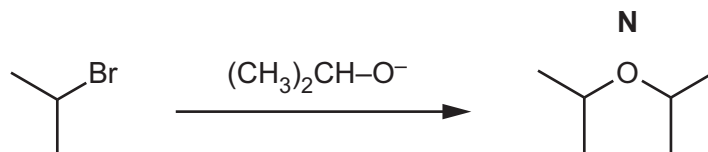
Fig. 3.1 shows some reactions of propan-2-ol.

**Fig. 3.1**

Context: part (b) (introduction)

- (b)** Propan-2-ol reacts with sodium to produce $(\text{CH}_3)_2\text{CH-O}^-$ anions.

These anions react with 2-bromopropane to form compound **N**, as shown in Fig. 3.3.

**Fig. 3.3**

- (i)** Write an equation for the reaction of propan-2-ol with sodium.

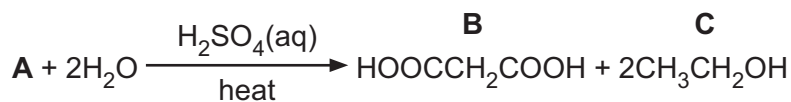
[1]

19. M/J 25/P21/Q5(c)(ii)

Context

- 5** Compound **A** contains the elements carbon, hydrogen and oxygen only.

When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

Context: part (c) (introduction)

- (c)** When aqueous Na_2CO_3 is added to separate samples of **B** and **C**, effervescence is observed with **B** only.

- (ii)** Suggest why **C** does **not** react with aqueous Na_2CO_3 in a similar type of reaction to **B**.

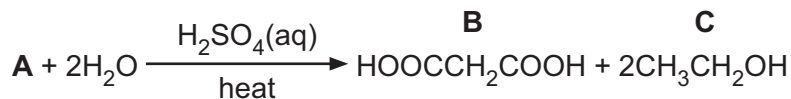
[1]

20. M/J 25/P21/Q5(d)(ii)

Context

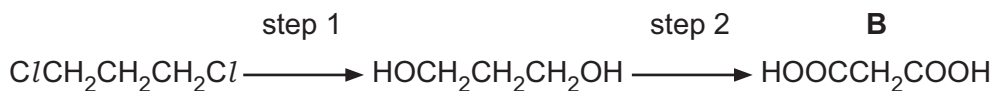
5 Compound **A** contains the elements carbon, hydrogen and oxygen only.

When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

Context: part (d) (introduction)

(d) A student suggests a two-step synthesis to produce **B**, as shown in Fig. 5.2.

**Fig. 5.2**

(ii) Identify the type of reaction that $\text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH}$ undergoes in step 2.

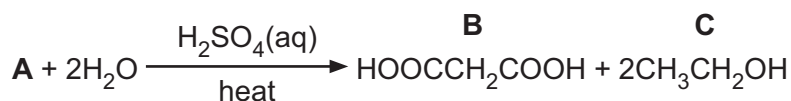
[1]

21. M/J 25/P21/Q5(e)(iii)

Context

5 Compound **A** contains the elements carbon, hydrogen and oxygen only.

When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

Context: part (e) (introduction)

(e) **B** reacts with an excess of reducing agent **R** to produce **D**.

An excess of PCl_5 is added to **D**.

- A vigorous reaction occurs.
- Misty fumes are seen.
- Organic compound **E** is produced.

(iii) Name organic compound **E**.

[1]

22. M/J 25/P21/Q6(c)(iii)

Context

- 6** Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.

Context: part (c) (introduction)

- (c) Separate samples of **F**, **G** and **H** are each tested with Tollens' reagent and with alkaline $I_2(aq)$. The observations are shown in Table 6.1.

Table 6.1

	Tollens' reagent	alkaline $I_2(aq)$
F	no observable change	pale yellow precipitate
G	silver mirror	pale yellow precipitate
H	silver mirror	no precipitate

- (iii) Compound **J** does **not** contain the same functional group as **F**, **G** and **H**. Compound **J** also reacts with alkaline $I_2(aq)$ to produce a pale yellow precipitate.

Suggest the structure of compound **J**.

[1]

23. M/J 25/P22/Q4(b)(i)

Context: part (b) (introduction)

- (b) Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, $SOCl_2$ is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with $NaOH(aq)$.

- (i) Name the type of reaction that occurs in step 1.

[1]

24. M/J 25/P22/Q4(c)(ii)-(iii)

Context: part (c) (introduction)

- (c) Compounds **A**, **B** and **C** belong to the alcohol homologous series. Each molecule of **A**, **B** and **C** contains four saturated carbon atoms.

- (ii) Identify the gas produced when $Na(s)$ is added to separate samples of each alcohol.

[1]

- (iii) Describe the role of $Na(s)$ when it reacts with alcohols.

[1]

25. M/J 25/P22/Q4(d)

(d) Table 4.1 shows the results of two tests on separate samples of **A**, **B** and **C**.

Table 4.1

compound	heat under reflux with acidified $K_2Cr_2O_7$	warm with alkaline $I_2(aq)$
A	remains orange	no visible change
B	orange to green	no visible change
C	orange to green	pale yellow precipitate

(i) Deduce the structure of **A**.

[1]

(ii) Deduce a possible name for **B**.

[1]

(iii) Deduce the identity of **C**.

[1]

26. O/N 24/P22/Q2(b)(iv)

Context

2 Oxygen is a Group 16 element.

Context: part (b) (introduction)

(b) H_2O reacts with both inorganic and organic compounds.

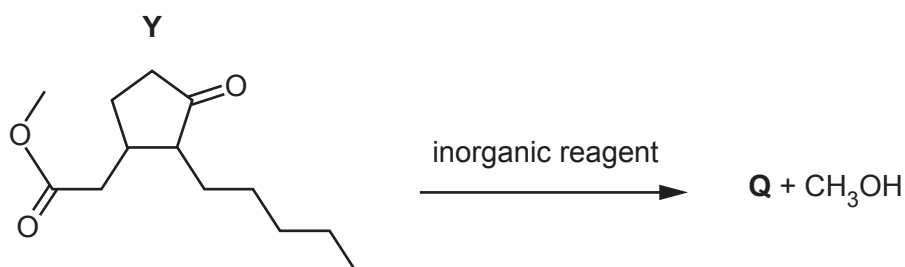
(iv) Explain why alcohols are less acidic than water.

[2]

27. M/J 24/P21/Q6(c)(iii)

Context: part (c) (introduction)

(c) Both functional groups in one molecule of **Y** react with an inorganic reagent to form one molecule of **Q** and one molecule of methanol, CH_3OH , as shown in Fig. 6.3.

**Fig. 6.3**

(iii) Under certain conditions, 0.0020 mol of **Q** reacts with an excess of sodium to produce a total of 44.8 cm^3 of gas at s.t.p.

Calculate the number of hydroxyl groups present in a molecule of **Q**.

Show your working.

number of hydroxyl groups =

[2]

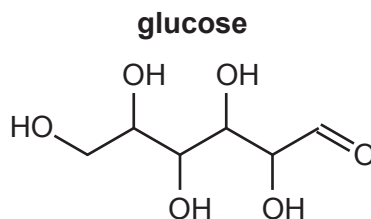
28. O/N 23/P21/Q3(e)(i)-(ii)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

Context: part (e) (introduction)

- (e)** The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.

**Fig. 3.2**

- (i)** Complete Table 3.1 to identify the number of primary, secondary and tertiary alcohol groups present in the structure shown in Fig. 3.2.

Table 3.1

type of alcohol group	primary	secondary	tertiary
number of groups			

[1]

- (ii)** Separate samples of aqueous glucose are tested with the reagents shown in Table 3.2.

Complete Table 3.2 with the observation for each reaction.

Write “no reaction” if applicable.

Table 3.2

reagent and conditions	observation with glucose
acidified $\text{KMnO}_4(\text{aq})$ and warm	
Fehling's reagent and warm	
alkaline $\text{I}_2(\text{aq})$ and warm	

[3]

29. O/N 23/P22/Q3(c)(ii)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (c) (introduction)

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

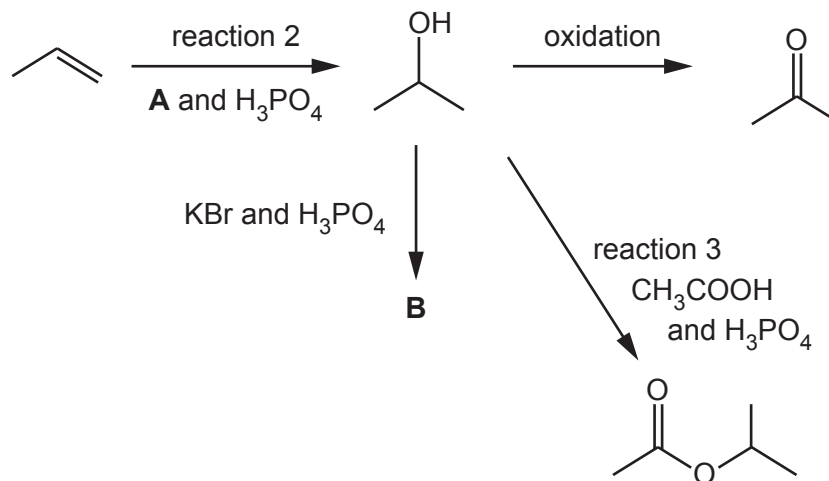


Fig. 3.1

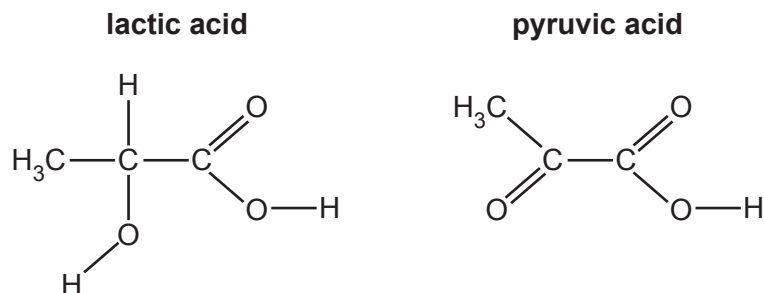
(ii) Draw the structure of **B**.

[1]

30. O/N 23/P22/Q4(b)(ii)

Context

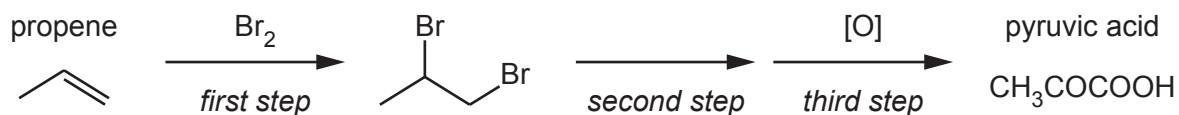
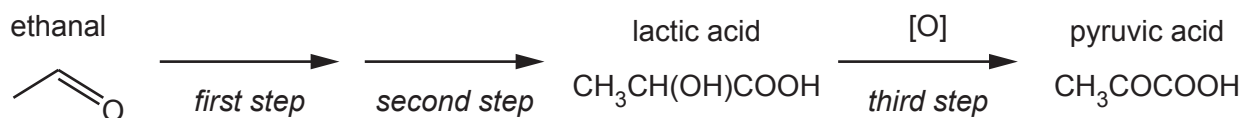
- 4** Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, $\text{CH}_3\text{COCO}\text{OH}$, both contain two functional groups.

**Fig. 4.1**

Context: part (b) (introduction)

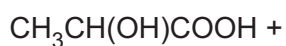
- (b)** Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

**Fig. 4.3****Fig. 4.4**

- (ii)** Write an equation for the oxidation of lactic acid to pyruvic acid, the third step of Fig. 4.4.

Use [O] to represent one atom of oxygen from an oxidising agent.



[1]

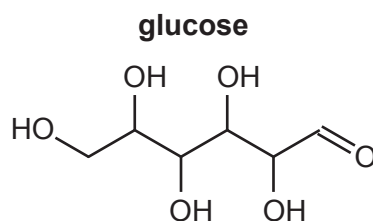
31. O/N 23/P23/Q3(e)(i)-(ii)

Context

- 3** Potassium chlorate, KClO_3 , is widely used as an oxidising agent and to make $\text{O}_2(\text{g})$.

Context: part (e) (introduction)

- (e)** The structure of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is shown in Fig. 3.2.

**Fig. 3.2**

- (i)** Complete Table 3.1 to identify the number of primary, secondary and tertiary alcohol groups present in the structure shown in Fig. 3.2.

Table 3.1

type of alcohol group	primary	secondary	tertiary
number of groups			

[1]

- (ii)** Separate samples of aqueous glucose are tested with the reagents shown in Table 3.2.

Complete Table 3.2 with the observation for each reaction.

Write “no reaction” if applicable.

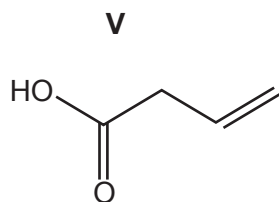
Table 3.2

reagent and conditions	observation with glucose
acidified $\text{KMnO}_4(\text{aq})$ and warm	
Fehling's reagent and warm	
alkaline $\text{I}_2(\text{aq})$ and warm	

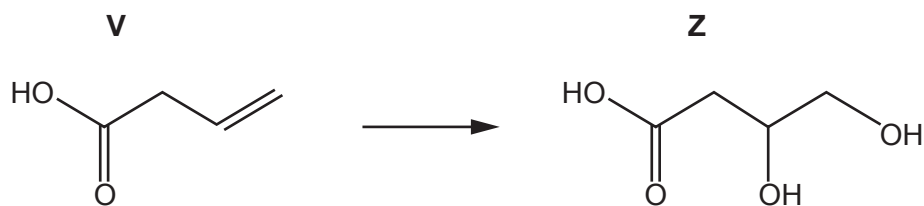
[3]

32. M/J 23/P21/Q4(b)(i)

Context

4 **V** is a colourless liquid.**Fig. 4.1**

Context: part (b) (introduction)

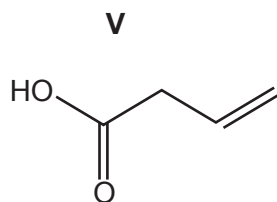
(b) **V** reacts to form **Z** in a single reaction, as shown in Fig. 4.2.**Fig. 4.2****(i)** Suggest the reagent and conditions needed to form **Z** from **V**.

[1]

33. M/J 23/P21/Q4(c)

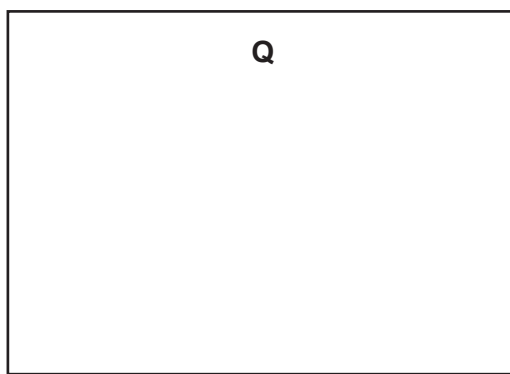
Context

4 **V** is a colourless liquid.

**Fig. 4.1**

- (c) **Q** contains the elements carbon, hydrogen and oxygen only. It is a saturated molecule with no branching in its carbon backbone.
Q contains only one functional group.
The relative molecular mass of **Q** is 88.
No effervescence is seen when Na_2CO_3 is added to **Q**.
Effervescence is seen when sodium is added to **Q**.
Q reacts with alkaline $\text{I}_2(\text{aq})$ to form a yellow precipitate.

Draw the structure of **Q** in the box.

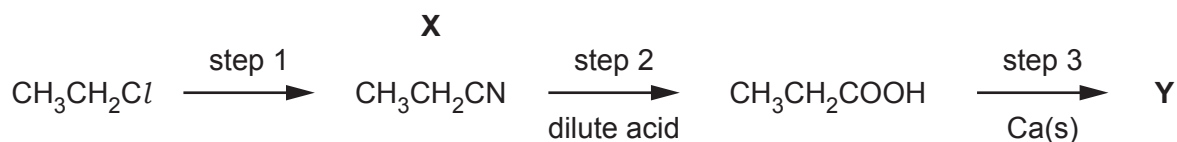


[2]

34. M/J 23/P22/Q3(b)

Context

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

- (b) $\text{CH}_3\text{CH}_2\text{COOH}$ can also be formed from propan-1-ol and potassium dichromate(VI).
State the conditions required.

[1]

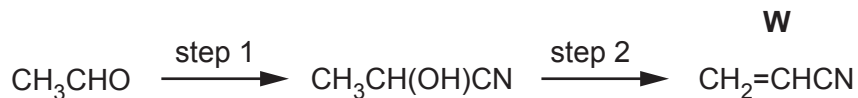
35. M/J 23/P23/Q6(b)(iii)

Context

- 6** Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

Context: part (b) (introduction)

- (b)** CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN .

**Fig. 6.1**

- (iii)** Suggest a suitable reagent and conditions for step 2 in Fig. 6.1.

[1]

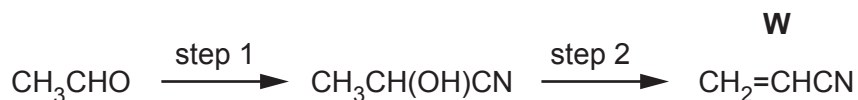
36. M/J 23/P23/Q6(f)(i)

Context

- 6** Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

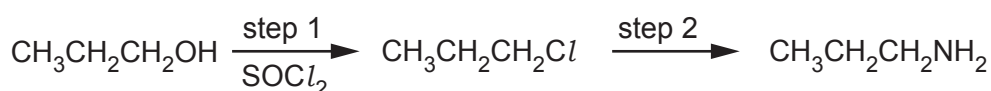
Context: part (b) (introduction)

- (b)** CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN .

**Fig. 6.1**

Context: part (f) (introduction)

- (f)** Propylamine can also be formed in a two-step synthesis from propan-1-ol, as shown in Fig. 6.4.

**Fig. 6.4**

- (i)** Name the type of reaction in step 1 in Fig. 6.4.

[1]

37. MAR 23/P22/Q4(a)(iii)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

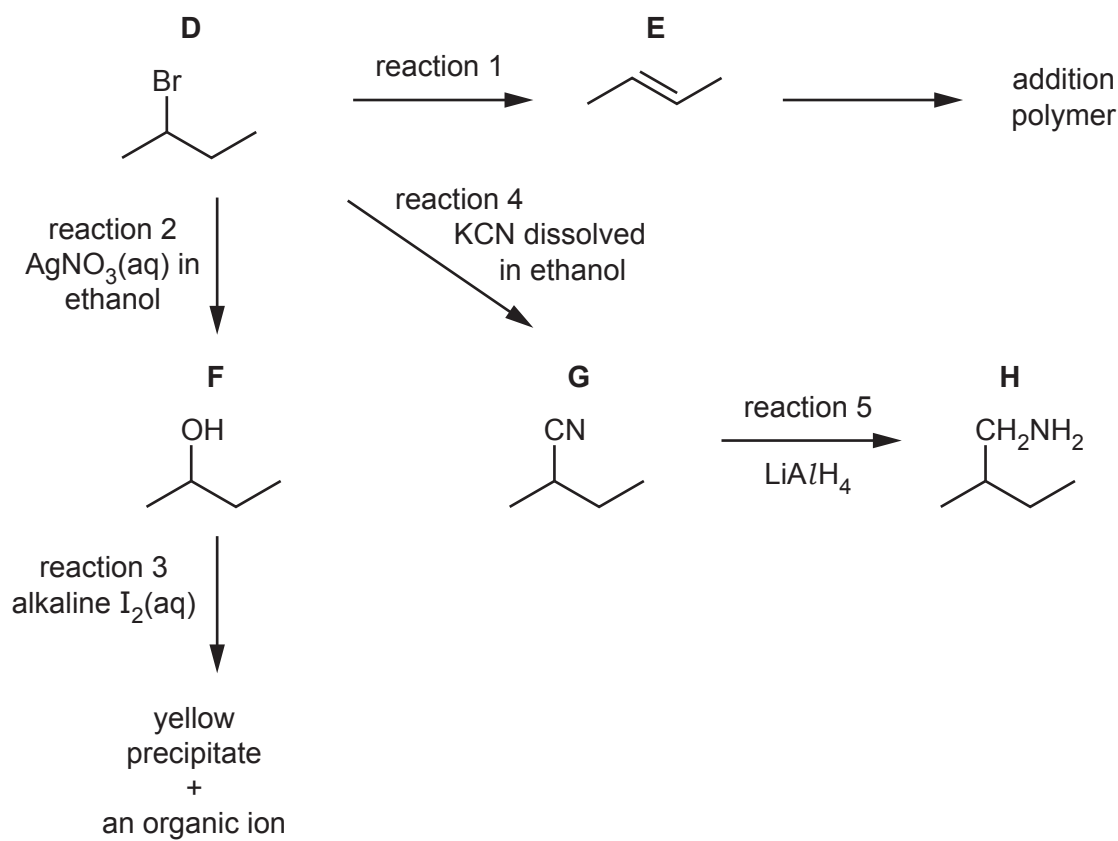


Fig. 4.1

(iii) **E** also forms when **F** is heated strongly in the presence of an Al_2O_3 catalyst.

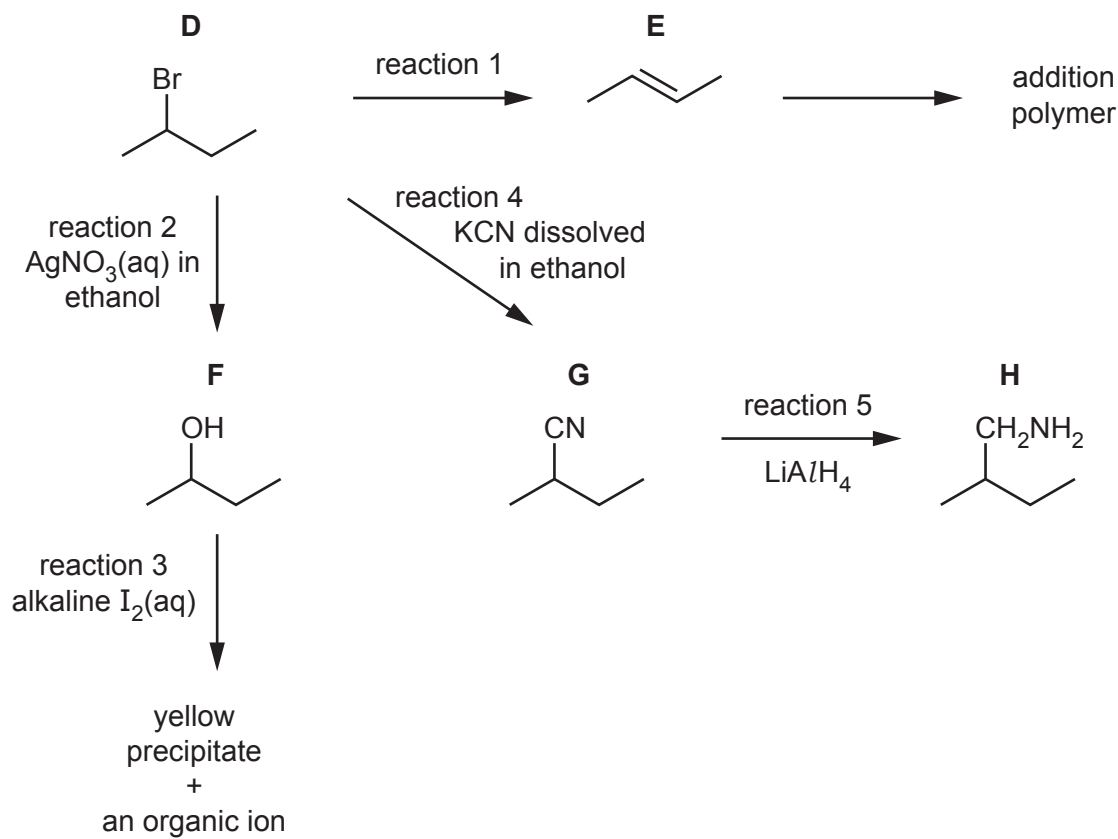
Write an equation for this reaction.

[1]

38. MAR 23/P22/Q4(b)(ii)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

**Fig. 4.1**

(ii) Identify the yellow precipitate and the organic ion formed in reaction 3.

yellow precipitate

organic ion

[2]

39. O/N 22/P21/Q3(c)

Context

3 Some of the common chlorides of Period 3 elements are shown in the list.



(c) PCl_5 reacts with alcohols to form chloroalkanes.

(i) Identify this type of reaction.

[1]

(ii) Draw the structure of the organic product formed in the reaction of an excess of PCl_5 with butane-1,3-diol.

[1]

40. O/N 22/P23/Q3(c)

Context

- 3** Some of the common chlorides of Period 3 elements are shown in the list.



- (c)** PCl_5 reacts with alcohols to form chloroalkanes.

- (i)** Identify this type of reaction.

[1]

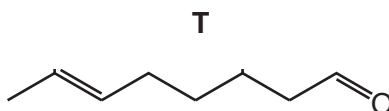
- (ii)** Draw the structure of the organic product formed in the reaction of an excess of PCl_5 with butane-1,3-diol.

[1]

41. M/J 22/P21/Q3(d)(i)-(ii)

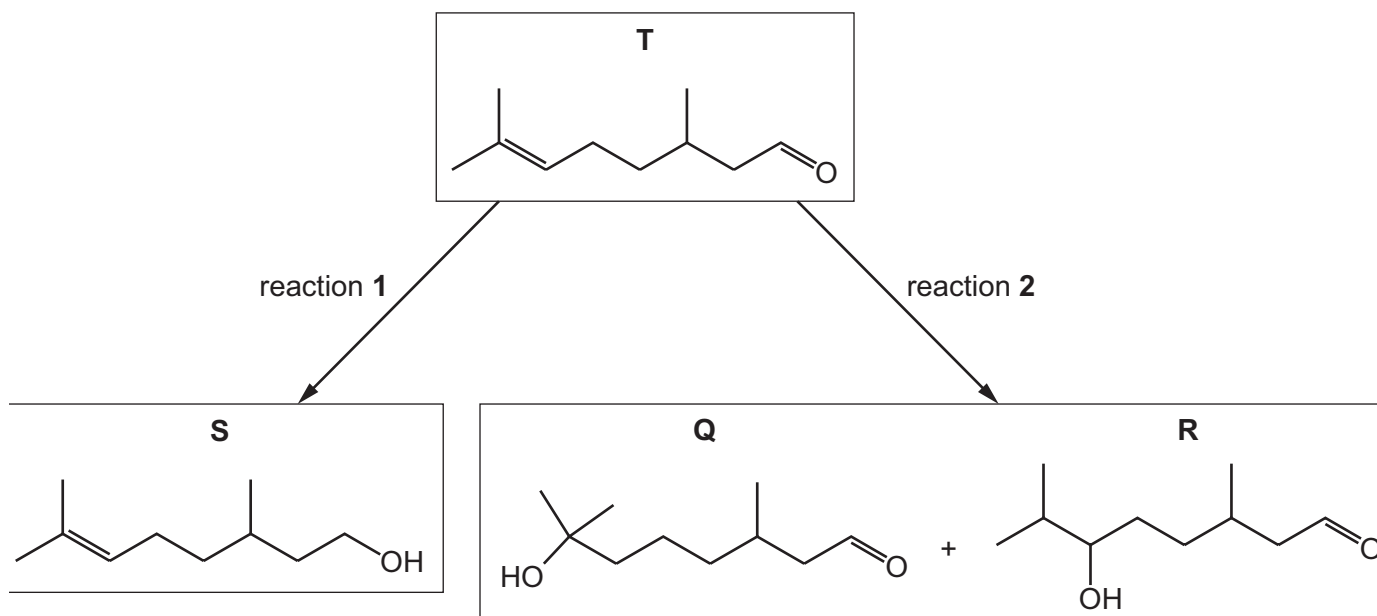
Context

- 3** Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

Context: part (c) (introduction)

- (c)** Fig. 3.2 shows two reactions of **T**.

**Fig. 3.2**

Context: part (d) (introduction)

- (d)** Separate samples of **Q** and **R** are added to separate test-tubes containing acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$ and heated.

Q**R**



Fig. 3.3

- (i) Predict the observations for each test-tube. Explain your answer in terms of the functional groups present in **Q** and **R**.

[3]

- (ii) When $\text{PCl}_5(\text{s})$ is added to separate samples of **Q** and **R** at room temperature, both react vigorously.

Complete the equation shown in Fig. 3.4 to describe the reaction that occurs when **R** reacts with $\text{PCl}_5(\text{s})$.

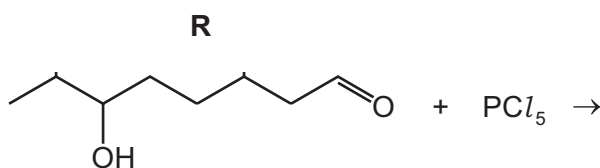


Fig. 3.4

[2]

42. M/J 22/P23/Q4(b)(iv)

Context: part (b) (introduction)

- (b) Two bottles labelled **Q** and **M** each contain a straight-chain halogenoalkane with molecular formula $\text{C}_4\text{H}_9\text{X}$, where **X** represents Cl, Br or I. A sample from each bottle is added to separate samples of equal amounts of aqueous silver nitrate in ethanol. In each reaction, the same organic product, **T**, and a precipitate are made, as shown in Fig. 4.2.



Fig. 4.2

Table 4.1 describes the colour of each of the precipitates made.

Table 4.1

halogenoalkane added to $\text{AgNO}_3(\text{aq})$ in ethanol	colour of precipitate
Q	white
M	yellow

- (iv) When pure **T** is added to alkaline $\text{I}_2(\text{aq})$, a yellow precipitate and an anion, **L**, are made.

Identify the anion **L**.

[1]

43. M/J 22/P23/Q5(a)(ii)

Context: part (a) (introduction)

5 (a) But-2-ene reacts with KMnO_4 to form organic product, **Y**.

Y does not react with Na_2CO_3 .

A gas is produced when an excess of Na is added to **Y**.

(ii) 24.0cm^3 of gas is produced when an excess of Na is added to 0.001mol of **Y**, when measured under room conditions.

Assume that 1mol of gas occupies 24.0dm^3 under room conditions.

Deduce a possible structure of **Y**. Explain your answer.

.....
[3]

Answers Section

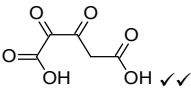
1. M/J 26/P21/Q4(a)(i)

4(a)(i)	(colour of mixture at start of reaction)orange AND (colour of mixture after the reaction)green	1	A any shade of orange A any shade of green I ppt(e) BOD blue / grey for green BOD brown for green (as dichromate is in excess)
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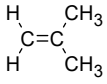
2. M/J 26/P21/Q4(b)(i)

4(b)(i)	reagent	observation with A	2	M1 I gas produced M1 I name of gas produced M1 I test for any gases M1 I solid disappears/ solid dissolves / colourless solution / flame M2 A (remains) orange solution / remains orange (solution) / no precipitate / no reaction / no observation M2 I 'turns orange'
	Na(s)	effervescence / fizzing / bubbling		
	2,4-dinitrophenylhydrazine (2,4-DNPH)	none		
	Fehling's reagent	none		

3. M/J 26/P21/Q4(b)(iii)

4(b)(iii)	$\text{HOOC}=\text{C}=\text{OCH}_2\text{COOH}$ ✓✓ OR  ✓✓ If 2 marks not awarded then max 1 for 2 of the following: • 5 carbon unbranched linear chain • 2 terminal carboxylic acid groups • 2 adjacent ketone groups only •✓	2	A any unambiguous structure I connectivity
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4. M/J 26/P22/Q4(a)

4(a)(i)	PCl_5 OR SOCl_2 OR HCl OR PCl_3 OR equivalent names	1	A thionyl chloride or sulfur oxide dichloride for SOCl_2 A phosphorus pentachloride or phosphorus trichloride BOD SOCl_2 (aq) A conc. HCl / concentrated hydrochloric acid A hydrogen chloride / HCl (g) I hydrochloric acid alone / HCl (aq) / dilute HCl I NaCl / KCl and conc. H_2SO_4 / conc. H_3PO_4 R PCl_5 (aq)
4(a)(ii)	catalyst OR dehydrating agent	1	A causes dehydration A speeds up / increases rate of reaction and remains unchanged owtte A lowers activation energy / E_A I heterogenous / homogenous I acid R redox
4(a)(iii)	$\text{H}_2\text{C}=\text{C}(\text{CH}_3)_2$ OR  OR 	1	A any unambiguous structure I C_4H_8

5. M/J 26/P22/Q4(b)(i)

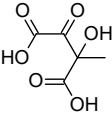
4(b)(i)	colour of mixture at start of reactionorange AND colour of mixture after the reactiongreen	1	A any shade of orange A any shade of green BOD blue / grey for green BOD brown for green (as dichromate is in excess)
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6. M/J 26/P23/Q4(a)(i)-(ii)

4(a)(i)		primary hydroxy groups	secondary hydroxy groups	tertiary hydroxy groups	2	•✓✓
	number of groups in one molecule of V	2	1	1		

4(a)(ii)			2	M1 I gas produced M1 I identity of gas M1 I test for gas M1 I solid disappears / dissolves OR colourless solution M1 I solid made M2 A (remains) orange (solution) / no precipitate / no reaction / no observation I turns orange
	reagent	observation with V		
	Na(s)	M1 effervescence / fizzing / bubbling		
	2,4-dinitrophenylhydrazine (2,4-DNPH)	M2 none		
	Fehling's reagent	none		

7. M/J 26/P23/Q4(b)

4(b)(i)	(colour of mixture at start of the reaction)orange (colour of mixture after the reaction)green	1	A any shade of orange at start A after reaction any shade of green BOD blue / grey for green BOD brown for green (dichromate is in excess)
4(b)(ii)	 OR $(\text{HOOC})\text{C}=\text{OC}(\text{COOH})(\text{OH})\text{CH}_3$	1	A any unambiguous structure I connectivity

8. M/J 26/P23/Q4(c)(iii)

4(c)(iii)	methanoic acid	1	A formic acid
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9. M/J 26/P23/Q4(d)(i)

4(d)(i)	(isomer) Y	1	A the structure of isomer Y
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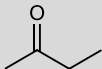
10. M/J 26/P24/Q4(a)

4(a)	tertiary / 3° (alcohol)	1	I names
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11. M/J 26/P24/Q4(b)

4(b)		3		
	reagent		observations	
	Na		M1 Bubbles / fizzing / effervescence	M1 I Na reacts / gets smaller
	acidified K ₂ Cr ₂ O ₇		M2 (Orange to) green (solution)	M2 I antiseptic smell
	alkaline I ₂ (aq)	M3 Yellow precipitate	M3 precipitate = ppt = solid = powder = crystals	

12. MAR 26/P22/Q3(e)

3(e)	reagent and conditions	observation with butan-2-ol	organic product(s)	3
	acidified $\text{K}_2\text{Cr}_2\text{O}_7$	M1 (orange solution) turns green		
	Na(s)	effervescence	M2 	
	M3 alkaline $\text{I}_2(\text{aq})$	yellow precipitate	 and 	

Answer for context part (d)

3(d)	M1 fastest D > C > B slowest M2 carbon—halogen gets weaker down the group (so breaks faster)	2
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13. O/N 25/P21/Q4(a)

4(a)(i)	reduction	1
4(a)(ii)	$\text{PCl}_5 / \text{PCl}_3 / \text{SOCl}_2$ OR KCl / NaCl AND conc. $\text{H}_2\text{SO}_4 / \text{conc. H}_3\text{PO}_4$	1
4(a)(iii)	M1 $\text{C}_2\text{H}_5\text{CH}_2\text{Cl} + \text{NaOH} \rightarrow \text{C}_3\text{H}_6 + \text{H}_2\text{O} + \text{NaCl}$ M2 ethanol (solvent) AND heat / reflux	2
4(a)(iv)	conc. H_2SO_4 OR conc. H_3PO_4 OR Al_2O_3 AND heat	1

14. O/N 25/P22/Q4(c)

4(c)(i)	orange to green	1
4(c)(ii)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + 2[\text{O}] \rightarrow \text{CH}_3\text{CH}_2\text{COOH} + \text{H}_2\text{O}$	1

15. O/N 25/P23/Q4(a)

4(a)(i)	reduction	1
4(a)(ii)	$\text{PCl}_5 / \text{PCl}_3 / \text{SOCl}_2$ OR KCl / NaCl AND conc. $\text{H}_2\text{SO}_4 / \text{conc. H}_3\text{PO}_4$	1
4(a)(iii)	M1 $\text{C}_2\text{H}_5\text{CH}_2\text{Cl} + \text{NaOH} \rightarrow \text{C}_3\text{H}_6 + \text{H}_2\text{O} + \text{NaCl}$ M2 ethanol (solvent) AND heat / reflux	2
4(a)(iv)	conc. H_2SO_4 OR conc. H_3PO_4 OR Al_2O_3 AND heat	1

16. O/N 25/P24/Q2(c)(ii)

2(c)(ii)	$2\text{CH}_3\text{OH} \rightarrow \text{CH}_3\text{OCH}_3 + \text{H}_2\text{O}$	1
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17. O/N 25/P24/Q3(a)(i)-(iii)

3(a)(i)		1
3(a)(ii)	yellow precipitate	1

3(a)(iii)	elimination	1
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18. O/N 25/P24/Q3(b)(i)

3(b)(i)	$(\text{CH}_3)_2\text{CHOH} + \text{Na} \rightarrow (\text{CH}_3)_2\text{CHO}^-(\text{Na}^+) + \frac{1}{2} \text{H}_2$	1
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19. M/J 25/P21/Q5(c)(ii)

5(c)(ii)	C is not as strong an acid as B	1
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20. M/J 25/P21/Q5(d)(ii)

5(d)(ii)	oxidation	1
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21. M/J 25/P21/Q5(e)(iii)

5(e)(iii)	1,3-dichloropropane	1
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22. M/J 25/P21/Q6(c)(iii)

6(c)(iii)	$\text{CH}_3\text{CH}(\text{OH})\text{-R}$ where R is represented as H or any alkyl group	1
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23. M/J 25/P22/Q4(b)(i)

4(b)(i)	substitution	1
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24. M/J 25/P22/Q4(c)(ii)-(iii)

4(c)(ii)	hydrogen / H_2	1
4(c)(iii)	(It / Na) reduces A / B / C / the alcohol (group) OR (it / Na behaves as) reducing agent OR (it / Na) donates electrons to A / B / C / the alcohol (group)	1

25. M/J 25/P22/Q4(d)

4(d)(i)	$(\text{CH}_3)_3\text{COH}$	1
4(d)(ii)	butan-1-ol OR (2) methylpropan-1-ol	1
4(d)(iii)	butan-2-ol OR $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$	1

26. O/N 24/P22/Q2(b)(iv)

2(b)(iv)	alkyl groups are electron donating / have positive inductive effect.	1
	strengthens the O—H bond / makes H^+ less likely to be donated	1

27. M/J 24/P21/Q6(c)(iii)

6(c)(iii)	$\text{ROH} + \text{Na} \rightarrow \text{RONa} + \frac{1}{2} \text{H}_2$ (0.002 mol Q produced 0.002 mol H_2 gas so) 2 OH groups M1 answer indicates that OH group(s) in Q react with Na to produce the H_2 in the ratio 1 mol OH : $\frac{1}{2}$ mol H_2 M2 uses data to show 2OH groups	2
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28. O/N 23/P21/Q3(e)(i)-(ii)

3(e)(i)	1 4 0	1
---------	-------	---

3(e)(ii)	Solution turns from purple to colourless	1
	Orange / red precipitate forms	1
	no reaction	1

29. O/N 23/P22/Q3(c)(ii)

3(c)(ii)		1
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30. O/N 23/P22/Q4(b)(ii)

4(b)(ii)	$\text{CH}_3\text{CH}(\text{OH})\text{COOH} + [\text{O}] \rightarrow \text{CH}_3\text{COCOOH} + \text{H}_2\text{O}$	1
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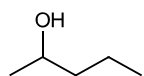
31. O/N 23/P23/Q3(e)(i)-(ii)

3(e)(i)	1 4 0	1
3(e)(ii)	Solution turns from purple to colourless	1
	Orange / red precipitate forms	1
	no reaction	1

32. M/J 23/P21/Q4(b)(i)

4(b)(i)	cold dilute acidified KMnO_4	1
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33. M/J 23/P21/Q4(c)

4(c)	 M1 molecule with 5C atoms and 1 alcohol group M2 molecule with $\text{—CH}(\text{CH}_3)\text{OH}$	2
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34. M/J 23/P22/Q3(b)

3(b)	acid (conditions) / $\text{H}^+(\text{aq})$ AND heat under reflux	1
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35. M/J 23/P23/Q6(b)(iii)

6(b)(iii)	heat + conc. sulfuric acid / conc. phosphoric acid OR heat + Al_2O_3	1
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36. M/J 23/P23/Q6(f)(i)

6(f)(i)	substitution	1
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37. MAR 23/P22/Q4(a)(iii)

4(a)(iii)	$\text{C}_4\text{H}_{10}\text{O} / \text{CH}_3\text{CH}(\text{OH})\text{C}_2\text{H}_5 \rightarrow \text{C}_4\text{H}_8 / \text{CH}_3\text{CHCHCH}_3 + \text{H}_2\text{O}$	1
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38. MAR 23/P22/Q4(b)(ii)

4(b)(ii)	CHI_3 / iodoform / tr(i)iodomethane	1
	$\text{CH}_3\text{CH}_2\text{CO}_2^-$ / propanoate	1

39. O/N 22/P21/Q3(c)

3(c)(i)	substitution	1
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3(c)(ii)		1
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40. O/N 22/P23/Q3(c)

3(c)(i)	substitution	1
3(c)(ii)		1

41. M/J 22/P21/Q3(d)(i)-(ii)

3(d)(i)	M1 Q orange → green M2 R orange → green M3 –CHO / aldehyde (in both) (and 2° / secondary alcohol in R) reacts / oxidised	3
3(d)(ii)	 M1 correct formula of organic product M2 correct inorganic products	2

42. M/J 22/P23/Q4(b)(iv)

4(b)(iv)	$\text{C}_2\text{H}_5\text{COO}^-$ OR $\text{CH}_3\text{CH}_2\text{CO}_2^-$ or propanoate (an)ion	1
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43. M/J 22/P23/Q5(a)(ii)

5(a)(ii)	M1 unbranched 4C structure AND any number of –OH in any position M2 0.001 mol H_2 made from 0.001 mol Y AND $\text{R-OH} + \text{Na} \rightarrow \text{RONa} + \frac{1}{2}\text{H}_2$ OR use of 1 OH (group) $\rightarrow \frac{1}{2}\text{H}_2$ M3 $\text{CH}_3\text{CH}(\text{OH})\text{CH}(\text{OH})\text{CH}_3$	3
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Unit 17

CARBONYL COMPOUNDS

30 items · 72 marks

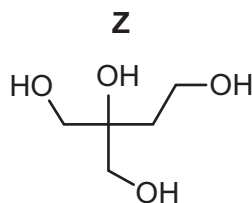
Syllabus sections in this unit

17.1 Aldehydes and ketones

Unit 17: Carbonyl compounds**1. M/J 26/P21/Q4(b)(ii)**

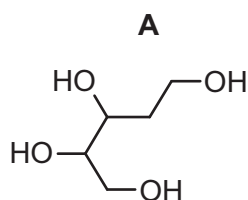
Context: part (a) (introduction)

- 4 (a)** Compound **Z** is oxidised to produce organic compound **W**, $\text{C}_5\text{H}_6\text{O}_7$, by heating under reflux with an excess of acidified $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$.

**Figure 4.1**

Context: part (b) (introduction)

- (b)** Compound **A** is a structural isomer of **Z**.

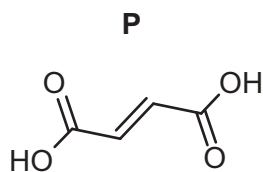
**Figure 4.2**

- (ii)** Suggest why Fehling's reagent does **not** react with **A** but does react with aldehyde functional groups.

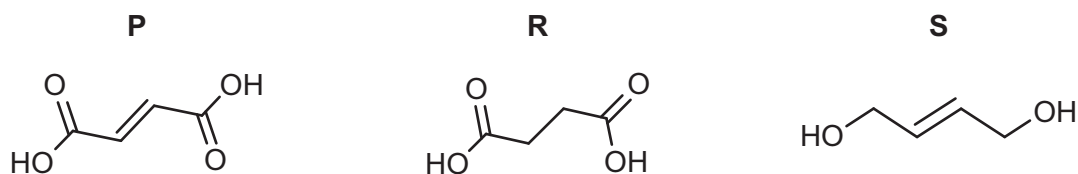
[1]

2. M/J 26/P22/Q6(c)(iii)

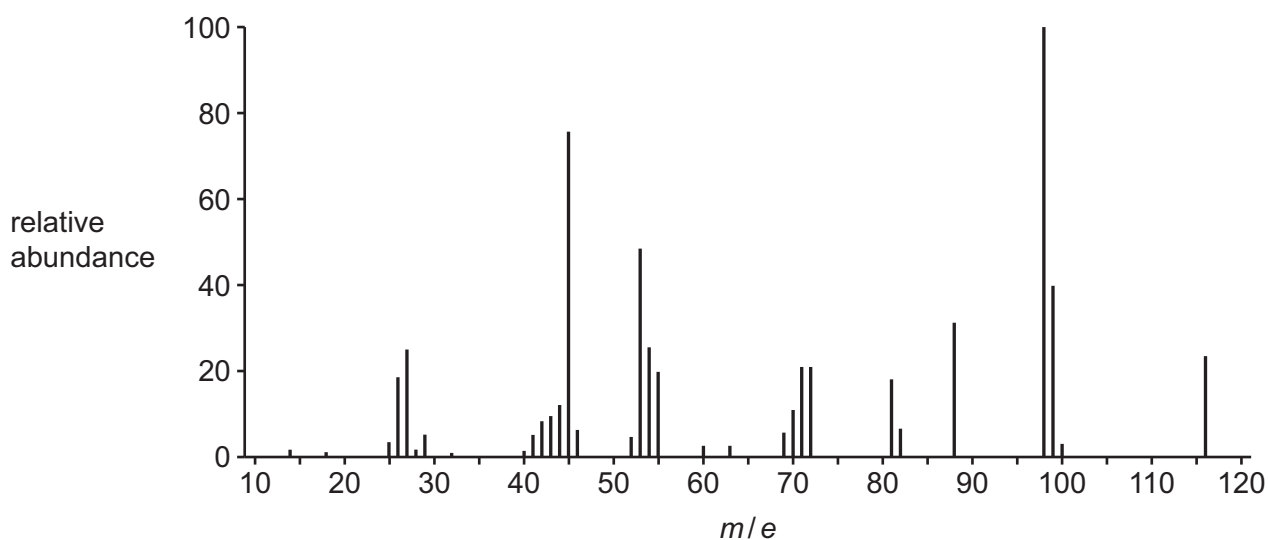
Context: part (a) (introduction)

6 (a) Figure 6.1 shows the structure of compound **P**.**Figure 6.1**

Context: part (c) (introduction)

(c) An unlabelled sample contains one of compounds **P**, **R** or **S**.**Figure 6.4**

The mass spectrum of the sample is shown in Figure 6.5.

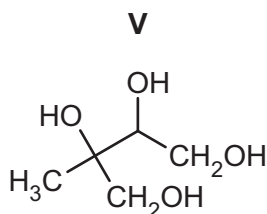
**Figure 6.5****(iii)** Suggest **one** reagent that is used to distinguish between samples of **R** and **S**. Include appropriate observations.

[2]

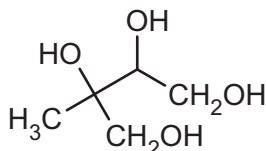
3. M/J 26/P23/Q4(a)(iii)

Context

- 4** The structure of compound **V** is shown in Figure 4.1.

**Figure 4.1**

Context: Fig. 4.1

**Figure 4.1**

- (iii) Suggest why Fehling's reagent does **not** react with **V** but does react with aldehyde functional groups.

[1]

4. M/J 25/P21/Q6(a)

Context

- 6** Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.
- (a) State the functional group in ethanal, propanal and propanone.

[1]

5. M/J 25/P21/Q6(b)

Context

- 6** Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.
- (b) State a reagent and the relevant observation that confirm that **F**, **G** and **H** have the same functional group.
- reagent
- observation

[2]

6. M/J 25/P21/Q6(c)(i)-(ii)

Context

- 6** Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.

Context: part (c) (introduction)

- (c)** Separate samples of **F**, **G** and **H** are each tested with Tollens' reagent and with alkaline $I_2(aq)$. The observations are shown in Table 6.1.

Table 6.1

	Tollens' reagent	alkaline $I_2(aq)$
F	no observable change	pale yellow precipitate
G	silver mirror	pale yellow precipitate
H	silver mirror	no precipitate

- (i)** Use Table 6.1 to name the organic compounds in bottles **F**, **G** and **H**.

F =**G** =**H** =

[2]

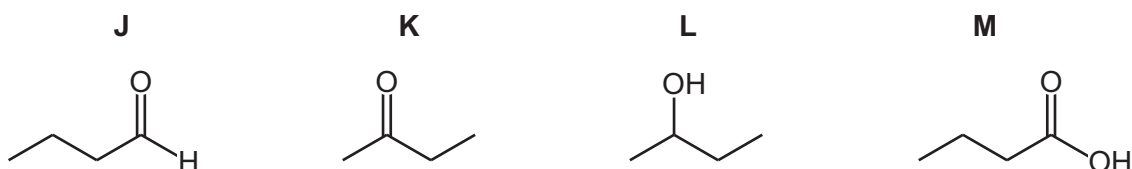
- (ii)** Identify the yellow precipitate produced when alkaline $I_2(aq)$ is added to separate samples of **F** and **G**.

[1]

7. MAR 25/P22/Q4(b)(i)

Context

- 4** Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

**Fig. 4.1**

Context: part (b) (introduction)

- (b)** **K** reacts with HCN in the presence of a KCN catalyst, forming **N**.

- (i)** Complete Fig. 4.2 to show the mechanism for this reaction.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

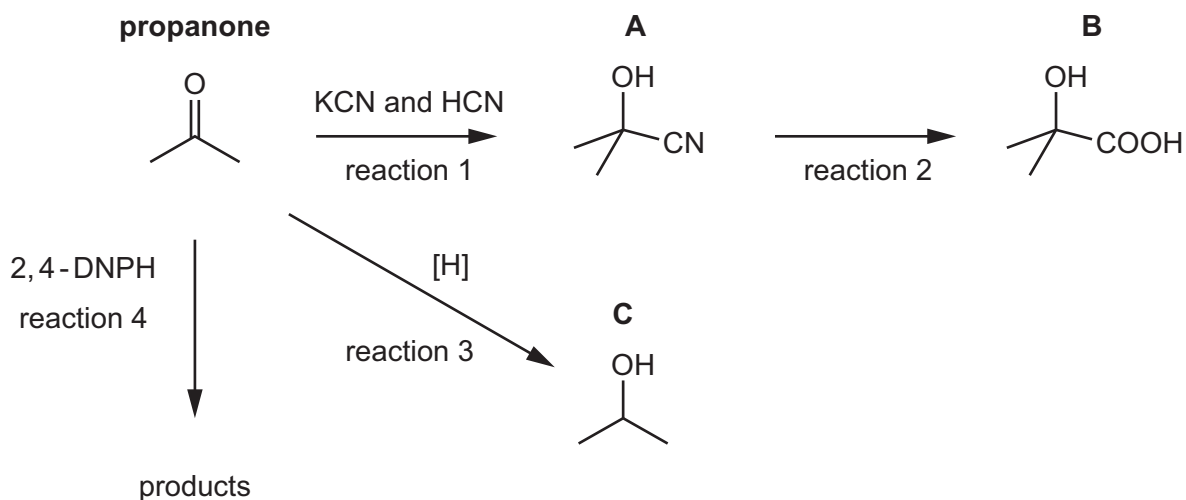
**Fig. 4.2**

[4]

8. O/N 24/P21/Q4(a)(i)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 is a nucleophilic addition reaction.

- (i)** Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

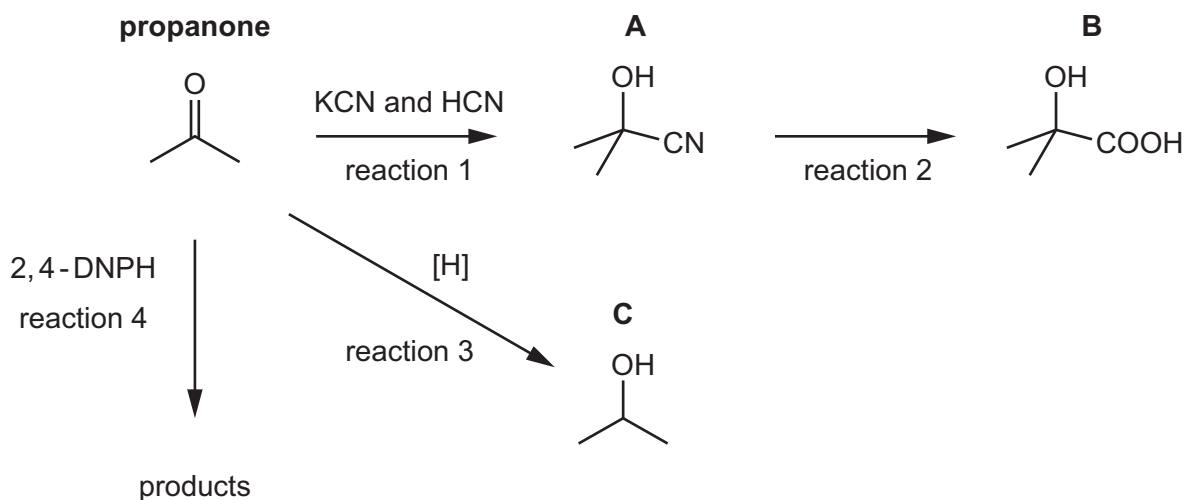
Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.

**Fig. 4.2**

9. O/N 24/P21/Q4(c)

Context

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

- (c) Reaction 3 is a reduction reaction.

- (i) Construct an equation to represent reaction 3.

Use [H] to represent one atom of hydrogen from the reducing agent.

[1]

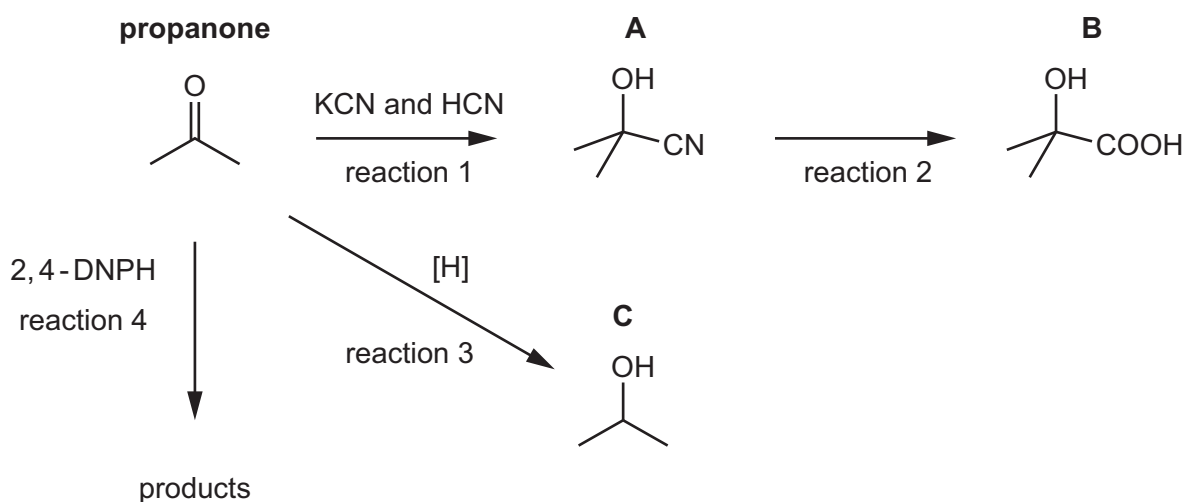
- (ii) Name **C**.

[1]

10. O/N 24/P21/Q4(d)

Context

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

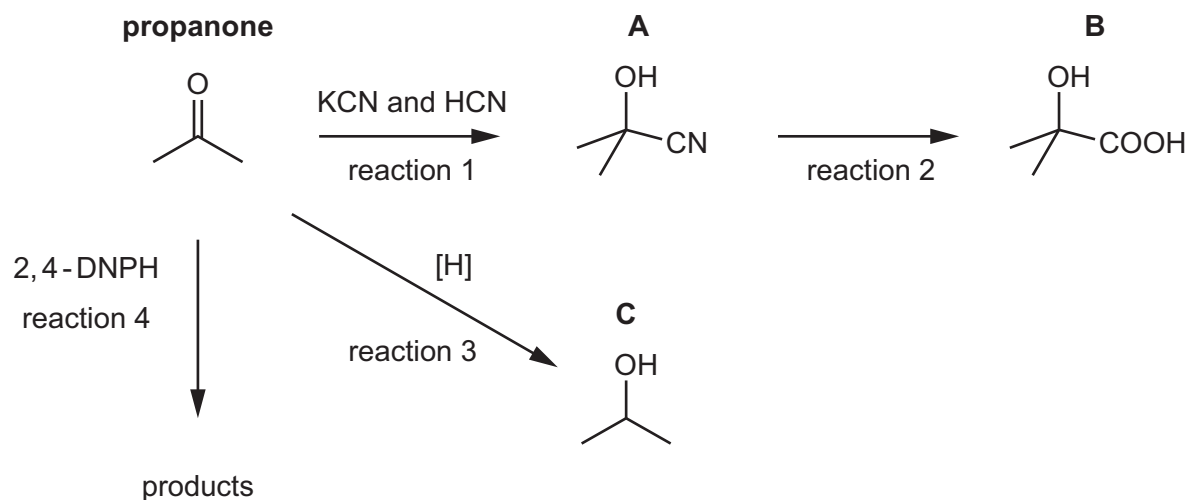
- (d) State what is observed in reaction 4.

[1]

11. O/N 24/P21/Q4(e)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

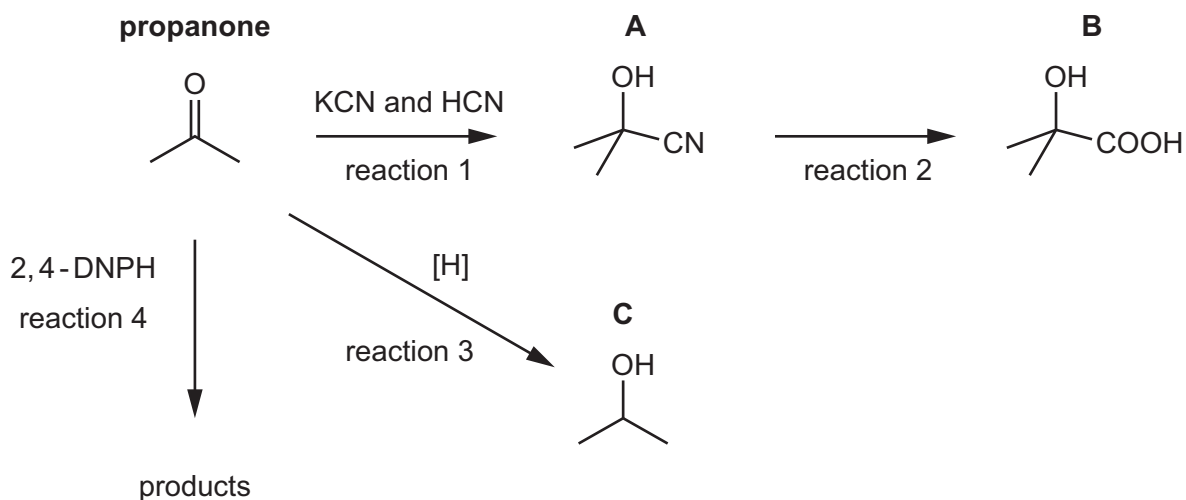
- (e)** Explain why Fehling's reagent does **not** react with propanone.

[1]

12. O/N 24/P23/Q4(a)(i)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

Context: part (a) (introduction)

- (a)** Reaction 1 is a nucleophilic addition reaction.

- (i)** Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

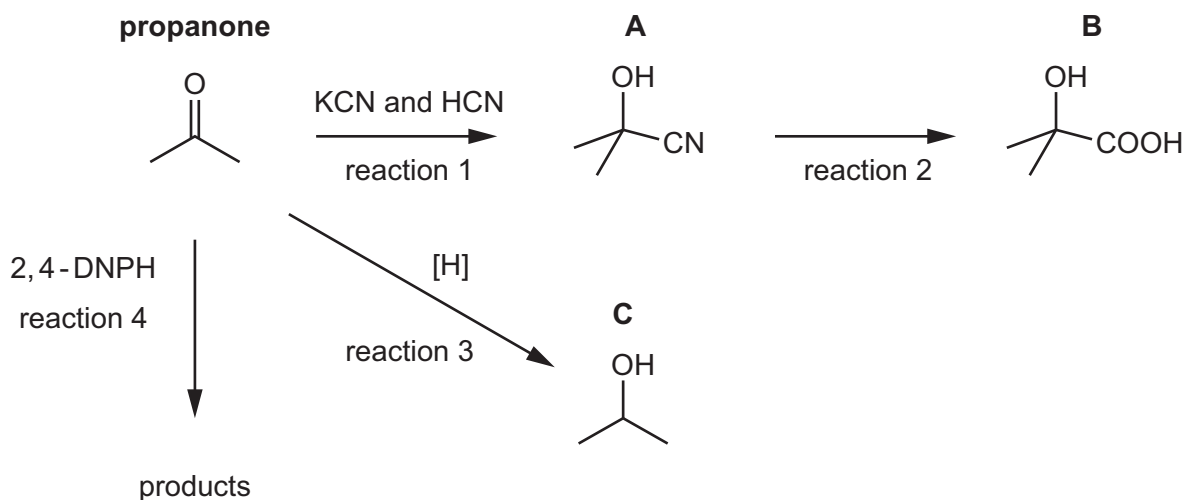
Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.

**Fig. 4.2**

13. O/N 24/P23/Q4(c)

Context

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

- (c) Reaction 3 is a reduction reaction.

- (i) Construct an equation to represent reaction 3.

Use $[\text{H}]$ to represent one atom of hydrogen from the reducing agent.

[1]

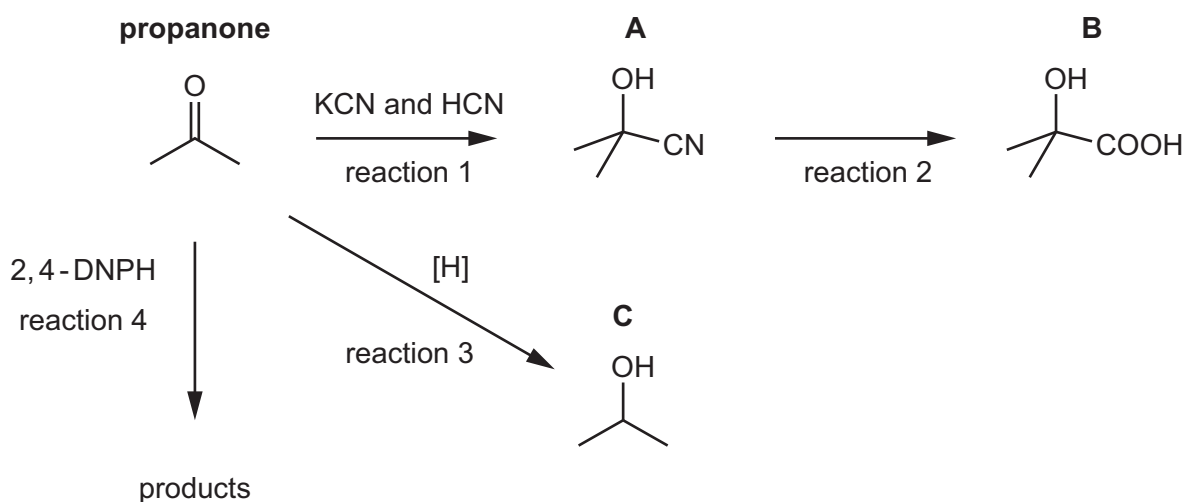
- (ii) Name **C**.

[1]

14. O/N 24/P23/Q4(d)

Context

- 4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

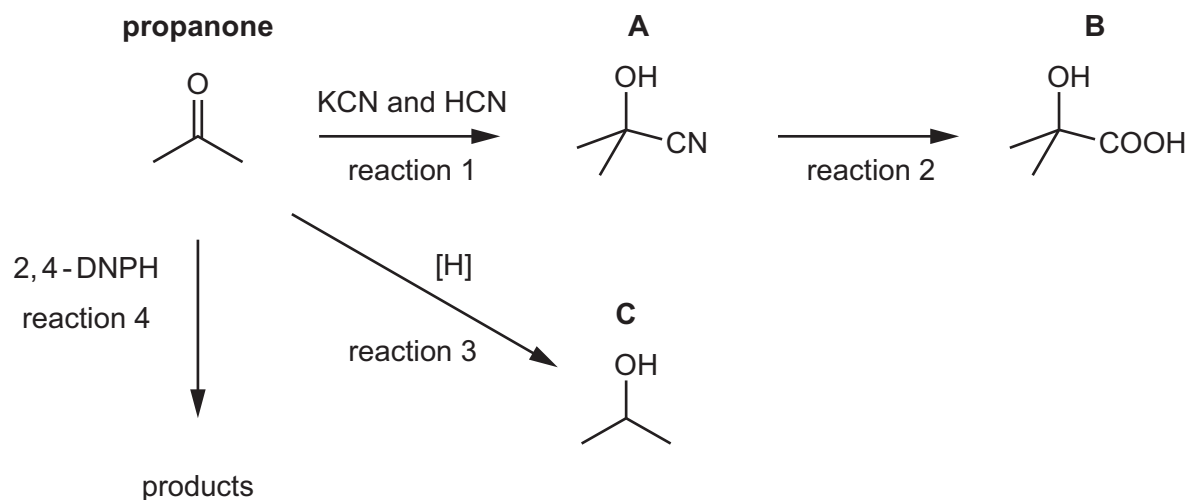
- (d) State what is observed in reaction 4.

[1]

15. O/N 24/P23/Q4(e)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

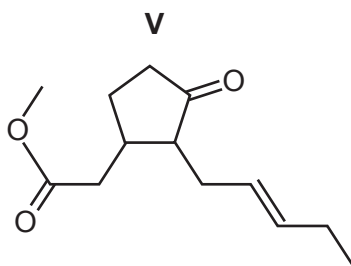
**Fig. 4.1**

- (e)** Explain why Fehling's reagent does **not** react with propanone.

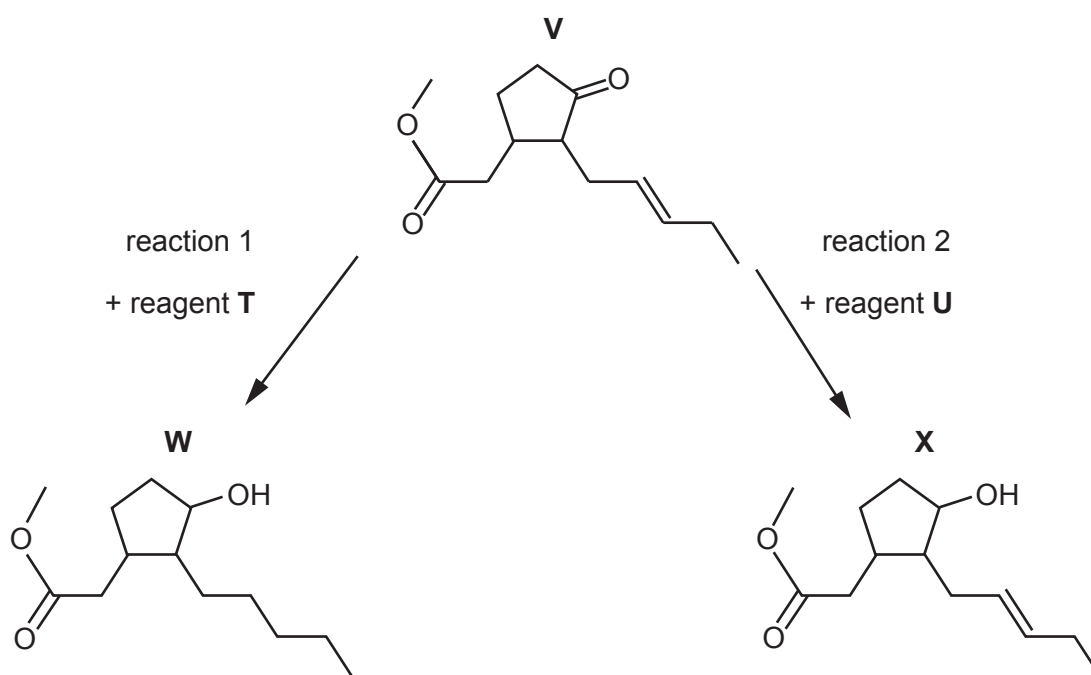
[1]

16. M/J 24/P21/Q6(b)(i)

Context: part (a) (introduction)

6 (a) V shows stereoisomerism.**Fig. 6.1**

Context: part (b) (introduction)

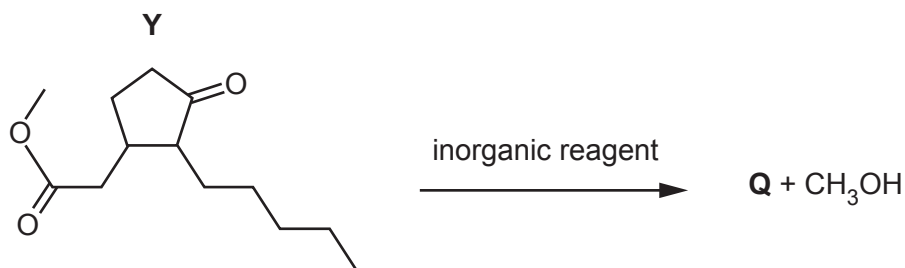
(b) Fig. 6.2 shows two reactions involving **V**.**Fig. 6.2****(i)** Identify the role of reagent **T** for each functional group that reacts in reaction 1.

[1]

17. M/J 24/P21/Q6(c)(ii)

Context: part (c) (introduction)

- (c) Both functional groups in one molecule of **Y** react with an inorganic reagent to form one molecule of **Q** and one molecule of methanol, CH_3OH , as shown in Fig. 6.3.

**Fig. 6.3**

- (ii) **Q** contains **only** hydroxyl functional groups.

Complete Table 6.1 to show the observations that occur when 2,4-dinitrophenylhydrazine (2,4-DNPH reagent) is added to separate samples of **Y** and **Q**.

Table 6.1

	observation on addition of 2,4-DNPH reagent
Y	
Q	

[1]

18. MAR 24/P22/Q1(d)(iii)-(iv)

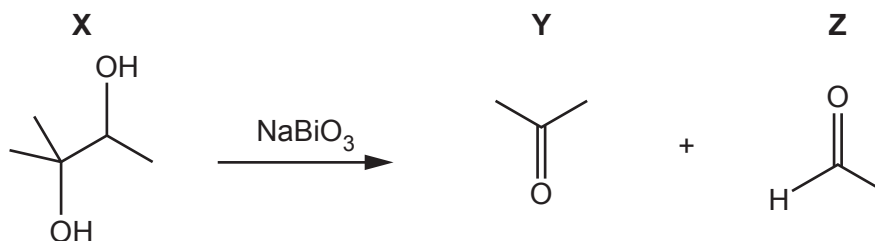
Context

- 1** Bismuth is an element in Group 15 of the Periodic Table.

Context: part (d) (introduction)

- (d)** NaBiO_3 is an oxidising agent with similar properties to KMnO_4 .

Fig. 1.1 shows an example of the use of NaBiO_3 as an oxidising agent.

**Fig. 1.1**

- (iii)** Complete Table 1.1 to show what is observed when compounds **Y** and **Z** react separately with the named reagents.

Table 1.1

reagent	observation with Y	observation with Z
$\text{Na}_2\text{CO}_3(\text{aq})$	no reaction	
alkaline $\text{I}_2(\text{aq})$		
2,4-dinitrophenylhydrazine (2,4-DNPH)		
Tollens' reagent		

[4]

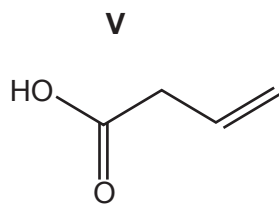
- (iv)** Construct an equation for the reaction of **Z** with NaBH_4 .

Use $[\text{H}]$ to represent an atom of hydrogen from the reducing agent.

[1]

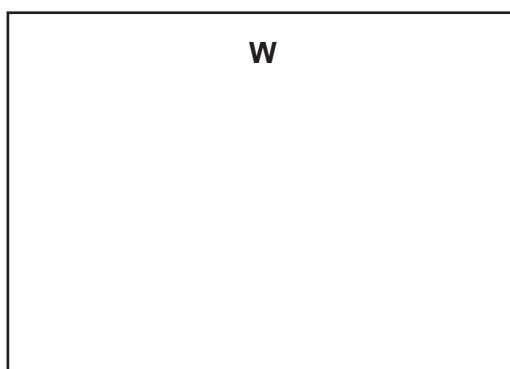
19. M/J 23/P21/Q4(a)

Context

4 **V** is a colourless liquid.**Fig. 4.1**

(a) **V** reacts with an excess of LiAlH_4 to form **W**.

(i) Draw the structure of **W** in the box.



[1]

(ii) Identify the role of LiAlH_4 in the reaction with **V**.

[1]

20. M/J 23/P22/Q5(b)

(b) **A** and **B** are structural isomers with molecular formula $C_5H_{10}O$.

They are both straight-chained molecules with only one functional group.

Table 5.1 describes observations when separate samples of **A** and **B** are added to different reagents.

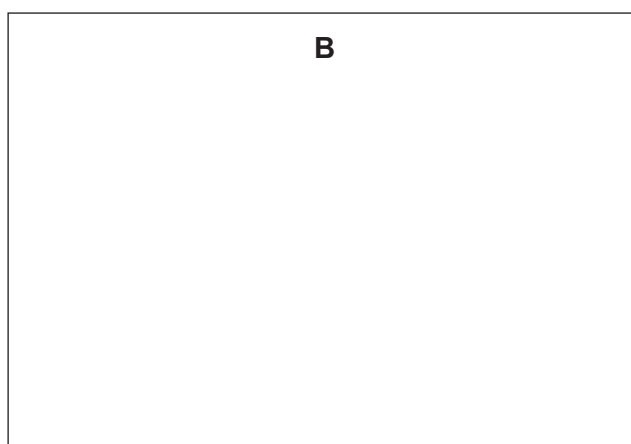
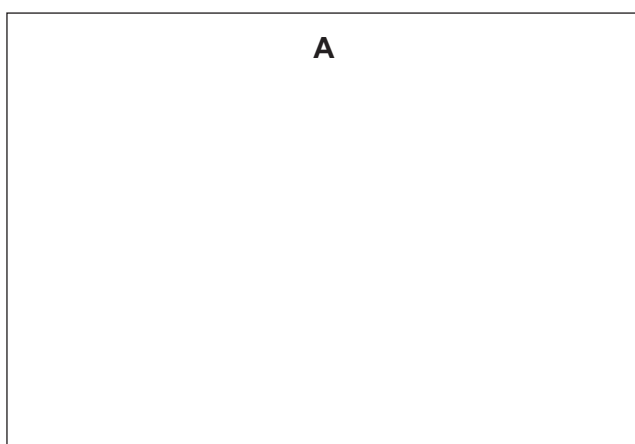
Table 5.1

reagent	A	B
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	orange precipitate appears	orange precipitate appears
Tollens' reagent	silver mirror appears	no reaction
alkaline $I_2(aq)$	no reaction	no reaction

(i) Name the functional group present in both **A** and **B**.

[1]

(ii) Draw the structures of **A** and **B** in the boxes.

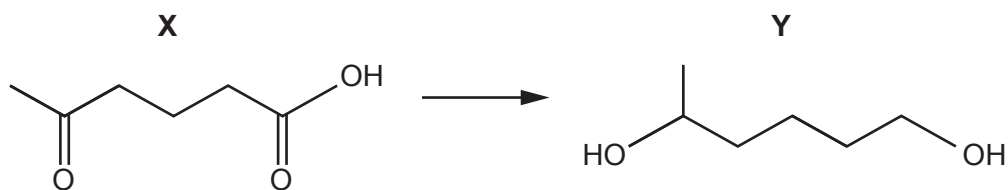


[2]

21. M/J 23/P23/Q5(b)

Context

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.

**Fig. 5.1**

(b) The formation of **Y** from **X** requires the addition of a suitable reducing agent.

- (i)** Construct an equation using molecular formulae and [H] for the reaction in Fig. 5.1. Use [H] to represent one atom of hydrogen from the reducing agent.

[1]

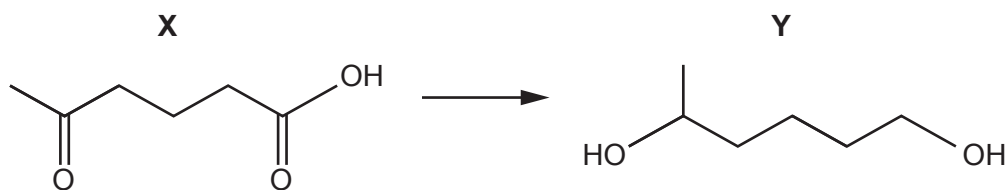
- (ii)** Identify a suitable non-gaseous reducing agent for the formation of **Y** from **X**.

[1]

22. M/J 23/P23/Q5(d)

Context

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.

**Fig. 5.1**

(d) Complete Table 5.2 with the expected observations that occur when the reagents shown are added to separate solutions of **X** and **Y**. Do **not** refer to temperature changes in your answer.

Table 5.2

reagent	observation on addition to X	observation on addition to Y
aqueous sodium carbonate		
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)		
alkaline aqueous iodine		

[3]

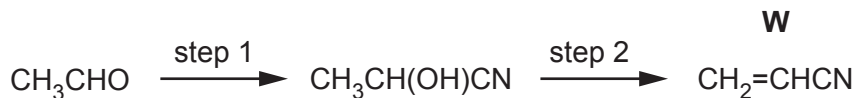
23. M/J 23/P23/Q6(b)(i)-(ii)

Context

- 6** Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

Context: part (b) (introduction)

- (b)** CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN .

**Fig. 6.1**

- (i)** Name the mechanism for the reaction in step 1 in Fig. 6.1.

[1]

- (ii)** Complete Fig. 6.2 to show the mechanism for the reaction in step 1.
Include all products, charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

**Fig. 6.2**

[3]

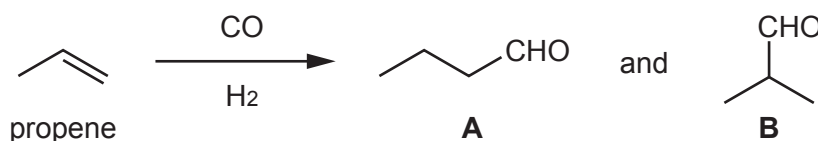
24. MAR 23/P22/Q3(b)

also Unit 13 (1 mark); this unit 8 marks

Context

- 3** Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2 .

**Fig. 3.1**

- (b)** **A** and **B** show reactions typical of aliphatic aldehydes.

- (i)** **A** undergoes a nucleophilic addition reaction with a mixture of HCN and KCN , forming compound **C**.

Complete the diagram to show the mechanism for this reaction.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

Draw the structure of the organic intermediate.



[4]

(ii) Table 3.1 shows information about three experiments involving **B**.

Complete Table 3.1.

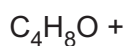
Table 3.1

experiment	reagents	observation with B
1		solution turns from orange to green
2		a silver mirror forms on the sides of the reaction vessel
3	$\text{Br}_2(\text{aq})$	

[3]

(iii) **B**, $\text{C}_4\text{H}_8\text{O}$, is oxidised by acidified potassium manganate(VII).

Complete the equation for this reaction. Use [O] to represent one atom of oxygen from the oxidising agent.



[1]

(iv) **C** is a chiral molecule.

Circle any chiral centres in the structure of **C** shown in Fig. 3.2.

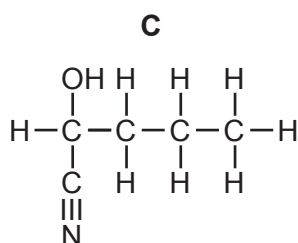


Fig. 3.2

[1]

25. O/N 22/P21/Q5(a)(ii)-(iii)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

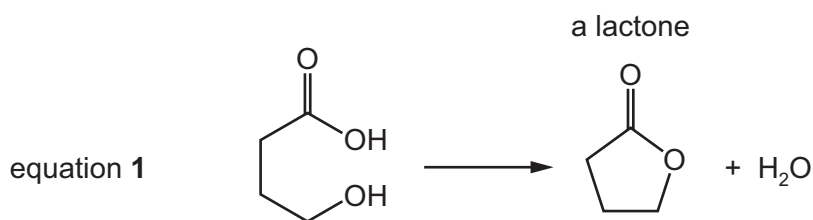


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

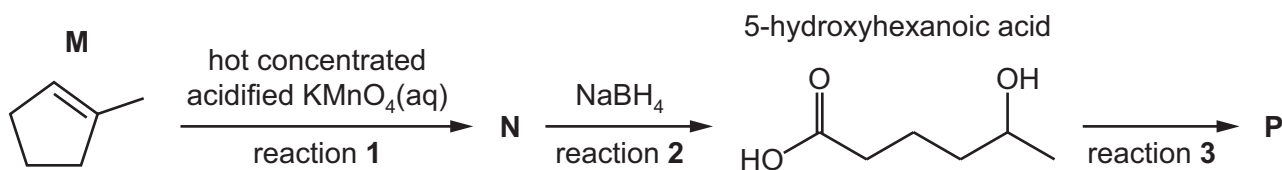


Fig. 5.1

- (ii) **N** is reduced by NaBH₄ to form 5-hydroxyhexanoic acid in reaction 2.

Construct an equation for reaction 2 using molecular formulae.

In the equation, use [H] to represent one atom of hydrogen from the reducing agent.

[1]

- (iii) Reaction 2 is a nucleophilic addition.

Suggest why reaction 2 creates a mixture of two organic compounds.

[2]

26. O/N 22/P23/Q5(a)(ii)-(iii)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

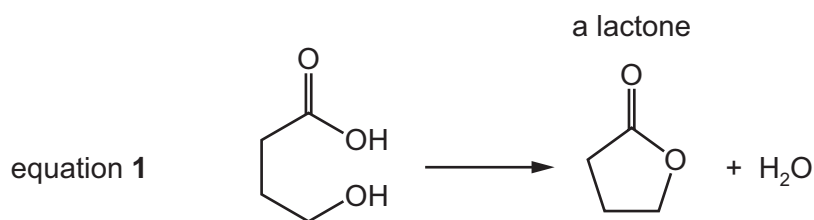
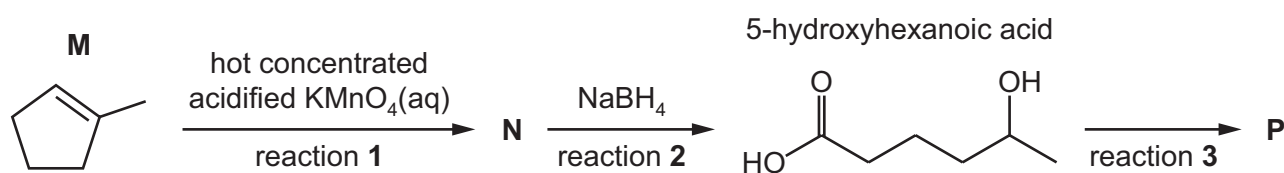


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

**Fig. 5.1**

- (ii)** **N** is reduced by NaBH₄ to form 5-hydroxyhexanoic acid in reaction 2.

Construct an equation for reaction 2 using molecular formulae.

In the equation, use [H] to represent one atom of hydrogen from the reducing agent.

[1]

- (iii)** Reaction 2 is a nucleophilic addition.

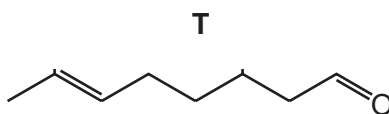
Suggest why reaction 2 creates a mixture of two organic compounds.

[2]

27. M/J 22/P21/Q3(c)(i)

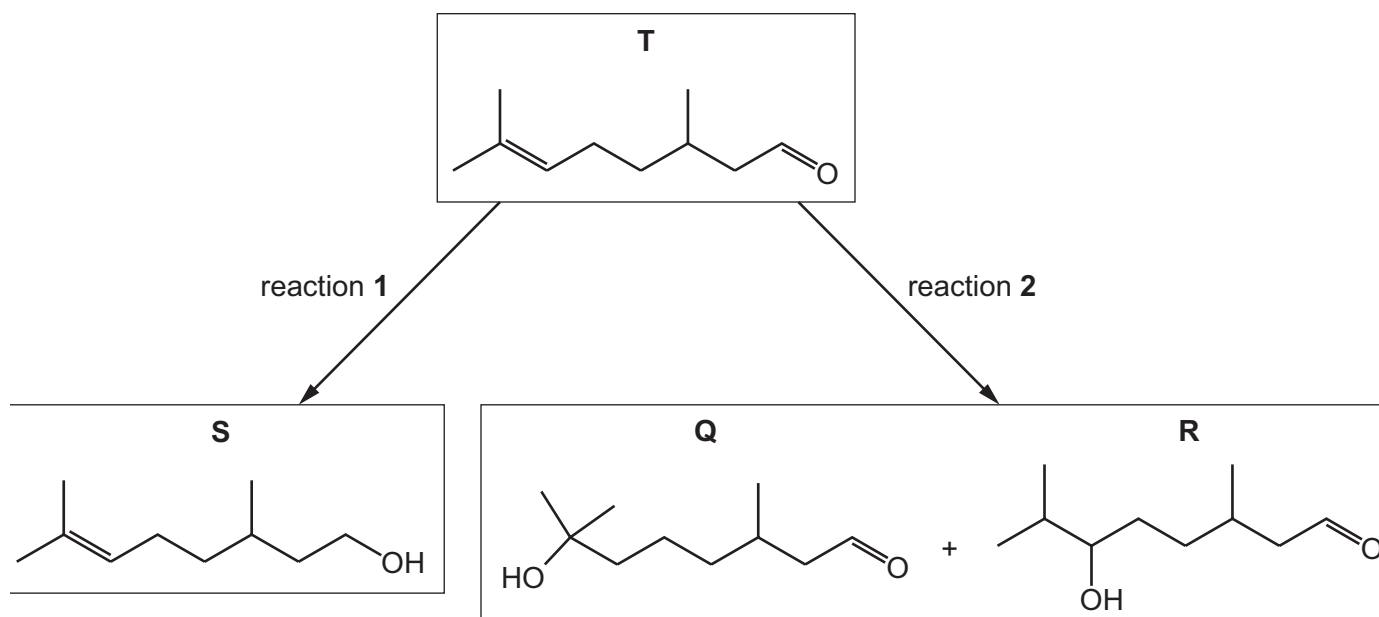
Context

3 Liquids that contain molecules of **T** smell like lemons.

**Fig. 3.1**

Context: part (c) (introduction)

(c) Fig. 3.2 shows two reactions of **T**.

**Fig. 3.2**

(i) Identify a suitable reagent for reaction 1.

[1]

28. M/J 22/P21/Q4(c)(i)-(ii)

Context

4 Compound **V** is a liquid.**V** contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.**V** has a relative molecular mass of 280.

Context: part (c) (introduction)

(c) **W**, **X** and **Y** have the same molecular formula, $C_5H_{10}O$.**W**, **X** and **Y** are added separately to different reagents. Observations for these reactions are described in Table 4.1.**Table 4.1**

	+ 2,4-dinitrophenylhydrazine	+ alkaline $I_2(aq)$	+ Fehling's reagent and warm
W	orange precipitate seen	no change	orange-red precipitate seen
X	orange precipitate seen	yellow precipitate seen	no change
Y	orange precipitate seen		

(i) **W**, **X** and **Y** each contain a common functional group.

Name the functional group that is present in all three compounds.

[1]

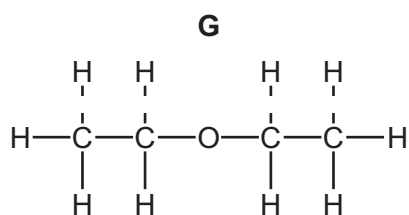
(ii) State the formula of the yellow precipitate produced when **X** is added to alkaline $I_2(aq)$.

[1]

29. M/J 22/P23/Q3(c)

Context

3 **G** belongs to a group of compounds called ethers.

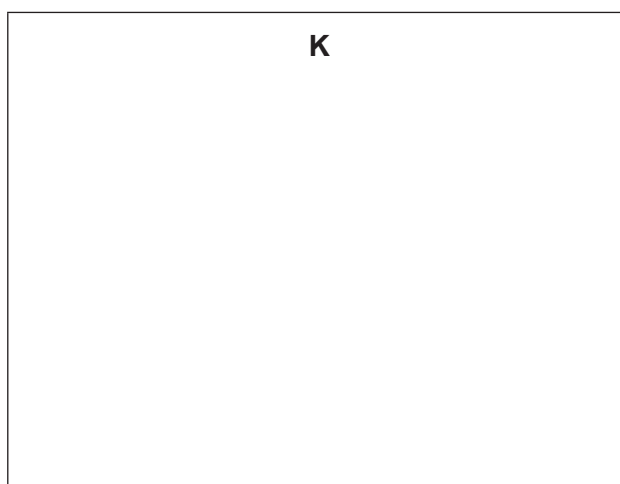
**Fig. 3.1**

(c) **K** has molecular formula $\text{C}_3\text{H}_6\text{O}$.

When **K** is added to 2,4-dinitrophenylhydrazine, an orange precipitate forms.

When **K** is warmed with Tollens' reagent, a silver mirror forms.

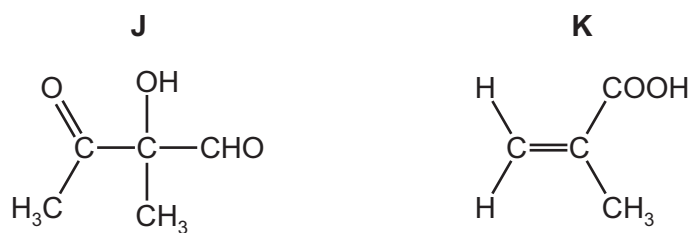
Draw the displayed formula of **K**.



[2]

30. MAR 22/P22/Q4(a)(i)

Context

4 Compounds **J** and **K** are found in plant oils.**Fig. 4.1**

(a) (i) Complete Table 4.1 to state what you would **observe** when **J** reacts with the reagents listed.

Table 4.1

reagent	observation with J
2,4-dinitrophenylhydrazine (2,4-DNPH)	
Tollens' reagent	
sodium metal	

[3]

Answers Section

1. M/J 26/P21/Q4(b)(ii)

4(b)(ii)	Alcohols / Compound A are / is not oxidised by Fehling's reagent (Aldehydes are oxidised by Fehling's reagent) OR Aldehydes are oxidised by Fehling's reagent (Alcohols / Compound A are / is not oxidised by Fehling's reagent)	1 A for alcohol groups = hydroxy = hydroxyl = OH group R for alcohol groups = hydroxide / OH⁻ group A alcohol groups are not a strong enough reducing agent (to reduce Fehling's) A Fehling's reagent is not a strong enough oxidising agent to oxidise the alcohol groups A Fehling's is a strong enough oxidising agent to oxidise aldehydes A Fehling's is too mild to oxidise alcohols A Fehling's isn't reduced by compound A / alcohols A Fehling's is reduced by aldehydes I Fehling's is a mild oxidising agent unqualified I aldehydes are easier to oxidise than alcohols I compound A has no aldehyde functional groups. I reacts with Fehling's I carbonyl R ketones are oxidised by Fehling's
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2. M/J 26/P22/Q6(c)(iii)

6(c)(iii)	M1 Br₂ (aq) / bromine water M2 orange / brown to colourless in S (not R) OR M1 Na₂CO₃ / sodium carbonate / NaHCO₃ / sodium hydrogencarbonate M2 fizzing / effervescence in R (not S)	2 R capitalisation once e.g. KMNO₄ M2 observation dependent on unambiguous attempt at M1 reagent e.g. M1 Br × M2 decolourises S ✓ OR M1 K₂Cr₂O₇ / Cr₂O₇²⁻ (no acid) ^ M2 (orange) to green ✓ OR M1 NaCO₃ × M2 bubbles in R ✓ etc M2 I alkene for S / carboxylic acid for R but A alcohol for S M1 A bromine / Br₂ M1 A aqueous iodine / aqueous I₂ / iodine water NOT iodine alone M1 R iodoform M2 A decolourises M2 A yellow to colourless M2 R red M2 A bubbles M2 I limewater (turns milky)
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6(c)(iii)		Alternatives: M1 acidified potassium dichromate / H^+ AND $\text{K}_2\text{Cr}_2\text{O}_7$ M1 BOD H^+ / $\text{Cr}_2\text{O}_7^{2-}$ M2 (orange) to green with S (not R) M1 potassium permanganate / KMnO_4 M1 BOD potassium manganate M1 BOD MnO_4^- M2 decolorises / (purple) to colourless (solution) / brown ppt with S (not R) M2 A green solution formed if alkaline KMnO_4 used M1 addition of named alcohol and (conc) sulfuric acid OR named alcohol and catalyst M2 sweet smell with R (not S) M1 addition of named carboxylic acid and (conc) sulfuric acid OR named carboxylic acid AND catalyst M2 sweet smell with S (and not R) M1 named acid-base indicator M2 appropriate colour change with R (and not S) M1 A Mg / magnesium M2 A faster bubbles with R (comparison required)
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3. M/J 26/P23/Q4(a)(iii)

4(a)(iii)	Alcohols / Compound V are / is not oxidised by Fehling's reagent (aldehydes are oxidised by Fehling's reagent) OR Aldehydes are oxidised by Fehling's reagent (alcohols / Compound V are / is not oxidised by Fehling's reagent)	1 A for alcohol groups = hydroxy = hydroxyl = OH group R for alcohol groups = hydroxide / OH^- group A alcohol groups are not a strong enough reducing agent (to reduce Fehling's) A Fehling's reagent is not a strong enough oxidising agent to oxidise the alcohol groups A Fehling's is a strong enough oxidising agent to oxidise aldehydes A Fehling's is too mild to oxidise alcohols A Fehling's is not reduced by V / alcohols A Fehling's is reduced by aldehydes I Fehling's is a mild oxidising agent (unqualified) I aldehydes are easier to oxidise than alcohols I compound V has no aldehyde functional groups. I reacts with Fehling's I carbonyl R ketones are oxidised by Fehling's
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4. M/J 25/P21/Q6(a)

6(a)	carbonyl	1
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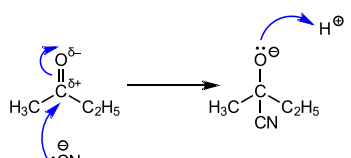
5. M/J 25/P21/Q6(b)

6(b)	M1 (2,4)-dinitrophenylhydrazine / (2,4)-DNPH M2 orange / red / yellow precipitate	2
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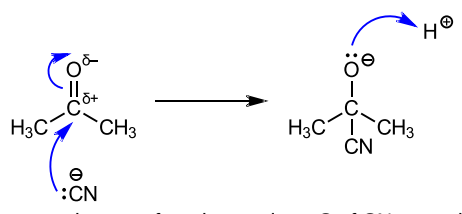
6. M/J 25/P21/Q6(c)(i)-(ii)

6(c)(i)	F = propanone G = ethanal H = propanal	2
6(c)(ii)	tri-iodomethane OR CHI_3	1

7. MAR 25/P22/Q4(b)(i)

4(b)(i)	 • curly arrow from lone pair on $:\text{CN}^-$ to $\text{C}(\delta^+)$ • curly arrow from $\text{C}=\text{O}$ to $\text{O}(\delta^-)$ AND correct dipole • correct intermediate • curly arrow from lone pair on O^- to H^+	4
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8. O/N 24/P21/Q4(a)(i)

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on $\text{C}=\text{O}$ • curly arrow from $\text{C}=\text{O}$ bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on O^- to (source of) H^+	1
		1
		1

9. O/N 24/P21/Q4(c)

4(c)(i)	$\text{CH}_3\text{COCH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	1
4(c)(ii)	propan-2-ol	1

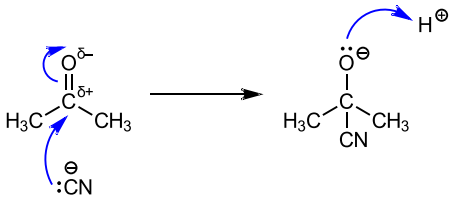
10. O/N 24/P21/Q4(d)

4(d)	red / orange / yellow ppt	1
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11. O/N 24/P21/Q4(e)

4(e)	Fehling's solution cannot oxidise ketones OR ketones are not easily oxidised	1
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12. O/N 24/P23/Q4(a)(i)

4(a)(i)	 • curly arrow from lone pair on C of CN⁻ to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O⁻ to (source of) H⁺	1
		1
		1

13. O/N 24/P23/Q4(c)

4(c)(i)	$\text{CH}_3\text{COCH}_3 + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	1
4(c)(ii)	propan-2-ol	1

14. O/N 24/P23/Q4(d)

4(d)	red / orange / yellow ppt	1
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15. O/N 24/P23/Q4(e)

4(e)	Fehling's solution cannot oxidise ketones OR ketones are not easily oxidised	1
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16. M/J 24/P21/Q6(b)(i)

6(b)(i)	reducing agent for C=O AND C=C	1
---------	---------------------------------------	---

17. M/J 24/P21/Q6(c)(ii)

6(c)(ii)		observation on addition of 2,4-DNPH	1
	Y	orange precipitate	
	Q	no precipitate	
	Both correct for one mark		

18. MAR 24/P22/Q1(d)(iii)-(iv)

1(d)(iii)	<table border="1"> <tr> <th>reagent</th><th>observation with Y</th><th>observation with Z</th></tr> <tr> <td>$\text{Na}_2\text{CO}_3(\text{aq})$</td><td>no reaction</td><td>• no reaction</td></tr> <tr> <td>alkaline $\text{I}_2(\text{aq})$</td><td>• yellow precipitate (ppt)</td><td>• yellow ppt</td></tr> <tr> <td>2,4-DNPH</td><td>• red / orange / yellow ppt</td><td>• red / orange / yellow ppt</td></tr> <tr> <td>Tollens' reagent</td><td>• no reaction</td><td>• silver mirror</td></tr> </table>	reagent	observation with Y	observation with Z	$\text{Na}_2\text{CO}_3(\text{aq})$	no reaction	• no reaction	alkaline $\text{I}_2(\text{aq})$	• yellow precipitate (ppt)	• yellow ppt	2,4-DNPH	• red / orange / yellow ppt	• red / orange / yellow ppt	Tollens' reagent	• no reaction	• silver mirror	4
reagent	observation with Y	observation with Z															
$\text{Na}_2\text{CO}_3(\text{aq})$	no reaction	• no reaction															
alkaline $\text{I}_2(\text{aq})$	• yellow precipitate (ppt)	• yellow ppt															
2,4-DNPH	• red / orange / yellow ppt	• red / orange / yellow ppt															
Tollens' reagent	• no reaction	• silver mirror															
1(d)(iv)	$\text{CH}_3\text{CHO} + 2[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{OH}$	1															

19. M/J 23/P21/Q4(a)

4(a)(i)		1
4(a)(ii)	reducing agent	1

20. M/J 23/P22/Q5(b)

5(b)(i)	carbonyl (group)	1
5(b)(ii)		2

21. M/J 23/P23/Q5(b)

5(b)(i)	$\text{C}_6\text{H}_{10}\text{O}_3 + [6\text{H}] \rightarrow \text{C}_6\text{H}_{14}\text{O}_2 + \text{H}_2\text{O}$	1
5(b)(ii)	LiAlH_4 (in dry ether)	1

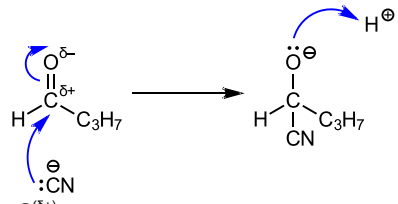
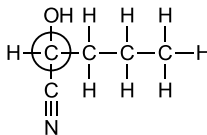
22. M/J 23/P23/Q5(d)

5(d)	reagent	observation on addition to X	observation on addition to Y	3
	sodium carbonate solution	effervescence	no change	
	2,4-DNPH	orange precipitate	no change	
	alkaline aqueous iodine	yellow precipitate	yellow precipitate	
	Two correct for one mark, four correct for two marks, six correct for three marks			

23. M/J 23/P23/Q6(b)(i)-(ii)

6(b)(i)	nucleophilic addition	1
6(b)(ii)	M1 correct dipole on C=O AND curly arrow from C=O bond to O AND from lone pair of electrons on C of :CN⁻ M2 correct intermediate (charge on O or overall) M3 curly arrow from lone pair of electrons on O to H⁺ AND correct organic product (+ CN⁻)	3

24. MAR 23/P22/Q3(b)

3(b)(i)	 curly arrow from lone pair on C of CN⁻ to C(^{δ+}) correct dipole on C=O AND curly arrow from C=O bond to O^{δ-} correct organic intermediate curly arrow from lone pair on alkoxide O⁽⁻⁾ to H⁺ / H of HCN / H of H₂O (to reform the alcohol)	1
3(b)(ii)	acidified K ₂ Cr ₂ O ₇	1
	Tollens' reagent	1
	no reaction	1
3(b)(iii)	C ₄ H ₈ O + [O] → C ₄ H ₈ O ₂	1
3(b)(iv)		1

25. O/N 22/P21/Q5(a)(ii)-(iii)

5(a)(ii)	C ₆ H ₁₀ O ₃ + 2[H] → C ₆ H ₁₂ O ₃	1
5(a)(iii)	M1 (ketone in) N is planar (so can be attacked from either side)	1
	M2 because different stereoisomers / optical isomers form	1

26. O/N 22/P23/Q5(a)(ii)-(iii)

5(a)(ii)	C ₆ H ₁₀ O ₃ + 2[H] → C ₆ H ₁₂ O ₃	1
5(a)(iii)	M1 (ketone in) N is planar (so can be attacked from either side)	1
	M2 because different stereoisomers / optical isomers form	1

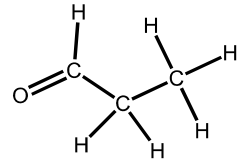
27. M/J 22/P21/Q3(c)(i)

3(c)(i)	NaBH ₄ / sodium borohydride / LiAlH ₄ / lithium tetrahydridoaluminate	1
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28. M/J 22/P21/Q4(c)(i)-(ii)

4(c)(i)	carbonyl	1
4(c)(ii)	CHI ₃	1

29. M/J 22/P23/Q3(c)

3(c)	displayed formula of propanal  M1 identification of propanal M2 correct use of displayed formula and correct formula used	2
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30. MAR 22/P22/Q4(a)(i)

4(a)(i)	red / orange / yellow precipitate / ppt / solid [1] silver mirror / silver / grey solid / precipitate / ppt [1] effervescence / bubbling / fizzing [1]	3
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Unit 18

CARBOXYLIC ACIDS AND DERIVATIVES

28 items · 56 marks

Syllabus sections in this unit

18.1 Carboxylic acids

18.2 Esters

Unit 18: Carboxylic acids and derivatives

1. M/J 26/P21/Q4(c)

(c) Compound **B** reacts with an excess of LiAlH_4 to produce organic compound **C**.

Complete the equation in Figure 4.3 to show the reaction of **B** with an excess of LiAlH_4 to produce **C**.

[H] represents one atom of hydrogen from the reducing agent, LiAlH_4 .

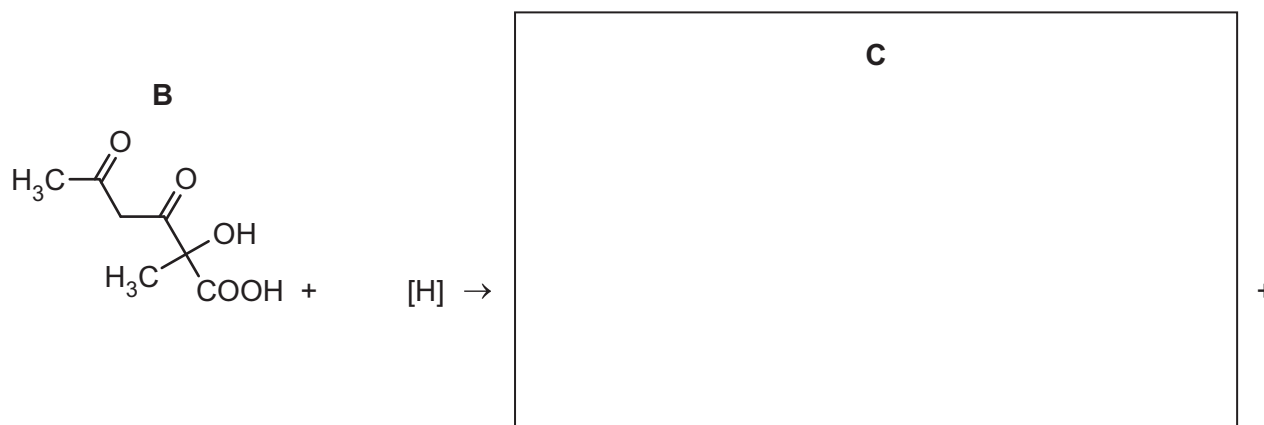


Figure 4.3

[2]

2. O/N 25/P22/Q4(d)(i)

Context

4 Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$.

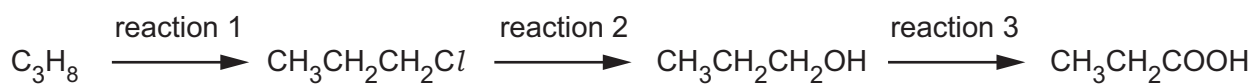


Fig. 4.1

Context: part (d) (introduction)

(d) $\text{CH}_3\text{CH}_2\text{COOH}$ reacts with an unsaturated alcohol **R** to form unsaturated ester **S**.

(i) State the **type** of reaction that forms **S**.

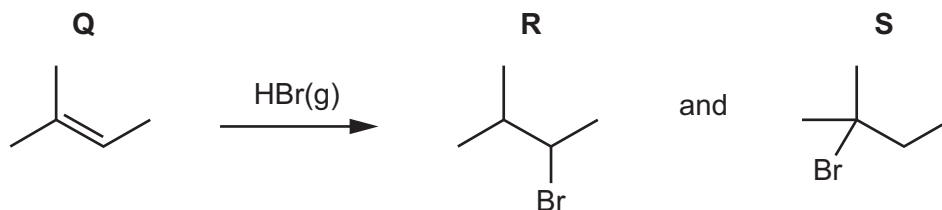
[1]

3. O/N 25/P24/Q4(b)(iii)

Context

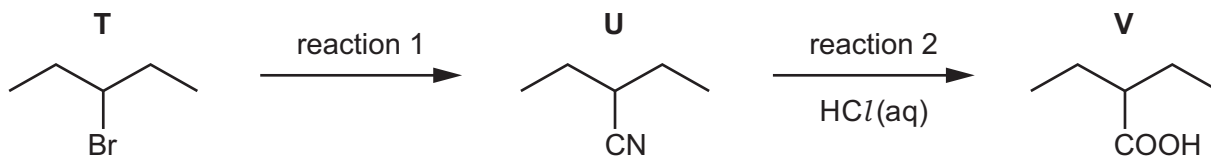
4 Compounds **P** and **Q** are structural isomers.**Fig. 4.1**

Context: part (a)(v)

(v) **Q** reacts with HBr(g) to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.**Fig. 4.2**State and explain why isomer **S** is the major product of the reaction.

[2]

Context: part (b) (introduction)

(b) Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.**Fig. 4.3****(iii)** **V** reacts with propan-2-ol in the presence of a catalytic amount of H_2SO_4 to form organic compound **W**.

Complete Table 4.1 to give details of this reaction.

Table 4.1

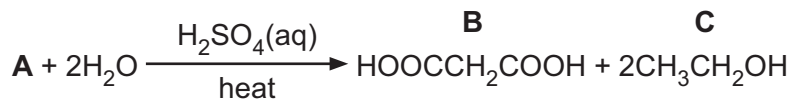
	reaction of V with propan-2-ol
type of reaction	
functional group formed	
molecular formula of organic product W	

[3]

4. M/J 25/P21/Q5(a)

Context

- 5** Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

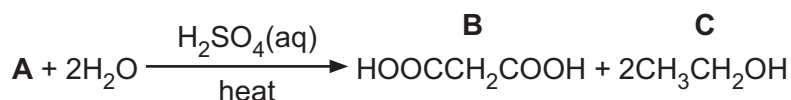
- (a) Draw the structure of **A**.

[2]

5. M/J 25/P21/Q5(b)

Context

- 5** Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

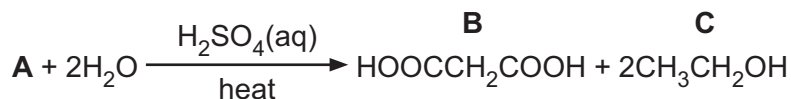
- (b) Name the type of reaction that occurs when **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$.

[1]

6. M/J 25/P21/Q5(c)(i)

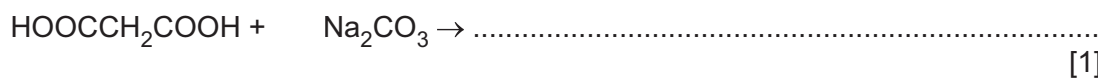
Context

- 5** Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

Context: part (c) (introduction)

- (c) When aqueous Na_2CO_3 is added to separate samples of **B** and **C**, effervescence is observed with **B** only.
- (i) Complete the equation to describe the reaction between **B** and an excess of aqueous Na_2CO_3 .



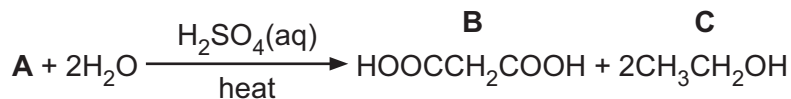
[1]

7. M/J 25/P21/Q5(e)(i)-(ii)

Context

5 Compound **A** contains the elements carbon, hydrogen and oxygen only.

When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

Context: part (e) (introduction)

(e) **B** reacts with an excess of reducing agent **R** to produce **D**.

An excess of PCl_5 is added to **D**.

- A vigorous reaction occurs.
- Misty fumes are seen.
- Organic compound **E** is produced.

(i) Identify the formula of the reducing agent **R**.

[1]

(ii) Complete the equation to describe the reaction of **B** with an excess of **R**.

Use [H] to represent one atom of hydrogen from **R**.



[2]

8. M/J 25/P22/Q4(a)

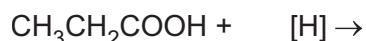
4 **(a)** Propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$, reacts with reducing agent **Q** to produce propan-1-ol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$.

(i) Suggest the formula of reducing agent **Q**.

[1]

(ii) Complete the equation to show the reduction of propanoic acid to propan-1-ol.

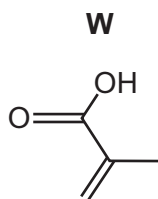
Use [H] to represent one atom of hydrogen from **Q**.



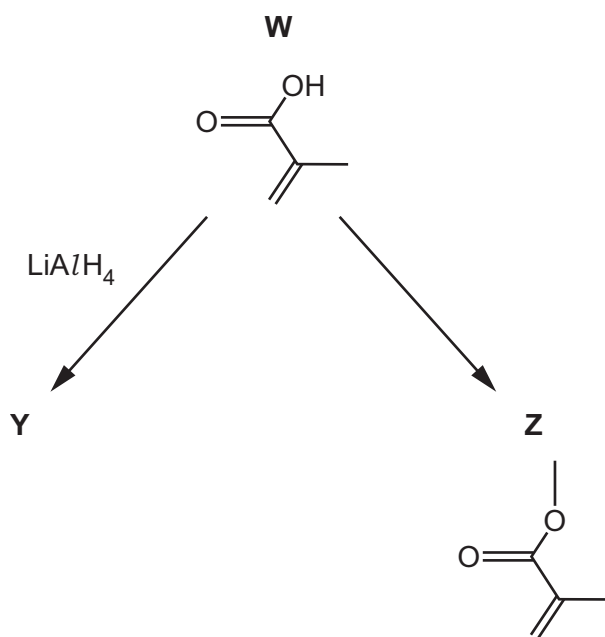
[1]

9. M/J 25/P23/Q5(b)(ii)-(iv)

Context: part (a) (introduction)

5 (a) W is a colourless liquid.**Fig. 5.1**

Context: part (b) (introduction)

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.**Fig. 5.2****(ii)** **W** reacts with LiAlH_4 to produce **Y**.Draw the structure of **Y**.

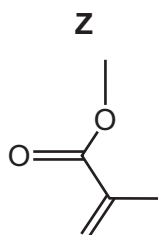
[1]

(iii) Identify the role of LiAlH_4 when it reacts with **W**.

[1]

(iv) Suggest the reagent and conditions required when **W** is converted into **Z**.

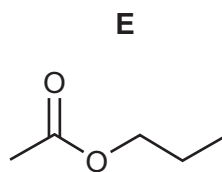
[1]

**Fig. 5.3**

10. M/J 25/P24/Q4(b)

Context

- 4** The skeletal formula of **E** is shown in Fig. 4.1.

**Fig. 4.1**

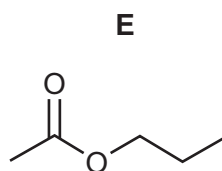
- (b)** Name **E**.

[1]

11. M/J 25/P24/Q4(c)

Context

- 4** The skeletal formula of **E** is shown in Fig. 4.1.

**Fig. 4.1**

- (c)** **E** is made when a carboxylic acid and an alcohol react together.

- (i)** Name the type of reaction that takes place during this esterification reaction.

[1]

- (ii)** Write an equation for the formation of **E** from a carboxylic acid and an alcohol.
Use structural formulae to represent the organic species.

[1]

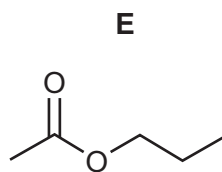
- (iii)** State the systematic name of the alcohol used to make **E**.

[1]

12. M/J 25/P24/Q4(d)(ii)

Context

- 4** The skeletal formula of **E** is shown in Fig. 4.1.

**Fig. 4.1**

Context: part (d) (introduction)

- (d)** **F** is an isomer of **E**.

F reacts with HCl(aq) to produce **G** and **H**.

G is a secondary alcohol.

H is also produced when HCl(aq) is added to CH_3CN .

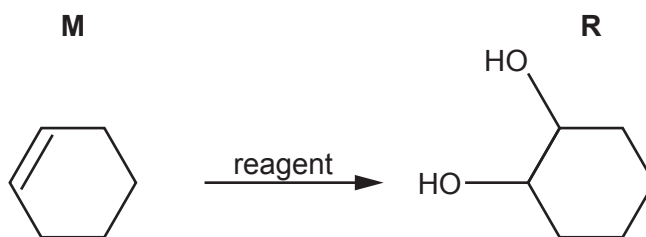
- (ii)** Draw the displayed formula of **G**.

[2]

13. M/J 24/P23/Q5(d)(iii)-(iv)

Context: part (a)

- 5** **(a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

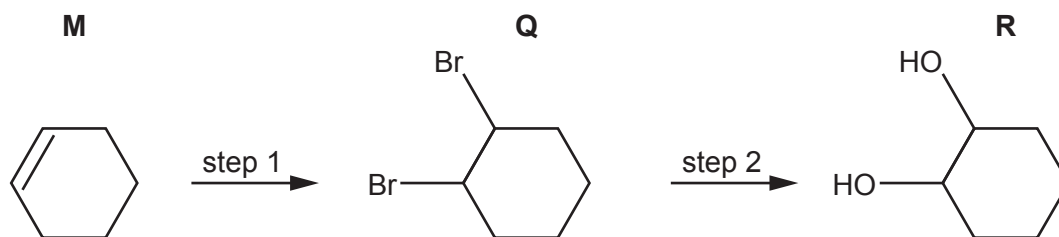
**Fig. 5.1**

Identify the reagent and conditions for this reaction.

[1]

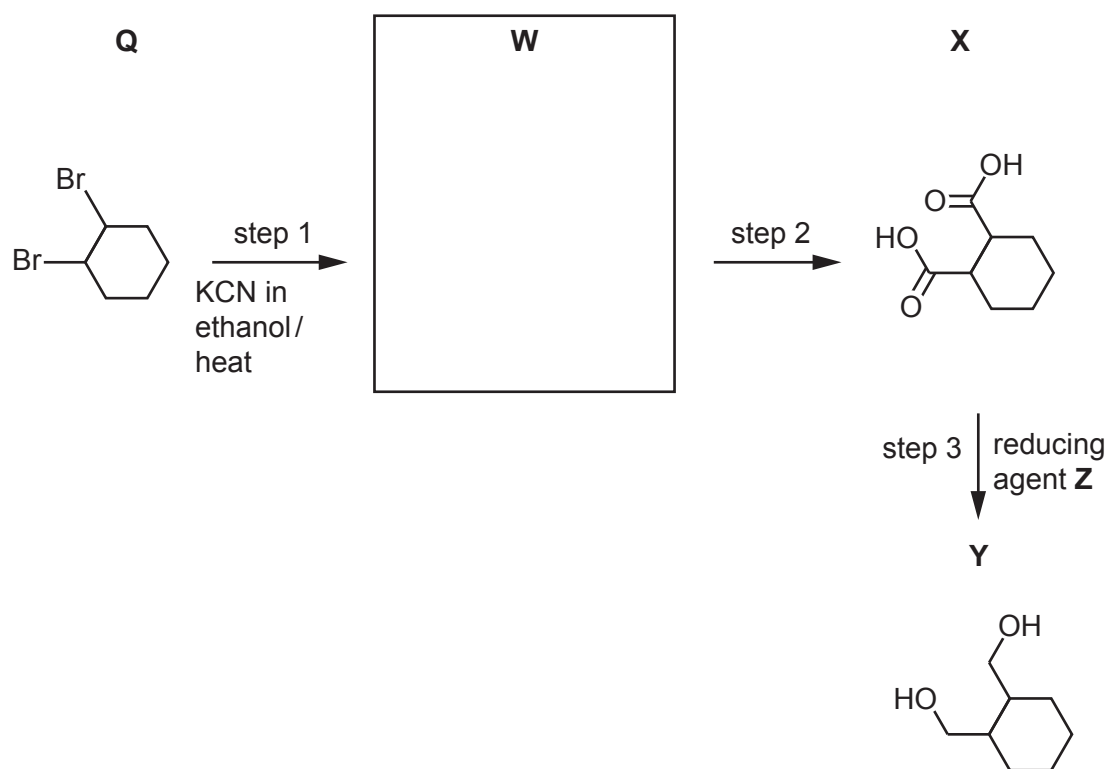
Context: part (b) (introduction)

- (b)** **R** is also made from **M** by two steps, as shown in Fig. 5.2.

**Fig. 5.2**

Context: part (d) (introduction)

- (d)** **Y** is made from **Q** in a three-step reaction.

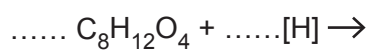
**Fig. 5.4**

(iii) In step 3, **X** reacts with reducing agent **Z** to produce **Y**.

Complete the equation for the reaction of **X** with **Z**.

Use a molecular formula to represent the organic product.

Use [H] to represent one atom of hydrogen from **Z**.



[1]

(iv) Identify **Z**.

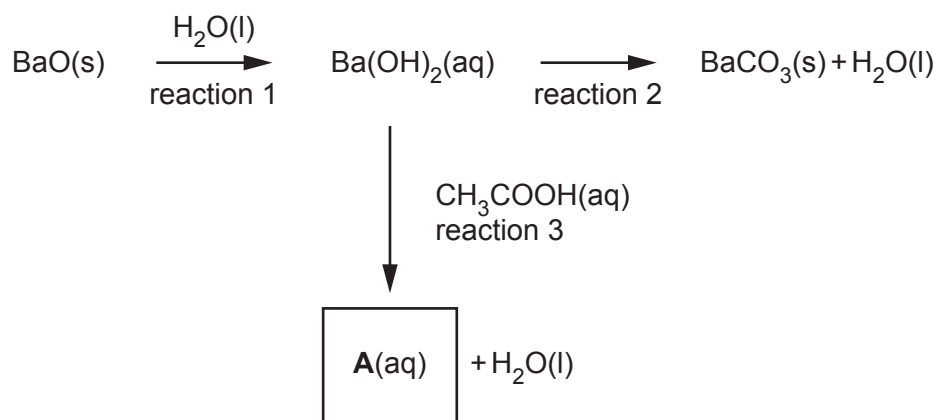
[1]

14. O/N 23/P21/Q2(c)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

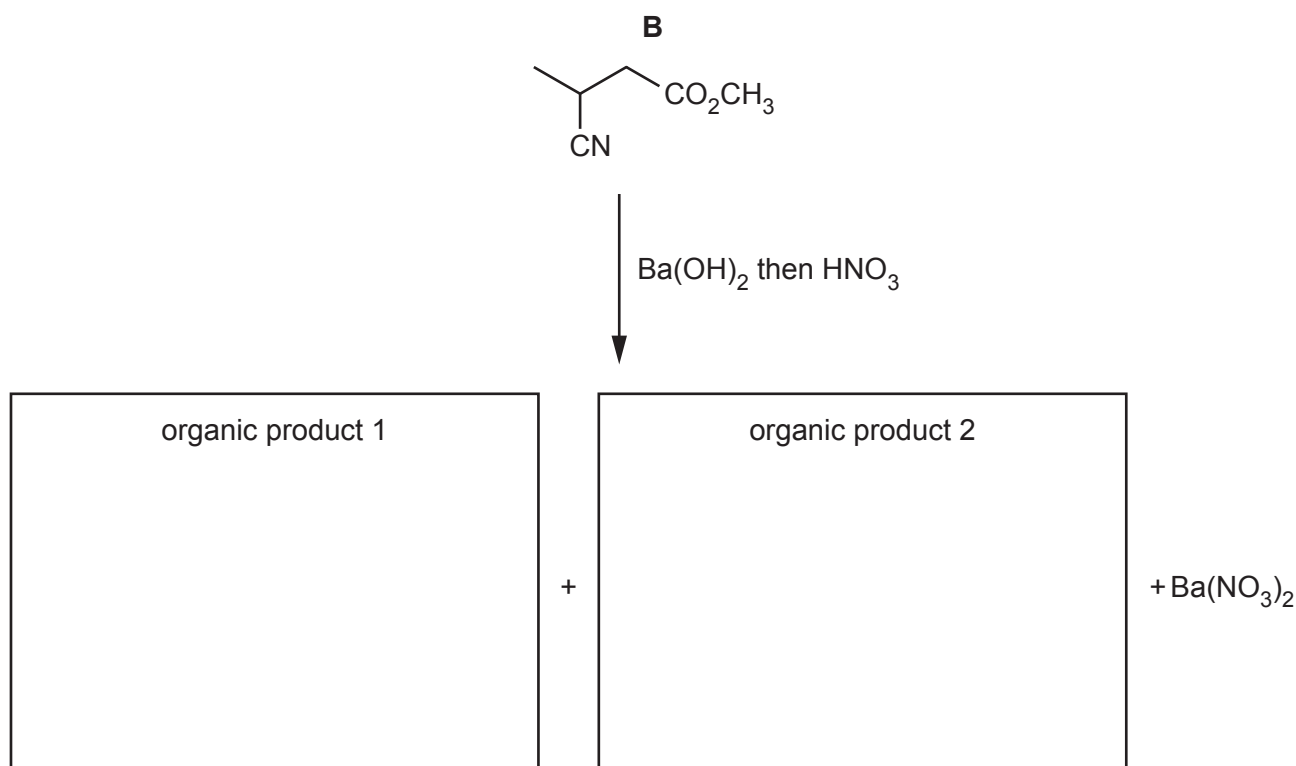
Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

**Fig. 2.1**

(c) $\text{Ba}(\text{OH})_2$ is used to hydrolyse organic compounds.

Fig. 2.2 shows the reaction of **B** with $\text{Ba}(\text{OH})_2$, followed by acidification.

Draw the structures of the organic products of the process shown in Fig. 2.2.

**Fig. 2.2**

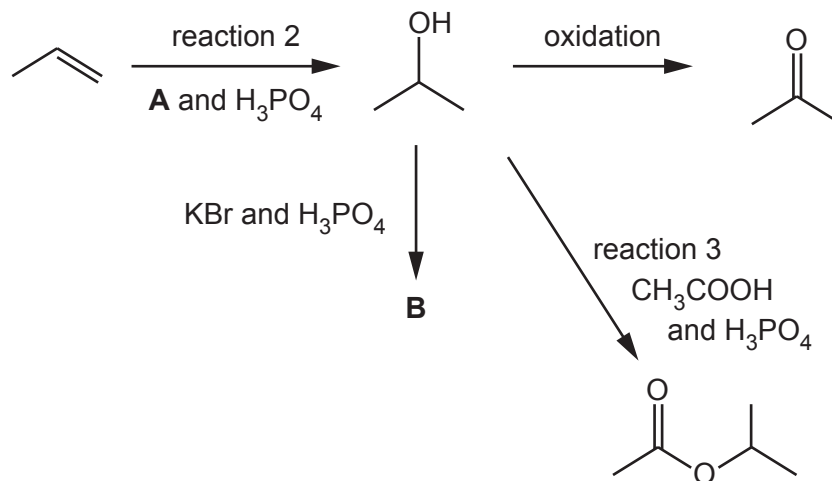
15. O/N 23/P22/Q3(c)(iii)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (c) (introduction)

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

**Fig. 3.1**

(iii) Name the type of reaction that occurs in reaction 3.

[1]

16. O/N 23/P22/Q3(d)(i)

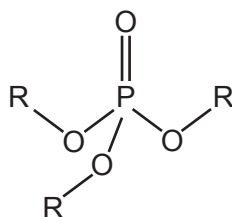
Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (d) (introduction)

(d) H_3PO_4 also reacts with alcohols to form organophosphates.

Organophosphates are compounds similar to esters. They have the general structure shown in Fig. 3.2.



R = alkyl group

Fig. 3.2

(i) Complete the equation to suggest the products of the reaction of H_3PO_4 with methanol, CH_3OH .



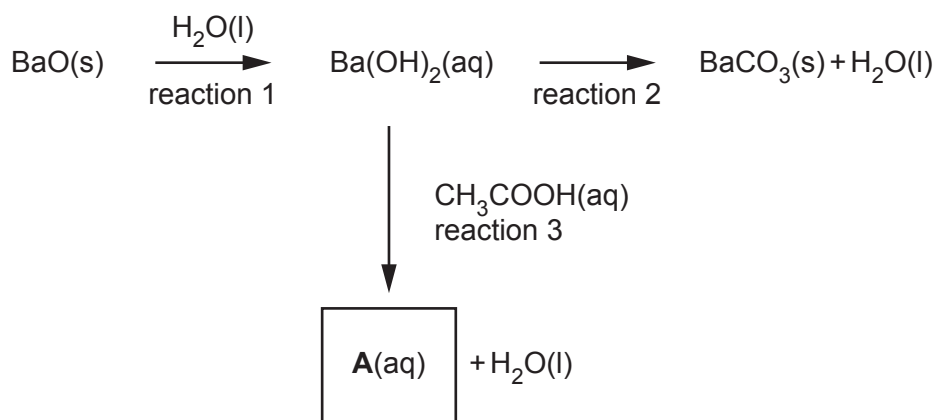
[1]

17. O/N 23/P23/Q2(c)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

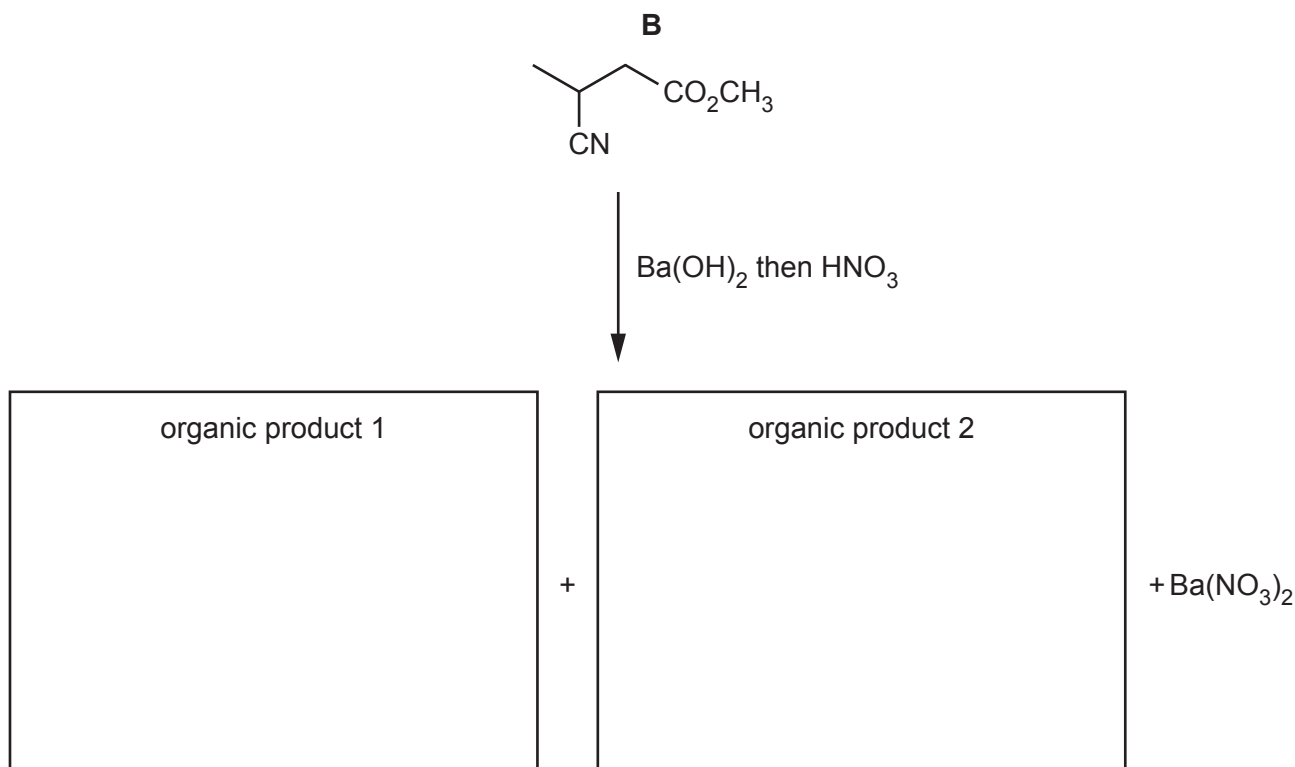
Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

**Fig. 2.1**

(c) $\text{Ba}(\text{OH})_2$ is used to hydrolyse organic compounds.

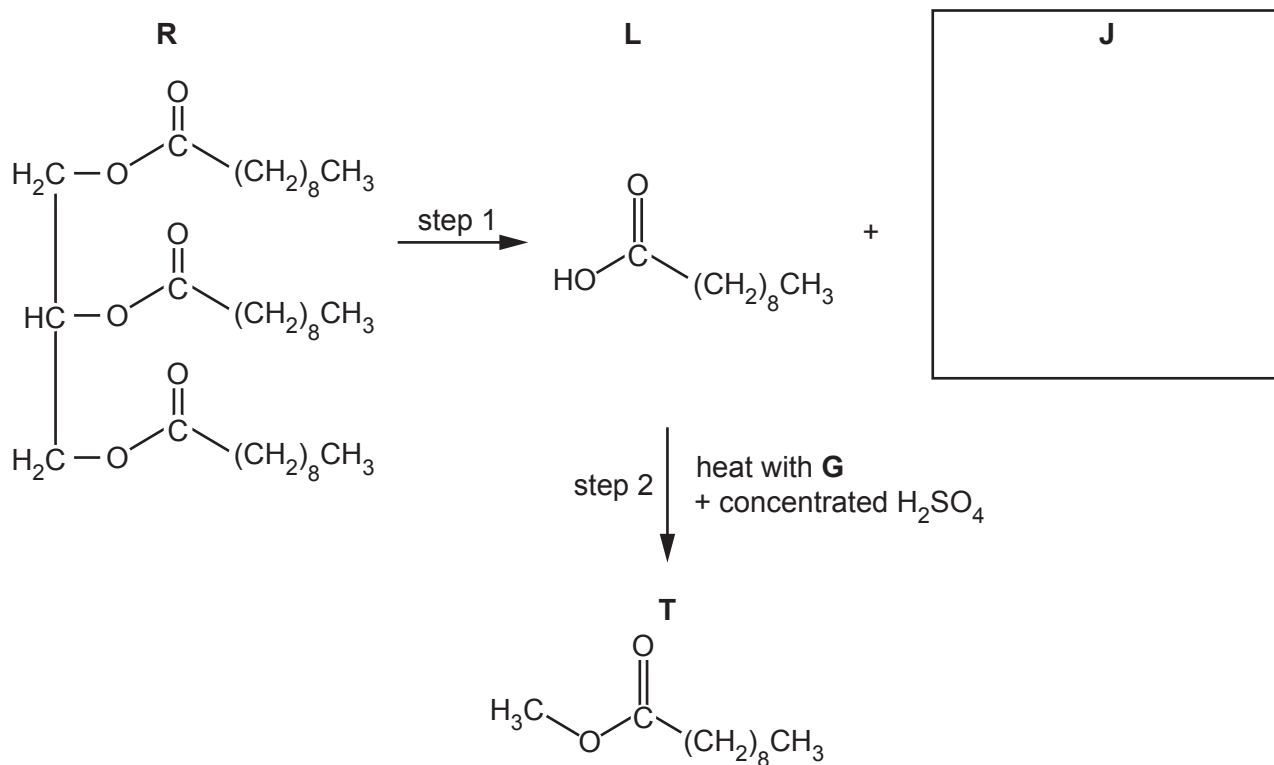
Fig. 2.2 shows the reaction of **B** with $\text{Ba}(\text{OH})_2$, followed by acidification.

Draw the structures of the organic products of the process shown in Fig. 2.2.

**Fig. 2.2**

18. M/J 23/P21/Q5(c)

(c) Biodiesel **T** is a fuel made from vegetable oil **R**. Fig. 5.1 shows the production of **T** from **R** in a two-step process.

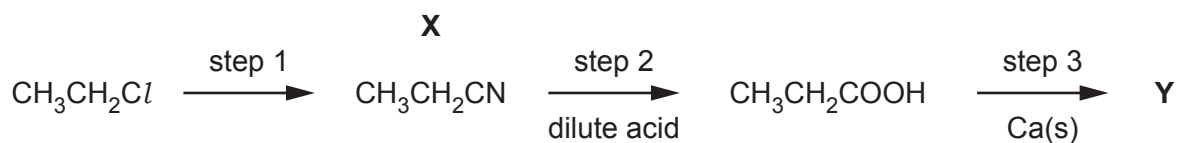
**Fig. 5.1**

- (i) In step 1 all three ester groups in **R** react. Suggest a suitable reagent and conditions for step 1. [1]
- (ii) Draw the structural formula of **J** in the box in Fig. 5.1. [1]
- (iii) Name the type of reaction that occurs in step 2. [1]
- (iv) Name organic reagent **G** used in step 2. [1]
- (v) **L** is called decanoic acid. Use systematic nomenclature to deduce the name of **T**. [1]

19. M/J 23/P22/Q3(a)(iv)

Context

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

- (iv)** In step 3, **Y** and a gas are produced.
Construct an equation for step 3.

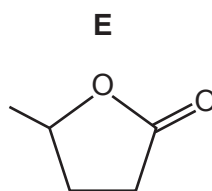
[2]

20. M/J 23/P22/Q5(d)(i)-(iii)

Context: part (d) (introduction)

- (d)** **D** reacts in the presence of a sulfuric acid catalyst to form **E** and water.

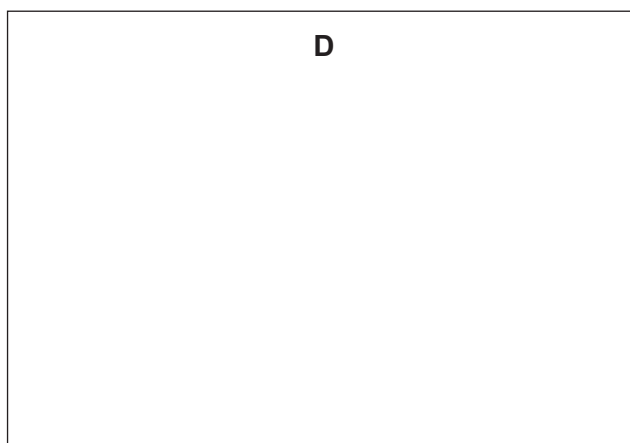
The structure of **E** is shown in Fig. 5.1.

**Fig. 5.1**

- (i)** Name the functional group present in **E**.
- (ii)** Identify the type of reaction that occurs when **D** reacts to form **E**.
- (iii)** Draw the structure of **D** in the box.

[1]

[1]

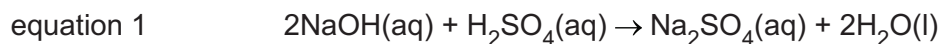


[1]

21. M/J 23/P23/Q3(c)(i)

Context

- 3** A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



Context: part (c) (introduction)

- (c)** The enthalpy change of neutralisation of CH₃COOH(aq) and NaOH(aq) is $-55.2 \text{ kJ mol}^{-1}$.

- (i)** Complete the equation for the reaction.

**22. O/N 22/P21/Q5(a)(iv)**

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

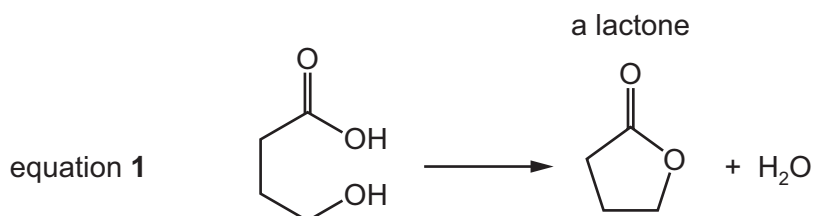
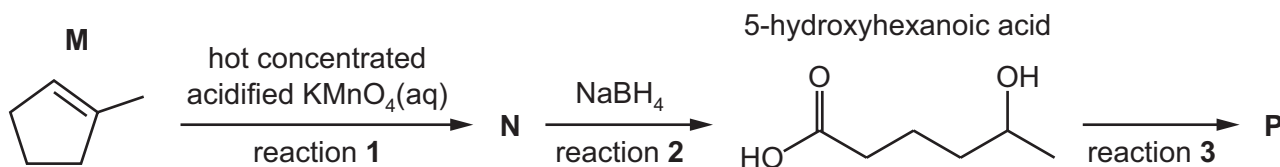


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

**Fig. 5.1**

- (iv)** Draw lactone **P**, the product of reaction 3.

[1]

23. O/N 22/P23/Q5(a)(iv)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

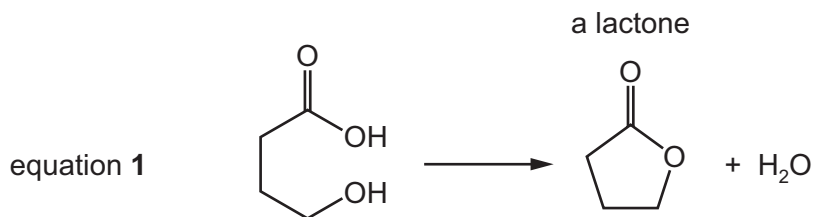
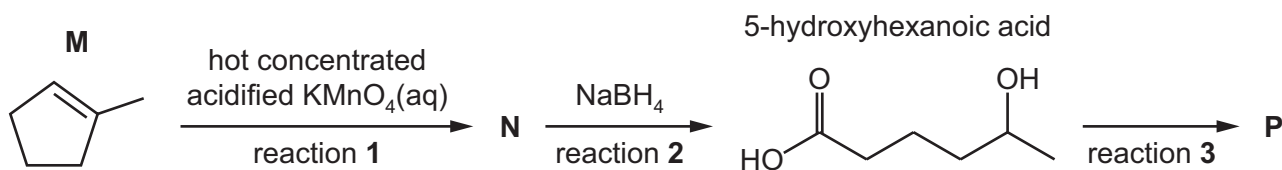


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

**Fig. 5.1**

- (iv)** Draw lactone **P**, the product of reaction 3.

[1]

24. M/J 22/P21/Q4(b)(i)

Context

- 4** Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

Context: part (b) (introduction)

- (b)** **V** contains two types of functional group: a carboxylic acid and an alkene.

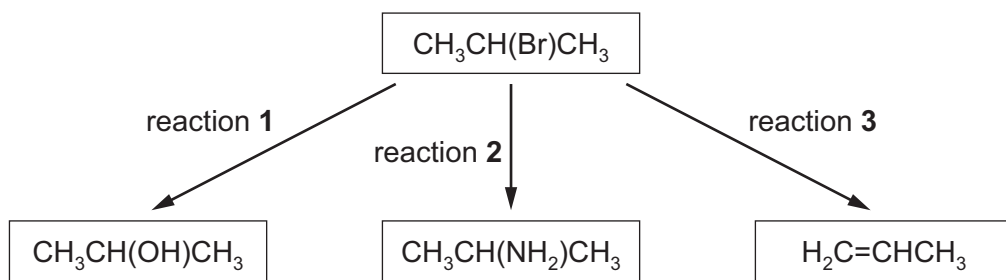
- (i)** Describe a chemical test and observation which confirms the presence of a carboxyl functional group.

[2]

25. M/J 22/P22/Q5(c)(iii)-(iv)

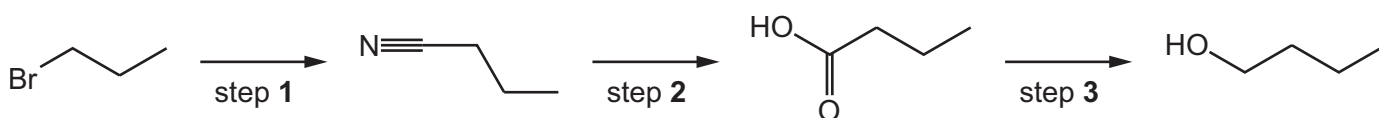
Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

**Fig. 5.1**

Context: part (c) (introduction)

(c) Fig. 5.2 shows how butan-1-ol can be made from 1-bromopropane in three steps.

**Fig. 5.2**

(iii) Step 3 is a reduction reaction.

Construct an equation for the reduction reaction in step 3. Use $[\text{H}]$ to represent one atom of hydrogen from the reducing agent.

[1]

(iv) State the identity of a suitable reducing agent in step 3.

[1]

26. M/J 22/P22/Q6(a)

Context

6 **Z** is a molecule which contains the elements carbon, hydrogen and oxygen only.

Z contains only alkene and carboxyl functional groups.

(a) Complete Table 6.1 by describing the observations that occur when two different reagents are added to separate samples of **Z**(aq).

Table 6.1

reagent added to Z (aq)	observation
$\text{Br}_2(\text{aq})$	
$\text{Na}_2\text{CO}_3(\text{s})$	

[2]

27. M/J 22/P22/Q6(c)(iii)

Context

6 **Z** is a molecule which contains the elements carbon, hydrogen and oxygen only.

Z contains only alkene and carboxyl functional groups.

Context: Table 6.1

Table 6.1

reagent added to Z (aq)	observation
Br ₂ (aq)	
Na ₂ CO ₃ (s)	

Context: part (b)

(b) Table 6.2 shows the percentage by mass of each element present in **Z**.

Table 6.2

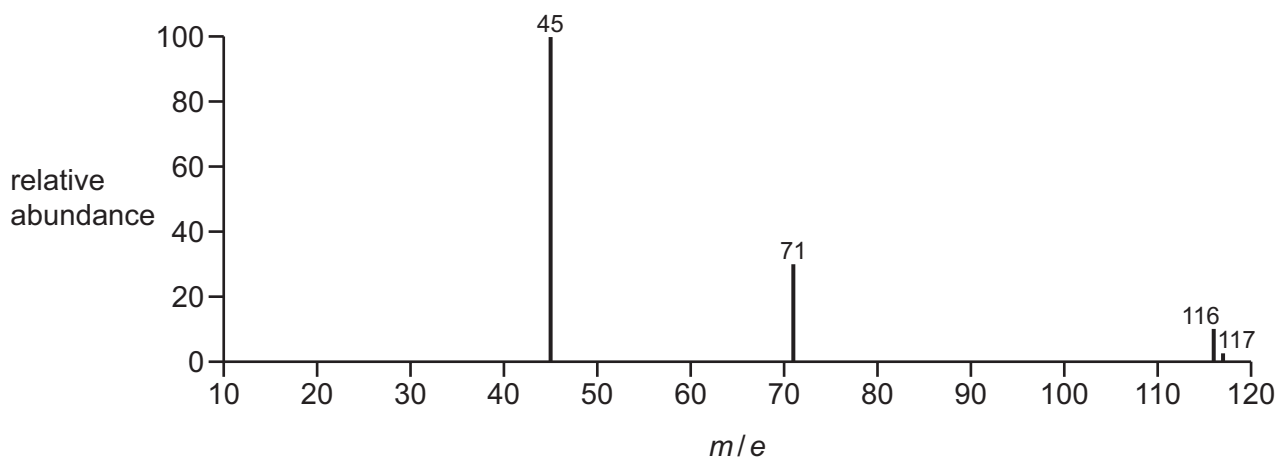
element	percentage by mass/%
carbon	41.38
hydrogen	3.45
oxygen	55.17

Using the data in Table 6.2, demonstrate that the empirical formula of **Z** is CHO.
Show your working.

[1]

Context: part (c) (introduction)

(c) Fig. 6.1 shows the mass spectrum of **Z**.

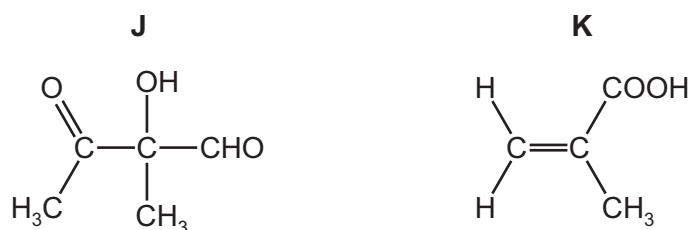
**Fig. 6.1**

(iii) Suggest the structure of **Z** using relevant information from Table 6.1, **(b)** and **(c)**.

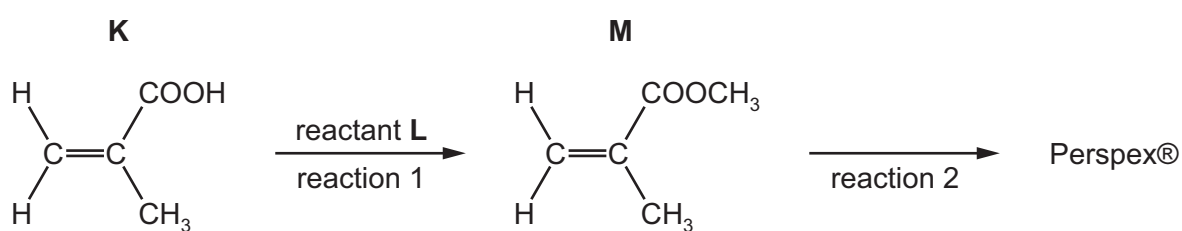
[1]

28. MAR 22/P22/Q4(b)(i)

Context

4 Compounds **J** and **K** are found in plant oils.**Fig. 4.1**

Context: part (b) (introduction)

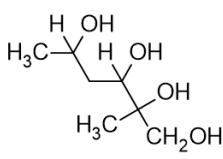
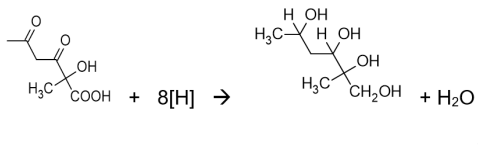
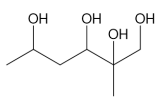
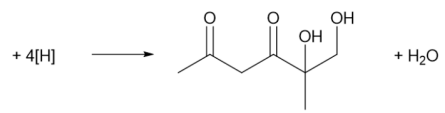
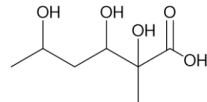
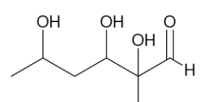
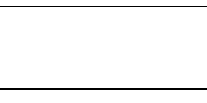
(b) **K** is used to make the addition polymer Perspex®. A synthesis of Perspex® is shown in Fig. 4.2.**Fig. 4.2****(i)** Identify **L**. State the conditions required for reaction 1.**L** =

conditions =

[2]

Answers Section

1. M/J 26/P21/Q4(c)

4(c)	M1  M2 equation fully correct  + 8[H] →  + H₂O ✓✓	2 M1 A any unambiguous structure of C M2 stoichiometry based on correct structure C A ecf on M2 if incorrect structure is correctly balanced – as long as structure contains 7 carbon atoms common ECFs for 1 mark  + 4[H] →  + H₂O + 4[H] →  + H₂O + 6[H] →  + H₂O
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2. O/N 25/P22/Q4(d)(i)

4(d)(i)	condensation	1
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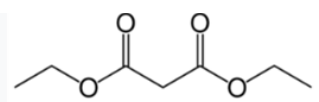
3. O/N 25/P24/Q4(b)(iii)

4(b)(iii)		reaction of V with propan-2-ol	3
	type of reaction	M1 condensation	
	functional group formed	M2 ester	
	molecular formula of organic product W	M3 C ₉ H ₁₈ O ₂	

Answer for context part (a)(v)

4(a)(v)	M1 (reaction proceeds via a) more stable intermediate / carbocation M2 because of greater positive inductive effect of more alkyl groups	2
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4. M/J 25/P21/Q5(a)

5(a)	M1 two ester links M2 rest of structure correct 	2
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5. M/J 25/P21/Q5(b)

5(b)	(acid) hydrolysis	1
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6. M/J 25/P21/Q5(c)(i)

5(c)(i)	$\text{HOOCCH}_2\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow \text{NaOOCCH}_2\text{COONa} + \text{CO}_2 + \text{H}_2\text{O}$	1
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7. M/J 25/P21/Q5(e)(i)-(ii)

5(e)(i)	LiAlH_4	1
5(e)(ii)	M1 correct unambiguous organic product M2 correct balanced equation $\text{HOOCCH}_2\text{COOH} + 8[\text{H}] \rightarrow \text{HOCH}_2\text{CH}_2\text{CH}_2\text{OH} + 2\text{H}_2\text{O}$	2

8. M/J 25/P22/Q4(a)

4(a)(i)	LiAlH_4	1
4(a)(ii)	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{H}_2\text{O}$	1

9. M/J 25/P23/Q5(b)(ii)-(iv)

5(b)(ii)		1
5(b)(iii)	reducing agent	1
5(b)(iv)	methanol / CH_3OH AND conc(entrated) (sulfuric) acid (catalyst)	1

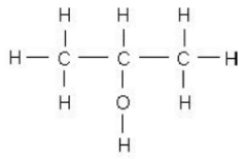
10. M/J 25/P24/Q4(b)

4(b)	propyl ethanoate	1
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11. M/J 25/P24/Q4(c)

4(c)(i)	condensation	1
4(c)(ii)	$\text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3 + \text{H}_2\text{O}$	1
4(c)(iii)	propan-1-ol	1

12. M/J 25/P24/Q4(d)(ii)

4(d)(ii)		2
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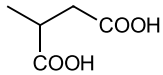
13. M/J 24/P23/Q5(d)(iii)-(iv)

5(d)(iii)	$\text{C}_8\text{H}_{12}\text{O}_4 + 8[\text{H}] \rightarrow \text{C}_8\text{H}_{16}\text{O}_2 + 2\text{H}_2\text{O}$	1
5(d)(iv)	LiAlH_4	1

Answer for context part (a)

5(a)	cold dilute KMnO_4 OR cold dilute potassium manganate(VII)	1
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14. O/N 23/P21/Q2(c)

2(c)	CH ₃ OH	1
	 one COOH group AND correct carbon framework (5C's) (each C has the correct valency)	1
	both COOH groups	1

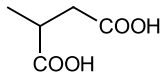
15. O/N 23/P22/Q3(c)(iii)

3(c)(iii)	condensation	1
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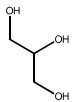
16. O/N 23/P22/Q3(d)(i)

3(d)(i)	$\text{H}_3\text{PO}_4 + 3\text{CH}_3\text{OH} \rightarrow (\text{CH}_3\text{O})_3\text{PO} + 3\text{H}_2\text{O}$	1
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17. O/N 23/P23/Q2(c)

2(c)	CH ₃ OH	1
	 one COOH group AND correct carbon framework (5C's) (each C has the correct valency)	1
	both COOH groups	1

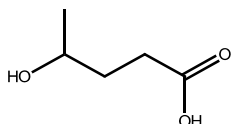
18. M/J 23/P21/Q5(c)

5(c)(i)	<i>suggest suitable reagent and conditions for step 1</i> heat AND any dilute named mineral acid e.g. dilute sulfuric acid / dilute hydrochloric acid	1
5(c)(ii)	 OR CH ₂ OHCH(OH)CH ₂ OH	1
5(c)(iii)	condensation	1
5(c)(iv)	methanol	1
5(c)(v)	methyl decanoate	1

19. M/J 23/P22/Q3(a)(iv)

3(a)(iv)	$2\text{CH}_3\text{CH}_2\text{COOH} + \text{Ca} \rightarrow (\text{CH}_3\text{CH}_2\text{COO})_2\text{Ca} + \text{H}_2$ M1 H ₂ as product M2 balanced equation based on correct formula of Y	2
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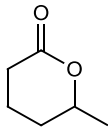
20. M/J 23/P22/Q5(d)(i)-(iii)

5(d)(i)	ester	1
5(d)(ii)	condensation	1
5(d)(iii)		1

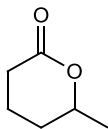
21. M/J 23/P23/Q3(c)(i)

3(c)(i)	$\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$	1
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22. O/N 22/P21/Q5(a)(iv)

5(a)(iv)		1
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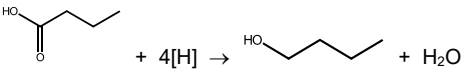
23. O/N 22/P23/Q5(a)(iv)

5(a)(iv)		1
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24. M/J 22/P21/Q4(b)(i)

4(b)(i)	M1 (add) group 1 carbonate / group 1 bicarbonate / Na_2CO_3 / NaHCO_3 etc. M2 effervescence / fizzing / bubbling	2
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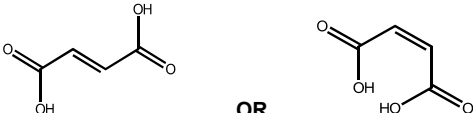
25. M/J 22/P22/Q5(c)(iii)-(iv)

5(c)(iii)	$\text{HOOC}(\text{CH}_2)_2\text{CH}_3 + 4[\text{H}] \rightarrow \text{HO}(\text{CH}_2)_3\text{CH}_3 + \text{H}_2\text{O}$ OR 	1
5(c)(iv)	LiAlH_4 OR lithium tetrahydridoaluminate (in ether)	1

26. M/J 22/P22/Q6(a)

6(a)	$\text{Br}_2(\text{aq})$	orange to colourless OR orange disappears	2
	$\text{Na}_2\text{CO}_3(\text{s})$	fizzing OR bubbles OR effervescence	

27. M/J 22/P22/Q6(c)(iii)

6(c)(iii)	 OR	1
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Answer for context part (b)

6(b)	<table><tr><td>C</td><td>:</td><td>H</td><td>:</td><td>O</td></tr><tr><td>41.38 / 12</td><td></td><td>3.45 / 1</td><td></td><td>55.17 / 16</td></tr><tr><td>3.45</td><td></td><td>3.45</td><td></td><td>3.45</td></tr></table> (so C ₍₁₎ H ₍₁₎ O ₍₁₎)	C	:	H	:	O	41.38 / 12		3.45 / 1		55.17 / 16	3.45		3.45		3.45	1
C	:	H	:	O													
41.38 / 12		3.45 / 1		55.17 / 16													
3.45		3.45		3.45													

28. MAR 22/P22/Q4(b)(i)

4(b)(i)	L = CH_3OH / methanol [1] conditions = acid(ic) / H^+ / H_2SO_4 AND (heat under) reflux [1]	2
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Unit 19

NITROGEN COMPOUNDS

17 items · 22 marks

Syllabus sections in this unit

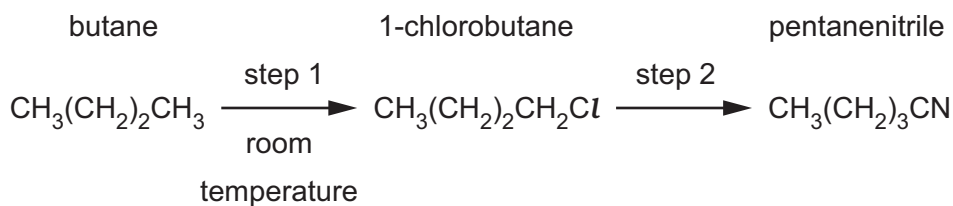
19.1 Primary amines

19.2 Nitriles and hydroxynitriles

Unit 19: Nitrogen compounds**1. M/J 26/P24/Q4(d)(ii)**

Context: part (d) (introduction)

(d) Butane is converted to pentanenitrile via a two-step synthesis, as shown in Figure 4.3.

**Figure 4.3**

(ii) Pentanenitrile reacts with dilute acid.

Draw the displayed formula of the organic product made in this reaction.

[1]

2. MAR 26/P22/Q4(a)(iv)

Context

4 Cyanoethanoic acid, $\text{CH}_2(\text{CN})\text{COOH}$, is a weak organic acid.

Context: part (a) (introduction)

(a) At room temperature, $\text{CH}_2(\text{CN})\text{COOH}$ undergoes an acid–base reaction with $\text{NaOH}(\text{aq})$.

(iv) Suggest the formula of the organic product when $\text{CH}_2(\text{CN})\text{COOH}(\text{aq})$ is heated under reflux with $\text{NaOH}(\text{aq})$.

[1]

3. O/N 25/P24/Q4(b)(ii)

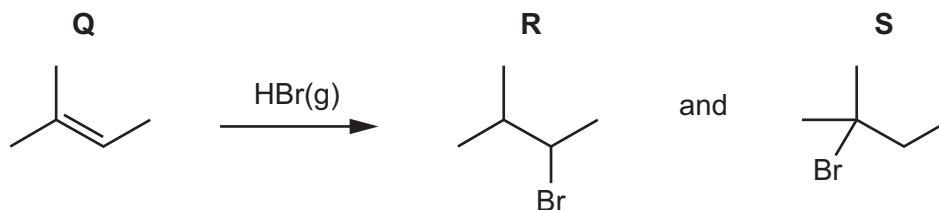
Context

- 4** Compounds **P** and **Q** are structural isomers.

**Fig. 4.1**

Context: part (a)(v)

- (v)** **Q** reacts with HBr(g) to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.

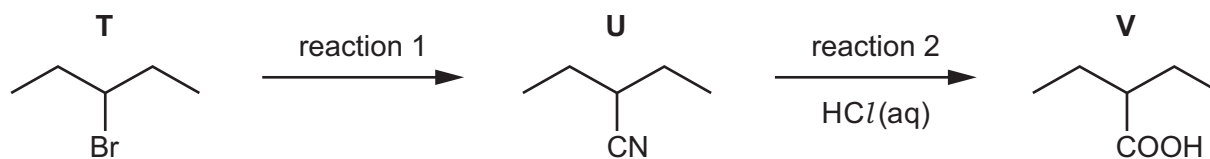
**Fig. 4.2**

State and explain why isomer **S** is the major product of the reaction.

[2]

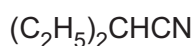
Context: part (b) (introduction)

- (b)** Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.

**Fig. 4.3**

- (ii)** Reaction 2 is a hydrolysis reaction.

Construct an equation for reaction 2.



[1]

4. M/J 25/P22/Q4(b)(iii)-(iv)

Context: part (b) (introduction)

- (b)** Propan-1-ol is converted to compound **T** in a three-step synthesis.

In step 1, SOCl_2 is added to propan-1-ol to produce compound **E**.

In step 2, **E** reacts with a suitable reagent to produce butanenitrile.

In step 3, butanenitrile is heated with NaOH(aq) .

- (iii)** Construct an equation to describe the reaction of butanenitrile with NaOH(aq) in step 3.

[2]

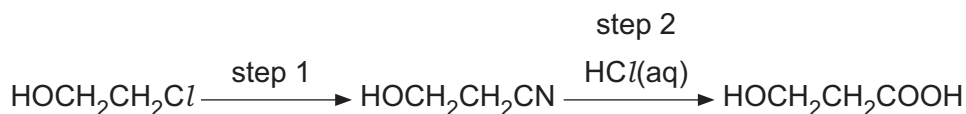
- (iv)** Name the type of reaction that occurs in step 3.

[1]

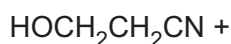
5. M/J 25/P23/Q6(c)(ii)-(iii)

Context: part (c) (introduction)

- (c) $\text{HOCH}_2\text{CH}_2\text{Cl}$ reacts in a two-step synthesis to produce $\text{HOCH}_2\text{CH}_2\text{COOH}$, as shown in Fig. 6.2.

**Fig. 6.2**

- (ii) Identify the type of reaction that occurs in step 2.
- (iii) Complete the equation to show the reaction in step 2.



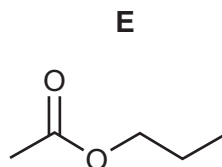
[1]

[1]

6. M/J 25/P24/Q4(d)(i)

Context

- 4 The skeletal formula of **E** is shown in Fig. 4.1.

**Fig. 4.1**

Context: part (d) (introduction)

- (d) **F** is an isomer of **E**.

F reacts with HCl(aq) to produce **G** and **H**.

G is a secondary alcohol.

H is also produced when HCl(aq) is added to CH_3CN .

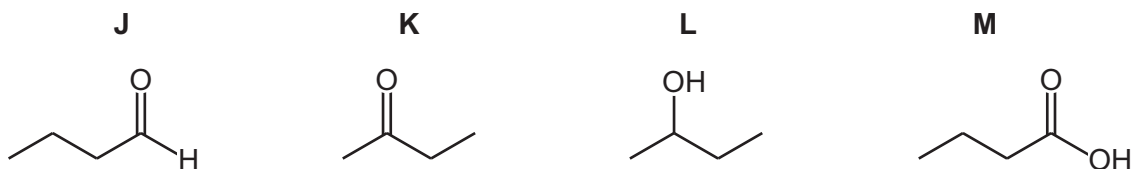
- (i) Construct an equation for the acid hydrolysis of CH_3CN with HCl(aq) .

[1]

7. MAR 25/P22/Q4(b)(ii)

Context

- 4 Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

**Fig. 4.1**

Context: part (b) (introduction)

- (b) **K** reacts with HCN in the presence of a KCN catalyst, forming **N**.

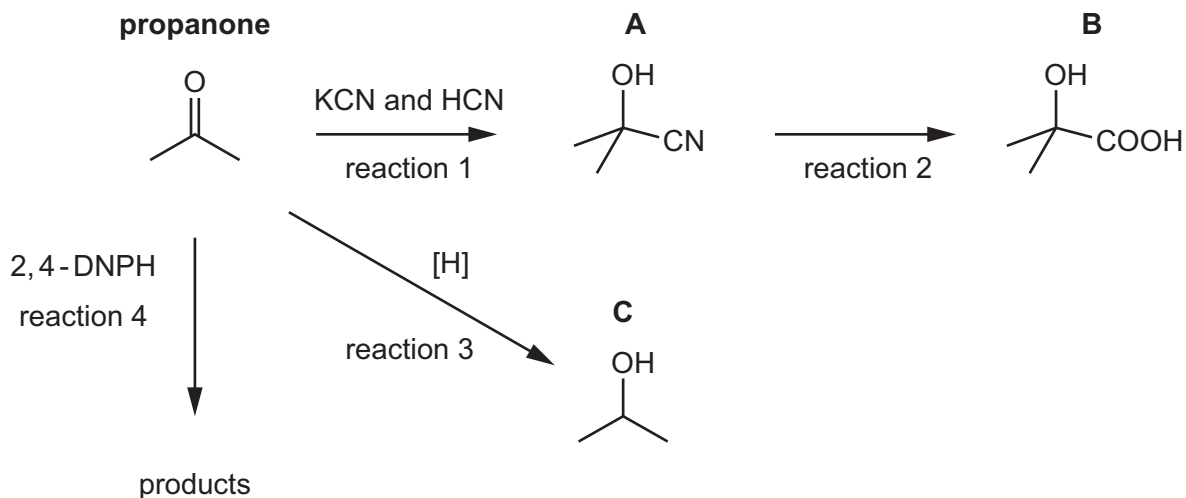
- (ii) Identify the class of compound of which **N** is a member.

[1]

8. O/N 24/P21/Q4(b)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

- (b)** Suggest the reagents and conditions for reaction 2.

[1]

9. O/N 24/P22/Q2(b)(ii)

Context

- 2** Oxygen is a Group 16 element.

Context: part (b) (introduction)

- (b)** H_2O reacts with both inorganic and organic compounds.

- (ii)** Write an equation for the reaction of CH_3CN with H_2O in acidic conditions.

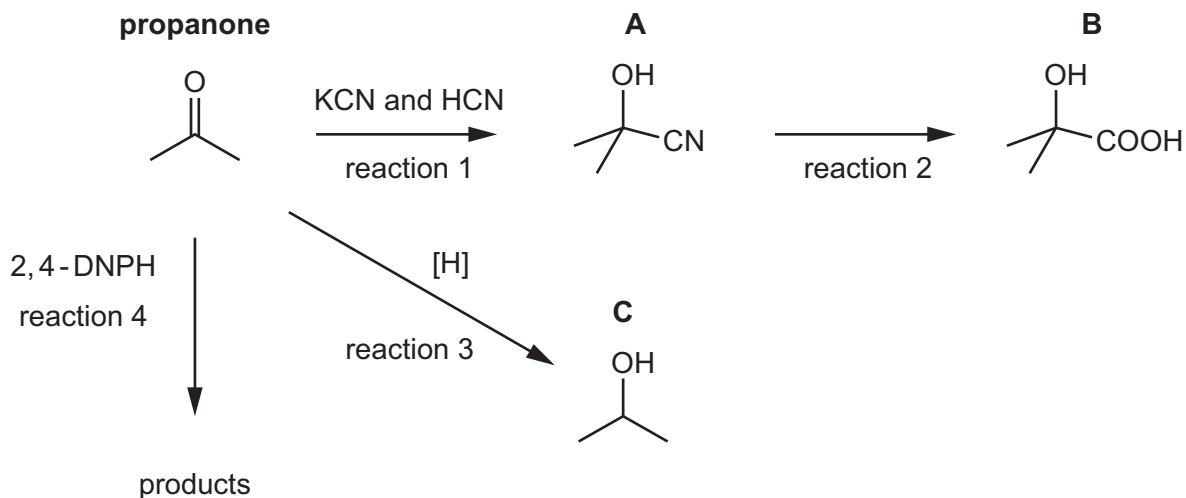


[1]

10. O/N 24/P23/Q4(b)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

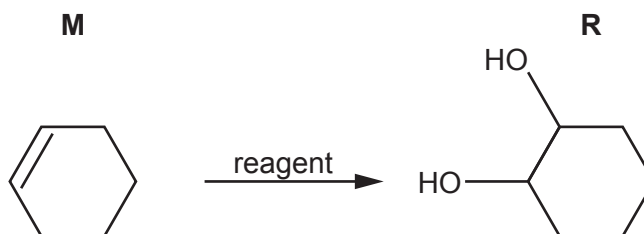
- (b)** Suggest the reagents and conditions for reaction 2.

[1]

11. M/J 24/P23/Q5(d)(i)-(ii)

Context: part (a)

- 5** **(a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

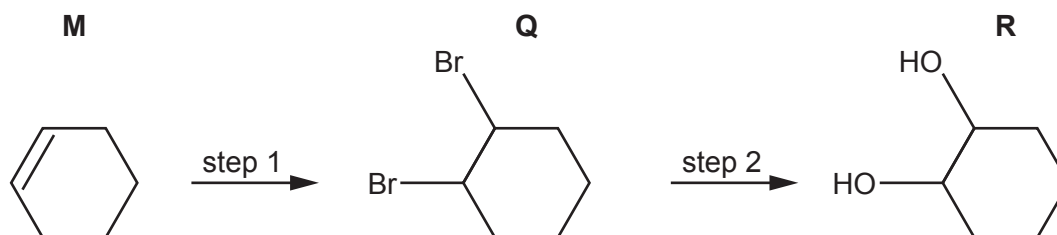
**Fig. 5.1**

Identify the reagent and conditions for this reaction.

[1]

Context: part (b) (introduction)

- (b)** **R** is also made from **M** by two steps, as shown in Fig. 5.2.

**Fig. 5.2**

Context: part (d) (introduction)

- (d)** **Y** is made from **Q** in a three-step reaction.

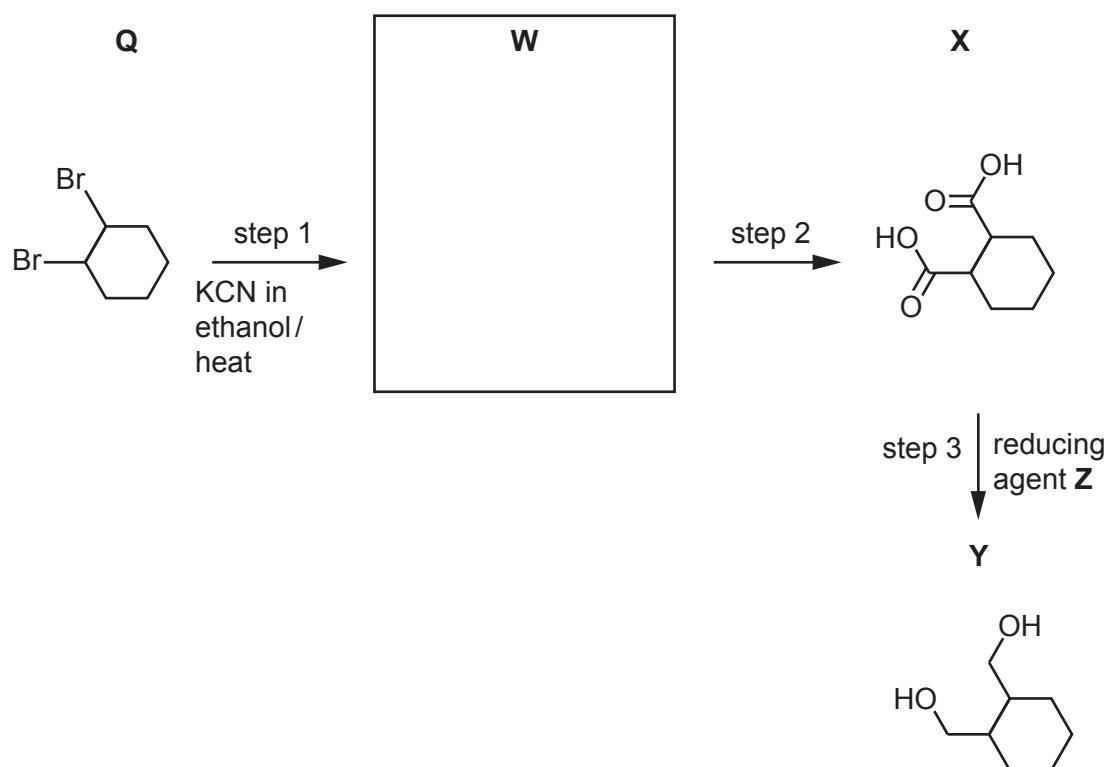


Fig. 5.4

- (i) Draw the structure of **W** in the box in Fig. 5.4. [1]
- (ii) In step 2, **W** is heated with HCl(aq) to produce **X** and an inorganic product. Identify the formula of the inorganic product. [1]

12. MAR 24/P22/Q4(b)(iii)

Context

- 4 Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

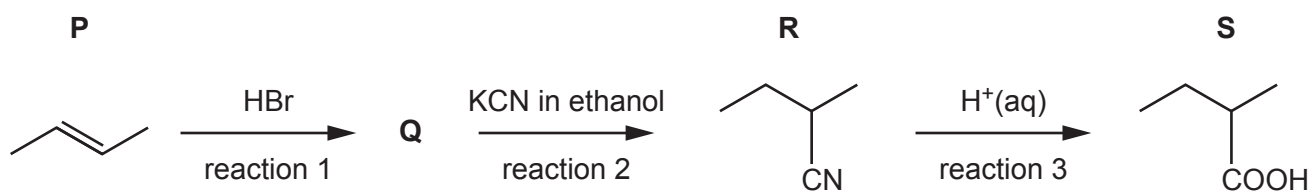
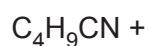


Fig. 4.1

- (iii) Complete the equation for reaction 3. **R** is represented as $\text{C}_4\text{H}_9\text{CN}$.

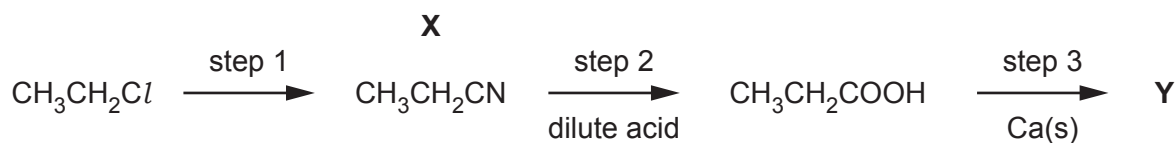


[1]

13. M/J 23/P22/Q3(a)(iii)

Context

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

- (iii) Identify the type of reaction that occurs when dilute acid is added to **X** in step 2.

[1]

14. M/J 23/P23/Q6(e)

Context

- 6** Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

- (e) Propanenitrile is heated with hydrogen gas and a platinum catalyst. The only product is propylamine.

Construct an equation for this reaction.

[1]

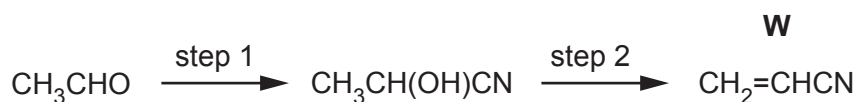
15. M/J 23/P23/Q6(f)(ii)

Context

- 6** Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

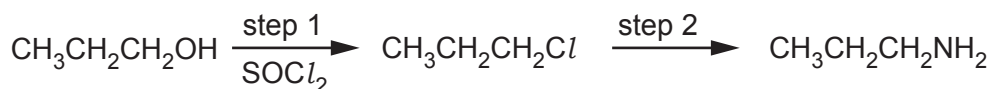
Context: part (b) (introduction)

- (b) CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN .

**Fig. 6.1**

Context: part (f) (introduction)

- (f) Propylamine can also be formed in a two-step synthesis from propan-1-ol, as shown in Fig. 6.4.

**Fig. 6.4**

- (ii) Identify the reagent and conditions for step 2 in Fig. 6.4.

[2]

16. MAR 23/P22/Q4(c)(ii)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

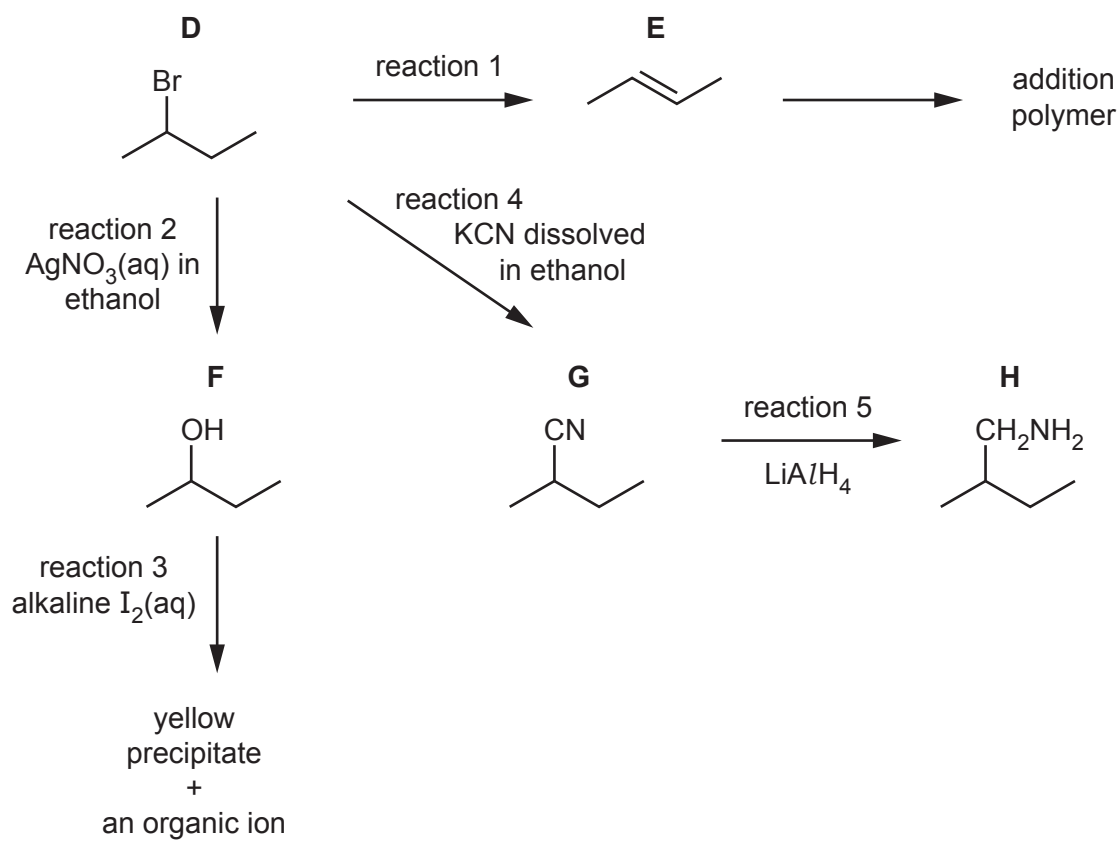


Fig. 4.1

(ii) Reaction 5 is similar to the reaction of LiAlH_4 with carboxylic acids to form alcohols.

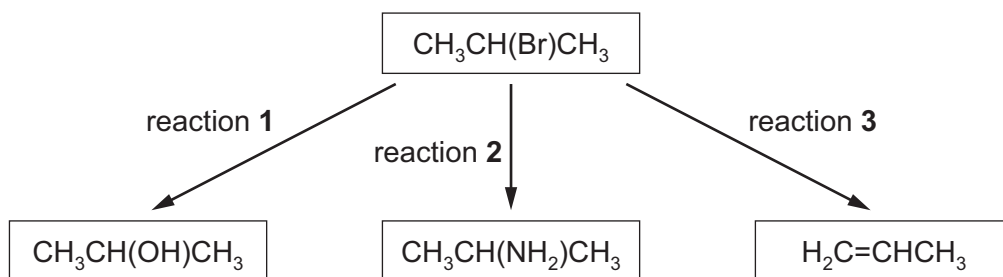
Suggest the role of LiAlH_4 in reaction 5.

[1]

17. M/J 22/P22/Q5(c)(ii)

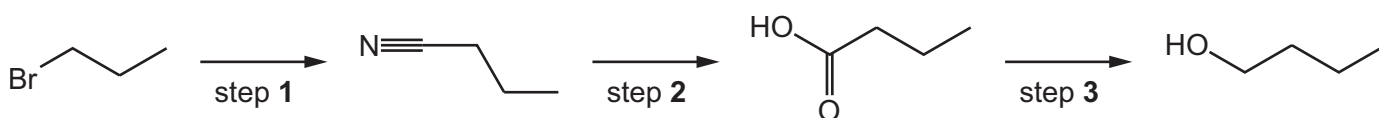
Context

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

**Fig. 5.1**

Context: part (c) (introduction)

(c) Fig. 5.2 shows how butan-1-ol can be made from 1-bromopropane in three steps.

**Fig. 5.2**

(ii) In step 2, butanenitrile is heated with $\text{HCl}(\text{aq})$. A hydrolysis reaction occurs.

Construct an equation for the reaction in step 2.

[1]

Answers Section

1. M/J 26/P24/Q4(d)(ii)

4(d)(ii)	$ \begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{H} & & \\ & & & & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{C} \\ & & & & & & & & & // \\ & \text{H} & & \text{H} & & \text{H} & & \text{H} & & \text{O} \\ & & & & & & & & & \\ & & & & & & & & & \text{O} - \text{H} \end{array} $	1	R-OH
----------	--	---	------

2. MAR 26/P22/Q4(a)(iv)

4(a)(iv)	CH ₂ (COO) ₂ Na ₂ OR CH ₂ (COONa) ₂	1
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3. O/N 25/P24/Q4(b)(ii)

4(b)(ii)	(C ₂ H ₅) ₂ CHCN + 2H ₂ O + HCl → (C ₂ H ₅) ₂ CHCOOH + NH ₄ Cl	1
----------	--	---

Answer for context part (a)(v)

4(a)(v)	M1 (reaction proceeds via a) more stable intermediate / carbocation M2 because of greater positive inductive effect of more alkyl groups	2
---------	---	---

4. M/J 25/P22/Q4(b)(iii)-(iv)

4(b)(iii)	CH ₃ CH ₂ CH ₂ CN + NaOH + H ₂ O → CH ₃ CH ₂ CH ₂ CO ₂ Na + NH ₃ OR CH ₃ CH ₂ CH ₂ CN + NaOH + 2H ₂ O → CH ₃ CH ₂ CH ₂ CO ₂ Na + NH ₄ OH M1 correct organic product (as molecule or anion) M2 rest of equation is correct	2
4(b)(iv)	(alkaline) hydrolysis	1

5. M/J 25/P23/Q6(c)(ii)-(iii)

6(c)(ii)	(acid) hydrolysis	1
6(c)(iii)	HOCH ₂ CH ₂ CN + 2H ₂ O + HCl → HOCH ₂ CH ₂ COOH + NH ₄ Cl	1

6. M/J 25/P24/Q4(d)(i)

4(d)(i)	CH ₃ CN + HCl + 2H ₂ O → CH ₃ COOH + NH ₄ Cl	1
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7. MAR 25/P22/Q4(b)(ii)

4(b)(ii)	hydroxynitrile	1
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8. O/N 24/P21/Q4(b)

4(b)	H ₂ SO ₄ (aq) or HCl(aq)	1
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9. O/N 24/P22/Q2(b)(ii)

2(b)(ii)	CH ₃ CN + 2H ₂ O + (1)H ⁺ → CH ₃ COOH + NH ₄ ⁺	1
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10. O/N 24/P23/Q4(b)

4(b)	H ₂ SO ₄ (aq) or HCl(aq)	1
------	--	---

11. M/J 24/P23/Q5(d)(i)-(ii)

5(d)(i)	W =	1
5(d)(ii)	NH_4Cl	1

Answer for context part (a)

5(a)	cold dilute KMnO_4 OR cold dilute potassium manganate(VII)	1
------	--	----------

12. MAR 24/P22/Q4(b)(iii)

4(b)(iii)	$\text{C}_4\text{H}_9\text{CN} + 2\text{H}_2\text{O} + \text{H}^+ \rightarrow \text{C}_4\text{H}_9\text{COOH} + \text{NH}_4^+$	1
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13. M/J 23/P22/Q3(a)(iii)

3(a)(iii)	(acid) hydrolysis	1
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14. M/J 23/P23/Q6(e)

6(e)	$\text{CH}_3\text{CH}_2\text{CN} + 2\text{H}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$	1
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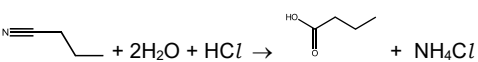
15. M/J 23/P23/Q6(f)(ii)

6(f)(ii)	M1 ammonia / NH_3 M2 heat + (M1 in ethanol) under pressure	2
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16. MAR 23/P22/Q4(c)(ii)

4(c)(ii)	reducing agent	1
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17. M/J 22/P22/Q5(c)(ii)

5(c)(ii)	$\text{CN}(\text{CH}_2)_2\text{CH}_3 + 2\text{H}_2\text{O} + \text{HCl} \rightarrow \text{HOOC}(\text{CH}_2)_2\text{CH}_3 + \text{NH}_4\text{Cl}$ OR 	1
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Unit 20

POLYMERISATION

14 items · 27 marks

Syllabus sections in this unit

20.1 Addition polymerisation

Unit 20: Polymerisation

1. M/J 26/P21/Q5(f)

Context

- 5 Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

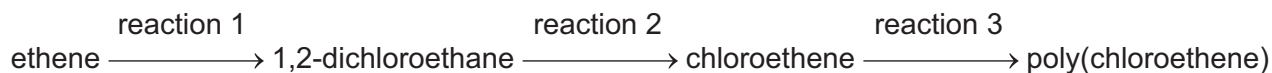


Figure 5.1

- (f) Draw the repeat unit of poly(chloroethene).

[1]

2. M/J 26/P24/Q5(b)(i)-(ii)

Context

- 5 Long-chain alkanes from crude oil can be cracked to make alkenes.

Context: part (b) (introduction)

- (b) Propene is an alkene that is made by cracking. Alkenes are used in addition polymerisation.

- (i) Draw **one** repeat unit of the addition polymer formed from propene.

[1]

- (ii) The disposal of poly(alkenes) by combustion produces harmful products.

State **one** other disadvantage associated with the disposal of poly(alkenes) such as poly(propene).

[1]

3. O/N 25/P21/Q4(b)(i)

Context

- 4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

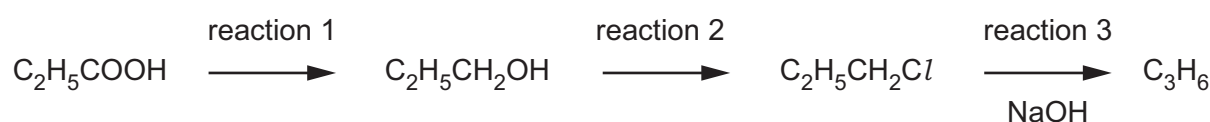


Fig. 4.1

Context: part (b) (introduction)

- (b) Under suitable conditions, propene polymerises to form poly(propene).

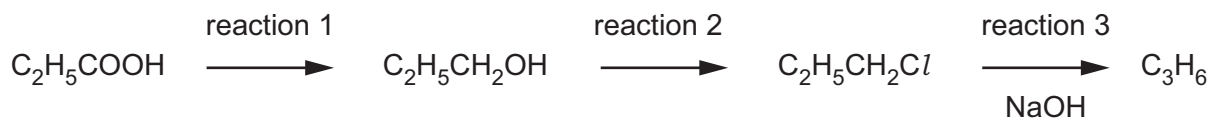
Poly(propene) exhibits stereoisomerism.

- (i) Identify the type of polymerisation that forms poly(propene) from propene.

[1]

4. O/N 25/P21/Q4(b)(iii)-(iv)

Context

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .**Fig. 4.1**

Context: part (b) (introduction)

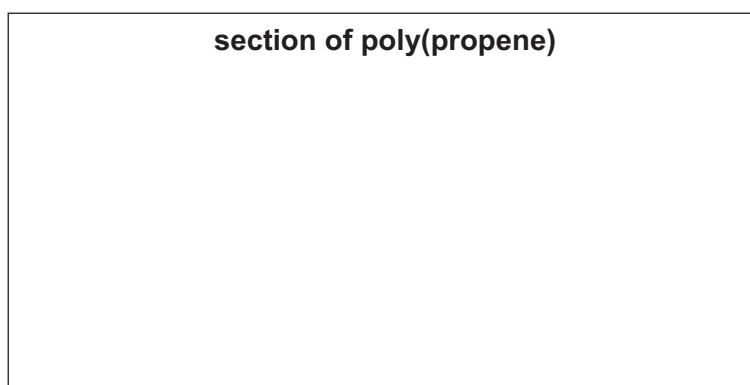
(b) Under suitable conditions, propene polymerises to form poly(propene).

Poly(propene) exhibits stereoisomerism.

(iii) Draw a section of poly(propene), showing **two** repeat units.

Use your diagram to identify the type of stereoisomerism shown by poly(propene).

Explain your answer.



[3]

(iv) State **two** difficulties associated with the disposal of poly(propene).

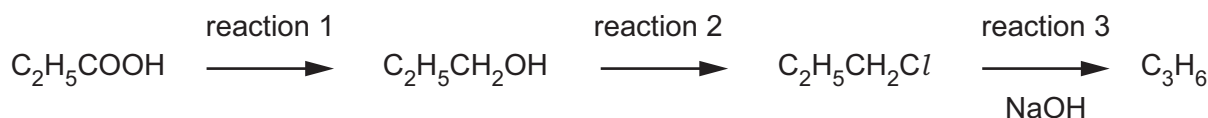
1

2

[2]

5. O/N 25/P23/Q4(b)(i)

Context

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .**Fig. 4.1**

Context: part (b) (introduction)

(b) Under suitable conditions, propene polymerises to form poly(propene).

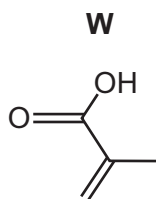
Poly(propene) exhibits stereoisomerism.

(i) Identify the type of polymerisation that forms poly(propene) from propene.

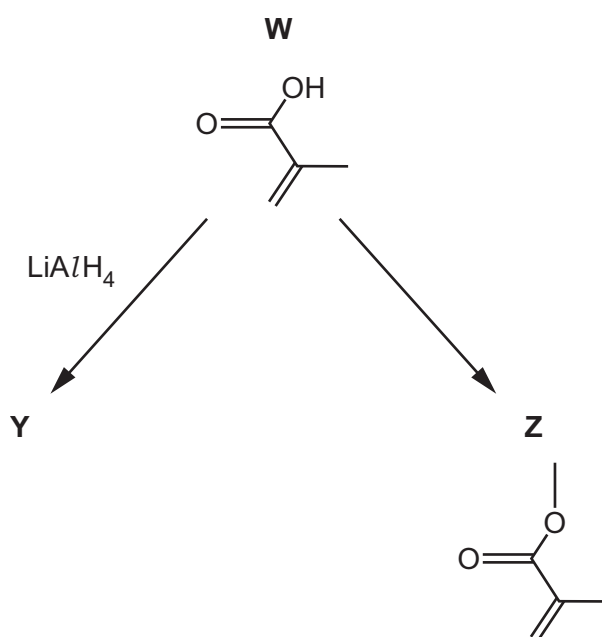
[1]

7. M/J 25/P23/Q5(b)(vi)

Context: part (a) (introduction)

5 (a) W is a colourless liquid.**Fig. 5.1**

Context: part (b) (introduction)

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.**Fig. 5.2****(vi)** **Z** is used to produce addition polymer **Q**.Draw the repeat unit of polymer **Q**.

[1]

8. M/J 25/P23/Q5(c)**(c)** Poly(ethene) is an addition polymer made from ethene.Explain why ethene reacts with electrophiles but poly(ethene) does **not**.

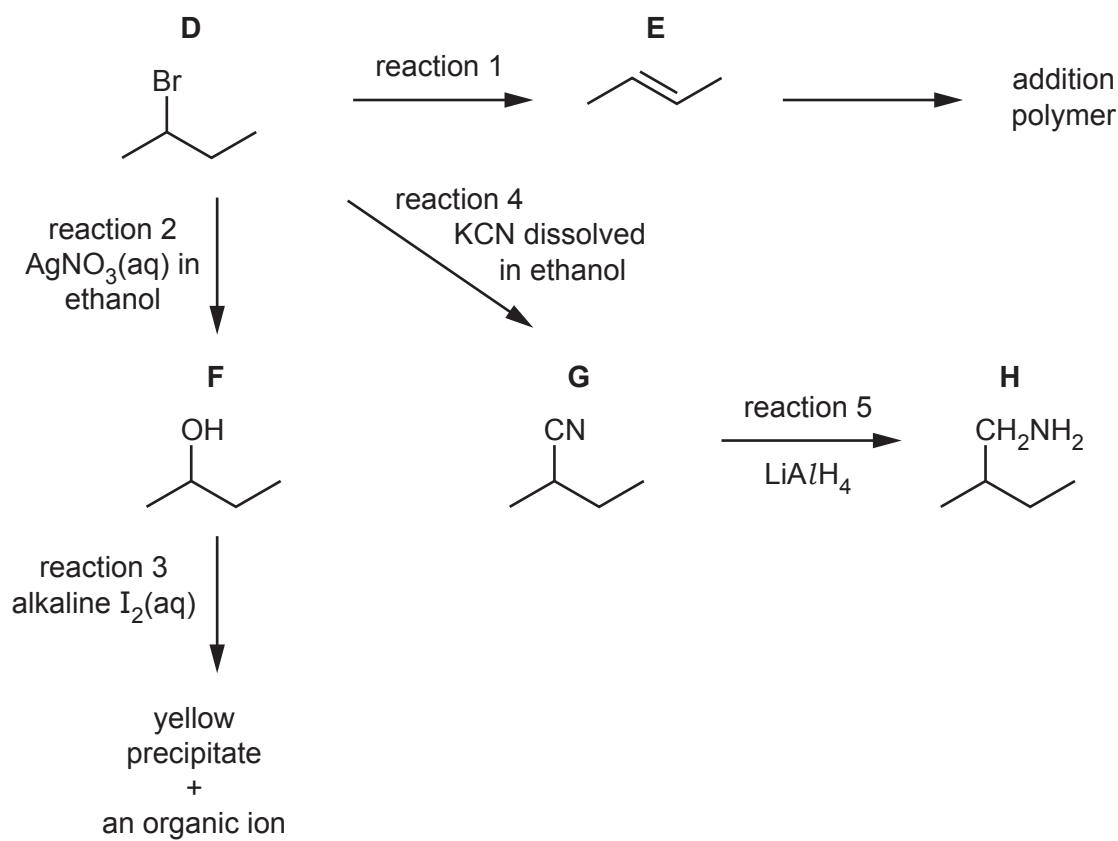
[1]

9. MAR 24/P22/Q2(d)(iii)

11. MAR 23/P22/Q4(a)(ii)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

**Fig. 4.1**

(ii) Draw the structure of **one** repeat unit of the addition polymer that forms from **E**.

[1]

12. O/N 22/P22/Q3(b)

Context

3 Organic compounds can be distinguished using chemical tests and analytical techniques.

(b) **D2** forms polymer **Z** when heated gently.

(i) Identify the type of polymer that forms from **D2**.

[1]

(ii) Draw one repeat unit of polymer **Z**.

[2]

13. M/J 22/P22/Q4(c)

(c) **X** is an addition polymer.

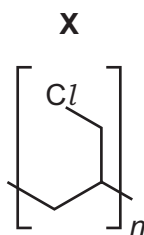


Fig. 4.3

(i) Draw the monomer of **X**.

[1]

(ii) Suggest **one** reason why the disposal of items made from **X** is difficult.

[1]

14. MAR 22/P22/Q4(b)(ii)

Context

4 Compounds **J** and **K** are found in plant oils.

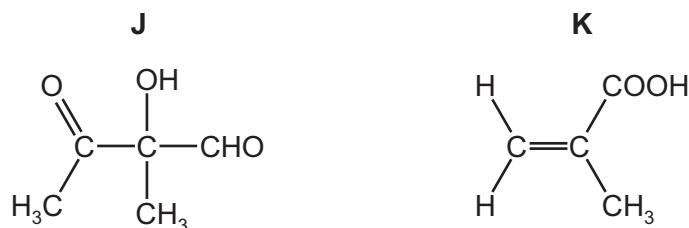


Fig. 4.1

Context: part (b) (introduction)

(b) **K** is used to make the addition polymer Perspex®. A synthesis of Perspex® is shown in Fig. 4.2.

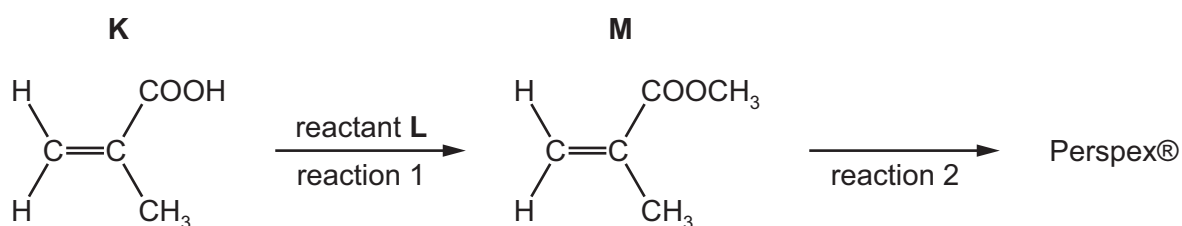


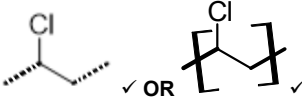
Fig. 4.2

(ii) Draw **one** repeat unit of the addition polymer Perspex®.

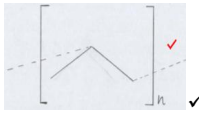
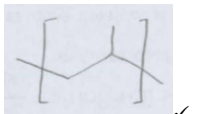
[2]

Answers Section

1. M/J 26/P21/Q5(f)

5(f)	$-\text{[CH}_2\text{-CHCl]}\text{-}$ OR <pre> H Cl - C - C - H H </pre> OR <pre> H H - C - C - Cl H </pre>	1 I brackets (any) I n A Cl on either C atom A dangling bonds / bracketed repeat unit If skeletal then dangling bonds are different (e.g. dashed, v. long) or go through brackets 
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2. M/J 26/P24/Q5(b)(i)-(ii)

5(b)(i)	<pre> CH3 H (C - C) H H n </pre>	1 I brackets I n A skeletal structure <u>with</u> hanging bonds / identifiable continuation bonds (i.e. dashed lines):  
5(b)(ii)	non-biodegradable	1 I produces harmful products / toxic gas I other environmental effects I disposal / cost A non-photodegradable A unreactive due to strong / non-polar bonds A phonetical spelling A BOD difficult / hard to decompose / does not decompose / does not break up / down (easily)

3. O/N 25/P21/Q4(b)(i)

4(b)(i)	addition	1
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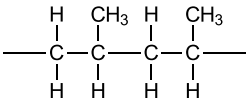
4. O/N 25/P21/Q4(b)(iii)-(iv)

4(b)(iii)	<pre> H CH3 H CH3 - C - C - C - C - H H H H </pre> M1 M2 optical isomerism M3 the methylated carbon has four different groups attached	3
4(b)(iv)	M1 non-biodegradability M2 harmful combustion products	2

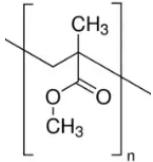
5. O/N 25/P23/Q4(b)(i)

4(b)(i)	addition	1
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6. O/N 25/P23/Q4(b)(iii)-(iv)

4(b)(iii)	 M1 M2 optical isomerism M3 the methylated carbon has four different groups attached	3
4(b)(iv)	M1 non-biodegradability M2 harmful combustion products	2

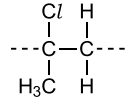
7. M/J 25/P23/Q5(b)(vi)

5(b)(vi)		1
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8. M/J 25/P23/Q5(c)

5(c)	ethene pi bond is a region of (high) electron density (above and below the C-C (sigma) bond)	1
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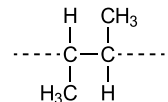
9. MAR 24/P22/Q2(d)(iii)

2(d)(iii)		1

10. M/J 23/P23/Q6(a)

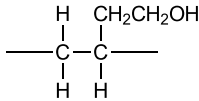
6(a)		1
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11. MAR 23/P22/Q4(a)(ii)

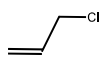
4(a)(ii)		1
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12. O/N 22/P22/Q3(b)

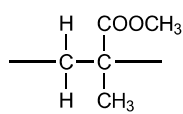
3(b)(i)	addition	1
---------	----------	----------

3(b)(ii)	 M1 correct carbon backbone of one repeat unit including 'dangling' bonds	1
	M2 rest of structure correct	1

13. M/J 22/P22/Q4(c)

4(c)(i)		1
4(c)(ii)	non-biodegradable OR poisonous / harmful <i>HCl</i> made when burnt	1

14. MAR 22/P22/Q4(b)(ii)

4(b)(ii)	 carbon backbone with 'dangling' bonds [1] rest of structure correct [1]	2
----------	--	----------

Unit 21

ORGANIC SYNTHESIS

10 items · 39 marks

Syllabus sections in this unit

21.1 Organic synthesis

Unit 21: Organic synthesis

1. M/J 26/P24/Q4(d)(i)

Context: part (d) (introduction)

(d) Butane is converted to pentanenitrile via a two-step synthesis, as shown in Figure 4.3.

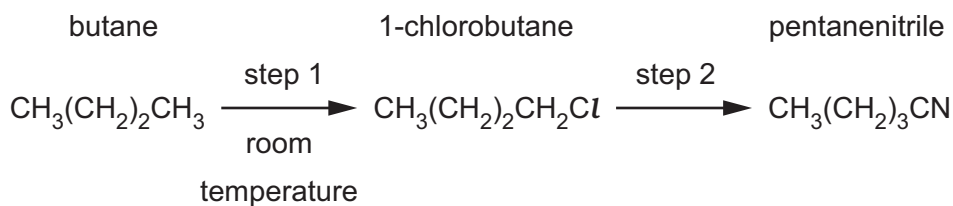


Figure 4.3

(i) Suggest the reagent and conditions for each step in Figure 4.3.

step 1

step 2

[2]

2. M/J 25/P21/Q5(d)(i)

Context

5 Compound **A** contains the elements carbon, hydrogen and oxygen only.

When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

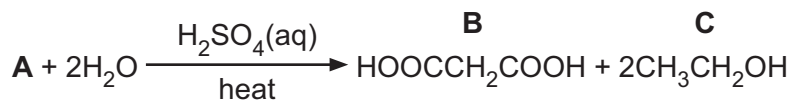


Fig. 5.1

Context: part (d) (introduction)

(d) A student suggests a two-step synthesis to produce **B**, as shown in Fig. 5.2.

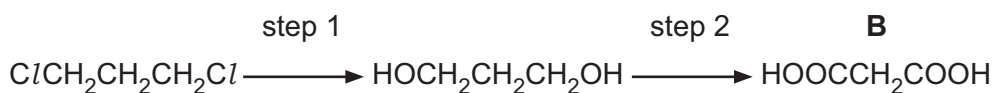


Fig. 5.2

(i) Identify the reagents and conditions required in steps 1 and 2.

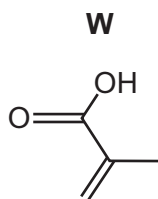
step 1

step 2

[2]

3. M/J 25/P23/Q5(a)(ii)

Context: part (a) (introduction)

5 (a) W is a colourless liquid.**Fig. 5.1**

(ii) Two different reagents are each added to separate samples of **W** as shown in Table 5.1.

Complete Table 5.1.

Table 5.1

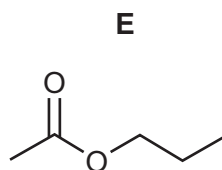
reagent and conditions	observation when reagent is added to W	structural formula of organic product when reagent is added to W
solid Na		
Br ₂ in absence of ultraviolet light		

[4]

4. M/J 25/P24/Q4(d)(iii)

Context

4 The skeletal formula of **E** is shown in Fig. 4.1.

**Fig. 4.1**

Context: part (d) (introduction)

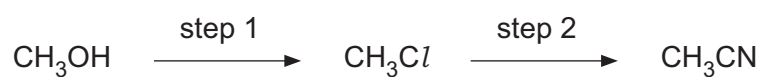
(d) **F** is an isomer of **E**.

F reacts with HCl(aq) to produce **G** and **H**.

G is a secondary alcohol.

H is also produced when HCl(aq) is added to CH_3CN .

(iii) CH_3OH is used to make CH_3CN in a two-step process, as shown in Fig. 4.2.

**Fig. 4.2**

State the reagents and conditions required for steps 1 and 2.

step 1

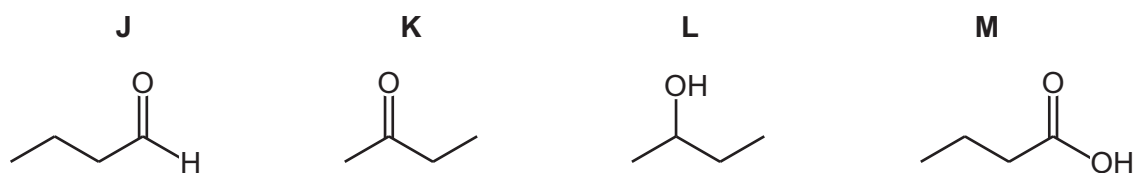
step 2

[3]

5. MAR 25/P22/Q4(a)

Context

4 Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

**Fig. 4.1**

(a) Table 4.1 gives details of tests on **J** to **M**. In each test, only **two** compounds give a positive result.

Complete Table 4.1.

Table 4.1

reagent	observation of positive result	compounds that give a positive result
acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$		J and L
alkaline $\text{I}_2(\text{aq})$	yellow precipitate	
	orange precipitate	J and K
$\text{Na}(\text{s})$		

[5]

6. M/J 24/P22/Q4(c)

Context

4 $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$ exists as a pair of stereoisomers.

(c) Separate samples of $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$, $\text{CH}_3(\text{CH}_2)_5\text{CH(OH)CH}_3$ and $\text{CH}_3(\text{CH}_2)_5\text{CHCH}_2$ are tested with different reagents.

Complete Table 4.1. If no reaction occurs, write $\times$ in the relevant box.

Table 4.1

reagent added	observation with $\text{CH}_3(\text{CH}_2)_5\text{CHBrCH}_3$	observation with $\text{CH}_3(\text{CH}_2)_5\text{CH(OH)CH}_3$	observation with $\text{CH}_3(\text{CH}_2)_5\text{CHCH}_2$
$\text{Br}_2(\text{l})$ in the dark			
$\text{PCl}_5(\text{s})$			
$\text{AgNO}_3(\text{aq})$			

[3]



7. O/N 23/P22/Q4(b)(iii)

Context

- 4 Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$, and pyruvic acid, $\text{CH}_3\text{COCO}\text{OH}$, both contain two functional groups.

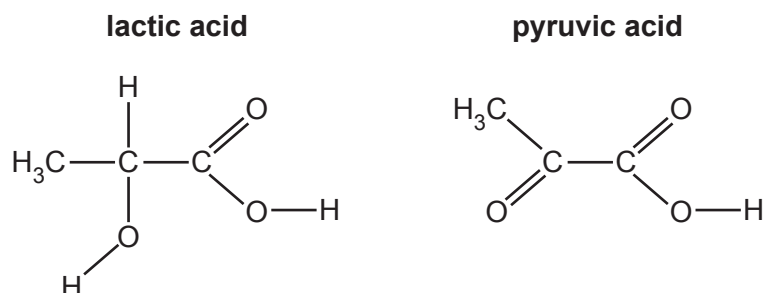


Fig. 4.1

Context: part (b) (introduction)

- (b) Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

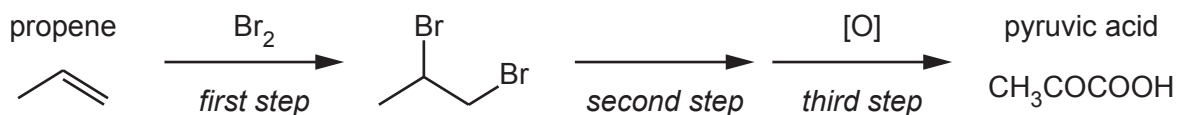


Fig. 4.3

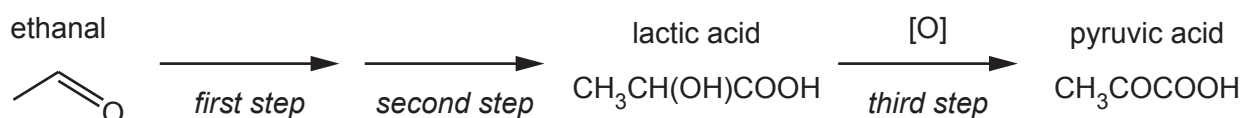


Fig. 4.4

- (iii) Complete Table 4.1 to give details of the reagents and conditions used in each of the two syntheses shown in Fig. 4.3 and Fig. 4.4.

Table 4.1

		synthesis from propene (shown in Fig. 4.3)	synthesis from ethanal (shown in Fig. 4.4)
reagents and conditions used	first step	Br_2	
	second step		
	third step		

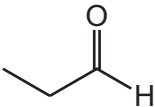
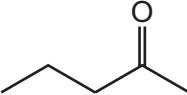
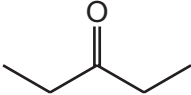
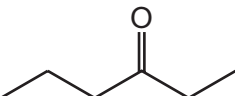
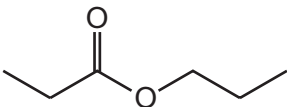
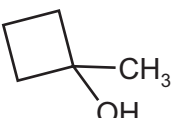
8. O/N 22/P22/Q3(a)(i)

Context

3 Organic compounds can be distinguished using chemical tests and analytical techniques.

Context: part (a) (introduction)

(a) Table 3.1 shows four pairs of organic compounds.**Table 3.1**

organic compounds		reagent	positive result of chemical test on identified compound
A1 	A2 		
B1 	B2 		
C1 	C2 		
D1 	D2 		

(i) Complete Table 3.1 to:

- identify a reagent which can distinguish between the compounds in each pair
- give the **positive** result of the chemical test **and** identify which compound shows this result.

Use a different reagent for each test.

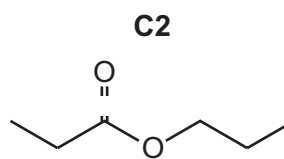
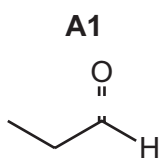
[8]

9. O/N 22/P22/Q3(d)

Context

3 Organic compounds can be distinguished using chemical tests and analytical techniques.

(d) **C2** can be synthesised using **A1** as a single organic reactant.

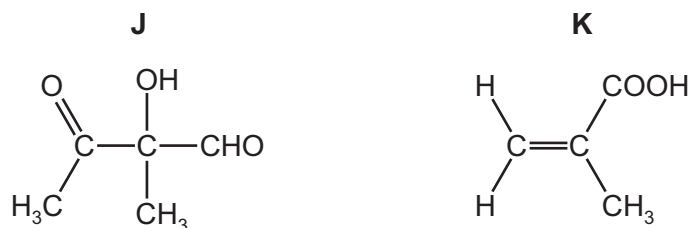


Devise a multi-step synthetic route to form **C2** from **A1**.
Identify relevant reagents and conditions, and state the organic products of each step.

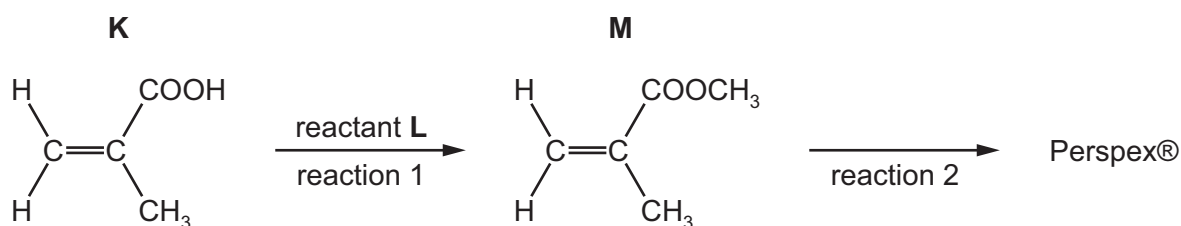
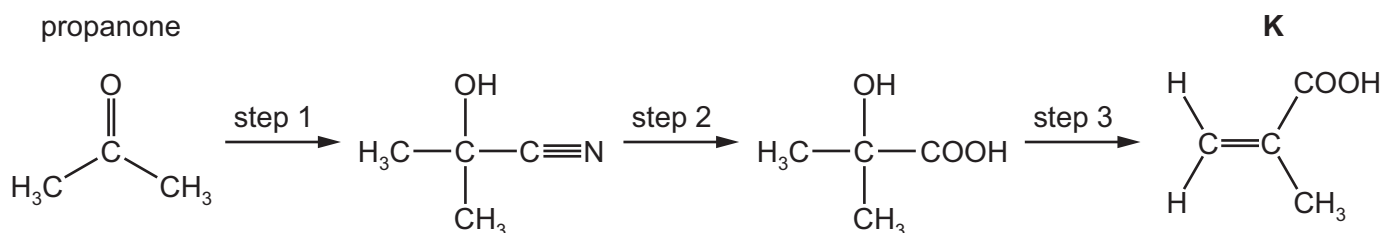
[3]

10. MAR 22/P22/Q4(b)(iv)

Context

4 Compounds **J** and **K** are found in plant oils.**Fig. 4.1**

Context: part (b) (introduction)

(b) **K** is used to make the addition polymer Perspex®. A synthesis of Perspex® is shown in Fig. 4.2.**Fig. 4.2****(iv)** **K** can be made from propanone in the three-step synthesis shown in Fig. 4.3.**Fig. 4.3**

Complete Table 4.3 to identify the reagent(s) used and the type of reaction in each step.

Table 4.3

step	reagent(s)	type of reaction
1		
2		
3	Al_2O_3	

Answers Section

1. M/J 26/P24/Q4(d)(i)

4(d)(i)	M1 (step 1) Cl_2 AND ultraviolet (light) / UV (light) M2 (step 2) KCN in ethanol AND heat (under reflux)	2	M2 A ethanolic KCN AND heat alcohol = (alc) A NaCN in ethanol AND heat R HCN R KCN in (aq) ethanol R ethanoic I warm high temp = heat = reflux = $> 60^\circ\text{C}$
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2. M/J 25/P21/Q5(d)(i)

5(d)(i)	M1 NaOH(aq) AND heat M2 acidified AND potassium dichromate(VI) AND heat under reflux	2
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3. M/J 25/P23/Q5(a)(ii)

5(a)(ii)	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 15%;">reagent</th><th style="width: 35%;">observation when reagent is added to W</th><th style="width: 30%;">structural formula of organic product when reagent is added to W</th></tr> </thead> <tbody> <tr> <td>Na</td><td>fizz / bubbles etc / OR solid / Na disappears / dissolves</td><td>$\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2\text{Na}$ OR $\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2^- \text{Na}^+$</td></tr> <tr> <td>$\text{Br}_2$ in the dark</td><td>brown to colourless / decolourise</td><td>$\text{CH}_2\text{BrCBr}(\text{CH}_3)\text{CO}_2\text{H}$</td></tr> </tbody> </table>	reagent	observation when reagent is added to W	structural formula of organic product when reagent is added to W	Na	fizz / bubbles etc / OR solid / Na disappears / dissolves	$\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2\text{Na}$ OR $\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2^- \text{Na}^+$	Br_2 in the dark	brown to colourless / decolourise	$\text{CH}_2\text{BrCBr}(\text{CH}_3)\text{CO}_2\text{H}$	4
reagent	observation when reagent is added to W	structural formula of organic product when reagent is added to W									
Na	fizz / bubbles etc / OR solid / Na disappears / dissolves	$\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2\text{Na}$ OR $\text{CH}_2\text{C}(\text{CH}_3)\text{CO}_2^- \text{Na}^+$									
Br_2 in the dark	brown to colourless / decolourise	$\text{CH}_2\text{BrCBr}(\text{CH}_3)\text{CO}_2\text{H}$									

4. M/J 25/P24/Q4(d)(iii)

4(d)(iii)	M1 step 1 PCl_3 and heat OR $\text{HCl}(\text{g})$ OR PCl_5 OR SOCl_2 OR NaCl / KCl and conc. H_2SO_4 / conc. H_3PO_4 M2 step 2 reagent KCN M3 conditions for step 2 dissolved in ethanol AND heat	3
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5. MAR 25/P22/Q4(a)

4(a)	<table border="1" style="width: 100%; border-collapse: collapse;"> <tbody> <tr> <td style="width: 30%;">acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$</td><td style="width: 30%;">(solution) turns (from orange to) green</td><td style="width: 40%; text-align: center;">J and L</td></tr> <tr> <td>alkaline $\text{I}_2(\text{aq})$</td><td>yellow precipitate forms</td><td style="text-align: center;">K and L</td></tr> <tr> <td>2,4-dinitrophenylhydrazine / 2,4-DNPH</td><td>orange precipitate forms</td><td style="text-align: center;">J and K</td></tr> <tr> <td>Na(s)</td><td>effervescence</td><td style="text-align: center;">L and M</td></tr> </tbody> </table>	acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$	(solution) turns (from orange to) green	J and L	alkaline $\text{I}_2(\text{aq})$	yellow precipitate forms	K and L	2,4-dinitrophenylhydrazine / 2,4-DNPH	orange precipitate forms	J and K	Na(s)	effervescence	L and M	5
acidified $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$	(solution) turns (from orange to) green	J and L												
alkaline $\text{I}_2(\text{aq})$	yellow precipitate forms	K and L												
2,4-dinitrophenylhydrazine / 2,4-DNPH	orange precipitate forms	J and K												
Na(s)	effervescence	L and M												

6. M/J 24/P22/Q4(c)

4(c)

Chemical test	observation with 2-bromooctane	observation with octan-2-ol	observation with $\text{CH}_3(\text{CH}_2)_5\text{CHCH}_2$
Br_2 in the dark	✗		orange / brown to colourless
PCl_5		steamy fumes	✗
$\text{AgNO}_3(\text{aq})$	cream precipitate OR off white precipitate	✗	

6 correct answers = 3 marks
4 or 5 correct answers = 2 marks
2 or 3 correct answers = 1 mark
0 or 1 correct answer = 0 marks

3

7. O/N 23/P22/Q4(b)(iii)

4(b)(iii)	$\text{HCN} \& \text{KCN}$ OR NaCN with H_2SO_4	1
	$\text{NaOH}(\text{aq})$ (1) dilute / aq(ueous) HCl / H_2SO_4 (1)	2
	acidified $\text{K}_2\text{Cr}_2\text{O}_7$	1

8. O/N 22/P22/Q3(a)(i)

3(a)(i)	M1 Tollens' OR Fehling's OR $\text{K}_2\text{Cr}_2\text{O}_7$ OR KMnO_4 OR alkaline $\text{I}_2(\text{aq})$ with	M2 A1 gives silver mirror OR brick-red ppt OR (orange solution) turns green OR (purple solution) turns colourless A2 gives yellow ppt	2
	M3 alkaline $\text{I}_2(\text{aq})$	M4 B1 gives yellow ppt	2
	M5 2,4-DNPH	M6 C1 gives red/orange/yellow ppt	2
	M7 $\text{Br}_2(\text{aq})$	M8 D2 turns it (orange to) colourless	2

9. O/N 22/P22/Q3(d)

3(d)	M1 one portion of A1 : NaBH_4 AND $\rightarrow$ propan-1-ol / $\text{CH}_3(\text{CH}_2)_2\text{OH}$	1
	M2 second portion of A1 : acidified $\text{K}_2\text{Cr}_2\text{O}_7$ AND $\rightarrow$ propanoic acid / $\text{C}_2\text{H}_5\text{CO}_2\text{H}$	1
	M3 two products: <u>conc</u> (entrated mineral) acid / H^+ / H_2SO_4 (catalyst)	1

10. MAR 22/P22/Q4(b)(iv)

4(b)(iv)	step 1 KCN / HCN OR $\text{NaCN} / \text{H}_2\text{SO}_4$ [1] step 2 $\text{H}^+ / \text{H}_2\text{SO}_4(\text{aq})$ [1] step 3	addition [1] hydrolysis / substitution [1] elimination / dehydration [1]	5
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Unit 22

ANALYTICAL TECHNIQUES

36 items · 92 marks

Syllabus sections in this unit

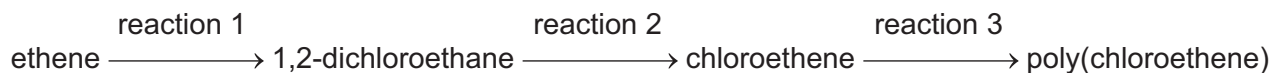
22.1 Infrared spectroscopy

22.2 Mass spectrometry

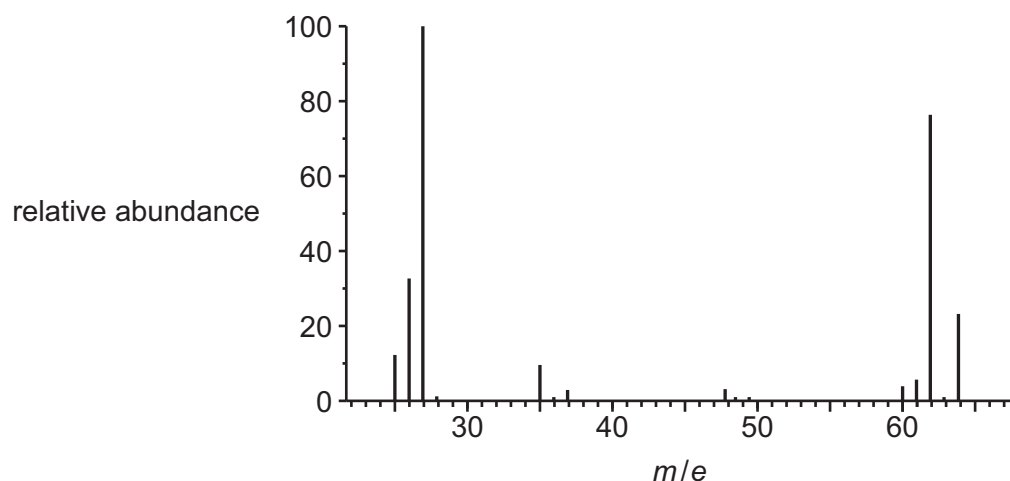
Unit 22: Analytical techniques**1. M/J 26/P21/Q5(g)**

Context

- 5** Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

**Figure 5.1**

- (g)** Figure 5.3 shows the mass spectrum of chloroethene.

**Figure 5.3**

- (i)** Identify the species responsible for the peaks at $m/e = 64$ and $m/e = 27$.

peak at $m/e = 64$

peak at $m/e = 27$

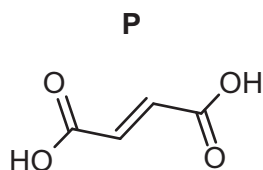
[2]

- (ii)** Explain the difference in relative abundance of the peaks at $m/e = 62$ and $m/e = 64$.

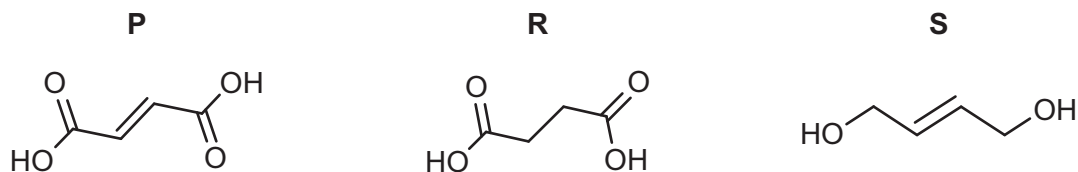
[2]

2. M/J 26/P22/Q6(c)(i)-(ii)

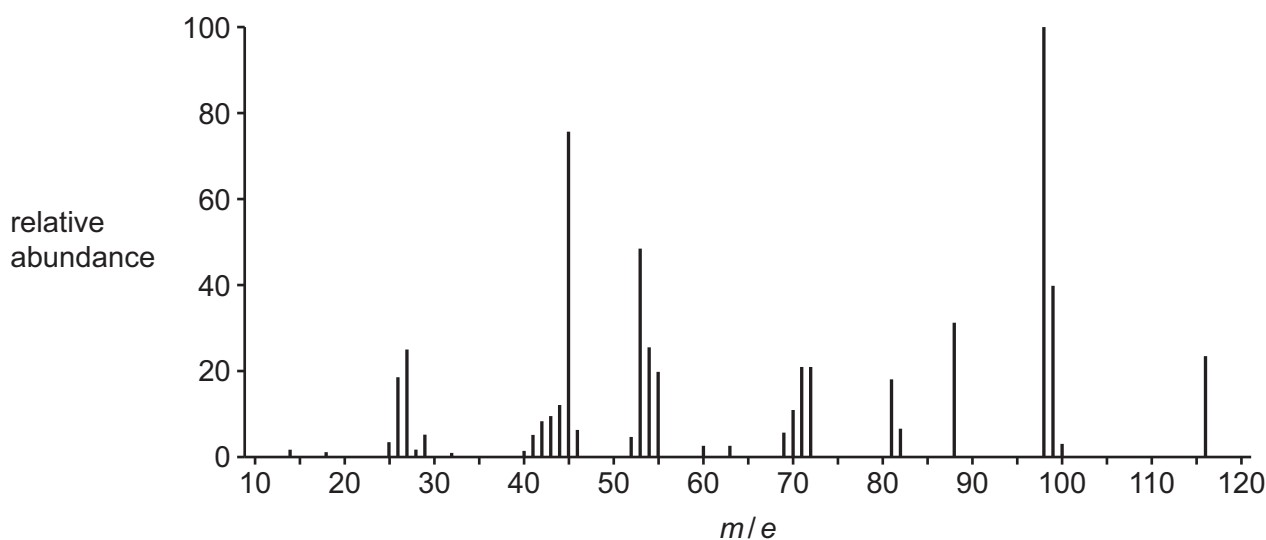
Context: part (a) (introduction)

6 (a) Figure 6.1 shows the structure of compound **P**.**Figure 6.1**

Context: part (c) (introduction)

(c) An unlabelled sample contains one of compounds **P**, **R** or **S**.**Figure 6.4**

The mass spectrum of the sample is shown in Figure 6.5.

**Figure 6.5****(i)** Identify the **one** peak on Figure 6.5 which confirms that the sample is **P**.

Explain your reasoning.

[2]

(ii) Suggest the structural formulae of the fragments responsible for the peaks at $m/e = 45$ and $m/e = 99$.structural formula of fragment at $m/e = 45$ structural formula of fragment at $m/e = 99$

[2]

3. M/J 26/P23/Q5(c)

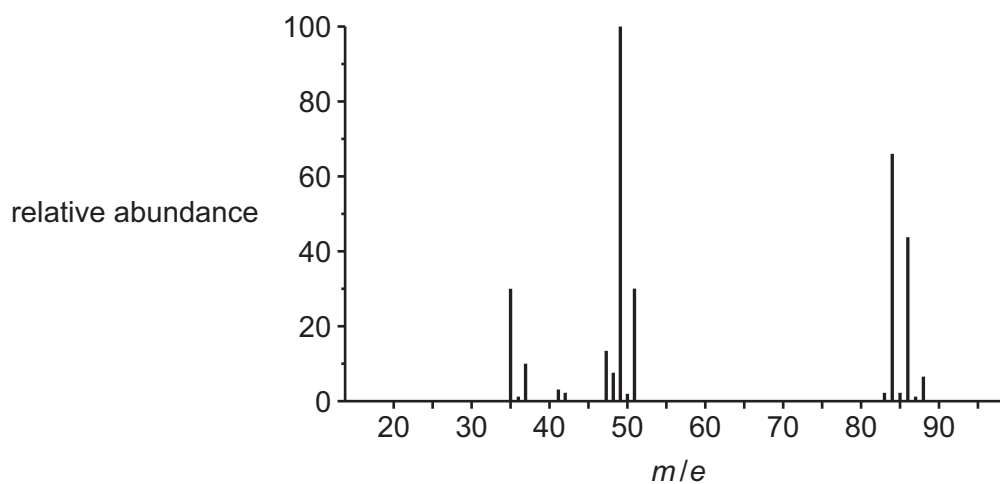
Context

- 5** Propane, $\text{CH}_3\text{CH}_2\text{CH}_3$, reacts with $\text{Cl}_2(\text{g})$ in the presence of ultraviolet light to produce different organic products, including those shown in Table 5.1.

Table 5.1

structure of product	systematic name of product
$\text{CH}_3(\text{CH}_2)_2\text{Cl}$	1-chloropropane
$\text{CH}_3\text{CHClCH}_3$	2-chloropropane
$(\text{CH}_3)_2\text{CHCH}(\text{CH}_3)_2$	2,3-dimethylbutane
	hexane
	2-methylpentane

- (c) Compound **A** has the molecular formula CH_2Cl_2 . Figure 5.2 shows the mass spectrum of **A**.

**Figure 5.2**

- (i) Explain the difference in relative abundance of the peaks at $m/e = 35$ and $m/e = 37$.

[2]

- (ii) Complete Table 5.2.

Table 5.2

m/e value of peak	molecular formula of species responsible for peak
84	
86	
88	

[2]

4. M/J 26/P24/Q4(c)

(c) Compound **W** is an aliphatic primary alcohol.

W is oxidised by acidified KMnO_4 when heated under reflux. Figure 4.2 shows the infrared spectrum of the resulting product.

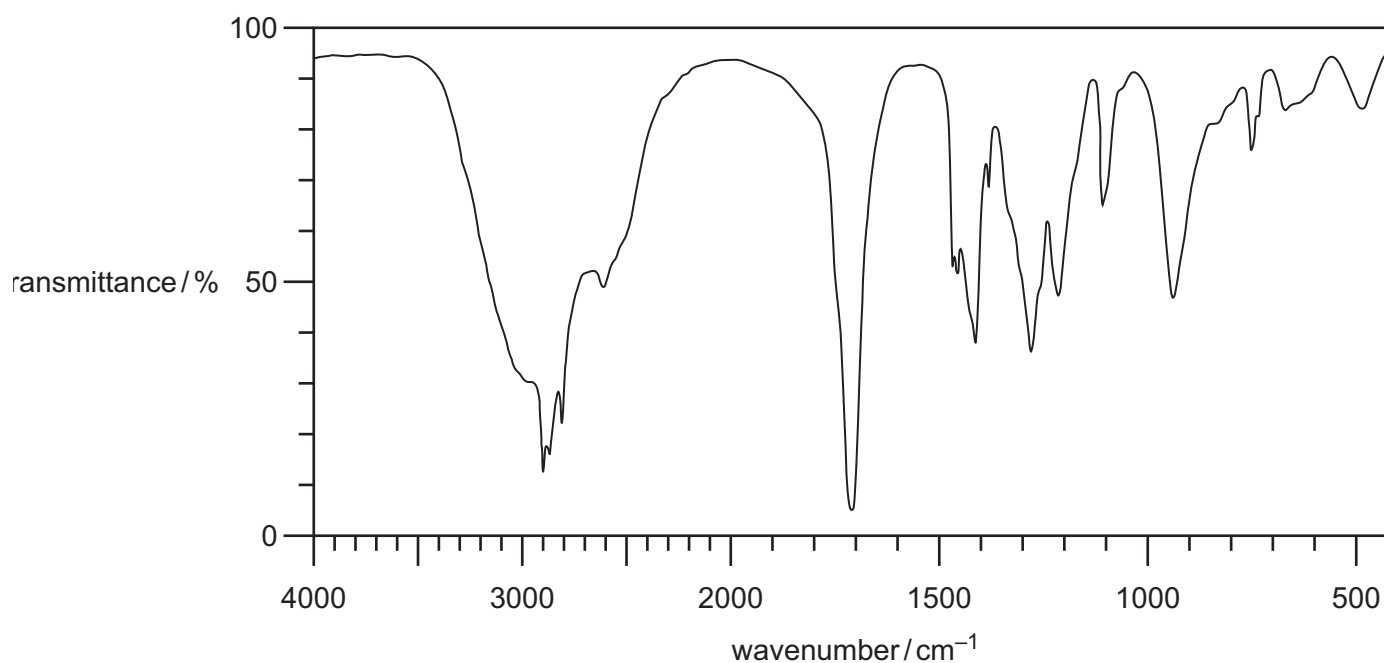


Figure 4.2

Table 4.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/ cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

The infrared spectrum in Figure 4.2 confirms that **W** is a primary alcohol.

Use Figure 4.2 to identify **two** absorbances that confirm **W** is a primary alcohol. State the bond and functional group containing the bond for each absorbance.

5. O/N 25/P21/Q3(e)(ii)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (e) (introduction)

- (e)** Cyclopentene, C_5H_8 , reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

- (ii)** The infrared spectrum of **W** is shown in Fig. 3.3.

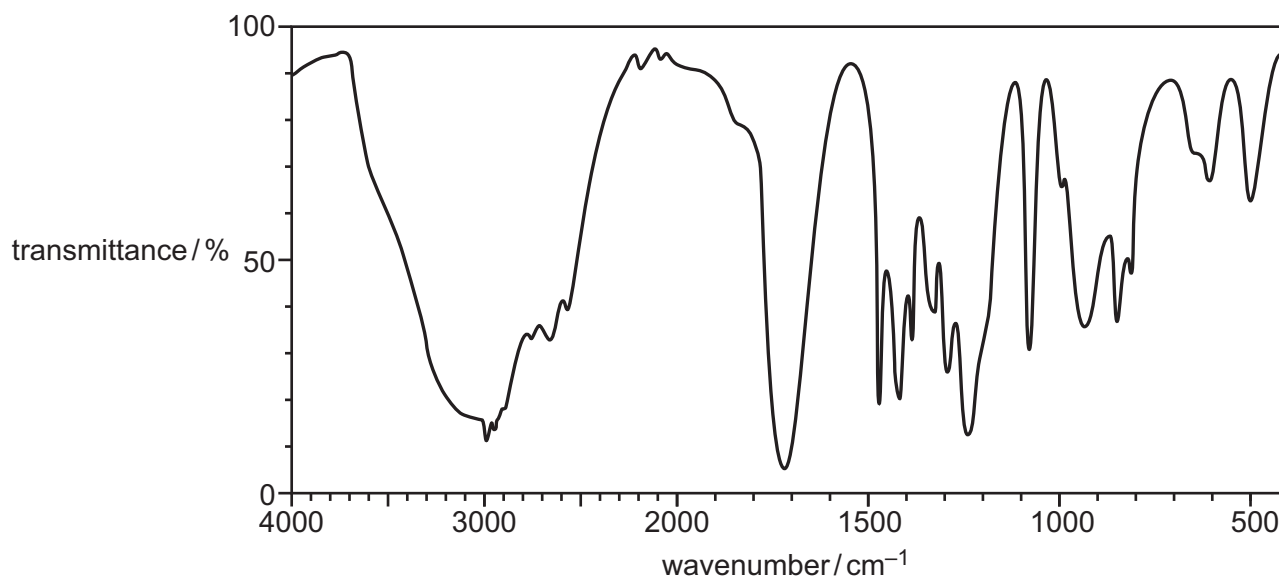


Fig. 3.3

Identify **two** absorptions in the infrared spectrum of **W** that would **not** be present in the infrared spectrum of cyclopentene.

- Write **1** or **2** on Fig. 3.3 against each of these two absorptions.
- Complete Table 3.2 to show which bond is responsible for each absorption that you have identified in Fig. 3.3.

Table 3.2

absorption	1	2
bond responsible		

Table 3.3

[2]

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

6. O/N 25/P21/Q4(c)(ii)-(iii)

Context

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

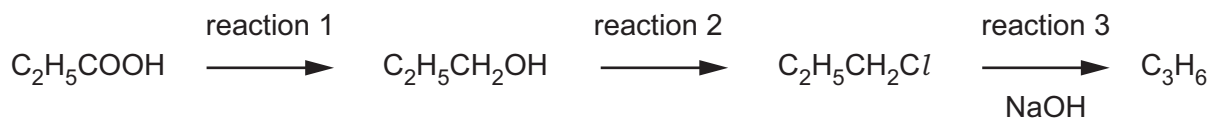


Fig. 4.1

Context: part (c) (introduction)

(c) Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.



(ii) Fig. 4.2 shows the mass spectrum of either compound **X** or compound **Y**.

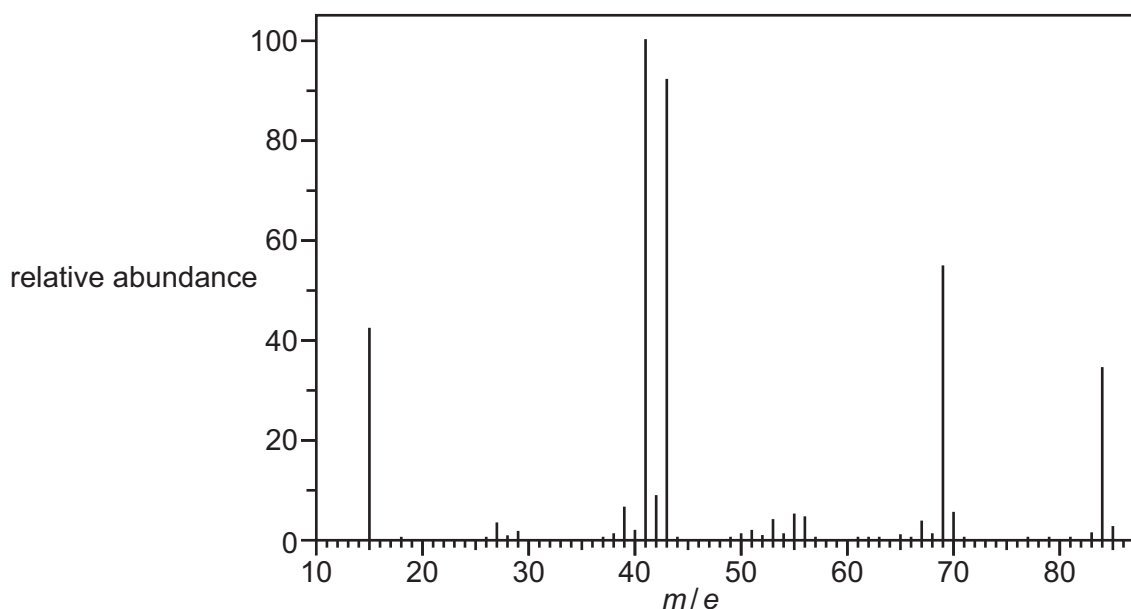


Fig. 4.2

Identify which of **X** and **Y** gives this mass spectrum.

Give **one** reason for your answer, referring to the fragmentation pattern.

[1]

(iii) The molecular ion peak in the spectrum in Fig. 4.2 has relative abundance 34.7.

Calculate the relative abundance of the $[M+1]^+$ peak in this spectrum.

relative abundance of $[M+1]^+$ peak =

[1]

7. O/N 25/P22/Q4(d)(ii)-(iii)

Context

4 Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, CH_3CH_2COOH .

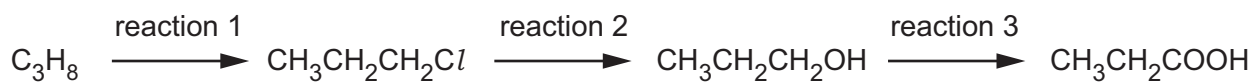


Fig. 4.1

Context: part (d) (introduction)

(d) $\text{CH}_3\text{CH}_2\text{COOH}$ reacts with an unsaturated alcohol **R** to form unsaturated ester **S**.

(ii) The infrared spectrum of **S** is shown in Fig. 4.2.

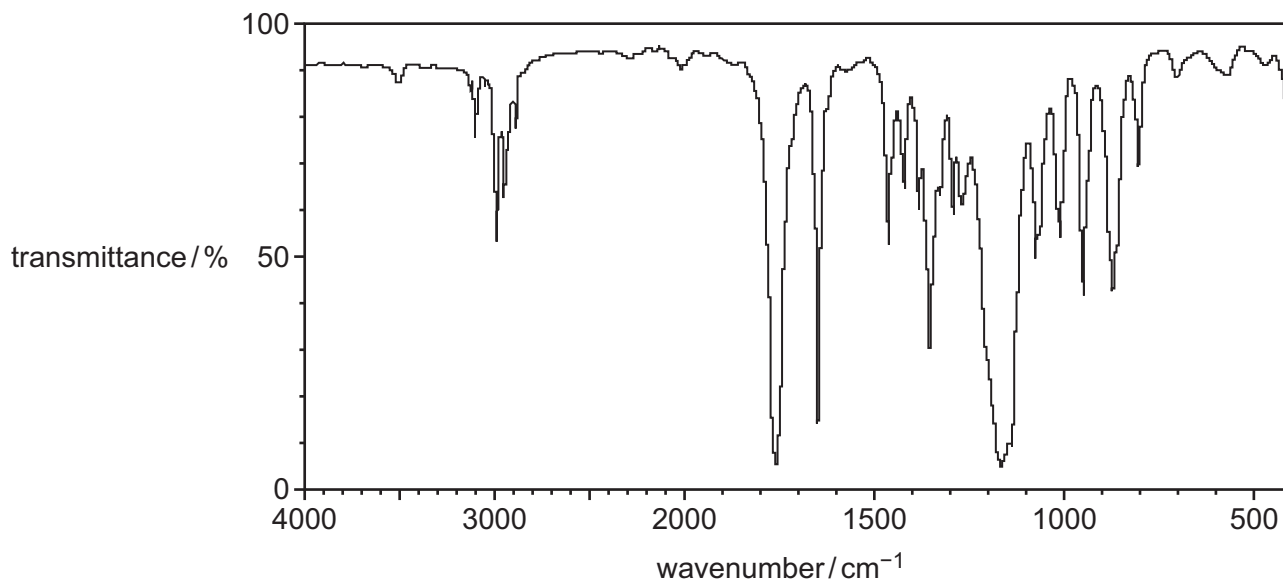


Fig. 4.2

Three absorptions in the infrared spectrum in Fig. 4.2 confirm that **S** is an ester and is unsaturated.

- Write **1**, **2** or **3** on Fig. 4.2 against each of these three absorptions.
- Complete Table 4.1 to show which bond is responsible for each absorption that you have identified in Fig. 4.2.

Table 4.1

absorption	1	2	3
bond responsible			

[3]

Table 4.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/ cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

(iii) The mass spectrum of **S** shows the following peaks.

Table 4.3

peak	relative abundance
M^+	4.7
$[M+1]^+$	0.31

Use Table 4.3 to calculate the number of carbon atoms in **S**.

number of carbon atoms in **S** =

[1]

8. O/N 25/P23/Q3(e)(ii)

Context

- 3** Cycloalkanes show similar chemical properties to alkanes but have the same empirical formula as alkenes.

Context: part (e) (introduction)

- (e) Cyclopentene, C_5H_8 , reacts with hot concentrated acidified $KMnO_4$ to form compound **W**, $C_5H_8O_4$.

- (ii) The infrared spectrum of **W** is shown in Fig. 3.3.

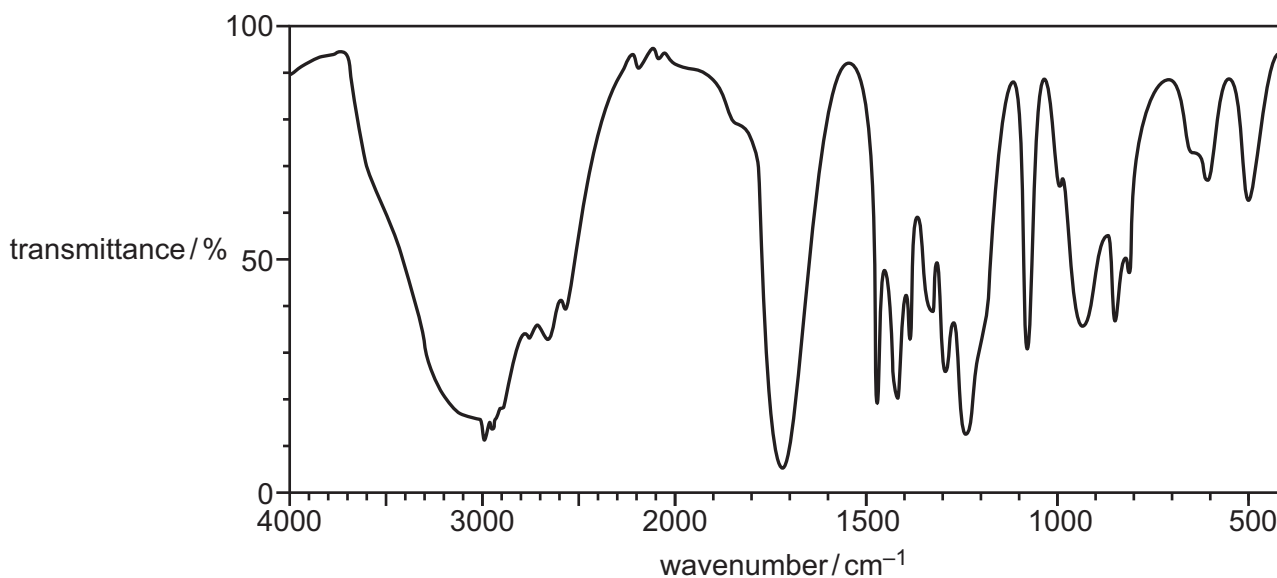


Fig. 3.3

Identify **two** absorptions in the infrared spectrum of **W** that would **not** be present in the infrared spectrum of cyclopentene.

- Write **1** or **2** on Fig. 3.3 against each of these two absorptions.
- Complete Table 3.2 to show which bond is responsible for each absorption that you have identified in Fig. 3.3.

Table 3.2

absorption	1	2
bond responsible		

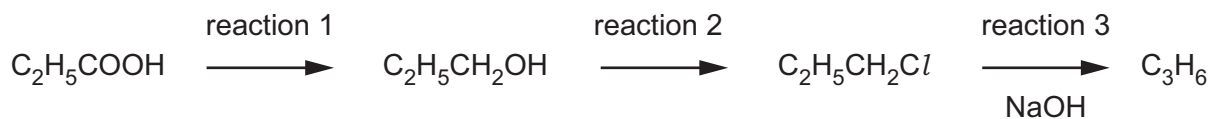
[2]

Table 3.3

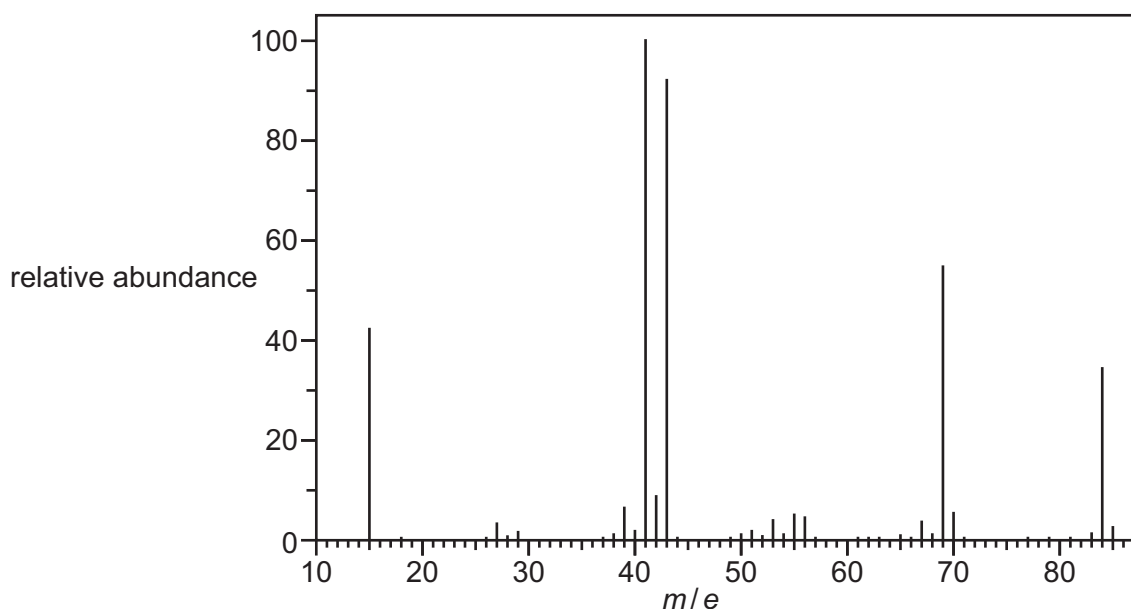
bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

9. O/N 25/P23/Q4(c)(ii)-(iii)

Context

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .**Fig. 4.1**

Context: part (c) (introduction)

(c) Under different conditions, two molecules of propene can combine to form compounds **X** and **Y**.**(ii)** Fig. 4.2 shows the mass spectrum of either compound **X** or compound **Y**.**Fig. 4.2**Identify which of **X** and **Y** gives this mass spectrum.Give **one** reason for your answer, referring to the fragmentation pattern.

[1]

(iii) The molecular ion peak in the spectrum in Fig. 4.2 has relative abundance 34.7.Calculate the relative abundance of the $[M+1]^+$ peak in this spectrum.relative abundance of $[M+1]^+$ peak =

[1]

10. M/J 25/P21/Q4(c)

Context

4 The structure of vitamin C is shown in Fig. 4.1.

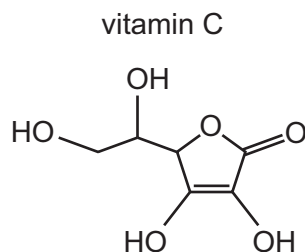


Fig. 4.1

Context: part (b) (introduction)

(b) The concentration of vitamin C is found by titration with $\text{I}_2(\text{aq})$.

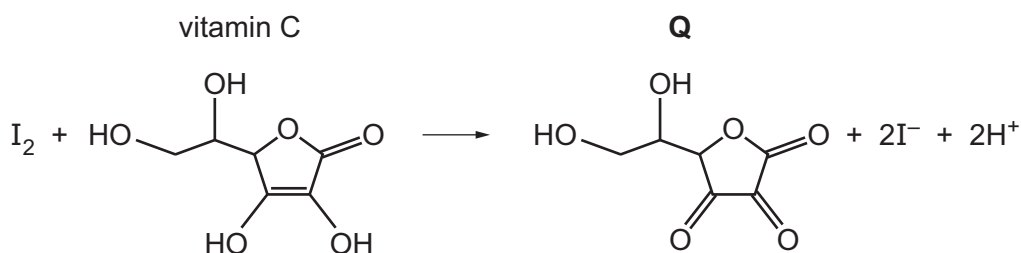


Fig. 4.2

A vitamin C tablet is dissolved in water to produce 200.0 cm³ of vitamin C solution. 5.00 cm³ of this vitamin C solution is added to a flask with approximately 150 cm³ of water and an indicator. Exactly 28.40 cm³ of 5.00 × 10⁻⁴ mol dm⁻³ I₂(aq) reacts with the sample of vitamin C solution in the flask.

[M_r : vitamin C, 176]

(c) Predict **two** absorptions that will be seen in the infrared spectra of both vitamin C and **Q**. Describe the relevant bond and the specific functional group that is responsible for each absorption identified.

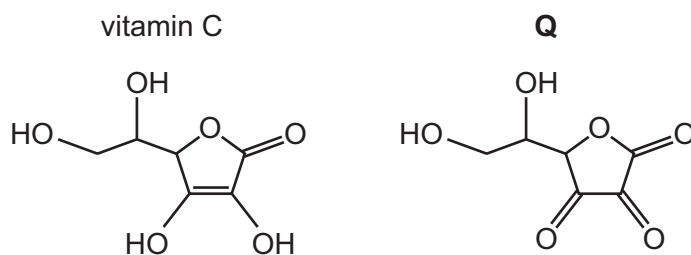


Fig. 4.3

[2]

11. M/J 25/P22/Q5(b)(iii)

Context

5 The structure of vitamin C is shown in Fig. 5.1.

vitamin C

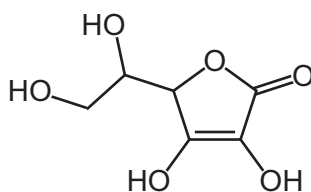


Fig. 5.1

Context: part (b) (introduction)

(b) The mass of vitamin C present in 150.0 g of lemon is found in an experiment.

stage 1 All the vitamin C in 150.0 g of lemon is extracted and dissolved in water to make 100.0 cm^3 of solution **L**.

stage 2 A 25.0 cm^3 sample of solution **L** is titrated with $5.00 \times 10^{-3}\text{ mol dm}^{-3}$ $\text{I}_2(\text{aq})$.

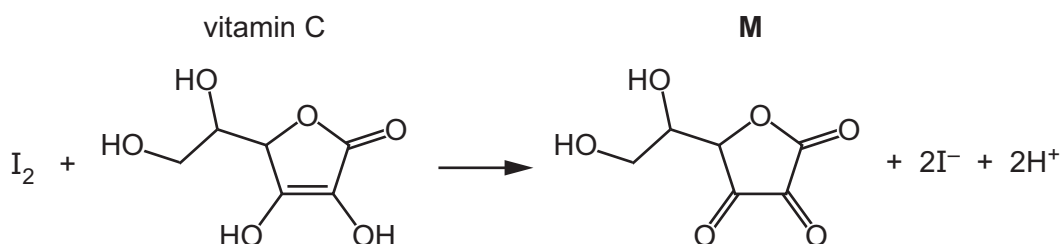


Fig. 5.2

Exactly 36.65 cm^3 of $\text{I}_2(\text{aq})$ reacts with the 25.0 cm^3 sample of solution **L**.

$[M_r: \text{vitamin C}, 176]$

(iii) The progress of the reaction of vitamin C with $\text{I}_2(\text{aq})$ to produce **M** is monitored using infrared spectroscopy.

Table 5.1 indicates the presence of some absorptions in the infrared spectrum of vitamin C. Complete Table 5.1 to predict which of these absorptions, if any, are present in the infrared spectrum of **M**.

Table 5.1

absorption/ cm^{-1}	present in spectrum of vitamin C	present in spectrum of M
1500–1680	✓	
2850–2950	✓	
3200–3650	✓	

Table 5.2

[1]

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

12. M/J 25/P23/Q5(b)(v)

Context: Fig. 5.3

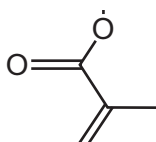


Fig. 5.3

Context: part (a) (introduction)

5 (a) W is a colourless liquid.

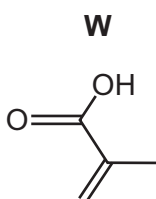


Fig. 5.1

Context: part (b) (introduction)

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.

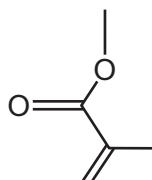
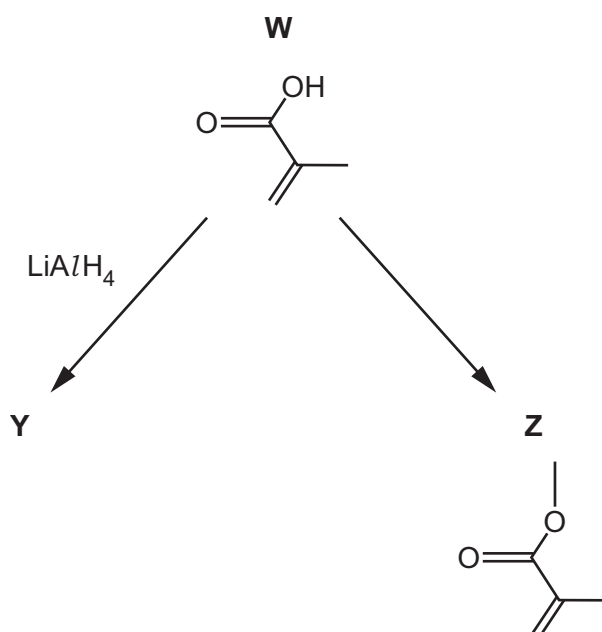
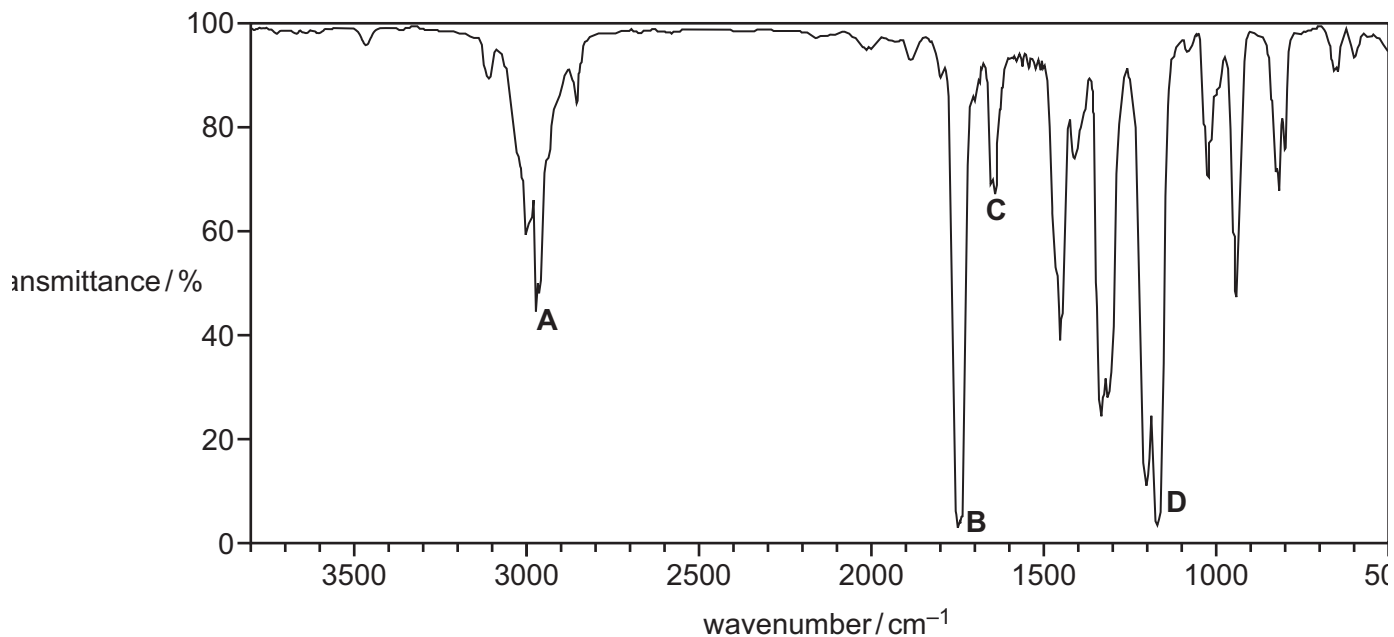


Fig. 5.2

(v) Fig. 5.3 shows the structure of **Z**.

Fig. 5.4 is the infrared spectrum of **Z**.

**Fig. 5.4**

Identify the bond and functional group responsible for each of the absorptions labelled **A**, **B**, **C** and **D** in Fig. 5.4.

A

B

C

D

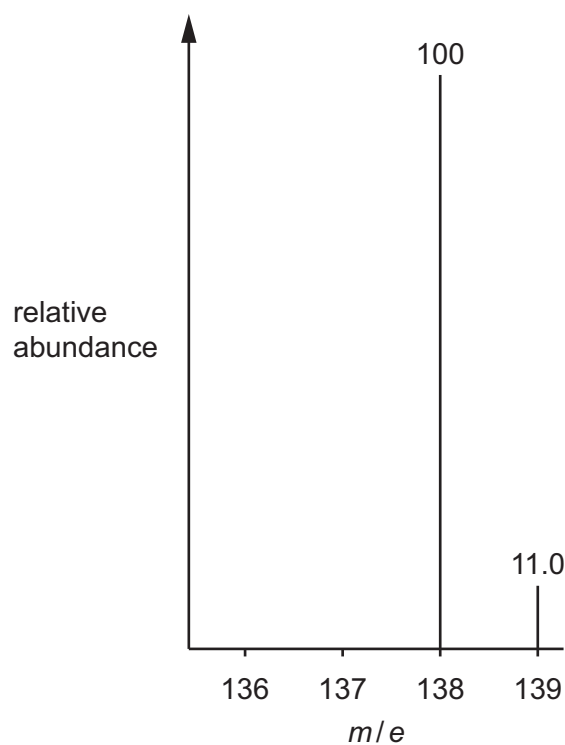
[2]

Table 5.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

13. M/J 25/P24/Q6(c)

Context

6 **P**, **Q** and **R** are three different hydrocarbon molecules.**(c)** The mass spectrum of hydrocarbon **R** is recorded.Information about the two peaks with m/e greater than 135 is shown in Fig. 6.1.**Fig. 6.1****(i)** Use Fig. 6.1 to deduce the number of carbon atoms in a molecule of **R**.

Show your working.

number of carbon atoms = [2]

(ii) Use Fig. 6.1 and your answer to **(c)(i)** to deduce the molecular formula of **R**.

[1]

(iii) Suggest the molecular formula of the fragment of **R** with $m/e = 57$.

[1]

14. MAR 25/P22/Q1(d)

Context

1 Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

(d) Compound **D** contains carbon, hydrogen and chlorine **only**.

Fig. 1.2 shows the mass spectrum of **D**.

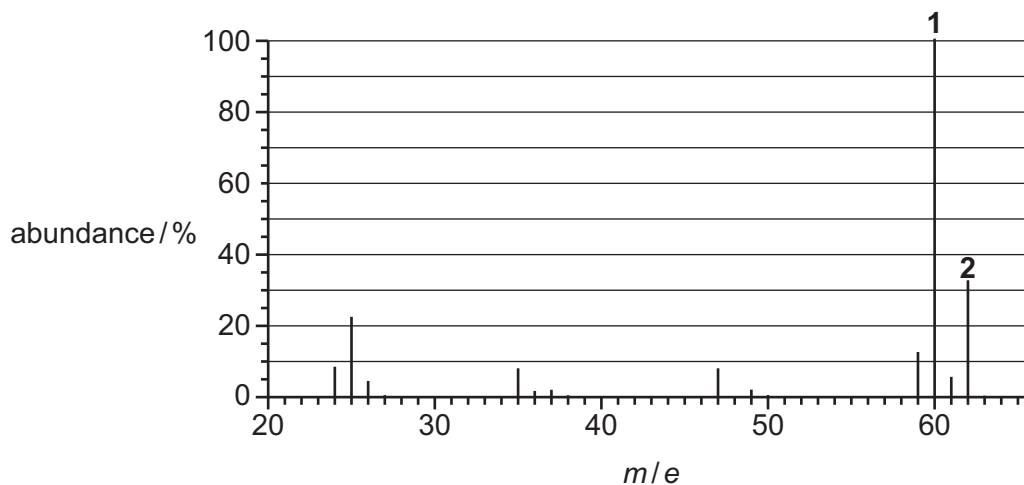


Fig. 1.2

(i) Explain the relative abundance of peaks **1** and **2** in the mass spectrum in Fig. 1.2.

[1]

(ii) Suggest the structure of **D**.

[1]

15. O/N 24/P21/Q4(f)

Context

4 Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

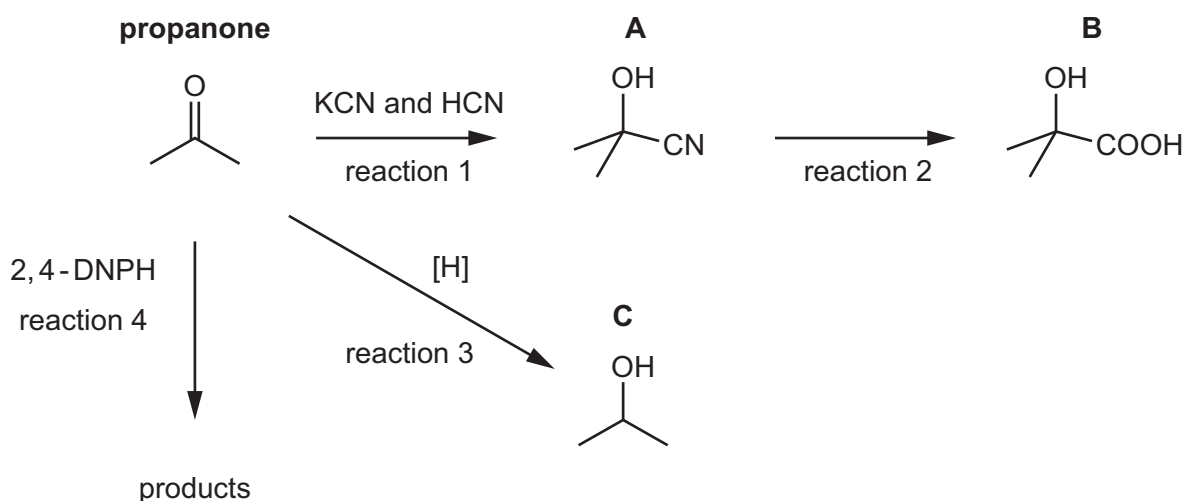


Fig. 4.1

(f) Compounds **A**, **B** and **C** can be distinguished using infrared spectroscopy.

A

B

C

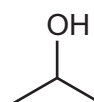
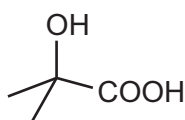
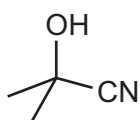


Fig. 4.3 shows the infrared spectrum of one of the compounds.

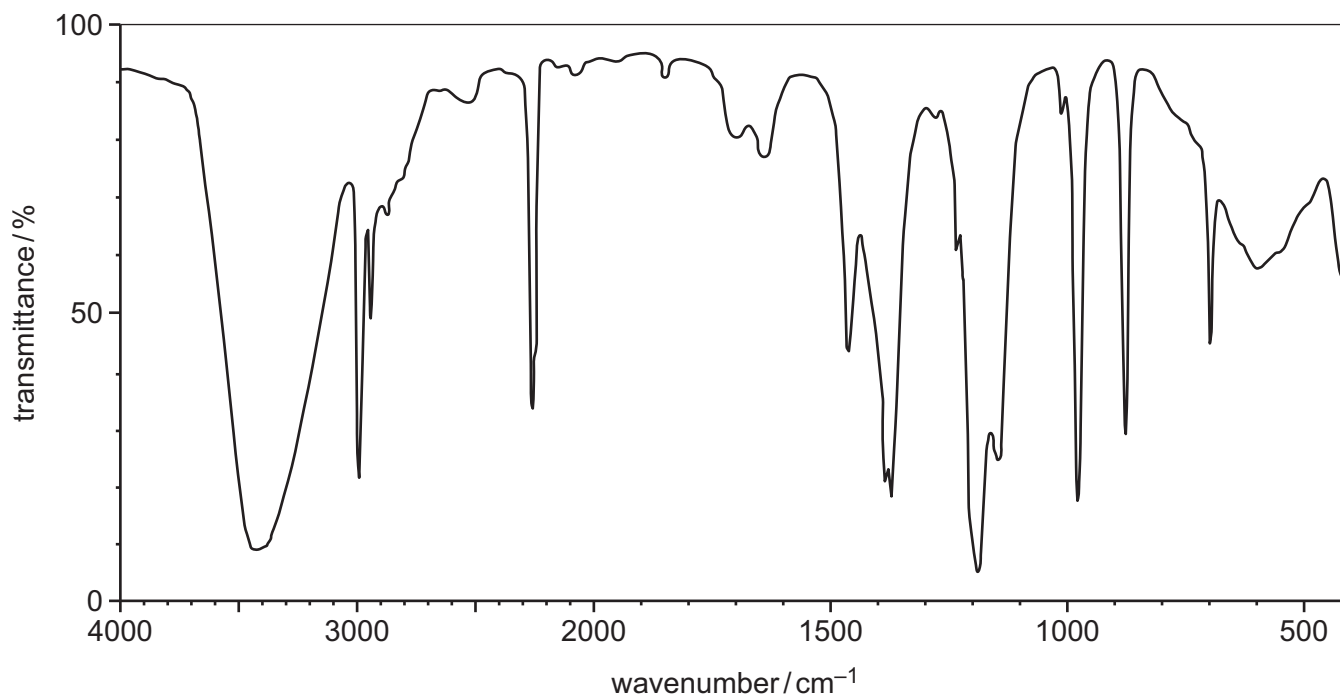


Fig. 4.3

Table 4.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

- (i) Explain why the absorptions at 2850–2950 cm⁻¹ are **not** useful to help determine which of the compounds **A**, **B** or **C** produces the infrared spectrum in Fig. 4.3.

Use Table 4.1 to answer this question.

[1]

- (ii) Identify which of compounds **A**, **B** or **C** produces the infrared spectrum in Fig. 4.3. Explain your answer.

compound

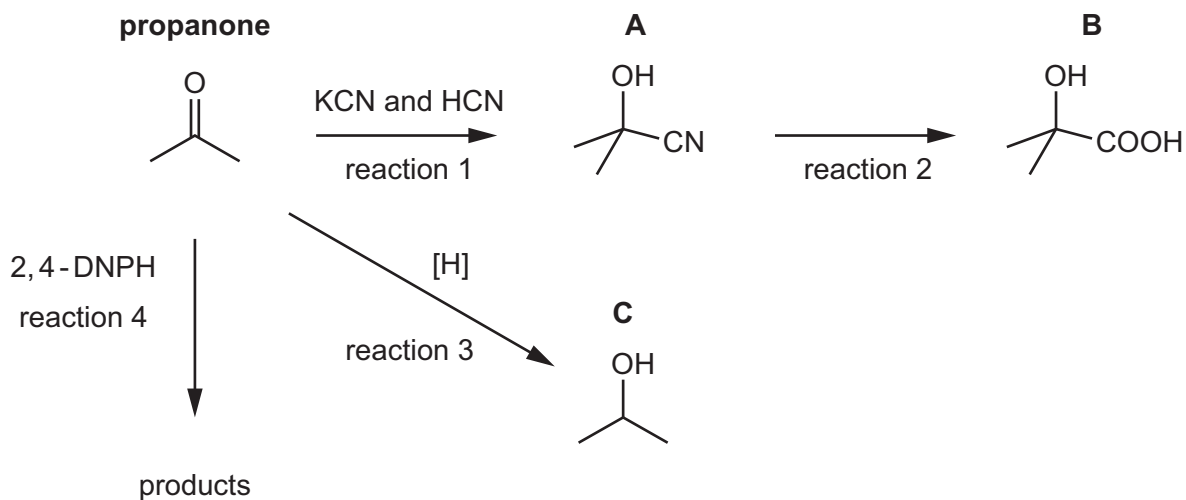
explanation

[1]

16. O/N 24/P23/Q4(f)

Context

- 4** Propanone, CH_3COCH_3 , is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

**Fig. 4.1**

- (f)** Compounds **A**, **B** and **C** can be distinguished using infrared spectroscopy.

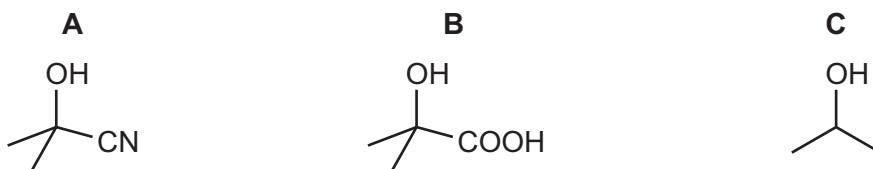


Fig. 4.3 shows the infrared spectrum of one of the compounds.

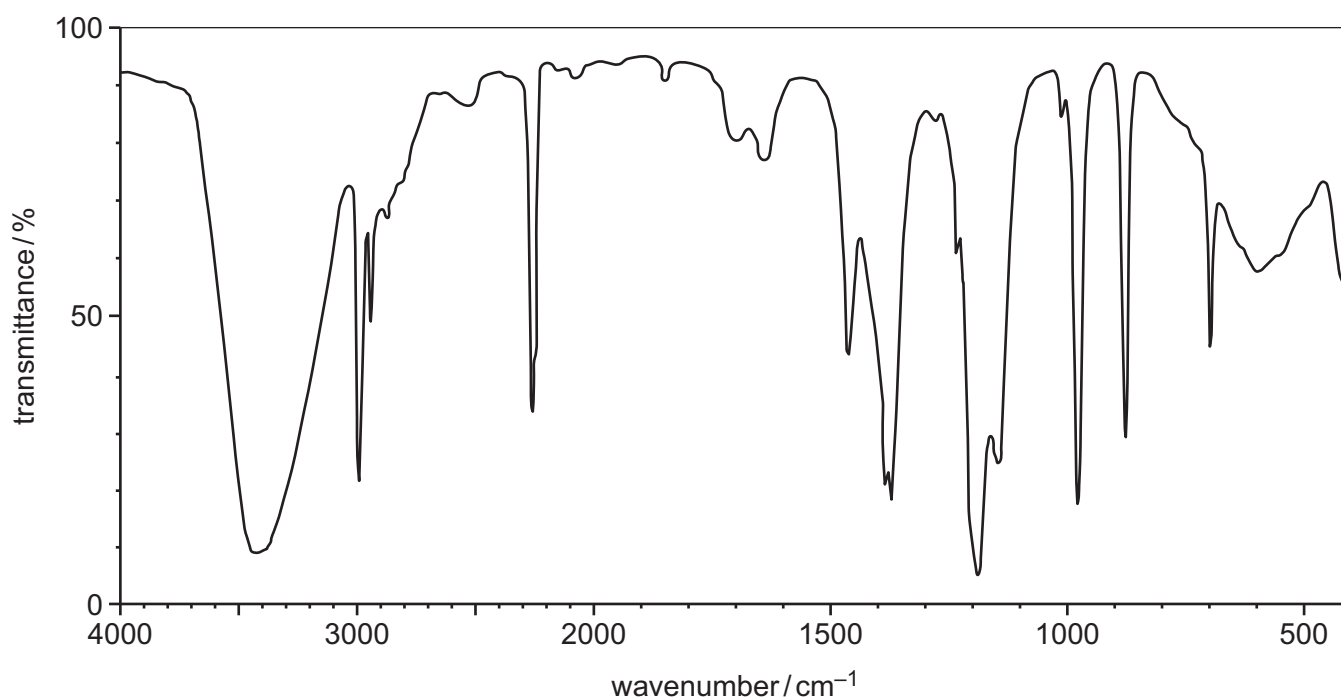


Fig. 4.3**Table 4.1**

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

- (i) Explain why the absorptions at 2850–2950 cm⁻¹ are **not** useful to help determine which of the compounds **A**, **B** or **C** produces the infrared spectrum in Fig. 4.3.

Use Table 4.1 to answer this question.

[1]

- (ii) Identify which of compounds **A**, **B** or **C** produces the infrared spectrum in Fig. 4.3. Explain your answer.

compound

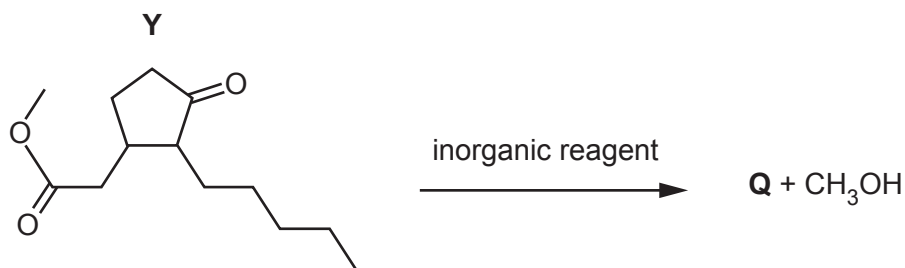
explanation

[1]

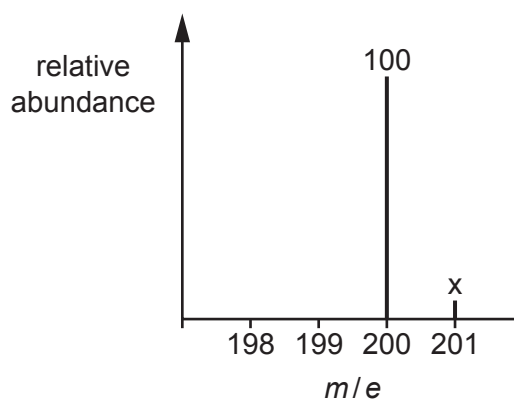
17. M/J 24/P21/Q6(c)(i)

Context: part (c) (introduction)

- (c) Both functional groups in one molecule of **Y** react with an inorganic reagent to form one molecule of **Q** and one molecule of methanol, CH_3OH , as shown in Fig. 6.3.

**Fig. 6.3**

- (i) Part of the mass spectrum for **Q** is shown in Fig. 6.4. Only peaks with m/e greater than 198 are shown.

**Fig. 6.4**

Calculate the relative abundance, x , of the peak at $m/e = 201$.

Show your working.

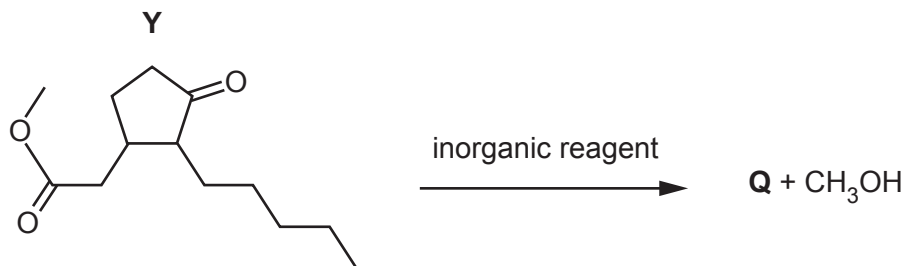
$x =$

[2]

18. M/J 24/P21/Q6(c)(iv)

Context: part (c) (introduction)

- (c) Both functional groups in one molecule of **Y** react with an inorganic reagent to form one molecule of **Q** and one molecule of methanol, CH_3OH , as shown in Fig. 6.3.

**Fig. 6.3**

- (iv) Use Table 6.2 to describe and explain **two** differences between the infrared spectrum of **Y** and **Q** in the region above 1500 cm^{-1} .

[2]

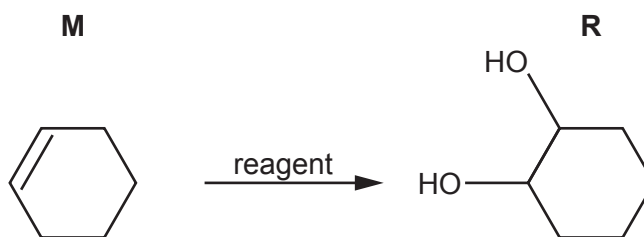
Table 6.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

19. M/J 24/P23/Q5(c)

Context: part (a)

- 5 (a) **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

**Fig. 5.1**

Identify the reagent and conditions for this reaction.

[1]

Context: part (b) (introduction)

(b) **R** is also made from **M** by two steps, as shown in Fig. 5.2.

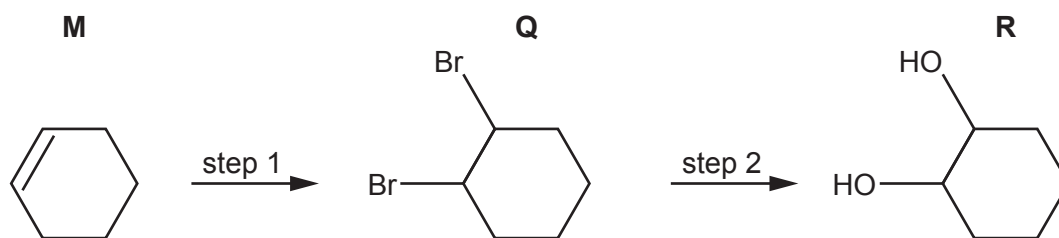


Fig. 5.2

(c) The infrared spectrum of **R** is shown in Fig. 5.3.

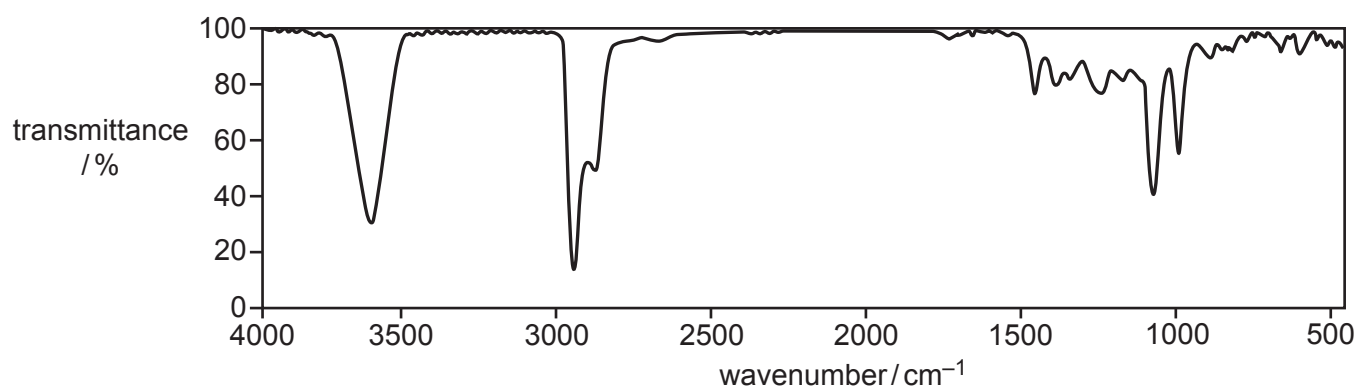


Fig. 5.3

Table 5.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Use the absorptions in the region above 1500 cm⁻¹ in Table 5.1 when answering this question.

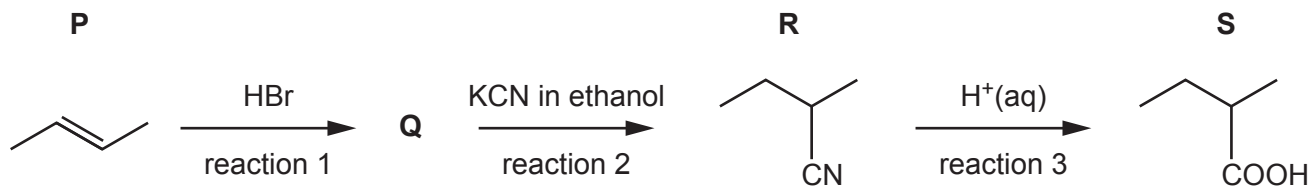
- Add **F** to Fig. 5.3 to identify the peak that is present in an infrared spectrum of both **Q** and **R**. Identify the bond that corresponds to the absorption for **F**.
- Add **G** to Fig. 5.3 to identify the peak that is **not** present in an infrared spectrum of **Q**. Identify the bond that corresponds to the absorption for **G**.

20. MAR 24/P22/Q4(c)

also Unit 18 (2 marks); this unit 4 marks

Context

- 4** Compound **S** is used in food flavourings. A possible synthesis of **S** is shown in Fig. 4.1.

**Fig. 4.1**

- (c) Compounds **S** and **T** react to form organic compound **U**, which has a single functional group.



Table 4.2 shows some data from the mass spectrum of **U**.

Table 4.2

peak	relative abundance
M^+	7.2
$[\text{M}+1]^+$	0.55

- (i) Use the data from Table 4.2 to show that **U** contains 7 carbon atoms.

Show your working.

[2]

- (ii) Fig. 4.2 shows the infrared spectrum of **U**.

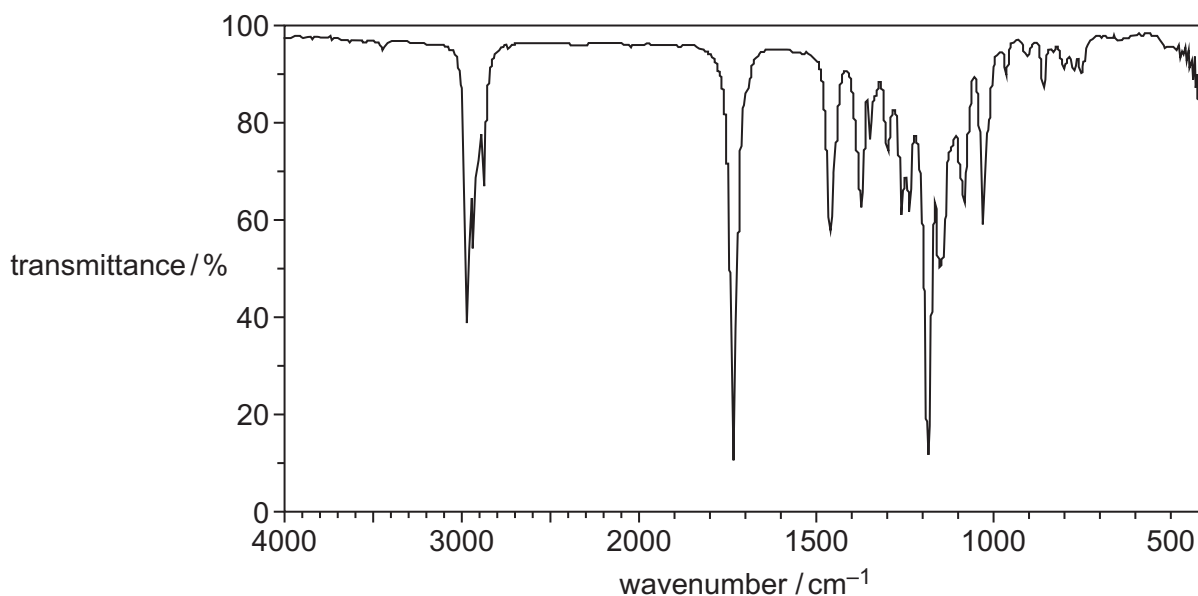


Fig. 4.2**Table 4.3**

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Use Fig. 4.2 and Table 4.3 to identify the functional group present in **U**.

Explain your answer fully.

functional group

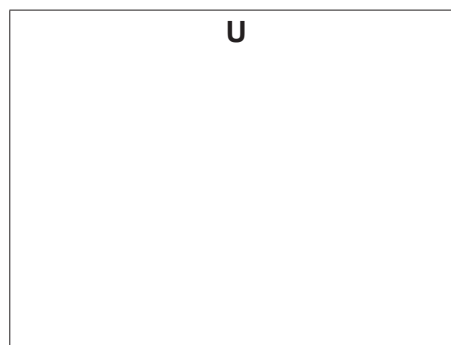
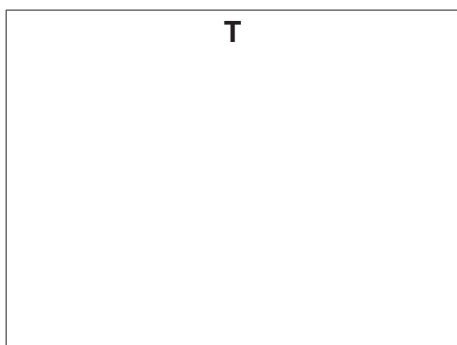
explanation

[2]

(iii) **T** also has a single functional group.

Use the information in (c)(i) and your answer to (c)(ii) to identify **T** and **U**.

Draw the structures of **T** and **U** in the boxes.



[2]

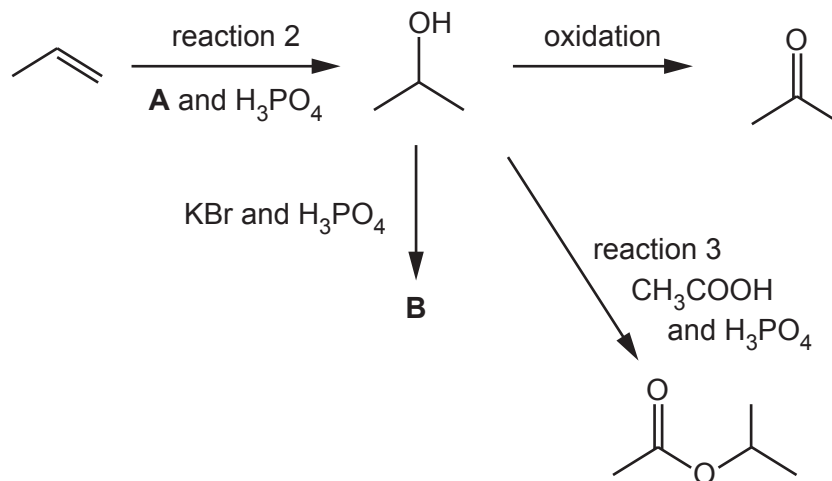
21. O/N 23/P22/Q3(c)(iv)

Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (c) (introduction)

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

**Fig. 3.1**

- (iv) Reaction 3 is monitored using infrared spectroscopy. It is not possible to use the O—H absorption frequency to monitor the reaction.

Use Table 3.2 to identify a suitable bond whose absorption frequency can be used to monitor the progress of reaction 3.

State the change you would see in the infrared spectrum during reaction 3.

bond

change in infrared spectrum

[2]

Table 3.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C—O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C—H	alkane	2850–2950

22. O/N 23/P22/Q3(d)(ii)

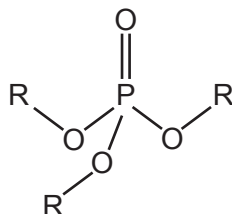
Context

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

Context: part (d) (introduction)

(d) H_3PO_4 also reacts with alcohols to form organophosphates.

Organophosphates are compounds similar to esters. They have the general structure shown in Fig. 3.2.



R = alkyl group

Fig. 3.2

(ii) Compound **T** is a simple organophosphate.

The mass spectrum of **T** shows a molecular ion peak at $m/e = 182$. This peak has a relative intensity of 12.7.

The relative intensity of the $M+1$ peak is 0.84.

Deduce the number of carbon atoms in **T**.
Hence suggest the molecular formula of **T**.

Assume that phosphorus and oxygen exist as single isotopes.

Show your working.

number of carbon atoms in **T** =

molecular formula of **T** =

[3]

23. M/J 23/P21/Q3(f)

Context

3 Chlorine is a very reactive element.**(f)** **X** is a product of the substitution reaction that occurs when CHClF_2 reacts with Br_2 .There is only one naturally occurring isotope of fluorine, ^{19}F .The mass spectrum of **X** shows molecular ion peaks at $m/e = 164$, 166 and 168.Complete Table 3.3 to show **all** the molecular ions responsible for each peak.**Table 3.3**

m/e	formulae of molecular ions
164	
166	
168	$(\text{CF}_2\ ^{37}\text{Cl}^{81}\text{Br})^+$

[2]

24. M/J 23/P22/Q5(d)(iv)

Context: part (c) (introduction)

(c) **C** is a structural isomer of **A** and **B**.**C** is straight chained and has two functional groups.**C** shows only one type of stereoisomerism.Table 5.2 describes observations when separate samples of **C** are added to different reagents.**Table 5.2**

reagent	C
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	no reaction
$\text{Br}_2(\text{aq})$	orange to colourless
alkaline $\text{I}_2(\text{aq})$	yellow precipitate appears

Context: part (d) (introduction)

(d) **D** reacts in the presence of a sulfuric acid catalyst to form **E** and water.The structure of **E** is shown in Fig. 5.1.**E**

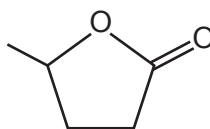


Fig. 5.1

(iv) The infrared spectrum of **E** is shown in Fig. 5.2.

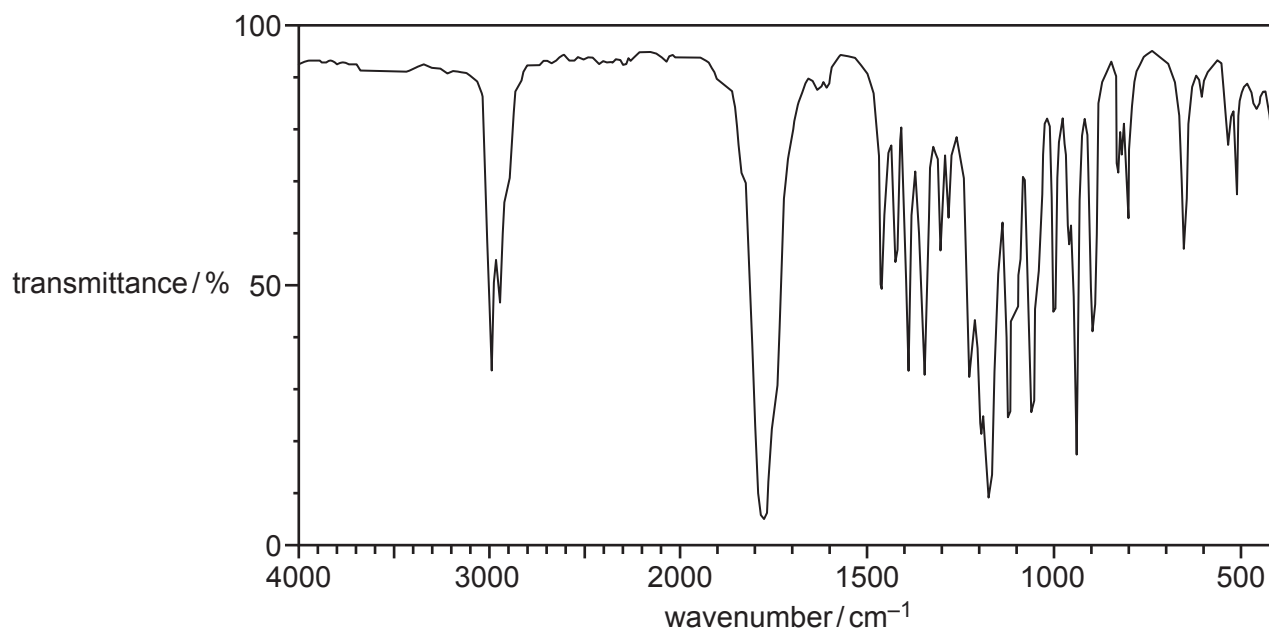


Fig. 5.2

Table 5.3

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

Use Fig. 5.2 and Table 5.3 to predict **two** differences in the absorptions above 1500 cm^{-1} of the infrared spectrum of **D** compared to **E**. Explain your answer.

[2]

25. M/J 23/P23/Q1(c)

Context

1 Copper is used in electrical equipment. It has a melting point of 1085 °C.

(c) The mass spectrum of a sample of pure copper is shown in Fig. 1.1.

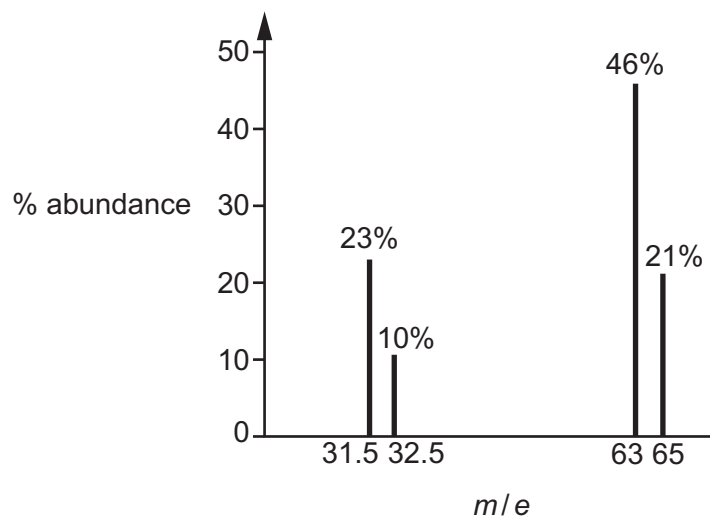


Fig. 1.1

Identify the ion with an abundance of 23% in the sample.

[1]

26. M/J 23/P23/Q6(b)(iv)

Context

6 Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

Context: part (b) (introduction)

(b) CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN .

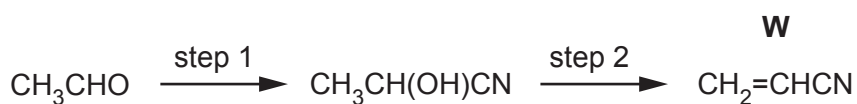


Fig. 6.1

(iv) Fig. 6.3 shows the infrared spectrum of **W**, $\text{CH}_2=\text{CHCN}$.

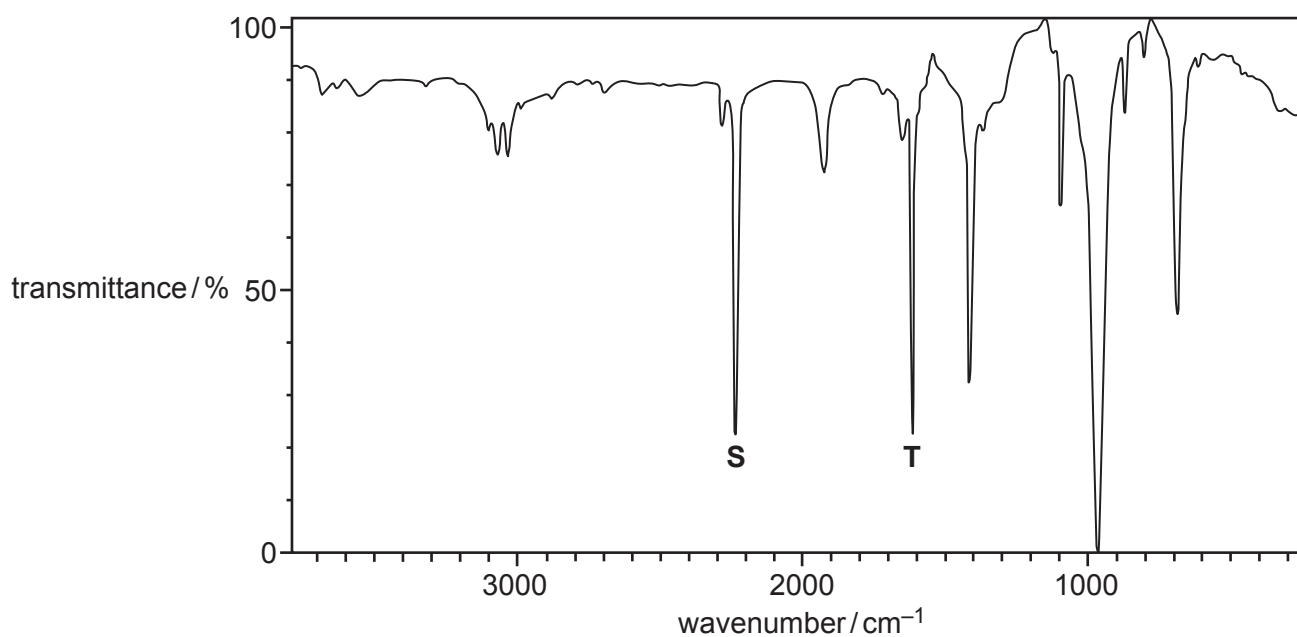


Fig. 6.3

Table 6.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

Use Table 6.1 to identify the bonds responsible for the absorptions marked **S** and **T** on Fig. 6.3.

S

T

27. MAR 23/P22/Q4(d)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

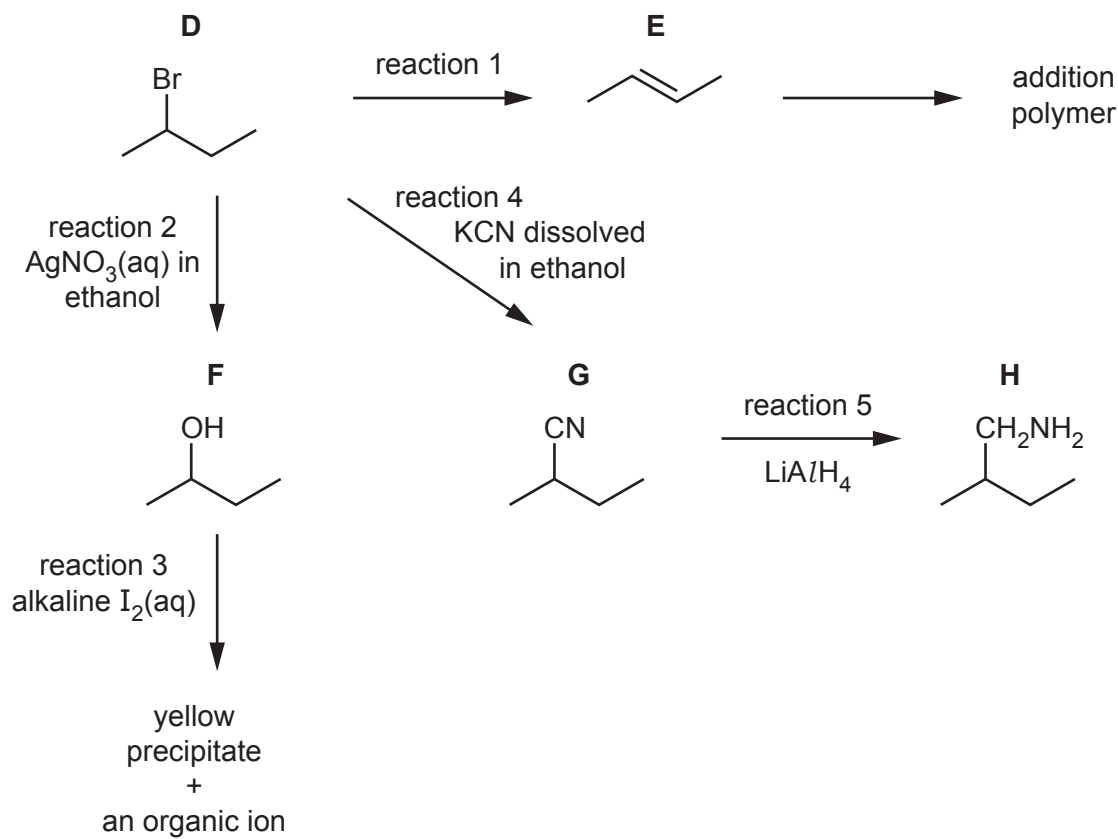


Fig. 4.1

(d) (i) Fig. 4.2 shows the infrared spectrum of one of the compounds **D**, **E**, **F**, **G** or **H**.

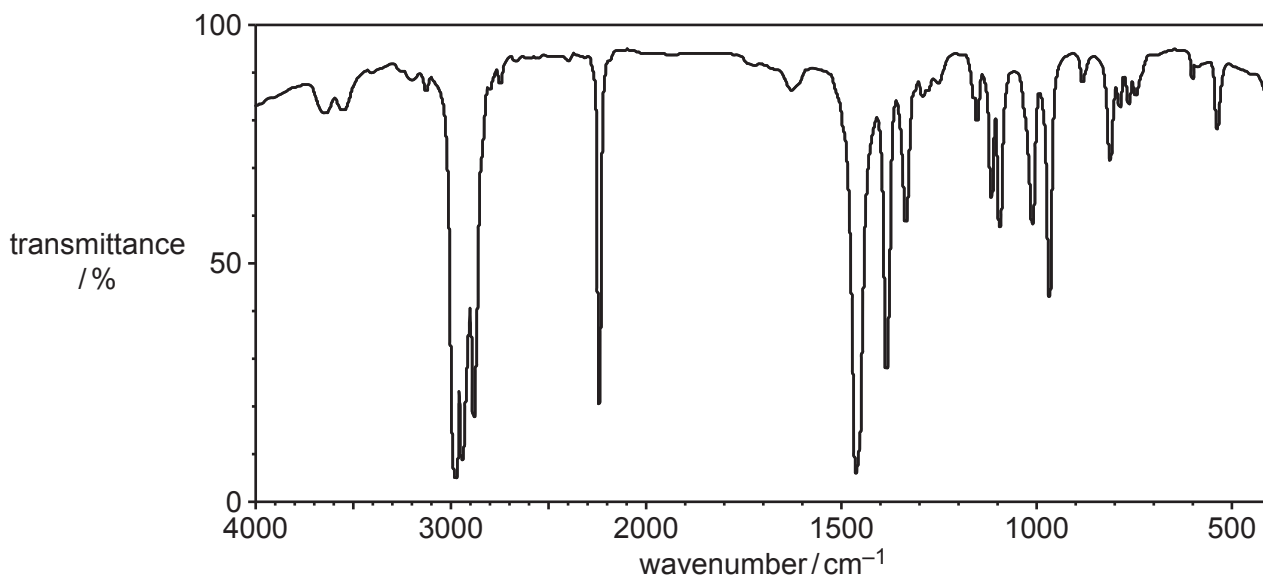


Fig. 4.2

Use information from Table 4.1 (on page 14) to identify which of the compounds **D**, **E**, **F**, **G** or **H** produces the infrared spectrum in Fig. 4.2.

Explain your answer.

[2]

Table 4.1

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3600

- (ii) In the mass spectrum of **D**, the relative abundance of the molecular ion peak is 3.4.

Predict the relative abundance of the $M+2$ peak for **D**.

Explain your answer.

[1]

28. O/N 22/P21/Q5(b)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

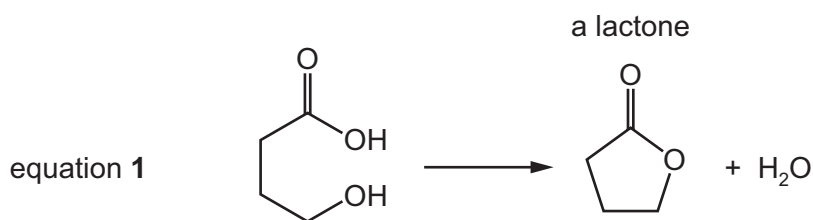
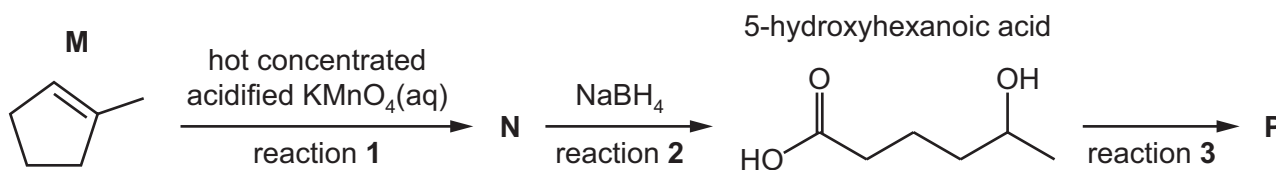


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

**Fig. 5.1**

- (b)** A student monitors the progress of reaction 2 using infrared spectroscopy.

Use Table 5.1 to suggest why it is difficult to distinguish between **N** and 5-hydroxyhexanoic acid using infrared spectroscopy.

[2]

Table 5.1

bond	functional group containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–3100
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

29. O/N 22/P21/Q5(c)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

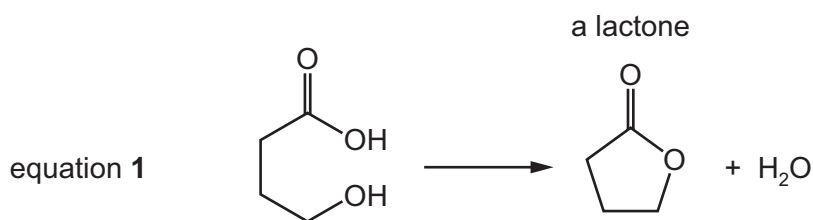
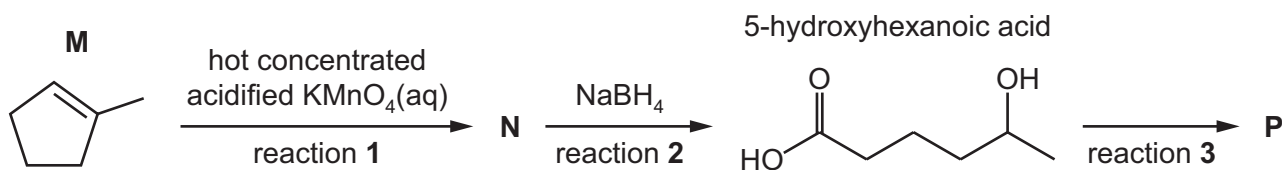


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

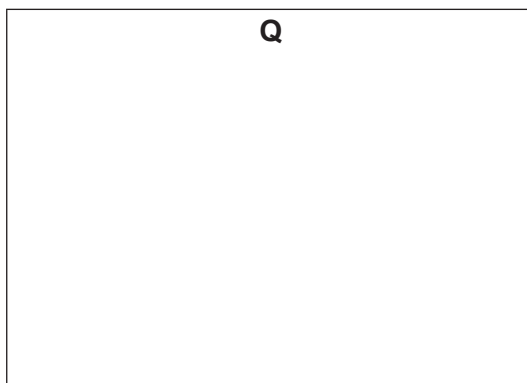
**Fig. 5.1**

- (c)** Unknown lactone **Q** is analysed using mass spectrometry.
Table 5.2 shows information from the mass spectrum.

Table 5.2

peak	m/e	abundance
M^+	72	95.5
$M+1$	73	3.15

Use these data to deduce the structure of **Q**.
Show your working.



[2]

30. O/N 22/P22/Q3(c)

also Unit 14 (1 mark); this unit 4 marks

Context

- 3** Organic compounds can be distinguished using chemical tests and analytical techniques.

(c) Organic compound **E** contains three carbon atoms.

E reacts with cold dilute acidified $\text{KMnO}_4(\text{aq})$ to form a single compound **F** with $M_r = 154.9$.

Fig. 3.1 shows the infrared spectrum of **E**.

Fig. 3.2 shows the infrared spectrum of **F**.

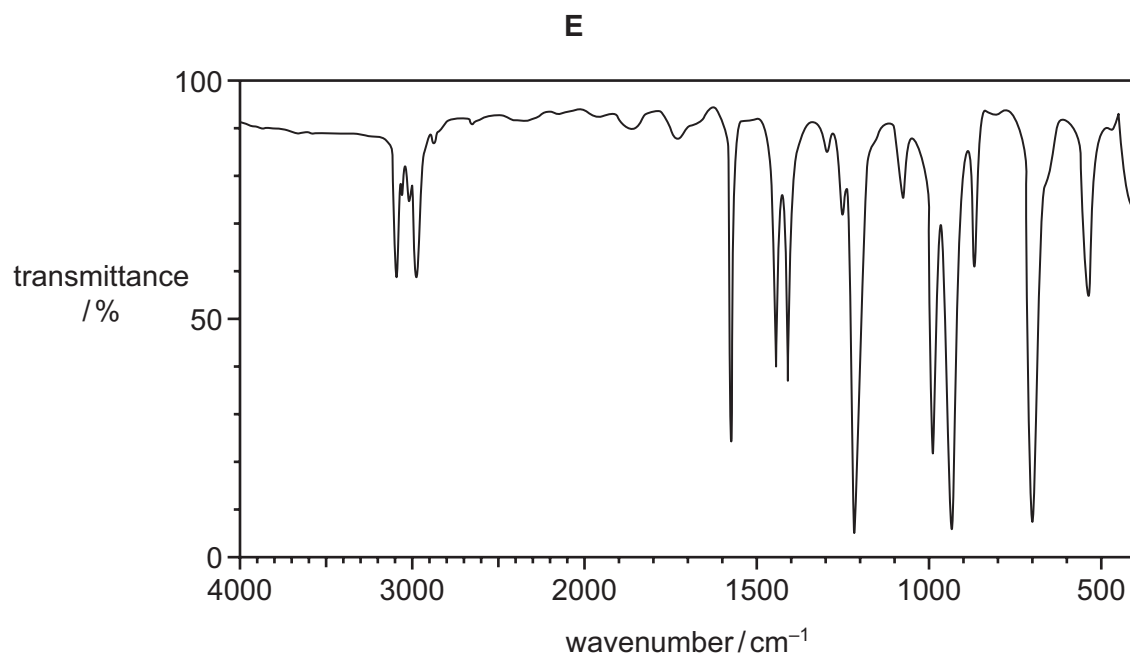


Fig. 3.1

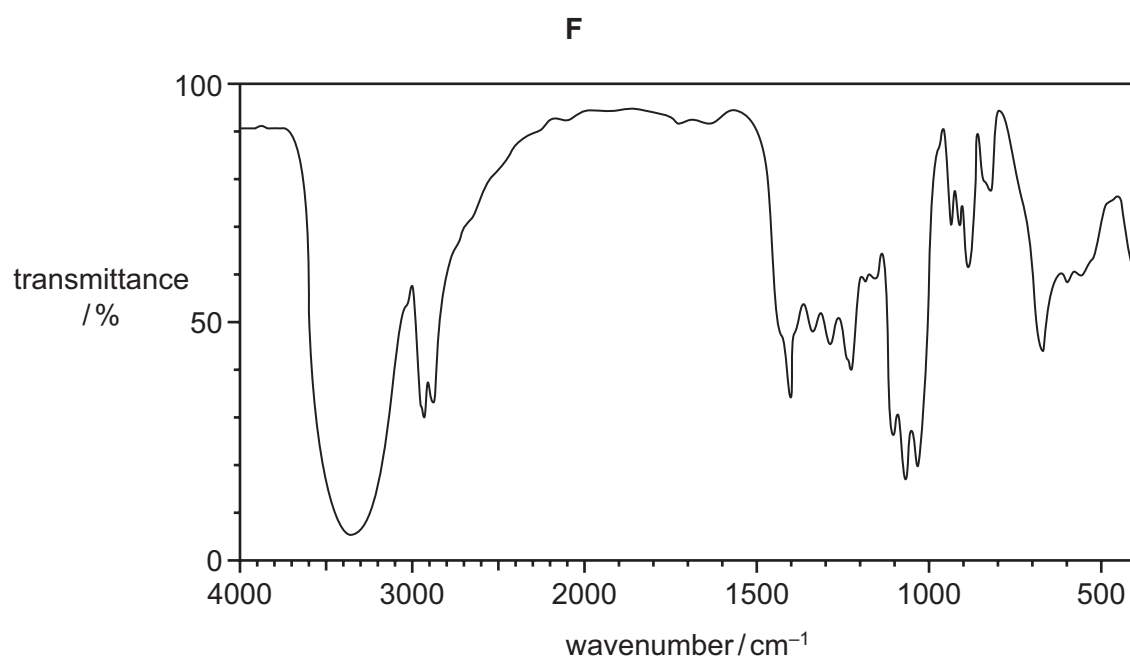


Fig. 3.2

Table 3.2

bond	functional group containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–3100
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Both spectra show absorptions between 2850 and 2950 cm⁻¹ owing to C–H bonds in each molecule.

- (i) Use the two infrared spectra and Table 3.2 to identify the functional group present only in **E**.

Explain your answer, referring only to absorptions at frequencies greater than 1500 cm⁻¹.

functional group

explanation

[1]

- (ii) Use the infrared spectrum of **F** to identify the functional group formed when **E** reacts with cold dilute acidified KMnO₄(aq).

Explain your answer, referring only to absorptions at frequencies greater than 1500 cm⁻¹.

functional group

explanation

[1]

- (iii) The mass spectrum of **E** shows a molecular ion peak and an M+2 peak of approximately equal abundance at $m/e = 120$ and 122.

Deduce the relative molecular mass, M_r , of **E**.

$M_r =$ [1]

- (iv) Use the information in 3(c) to suggest a structure for **E**.

[1]

- (v) Complete the equation for the reaction of **E** with cold dilute acidified KMnO₄(aq) to form **F**. In the equation, [O] represents cold dilute acidified KMnO₄(aq).

H₂O + [O] + →

[1]

31. O/N 22/P23/Q5(b)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

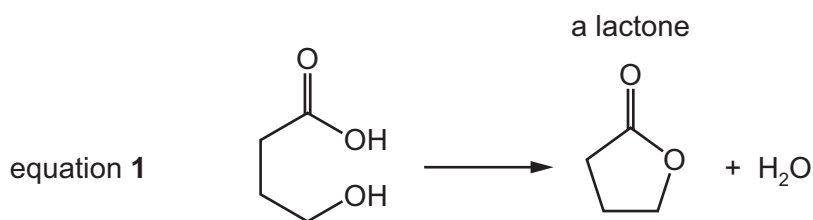
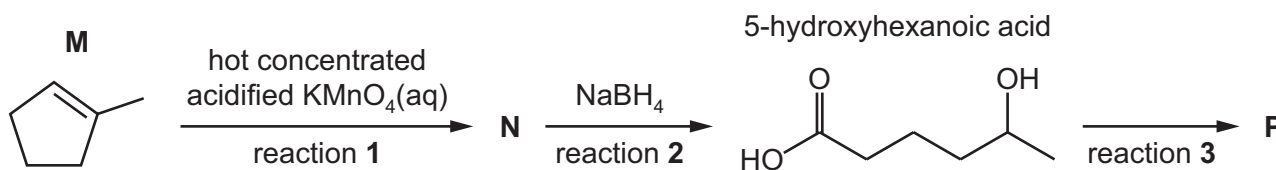


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

**Fig. 5.1**

- (b)** A student monitors the progress of reaction 2 using infrared spectroscopy.

Use Table 5.1 to suggest why it is difficult to distinguish between **N** and 5-hydroxyhexanoic acid using infrared spectroscopy.

[2]

Table 5.1

bond	functional group containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–3100
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

32. O/N 22/P23/Q5(c)

Context

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

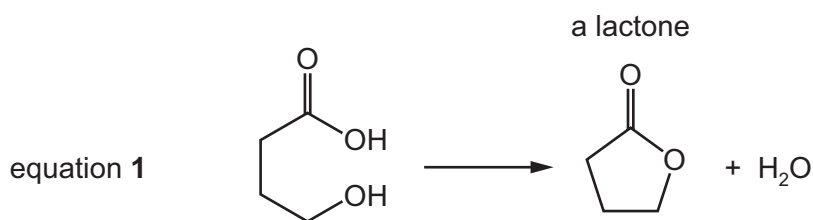
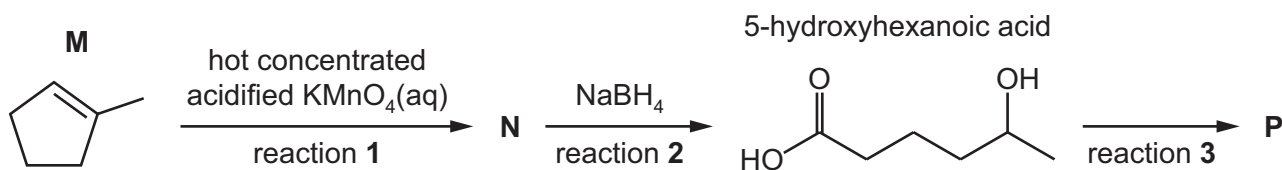


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

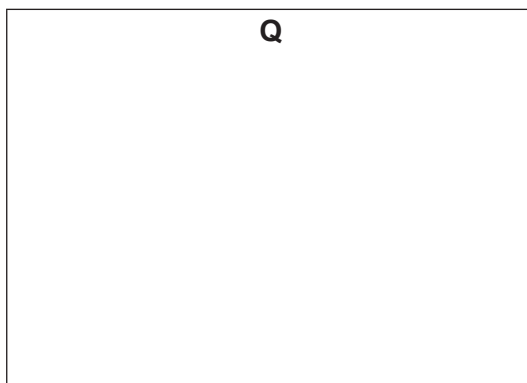
**Fig. 5.1**

- (c)** Unknown lactone **Q** is analysed using mass spectrometry.
Table 5.2 shows information from the mass spectrum.

Table 5.2

peak	m/e	abundance
M^+	72	95.5
$M+1$	73	3.15

Use these data to deduce the structure of **Q**.
Show your working.



33. M/J 22/P21/Q4(d)

Context

4 Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

(d) Fig. 5.1 shows the mass spectrum of ketone **Z**, $\text{C}_5\text{H}_{10}\text{O}$.

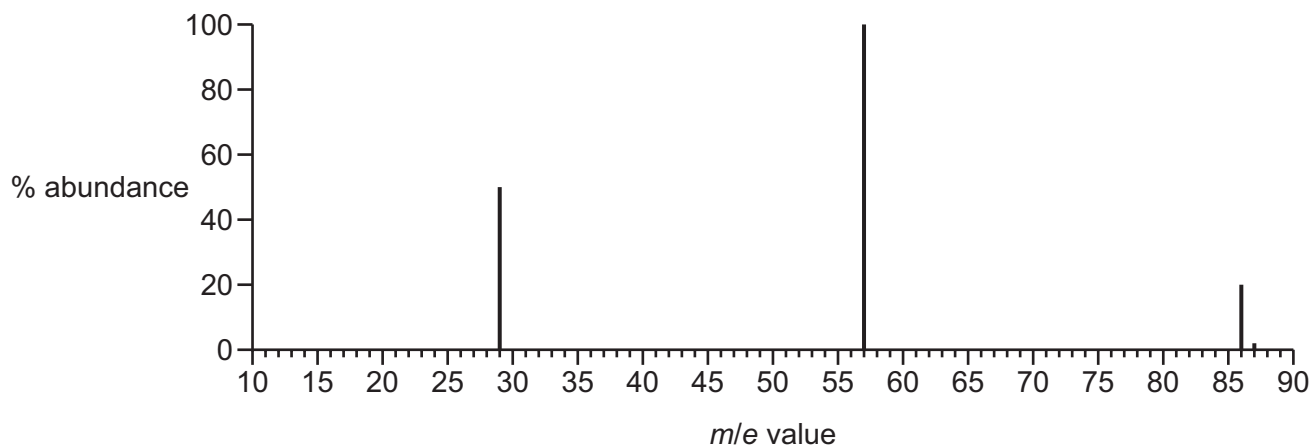


Fig. 5.1

Use the information in Fig. 5.1 to suggest the formulae of the fragments with m/e peaks at 29 and 57. Deduce the identity of **Z**.

$m/e = 29$

$m/e = 57$

identity of **Z**

[3]

34. M/J 22/P22/Q6(c)(i)-(ii)

Context

6 **Z** is a molecule which contains the elements carbon, hydrogen and oxygen only.

Z contains only alkene and carboxyl functional groups.

Context: part (c) (introduction)

(c) Fig. 6.1 shows the mass spectrum of **Z**.

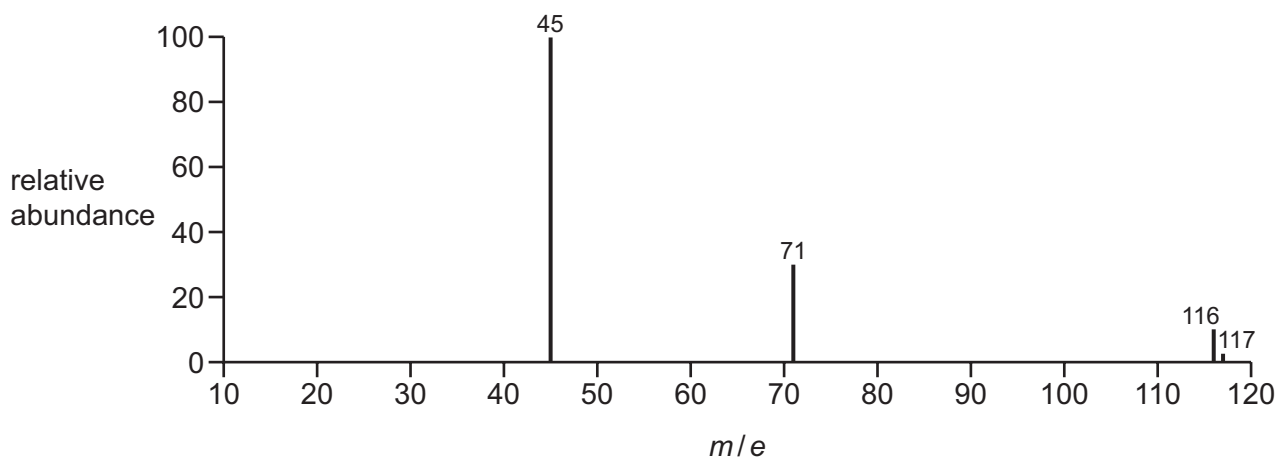


Fig. 6.1

- (i)** Deduce the molecular formula of **Z**. Explain your answer by referring to the molecular ion peak in Fig. 6.1 and the empirical formula of **Z**.

[1]

- (ii)** Use Fig. 6.1 to suggest the formulae of the fragments with m/e peaks at 45 and at 71.

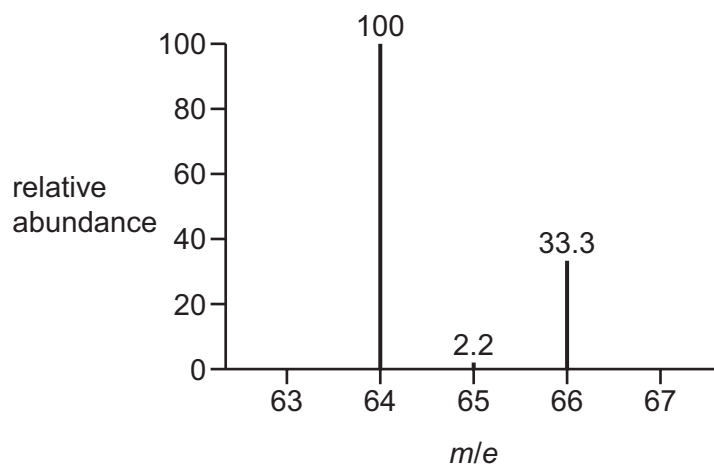
m/e 45

m/e 71

[2]

35. M/J 22/P23/Q5(b)

- (b) **Z** contains three types of atom: carbon, hydrogen and a halogen. The mass spectrum of **Z** is recorded. Fig. 5.1 shows a section of the mass spectrum at m/e greater than 63. The fragment at $m/e = 64$ is the molecular ion peak.

**Fig. 5.1**

- (i) Deduce the number of carbon atoms present in a molecule of **Z** using Fig. 5.1. Show your working.

[1]

- (ii) Deduce which halogen is present in **Z** using Fig. 5.1. Explain your answer.

[1]

- (iii) There are also peaks at $m/e = 29$ and $m/e = 49$.

Suggest the formulae of these fragments.
Deduce the name of **Z**.

$m/e = 29$

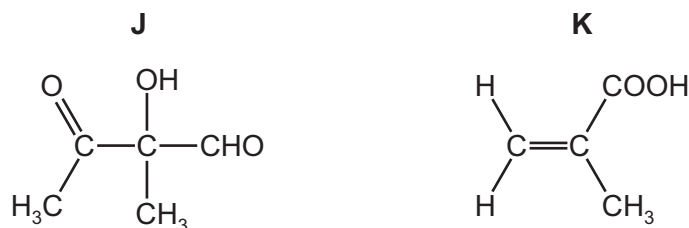
$m/e = 49$

name of **Z**

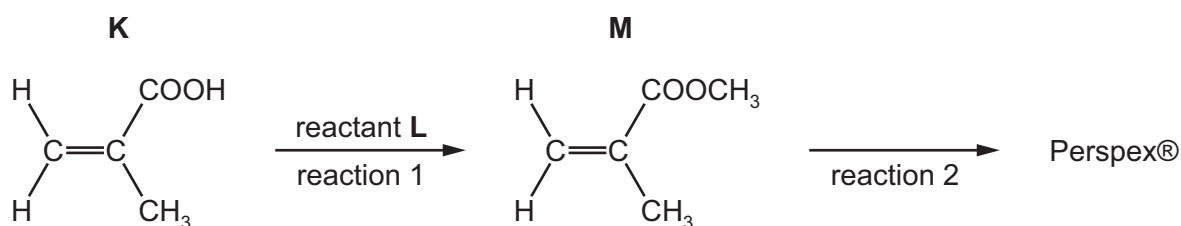
[3]

36. MAR 22/P22/Q4(b)(iii)

Context

4 Compounds **J** and **K** are found in plant oils.**Fig. 4.1**

Context: part (b) (introduction)

(b) **K** is used to make the addition polymer Perspex®. A synthesis of Perspex® is shown in Fig. 4.2.**Fig. 4.2****(iii)** Use information from Table 4.2 to suggest how the infrared spectra of **M** and Perspex® would differ. Explain your answer.

[1]

Table 4.2

bond	functional group containing the bond	characteristic infrared absorption range (in wavenumbers)/cm ⁻¹
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–3100
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

Answers Section

1. M/J 26/P21/Q5(g)

5(g)(i) M1 $[\text{CH}_2\text{CH}^{37}\text{Cl}]^+$ M2 $[\text{CH}_2\text{CH}]^+$	2 M1 A $\text{C}_2\text{H}_3\text{Cl}^+$ where $\text{Cl} = \text{Cl-}^{37}$ A $\text{CH}_2\text{CHCl}^{37+}/[\text{CH}_2\text{CHCl}^{37}]^+$ A $\text{CH}_2=\text{CH}^{37}\text{Cl}^+$ I lack of brackets I dots M2 A C_2H_3^+ A $\text{CH}_2=\text{CH}^+$ A molecular formulas for both A any unambiguous formula for both A any unambiguous representation of ^{37}Cl Max 1 both without charges OR both negative charges OR both same incorrect positive charge (all else correct).
5(g)(ii) M1 2 chlorine isotopes (exist) M2 ^{35}Cl is 3 times more abundant than ^{37}Cl	2 75% ^{35}Cl and 25% ^{37}Cl ✓✓ ^{35}Cl is 3 times more abundant than ^{37}Cl ✓✓ ^{35}Cl is more abundant than ^{37}Cl ✓ M1 A m/e 64 is $[\text{M}+2]^+$ ion with ^{37}Cl M1 A peak at m/e 64 is due to ^{37}Cl AND peak at 62 is due to ^{35}Cl

2. M/J 26/P22/Q6(c)(i)-(ii)

6(c)(i)	M1 (molecular ion peak at) 116 M2 has the same (116) molecular mass (as P)	2 M1 A the peak to be identified on the spectrum A the peak furthest to the right / the peak with the highest m/e value I an unquantified peak between 110 and 120 I 99 for HOOCCH=CHCO⁺ I 71 for C₃H₃O₂⁺ R any other peak except 99 and 71 M2 look at structures at the top of the page for evidence of the Mr of P = 116 I AMU M⁺ ion peak = molecular ion peak = M peak Molecular mass = M_r = RFM = molar mass A The molecular mass (of P) is 116 ✓✓ unless contradicted R relative atomic mass / A_r
6(c)(ii)	M1 (m / e = 45) COOH⁺ M2 (m / e = 99) HOOCCHCHCO⁺ / HOOCCH=CHCO⁺	2 Max 1 for both correct structures with no + charge OR – charge OR • OR no charges OR dots BOD sticks on fragment e.g. –COOH⁺ A CO₂H⁺ / CHO₂⁺ I C₄H₃O₃

3. M/J 26/P23/Q5(c)

5(c)(i)	M1 two Cl isotopes M2 ^{35}Cl is 3 times more abundant than ^{37}Cl	2 M1 A (2) different Cl isotopes M1 BOD peaks are the isotopes of Cl 75% ^{35}Cl and 25% ^{37}Cl ✓✓ ^{35}Cl is 3 times more abundant than ^{37}Cl ✓✓ ^{35}Cl is more abundant than ^{37}Cl ✓^ ^{35}Cl : ^{37}Cl is 3 : 1 ✓✓
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5(c)(ii)	m / e value of peak	molecular formula of species responsible for peak	2	max 1 for (CH ₂ ³⁵ C _{l₂}) (CH ₂ ³⁵ C ^{β³⁷} Cl) (CH ₂ ³⁷ C _{l₂}) ... i.e. all else correct but no charge OR negative charge OR shown as free radicals I lack of brackets I dots I position of positive charge
	84	(CH ₂ ³⁵ C _{l₂}) ⁺ BOD (C ³⁷ C ^{β³⁵} Cl) ⁺		
	86	(CH ₂ ³⁵ C ^{β³⁷} Cl) ⁺ BOD (C ³⁷ C _{l₂}) ⁺		
	88	(CH ₂ ³⁷ C _{l₂}) ⁺		

4. M/J 26/P24/Q4(c)

4(c)	M1 bond: C=O (functional group: carboxyl / carbonyl) AND absorbance: 1670–1740 M2 bond: O–H AND absorbance: 2500–3500 AND functional group: carboxyl(ic acid)	2	M1 I references to C–H / C–O bonds A wavenumbers in the absorption range of 1670–1740 e.g. 1700 R functional group: amide / ester M2 A wavenumbers in the absorption range of 2500–3500 e.g. 2700 I OH A C=O group / O – H group R – OH and – O – H
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5. O/N 25/P21/Q3(e)(ii)

3(e)(ii)	M1 absorption labelled at approx. 3000 cm ⁻¹ identified as O–H M2 absorption labelled at approx. 1720 cm ⁻¹ identified as C=O	2
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6. O/N 25/P21/Q4(c)(ii)-(iii)

4(c)(ii)	(The spectrum is of) X AND 41 is [CH ₂ =CCH ₃] ⁺ / 41 is C ₃ H ₅ ⁺ OR 43 is [CH ₃ CHCH ₃] ⁺ / 43 is C ₃ H ₇ ⁺ OR X would NOT have significant peak at 54 / [(CH ₃) ₂ C=C ⁺] / C ₄ H ₆ ⁺ OR X would NOT have significant peak at 42 due to [(CH ₃) ₂ C ⁺] / C ₃ H ₆ ⁺	1
4(c)(iii)	([M+1] ⁺ / 34.7 × 100 / 1.1 = 6) [M+1] ⁺ = 2.29	1

7. O/N 25/P22/Q4(d)(ii)-(iii)

4(d)(ii)	M1 absorption labelled at approx. 1750 cm ⁻¹ identified as C=O M2 absorption labelled at approx. 1650 cm ⁻¹ identified as C=C M3 absorption labelled at approx. 1170 cm ⁻¹ identified as C—O	3
4(d)(iii)	(0.31 / 4.7 × 100 / 1.1) = 6 carbon atoms	1

8. O/N 25/P23/Q3(e)(ii)

3(e)(ii)	M1 absorption labelled at approx. 3000 cm ⁻¹ identified as O–H M2 absorption labelled at approx. 1720 cm ⁻¹ identified as C=O	2
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9. O/N 25/P23/Q4(c)(ii)-(iii)

4(c)(ii)	(The spectrum is of) X AND 41 is $[\text{CH}_2=\text{CCH}_3]^+$ / 41 is C_3H_5^+ OR 43 is $[\text{CH}_3\text{CHCH}_3]^+$ / 43 is C_3H_7^+ OR X would NOT have significant peak at 54 / $[(\text{CH}_3)_2\text{C}=\text{C}^+]$ / C_4H_6^+ OR X would NOT have significant peak at 42 due to $[(\text{CH}_3)_2\text{C}^+]$ / C_3H_6^+	1
4(c)(iii)	$([\text{M}+1]^+ / 34.7 \times 100 / 1.1 = 6)$ $[\text{M}+1]^+ = 2.29$	1

10. M/J 25/P21/Q4(c)

4(c)	Any two complete rows from:	2															
	<table border="1"> <thead> <tr> <th>wavenumber</th><th>bond</th><th>functional group</th></tr> </thead> <tbody> <tr> <td>1040 – 1300</td><td>C–O</td><td>ester / hydroxy</td></tr> <tr> <td>3200 – 3650</td><td>O–H</td><td>hydroxy / alcohols</td></tr> <tr> <td>2850 – 2950</td><td>C–H</td><td>alkane</td></tr> <tr> <td>1710 – 1750</td><td>C=O</td><td>ester</td></tr> </tbody> </table>	wavenumber	bond	functional group	1040 – 1300	C–O	ester / hydroxy	3200 – 3650	O–H	hydroxy / alcohols	2850 – 2950	C–H	alkane	1710 – 1750	C=O	ester	
wavenumber	bond	functional group															
1040 – 1300	C–O	ester / hydroxy															
3200 – 3650	O–H	hydroxy / alcohols															
2850 – 2950	C–H	alkane															
1710 – 1750	C=O	ester															

11. M/J 25/P22/Q5(b)(iii)

5(b)(iii)	<table border="1"> <thead> <tr> <th>absorption / cm^{-1}</th><th>present in spectrum of vitamin C</th><th>present in spectrum of M</th></tr> </thead> <tbody> <tr> <td>1500–1680</td><td>✓</td><td>✗ / – / blank</td></tr> <tr> <td>2850–2950</td><td>✓</td><td>✓</td></tr> <tr> <td>3200–3650</td><td>✓</td><td>✓</td></tr> </tbody> </table>	absorption / cm^{-1}	present in spectrum of vitamin C	present in spectrum of M	1500–1680	✓	✗ / – / blank	2850–2950	✓	✓	3200–3650	✓	✓	1
absorption / cm^{-1}	present in spectrum of vitamin C	present in spectrum of M												
1500–1680	✓	✗ / – / blank												
2850–2950	✓	✓												
3200–3650	✓	✓												

12. M/J 25/P23/Q5(b)(v)

5(b)(v)	A C–H alkane B C=O ester C C=C alkene D C–O ester 2 points = 1 mark, all 4 points = 2 marks	2
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13. M/J 25/P24/Q6(c)

6(c)(i)	M1 $\frac{100}{1.1} \times \frac{11}{100}$ M2 10	2
6(c)(ii)	$\text{C}_{10}\text{H}_{18}$	1
6(c)(iii)	$(\text{C}_4\text{H}_9)^+$	1

14. MAR 25/P22/Q1(d)

1(d)(i)	(relative) abundance is 3:1 / 100:33 AND link to two isotopes of <i>Cl</i>	1
1(d)(ii)	D is $\text{Cl}-\text{C}\equiv\text{C}-\text{H}$	1

15. O/N 24/P21/Q4(f)

4(f)(i)	All three have a C—H OR CH bond	1
4(f)(ii)	compound A AND absorption at 2200–2250 cm ⁻¹ indicates C≡N	1

16. O/N 24/P23/Q4(f)

4(f)(i)	All three have a C—H OR CH bond	1
4(f)(ii)	compound A AND absorption at 2200–2250 cm ⁻¹ indicates C≡N	1

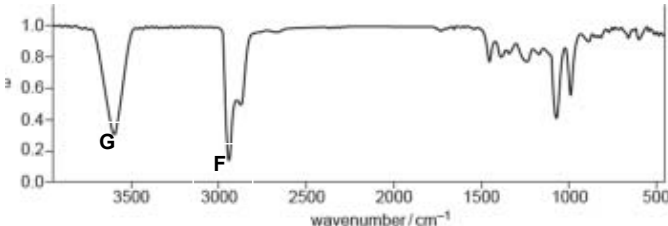
17. M/J 24/P21/Q6(c)(i)

6(c)(i)	M1 deduce $n = 12$ (from $Y - 1C$ (in methanol)) M2 $(100 \times x) \div (1.1 \times 100) = 12$ (so) $x = 13.2$	2
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18. M/J 24/P21/Q6(c)(iv)

6(c)(iv)	M1 Y will have absorption / peak / trough between 1670–1740 due to C=O (Q will not) M2 Q will have absorption / peak / trough between 3200–3600 due to O—H (Y will not)	2
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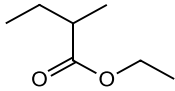
19. M/J 24/P23/Q5(c)

5(c)	 M1 peak shown around 2850–2950 labelled F AND (corresponds to) C—H M2 peak shown around 3200–3600 labelled G AND (corresponds to) O—H (in hydroxy(l) / alcohol)	2
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Answer for context part (a)

5(a)	cold dilute KMnO ₄ OR cold dilute potassium manganate(VII)	1
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20. MAR 24/P22/Q4(c)

4(c)(i)	$n = 100 / 1.1 \times 0.55 / 7.2$	1
	$= 6.94$ (so 7 carbon atoms)	1
4(c)(ii)	FG = ester / (R)COOR(')	1
	(absorption at) 1040–1300 cm ⁻¹ indicates C—O (stretch) AND (absorption at) 1710–1750 cm ⁻¹ indicates (C=O stretch)	1
4(c)(iv)	T 	1
	U 	1

21. O/N 23/P22/Q3(c)(iv)

3(c)(iv)	C=O	1
	absorption changes to frequency / wavenumber / from 1670–1740 to 1710–1750 cm ⁻¹	1

22. O/N 23/P22/Q3(d)(ii)

3(d)(ii)	$\frac{0.84}{12.7} \times \frac{100}{1.1}$ (= 6.013)	1
	no. of carbon atoms = 6	1
	molecular formula = C ₆ H ₁₅ O ₄ P	1

23. M/J 23/P21/Q3(f)

3(f)	M1 <i>m/e</i> 164 (CF ₂ ³⁵ Cl ⁷⁹ Br) ⁺ M2 <i>m/e</i> 166 (CF ₂ ³⁷ Cl ⁷⁹ Br) ⁺ AND (CF ₂ ³⁵ Cl ⁸¹ Br) ⁺	2
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24. M/J 23/P22/Q5(d)(iv)

5(d)(iv)	Any two of the following differences in spectra D compared to E :	2																				
	<table><tr><th></th><th>bond</th><th>functional group</th><th>value or range of values of absorbance inclusive (cm⁻¹)</th></tr><tr><td>E</td><td>C=O / C double bond to O</td><td>(ester)</td><td>1710–1750</td></tr><tr><td>D</td><td>C=O</td><td>(carboxy(l))</td><td>1670–1740</td></tr><tr><td>D</td><td>O-H OR OH bond</td><td>(carboxy(l))</td><td>2500–3000</td></tr><tr><td>D</td><td>O-H OR OH bond</td><td>(hydroxy(l) OR alcohol)</td><td>3200–3600</td></tr></table>		bond	functional group	value or range of values of absorbance inclusive (cm ⁻¹)	E	C=O / C double bond to O	(ester)	1710–1750	D	C=O	(carboxy(l))	1670–1740	D	O-H OR OH bond	(carboxy(l))	2500–3000	D	O-H OR OH bond	(hydroxy(l) OR alcohol)	3200–3600	
	bond	functional group	value or range of values of absorbance inclusive (cm ⁻¹)																			
E	C=O / C double bond to O	(ester)	1710–1750																			
D	C=O	(carboxy(l))	1670–1740																			
D	O-H OR OH bond	(carboxy(l))	2500–3000																			
D	O-H OR OH bond	(hydroxy(l) OR alcohol)	3200–3600																			

25. M/J 23/P23/Q1(c)

1(c)	⁶³ Cu ²⁺	1
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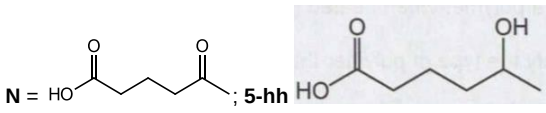
26. M/J 23/P23/Q6(b)(iv)

6(b)(iv)	S C≡N AND T C=C	1
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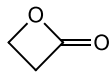
27. MAR 23/P22/Q4(d)

4(d)(i)	G	1
	(absorption at) 2200–2250 (cm ⁻¹) AND C≡N	1
4(d)(ii)	3.4 AND relative abundance of ⁷⁹ Br: ⁸¹ Br ≈ 50:50 OR 1:1	1

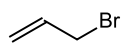
28. O/N 22/P21/Q5(b)

5(b)	 N = HO-C(=O)-CH₂-CH₂-C(=O)-CH₃; 5-hh HO-C(=O)-CH₂-CH₂-CH₂-CH₂-OH M1 absorptions will overlap / be similar / the same / indistinguishable M2 both have some bonds in similar environments owtte	1
		1

29. O/N 22/P21/Q5(c)

5(c)	$100 / 1.1 \div 95.5 / 3.15 = 3$ carbon atoms	1
		1

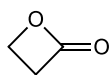
30. O/N 22/P22/Q3(c)

3(c)(i)	C=C (bond) / alkene AND (absorption within) 1500–1680 cm ⁻¹ (present in E 's IR spectrum only)	1
3(c)(ii)	hydroxyl / alcohol AND (broad absorption within) 3200–3650 cm ⁻¹ (present in F 's IR spectrum only)	1
3(c)(iii)	120.9 / 121	1
3(c)(iv)		1
3(c)(v)	$(1)\text{H}_2\text{O} + (1)[\text{O}] + \text{C}_3\text{H}_5\text{Br} \rightarrow \text{C}_3\text{H}_7\text{BrO}_2$	1

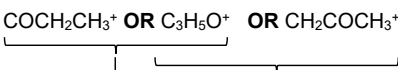
31. O/N 22/P23/Q5(b)

5(b)	 M1 absorptions will overlap / be similar / the same / indistinguishable M2 both have some bonds in similar environments owtte	1
		1

32. O/N 22/P23/Q5(c)

5(c)	$100 / 1.1 \div 95.5 / 3.15 = 3$ carbon atoms	1
		1

33. M/J 22/P21/Q4(d)

4(d)	29 C₂H₅⁺ 57 COCH₂CH₃⁺ OR C₃H₅O⁺ OR CH₂COCH₃⁺  identity of Z pentan-3-one pentan-2-one / CH₃CH₂COCH₂CH₃ / CH₃COCH₂CH₂CH₃	3
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34. M/J 22/P22/Q6(c)(i)-(ii)

6(c)(i)	Look for some reference to 116 as the mass of the molecular ion AND mass of C ₍₁₎ H ₍₁₎ O ₍₁₎ = 29 to conclude molecular formula is C ₄ H ₄ O ₄ 116 / 29 = 4 so C ₄ H ₄ O ₄	1
6(c)(ii)	M1 <i>m/e</i> 45: ⁺ COOH OR ⁺ CHO ₂ M2 <i>m/e</i> 71: C ₃ H ₃ O ₂ ⁺	2

35. M/J 22/P23/Q5(b)

5(b)(i)	$100 \times 2.2 / 1.1 \times 100 = 2$	1
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5(b)(ii)	chlorine / Cl AND peak at $M+2$ represents the molecular ion with $37-Cl$ (rather than $35-Cl$ as relative abundance of (peaks) $M: M+2$ is 100:33 / 3:1) OR relative abundance of (peaks) $M: M+2$ is 100:33.3 / 3:1 (so peak at $M+2$ contains $37-Cl$)	1
5(b)(iii)	$m/e = 29$: $C_2H_5^+$ $m/e = 49$: CH_2Cl^+ name of Z: chloroethane	3

36. MAR 22/P22/Q4(b)(iii)

4(b)(iii)	Perspex® would not have absorption $1500-1680\text{ cm}^{-1}$ AND Perspex® does not have $C=C$	1
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Topic index: where each part was filed

Reference	Unit	Marks	Page	Also / context
M/J 26/P21/Q1(a)	3	2	74	
M/J 26/P21/Q1(b)	1	5	5	
M/J 26/P21/Q1(c)	3	4	74	
M/J 26/P21/Q1(d)	3	3	75	
M/J 26/P21/Q1(e)	2	3	48	
M/J 26/P21/Q2(a)	12	1	327	
M/J 26/P21/Q2(b)(i)	7	1	180	
M/J 26/P21/Q2(b)(ii)	8	2	215	
M/J 26/P21/Q2(c)(i)	12	1	327	
M/J 26/P21/Q2(c)(ii)	4	4	117	
M/J 26/P21/Q3(a)	11	2	296	
M/J 26/P21/Q3(b)-(c)	11	2	297	
M/J 26/P21/Q3(d)	6	2	165	
M/J 26/P21/Q3(e)	11	1	298	ctx (b)
M/J 26/P21/Q3(f)	11	3	299	
M/J 26/P21/Q4(a)(i)	16	1	461	
M/J 26/P21/Q4(a)(ii)	6	1	165	
M/J 26/P21/Q4(b)(i)	16	2	462	ctx (a) intro
M/J 26/P21/Q4(b)(ii)	17	1	497	ctx (a) intro
M/J 26/P21/Q4(b)(iii)	16	2	463	ctx (a) intro
M/J 26/P21/Q4(c)	18	2	530	
M/J 26/P21/Q5(a)	13	2	339	
M/J 26/P21/Q5(b)	14	1	389	
M/J 26/P21/Q5(c)	13	2	339	
M/J 26/P21/Q5(d)	14	4	389	
M/J 26/P21/Q5(e)	15	1	434	
M/J 26/P21/Q5(f)	20	1	565	
M/J 26/P21/Q5(g)	22	4	588	
M/J 26/P22/Q1(a)	1	2	5	
M/J 26/P22/Q1(b)	3	2	75	
M/J 26/P22/Q1(c)	9	4	237	
M/J 26/P22/Q1(d)	9	1	237	
M/J 26/P22/Q1(e)(i)	7	2	180	
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M/J 26/P22/Q2(a)	1	2	6	
M/J 26/P22/Q2(b)	1	4	6	
M/J 26/P22/Q3(a)	7	3	181	also U11:1
M/J 26/P22/Q3(b)-(d)	11	5	300	ctx (a)(i)
M/J 26/P22/Q4(a)	16	3	464	
M/J 26/P22/Q4(b)(i)	16	1	464	
M/J 26/P22/Q4(b)(ii)-(iii)	2	2	48	
M/J 26/P22/Q4(c)	6	4	166	also U17:2; ctx (b) intro
M/J 26/P22/Q5(a)	13	2	340	
M/J 26/P22/Q5(b)(i)-(ii)	13	2	341	
M/J 26/P22/Q5(b)(iii)-(iv)	14	4	390	
M/J 26/P22/Q5(c)	14	1	391	
M/J 26/P22/Q6(a)	13	3	341	
M/J 26/P22/Q6(b)(i)	14	2	392	ctx (a) intro
M/J 26/P22/Q6(b)(ii)	13	2	342	ctx (a) intro
M/J 26/P22/Q6(c)(i)-(ii)	22	4	589	ctx (a) intro
M/J 26/P22/Q6(c)(iii)	17	2	498	ctx (a) intro
M/J 26/P23/Q1(a)	9	4	237	
M/J 26/P23/Q1(b)	11	2	300	
M/J 26/P23/Q1(c)	11	4	301	
M/J 26/P23/Q1(d)	11	2	301	
M/J 26/P23/Q2(a)	1	2	7	
M/J 26/P23/Q2(b)	4	3	117	
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M/J 26/P23/Q3(c)	7	3	182	
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Reference	Unit	Marks	Page	Also / context
M/J 26/P23/Q4(a)(i)-(ii)	16	4	465	ctx Fig. 4.1
M/J 26/P23/Q4(a)(iii)	17	1	499	ctx Fig. 4.1
M/J 26/P23/Q4(b)	16	2	466	
M/J 26/P23/Q4(c)(i)-(ii)	13	4	343	ctx Fig. 4.1
M/J 26/P23/Q4(c)(iii)	16	1	467	ctx Fig. 4.1
M/J 26/P23/Q4(d)(i)	16	1	468	ctx (c) intro, Fig. 4.1
M/J 26/P23/Q4(d)(ii)	6	1	167	ctx (c) intro, Fig. 4.1
M/J 26/P23/Q5(a)(i)	13	2	344	
M/J 26/P23/Q5(a)(ii)	14	1	392	
M/J 26/P23/Q5(b)(i)	1	1	7	
M/J 26/P23/Q5(b)(ii)-(iii)	14	4	393	
M/J 26/P23/Q5(c)	22	4	590	
M/J 26/P24/Q1(a)	1	3	8	
M/J 26/P24/Q1(b)	2	1	48	
M/J 26/P24/Q1(c)	3	1	76	ctx Fig. 1.3
M/J 26/P24/Q1(d)	3	3	77	
M/J 26/P24/Q2(a)(i)	3	2	77	
M/J 26/P24/Q2(a)(ii)	12	2	327	
M/J 26/P24/Q2(b)(i)	8	1	216	
M/J 26/P24/Q2(b)(ii)	7	2	183	
M/J 26/P24/Q2(c)	7	9	184	
M/J 26/P24/Q2(d)	8	3	216	
M/J 26/P24/Q2(e)	8	2	217	
M/J 26/P24/Q3(a)	9	3	238	
M/J 26/P24/Q3(b)	9	4	238	
M/J 26/P24/Q3(c)	10	2	279	
M/J 26/P24/Q4(a)	16	1	468	
M/J 26/P24/Q4(b)	16	3	469	
M/J 26/P24/Q4(c)	22	2	591	
M/J 26/P24/Q4(d)(i)	21	2	576	
M/J 26/P24/Q4(d)(ii)	19	1	553	
M/J 26/P24/Q5(a)	14	1	393	
M/J 26/P24/Q5(b)(i)-(ii)	20	2	565	
M/J 26/P24/Q5(b)(iii)	14	1	394	
M/J 26/P24/Q5(c)	13	1	344	
M/J 26/P24/Q5(d)	14	5	394	
MAR 26/P22/Q1(a)	1	6	9	
MAR 26/P22/Q1(b)	10	4	279	
MAR 26/P22/Q1(c)	10	2	279	
MAR 26/P22/Q1(d)	11	2	301	
MAR 26/P22/Q2(a)(i)	13	1	344	
MAR 26/P22/Q2(a)(ii)	3	1	78	
MAR 26/P22/Q2(b)	4	4	118	
MAR 26/P22/Q2(c)	12	4	328	
MAR 26/P22/Q2(d)(i)	5	2	140	
MAR 26/P22/Q2(d)(ii)	8	1	218	ctx Fig. 2.3
MAR 26/P22/Q3(a)	15	1	434	
MAR 26/P22/Q3(b)	13	5	345	
MAR 26/P22/Q3(c)	14	3	395	
MAR 26/P22/Q3(d)	15	2	435	
MAR 26/P22/Q3(e)	16	3	469	ctx (d)
MAR 26/P22/Q4(a)(i)-(iii)	7	5	185	
MAR 26/P22/Q4(a)(iv)	19	1	553	
MAR 26/P22/Q4(b)	3	3	78	
MAR 26/P22/Q4(c)	15	4	436	
MAR 26/P22/Q4(d)	13	6	345	also U18:1, U20:1, U22:2
O/N 25/P21/Q2(a)	9	4	239	
O/N 25/P21/Q2(b)	9	3	239	
O/N 25/P21/Q2(c)	9	2	240	
O/N 25/P21/Q2(d)	10	2	279	
O/N 25/P21/Q3(a)	2	1	49	
O/N 25/P21/Q3(b)	13	2	347	
O/N 25/P21/Q3(c)(i)	13	1	348	
O/N 25/P21/Q3(c)(ii)-(iii)	14	3	395	
O/N 25/P21/Q3(d)(i)	14	1	396	
O/N 25/P21/Q3(d)(ii)	5	2	141	

Reference	Unit	Marks	Page	Also / context
O/N 25/P21/Q3(e)(i)	14	1	396	
O/N 25/P21/Q3(e)(ii)	22	2	592	
O/N 25/P21/Q4(a)	16	5	471	also U18:1, U15:2
O/N 25/P21/Q4(b)(i)	20	1	565	
O/N 25/P21/Q4(b)(ii)	13	2	348	
O/N 25/P21/Q4(b)(iii)-(iv)	20	5	566	
O/N 25/P21/Q4(c)(i)	13	1	348	
O/N 25/P21/Q4(c)(ii)-(iii)	22	2	594	
O/N 25/P22/Q2(a)-(c)	9	8	240	
O/N 25/P22/Q2(d)	9	2	241	
O/N 25/P22/Q2(e)	7	6	186	
O/N 25/P22/Q2(f)	10	4	280	
O/N 25/P22/Q3(a)	13	2	349	
O/N 25/P22/Q3(b)	14	2	396	
O/N 25/P22/Q3(c)	14	1	397	
O/N 25/P22/Q3(d)(i)	13	2	349	
O/N 25/P22/Q3(d)(ii)	3	3	79	
O/N 25/P22/Q3(e)(i)	14	1	397	
O/N 25/P22/Q3(e)(ii)	5	2	142	
O/N 25/P22/Q3(e)(iii)	14	1	398	
O/N 25/P22/Q4(a)(i)-(ii)	14	4	398	
O/N 25/P22/Q4(a)(iii)-(v)	13	4	350	
O/N 25/P22/Q4(b)	15	1	436	
O/N 25/P22/Q4(c)	16	2	471	
O/N 25/P22/Q4(d)(i)	18	1	530	
O/N 25/P22/Q4(d)(ii)-(iii)	22	4	594	
O/N 25/P23/Q2(a)	9	4	242	
O/N 25/P23/Q2(b)	9	3	242	
O/N 25/P23/Q2(c)	9	2	243	
O/N 25/P23/Q2(d)	10	2	280	
O/N 25/P23/Q3(a)	2	1	49	
O/N 25/P23/Q3(b)	13	2	351	
O/N 25/P23/Q3(c)(i)	13	1	351	
O/N 25/P23/Q3(c)(ii)-(iii)	14	3	399	
O/N 25/P23/Q3(d)(i)	14	1	399	
O/N 25/P23/Q3(d)(ii)	5	2	143	
O/N 25/P23/Q3(e)(i)	14	1	400	
O/N 25/P23/Q3(e)(ii)	22	2	596	
O/N 25/P23/Q4(a)	16	5	472	also U18:1, U15:2
O/N 25/P23/Q4(b)(i)	20	1	566	
O/N 25/P23/Q4(b)(ii)	13	2	351	
O/N 25/P23/Q4(b)(iii)-(iv)	20	5	567	
O/N 25/P23/Q4(c)(i)	13	1	352	
O/N 25/P23/Q4(c)(ii)-(iii)	22	2	598	
O/N 25/P24/Q1(a)(i)	11	1	302	
O/N 25/P24/Q1(a)(ii)	9	1	243	
O/N 25/P24/Q1(a)(iii)	11	1	302	
O/N 25/P24/Q1(b)(i)-(ii)	11	3	302	
O/N 25/P24/Q1(b)(iii)-(iv)	5	4	144	
O/N 25/P24/Q1(b)(v)	3	2	79	
O/N 25/P24/Q1(c)	11	6	303	
O/N 25/P24/Q1(d)	9	4	243	
O/N 25/P24/Q2(a)	4	3	118	
O/N 25/P24/Q2(b)	7	5	187	
O/N 25/P24/Q2(c)(i)	8	1	218	ctx (b) intro
O/N 25/P24/Q2(c)(ii)	16	1	472	ctx (b) intro
O/N 25/P24/Q2(c)(iii)	9	1	244	ctx (b) intro
O/N 25/P24/Q3(a)(i)-(iii)	16	3	473	
O/N 25/P24/Q3(a)(iv)	13	1	352	
O/N 25/P24/Q3(b)(i)	16	1	474	
O/N 25/P24/Q3(b)(ii)-(iii)	15	5	437	
O/N 25/P24/Q3(b)(iv)	2	1	49	
O/N 25/P24/Q4(a)	14	11	400	also U13:4, U22:2
O/N 25/P24/Q4(b)(i)	15	1	438	ctx (a)(v)
O/N 25/P24/Q4(b)(ii)	19	1	554	ctx (a)(v)
O/N 25/P24/Q4(b)(iii)	18	3	531	ctx (a)(v)

Reference	Unit	Marks	Page	Also / context
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M/J 25/P21/Q1(a)(ii)	3	1	79	
M/J 25/P21/Q1(a)(iii)	4	1	119	
M/J 25/P21/Q1(b)	9	4	244	
M/J 25/P21/Q1(c)	9	2	245	
M/J 25/P21/Q1(d)	4	2	119	
M/J 25/P21/Q2(a)	1	1	9	
M/J 25/P21/Q2(b)(i)	2	2	50	
M/J 25/P21/Q2(b)(ii)	1	2	10	
M/J 25/P21/Q2(c)	1	1	10	
M/J 25/P21/Q2(d)	1	1	10	
M/J 25/P21/Q2(e)	1	4	11	
M/J 25/P21/Q3(a)	8	6	219	
M/J 25/P21/Q3(b)	7	2	187	
M/J 25/P21/Q3(c)	7	2	188	
M/J 25/P21/Q4(a)	2	1	50	
M/J 25/P21/Q4(b)(i)-(ii)	2	3	51	
M/J 25/P21/Q4(b)(iii)	6	1	168	
M/J 25/P21/Q4(b)(iv)	14	2	402	
M/J 25/P21/Q4(c)	22	2	599	ctx (b) intro
M/J 25/P21/Q5(a)	18	2	532	
M/J 25/P21/Q5(b)	18	1	532	
M/J 25/P21/Q5(c)(i)	18	1	532	
M/J 25/P21/Q5(c)(ii)	16	1	474	
M/J 25/P21/Q5(d)(i)	21	2	576	
M/J 25/P21/Q5(d)(ii)	16	1	475	
M/J 25/P21/Q5(e)(i)-(ii)	18	3	533	
M/J 25/P21/Q5(e)(iii)	16	1	475	
M/J 25/P21/Q6(a)	17	1	499	
M/J 25/P21/Q6(b)	17	2	499	
M/J 25/P21/Q6(c)(i)-(ii)	17	3	500	
M/J 25/P21/Q6(c)(iii)	16	1	476	
M/J 25/P22/Q1(a)	4	3	119	
M/J 25/P22/Q1(b)	9	6	245	
M/J 25/P22/Q1(c)	9	2	245	
M/J 25/P22/Q1(d)	4	5	120	
M/J 25/P22/Q2(a)	2	4	52	
M/J 25/P22/Q2(b)	1	1	11	
M/J 25/P22/Q2(c)	1	1	11	
M/J 25/P22/Q2(d)	1	2	11	
M/J 25/P22/Q2(e)	1	2	11	
M/J 25/P22/Q3(a)(i)	13	1	353	
M/J 25/P22/Q3(a)(ii)-(iii)	14	6	403	
M/J 25/P22/Q3(b)	8	2	220	
M/J 25/P22/Q3(c)	5	2	144	
M/J 25/P22/Q3(d)	8	3	221	ctx (c)
M/J 25/P22/Q3(e)	14	2	403	
M/J 25/P22/Q4(a)	18	2	533	
M/J 25/P22/Q4(b)(i)	16	1	476	
M/J 25/P22/Q4(b)(ii)	15	1	438	
M/J 25/P22/Q4(b)(iii)-(iv)	19	3	554	
M/J 25/P22/Q4(c)(i)	13	1	353	
M/J 25/P22/Q4(c)(ii)-(iii)	16	2	476	
M/J 25/P22/Q4(d)	16	3	477	
M/J 25/P22/Q5(a)	2	1	52	
M/J 25/P22/Q5(b)(i)-(ii)	2	3	53	
M/J 25/P22/Q5(b)(iii)	22	1	599	
M/J 25/P23/Q1(a)	1	3	12	
M/J 25/P23/Q1(b)	1	7	12	
M/J 25/P23/Q2(a)	9	1	245	
M/J 25/P23/Q2(b)	9	1	245	
M/J 25/P23/Q2(c)(i)	9	1	246	
M/J 25/P23/Q2(c)(ii)	4	2	121	
M/J 25/P23/Q2(d)	7	1	188	
M/J 25/P23/Q2(e)	9	2	246	
M/J 25/P23/Q3(a)(i)	14	1	403	

Reference	Unit	Marks	Page	Also / context
M/J 25/P23/Q3(a)(ii)	12	2	328	
M/J 25/P23/Q3(b)(i)	14	1	403	
M/J 25/P23/Q3(b)(ii)-(iii)	8	3	222	
M/J 25/P23/Q4(a)	8	7	222	
M/J 25/P23/Q4(b)	3	4	80	
M/J 25/P23/Q5(a)(i)	2	1	54	
M/J 25/P23/Q5(a)(ii)	21	4	577	
M/J 25/P23/Q5(b)(i)	13	2	353	ctx (a) intro
M/J 25/P23/Q5(b)(ii)-(iv)	18	3	534	ctx (a) intro
M/J 25/P23/Q5(b)(v)	22	2	601	ctx (a) intro, Fig. 5.3
M/J 25/P23/Q5(b)(vi)	20	1	568	ctx (a) intro
M/J 25/P23/Q5(c)	20	1	568	
M/J 25/P23/Q6(a)(i)	13	1	354	
M/J 25/P23/Q6(a)(ii)-(iii)	11	2	303	
M/J 25/P23/Q6(b)	14	4	404	
M/J 25/P23/Q6(c)(i)	15	1	439	
M/J 25/P23/Q6(c)(ii)-(iii)	19	2	555	
M/J 25/P24/Q1(a)	4	3	121	
M/J 25/P24/Q1(b)	9	2	247	
M/J 25/P24/Q1(c)	1	7	12	
M/J 25/P24/Q1(d)	9	2	247	
M/J 25/P24/Q2(a)	3	2	81	
M/J 25/P24/Q2(b)(i)	9	2	247	
M/J 25/P24/Q2(b)(ii)	6	2	168	
M/J 25/P24/Q2(b)(iii)	9	1	248	
M/J 25/P24/Q3(a)	2	3	54	
M/J 25/P24/Q3(b)	7	3	188	
M/J 25/P24/Q3(c)	7	1	189	
M/J 25/P24/Q3(d)(i)	3	1	81	
M/J 25/P24/Q3(d)(ii)-(iii)	11	3	304	ctx (c)
M/J 25/P24/Q4(a)	13	1	354	
M/J 25/P24/Q4(b)	18	1	535	
M/J 25/P24/Q4(c)	18	3	535	
M/J 25/P24/Q4(d)(i)	19	1	555	
M/J 25/P24/Q4(d)(ii)	18	2	536	
M/J 25/P24/Q4(d)(iii)	21	3	578	
M/J 25/P24/Q5(a)-(b)	13	4	354	
M/J 25/P24/Q5(c)	13	1	355	
M/J 25/P24/Q5(d)	14	2	404	
M/J 25/P24/Q6(a)(i)-(ii)	2	3	55	
M/J 25/P24/Q6(a)(iii)	13	1	355	
M/J 25/P24/Q6(b)	4	2	121	
M/J 25/P24/Q6(c)	22	4	603	
MAR 25/P22/Q1(a)	6	8	169	also U7:2, U2:2, U4:2
MAR 25/P22/Q1(b)(i)	9	1	248	
MAR 25/P22/Q1(b)(ii)	3	1	82	
MAR 25/P22/Q1(b)(iii)-(v)	9	3	249	
MAR 25/P22/Q1(c)(i)	13	1	355	
MAR 25/P22/Q1(c)(ii)	15	1	439	
MAR 25/P22/Q1(d)	22	2	604	
MAR 25/P22/Q2(a)	3	2	82	
MAR 25/P22/Q2(b)	1	4	14	
MAR 25/P22/Q2(c)	1	7	15	
MAR 25/P22/Q2(d)	10	4	280	
MAR 25/P22/Q3(a)	11	3	304	
MAR 25/P22/Q3(b)	11	2	305	
MAR 25/P22/Q3(c)	11	3	305	
MAR 25/P22/Q3(d)	14	7	405	also U3:1, U13:1
MAR 25/P22/Q4(a)	21	5	579	
MAR 25/P22/Q4(b)(i)	17	4	500	
MAR 25/P22/Q4(b)(ii)	19	1	555	
MAR 25/P22/Q4(b)(iii)	13	1	355	
O/N 24/P21/Q1(a)	1	3	16	
O/N 24/P21/Q1(b)	1	3	16	
O/N 24/P21/Q1(c)	2	4	55	
O/N 24/P21/Q1(d)(i)	2	2	56	

Reference	Unit	Marks	Page	Also / context
O/N 24/P21/Q1(d)(ii)-(iv)	13	4	356	
O/N 24/P21/Q2(a)	11	4	306	
O/N 24/P21/Q2(b)(i)	13	1	356	
O/N 24/P21/Q2(b)(ii)-(iii)	14	3	406	
O/N 24/P21/Q2(c)(i)	11	1	306	
O/N 24/P21/Q2(c)(ii)-(iii)	6	2	170	
O/N 24/P21/Q2(c)(iv)	15	2	439	
O/N 24/P21/Q3(a)(i)-(ii)	4	4	122	
O/N 24/P21/Q3(a)(iii)	3	1	82	
O/N 24/P21/Q3(b)(i)-(ii)	10	2	281	
O/N 24/P21/Q3(b)(iii)	7	2	189	
O/N 24/P21/Q3(c)	1	3	17	
O/N 24/P21/Q3(d)	3	3	83	
O/N 24/P21/Q3(e)(i)-(ii)	9	2	249	
O/N 24/P21/Q3(e)(iii)	4	2	122	
O/N 24/P21/Q3(f)	9	1	249	
O/N 24/P21/Q4(a)(i)	17	3	501	
O/N 24/P21/Q4(a)(ii)	13	1	357	
O/N 24/P21/Q4(b)	19	1	556	
O/N 24/P21/Q4(c)	17	2	502	
O/N 24/P21/Q4(d)	17	1	502	
O/N 24/P21/Q4(e)	17	1	503	
O/N 24/P21/Q4(f)	22	2	604	
O/N 24/P22/Q1(a)	1	3	18	
O/N 24/P22/Q1(b)(i)	1	2	18	
O/N 24/P22/Q1(b)(ii)-(iii)	2	4	56	ctx Table 1.1
O/N 24/P22/Q2(a)(i)	9	2	250	
O/N 24/P22/Q2(a)(ii)	3	1	83	
O/N 24/P22/Q2(a)(iii)	9	2	250	
O/N 24/P22/Q2(b)(i)	9	2	250	
O/N 24/P22/Q2(b)(ii)	19	1	556	
O/N 24/P22/Q2(b)(iii)	14	2	406	
O/N 24/P22/Q2(b)(iv)	16	2	477	
O/N 24/P22/Q2(c)	3	3	84	
O/N 24/P22/Q3(a)(i)-(iii)	12	4	328	
O/N 24/P22/Q3(a)(iv)-(v)	10	2	281	
O/N 24/P22/Q3(b)(i)-(ii)	6	2	170	
O/N 24/P22/Q3(b)(iii)	12	1	329	
O/N 24/P22/Q3(b)(iv)	3	1	84	
O/N 24/P22/Q3(c)	1	3	19	
O/N 24/P22/Q3(d)	4	2	123	
O/N 24/P22/Q3(e)	4	1	123	
O/N 24/P22/Q3(f)(i)	3	2	84	
O/N 24/P22/Q3(f)(ii)	12	1	329	
O/N 24/P23/Q1(a)	1	3	20	
O/N 24/P23/Q1(b)	1	3	20	
O/N 24/P23/Q1(c)	2	4	57	
O/N 24/P23/Q1(d)(i)	2	2	57	
O/N 24/P23/Q1(d)(ii)-(iv)	13	4	358	
O/N 24/P23/Q2(a)	11	4	306	
O/N 24/P23/Q2(b)(i)	13	1	358	
O/N 24/P23/Q2(b)(ii)-(iii)	14	3	406	
O/N 24/P23/Q2(c)(i)	11	1	306	
O/N 24/P23/Q2(c)(ii)-(iii)	6	2	170	
O/N 24/P23/Q2(c)(iv)	15	2	439	
O/N 24/P23/Q3(a)(i)-(ii)	4	4	123	
O/N 24/P23/Q3(a)(iii)	3	1	85	
O/N 24/P23/Q3(b)(i)-(ii)	10	2	281	
O/N 24/P23/Q3(b)(iii)	7	2	189	
O/N 24/P23/Q3(c)	1	3	21	
O/N 24/P23/Q3(d)	3	3	85	
O/N 24/P23/Q3(e)(i)-(ii)	9	2	250	
O/N 24/P23/Q3(e)(iii)	4	2	124	
O/N 24/P23/Q3(f)	9	1	251	
O/N 24/P23/Q4(a)(i)	17	3	504	
O/N 24/P23/Q4(a)(ii)	13	1	359	

Reference	Unit	Marks	Page	Also / context
O/N 24/P23/Q4(b)	19	1	557	
O/N 24/P23/Q4(c)	17	2	505	
O/N 24/P23/Q4(d)	17	1	505	
O/N 24/P23/Q4(e)	17	1	506	
O/N 24/P23/Q4(f)	22	2	606	
M/J 24/P21/Q1(a)	11	1	307	
M/J 24/P21/Q1(b)	11	3	307	
M/J 24/P21/Q1(c)	11	2	307	
M/J 24/P21/Q1(d)	11	4	307	
M/J 24/P21/Q2(a)	9	2	251	
M/J 24/P21/Q2(b)	3	4	86	
M/J 24/P21/Q2(c)	7	4	190	also U6:1, U10:1
M/J 24/P21/Q2(d)(i)	2	1	58	
M/J 24/P21/Q2(d)(ii)	4	1	124	
M/J 24/P21/Q3(a)	7	2	190	
M/J 24/P21/Q3(b)-(c)	7	7	191	
M/J 24/P21/Q4(a)	5	2	144	
M/J 24/P21/Q4(b)	5	4	145	
M/J 24/P21/Q5(a)	3	1	86	
M/J 24/P21/Q5(b)	13	3	359	
M/J 24/P21/Q5(c)(i)	13	1	359	
M/J 24/P21/Q5(c)(ii)	14	4	407	
M/J 24/P21/Q6(a)	13	5	360	
M/J 24/P21/Q6(b)(i)	17	1	507	ctx (a) intro
M/J 24/P21/Q6(b)(ii)	14	1	408	ctx (a) intro
M/J 24/P21/Q6(c)(i)	22	2	608	
M/J 24/P21/Q6(c)(ii)	17	1	508	
M/J 24/P21/Q6(c)(iii)	16	2	477	
M/J 24/P21/Q6(c)(iv)	22	2	609	
M/J 24/P22/Q1(a)	1	2	21	
M/J 24/P22/Q1(b)	1	3	22	
M/J 24/P22/Q1(c)	3	1	86	
M/J 24/P22/Q1(d)-(e)	9	7	252	
M/J 24/P22/Q1(f)	9	2	253	
M/J 24/P22/Q2(a)	5	2	146	
M/J 24/P22/Q2(b)	5	6	147	
M/J 24/P22/Q2(c)	10	2	281	
M/J 24/P22/Q3(a)	7	2	191	
M/J 24/P22/Q3(b)	7	7	192	
M/J 24/P22/Q3(c)	1	1	22	
M/J 24/P22/Q3(d)	3	2	87	
M/J 24/P22/Q4(a)	13	2	360	
M/J 24/P22/Q4(b)	15	3	440	
M/J 24/P22/Q4(c)	21	3	580	
M/J 24/P22/Q4(d)	15	2	440	
M/J 24/P22/Q4(e)	13	5	361	ctx (d) intro
M/J 24/P23/Q1(a)(i)	12	2	329	
M/J 24/P23/Q1(a)(ii)	3	4	87	
M/J 24/P23/Q1(b)	9	2	253	
M/J 24/P23/Q1(c)	9	3	253	
M/J 24/P23/Q1(d)(i)	6	1	171	
M/J 24/P23/Q1(d)(ii)	10	1	282	
M/J 24/P23/Q2(a)	7	1	193	
M/J 24/P23/Q2(b)(i)	7	1	193	
M/J 24/P23/Q2(b)(ii)	8	2	224	
M/J 24/P23/Q2(c)	7	5	194	
M/J 24/P23/Q2(d)	12	5	330	
M/J 24/P23/Q3(a)	5	2	147	
M/J 24/P23/Q3(b)	5	2	148	
M/J 24/P23/Q3(c)(i)	4	1	124	
M/J 24/P23/Q3(c)(ii)	2	2	58	
M/J 24/P23/Q4(a)	3	2	87	
M/J 24/P23/Q4(b)(i)	7	1	194	
M/J 24/P23/Q4(b)(ii)	12	1	330	
M/J 24/P23/Q4(c)	15	2	440	
M/J 24/P23/Q4(d)	15	2	441	

Reference	Unit	Marks	Page	Also / context
M/J 24/P23/Q4(e)(i)-(ii)	15	4	442	
M/J 24/P23/Q4(e)(iii)	13	1	361	
M/J 24/P23/Q4(f)	15	3	442	
M/J 24/P23/Q5(a)	14	1	409	
M/J 24/P23/Q5(b)(i)	15	2	443	ctx (a)
M/J 24/P23/Q5(b)(ii)	14	1	409	ctx (a)
M/J 24/P23/Q5(c)	22	2	609	ctx (a), (b) intro
M/J 24/P23/Q5(d)(i)-(ii)	19	2	557	ctx (a), (b) intro
M/J 24/P23/Q5(d)(iii)-(iv)	18	2	536	ctx (a), (b) intro
MAR 24/P22/Q1(a)	3	1	88	
MAR 24/P22/Q1(b)(i)	6	1	171	
MAR 24/P22/Q1(b)(ii)	4	2	125	
MAR 24/P22/Q1(c)	6	2	171	
MAR 24/P22/Q1(d)(i)	6	1	172	
MAR 24/P22/Q1(d)(ii)	14	1	410	
MAR 24/P22/Q1(d)(iii)-(iv)	17	5	509	
MAR 24/P22/Q1(e)	2	3	59	
MAR 24/P22/Q2(a)	9	3	254	
MAR 24/P22/Q2(b)	11	5	308	
MAR 24/P22/Q2(c)(i)	3	2	88	
MAR 24/P22/Q2(c)(ii)	9	2	255	
MAR 24/P22/Q2(d)(i)	14	4	411	
MAR 24/P22/Q2(d)(ii)	15	1	444	
MAR 24/P22/Q2(d)(iii)	20	1	569	
MAR 24/P22/Q3(a)	12	2	330	
MAR 24/P22/Q3(b)	12	3	331	
MAR 24/P22/Q3(c)(i)	5	2	148	
MAR 24/P22/Q3(c)(ii)-(iii)	8	3	224	
MAR 24/P22/Q3(d)	3	5	89	
MAR 24/P22/Q4(a)	13	2	362	
MAR 24/P22/Q4(b)(i)	14	1	412	
MAR 24/P22/Q4(b)(ii)	15	1	445	
MAR 24/P22/Q4(b)(iii)	19	1	558	
MAR 24/P22/Q4(c)	22	6	611	also U18:2
O/N 23/P21/Q1(a)	1	7	22	
O/N 23/P21/Q1(b)	1	3	23	
O/N 23/P21/Q1(c)	11	3	309	
O/N 23/P21/Q1(d)(i)	4	2	125	
O/N 23/P21/Q1(d)(ii)	9	1	256	
O/N 23/P21/Q1(d)(iii)	3	2	90	
O/N 23/P21/Q1(e)	5	4	149	
O/N 23/P21/Q2(a)	10	5	282	
O/N 23/P21/Q2(b)	10	2	283	
O/N 23/P21/Q2(c)	18	3	538	
O/N 23/P21/Q3(a)	6	1	172	
O/N 23/P21/Q3(b)	8	2	225	
O/N 23/P21/Q3(c)	6	1	172	
O/N 23/P21/Q3(d)	4	3	126	
O/N 23/P21/Q3(e)(i)-(ii)	16	4	478	
O/N 23/P21/Q3(e)(iii)	13	1	362	
O/N 23/P22/Q1(a)	9	4	256	
O/N 23/P22/Q1(b)	1	9	24	also U9:1, U3:1
O/N 23/P22/Q2(a)	7	1	195	
O/N 23/P22/Q2(b)	7	2	195	
O/N 23/P22/Q2(c)	7	3	196	
O/N 23/P22/Q2(d)	12	2	331	
O/N 23/P22/Q2(e)(i)	13	1	363	
O/N 23/P22/Q2(e)(ii)-(iii)	12	3	332	
O/N 23/P22/Q2(f)	10	3	284	
O/N 23/P22/Q3(a)(i)	9	2	257	
O/N 23/P22/Q3(a)(ii)	7	2	196	
O/N 23/P22/Q3(b)(i)-(ii)	5	4	150	
O/N 23/P22/Q3(b)(iii)	6	2	173	
O/N 23/P22/Q3(c)(i)	14	1	412	
O/N 23/P22/Q3(c)(ii)	16	1	479	
O/N 23/P22/Q3(c)(iii)	18	1	539	

Reference	Unit	Marks	Page	Also / context
O/N 23/P22/Q3(c)(iv)	22	2	613	
O/N 23/P22/Q3(d)(i)	18	1	539	
O/N 23/P22/Q3(d)(ii)	22	3	614	
O/N 23/P22/Q4(a)(i)-(ii)	13	2	363	
O/N 23/P22/Q4(a)(iii)	3	3	91	
O/N 23/P22/Q4(b)(i)	14	3	413	
O/N 23/P22/Q4(b)(ii)	16	1	480	
O/N 23/P22/Q4(b)(iii)	21	4	581	
O/N 23/P23/Q1(a)	1	7	25	
O/N 23/P23/Q1(b)	1	3	26	
O/N 23/P23/Q1(c)	11	3	310	
O/N 23/P23/Q1(d)(i)	4	2	127	
O/N 23/P23/Q1(d)(ii)	9	1	257	
O/N 23/P23/Q1(d)(iii)	3	2	92	
O/N 23/P23/Q1(e)	5	4	151	
O/N 23/P23/Q2(a)	10	5	285	
O/N 23/P23/Q2(b)	10	2	286	
O/N 23/P23/Q2(c)	18	3	540	
O/N 23/P23/Q3(a)	6	1	173	
O/N 23/P23/Q3(b)	8	2	225	
O/N 23/P23/Q3(c)	6	1	173	
O/N 23/P23/Q3(d)	4	3	128	
O/N 23/P23/Q3(e)(i)-(ii)	16	4	481	
O/N 23/P23/Q3(e)(iii)	13	1	364	
M/J 23/P21/Q1(a)	1	3	26	
M/J 23/P21/Q1(b)	1	1	26	
M/J 23/P21/Q1(c)	1	1	27	
M/J 23/P21/Q1(d)	1	4	27	
M/J 23/P21/Q1(e)	3	2	93	
M/J 23/P21/Q1(f)	9	2	258	
M/J 23/P21/Q2(a)	5	2	151	
M/J 23/P21/Q2(b)(i)	2	1	59	
M/J 23/P21/Q2(b)(ii)	5	3	152	
M/J 23/P21/Q2(c)(i)	10	2	286	
M/J 23/P21/Q2(c)(ii)	5	1	152	
M/J 23/P21/Q3(a)	9	2	258	
M/J 23/P21/Q3(b)	9	4	258	
M/J 23/P21/Q3(c)	11	2	310	
M/J 23/P21/Q3(d)	14	4	414	
M/J 23/P21/Q3(e)	15	1	445	
M/J 23/P21/Q3(f)	22	2	615	
M/J 23/P21/Q4(a)	17	2	510	
M/J 23/P21/Q4(b)(i)	16	1	482	
M/J 23/P21/Q4(b)(ii)	2	1	60	
M/J 23/P21/Q4(b)(iii)	13	2	364	
M/J 23/P21/Q4(c)	16	2	483	
M/J 23/P21/Q5(a)	14	4	414	
M/J 23/P21/Q5(b)	12	6	332	ctx (a) intro
M/J 23/P21/Q5(c)	18	5	541	
M/J 23/P22/Q1(a)(i)-(ii)	4	2	128	
M/J 23/P22/Q1(a)(iii)	9	2	259	
M/J 23/P22/Q1(b)	3	8	94	
M/J 23/P22/Q2(a)	9	2	259	
M/J 23/P22/Q2(b)	9	2	259	
M/J 23/P22/Q2(c)	9	2	260	
M/J 23/P22/Q2(d)	11	2	311	
M/J 23/P22/Q2(e)(i)	11	1	311	
M/J 23/P22/Q2(e)(ii)	6	1	174	
M/J 23/P22/Q2(e)(iii)	2	3	60	
M/J 23/P22/Q3(a)(i)	15	1	445	
M/J 23/P22/Q3(a)(ii)	13	1	365	
M/J 23/P22/Q3(a)(iii)	19	1	559	
M/J 23/P22/Q3(a)(iv)	18	2	542	
M/J 23/P22/Q3(b)	16	1	483	
M/J 23/P22/Q3(c)	13	2	365	
M/J 23/P22/Q4(a)	5	2	153	

Reference	Unit	Marks	Page	Also / context
M/J 23/P22/Q4(b)	7	2	197	
M/J 23/P22/Q4(c)(i)	7	1	197	
M/J 23/P22/Q4(c)(ii)-(iii)	14	3	415	
M/J 23/P22/Q4(c)(iv)-(v)	8	3	226	ctx Fig. 4.2
M/J 23/P22/Q4(c)(vi)	14	2	416	
M/J 23/P22/Q4(d)	13	2	366	
M/J 23/P22/Q5(a)	13	1	366	
M/J 23/P22/Q5(b)	17	3	511	
M/J 23/P22/Q5(c)	13	3	367	
M/J 23/P22/Q5(d)(i)-(iii)	18	3	542	
M/J 23/P22/Q5(d)(iv)	22	2	615	ctx (c) intro
M/J 23/P23/Q1(a)(i)	4	1	129	
M/J 23/P23/Q1(a)(ii)	3	1	94	
M/J 23/P23/Q1(b)	2	3	61	
M/J 23/P23/Q1(c)	22	1	617	
M/J 23/P23/Q1(d)(i)-(ii)	6	2	174	
M/J 23/P23/Q1(d)(iii)	1	1	27	
M/J 23/P23/Q2(a)	9	4	260	
M/J 23/P23/Q2(b)	10	1	287	
M/J 23/P23/Q2(c)	9	5	261	
M/J 23/P23/Q3(a)	5	2	153	
M/J 23/P23/Q3(b)(i)	2	1	61	
M/J 23/P23/Q3(b)(ii)	5	3	154	
M/J 23/P23/Q3(c)(i)	18	1	543	
M/J 23/P23/Q3(c)(ii)	5	1	154	
M/J 23/P23/Q4(a)	11	3	311	
M/J 23/P23/Q4(b)	7	5	198	
M/J 23/P23/Q4(c)	7	2	198	
M/J 23/P23/Q5(a)	2	1	62	
M/J 23/P23/Q5(b)	17	2	512	
M/J 23/P23/Q5(c)	13	2	368	
M/J 23/P23/Q5(d)	17	3	513	
M/J 23/P23/Q6(a)	20	1	569	
M/J 23/P23/Q6(b)(i)-(ii)	17	4	514	
M/J 23/P23/Q6(b)(iii)	16	1	484	
M/J 23/P23/Q6(b)(iv)	22	1	617	
M/J 23/P23/Q6(c)	13	3	368	
M/J 23/P23/Q6(d)	13	1	369	
M/J 23/P23/Q6(e)	19	1	559	
M/J 23/P23/Q6(f)(i)	16	1	484	ctx (b) intro
M/J 23/P23/Q6(f)(ii)	19	2	559	ctx (b) intro
MAR 23/P22/Q1(a)	3	3	95	
MAR 23/P22/Q1(b)	3	3	95	
MAR 23/P22/Q1(c)	1	3	28	
MAR 23/P22/Q1(d)(i)-(ii)	9	2	261	
MAR 23/P22/Q1(d)(iii)	10	1	287	
MAR 23/P22/Q1(e)(i)	3	2	96	
MAR 23/P22/Q1(e)(ii)	4	3	129	ctx Table 1.2
MAR 23/P22/Q2(a)	3	3	96	
MAR 23/P22/Q2(b)	10	1	287	
MAR 23/P22/Q2(c)	10	2	287	
MAR 23/P22/Q2(d)(i)	11	2	311	
MAR 23/P22/Q2(d)(ii)	2	3	62	
MAR 23/P22/Q3(a)(i)-(iii)	13	3	369	
MAR 23/P22/Q3(a)(iv)	2	1	62	
MAR 23/P22/Q3(b)	17	9	514	also U13:1
MAR 23/P22/Q3(c)	13	2	370	
MAR 23/P22/Q3(d)(i)	8	2	227	
MAR 23/P22/Q3(d)(ii)	7	1	199	
MAR 23/P22/Q3(d)(iii)	5	2	155	
MAR 23/P22/Q3(d)(iv)	3	1	97	
MAR 23/P22/Q4(a)(i)	15	1	446	
MAR 23/P22/Q4(a)(ii)	20	1	570	
MAR 23/P22/Q4(a)(iii)	16	1	485	
MAR 23/P22/Q4(b)(i)	15	1	447	
MAR 23/P22/Q4(b)(ii)	16	2	486	

Reference	Unit	Marks	Page	Also / context
MAR 23/P22/Q4(c)(i)	15	1	448	
MAR 23/P22/Q4(c)(ii)	19	1	560	
MAR 23/P22/Q4(d)	22	3	619	
O/N 22/P21/Q1(a)(i)	1	2	28	
O/N 22/P21/Q1(a)(ii)-(iii)	2	4	63	
O/N 22/P21/Q1(b)	1	4	29	
O/N 22/P21/Q2(a)	3	2	97	
O/N 22/P21/Q2(b)	10	4	288	
O/N 22/P21/Q2(c)	11	2	311	
O/N 22/P21/Q2(d)	11	1	312	
O/N 22/P21/Q3(a)	9	4	262	
O/N 22/P21/Q3(b)	11	1	312	
O/N 22/P21/Q3(c)	16	2	486	
O/N 22/P21/Q3(d)	3	11	97	also U4:2, U5:2, U7:1, U13:2
O/N 22/P21/Q5(a)(i)	14	1	417	
O/N 22/P21/Q5(a)(ii)-(iii)	17	3	516	
O/N 22/P21/Q5(a)(iv)	18	1	543	
O/N 22/P21/Q5(b)	22	2	621	
O/N 22/P21/Q5(c)	22	2	622	
O/N 22/P22/Q1(a)	3	2	98	
O/N 22/P22/Q1(b)	1	3	29	
O/N 22/P22/Q1(c)(i)	7	1	199	
O/N 22/P22/Q1(c)(ii)	12	1	333	
O/N 22/P22/Q1(c)(iii)	3	2	99	
O/N 22/P22/Q1(d)	2	3	63	
O/N 22/P22/Q1(e)	15	5	449	
O/N 22/P22/Q2(a)	9	4	262	
O/N 22/P22/Q2(b)	9	3	263	
O/N 22/P22/Q2(c)(i)	3	2	100	
O/N 22/P22/Q2(c)(ii)	7	1	200	
O/N 22/P22/Q2(d)	4	3	130	
O/N 22/P22/Q2(e)(i)	11	2	312	
O/N 22/P22/Q2(e)(ii)	3	2	100	
O/N 22/P22/Q2(f)(i)	3	2	101	
O/N 22/P22/Q2(f)(ii)	4	2	130	
O/N 22/P22/Q3(a)(i)	21	8	582	
O/N 22/P22/Q3(a)(ii)-(iv)	13	3	371	
O/N 22/P22/Q3(b)	20	3	570	
O/N 22/P22/Q3(c)	22	5	622	also U14:1
O/N 22/P22/Q3(d)	21	3	583	
O/N 22/P23/Q1(a)(i)	1	2	30	
O/N 22/P23/Q1(a)(ii)-(iii)	2	4	64	
O/N 22/P23/Q1(b)	1	4	30	
O/N 22/P23/Q2(a)	3	2	101	
O/N 22/P23/Q2(b)	10	4	289	
O/N 22/P23/Q2(c)	11	2	312	
O/N 22/P23/Q2(d)	11	1	313	
O/N 22/P23/Q3(a)	9	4	263	
O/N 22/P23/Q3(b)	11	1	313	
O/N 22/P23/Q3(c)	16	2	487	
O/N 22/P23/Q3(d)	3	11	101	also U4:2, U5:2, U7:1, U13:2
O/N 22/P23/Q5(a)(i)	14	1	418	
O/N 22/P23/Q5(a)(ii)-(iii)	17	3	517	
O/N 22/P23/Q5(a)(iv)	18	1	544	
O/N 22/P23/Q5(b)	22	2	625	
O/N 22/P23/Q5(c)	22	2	626	
M/J 22/P21/Q1(a)	1	1	31	
M/J 22/P21/Q1(b)	10	2	289	
M/J 22/P21/Q1(c)	10	1	290	
M/J 22/P21/Q1(d)	1	2	31	
M/J 22/P21/Q1(e)(i)-(ii)	2	3	64	
M/J 22/P21/Q1(e)(iii)	1	2	31	
M/J 22/P21/Q1(f)(i)	9	1	264	
M/J 22/P21/Q1(f)(ii)	5	3	156	
M/J 22/P21/Q1(g)	8	2	227	
M/J 22/P21/Q2(a)	3	3	102	

Reference	Unit	Marks	Page	Also / context
M/J 22/P21/Q2(b)	12	3	333	
M/J 22/P21/Q2(c)	12	1	333	
M/J 22/P21/Q2(d)	7	5	201	
M/J 22/P21/Q3(a)	13	2	372	
M/J 22/P21/Q3(b)	14	2	419	
M/J 22/P21/Q3(c)(i)	17	1	518	
M/J 22/P21/Q3(c)(ii)-(iii)	14	5	420	
M/J 22/P21/Q3(d)(i)-(ii)	16	5	487	ctx (c) intro
M/J 22/P21/Q3(d)(iii)	9	2	265	ctx (c) intro
M/J 22/P21/Q4(a)	2	3	65	
M/J 22/P21/Q4(b)(i)	18	2	544	
M/J 22/P21/Q4(b)(ii)	2	1	65	
M/J 22/P21/Q4(c)(i)-(ii)	17	2	519	
M/J 22/P21/Q4(c)(iii)	13	3	373	
M/J 22/P21/Q4(d)	22	3	627	
M/J 22/P22/Q1(a)	4	2	131	
M/J 22/P22/Q1(b)(i)-(ii)	6	2	174	
M/J 22/P22/Q1(b)(iii)-(iv)	5	5	156	
M/J 22/P22/Q2(a)	4	1	131	
M/J 22/P22/Q2(b)	3	1	102	
M/J 22/P22/Q2(c)	10	6	290	
M/J 22/P22/Q2(d)	11	3	313	
M/J 22/P22/Q3(a)	7	6	202	
M/J 22/P22/Q3(b)	5	2	157	ctx (a) intro
M/J 22/P22/Q3(c)	7	1	202	
M/J 22/P22/Q3(d)	11	3	314	
M/J 22/P22/Q4(a)	13	3	374	
M/J 22/P22/Q4(b)(i)-(ii)	13	2	374	
M/J 22/P22/Q4(b)(iii)	1	1	32	
M/J 22/P22/Q4(c)	20	2	571	
M/J 22/P22/Q5(a)	15	6	450	
M/J 22/P22/Q5(b)	15	2	451	
M/J 22/P22/Q5(c)(i)	15	2	452	
M/J 22/P22/Q5(c)(ii)	19	1	561	
M/J 22/P22/Q5(c)(iii)-(iv)	18	2	545	
M/J 22/P22/Q6(a)	18	2	545	
M/J 22/P22/Q6(b)	2	1	65	
M/J 22/P22/Q6(c)(i)-(ii)	22	3	628	
M/J 22/P22/Q6(c)(iii)	18	1	546	ctx (b), Table 6.1
M/J 22/P23/Q1(a)	1	2	32	
M/J 22/P23/Q1(b)	1	1	32	
M/J 22/P23/Q1(c)	1	2	32	
M/J 22/P23/Q1(d)	1	2	33	
M/J 22/P23/Q1(e)	1	3	33	
M/J 22/P23/Q1(f)(i)	2	1	66	
M/J 22/P23/Q1(f)(ii)	4	2	131	
M/J 22/P23/Q2(a)	9	6	266	
M/J 22/P23/Q2(b)	11	4	314	
M/J 22/P23/Q3(a)(i)	3	1	103	
M/J 22/P23/Q3(a)(ii)	5	2	157	
M/J 22/P23/Q3(a)(iii)	7	2	203	
M/J 22/P23/Q3(a)(iv)-(v)	8	4	228	
M/J 22/P23/Q3(b)	13	5	374	
M/J 22/P23/Q3(c)	17	2	520	
M/J 22/P23/Q4(a)	14	5	421	
M/J 22/P23/Q4(b)(i)-(iii)	15	4	453	
M/J 22/P23/Q4(b)(iv)	16	1	488	
M/J 22/P23/Q4(b)(v)	15	2	454	
M/J 22/P23/Q5(a)(i)	14	1	421	
M/J 22/P23/Q5(a)(ii)	16	3	489	
M/J 22/P23/Q5(b)	22	5	629	
MAR 22/P22/Q1(a)	1	1	33	
MAR 22/P22/Q1(b)	1	5	34	
MAR 22/P22/Q1(c)	1	3	35	
MAR 22/P22/Q2(a)(i)	6	1	175	
MAR 22/P22/Q2(a)(ii)	9	1	266	

Reference	Unit	Marks	Page	Also / context
MAR 22/P22/Q2(b)(i)-(ii)	9	2	267	
MAR 22/P22/Q2(b)(iii)	8	3	229	
MAR 22/P22/Q2(c)(i)	4	2	131	
MAR 22/P22/Q2(c)(ii)	6	1	175	
MAR 22/P22/Q2(c)(iii)	5	1	158	
MAR 22/P22/Q2(c)(iv)	9	1	267	
MAR 22/P22/Q2(d)(i)-(ii)	12	2	334	
MAR 22/P22/Q2(d)(iii)	7	2	204	
MAR 22/P22/Q3(a)	11	2	315	
MAR 22/P22/Q3(b)(i)	9	1	267	
MAR 22/P22/Q3(b)(ii)	11	3	315	
MAR 22/P22/Q3(c)	11	1	316	
MAR 22/P22/Q3(d)	7	5	205	
MAR 22/P22/Q3(e)(i)-(ii)	14	6	422	
MAR 22/P22/Q3(e)(iii)	15	2	454	
MAR 22/P22/Q4(a)(i)	17	3	521	
MAR 22/P22/Q4(a)(ii)	13	2	376	
MAR 22/P22/Q4(b)(i)	18	2	547	
MAR 22/P22/Q4(b)(ii)	20	2	571	
MAR 22/P22/Q4(b)(iii)	22	1	630	
MAR 22/P22/Q4(b)(iv)	21	5	584	

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Group

Group																							
1	2											13	14	15	16	17	18						
						1 H hydrogen 1.0												2 He helium 4.0					
		Key atomic number atomic symbol name relative atomic mass																					
3 Li lithium 6.9	4 Be beryllium 9.0											5 B boron 10.8	6 C carbon 12.0	7 N nitrogen 14.0	8 O oxygen 16.0	9 F fluorine 19.0	10 Ne neon 20.2						
11 Na sodium 23.0	12 Mg magnesium 24.3	3	4	5	6	7	8	9	10	11	12	13 Al aluminium 27.0	14 Si silicon 28.1	15 P phosphorus 31.0	16 S sulfur 32.1	17 Cl chlorine 35.5	18 Ar argon 39.9						
19 K potassium 39.1	20 Ca calcium 40.1	21 Sc scandium 45.0	22 Ti titanium 47.9	23 V vanadium 50.9	24 Cr chromium 52.0	25 Mn manganese 54.9	26 Fe iron 55.8	27 Co cobalt 58.9	28 Ni nickel 58.7	29 Cu copper 63.5	30 Zn zinc 65.4	31 Ga gallium 69.7	32 Ge germanium 72.6	33 As arsenic 74.9	34 Se selenium 79.0	35 Br bromine 79.9	36 Kr krypton 83.8						
37 Rb rubidium 85.5	38 Sr strontium 87.6	39 Y yttrium 88.9	40 Zr zirconium 91.2	41 Nb niobium 92.9	42 Mo molybdenum 95.9	43 Tc technetium —	44 Ru ruthenium 101.1	45 Rh rhodium 102.9	46 Pd palladium 106.4	47 Ag silver 107.9	48 Cd cadmium 112.4	49 In indium 114.8	50 Sn tin 118.7	51 Sb antimony 121.8	52 Te tellurium 127.6	53 I iodine 126.9	54 Xe xenon 131.3						
55 Cs caesium 132.9	56 Ba barium 137.3	57–71 lanthanoids	72 Hf hafnium 178.5	73 Ta tantalum 180.9	74 W tungsten 183.8	75 Re rhenium 186.2	76 Os osmium 190.2	77 Ir iridium 192.2	78 Pt platinum 195.1	79 Au gold 197.0	80 Hg mercury 200.6	81 Tl thallium 204.4	82 Pb lead 207.2	83 Bi bismuth 209.0	84 Po polonium —	85 At astatine —	86 Rn radon —						
87 Fr francium	88 Ra radium	89–103 actinoids	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Nh nihonium	114 Fl flerovium	115 Mc moscovium	116 Lv livermorium	117 Ts tennessine	118 Og oganeson						

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actinoids

57 La lanthanum 138.9	58 Ce cerium 140.9	59 Pr praseodymium 140.9	60 Nd neodymium 144.2	61 Pm promethium	62 Sm samarium 150.4	63 Eu europium 152.0	64 Gd gadolinium 157.3	65 Tb terbium 158.9	66 Dy dysprosium 162.5	67 Ho holmium 164.9	68 Er erbium 167.3	69 Tm thulium 168.9	70 Yb ytterbium 173.1	71 Lu lutetium 175.0
89 Ac actinium 227.0	90 Th thorium 232.0	91 Pa protactinium 231.0	92 U uranium 238.0	93 Np neptunium	94 Pu plutonium	95 Am americium	96 Cm curium	97 Bk berkelium	98 Cf californium	99 Es einsteinium	100 Fm fermium	101 Md mendelevium	102 No nobelium	103 Lr lawrencium

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